THE QUANTUM RULES

What They Allow Us to Know, Communicate, and Compute

Revised edition · v1.2

A note to the reader

How to read this book

The book begins with an experiment and develops the mathematics where it becomes useful. Follow the main argument in order on a first reading. At each stage, ask what was prepared, what the description assumes, what the calculation predicts, and what observations can establish. The recurring examples acquire new meanings as preparation, interference, joint states, computation, and physical implementation become connected. A useful test of understanding is to predict what would change if one of the stated conditions changed.

The order follows the ideas we need, rather than the chronology of discovery. Observations can motivate hypotheses; established theories can also predict phenomena before they are observed. A calculation establishes consequences of stated assumptions. Experimental evidence establishes how well those assumptions and consequences describe the physical world. Quantum descriptions must remain accountable to that evidence. Later demonstrations often help us examine rules developed earlier.

Equations and illustrations

An equation states a relation under the conditions named in the surrounding text. A circuit wire identifies a register through operations, not a particle track. Figures distinguish ideal predictions and constructed examples from reported experimental data. The explanatory captions are part of the argument. The notation pages that follow are a reference to consult as terms arise; their concepts are developed in the main text.

Notes and references

Each chapter, part opening, prologue and epilogue has its own numbered notes. These distinguish theory, estimates and experiment, with source versions identified through 16 October 2026. The accompanying v1.2 Revision Supplement provides figure data, editable sources, plotting programs and verification calculations. Its README.html contains instructions, an equation-and-figure map, and a record distinguishing checked calculations, reported experimental estimates and material not independently reproduced.

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Notation and reading conventions

States and outcomes

A ket such as $|0\rangle$ names a reference state, not the absence of a physical carrier. An amplitude is a complex number; its squared magnitude contributes to a measurement probability. A density matrix describes pure or mixed states, including a system considered without its partner.

Operators and order

H can name a Hadamard gate or a Hamiltonian; the local definition specifies the role. X, Y, and Z can denote gates or measured observables. Diagrams run left to right, while the rightmost operator in a product acts first. A dagger denotes an adjoint, which is the inverse for a unitary.

Notation and reading conventions, continued

Probabilities and information

The entropy of a state is not generally the entropy of one measurement record. Conditioning on a reported result is not the same as averaging over all results. A successful detection, a correct retained output, and a successful attempt have different denominators.

Physical scales

Angular frequency and cycles per second differ by a factor of two pi. Molar and per-particle energies must not be interchanged. Every precision statement needs its units, error convention, and resource count. Model assumptions and sampling uncertainties are not interchangeable.

PROLOGUE

The Detector That Stayed Dark

A light detector can receive a photon by either of two available routes. Close one route without changing the source. Before calculating anything, make a prediction: could that detector now respond more often?

A simple particle picture suggests that it cannot: blocking a route removes possible arrivals rather than adding them. Keep that prediction in mind. It assumes that each particle of light selects a route and continues independently of whatever happens along the other one.

Let us build an arrangement in which we can test that expectation.

Imagine a source that prepares a single photon—a quantum of light—on each trial. Ahead of it is a partly reflecting surface called a beam splitter. Light approaching this surface has two possible ways onward: through it or away from it by reflection. Mirrors guide these two routes toward a second beam splitter, where they meet again. Beyond that second surface are two detectors, which we will call A and B. This split-and-recombine arrangement is an interferometer.[1]

There is no need to memorize the name. Keep the arrangement in mind: a source, two routes, a place where they recombine, and two possible destinations.

For the moment, give us ideal equipment. The source prepares exactly one photon at a time. The optical components introduce no unwanted losses. The detectors register every photon that reaches them. These assumptions let us examine a precise prediction before dealing with the complications of building the apparatus.

We choose beam splitters that give equal probabilities for their two outputs when a photon arrives along just one input. We then adjust the two routes so that their contributions reinforce one another at A and cancel at B. Nothing records which route was taken, and the contributions arrive matched in their other physical properties.

With both routes open, the prediction is definite: every photon reaches A. None reaches B. Figure P.1 shows the arrangement.

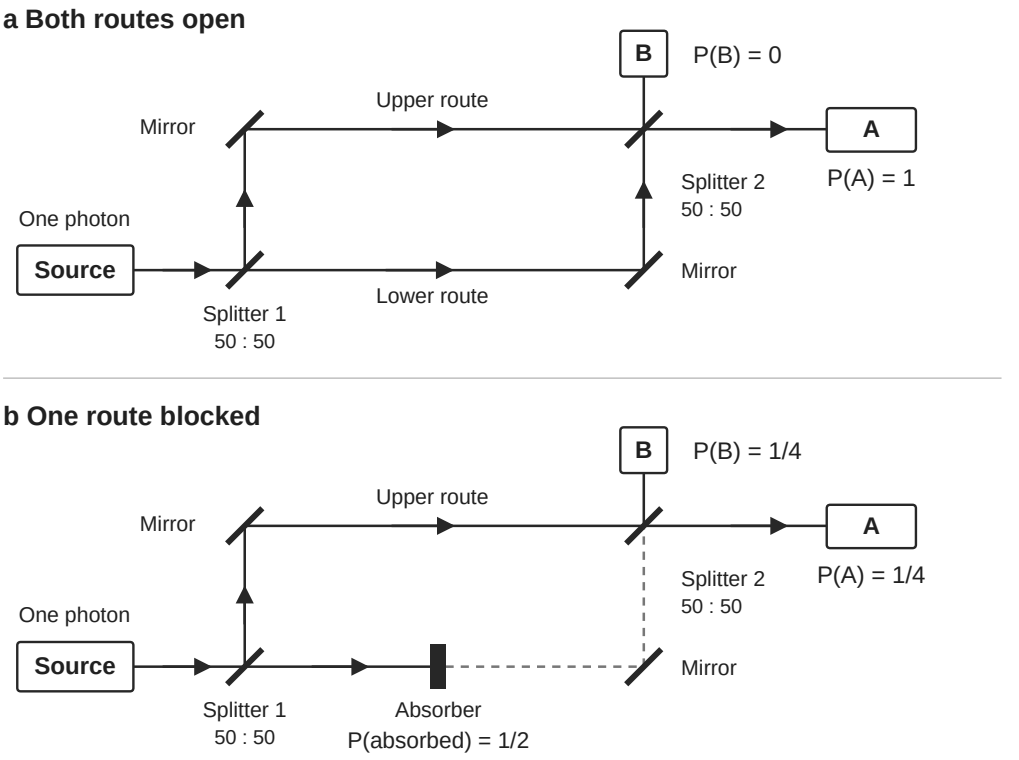


Figure P.1. Blocking one route can brighten a dark output. (a) Both routes are open and adjusted so that all photons are detected at A. (b) An absorber blocks the lower route. Half the emitted photons are absorbed on average; one-quarter reach each detector. All probabilities are per photon emitted by the source, not conditional on survival. Dashed segments mark the blocked continuation of a route. Lines show optical routes, not measured photon tracks. The apparatus is idealized and not drawn to scale.

Now place an absorbing stop in the lower route.

Each photon has a one-half probability of being absorbed. A photon that survives reaches the second beam splitter and has equal chances of being detected at A or B. Per photon emitted by the source, the probabilities are therefore one-half for absorption, one-quarter for detection at A, and one-quarter for detection at B.

B has gone from receiving no photons to receiving, on average, one in four.

There is less light reaching the detectors altogether. Nothing has been created. But the particular detector that previously received nothing now receives some.

To understand this result, we need to examine how we calculated the chances.

An ordinary probability is never negative. A one-quarter chance of arrival by one route cannot cancel a one-quarter chance of arrival by another. Simply adding those two probabilities would give one-half, not zero.

Quantum theory does not tell us to subtract probabilities. It tells us that, when the two routes contribute coherently to the same outcome, we must first combine their amplitudes

and only then calculate the probability. Amplitudes can reinforce or cancel; the resulting probabilities remain nonnegative.[2]

For our carefully adjusted apparatus, we can represent the two contributions to B by $+½$ and $-½$. These are not two chances. They are the ingredients from which the chance is calculated. Their signs include the effects of reflection and propagation along the routes.

Let $P(B)$ mean the probability that a photon emitted by the source is detected at B. With both routes open, we add the amplitudes and then square the magnitude of their sum:

$$P(B)=\left|\frac{1}{2}-\frac{1}{2}\right|^2=0 \tag{P.1}$$

With one route blocked, only one contribution remains:

$$P(B)=\left|\frac{1}{2}\right|^2=\frac{1}{4} \tag{P.2}$$

The vertical bars mean magnitude. Squaring that magnitude gives the probability. For these real numbers, this is just ordinary squaring after taking the absolute value.

That is the origin of the one-in-four prediction. The relative minus sign in the first calculation represents a difference in phase: a relationship between the contributions that determines how they combine. More generally, amplitudes are complex numbers, which can be pictured as arrows with different lengths and directions. The rule remains the same: combine the amplitudes, then take the squared magnitude.

We will develop that rule carefully later. Already, however, it has done something useful. It has turned a surprising statement into a calculation we can check.

There is an important qualification.

Ordinary waves also interfere. Two water waves can reinforce each other where their crests coincide and oppose each other where a crest meets a trough. The bright and dark outputs of an optical interferometer can likewise be explained using classical wave optics. A dark detector, by itself, is not proof that light is quantum.[3]

The quantum problem is more demanding. Our source prepares one photon at a time, and ideal photodetectors register individual absorption events. One photon is detected at one output, not as half a photon at each. Yet the probabilities of these individual events depend on the same kind of reinforcement and cancellation that we associate with waves. A satisfactory description must account for both features.[4]

The tempting picture of a tiny pellet making an independent random choice at each beam splitter does not do that.

Nor does saying “light is strange” help us predict anything.

What helps is keeping track of something that an ordinary list of chances leaves out. Two preparations can give identical probabilities for one measurement and still respond differently to the same subsequent operation. Their amplitudes contain a relationship that becomes visible when the contributions are brought together.[5]

This is where our experiment begins to lead beyond a puzzle about light.

So far, we have asked what the apparatus does. We can also ask what we could make it do. Could we deliberately change the relationship between the two routes? Could the detector results reveal that change? Could we arrange a sequence of operations so that the final result answers a question we care about?

That progression—from preparing a state, through controlled transformations, to a measured outcome—is also the basic organization of a quantum circuit. The optical experiment gives us a physical example of ideas that can be expressed as a small quantum program.[6]

It is not, on its own, evidence of a computational speed advantage. We have just calculated its behavior with a few lines of arithmetic. Understanding when quantum operations provide a useful advantage will require considerably more than showing that interference occurs.

But we now have a starting point.

We can ask what happens when two quantum systems interact. What information can be obtained from one of them? What can be learned only by comparing them? Can an unknown state be copied? How might a calculation exploit these rules, and how could it survive the disturbances of a real machine?

We can also turn back toward matter. What do these rules tell us about an atom, a chemical bond, or a reaction? Which parts of those problems can we already calculate well, and which remain difficult?

These questions will take us through physics, chemistry, mathematics, and computation. We will introduce each subject where it helps resolve a problem already in view.

For now, consider the detector.

A click is a physical event. It is also a record from which someone can learn something. In our experiment, even the absence of clicks becomes informative—but only after we understand the source, the apparatus, and the alternatives.

Before asking what makes information quantum, we should begin with a more familiar question.

What must happen in the physical world for there to be information at all?

Notes and sources

1. Edmund Optics. *“Building a Mach–Zehnder Interferometer.”* The figure in this prologue is an original schematic; its probabilities are derived for the idealized conditions stated in the text.

2. Richard P. Feynman, Robert B. Leighton, and Matthew Sands. *The Feynman Lectures on Physics, Vol. III, Chapter 3, “Probability Amplitudes.”* See the rules for combining amplitudes for alternative and successive events.

3. Feynman, Leighton, and Sands. *The Feynman Lectures on Physics, Vol. III, Chapter 1, “Quantum Behavior,” §§1–3 to 1–5.*

4. V. Jacques, E. Wu, T. Toury, F. Treussart, A. Aspect, P. Grangier, and J.-F. Roch. *“Single photon wavefront-splitting interference: An illustration of the light quantum in action.”* European Physical Journal D 35, 561–565 (2005), doi:10.1140/epjd/e2005-00201-y; archived as arXiv:2011.12664 in 2020. Reports single-photon interference and a separate detection-anticorrelation test with a Fresnel biprism, not the book’s ideal two-beam-splitter apparatus.

5. IBM Quantum Learning. *Basics of Quantum Information, “Single systems: Quantum information.”*

6. IBM Quantum Learning. *Basics of Quantum Information, “Quantum circuits: Circuits.”*

PART I

A New Way to Describe the World

The detector in the prologue stayed dark until one route was blocked. We could calculate the result, but a calculation is only the beginning of an explanation. What did its numbers describe? Which details of the apparatus made them the right numbers? And what would change if we asked a different question?

Those questions will guide our first steps. We will begin not with an unfamiliar particle but with a familiar message: a distinction carried from one place to another. A mark on paper, a voltage, and a pulse of light can all carry a message. To understand any of them, we have to connect what was prepared with what can be read. That connection gives information a physical setting.

The interferometer will then make the setting less familiar. We will compare preparations that agree in one test and disagree in another. We will learn why a list of ordinary chances can leave out something essential, and why numbers with a direction can restore it. The mathematics will enter where it helps us predict a result we already care about.

The destination is the qubit. Rather than begin with its name and a collection of claims about its power, we will assemble its description from experiments, amplitudes, and choices of measurement. By the time we draw its state space, the drawing should have a meaning: each direction will connect to a question that an apparatus can ask.

Our first task is to learn what a quantum description contains—and what one observation does not reveal. The distinction will matter later when we try to build a computer, understand a molecule, or send information to someone else. Before using the rules, we need to know what their statements mean.

CHAPTER 1

A Message Must Live Somewhere

A friend is waiting for an answer. Will you bring the book?

You have decided that you will. You could write “yes” on a slip of paper. You could send an electrical signal along a wire. Or, with a lamp and a clear view across a courtyard, you could send a flash of light. The paper, the wire, and the light have very different physical properties. Yet each could carry the same answer.

Now imagine that the answer travels through all three. You mark the paper. Someone reads it and operates a switch. At the far end of a wire, a circuit turns on a lamp. Your friend sees the flash and writes “yes” in a notebook. We can follow the message from one place to another, but we will not find a particular speck of matter that has made the entire journey. The ink has stayed on the first sheet. The electrical signal has been converted into light. The light has been detected, and a new mark has been made.

What, then, has arrived?

Not the original ink. Not a miniature word concealed inside the wire. What has arrived is an answer that can be recovered from a different physical arrangement. The important achievement is that the final record distinguishes the same alternative that you selected at the beginning.

There is a condition hidden in that description. Your friend must know what the flash means. You could agree that a flash in the first of two time slots means no, while a flash in the second means yes. You must also agree when those slots begin. Without that arrangement, the flash might be a greeting, a test of the lamp, or an accident. A physical event does not come with an explanation attached.

This modest exchange contains the beginnings of information science. There are alternatives, a way of representing them, a process that carries the representation, and a way of reading the result. Each step can succeed or fail. Each step occurs in the physical world. The same abstract message can use different carriers; an actual transmission cannot dispense with a carrier altogether.[1]

Quantum information begins by taking that last fact seriously. Before we ask what quantum physics changes, we should see how much is already required to send an ordinary yes.

What survives the journey

Suppose the slip of paper says “yes” in blue ink, while your friend writes the answer in black. We normally call this a faithful copy. We do not demand the same color, handwriting, paper, or arrangement of individual molecules. We require agreement about the answer.

Change the task, however, and the verdict changes. An archivist trying to reproduce the original document may care about the color. Someone examining a signature may care about the shape of each stroke. What counts as successful communication depends on which distinctions the receiver needs to recover. For our exchange, the relevant distinction is between yes and no. Other features of the carrier can vary without changing that answer.

Let us make the agreement explicit. We will use the label 0 for no and the label 1 for yes. Those labels do not say how the answer must be embodied. A card might have two boxes, with a mark in the left box for 0 and a mark in the right for 1. A circuit might use a low voltage for 0 and a high voltage for 1. Our optical signal might use an early flash for 0 and a late flash for 1. Figure 1.1 shows these three arrangements.

The rules that assign representations to alternatives constitute a code. In this sense, a code need not be secret. It is simply an agreement that makes a physical difference interpretable. We could reverse all our assignments and communicate just as successfully, provided that the sender and receiver reversed them together.

Notice what that reversal would not do. It would not alter the chemistry of the ink or the way light propagates. The meaning assigned to a signal is conventional; the behavior of the signal is not something we can change by agreement. Calling the late flash 0 rather than 1 does not make it arrive earlier.

This separation was central to Claude Shannon’s 1948 paper, “A Mathematical Theory of Communication.” Shannon treated the engineering problem of recovering a message selected from a set of possible messages, setting aside what the message meant to its recipient. That was not a claim that meaning is unimportant. It made a particular problem precise enough to analyze.[2]

Our friend may be delighted that the book is coming. A receiving circuit need not share that delight, or understand books, to distinguish the two signals. We can therefore ask how reliably the apparatus works before asking why the answer matters. Conversely, a perfectly transmitted answer can still be mistaken or dishonest. Faithful transmission and truth are different achievements.

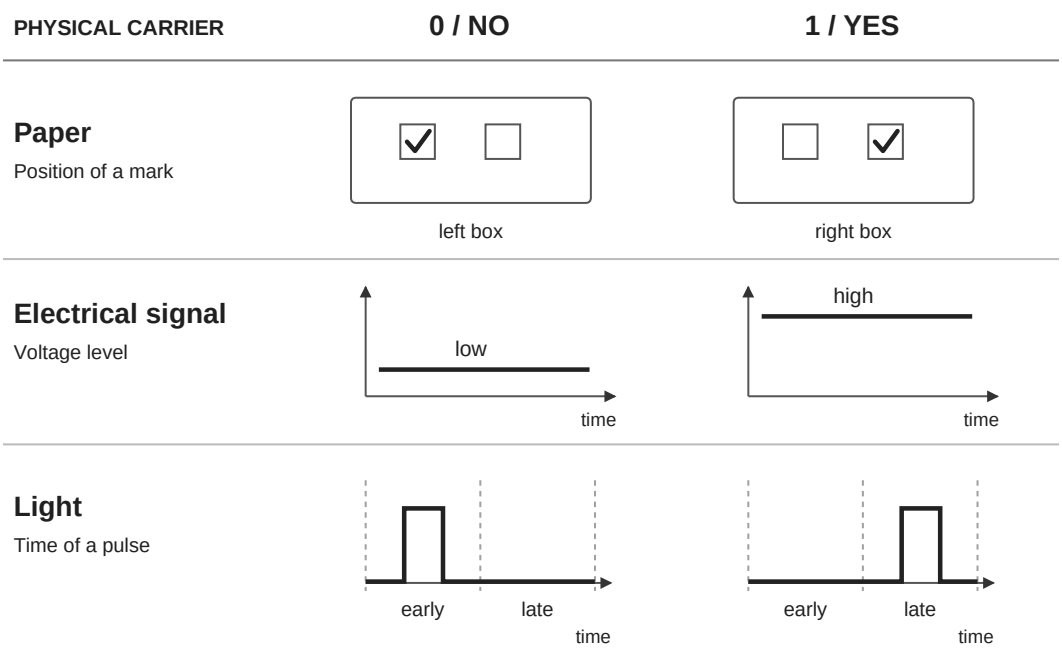


Figure 1.1. One distinction, three physical encodings. Each row represents the same two alternatives: 0 means no and 1 means yes. The paper uses the position of a mark; the electrical signal uses a voltage level; the light signal uses the time of a pulse. The two optical time slots have agreed boundaries. Pulse height is only schematic, not a claim about brightness or energy. The receiver must know which encoding is in use.

A code also cannot manufacture an unlimited message from a single distinction. We could agree that 1 means “I will bring the book tomorrow afternoon,” instead of merely yes. The signal would then select that longer statement from an agreed menu. The extra detail would come from the shared agreement. The same one-place signal could not select arbitrarily among three new answers without some additional distinguishable feature, such as another pulse or another time slot.

A difference you can read

Writing down a code is easier than building an apparatus that can use it. Imagine assigning no and yes to two shades of gray so similar that your receiver cannot tell them apart. There are two labels in the instructions, but the instructions have not made the physical task possible.

To judge the arrangement, send known examples of each alternative and observe what the receiver reports. If it responds differently in a reliable way, the distinction is usable. If it produces the same response regardless of what was sent, it has not recovered this message. Between those extremes lie systems that sometimes give the wrong answer. The performance belongs to the combination of preparation, transmission, and reading, not to the written labels alone.

We will often describe such arrangements using the word state. For the card, we might specify whether the left or right box is marked. For the signal in a circuit, we might

specify the relevant voltage. These are descriptions chosen for the problem at hand. Neither is a complete inventory of the physical object. A card also has a temperature, a mass, a texture, and innumerable details that our two-label description ignores.

That omission is useful. When we call a device a bit—a binary digit—we describe it at a level where two alternatives, conventionally 0 and 1, are the ones being represented. We are not claiming that its material has only two possible microscopic configurations. The ideal classical description treats those two logical alternatives as distinguishable.[3]

Consider a deliberately simple electrical example. Suppose the sender aims for either 1 volt or 4 volts, and the receiver reports 0 for a measured value below 2.5 volts and 1 for a value at or above it. These numbers describe our invented device, not a specification for a commercial component. A signal intended as 4 volts may arrive as 3.9 or 4.1 volts without changing the reported answer. The receiver groups many physical values into the same logical category.

Now let a disturbance push that high signal down to 2.4 volts. The same reading rule produces the wrong answer. Calling the signal digital has not prevented a physical error. It has given us a clear way to decide which physical changes matter to the message.

The paper works similarly. A stroke can thicken or fade a little and remain recognizably inside the same box. Small changes need not change the answer. If the mark spreads into both boxes, however, the receiver may no longer know what to report. The distinction has become harder to read, even though plenty of ink remains.

These examples explain why abstract information can be independent of a particular material without being independent of physics. The symbols 0 and 1 allow us to discuss what many devices have in common. The materials determine how faithfully a particular device can realize those symbols, how it can be controlled, and how long its record can last.

Two choices, then four

One binary position offers two alternatives. Give the message a second position and the available strings are 00, 01, 10, and 11. We have not merely added two more symbols to a bag. We have made four ordered combinations: the first position and the second can each carry a choice.

A third position doubles the number again. Each of the four strings can be followed by either 0 or 1, giving eight strings. Continue the same argument and every extra position doubles the number of possible messages. If n is the number of binary positions and N is the number of distinct strings, then

$$N = 2^n \tag{1.1}$$

The small raised n tells us to multiply together n copies of 2. For three positions, that gives $2 \times 2 \times 2 = 8$. For ten positions, it gives 1,024. Nothing specifically quantum has entered this calculation. It is a count of the combinations in an ordinary binary alphabet.[4]

The conditions matter. The receiver must know how many positions to read and their order, and every combination being counted must be allowed. If we require the two positions always to agree, only 00 and 11 remain. There are then two permitted messages, not four. Duplicating the answer may serve another purpose, but it does not add a new independently selectable answer.

Nor do ten binary positions mean ten physical objects. We could send ten successive pulses through one wire, use ten marked positions on one sheet, or choose a physical device with enough distinguishable settings to represent the whole string. The count concerns the alternatives in the representation. The hardware is a separate question.

It is equally important not to confuse the number of available strings with the importance of the string we actually receive. A single yes or no can settle a matter that has occupied someone for years. A long file can contain material they already know. Our counting rule measures neither significance nor surprise. Later we will ask how probabilities and prior knowledge change the amount learned from a message. For now, we are counting the alternatives the system can represent.

There is a tempting shortcut. Why use ten positions at all? Why not store a whole message in one voltage, giving each possible message a slightly different value? In an ideal mathematical description, even a short interval contains endlessly many distinct real numbers. Could we assign one to every book ever written?

We could write such an assignment on paper. To use it, however, we would have to prepare the voltage finely enough, keep it from drifting, and read it with enough resolution to distinguish the assigned values. Adding more decimal places to the label does not improve the apparatus. As the proposed values crowd together, the required discrimination becomes more demanding. A mathematical continuum is not, by itself, a demonstration of an unlimited physical memory.

This is a recurring lesson. Whenever someone proposes to store or extract a remarkable amount of information, ask which alternatives can actually be prepared and distinguished, with what equipment, in what time, and with what probability of error. We do not yet need a universal bound to recognize that these questions cannot be skipped.

How a record earns its name

Return to the lamp. Your friend could watch it directly, but suppose instead that a receiver registers the pulse and writes 0 or 1 into a memory. After the pulse has passed, the memory remains. A short-lived signal has produced a longer-lived record.

The distinction between signal and record is about their roles, not two fundamentally different kinds of substance. The marked card can itself be a record while it waits on a desk, then become a message carrier when someone delivers it. A stored electrical state can later control another transmitted signal. Information processing links physical systems by arranging how changes in one affect another.

A detector need not produce a word or a sound. It must respond in a way that allows the relevant alternatives to be told apart. An electrical pulse, a pointer position, or a mark on a screen can serve. In physical models of measurement, the interaction connects the measured system with distinguishable states of the apparatus. A human may inspect the resulting record much later.[5]

How do we know that the record deserves our trust? Looking at it alone may not be enough. A receiver that displays an attractive stream of zeros and ones could be responding to the sender, or it could be producing unrelated output. We need to compare the records with what was actually sent during a test.

Figure 1.2 presents two constructed sets of test results. Each contains two hundred trials: one hundred 0s sent and one hundred 1s sent. In the first, the reported symbol agrees with the sent symbol in one hundred and eighty trials. In the second, the receiver is right in only one hundred trials. The marginal counts—the totals viewed at either end separately—are identical: one hundred of each symbol sent and one hundred of each reported.

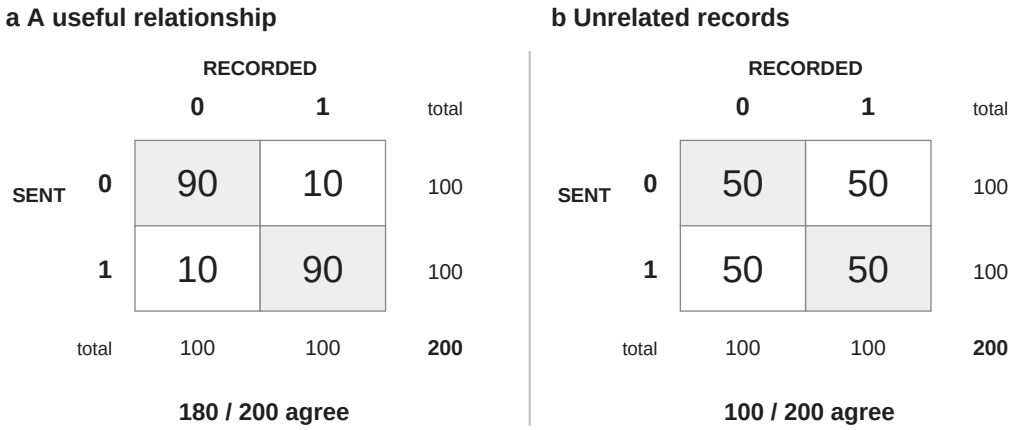


Figure 1.2. A convincing-looking output is not enough. Rows show the symbol sent; columns show the symbol recorded. Each panel contains 200 constructed trials, with 100 of each input. Panel (a) has 180 agreements; panel (b) has 100, and the record is independent of the input in these proportions. Both panels have identical input and output totals. The difference appears when the records are paired with their inputs. These are illustrative counts, not measured experimental data.

The first receiver provides useful evidence about the input. The second does not in the proportions shown: either recorded symbol leaves the two possible inputs equally represented. What matters is the relationship between the ends of the experiment. This is an example of statistical dependence, often described as correlation.[6]

Agreement is not the only useful relationship. A receiver that always reports 1 when we send 0, and 0 when we send 1, could still communicate perfectly once we recognize the reversal. We would simply reverse the decoding rule. By contrast, a display that always shows 1 cannot distinguish our two inputs at all. Reliability concerns a known, recoverable relationship, not the visual appearance of the output.

The same reasoning helps distinguish a record of the signal from a record of something else. A detector might respond to the intended lamp, to a background light, or to an internal fault. Its output can be physically real in all three cases. What needs testing is the interpretation we assign to it.

A record must also persist long enough for its purpose. Ink on a sheet is not motionless at the molecular level. Solids contain atoms that move, and the physical behavior of a material depends on temperature and interactions among its constituents.[7] Keeping a readable mark does not require freezing every microscopic detail. It requires that those changes do not destroy the distinction we need before we use it.

For a handwritten answer, that may be easy. For a rapidly changing device, preserving a state even for the duration of a calculation can be a substantial task. Storage is therefore an achievement of the material and its operation, not a consequence of calling something a memory.

When nothing arrives

Our early-or-late flash has one practical virtue: both answers are represented by an intended pulse. In a working receiver, failure to see either pulse can be flagged rather than automatically interpreted as one of the answers. This does not remove every possible error, but it distinguishes an absent signal from the two intended signals.

There is an apparently simpler code. Turn on the lamp for yes and leave it off for no.

This can work. But what makes the darkness a message? Your friend must know when to look, how long to wait, and that a yes signal would probably have reached the receiver. Without that context, darkness could mean no, a failed lamp, an obstructed route, a missed appointment, or a receiver that was never switched on.

The point is not that silence is meaningless. It is that an absence becomes informative through the alternatives it helps rule out. Even the decision “nothing arrived” needs a physical procedure: a clock or other timing reference, a receiver operating during an interval, and a record of what happened within that interval.

Let us give this a numerical example. Suppose yes and no are equally likely. For yes, the sender emits a pulse. For no, the sender emits none. A working receiver registers nine out of every ten emitted pulses on average, and never produces a false click when no pulse was sent. We are stipulating a simple model, not reporting the performance of a particular instrument.

Across many trials, the model predicts that, per two hundred transmissions on average, one hundred contain no pulse and all one hundred give no click. Of the hundred with a pulse, ninety give a click and ten do not. There are therefore one hundred and ten no-click records: one hundred associated with no pulse and ten associated with a missed pulse.

After a no-click record, the probability that no pulse was sent is thus

$$P(\text{no pulse} \mid \text{no click}) = \frac{100}{110} = \frac{10}{11} \approx 0.909 \tag{1.2}$$

Here *P* means probability, and the vertical bar means “given that.” Read the left side as “the probability that no pulse was sent, given that no click was recorded.” The fraction uses only the no-click cases, because those are the cases compatible with the observation. Under our assumptions, a no-click record supports no pulse with a probability of about 91 percent. It does not establish it with certainty.[8]

The distinction between the two directions of the question is essential. We assumed that no pulse always gives no click. It does not follow that no click always means no pulse. A consequence can have more than one cause. The receiver’s missed detections supply the second cause in this example.

Change what the sender usually does and the inference changes as well. If pulses are sent much more often, missed pulses become a larger share of the no-click records. Improve the receiver, and they become a smaller share. Permit false clicks, and a click itself must also be interpreted with care. The probability assigned to the input after observing the event depends on the preparation and detection model.

At the extreme, cover the detector so that no light reaches it. The receiver now records no click whatever the sender does. The output is perfectly repeatable but tells us nothing about which answer was chosen. This is why two silent receivers need not provide equivalent evidence, even when their displays look the same.

We have reached back to the dark detector in the prologue. There, darkness was a prediction for a specified, functioning apparatus under stated conditions. In a laboratory, an unexpected silence would invite checks of the source, the optical routes, the timing, and the detector. Knowing the theory does not remove the need to know whether the equipment is working.

From a number to a claim

The answer about the book was deliberately selected by a person. Scientific measurements extend the same chain of physical representation to systems that are not trying to tell us anything.

A thermometer does not receive a sentence from a warm object. It interacts with the object and produces an indication that can be related to temperature. A light detector does not receive a label saying what produced the light. It responds, and we use the arrangement of the experiment to interpret that response. To draw a conclusion, we must establish how an observed record is related to the quantity or event of interest.

Calibration is one part of this work. Under specified conditions, it relates an instrument's indications to reference quantities and their uncertainties, allowing readings to be used in obtaining measurement results. An indication and a justified measurement result are therefore not interchangeable.[9]

Imagine that our electrical receiver correctly preserves every digit of a voltage reading. That success would not show that the instrument's voltage scale is correct. A wrong scale can produce a precisely stored wrong number. Equally, an excellent instrument can make a good measurement whose digital record is later corrupted. Measurement quality and communication quality enter at different places in the chain.

This matters especially when the final claim is distant from the immediate observation. A record of detected light may help us estimate a property of a sample. That estimate may enter a calculation about the sample's structure. A conclusion about structure may then support an explanation of a chemical reaction. Each transition needs a relationship we can justify. Adding decimal places at the end does not repair an unsupported step near the beginning.

Computers do not exempt us from this requirement. A calculation takes an encoded problem, performs operations on its representation, and produces another representation. The output can be correctly computed and correctly read while answering a question different from the one we meant to ask. To use the result, we must connect the symbols back to the physical situation.

The distinction is not a reason to distrust measurements or calculations indiscriminately. It tells us where trust comes from: tested relationships, stated conditions, and an account of the errors that could change the conclusion. A useful record is part of an explanation, not a replacement for one.

What the carrier permits

We can now separate three things that were combined in our everyday phrase "send an answer." There is the message selected from alternatives. There is the physical representation that carries it. And there is the record from which the receiver tries to

recover it. A successful communication keeps the necessary distinctions intact across that chain.

Within the abstract description, we are free to call the alternatives 0 and 1. We can count strings without specifying whether they are written in ink or sent as light. That freedom is powerful. But when we ask which representations can be prepared, which can be distinguished, and how they can be transformed, our freedom ends at the behavior of the physical systems.[10]

So far, we have worked with an ordinary classical picture. We selected a definite logical alternative, allowed imperfections to interfere with its transmission, and treated the final output as uncertain evidence about the input. The imperfections could be serious, but they did not require a new rule for describing the alternatives themselves.

Quantum theory does. It permits states that cannot be treated simply as an unread choice from an ordinary list. It also places limits on distinguishing different preparations: some distinct quantum states cannot be identified without error on every trial from a single supplied system, even when the alternatives are known and the measurement is ideal. That restriction is not merely a complaint about inadequate equipment.[11]

We should not accept that statement on the strength of its unfamiliarity. We will build the description and examine what it predicts. The experiment in the prologue has already given us a reason to begin: closing one route changed a detection probability in a way that a simple random choice of route did not explain.

Nor should we think that ordinary computers use nonquantum matter while quantum computers use a special substance. The distinction concerns how physical systems are prepared, controlled, and described for the task. Ordinary devices also inhabit the quantum world. To find out what more its rules permit, we have to move beyond the two labels without losing the connection to what an apparatus can actually do.

Think once more of the lamp across the courtyard. A person chose an answer, a physical process carried it, and another physical process left a record. By understanding the apparatus, we understood what the record could tell us. That is the route we will follow into the quantum world as well.

This time, however, we will let the light take two routes before we ask for the answer.

Notes and sources

1. John Preskill, Lecture Notes for Physics 219/Computer Science 219, Chapter 1, “Introduction and Overview,” §1.1, “Physics of information.” The opening paper–wire–light exchange is an illustrative construction, not a historical episode. Source
2. Claude E. Shannon, “A Mathematical Theory of Communication,” Bell System Technical Journal 27 (1948): 379–423 and 623–656, introduction. This chapter uses the distinction between a selected message and its meaning, not the later quantitative theory of entropy. Source
3. John Watrous, IBM Quantum Learning, Basics of Quantum Information, “Single systems: Classical information,” especially “Classical states and probability vectors.” The voltage thresholds and marked-card examples here are illustrative encodings, not hardware specifications. Source
4. The counting argument and examples following Eq. (1.1) are elementary combinatorial derivations for fixed-length binary strings with every combination allowed. See also Watrous, “Multiple systems: Classical information,” on joint classical state sets. Source
5. Preskill, Lecture Notes, Chapter 3, “Foundations II: Measurement and Evolution” (version updated October 2018), §3.1.1, on correlating a measured system with distinguishable apparatus states. The discussion here describes the role of a record; it does not claim to resolve the quantum measurement problem. Source
6. Watrous, “Multiple systems: Classical information,” discussion of independence and correlation. Figure 1.2 contains constructed counts. It illustrates dependence; it is not an estimate of an actual receiver’s error rate. Source
7. Richard P. Feynman, Robert B. Leighton, and Matthew Sands, The Feynman Lectures on Physics, Vol. I, Chapter 1, “Atoms in Motion,” §1–2. Microscopic motion need not destroy a macroscopic distinction on the timescale relevant to a record. Source
8. Equation (1.2) is an original elementary conditional-probability example. It assumes equally probable yes/no messages, one intended pulse for yes, no intended pulse for no, 90% registration conditional on a sent pulse, no false clicks, and known trial boundaries. Counts in the prose are expected counts under this stipulated model, not observations. The numerical conclusion changes if the assumptions change.
9. Joint Committee for Guides in Metrology, International Vocabulary of Metrology—Basic and General Concepts and Associated Terms, JCGM 200:2012, entries 2.9 (measurement result), 2.26 (measurement uncertainty), 2.39 (calibration), and 4.1 (indication). Source
10. Preskill, Introduction and Overview, §1.1, “Physics of information,” on the physical encoding and manipulation of information. The chapter’s distinction between labels and their implementation is operational; it is not a claim that information is a separate material substance. Source
11. For the contrast between quantum states and ordinary probability distributions, see Watrous, “Single systems: Quantum information.” For imperfect distinguishability even with ideal measurements, see Preskill, Chapter 3, exercises 3.1, “Which state did Alice make?,” and 3.2, “Eavesdropping and disturbance.” Here “a single supplied system” excludes access to a separate record of the sender’s choice. Source; state-discrimination exercises

CHAPTER 2

The Experiment That Changes the Rules

The light reaches a fork. Before it arrives at the other end of the apparatus, we have a decision to make.

We can place a detector in each of the two routes and find out which one registers the photon. Or we can leave the routes open, bring them together, and detect the photon afterward. The two arrangements seem to offer different views of the same journey: an inspection halfway through, or an inspection at the end.

It is tempting to suppose that the first inspection merely supplies a missing detail. Surely the photon has a route, whether we check it or not. If the check could be made gently enough, why should it affect the final result?

To answer, we need something more exact than a picture of a small object traveling through space. We need a prediction that distinguishes that picture from the quantum description. Our two routes will provide one. The crucial comparison will be between a coherent preparation and an ordinary random choice, arranged to give the same route counts.

Two routes, two questions

Return to the interferometer from the prologue. A source prepares one photon. A half-reflecting beam splitter gives access to an upper route, U, and a lower route, L. Mirrors bring the routes to a second beam splitter, beyond which are detectors A and B. We will call the second beam splitter the recombiner. The new name describes its job, not a new kind of component.

For the calculation, keep our ideal conditions: one photon per trial, no unwanted absorption, and detectors that register every photon reaching them. The two beam splitters are balanced. A photon arriving along one input of either splitter has equal probabilities for its two outputs. We also arrange the light arriving by the two routes to match in its timing, polarization, and other properties, apart from the controlled phase difference we will soon investigate. Nothing outside the two-route system retains a record distinguishing U from L.[1]

First, intercept the light in the routes, before recombination. Place an absorbing detector on U and another on L, as in Figure 2.1a. The prediction is one detection per trial, with equal probabilities for the two detectors. Over many trials, the counts approach equal proportions. There is no requirement that every small batch divide exactly in half.

These detectors answer a well-defined question about where absorption occurred. They do not show that the undetected photon in a different arrangement would have followed a little track identical to a detected one. Nor do these particular detectors let us continue the same trial to the recombiner. Absorption ends that photon's passage through our optical apparatus.

Now remove the route detectors. Let the two routes reunite, as in Figure 2.1b. At the adjustment used in the prologue, every photon is predicted to reach A, while B remains dark. The beam splitters have not ceased to be balanced. What has changed is that the recombiner now receives a coherent contribution from both routes rather than a photon prepared in just one of its inputs.

The first arrangement gave a random U or L. The second gives a definite A. We should not combine these into a single imagined record—"U, then A" or "L, then A"—without asking whether a physical procedure can produce that record while preserving the A-only result.

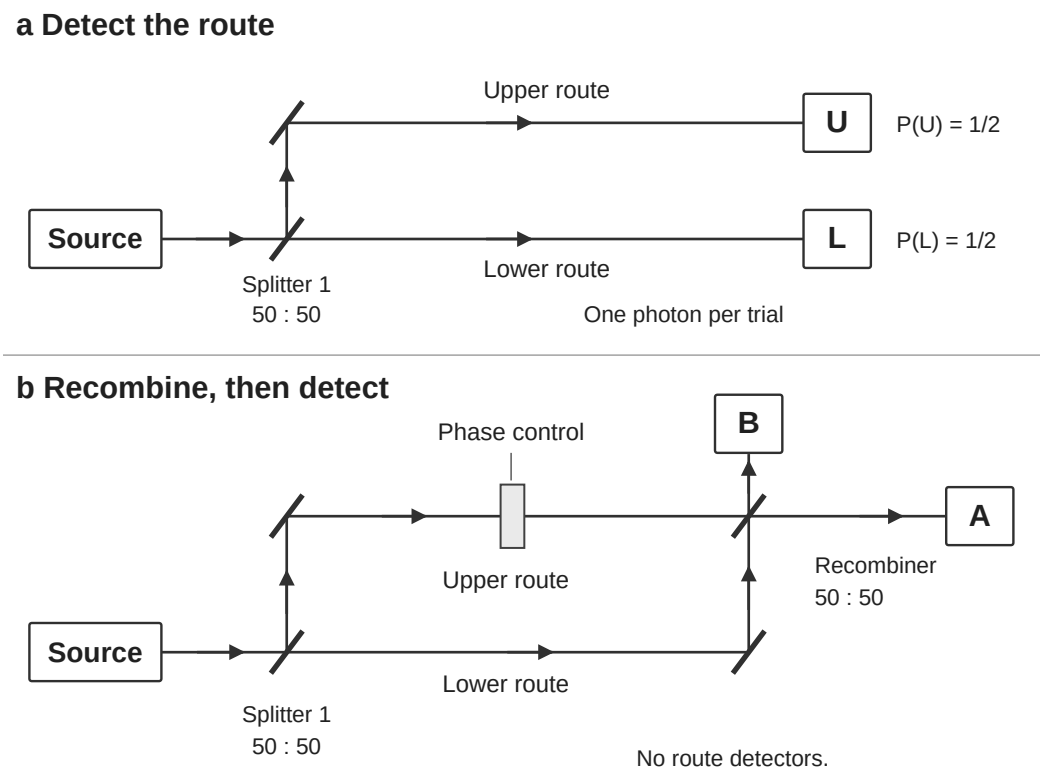


Figure 2.1. Two ways to read the two-route system. (a) Absorbing route detectors register U or L with equal probability. (b) Those detectors are removed and the routes recombine; the probabilities at A and B depend on the relative phase. These are separate experimental arrangements, not successive readings on one photon. Unlabeled diagonal strokes are mirrors; the labeled beam splitters are balanced. Lines show optical routes, not measured photon tracks. Probabilities are per source photon. The apparatus is idealized and not drawn to scale.

The prediction of an unread coin

Before trying to record the route without absorbing the photon, let us build the ordinary random alternative explicitly.

Put the first beam splitter aside. On each trial, toss a fair coin and use the result to prepare one photon in U or in L, just before the fixed recombiner. Prepare otherwise matching photons and tell nobody the coin result. Half the preparations use U on average; half use L. This is a physical lottery, not just a description of our uncertainty.

For a U input alone, the balanced recombiner sends the photon to A or B with equal probability. For an L input alone, it does the same. Therefore the probability of A is the probability of selecting U and then reaching A, plus the probability of selecting L and then reaching A:

$$P_{\text{mix}}(A) = \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} = \frac{1}{2} \tag{2.1}$$

The subscript “mix” labels this random mixture of preparations. The multiplication follows each of two possible sequences; the addition combines their mutually exclusive alternatives. This is the same ordinary probability reasoning we used in Chapter 1. B likewise has probability one-half.

The result remains the same if the person holding the coin record leaves the room, burns the notebook, or never writes the result down. Keeping us ignorant does not turn a lottery over U and L into a preparation that directs every photon to A. The distinction is already measurable without consulting the coin.

This gives us a sharper test than the phrase “the photon could be on either route.” That phrase fits both preparations if it means only that a route measurement has two possible outcomes. Yet their output predictions differ: half at A for the lottery, all at A for the coherent preparation. The route probabilities alone are not a complete description of what we have prepared.[2]

The failure is not in probability theory. Equation (2.1) is correct for the lottery. The mistake would be to assume that every preparation giving equal U and L counts must be that lottery. We have mistaken agreement on one test for agreement on every test.

Unlike blocking a route, neither preparation removes photons: both supply one to the recombiner on every trial.

A change that the route counters cannot see

There is a further surprise. We can change which output is certain without changing the probabilities of the two route detections.

Place an adjustable, transparent element in one route. Its task is to change the relative phase of the two contributions. For now, think of phase as a position within a repeating

cycle. Two regular oscillations can rise and fall together, or one can be displaced relative to the other. A full-cycle change restores their original relationship. A half-cycle change reverses it.[3]

We are using that familiar wave picture to introduce a mathematical relationship. We are not claiming that a photon is a tiny bead oscillating up and down, or that a single-photon state is simply a weak classical electric field. The testable point is that the relative phase changes how the two contributions combine.

Begin with the setting that sends every photon to A. Change the relative phase by half a cycle while preserving the other conditions. The prediction reverses: every photon now reaches B. Between these settings, the output probabilities vary smoothly. Another half-cycle brings the original A-only prediction back.

Check the routes instead, using fresh repetitions of each preparation, and the counts are still evenly divided between U and L. The phase adjustment has not changed their probabilities. It changes a relationship that this particular measurement does not reveal.

We now have three preparations with identical route probabilities: the coherent A-setting, the coherent B-setting, and the coin-selected mixture. The same recombiner distinguishes them through different output statistics. Figure 2.2 puts those predictions side by side. The experiment does not require us to know the name of the hidden difference before we can establish that a difference exists.

“Superposition” is the word quantum theory uses for a state built by combining states with amplitudes. In this example, a coherent superposition involves U and L with a specified relative phase. It is not shorthand for our having forgotten which route was selected. The two coherent preparations and the lottery already tell us why those descriptions must be kept separate.[4]

This is the first substantial step beyond the classical bit of Chapter 1. We still have two possible route outcomes. But the pair of numbers one-half and one-half no longer says everything needed to predict what a device will do to the prepared system.

PREPARATION	ROUTE DETECTION		AFTER RECOMBINATION	
	U	L	A	B
Coherent A-setting	1/2	1/2	1	0
Coherent B-setting	1/2	1/2	0	1
Coin-selected mixture	1/2	1/2	1/2	1/2

Figure 2.2. Equal route probabilities do not specify the preparation. The three preparations give the same U/L probabilities but different A/B probabilities after the same fixed recombiner. Route and output columns refer to separate trials in the corresponding arrangements, not consecutive measurements on one photon. All entries are ideal theoretical probabilities per source photon, not measured counts.

What adds before the squaring

We can now explain the arithmetic that appeared in the prologue, including where its halves came from.

A probability amplitude is a number used to calculate a probability. For an outcome described by an amplitude, the probability is the squared magnitude of that amplitude. When only real numbers are needed, magnitude means ordinary absolute value. Thus +1/2 and −1/2 each give a probability of one-quarter when taken separately. Their signs matter when amplitudes are combined, before the squaring.

At a balanced splitter, each output probability is one-half. The corresponding amplitude magnitude is therefore not one-half. It is $1/\sqrt{2}$, because squaring $1/\sqrt{2}$ gives one-half. The square root sign asks for the positive number whose square is 2; $\sqrt{2}$ is approximately 1.414, so its reciprocal is approximately 0.707.

For a contribution that passes through two balanced splitters in succession, the amplitude magnitudes multiply:

$$\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{1}{2} \tag{2.2}$$

Each of the two route contributions to a chosen final detector consequently has magnitude one-half. Their relative phase depends on the reflections, propagation, and controlled adjustment along the routes. We can choose a consistent mathematical convention in which, at the A-setting, the contributions to A are +1/2 and +1/2, while the contributions to B are +1/2 and −1/2. The signs here include the beam splitters' effects; they are not labels arbitrarily attached to physical routes.[5]

Add the contributions for each final output, then square the magnitude:

$$P(A) = \left|\frac{1}{2} + \frac{1}{2}\right|^2 = 1$$
$$P(B) = \left|\frac{1}{2} - \frac{1}{2}\right|^2 = 0 \tag{2.3}$$

A has probability one and B has probability zero. Their probabilities add to one, as they must in our lossless arrangement. At the B-setting, the half-cycle shift reverses the contribution from one route. The roles of the two detectors interchange.

Why does the rule differ from the coin calculation? In that calculation, the physical preparation selected one input or the other. Here the state retains a coherent relationship between the inputs. Recombination makes that relationship matter. Squaring each contribution first would discard the very sign or phase relationship that the output experiment detects.

We have not derived the probability rule from common sense. We have stated a rule of quantum theory, often called the Born rule, and combined it with the rules for coherent alternatives and successive transformations. The resulting predictions can be compared with experiments. Successful examples give us reason to use the rule; they do not turn an experimental science into a deduction from arithmetic alone.[6]

Notice how little mystery the arithmetic itself contains. Opposite contributions cancel before a probability is formed. No detector reports a negative number of photons. Nor does cancellation destroy energy in the lossless apparatus. The probabilities of the two possible destinations change together.

The difficulty lies in choosing the right mathematical objects. Route probabilities were insufficient; amplitudes supply the structure the experiment requires.

One arrival at a time

So far we have compared predictions for an ideal single-photon apparatus. Which part of this demands quantum physics rather than ordinary optics?

Interference itself is older than quantum theory. Classical electromagnetic waves can split, acquire a relative phase, and recombine into bright and dark outputs. In that description, the electric and magnetic fields combine, and the energy flow depends on their combined values. Ordinary wave optics therefore explains the bright-and-dark pattern of a well-adjusted interferometer illuminated by a laser.[7]

A quantum account must also describe how light is prepared and detected. An ideal single-photon preparation produces one absorbing detection, not two half-photon absorptions at our two outputs. At settings with two possible outcomes, the individual

result is not a miniature copy of the probability distribution. It is one event. A distribution becomes visible through repetitions.

A source described as “very dim” does not automatically satisfy the single-photon assumption. A weak laser pulse, for example, can contain no photon, one photon, or more than one. Most trials can be empty while a small fraction still contain several. And a detector can turn a continuous classical light field into discrete-looking electrical records. A sparse set of clicks, by itself, is therefore not enough to establish the character of the source.

This is why a single-photon experiment needs an additional test. Split the source's light toward two independent detectors and compare coincidences—detections associated with the same source trial—with the rates at which the detectors respond separately. A properly normalized suppression of coincidences can distinguish single-photon behavior from the predictions of classical-field photodetection models. Low efficiency alone would make both individual counts and coincidences scarce; the comparison has to account for that.

Vincent Jacques and colleagues made these distinctions explicit in work published in 2005. Their source used an optically excited defect in a diamond nanocrystal. One arrangement recorded the gradual buildup of interference fringes with a camera. A separate arrangement measured correlations between two detectors to test the source's single-photon behavior. They also compared the coincidence result with faint laser pulses. The two tests addressed different claims: interference, and the statistics of the light supplied to the apparatus.[8]

To see what the test measures, keep four counts. Let N_T be the number of triggered trials, including trials with no detection. Let N_1 and N_2 count the responses of the two detectors, and let N_C count trials in which both respond within the specified coincidence window. The normalized estimate is

$$\hat{\alpha} = \frac{N_C / N_T}{\left(N_1 / N_T\right)\left(N_2 / N_T\right)} = \frac{N_C N_T}{N_1 N_2}.$$

The hat marks an estimate obtained from a finite record. The numerator is the observed frequency of joint detections. The denominator multiplies the two individual detection frequencies: the coincidence frequency expected if the detector records were independent. A value of one means agreement with that reference; a smaller value means fewer coincidences. The denominator must include all triggered trials. Replacing N_T by the number of successful detections would ask a different question.

Why does the classical comparison give a lower bound? Imagine that each pulse carries a nonnegative amount of classical light, W, which may fluctuate between trials. Fixed fractions reach the two detectors. In the weak-response regime, each detection probability is proportional to W. At fixed W, assume independent detector responses, so their probabilities multiply. Averaging over trials then makes the coincidence probability proportional to the average of W squared. The fixed splitting and detection factors cancel in the normalized ratio. If angle brackets mean an average over trials, the underlying parameter is therefore[9]

$$\alpha = \frac{\langle W^2 \rangle}{\langle W \rangle^2} = 1 + \frac{\langle (W - \langle W \rangle)^2 \rangle}{\langle W \rangle^2} \geq 1.$$

The last numerator is an average of squares, so it cannot be negative. This is why merely dimming classical light does not produce the required anticorrelation. Pulse fluctuations can make the ratio larger; they cannot make this model's underlying ratio smaller than one. Constant pulse energy gives the limiting value one. The conclusion assumes a nonzero mean signal, fixed splitting, and the stated detection response. It is a prediction of a physical model, not a theorem about arbitrary pairs of devices.

Now inspect a published count record. In one 5.138-second run, Jacques and colleagues reported N_1 = 49,135, N_2 = 50,865, and N_C = 28, with a normalized estimate of 0.132. Their count analysis used a 100-nanosecond coincidence window and describes a 436-nanosecond excitation interval. Dividing the run time by that rounded interval gives approximately 11.78 million triggered trials. These values reproduce the estimate:[8]

$$\hat{\alpha} \approx \frac{28 \times \left(5.138 / \left(436 \times 10^{-9}\right)\right)}{49,135 \times 50,865} \approx 0.132.$$

At these measured individual rates, the estimated independent reference is about 212 coincidences; 28 were recorded. This comparison explains what the small ratio means. It also shows why twenty-eight coincidences cannot be assessed in isolation: without the individual counts and the number of opportunities, we would not know whether twenty-eight was unexpectedly small. The reconstructed trigger total is approximate, because the table does not list the exact trigger count; it is enough to follow the normalization, not to reconstruct the acquisition record.

Their first faint-laser run instead recorded 49,448 and 50,552 individual counts and 269 coincidences over 4.780 seconds, giving 1.180. Across ten runs per source, the reported summaries were 1.00 ± 0.06 for the laser and 0.13 ± 0.01 for the single emitter, with 95-percent confidence intervals. Table 2.1 gives the comparison; the accompanying plot shows the published run estimates.[8]

Some finite-run laser estimates fall below one. They do not individually overturn the classical bound, which concerns the underlying parameter rather than every possible sample. Conversely, an uncertainty interval addresses variation in the estimate under its statistical assumptions; it does not automatically cover a miscalibrated detector or an

inappropriate timing rule. An experimental conclusion needs both the count comparison and the checks that make the detector model applicable.

The source result establishes a departure from this classical counting model. It does not establish perfect one-photon delivery on every excitation pulse, and it does not measure an interference fringe. Those are separate questions. The strength of the combined experiment is that the preparation is checked through one set of records and coherent recombination through another, using the same source.

Table 2.1. A source test compares measurements with a specified prediction.

Preparation or comparator	Normalized coincidence parameter	Standing
Classical-field detection model	At least 1	Bound on the underlying parameter
Ideal single-photon input	0	Ideal prediction
Diamond single-emitter source	0.13 ± 0.01	Reported measurement
Faint laser pulses	1.00 ± 0.06	Reported measurement

The two measured values are the summaries reported by Jacques and colleagues (note 8), with their stated 95-percent confidence intervals. The normalization compares coincidences with individual-detection rates for triggered trials; the two data sets used comparable mean detected counts per pulse. This table is not the interferometer's phase curve, and it does not assert one delivered photon on every excitation pulse.

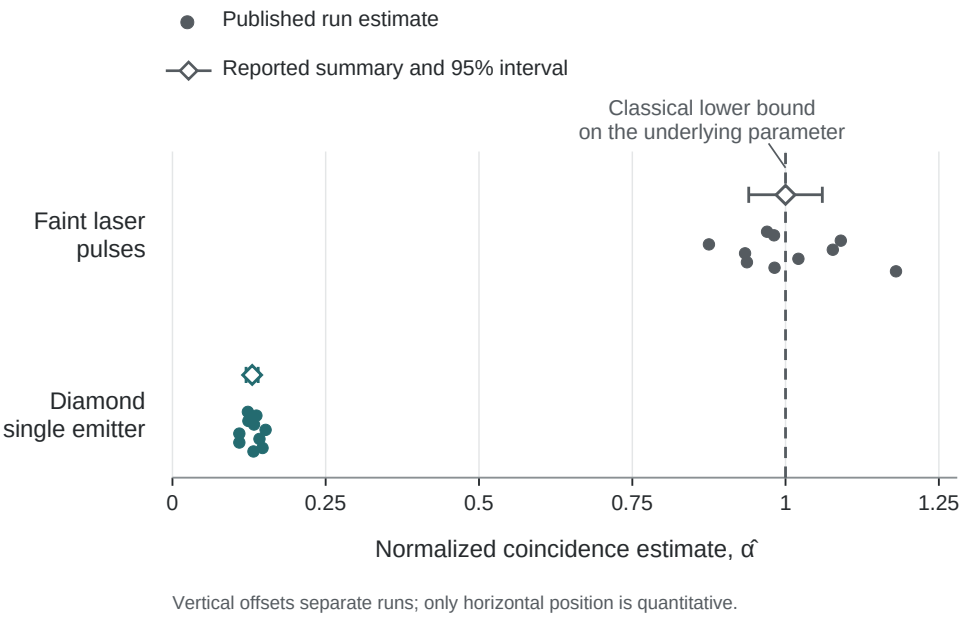


Figure 2.3. Coincidence estimates for two preparations. Small points show the ten published runs per source; diamonds and bars show reported summaries and 95-percent intervals. The dashed line marks the classical model's bound on the underlying parameter. Original replot of Jacques and colleagues' Table I, without individual-run error bars or a new analysis of detector timestamps.[8]

Their apparatus used a prism to split and overlap light, rather than our pair of beam splitters. The physical lesson does not depend on preserving our particular drawing. What matters is that coherent alternatives contribute to detection events, while the preparation and detector statistics are checked independently.

Matter provides another route to the same lesson, and its history shows why discovery cannot follow a single order. In his 1924 thesis, Louis de Broglie proposed associating waves with material particles. In modern notation the relation is $\lambda = h / p$: lambda is the wavelength, p is the magnitude of momentum, and h is Planck's constant. Momentum measures motion; for a slow electron it is its mass multiplied by its speed. The relation made the wave scale calculable before electron diffraction had been established experimentally.[10]

In 1927, Clinton Davisson and Lester Germer measured how electrons scattered from a nickel crystal as the incident speed and observation direction changed. Strong peaks occurred at selected combinations. Wavelengths inferred from the diffraction pattern agreed with the matter-wave relation within their analysis. The inference linked three things: controlled electron motion, the crystal's spatial order, and measured scattering directions. A bright patch alone would not have established that connection. Their paper also recorded unexplained and missing beams; the achievement was a testable correspondence, not a claim that every experimental detail had been exhausted.[11]

A different arrangement produced corroborating evidence that year: George Paget Thomson and Alexander Reid observed rings after electrons passed through a thin film. Thus a theoretical proposal acquired support from materially different experiments. Such agreement tests a prediction; it does not mean that the hypothesis was read directly from an image. Modern experiments make other parts of the reasoning visible, including the accumulation of a pattern from individual arrivals.[12]

In a study published in 2013, Roger Bach and colleagues used electrons and two narrow slits. A movable mask let them compare the patterns obtained with either slit available and with both available. Recording individual electron arrivals allowed them to watch the interference pattern build up. Their displayed sequence progressed from just two detected electrons to thousands.[13]

The early dots did not each contain a little fringe pattern. The spatial structure emerged from their accumulated positions. Unlike our two-output interferometer, the screen offered many possible detection locations. The probabilities varied from place to place, making bands of frequent and infrequent arrivals. In either apparatus, the wave-like structure belongs to the predictions for detection, while a particular detection is localized.

We should not force that behavior into a choice between a familiar water wave and a familiar pellet. Neither picture, carried over unchanged, supplies the full description. Quantum theory gives us a state, rules for its transformation, and probabilities for specified measurements. That is enough to make quantitative predictions without first inventing a microscopic movie that resembles everyday motion.

A record that changes the experiment

We can now return to the question we postponed. Could we record the route, let the photon continue, and still preserve the interference?

For an idealized test, suppose an additional physical system can serve as a marker. Its initial state is the same on every trial. An interaction associates U with one final marker state and L with another. Make these two states perfectly distinguishable: a later measurement of the marker can identify the corresponding route without error. Assume also that the interaction does not absorb the photon or move it from one route to the other.

That describes the job required of the marker, not a claim that any ordinary camera can perform it. Quantum theory allows such a controlled correlation. Some implementations encode the marking in another property of the same particle, such as polarization, rather than in a separate object.[14]

Now let the light reach the recombiner and keep the marker. The prediction is equal probabilities for A and B, even at the former A-setting. The route probabilities are unchanged. The interference in the output counts is gone.

The reason appears when we specify the full alternatives. Previously, the two contributions to B led to the same final situation: a detection at B, with nothing else distinguishing the routes. Now one leads to B together with the U marker, and the other to B together with the L marker. Those are distinguishable joint outcomes. The two signed numbers cannot cancel as though the accompanying marker states were identical.

For the ideal marking just described, each joint outcome at B has probability one-quarter. If we record only B and set the marker record aside, we must add those two probabilities:

$$P_{\text{marked}}(B) = \left|\frac{1}{2}\right|^2 + \left|-\frac{1}{2}\right|^2 = \frac{1}{2} \tag{2.4}$$

Here “marked” labels the experiment with the route-correlated marker retained. The notation makes the contrast with Eq. (2.3) visible: the magnitudes are squared separately because the contributions finish with distinguishable marker states. A has probability one-half by the same reasoning.

The cancellation has not been defeated by a person's curiosity. It is absent from these counts because the physical correlations have changed. Nobody needs to read the marker. Conversely, merely planning to install a marker does nothing: an interaction must actually establish the route-dependent states.

This gives Chapter 1's discussion of a record a new consequence. There, a useful record required a reliable relationship between systems. Here, making such a relationship can change which interference is accessible in one system considered on its own. A marker that remains in exactly the same state for both routes carries no route information. It need not spoil interference, provided it also preserves the other conditions.[15]

It is tempting to explain every lost pattern by saying that a measurement kicked the particle out of position. That can happen, but it is not the whole account. In a 1998 atom-interferometer experiment, Stephan Dürr, Thomas Nonn, and Gerhard Rempe stored route information in internal atomic states. They found that the momentum disturbance was too small to explain the loss of interference. The correlations between the route and the internal state were essential to the explanation.[16]

We must also distinguish failing to read a record from physically undoing the marking interaction. In an ideal controlled system, an appropriate reversal can remove the route correlation before recombination and restore interference. Deleting a line in a computer log is not generally that reversal; copies of the distinction may remain elsewhere. Experiments that recover interference by selecting particular marker outcomes involve still another procedure. They do not mean that forgetting an observation changes earlier detector records.

For this balanced interferometer, perfect route marking retained through recombination removes the phase dependence of the unsorted A and B counts.

When a pattern fades

Real fringes rarely alternate between perfect brightness and perfect darkness. Imperfect overlap, unequal contributions, background counts, and unstable phase can all reduce the contrast. Partial route marking can also reduce it. The contrast alone does not identify which cause is responsible.

Figure 2.4 shows ideal predictions while the phase setting is swept through a complete cycle. With matching, coherent route contributions, the probability of A rises and falls between zero and one. With the perfect marker retained, A remains at one-half whatever phase we choose. The intermediate curve shows a stipulated test in which a perfect marker is used on half of the trials and is absent on the other half. Pooling the results halves the contrast. None of these curves is a fit to measured data.[17]

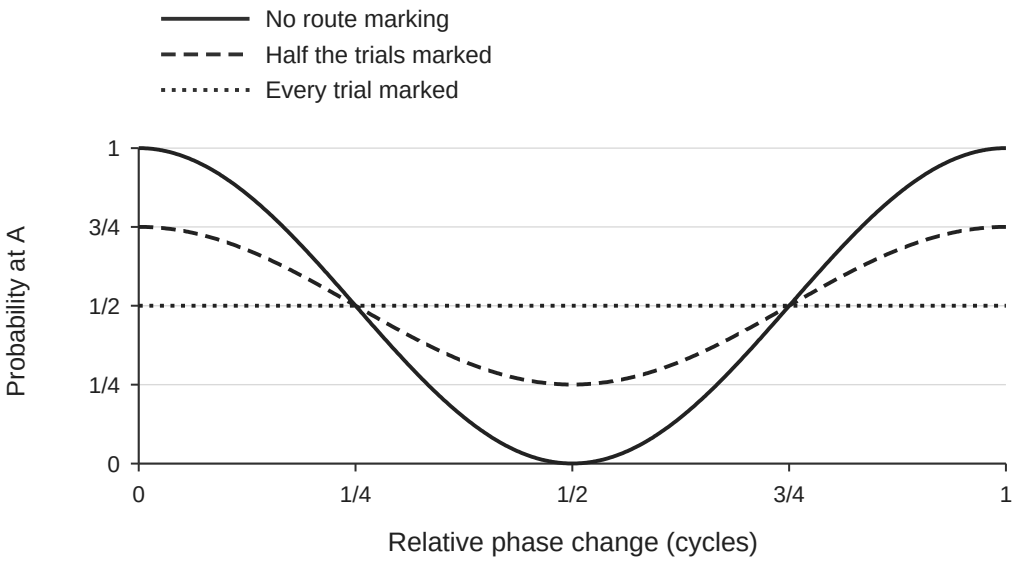


Figure 2.4. How the output responds to a phase change. The curves are theoretical probabilities, not experimental data. Zero cycles is the A-setting; half a cycle is the B-setting. A perfect route marker retained on every trial removes the phase dependence of the unsorted output counts. Marking a randomly selected half of the trials, independently of phase, and pooling all results gives the intermediate curve. No trials are discarded. In this lossless model the probability of B is one minus the probability of A.

A fading pattern need not mean that a usable route record has been created. Suppose instead that the relative phase wanders unpredictably from trial to trial. On each trial there can still be a coherent preparation, but the bright-output setting is no longer the same across trials. Pooling their results can flatten the observed variation. At the extreme of a uniformly random phase, the averaged output probabilities are one-half and one-half without anyone obtaining route information.[18]

This distinction matters experimentally. A route marker and a drifting phase can produce the same output proportions, yet they point to different physical causes and different remedies. We might stabilize the optical paths to address drift. We would have to prevent, reverse, or otherwise handle the route correlation to address marking. Neither diagnosis follows just from a detector reading fifty percent.

Nor does a fifty-fifty output at one fixed setting demonstrate that coherence has been lost. An entirely coherent preparation can give those probabilities at an intermediate phase. We learn more by changing a control and observing the response than by staring longer at one pair of totals.

Our earlier comparison now has a practical form. Ask which preparation was made, which transformation was applied, and which records were retained. These details determine whether a probability table is a complete description of the experiment or only one view of it.

More than an unread answer

We began with what looked like a modest addition to an ordinary journey: inspect the route as well as the destination.

The experiment required us to separate three ideas that everyday language tends to blend. A random choice can leave us ignorant of a route that was selected. A coherent preparation can produce the same route probabilities while giving different results after recombination. A route-marking interaction can preserve those route probabilities while changing the correlations and the interference.

The distinctions are not verbal. Each has a preparation procedure and a predicted set of records. The new description earns its place by explaining why those records differ.

What, then, should we say happened between source and detector? We can say that quantum theory represents the unmarked state with amplitudes for both routes, and that their relative phase affects the later detection probabilities. Saying only “it took one route; we do not know which” leaves out experimentally relevant structure.

That does not by itself settle every possible account of microscopic reality. Our calculation rules out the simple route lottery followed by the fixed single-input response. It is not a proof that every conceivable theory containing trajectories is impossible. A more elaborate account would still have to reproduce the phase dependence, the detection statistics, and the effects of marking. The experiment imposes those requirements before we choose a story.

For quantum information, the immediate consequence is already substantial. A physical representation can contain relationships that one measurement does not reveal but a later transformation makes visible. To preserve and control the representation, we must account for those relationships, not just the frequencies of two labels.

A half-cycle change can reverse an output while leaving the route chances untouched. To describe every intermediate change just as precisely, we need numbers that retain more than size. The next step is to give those numbers a direction.

Notes and sources

1. Richard P. Feynman, Robert B. Leighton, and Matthew Sands, The Feynman Lectures on Physics, Vol. III, Chapter 3, “Probability Amplitudes.” Figure 2.1 is an original schematic. Its predictions assume one photon per trial, balanced lossless optics, matched route contributions apart from controlled phase, ideal detection, and no retained route marker. Source

2. John Preskill, Lecture Notes for Physics 219/Computer Science 219, Chapter 2, “Foundations I: States and Ensembles,” on pure states and statistical ensembles. Equation (2.1) is the elementary probability calculation for the stipulated fair, coin-selected mixture; each selected input gives equal output probabilities at the fixed recombiner. Source

3. Feynman, Leighton, and Sands, The Feynman Lectures on Physics, Vol. I, Chapter 29, “Interference,” on phase and the addition of coherent contributions. The phase control in our idealized apparatus changes relative phase without introducing loss, route marking, or other distinguishability. Source

4. John Watrous, IBM Quantum Learning, Basics of Quantum Information, “Single systems: Quantum information,” especially quantum states, the standard-basis measurement, and the Hadamard operation. Coherent equal-weight states with opposite relative signs have the same standard-basis probabilities but different transformed outcomes. Source

5. Feynman, Leighton, and Sands, Vol. III, Chapter 3: multiply amplitudes for successive stages and add amplitudes for coherent alternatives leading to the same final outcome. Equations (2.2)–(2.3) use one consistent choice of phases for the balanced splitters and routes; reflection and propagation phases are already included. Source

6. Preskill, “Foundations I: States and Ensembles,” §2.1, and Watrous, “Single systems: Quantum information,” on quantum measurement probabilities. The chapter states the amplitude-to-probability rule as part of the quantum model rather than deriving it from classical probability alone. Preskill; Watrous

7. Feynman, Leighton, and Sands, Vol. I, Chapters 29, “Interference,” and 30, “Diffraction.” Classical wave interference by itself is not a test of a single-photon source. For the contrast between interference and individual detection events, see Vol. III, Chapter 1, “Quantum Behavior.” Interference; Quantum behavior

8. V. Jacques, E. Wu, T. Toury, F. Treussart, A. Aspect, P. Grangier, and J.-F. Roch, ‘Single photon wavefront-splitting interference: An illustration of the light quantum in action,’ European Physical Journal D 35, 561–565 (2005). Section III, Eq. (1), and Table I supply the definition, published counts, run estimates, and reported 95-percent intervals. The worked example uses Table I(b), row 1. Its approximate trigger total uses the 436 ns interval in the count-analysis text; Figure 3 separately labels its histogram with 444 ns. Exact trigger totals and event timestamps are not reconstructed. The new figure replots published estimates without new error bars. Published article. Author text.

9. P. Grangier, G. Roger, and A. Aspect, ‘Experimental Evidence for a Photon Anticorrelation Effect on a Beam Splitter: A New Light on Single-Photon Interferences,’ Europhysics Letters 1, 173–179 (1986), especially pp. 174–175. The text gives an elementary weak-response version of the classical bound, using pulse energy and the nonnegativity of variance. Source.

10. Louis de Broglie, Recherches sur la théorie des quanta, doctoral thesis, University of Paris (1924), published in Annales de Physique 3, 22–128 (1925). See Chapters 1–2; English translation by A. F. Kracklauer, On the Theory of Quanta (2004). The book uses modern momentum notation. Source.

11. C. Davisson and L. H. Germer, ‘Diffraction of Electrons by a Crystal of Nickel,’ Physical Review 30, 705 –740 (1927). The primary report describes both wavelength agreement and unresolved features. Source.

12. G. P. Thomson and A. Reid, ‘Diffraction of Cathode Rays by a Thin Film,’ Nature 119, 890 (1927). Source.

13. Roger Bach, Damian Pope, Sy-Hwang Liou, and Herman Batelaan, “Controlled double-slit electron diffraction,” New Journal of Physics 15, 033018 (2013), doi:10.1088/1367-2630/15/3/033018. The

experiment controls slit availability with a mask and records interference buildup from individual detections. The linked preprint was first posted in 2012. Author version

14. Peter D. D. Schwindt, Paul G. Kwiat, and Berthold-Georg Englert, “Quantitative wave-particle duality and nonerasing quantum erasure,” Physical Review A 60, 4285 (1999), doi:10.1103/PhysRevA.60.4285. Their interferometer uses polarization to label paths and single-photon counting with strongly attenuated illumination. Sources included a diode laser and, for a mixed-input test, a filtered lamp. Section I distinguishes this arrangement from a direct test of the quantum character of light. The route-marker model in this chapter is an ideal limit, not a reproduction of their full apparatus. Author version

15. Preskill, Lecture Notes, Chapter 3, “Foundations II: Measurement and Evolution,” §3.1.1 and §3.4.2, on correlations, measurement, and dephasing. Ignoring a correlated marker, reversing its interaction, and selecting particular marker outcomes are different procedures. For an experimental treatment of conditional interference, see Schwindt, Kwiat, and Englert (note 14). Lecture notes; Experiment

16. S. Dürr, T. Nonn, and G. Rempe, “Origin of quantum-mechanical complementarity probed by a ‘which-way’ experiment in an atom interferometer,” Nature 395, 33–37 (1998), doi:10.1038/25653. The reported momentum back action is too small to explain the observed loss of interference; correlations with internal atomic states provide the explanation discussed in the paper. Source

17. Figure 2.4 contains original theoretical curves. For relative phase x measured in cycles, the unmarked probability is one-half times $[1 + \cos(2\pi x)]$; the fully marked probability is one-half. The half-marked curve is their equal-weight average. This means marking half the trials, not guessing the route with 50% accuracy. More general limits on contrast and route distinguishability are treated by B.-G. Englert, “Fringe Visibility and Which-Way Information: An Inequality,” Physical Review Letters 77, 2154–2157 (1996). Further theory

18. Preskill, “Foundations II: Measurement and Evolution,” §3.4.2, “Dephasing channel,” and §3.6.1, “Gaussian phase noise.” Averaging over a phase distribution can suppress interference. A flat unsorted output distribution does not by itself establish that accessible route information has been obtained. Source

CHAPTER 3

Numbers That Carry Direction

The detector is no longer dark, but it is not yet certain to click. We have turned the phase control partway between the two settings that sent every photon to A or every photon to B. What should happen here, between the extremes?

Chapter 2 supplied two numbers for each output: contributions of magnitude one-half, carrying either the same sign or opposite signs. Those signs explained perfect reinforcement and perfect cancellation. They cannot, by themselves, describe a smooth change from one to the other while the contribution magnitudes stay fixed. Something needs room to turn.

Keep the same ideal apparatus: one photon on each trial, balanced and lossless optics, matching contributions at the recombiner, and no retained route marker. Our task is now mathematical, but the question is still physical. Given a particular adjustment, what fraction of many trials should end at each detector?

We will answer by drawing arrows. Their lengths will keep track of amplitude magnitudes; their directions will keep track of phase. Then we will give the arrows a language precise enough to calculate with. That language is the arithmetic of complex numbers.

Room to turn

On a number line, positive one-half lies to the right of zero and negative one-half lies equally far to the left. Think of each as an arrow starting at zero. Changing the sign reverses the arrow without changing its length. What we called a half-cycle phase change already has a geometric picture: a half-turn.

Now allow the tip to leave the line. The arrow can turn through a quarter-circle, or a third of a circle, or any smaller angle. Its length need not change. The old positive and negative numbers become two particular directions among many.

An arrow on this plane has two coordinates. Let x tell us how far its tip lies horizontally from zero, and let y tell us how far it lies vertically. Negative coordinates point left or down. To write the two coordinates as one number, introduce a symbol i and write $z = x + i\,y$. The horizontal coordinate is called the real part; the vertical coordinate, y, is called the imaginary part.[1]

These names do not divide the diagram into a real world and an imaginary one. They distinguish two parts of a mathematical representation. An amplitude pointing upward is

no less usable in a calculation than one pointing right. The symbol i will also tell us how multiplication works; it is more than a label attached to the vertical axis.

For now, find the arrow's length. The horizontal and vertical displacements form the two short sides of a right triangle. The familiar rule for such a triangle says that the square of the long side is the sum of their squares. The magnitude of z, written $|z|$, is that long side:

$$z = x + i\,y\,, \qquad |z|^2 = x^2 + y^2 \tag{3.1}$$

Take $z = 3/10 + 4\,i/10$, shown in Figure 3.1a. Its squared magnitude is $9/100 + 16/100 = 25/100 = 1/4$. Its magnitude is therefore one-half. If this number is the total amplitude for a specified discrete outcome, the probability of that outcome is one-quarter. Both coordinates enter the calculation.[2]

There is no additional outcome called "imaginary." We have not split the detector into a horizontal detector and a vertical detector. The two coordinates describe one amplitude, and its squared length supplies one probability.

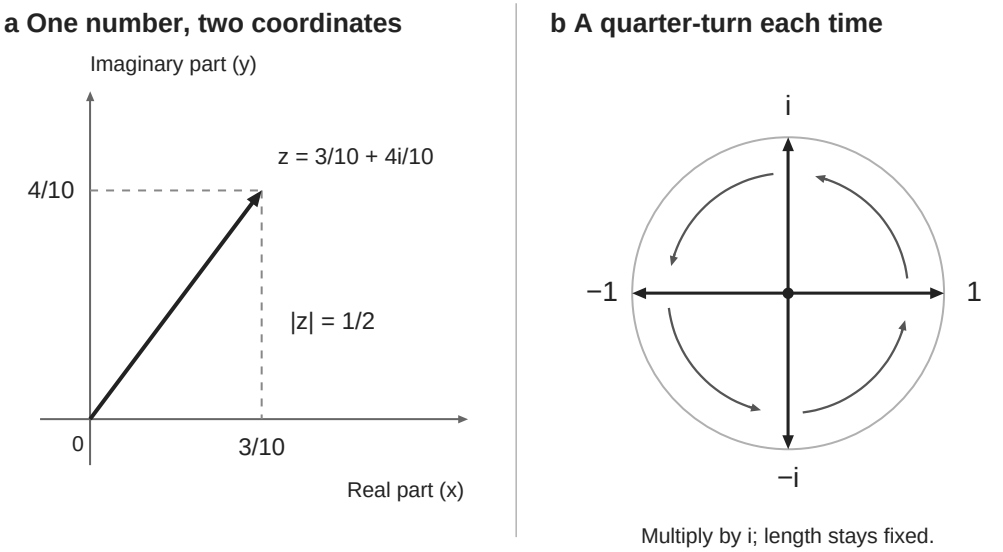


Figure 3.1. A plane for the amplitude. (a) The number z = 3/10 + 4i/10 is represented by an arrow from zero to its two coordinates. Dashed lines show its horizontal and vertical components; its length is 1/2. (b) Multiplication by i turns any nonzero arrow counterclockwise through a quarter-circle. The four unit arrows show successive products starting from 1. The two panels use different scales. These are diagrams of complex numbers, not directions of photon motion.

A quarter-turn called i

We know what multiplying an ordinary number by -1 does to its arrow: it reverses the direction. Let multiplication by i mean a counterclockwise quarter-turn, with the length unchanged. Two such turns produce a reversal. The new rule is therefore $i^2 = -1$.[3]

This explains why i cannot be an ordinary real number. The square of any real number is nonnegative. But our new operation has a consistent meaning on the plane: turn a quarter-circle twice, and the result points the other way.

Start at 1 and keep multiplying by i. The arrow visits i, then -1, then $-i$, then returns to 1, as Figure 3.1b shows. Four quarter-turns restore the original arrow. Multiplying by $-i$ instead makes a clockwise quarter-turn.

The same rule works away from the axes. Using ordinary multiplication and $i^2 = -1$, we get $i(x + i\,y) = -y + i\,x$. The new horizontal coordinate is $-y$ and the new vertical coordinate is x. Our example $3/10 + 4i/10$ becomes $-4/10 + 3i/10$: left four-tenths and up three-tenths. It has turned without growing.

More generally, multiplying two nonzero complex numbers multiplies their lengths and adds their angles. A number with length two and angle a quarter-turn both doubles an arrow and turns it. A number of length one changes only its direction. These operations keep the same addition and multiplication rules we already use, including distributing multiplication over a sum.[4]

This is just the combination quantum amplitudes need. For a specified sequence of stages, we multiply the stage amplitudes. The magnitudes multiply, while the phase changes accumulate. If two stages contribute factors $i/\sqrt{2}$, their product is $-1/2$: two factors of $1/\sqrt{2}$ supply the length one-half, and two quarter-turns supply the minus sign. The example illustrates the arithmetic, not a new convention for every reflection in our interferometer.

For different coherent alternatives leading to the same final outcome, we add their complete contributions. These are different jobs: multiply along a sequence, add across its alternatives. The apparatus tells us which job is required.[5]

Adding without losing the angle

To add two arrows, place the tail of the second at the tip of the first without turning either one. The sum is the arrow from the original starting point to the final tip. Moving an arrow across the drawing does not change the number it represents; its length and direction stay the same.

This construction is the geometric form of adding horizontal coordinates together and vertical coordinates together. It also makes clear why adding lengths alone cannot work. Two walks of equal length can end far from the starting point or bring us back to it, depending on their directions. Here the “walk” is arithmetic on a diagram, not a proposed photon trajectory.

At the A-setting, each route contributes $+1/2$ to A. The two arrows point right. Placed head to tail, they give an arrow of length one. Squaring that length gives probability one.

Now change the lower-route phase by a quarter-cycle. Its contribution to A becomes $i/2$, while the upper-route contribution stays $1/2$. The sum goes right by one-half and up by one-half. Its squared length is:

$$P(A) = \left|\frac{1}{2} + \frac{i}{2}\right|^2 = \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{2} \tag{3.2}$$

B also has probability one-half at this setting. An additional quarter-turn makes the lower contribution to A equal $-1/2$. The two arrows then close back on their starting point, and A has probability zero. Figure 3.2 follows these three cases.

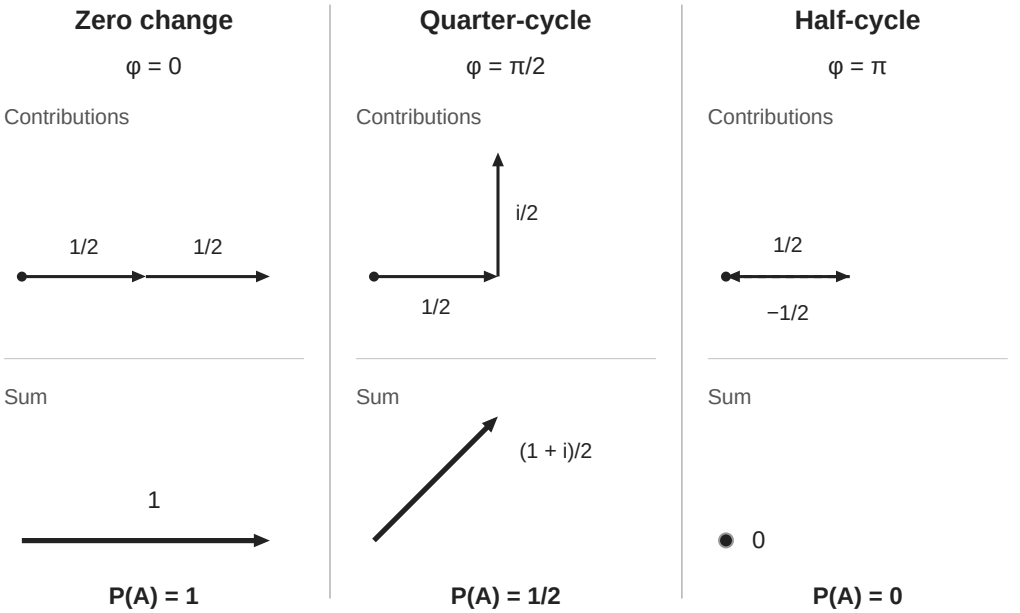


Figure 3.2. Three settings, one addition rule. The upper row adds the two complete route contributions to A head to tail. Each has magnitude 1/2; the second turns as the lower-route phase is changed. In the half-cycle panel, the arrows retrace the same segment. The lower row redraws the resulting sum. Its squared length gives P(A): 1, 1/2, or 0. Arrow scales are identical across panels. These are ideal amplitude calculations for the unmarked, lossless apparatus, not experimental records.

At the quarter-cycle setting, the answer happens to equal the sum of the two separate squared magnitudes. That agreement at one angle does not turn the preparation into a random route choice. Another quarter-turn changes the coherent result to zero; the coin-selected mixture remains at one-half. A calculation that discards directions cannot reproduce both settings.

Notice also what we did not square. The algebraic square of $1/2 + i/2$ is $i/2$, which is not a probability. The squared magnitude is the real, nonnegative number one-half. The vertical bars in Eq. (3.2) are essential.

A useful way to calculate a squared magnitude is to multiply a number by its mirror image across the horizontal axis. That reflected number, $z^* = x - iy$, is called its complex

conjugate. Multiplication gives $z\,z^{*} = x^2 + y^2$: the imaginary terms cancel. This is the same squared length as before, written algebraically.

An angle for every setting

We have used whole, half, and quarter-cycles. To describe an arbitrary setting, let ϕ, the Greek letter phi, denote the lower-route phase change relative to the A-setting. A positive change will turn its contribution counterclockwise in our drawing. The sign is a convention, used consistently throughout the calculation.

Angles can be measured in degrees, cycles, or radians. An angle in radians is the corresponding circular arc length divided by the radius. Because a circle's circumference is 2π times its radius, one complete turn is 2π radians. A half-turn is π radians, and a quarter-turn is $\pi/2$. These are different units for the same turn, not different kinds of phase.

Draw an arrow of length one at angle ϕ. Its horizontal coordinate is called $\cos\phi$, and its vertical coordinate is called $\sin\phi$. This is enough trigonometry for our calculation. At zero angle the coordinates are one and zero; at a quarter-turn they are zero and one. At every angle, their squares add to one because the arrow has unit length.

There is a compact notation for this unit arrow:

$$e^{i\phi} = \cos\phi + i\sin\phi \tag{3.3}$$

This is Euler's formula. The constant e, approximately 2.71828, is the same number that appears in the mathematics of continuous growth. Extending the exponential function to imaginary arguments produces a rotation instead of real exponential growth. In this expression ϕ is measured in radians.[6]

For the present calculation, read $e^{i\phi}$ as "the unit arrow turned through ϕ." The equality on the right tells us its two coordinates. Nothing in the next step requires guessing what an imaginary power looks like.

The notation also makes successive turns easy to combine: $e^{i\phi}e^{i\theta} = e^{i(\phi+\theta)}$. Turning through ϕ and then through θ adds the angles, just as the exponent rule says. The number $e^{-i\phi}$ reverses the turn.

Our lower-route contribution to A is therefore $e^{i\phi}/2$. The upper contribution stays $1/2$. Using the same splitter convention as Chapter 2, the lower contribution to B has the extra minus sign. Let α_A and α_B name the total output amplitudes; alpha is simply a letter, and its subscript identifies the detector. We have:

$$\alpha_A = \frac{1 + e^{i\phi}}{2}, \qquad \alpha_B = \frac{1 - e^{i\phi}}{2} \tag{3.4}$$

These expressions already include the first splitter, the route contributions, and the fixed recombiner. They are amplitudes per photon emitted by the source, not probabilities

conditional on a particular route. We are changing only the controlled lower-route phase, not the convention for the optical elements midway through the argument.[7]

Substitute Euler's formula into the first expression. The horizontal coordinate of α_A is $(1+\cos\phi)/2$; its vertical coordinate is $\sin\phi/2$. Square the coordinates and add. Applying the same procedure to B gives:

$$P(A) = \frac{(1+\cos\phi)^2 + \sin^2\phi}{4} = \frac{1+\cos\phi}{2}$$
$$P(B) = \frac{1-\cos\phi}{2}. \qquad (3.5)$$

The simplification uses only $\cos^2\phi + \sin^2\phi = 1$. It recovers the three arrow diagrams. At zero angle, cosine is one and A is certain. At a quarter-turn, cosine is zero and the outputs are equally probable. At a half-turn, cosine is minus one and B is certain. After a complete turn, the original prediction returns.

We can check an intermediate setting without drawing another apparatus. At one-sixth of a cycle, or $\pi/3$ radians, the horizontal coordinate of a unit arrow is one-half. Equation (3.5) therefore predicts $P(A) = 3/4$ and $P(B) = 1/4$. That is a new quantitative prediction, not one of the endpoints used to introduce the rule.

For every setting, the two probabilities add to one and each stays between zero and one. Destructive interference at one output is accompanied by increased probability at the other. In this model no photon disappears when a contribution rotates.

We have now calculated the smooth unmarked curve shown in Chapter 2. The geometry did not derive the Born rule from ordinary experience. We used that quantum rule, together with the optical model, and derived its consequences for every phase setting.

When all the arrows turn together

So far, only one route's contribution has turned. Suppose instead that we turn both contributions by the same angle. Would the counters notice?

At the A-setting the two contributions to A are $1/2$ and $1/2$. Multiply both by i. They become $i/2$ and $i/2$, and their sum is i. Its magnitude is still one, so A is still certain. The contributions to B become $i/2$ and $-i/2$, and still cancel.

Figure 3.3 contrasts this common quarter-turn with a quarter-turn applied to just one contribution. The common turn changes the direction of the sum but not its length. The one-sided turn changes the length because it changes the angle between the contributions.

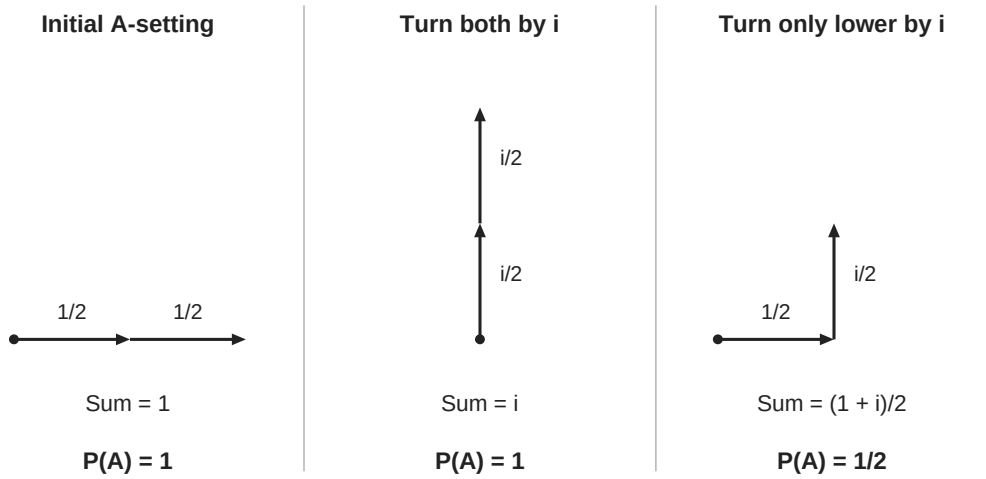


Figure 3.3. A common turn and a relative turn. Starting from the A-setting, multiplying both route contributions by i rotates their sum without changing its length: P(A) remains 1. Multiplying only the lower-route contribution by i changes the relative phase, giving P(A) = 1/2. The diagrams show contributions to A after all fixed optical factors have been included. The common-turn example leaves B dark; the relative-turn example gives P(B) = 1/2. These are theoretical predictions under the chapter's ideal assumptions.

The distinction is called global phase versus relative phase. A common phase factor multiplying every amplitude of a complete state is a global phase. It does not change the state's measurement probabilities. A change between its components is relative and can affect an interference experiment.[8]

For any output amplitude z, multiplying by a unit arrow gives:

$$\left|e^{i\chi}z\right|^2=\left|z\right|^2 \tag{3.6}$$

Here χ, the Greek letter chi, names the common rotation. The statement is not restricted to our present detector arrangement. In later linear transformations, the common factor can be pulled outside every amplitude sum. It remains a common factor in every resulting amplitude, and its magnitude is still one.

This tells us which part of an arrow direction matters physically. The shared orientation of the entire description is redundant; the relative orientations can change predictions. Choosing a horizontal reference is not the same as measuring a hidden absolute direction.

The word "complete" matters. Applying a phase change to one branch of a larger experiment while leaving another branch unchanged is a relative change in that larger system. Calling the altered branch "the whole system" would hide the reference with which it can interfere.

Ask a second question

Consider two preparations of our equal-weight route superposition. In one, the relative phase is a positive quarter-turn; in the other, it is a negative quarter-turn. The lower-route factors are i and $-i$. Equation (3.5) gives equal A/B probabilities for both.

At this fixed setting, even a vast collection of output counts cannot distinguish them. More repetitions estimate the same probability more accurately. They do not supply information that this measurement does not contain.

But we have a control. Add the same known negative quarter-turn to the lower route before recombination in fresh trials of either preparation. For the first, the two quarter-turns cancel and the total relative phase becomes zero: A is certain. For the second, the negative quarter-turns add to a negative half-turn: B is certain.

The initial equality of probabilities concealed a measurable difference. The known control converted that difference into a different final record. This is a simple example of choosing an operation before measurement so that the measurement answers the question of interest.[9]

We specified two particular candidate preparations. Under our ideal assumptions, that specially chosen measurement distinguishes them in one trial. It does not follow that one detection can reveal an arbitrary, initially unknown phase. Estimating a general phase requires statistics from repeated preparations, with appropriate settings and an account of uncertainty.

There is another distinction hidden in the fifty-fifty result. A source that keeps the same quarter-turn on every trial is not equivalent to a source that randomly chooses zero or a half-turn with equal probabilities. Their initial A/B totals agree, but their responses to the added control differ. The fixed quarter-turn becomes a certain A; the random source still gives one-half at A after pooling its two possible preparations.

The arithmetic for that pooling is ordinary probability arithmetic. With zero phase, the original output amplitude at A is one and its probability is one. With half-cycle phase, that amplitude is zero and its probability is zero. Choosing these preparations equally often gives an average probability of one-half. Averaging their amplitudes first would give one-half, whose square is one-quarter. That would be the wrong experiment.[10]

Arrows from coherent alternatives within a trial may add. Detector statistics from distinct, classically selected trials are combined by averaging their probabilities. A notebook does not create interference between separate photon emissions by placing their calculated amplitudes in the same column.

What turns in the laboratory?

Nothing in the arrow diagram requires a photon to move around a little circle. The angle belongs to a probability amplitude. To change it physically, the apparatus must change the transformation experienced by the state.

Propagation provides an example. For light of a definite wavelength, increasing the propagation distance by one wavelength advances the phase by a full cycle in an appropriate sign convention. A quarter-wavelength corresponds to a quarter-cycle. In a real experiment we use a sufficiently narrow wavelength range and maintain the overlap

needed at recombination; an uncontrolled delay that makes the contributions distinguishable is not a pure phase adjustment.[11]

As an illustrative scale, choose a vacuum wavelength of 600 nanometers. A nanometer is one-billionth of a meter. An extra 150 nanometers of vacuum propagation corresponds to a quarter-cycle phase change. This is a chosen numerical example, not a report from a particular experiment. The distance is the added optical route length; it need not equal the displacement of a mirror, since moving a mirror can change more than one segment of the route.

The smallness of that distance explains why controlling relative phase can demand careful engineering. It also suggests a use: a physical change that shifts the relative phase can alter detector probabilities. To learn the change, we need a model linking it to phase, a controlled comparison, and enough observations. The arrow itself is not an extra measuring instrument.

Complex numbers are used in classical wave calculations too. Their usefulness here does not, by itself, distinguish a single photon from a classical light field. That distinction still rests on the preparation and detection tests discussed in Chapter 2. Nor does the notation identify an imaginary number with a material substance. We can perform the same calculations with pairs of real coordinates, provided we preserve the full addition, multiplication, and probability rules.[12]

What we cannot do is keep only an arrow's horizontal projection and discard the rest. A vertical amplitude can have nonzero magnitude. A subsequent rotation can bring its contribution onto the horizontal axis. The direction carries information relevant to later transformations even when the present detector statistics do not reveal it.

From an amplitude to a state

One final distinction will help us move forward. The arrow representing a complex number is not, on its own, a picture of an entire qubit.

Between the splitters, our coherent preparation requires an amplitude for U and an amplitude for L. In the convention used here they are $1/\sqrt{2}$ and $e^{i\phi}/\sqrt{2}$. Each can be represented by an arrow on the complex plane. Together, with the meaning of U and L specified, they describe this coherent preparation. Their squared magnitudes add to one. A state described by one such normalized list of amplitudes is called a pure state.[13]

After the recombiner, α_A and α_B are the amplitudes for its two output alternatives. In obtaining either one, we combined the contributions leading to that same output. We did not add the amplitudes for distinguishable detector outcomes and square the result to find the total detection probability. For that, we add $P(A)$ and $P(B)$.

This is why a drawing must say what its arrows represent. Two coordinates can describe a single complex number. Two complex numbers can describe the components of a two-

state pure quantum state. Two squared magnitudes can describe the outcome probabilities of one measurement. The objects are related, but they are not interchangeable.

We began with a detector partway between darkness and certainty. We can now predict its response, follow how a phase adjustment changes that prediction, and distinguish a change in a coherent relationship from a lottery over preparations. We have also found a freedom in the description: rotating the whole set of amplitudes together changes no observable result.

The next question is no longer how to add the arrows. It is how a choice of measurement turns a state into a particular set of possible answers. Our photon already offers two different questions: detect the route, or recombine and detect the output. Understanding that choice will take us from the two routes to the qubit itself.

Notes and sources

1. Richard P. Feynman, Robert B. Leighton, and Matthew Sands, The Feynman Lectures on Physics, Vol. I, Chapter 22, “Algebra,” §§22–5 and 22–6. The complex plane represents one complex number by two real coordinates; the imaginary part of x + iy is the real coefficient y. Source

2. John Watrous, IBM Quantum Learning, Basics of Quantum Information, “Single systems: Quantum information,” section “Measuring quantum states.” The amplitude-to-probability statement here concerns discrete outcomes. The 3/10 + 4i/10 example and Figure 3.1 are original calculations and artwork. Source

3. Feynman, Leighton, and Sands, Vol. I, Chapter 22, §22–5. The relation $i^2 = -1$ defines the imaginary unit; choosing counterclockwise as the positive quarter-turn fixes our diagram convention. Source

4. Feynman, Leighton, and Sands, Vol. I, Chapter 23, “Resonance,” §§23–1 and 23–2, on complex-number representation, magnitude, conjugation, and multiplication. Angles add for nonzero factors; the zero complex number has no defined phase. Source

5. Feynman, Leighton, and Sands, Vol. III, Chapter 3, “Probability Amplitudes,” §3–1. Successive-stage amplitudes multiply; coherent alternatives contributing to the same complete final outcome add. Retained distinguishing markers require a description of the full system, as explained in Chapter 2. Source

6. Feynman, Leighton, and Sands, Vol. I, Chapter 22, §22–6, especially Eq. (22.9), for Euler’s formula. The elementary derivation of the interferometer probabilities uses the unit-circle identity, not any claim that complex-number arithmetic alone entails quantum physics. Source

7. Equations (3.4)–(3.5) are derived for this book’s ideal balanced interferometer. They use the real balanced-splitter convention from Chapter 2: an equal positive route preparation and a recombiner taking the normalized sum and difference. Fixed optical phases are absorbed into that convention. The same algebra is given by a Hadamard operation, a relative phase operation, and a second Hadamard operation; see Watrous, “Single systems: Quantum information,” section “Unitary operations.” Figures 3.2–3.3 show theoretical amplitudes, not measured data. Source

8. Watrous, IBM Quantum Learning, General Formulation of Quantum Information, “Density matrices: Density matrix basics,” section “Connection to quantum state vectors,” on the equivalence of state vectors differing only by global phase. The complete state, not an isolated branch relative to another branch, is at issue. Source

9. The positive- and negative-quarter-cycle discrimination is an original worked example using Eq. (3.5). In qubit notation the candidate states are $(|0\rangle + i|1\rangle)/\sqrt{2}$ and $(|0\rangle - i|1\rangle)/\sqrt{2}$. They are orthogonal, which permits ideal single-trial discrimination after a suitable operation; this does not permit exact recovery of an arbitrary unknown state from one copy. For the operations and measurement rule, see Watrous, Basics of Quantum Information, “Single systems: Quantum information.” Source

10. The two-setting mixture is a classical random selection of preparations, not a coherent superposition of preparation devices. Probabilities must be averaged with the selection weights. The numerical comparison is an original application of the law of total probability. A general formalism for describing mixed as well as pure quantum states is introduced by Watrous, General Formulation of Quantum Information, “Density matrices: Density matrix basics.” Source

11. Feynman, Leighton, and Sands, Vol. I, Chapter 29, “Interference,” §29–3, defines the wave number and wavelength relation $k\lambda = 2\pi$. The 600-nanometer and 150-nanometer values here are illustrative. The narrowband, matched-mode qualification is necessary to model a physical path change solely as a scalar relative-phase factor. Source

12. Feynman, Leighton, and Sands, Vol. I, Chapter 29, §29–5, develops complex-number methods for classical wave interference. Replacing a complex number by two real coordinates preserves its information only when the corresponding operations are retained. This observation does not assert that all alternative real-number formulations of quantum theory have identical structures. Source

13. Feynman, Leighton, and Sands, Vol. III, Chapter 8, “The Hamiltonian Matrix,” §8–1, distinguishes a state vector from its amplitude components. The present pair describes a pure state of the two-route degree of freedom under the stated assumptions, not a general mixture or the full set of physical degrees of freedom of a photon. Source

CHAPTER 4

A Qubit and the Questions We Ask

A detector clicks, and we write a zero in the notebook. What have we learned?

In one arrangement, zero means that the photon was absorbed on the upper route. In another, it labels an output after the two routes have recombined. The marks on the page can look identical while answering different physical questions. To interpret the record, we need to know what stood between the source and the detector.

The previous chapters gave us a way to calculate the answers. We can now turn the problem around: which measurements would let us learn what was prepared? Not simply which detector happened to click, but which description would correctly predict the results of other experiments on the same preparation.

Our photon is already enough to explore this question. We do not need a larger computer or a new kind of particle. We need to understand the smallest quantum information system: a qubit.

The qubit already on the table

Keep the two routes, but give their reference states shorter names. A photon prepared in the upper route will be called $|0\rangle$; one prepared in the lower route will be called $|1\rangle$. Read these symbols as “the zero state” and “the one state.” The vertical line and angled bracket form a ket, a notation for a state vector. The zero inside the bracket is a label, not an assertion that there is no photon.

Between preparation and readout, a pure state of this two-route system has the form

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, |\alpha|^2 + |\beta|^2 = 1. \quad (4.1)$$

The letter ψ, pronounced “psi,” names the state. The numbers α and β are its two complex amplitudes. Their squared magnitudes give the probabilities of the two route outcomes. The second part of the equation says that these probabilities add to one. It is the normalization condition we have already been using.[1]

Two reference states do not mean that only two states can be prepared. The amplitudes can vary, and their relative phase can vary. What stays fixed is that two complex components suffice for a pure state once the reference pair has been specified. Mathematically, the state vectors occupy a space with two complex dimensions. This is what makes the system a qubit—a quantum bit.

There is also only one qubit here, not one on each route. We have restricted the experiment to exactly one photon distributed over two modes, with its other properties

fixed. Allowing zero photons, several photons, or additional independently controlled properties would enlarge the description. Counting drawn paths is not a way to count qubits.

The same two-dimensional structure can describe two polarization states of a photon or two selected internal states of an atom. The physical operations are different, but the amplitude rules are the same. A real atom can have many other states; using it as a qubit requires controlling a chosen pair and limiting unwanted excursions outside them.[2]

For this chapter, the routes remain our working example. They keep preparation and readout visible: we can draw where a device goes and say what it does. The qubit is not an object added to the apparatus. It is the quantum information carried by the degree of freedom we have chosen to use.

A basis is a choice of reference

The pair $|0\rangle, |1\rangle$ is a basis for this state space. Each state has unit length, and the two are orthogonal: the ideal route measurement distinguishes them with certainty. Any pure qubit state can be expressed using their amplitudes. “Basis” names this role as a complete pair of reference states, not a uniquely required choice.

There is another pair already familiar from the interferometer:

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}. \tag{4.2}$$

The plus state is the equal-weight preparation that our fixed recombiner sends to A. The minus state is the equal-weight preparation it sends to B. Each gives equal probabilities in a route measurement. Yet they are perfectly distinguishable by recombining before detection. They form another basis.[3]

Call direct route detection a Z measurement. Call recombination followed by detection an X measurement, labeling A as + and B as -. These conventional letters identify two measurement settings. They do not mean that the upper and lower routes have become three-dimensional coordinate axes.

For the general input in Eq. (4.1), the fixed recombiner takes the normalized sum and difference of the route amplitudes. Therefore

$$P_X(+) = \left|\frac{\alpha + \beta}{\sqrt{2}}\right|^2, P_X(-) = \left|\frac{\alpha - \beta}{\sqrt{2}}\right|^2. \tag{4.3}$$

The subscript identifies the measurement; the sign identifies its outcome. By contrast, the Z probabilities are $P_Z(0) = |\alpha|^2$ and $P_Z(1) = |\beta|^2$. The same input amplitudes enter both calculations. What changes is the operation that turns them into the final detection amplitudes.

We can also express the input state itself in the plus/minus basis. Its two coefficients are $(\alpha+\beta)/\sqrt{2}$ and $(\alpha-\beta)/\sqrt{2}$. This is a change of description: the state has not physically changed merely because we have rewritten it. Inserting a recombiner is a physical operation that makes this choice of measurement possible. Keeping those two ideas separate prevents a good deal of confusion.

A state that is a superposition in one basis can be a single basis state in another. The plus state is the simplest example. Asking whether something is "in a superposition," without specifying the alternatives, leaves part of the question unstated. But the difference between that pure preparation and a random choice of upper or lower route remains measurable; a change of notation cannot erase it.

Check the prediction

If two sources give equal probabilities for zero and one in Z, must they prepare the same state? What additional observation would distinguish a plus preparation, a minus preparation, and a fair lottery over the two routes?

No amount of additional Z data can distinguish these three ideal preparations: each gives zero and one with equal probability. Use fresh preparations and change the apparatus to X instead. The plus source then gives the plus outcome with certainty, the minus source gives it with probability zero, and the route lottery gives it with probability one-half. The first two answers follow by adding the route amplitudes before squaring; the lottery requires averaging the probabilities of its separately prepared alternatives.

A single plus result excludes the minus source in this ideal model, but does not distinguish the plus source from the lottery. Repeated X trials estimate the relevant probability. The lesson is that more observations of one setting cannot supply a distinction that setting does not measure; choosing a different physical test can.

The rule behind the questions

There is one compact calculation behind these different measurements. Let a normalized reference state $|q\rangle$ have amplitudes u and v in the zero/one basis. To find the amplitude associated with that reference, conjugate its coefficients, multiply them by the corresponding input coefficients, and add:

$$\langle q|\psi\rangle = u^{*}\alpha + v^{*}\beta, P(q \mid \psi) = |\langle q|\psi\rangle|^{2}. \tag{4.4}$$

The star means complex conjugation: reverse the sign of the imaginary part, as in Chapter 3. The expression $\langle q|\psi\rangle$ is an inner product, often called the overlap. It is a complex number, not a new state. Its squared magnitude gives the probability of the q-labeled outcome in a measurement that uses this state as one of its two references.[4]

Choose $|q\rangle = |0\rangle$, and only α remains. Choose $|q\rangle = |+\rangle$, and the normalized sum appears. The earlier calculations are special cases of this rule, not separate instructions invented for different pieces of equipment.

Orthogonal states have zero overlap. For the plus and minus states, the two contributions to their inner product are $1/2$ and $-1/2$, which cancel. A measurement aligned with one never confuses the other with it. For a state compared with itself, the overlap is one because of normalization.

An intermediate overlap has an important meaning. The input $|0\rangle$ has overlap $1/\sqrt{2}$ with $|+\rangle$, so it can produce a plus result with probability one-half. A plus click therefore does not certify that the incoming state was the plus state. It identifies the recorded outcome of a specified test. Inferring the preparation requires considering the alternatives that could have produced it.

Our word “question” is shorthand for this physical test. It does not mean asking a qubit to report a prewritten entry in a notebook. The apparatus determines the outcome alternatives and the amplitudes relevant to them.

The missing direction

Would the Z and X settings together tell us everything? Chapter 3 supplied a reason to suspect otherwise. Equal route weights with a positive quarter-cycle phase and equal weights with a negative quarter-cycle phase give the same probabilities at both settings.

Give these two preparations names:

$$|+_y\rangle = \frac{|0\rangle + i|1\rangle}{\sqrt{2}}, \quad |-_y\rangle = \frac{|0\rangle - i|1\rangle}{\sqrt{2}}. \tag{4.5}$$

Both give one-half for each Z outcome and one-half for each X outcome. Those two measurements cannot distinguish them even with arbitrarily many independent repetitions. The problem is not too few clicks. It is a missing measurement setting.

Use the same remedy as before: apply a negative quarter-cycle phase change to the lower route, then recombine. Multiplying the lower component by $-i$ turns $|+_y\rangle$ into $|+\rangle$ and $|-_y\rangle$ into $|-\rangle$. The first now reaches A with certainty; the second reaches B. Call this the Y measurement, with outcomes $+_y$ and $-_y$.[5]

For an arbitrary input, its A amplitude is $(\alpha - i\beta)/\sqrt{2}$. That minus sign is also what the conjugation rule in Eq. (4.4) supplies when the reference state is $|+_y\rangle$. Geometry, the optical control, and the inner product give the same prediction.

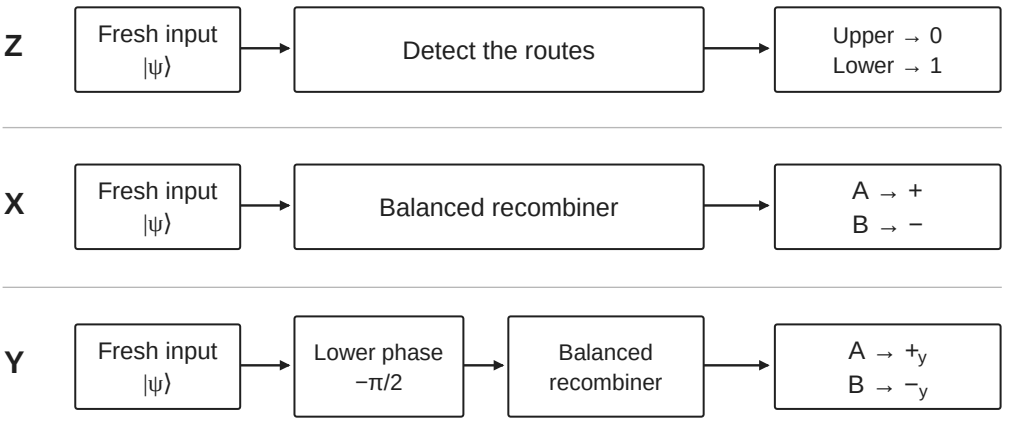


Figure 4.1. Three questions, three arrangements. Each row uses a fresh preparation of the same input state. Z detects the routes directly. X uses the fixed balanced recombiner. Y first multiplies the lower-route amplitude by $-i$, then uses that recombiner. In X and Y, the labels at the right refer to the same physical output detectors A and B; their meanings depend on the preceding operations. Rows are alternative experiments, not three copies made from one photon. Schematics are functional diagrams, not optical layouts.

Table 4.1 collects the predictions for six reference states and the fair route mixture. Only one probability per setting is needed; the other is one minus the displayed value. Every entry follows from the amplitude rules, or from averaging probabilities for the mixture. These are ideal predictions, not experimental counts.

Table 4.1. The same preparation, three measurement settings.

Preparation	$P_Z(0)$	$P_X(+)$	$P_Y(+_y)$
$\|0\rangle$	1	1/2	1/2
$\|1\rangle$	0	1/2	1/2
$\|+\rangle$	1/2	1	1/2
$\|-\rangle$	1/2	0	1/2
$\|+_y\rangle$	1/2	1/2	1
$\|-_y\rangle$	1/2	1/2	0
Fair route mixture	1/2	1/2	1/2

Each column refers to separate trials with the indicated arrangement. “Fair route mixture” means an independent, equal-probability choice of $|0\rangle$ or $|1\rangle$ on each trial, with the choice record unavailable. The complementary outcome probability in each setting is one minus the entry shown.

The Y column separates two preparations that both of the earlier columns left indistinguishable. It also separates $|+_y\rangle$ from the fair route mixture. A fifty-fifty result is not a diagnosis of randomness in the source: its meaning depends on which test produced it.

A map made from probabilities

The three settings suggest a useful way to draw the state. For each setting, subtract the probability of the second outcome from that of the first. Call the resulting differences x, y, and z:

$$\begin{aligned} x &= 2P_X(+) - 1, \\ y &= 2P_Y(+_y) - 1, \\ z &= 2P_Z(0) - 1. \end{aligned} \tag{4.6}$$

Each coordinate runs from minus one to plus one. Zero means equal probabilities. Plus one means certainty for the first outcome; minus one means certainty for the second. These are differences between probabilities, not extra results that appear in a single detection.

The zero state has coordinates $(0, 0, 1)$. The one state lies at $(0, 0, -1)$. The plus and minus states sit at the two ends of the x axis, and the two quarter-cycle states at the ends of the y axis. Three measurement settings have given us a three-dimensional map.

For a pure state, substituting the amplitude formulas gives $x = 2\,\mathrm{Re}(\alpha^* \beta)$, $y = 2\,\mathrm{Im}(\alpha^* \beta)$, and $z = |\alpha|^2 - |\beta|^2$. Here Re and Im mean the real and imaginary parts. The squared-magnitude rule of Chapter 3 gives $x^2 + y^2 = 4|\alpha|^2|\beta|^2$. Adding the square of the population difference then gives

$$x^2 + y^2 + z^2 = \left(|\alpha|^2 + |\beta|^2\right)^2 = 1. \tag{4.7}$$

That is the equation of a sphere of radius one. The surface is the Bloch sphere, a map of pure qubit states. Its geometry has emerged from the probability rules rather than being imposed as a picture of the particle.[6]

The map leaves out the irrelevant global phase. Two amplitude lists that differ only by a common phase factor give the same coordinates. The relative phase still matters: our equal-weight route superpositions move around the equator as their relative phase changes. Zero relative phase gives the positive x point; a positive quarter-cycle gives the positive y point.

Figure 4.2 marks these landmarks. Unlike a complex-plane arrow in Chapter 3, a point here represents a whole qubit state. Its three coordinates are probability differences. The sphere is not the surface of a tiny photon, and its axes are not directions in which our optical routes must point.

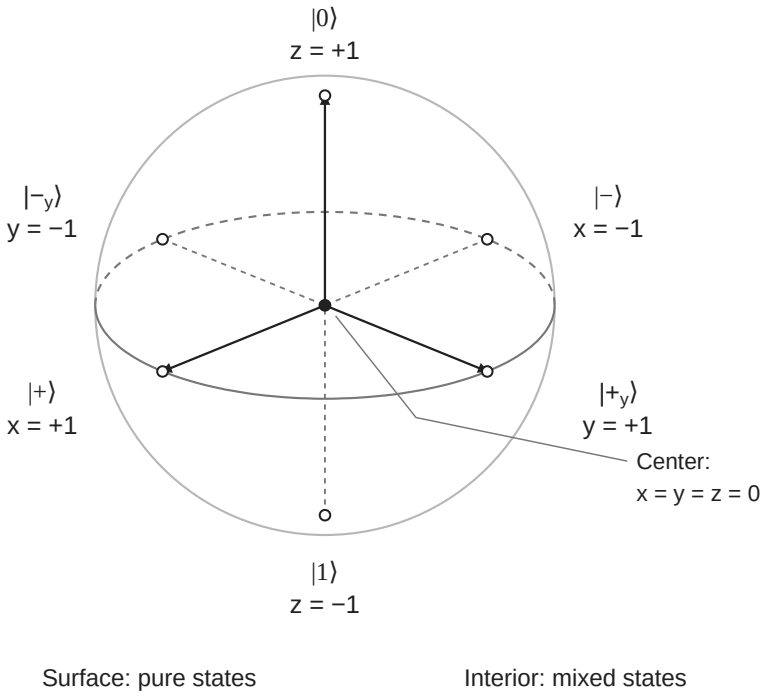


Figure 4.2. A sphere for the state, not the particle. The six labeled surface points are the pure reference states in Table 4.1. Coordinates are the X, Y, and Z probability differences in Eq. (4.6). The equator contains pure equal-weight route superpositions; its rear arc is dashed. The center represents the maximally mixed qubit state. Orthogonal pure states lie at opposite ends of a diameter. The depth convention is purely geometric; no physical position, radius, or photon trajectory is depicted.

Orthogonal state vectors appear at opposite ends of a diameter, not at right angles on this sphere. For example, $|0\rangle$ and $|1\rangle$ are opposite poles. States on perpendicular Bloch axes, such as $|0\rangle$ and $|+\rangle$, instead have squared overlap one-half. The state-vector geometry and the three-dimensional map are related but are not the same drawing.

What about a mixture? The fair route lottery has zero for all three coordinates. It belongs at the center. Other mixtures lie inside the sphere: their probability differences are weighted averages of the coordinates of their constituent pure preparations. The general condition is $x^2 + y^2 + z^2 \leq 1$.[7]

Thus the surface describes pure states, while the interior describes mixed states of one qubit. At the center, every two-outcome projective measurement gives equal probabilities. This does not mean that every conceivable device must report fifty-fifty; we are referring to the complete, two-state basis measurements used here.

There is more than one way to reach that center. Randomly choose the plus or minus state with equal probabilities, and every entry in the last row of the table is reproduced. The route lottery and the plus/minus lottery are different preparation procedures, but no measurement on the delivered qubit alone distinguishes them. Additional records of the preparation can distinguish the procedures; the qubit by itself cannot.[8]

We have reached a useful boundary. A state specifies what can be predicted from the system under the quantum rules. It need not identify a unique history of how the system was prepared. Later, when systems interact, a mixed state will also be needed to describe one part without keeping the full description of its partners.

Pure does not mean predictable under every measurement. The zero state is pure, yet X and Y both give random outcomes. A pure qubit permits a basis measurement with a certain answer; it does not make every possible question certain.

The Bloch ball makes this limit precise. If Z has a certain outcome, z is plus or minus one. The sum of the three squared coordinates cannot exceed one, so x and y must both be zero: X and Y each give equal probabilities. No allowed qubit state can make both X and Z certain. This is a preparation uncertainty constraint. It concerns the distributions predicted for separate measurements on identically prepared systems, even with ideal apparatus. It is distinct from uncertainty in our estimates of those probabilities and from the change caused by an earlier measurement. Preparing the qubit at an X pole makes X certain, but then Z gives equal probabilities.

Learning the state without pretending to see it

Suppose a source repeatedly prepares the same, initially unknown qubit state. Divide its independently prepared outputs among the Z, X, and Y settings. Estimate the three probabilities from their respective counts, then obtain x, y, and z. This is a simple form of quantum state tomography: reconstructing a state description from measurements on many preparations.[9]

To see the arithmetic, stipulate exact probabilities $P_Z(0) = 9/10$, $P_X(+) = 4/5$, and $P_Y(+_y) = 1/2$. Equation (4.6) gives $(x, y, z) = (3/5, 0, 4/5)$. Their squared sum is $9/25 + 16/25 = 1$, so this ideal example is pure. One amplitude description is $(3|0\rangle + |1\rangle)/\sqrt{10}$. Squaring its zero component gives $9/10$; taking the normalized sum of its components and squaring gives $4/5$. The reconstructed description passes the tests that produced it.

The three coordinates also predict measurements not yet performed. Let a new basis measurement point along a unit direction with coordinates n_x, n_y, and n_z on the Bloch sphere. Its positive reference state lies at that surface point. Then

$$P_n(+) = \frac{1 + x\,n_x + y\,n_y + z\,n_z}{2}. \tag{4.8}$$

The products in the numerator measure how the state direction lines up with the measurement direction. The formula reproduces Z, X, and Y by selecting their axes. For the example just reconstructed, choose the direction $(3/5, 0, 4/5)$ itself. The probability becomes $(1 + 9/25 + 16/25)/2 = 1$. A suitable measurement would have a certain outcome, although none of the three settings used to learn the state did.

That is what a successful state description buys us: predictions beyond the observations used to construct it. The formula applies to interior points as well as surface points. Its more general justification uses density matrices, which package the same single-qubit information into a small array of numbers; we can use this geometric form without introducing that machinery here.

Real counts give estimates, not the exact probabilities stipulated above. Finite samples fluctuate. Imperfect controls and unequal detector responses can introduce systematic errors. If estimated coordinates land just outside the sphere, that need not reveal a new kind of state: statistical fluctuations or a faulty measurement model can produce such a result. Estimation must respect the allowed state space and report uncertainty rather than treating raw frequencies as exact coordinates.[10]

The repeated-preparation assumption matters too. Changing the source between settings would combine answers about different inputs. A stable, calibrated comparison—and, where appropriate, interleaving the settings—is part of the experiment, not a bookkeeping detail.

Most importantly, three settings do not mean three individual detections are enough. Each probability is inferred from many trials. The result is a description learned about the preparation, not a photograph extracted from a single qubit.

What remains after an answer

Could we avoid preparing all those fresh systems by asking one qubit question after question? Our absorbing photodetectors cannot do that: once the photon is absorbed, it is no longer available. But absorption is not the only possible kind of readout.

In an experiment reported in 2007, David Hume, Till Rosenband, and David Wineland used an aluminum ion as the primary system and a beryllium ion as an auxiliary system. Repeated information transfer and readout improved their ability to identify the measured state. A qubit measurement need not destroy the carrier in order to produce a record. Here repeated readout preserved the distinction being measured, not every superposition that could have been supplied.[11]

For a clean test, consider an ideal repeatable projective measurement that leaves the qubit available afterward. If its outcome is q, the resulting qubit state is $|q\rangle$. Repeating the same measurement immediately, with no intervening change, returns that outcome with certainty. This is a specified measurement model, not a claim about every possible instrument.[12]

Begin with $|0\rangle$ and measure Z. The result is zero, and a second Z measurement also gives zero. Now repeat the experiment on a fresh $|0\rangle$, but insert an X measurement between the two Z measurements.

The first Z answer is still zero. The X measurement then yields plus or minus, each with probability one-half, leaving the corresponding plus or minus state. Either of those states

gives equal probabilities in the final Z measurement. Therefore the final zero probability is $(1/2)(1/2)+(1/2)(1/2)=1/2$. The inserted measurement has changed the prediction.

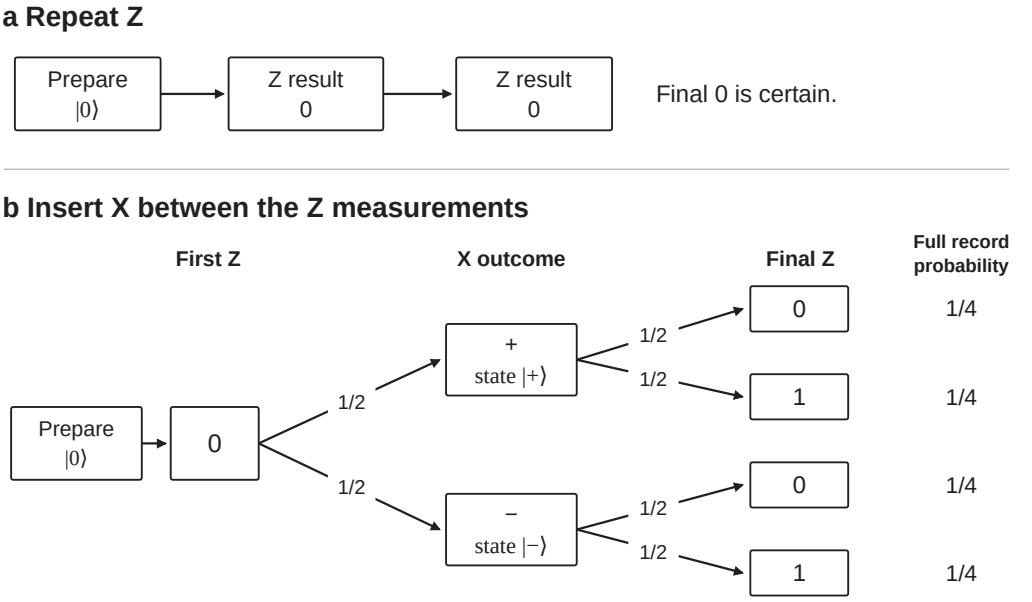


Figure 4.3. An intervening question changes the later answer. Both experiments begin with $|0\rangle$. The first repeats Z with no intervening operation. The second inserts X between two Z measurements. Branches denote possible measurement records, not duplicated qubits. Labels on branches are conditional probabilities; each complete record in the lower experiment has probability 1/4. No outcomes are discarded. The model uses ideal repeatable, qubit-preserving projective measurements and no other evolution. It is not a sequence of absorbing photon detectors.

Figure 4.3 retains every branch. Nothing is explained by silently throwing away inconvenient results. Nor does the person reading the notebook cause the change. The X measurement is a physical process establishing an outcome record, whether or not anyone later consults it.

Notice the contrast with rewriting a state in a different basis. Rewriting leaves every physical prediction unchanged. Performing the intermediate measurement changes the preparation available to the next device. It can remove a certainty that the earlier preparation supplied.

It would also be wrong to conclude that measurement always makes a state less definite. A plus state remains a plus state under an ideal repeatable X measurement. The same plus state, measured in Z, yields a definite zero or one state conditional on the result. Which property is predictable depends on both the state and the next question.

This is why a long record from one repeatedly measured qubit is not interchangeable with a collection of fresh, identically prepared qubits. Later answers describe the state left by earlier operations. Tomography needs the probabilities associated with the input preparation, not a succession of altered inputs treated as though they were unchanged.

A continuum is not an unlimited message

The sphere contains infinitely many pure states. It is tempting to imagine using them as an enormous alphabet: choose a finely specified point, send one qubit, and have the receiver read off the coordinates.

The overlap rule exposes the obstacle. Different states need not yield disjoint sets of measurement records. The zero and plus states, for instance, can both produce a zero in Z and can both produce a plus in X. Their distinct coordinates do not make them perfectly distinguishable messages.

These two calculations show only that those particular measurements fail. Quantum theory makes a stronger statement: two specified pure states can be distinguished without error, with a conclusive answer on every trial, from one copy exactly when their overlap is zero. For nonorthogonal states, changing to a more elaborate measurement cannot remove both the possibility of error and the possibility of an inconclusive result.[13]

A qubit admits a pair of orthogonal states, but not three mutually orthogonal states. Accordingly, a single qubit delivered on its own cannot carry an alphabet of three or more messages that are all read with certainty in one use. This statement assumes the receiver has only that qubit, without additional message-bearing systems or a previously shared entangled resource.

More general measurements can have more than two recorded outcomes. Adding outcome labels, however, does not turn nonorthogonal input states into perfectly distinguishable ones. Counting lights on the readout panel is no substitute for calculating the relationship between the encoded message and the record.

There is no contradiction between a continuously variable state and limited information obtainable from one system. The detailed state description predicts probabilities across possible experiments. A particular measurement produces a particular record. Learning the description and reading one record are different tasks.

The state and the next question

We began with an unadorned zero in a notebook. It now has a context: an input preparation, a measurement setting, an outcome rule, and, when needed, a description of what the measurement leaves behind.

The two routes have become one example of a broader structure. We can specify a qubit with amplitudes, choose a basis, calculate overlaps, and reconstruct its state from repeated preparations. We can also see why a state cannot be replaced by the probability table for just one question. Another arrangement can reveal distinctions that the first one misses.

This gives quantum information both its possibilities and its constraints. Relative phase can affect a later result without appearing in a route count. Yet the amplitudes are not a

collection of numbers that one detector will print for us. They must be used through controlled physical operations and interpreted through the resulting statistics.

One more question is now unavoidable. We have changed a state by inserting optical elements and by making measurements. What determines how it changes while the apparatus simply waits? To answer, we must connect the state to energy and time. The next part of the story concerns not just the questions we can ask of matter, but the rules by which matter evolves before we ask them.

Notes and sources

1. John Watrous, IBM Quantum Learning, Basics of Quantum Information, “Single systems: Quantum information,” sections “Quantum state vectors” and “Measuring quantum states.” Equation (4.1) concerns a pure state in a chosen two-dimensional space. It is not a general description of a mixed state. Quantum states and the Born rule.

2. John Preskill, Lecture Notes for Physics 219/Computer Science 219, Chapter 2, “Foundations I: States and Ensembles,” §2.2. The route encoding used here is restricted to the single-photon subspace of two modes. Extra photon-number sectors, polarization, or unwanted atomic levels are not included in that encoding. Qubit realizations.

3. Richard P. Feynman, Robert B. Leighton, and Matthew Sands, The Feynman Lectures on Physics, Vol. III, Chapter 8, §§8–1 and 8–2, on states, bases, and amplitudes. The recombiner convention is unchanged from Chapters 2–3 of this book; in matrix language it is the Hadamard operation. Re-expressing a vector and physically transforming it are distinct operations. Amplitudes and vectors; Hadamard operation.

4. Feynman, Leighton, and Sands, Vol. III, Chapter 8, §8–2, especially Eq. (8.15). Equation (4.4) applies to the q outcome of a rank-one projective measurement containing the normalized state $|q\rangle$, with the input state specified at the measurement boundary. Its other basis vector is orthogonal to $|q\rangle$. Inner products.

5. Figure 4.1 and Table 4.1 are original presentations of the stipulated two-route model. The Y analyzer multiplies the lower component by $-i$, then applies the fixed Hadamard recombiner, giving output amplitudes $(\alpha - i\beta)/\sqrt{2}$ and $(\alpha + i\beta)/\sqrt{2}$. The states called $|\pm_y\rangle$ here are often denoted $|\pm i\rangle$. All settings use matched modes, lossless ideal operations, and ideal detection. Phase and Hadamard operations.

6. Preskill, “Foundations I: States and Ensembles,” §2.3.2, on the Bloch representation. The coordinates in Eq. (4.6) are expectations of the three Pauli measurements. Equation (4.7) follows by substituting the normalized pure-state amplitudes into these coordinates. Figure 4.2 is an original diagram of these coordinates, not a spatial representation of a photon. Bloch sphere and measurement coordinates.

7. Watrous, General Formulation of Quantum Information, “Density matrices: Density matrix basics,” for the general state description. The three Bloch coordinates determine the qubit density matrix. Nonnegative eigenvalues require $x^2 + y^2 + z^2 \le 1$. A rank-one projective measurement along a unit direction has the positive-outcome probability given in Eq. (4.8). Density matrices; Bloch-coordinate derivation.

8. Preskill, “Foundations I: States and Ensembles,” §2.5.2, “Ensemble preparation.” The two fair mixtures have the same density matrix, $I/2$. Their equivalence concerns access to the qubit alone, not access to correlated preparation records. Independently repeating either mixture gives indistinguishable delivered ensembles as well. Nonunique ensemble decompositions.

9. D. F. V. James, P. G. Kwiat, W. J. Munro, and A. G. White, “Measurement of qubits,” Physical Review A 64, 052312 (2001), doi:10.1103/PhysRevA.64.052312, especially §II.A on single-qubit tomography. The author version is titled “On the Measurement of Qubits.” The (3/5, 0, 4/5) example is an original ideal calculation, not reported data. Author version.

10. James and colleagues, “Measurement of qubits,” §§IV–V, on physical state estimation and experimental uncertainty. Also see Watrous, “General measurements: Quantum state discrimination and tomography,” for tomography using independent copies and Pauli measurements. The chapter assumes a calibrated two-dimensional model; observed frequencies alone do not establish calibration, stationarity, or independence. State estimation and errors; Pauli-measurement tomography.

11. D. B. Hume, T. Rosenband, and D. J. Wineland, “High-Fidelity Adaptive Qubit Detection through Repetitive Quantum Nondemolition Measurements,” Physical Review Letters 99, 120502 (2007), doi:10.1103/PhysRevLett.99.120502. The experiment uses $^{27}Al^+$ and $^{9}Be^+$ as primary and ancillary systems. Its repeated readout protects the measured distinction; it does not preserve every possible input superposition unchanged. Author version.

12. Preskill, Lecture Notes, Chapter 3, “Foundations II: Measurement and Evolution,” §3.1.1, on orthogonal measurements and outcome-conditioned states. The sequence in Figure 4.3 uses the ideal repeatable projective instrument, with no additional evolution or feedback. A general measurement instrument can have different effects even when some outcome probabilities agree. Measurement model.
13. Watrous, “Quantum state discrimination and tomography,” on optimal state discrimination. For distinct pure states with nonzero prior probabilities, perfect deterministic single-copy discrimination requires orthogonality. The discussion excludes additional message-bearing systems and previously shared entanglement; it does not claim that all qubit measurements have exactly two outcomes. More elaborate measurements may include inconclusive outcomes without violating this limit. Discrimination and more general measurements.

PART II

Matter, Information, and Computation

We can now describe a qubit and calculate the probabilities of different measurements. But knowing how to read a state leaves another question unanswered: what makes it change? Its description alone does not tell us what happens while we wait, why atoms bind, or how a machine performs a calculation.

This part connects the description to the physical world that carries it. The Hamiltonian will specify how a state changes. The same mathematics that describes a controlled atomic transition will help us ask where molecular energies come from. We will move from levels supplied as assumptions to levels produced by interactions.

A molecule is a demanding test. We must specify the particles, count their attractions and repulsions, construct a state, and compare its energy with a defined alternative. We will also see why an approximate calculation can teach us something real without becoming the final answer. Keeping those possibilities separate is part of reasoning with a model.

Then we will turn toward information itself. A detailed description is not the same as a large amount of recoverable information. To compare messages and devices, we will need a measure of uncertainty and an account of which distinctions can be preserved.

Computation brings these strands together. A procedure can be written abstractly, but carrying it out requires physical changes. We must ask whether a device can produce an answer reliably, with resources we can specify.

Matter, messages, and machines each ask us to connect a description with a physical process and an observable result. We begin with the smallest possible change in the experiment: prepare the system, leave it alone, and wait.

CHAPTER 5

The Rules of Change

Leave a well-prepared atom alone for a while. Then ask the same question you would have asked immediately after preparing it. Can waiting change the answer?

Not merely by allowing the atom to collide with something, absorb stray light, or escape its trap. Set those complications aside. Keep the surrounding conditions fixed. Can the quantum state itself change while nothing in the apparatus is being adjusted?

Yes. More surprisingly, a measurement of energy can miss the change entirely. At every waiting time, the same two energies may appear with the same probabilities. Another measurement can nevertheless go from certainty for one outcome to certainty for the other. The difference is in the relative phase.

The previous chapter taught us to ask different questions of one preparation. Here we will ask a fixed question at different times. That will connect the arrows of Chapter 3 to energy, and the state of Chapter 4 to a law of motion. Eventually, waiting will become something useful: part of a way to measure a frequency, control a qubit, and understand how matter changes.

Two levels, not two places

For this chapter, move from the photon's two routes to two selected internal states of an atom. Keep the labels $|0\rangle$ and $|1\rangle$, but change their physical meaning. They now denote states with energies E_0 and E_1, with $E_1 > E_0$. A measurement that distinguishes them is our new Z measurement. An appropriate control pulse followed by readout can implement an X or Y measurement of their superpositions.

This change of carrier does not change the qubit mathematics. It does change what must be built. A beam splitter does not mix atomic internal states simply because it mixed optical routes. We need an interaction that couples the chosen levels. Later in the chapter, an oscillating electromagnetic field will provide one.[1]

An energy level is not a shelf inside the atom, and the two states are not miniature planetary orbits. The labels specify quantum states and the energy values an ideal measurement would return. Energy is the physical quantity accounted for when a system does work, absorbs radiation, or transfers energy to another system. In quantum theory, its possible measurement values and its role in time evolution belong to the same mathematical description.

For now, assume that other atomic states remain irrelevant, that the energies stay fixed, and that no uncontrolled interaction leaves a record outside our two-state system. Treat the atom's position and motion as spectators. This is an effective two-level model, not a claim that a real atom has only two states or can be isolated forever.

The distinction between the two energies will matter repeatedly. Write it as $\Delta E = E_1 - E_0$, where the triangle-shaped delta means “difference.” This gap is not an uncertainty or an error bar. It is a property of the model we have specified.

How experiments reveal energy levels

Where did those two energies come from? A spectrometer begins with light, not with a picture of levels inside an atom. It separates radiation by wavelength and records how much arrives in each interval. Under suitable conditions, excited atomic hydrogen produces narrow spectral features near 656, 486, 434, and 410 nanometres. Those are rounded locations of observed groups of lines; their finer components require higher resolution. The reproducible pattern is the evidence to be explained.[2]

In 1885, Johann Balmer found a simple mathematical relation among hydrogen wavelengths. That relation organized measurements before there was a satisfactory theory of the atom. In 1913, Niels Bohr connected spectral frequencies with differences between allowed atomic energies. His construction was an important step, but its special assumptions about orbits were not the spatial wave mechanics we will use in Chapter 6. Observation supplied a pattern; successive theories proposed different explanations and further tests.[3]

Neglecting atomic recoil and treating the transition as sharply defined, the energy accounting for emission can be written as follows:

$$E_{upper} - E_{lower} = hf\,, f = \frac{c}{\lambda_{vac}}.$$

Here f is the emitted light’s frequency, h is Planck’s constant, c is the vacuum speed of light, and the wavelength is explicitly the vacuum wavelength. An absorbed photon supplies the same energy difference for the reverse transition. Repeated lines sharing initial or final states can therefore constrain a consistent network of level differences. They do not measure an absolute energy zero or photograph an electron’s route.

Energy matching is necessary, but it does not make every possible line strong or even detectable. The interaction must connect the two states. Its coupling determines transition strengths and symmetry restrictions, while state populations and the apparatus affect the recorded intensities. The restrictions on which states an interaction can connect are called selection rules. A missing line need not mean that a level is absent, and a finite spectrum does not uniquely reconstruct an arbitrary Hamiltonian.[4]

This is the status of our two-level model: it is a selected part of an experimentally tested atomic description. We now ask what the same energy gap predicts about a prepared superposition while we wait. Agreement with that new experiment tests more than the location of a spectral line.

An equation for the next instant

The rule governing coherent evolution is the time-dependent Schrödinger equation:

$$i\hbar \frac{d}{dt}|\psi(t)\rangle = H(t)|\psi(t)\rangle . \tag{5.1}$$

The argument t says that the state depends on time. The derivative d/dt asks how rapidly it is changing at that instant. A speedometer reports a rate of change of distance; here the rate concerns the complex components of a state. We do not need a course in calculus to use the particular solutions that follow.

The symbol i is the quarter-turn from Chapter 3. The symbol $\hbar$, pronounced "h-bar," is Planck's constant divided by 2π. Planck's constant h has units of energy multiplied by time. Accordingly, energy divided by $\hbar$ has the units needed for a phase-change rate. Both sides of Eq. (5.1) have units of energy times a state vector.[5]

The equation is a physical law tested through its predictions, not a consequence of the notation.

The new object is H, the *Hamiltonian*. It is an operator: a rule that acts on a state vector. It contains the energy terms and interactions in the model. Specifying an initial state is not enough to predict the future; we must also specify this rule. The possible energy-measurement values are its eigenvalues, a term we can understand through the simplest example.[6]

With no field coupling our two levels, call the Hamiltonian H_0. In the zero/one basis, its action is

$$H_0 = \begin{pmatrix} E_0 & 0 \\ 0 & E_1 \end{pmatrix}, H_0 \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} E_0\alpha \\ E_1\beta \end{pmatrix}. \tag{5.2}$$

The square array is a matrix. To produce the upper output, multiply the entries of its top row by the two input components and add; do the same with the bottom row for the lower output. Here the zeros make the task especially simple. Each amplitude is multiplied by its own level's energy. There is no term mixing the components.

In particular, $H_0|0\rangle = E_0|0\rangle$: acting on the zero state gives the same vector multiplied by a number. That makes $|0\rangle$ an eigenstate and E_0 its eigenvalue. The one state obeys the corresponding equation with E_1. These two relations identify states of definite energy for this Hamiltonian.

A Hamiltonian matrix is not a table of transition probabilities. Its entries have energy units, can include negative or complex numbers, and need not sum to one. It enters the evolution equation; only afterward do we calculate measurement probabilities from the resulting state. Changing basis changes the matrix's entries without changing the physical operator.

In this closed two-state model, the Hamiltonian is Hermitian: exchanging its rows and columns and taking the complex conjugate returns the same matrix, $H = H^{\dagger}$. The dagger denotes that combined operation, called the adjoint. Hermiticity ensures real energy eigenvalues and evolution that preserves total probability; it does not make the matrix entries probabilities.

A state that changes only on paper

Begin with an atom prepared in a single energy eigenstate. The solution of Eq. (5.1) is particularly simple. For either level, labeled $j = 0$ or $j = 1$,

$$|\psi_j(t)\rangle = e^{-iE_j t/\hbar}|j\rangle. \tag{5.3}$$

The exponential is the unit arrow we learned to read in Chapter 3. Its angle is $-E_j t/\hbar$. As time passes, this amplitude turns without changing its length. The minus sign fixes the direction in our convention.

We can check the solution directly. A factor $e^{-iEt/\hbar}$ changes at the rate $-iE/\hbar$ times itself. Multiplying that rate by $i\hbar$, as the left side of the Schrödinger equation requires, gives E times the original factor. That is exactly what the Hamiltonian does to an energy eigenstate.

But there is only one factor multiplying the entire state. It is a global phase. Every probability for a fixed measurement remains unchanged. This is a *stationary state*: stationary in its physical predictions, even though its chosen vector representative acquires a time-dependent phase.[7]

Stationary does not mean that an electron sits motionless at a point. The claim concerns a quantum state and its measurement statistics, not a little object's hidden itinerary. Nor must every stationary preparation have a definite energy: a classical mixture of energy eigenstates is stationary too. Its components' weights do not change under H_0.

This also tells us why the zero of energy is arbitrary within this fixed, nongravitational model. Add the same constant C to both levels, and every component acquires the same extra factor $e^{-iCt/\hbar}$. It changes no measurement probability. Raising only one level would be different: that would change the gap, not merely the reference from which energy is counted.

A very large common energy does not, by itself, make the qubit race through distinguishable states. The observable change we are about to calculate depends on a difference between energies.

Waiting becomes a phase control

Now prepare the plus state, $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Each energy component evolves with its own phase factor. Because the evolution is linear, we can evolve them separately and combine the results:

$$|\psi(t)\rangle = e^{-iE_0 t/\hbar}\frac{|0\rangle + e^{-i\omega_0 t}|1\rangle}{\sqrt{2}}, \omega_0 = \frac{\Delta E}{\hbar}. \tag{5.4}$$

The first factor is global and can be omitted when calculating probabilities. The second phase is relative. The Greek letter omega, ω_0, names the angular frequency associated with the gap. It measures radians per second. The corresponding frequency in cycles per second is $f_0 = \Delta E / h$, so $\omega_0 = 2\pi f_0$.

This is the phase control from the interferometer, now supplied by a waiting interval. The two alternatives are internal energy states rather than spatial routes. Their relative phase turns clockwise in our convention because E_1 exceeds E_0.

Use the same X references as before, the plus and minus states. Taking the overlap with $|+\rangle$ gives an amplitude proportional to $1 + e^{-i\omega_0 t}$. The probability is

$$P_X(+;t) = \frac{1+\cos(\omega_0 t)}{2}, P_Z(1;t) = \frac{1}{2}. \tag{5.5}$$

The semicolon separates the outcome label from the waiting time. These are predictions for separate trials: prepare the same input, wait the chosen interval, and then make one measurement. We are not repeatedly measuring a single undisturbed atom.

At zero time, X gives plus with certainty. After a quarter-cycle, it gives equal probabilities. After a half-cycle, the state is the minus state, apart from its irrelevant global phase, and X gives minus with certainty. After a full cycle, the initial state returns. At every one of these times, the energy measurement still gives each level with probability one-half.

For a numerical illustration, choose a gap for which $\Delta E / h$ is one million cycles per second, or one megahertz. A full relative-phase cycle takes one microsecond, a millionth of a second. The X-plus probability is one-half after a quarter of a microsecond and zero after half a microsecond. These are chosen model values, not the measured transition of a named atom.[8]

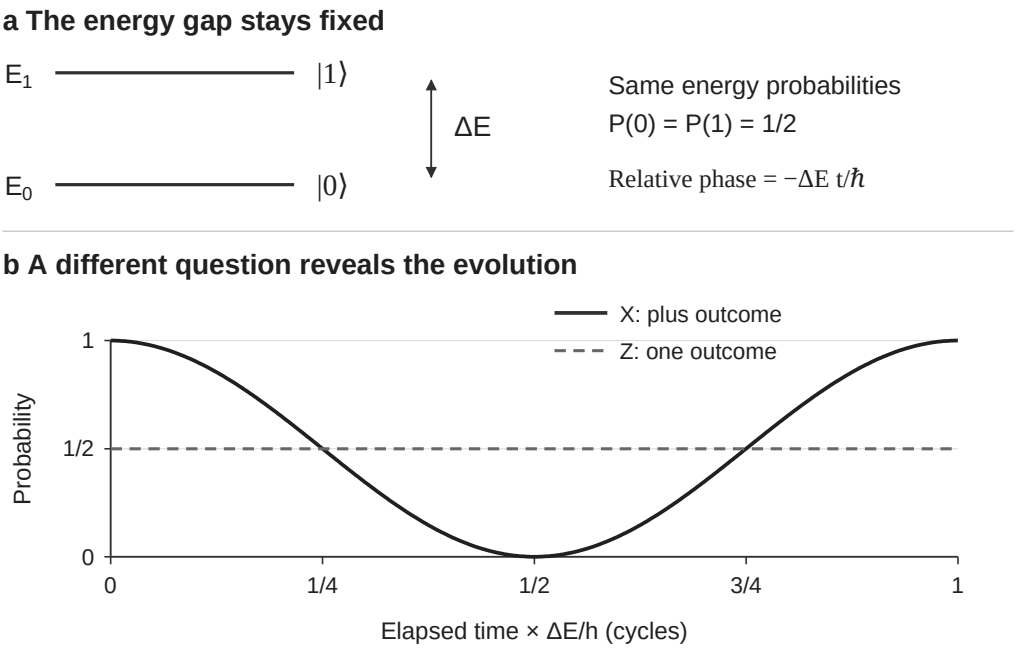


Figure 5.1. **Waiting changes one answer while leaving another unchanged.** The upper panel identifies the fixed energy gap of the two-level model; vertical spacing represents energy, not position. The lower panel shows the theoretical probabilities for fresh preparations of the plus state, followed by free evolution and either X or Z measurement. Time is measured in relative-phase cycles, $\Delta E\,t\,/\,h$. The solid curve is X-plus; the dashed line is the excited-level probability. No drive, decay, or intermediate measurement is included. The graph is not experimental data.

There is no violation of energy conservation. The average energy remains $\left(E_0+E_1\right)/2$, and the entire energy-outcome distribution is unchanged. What evolves is a relationship between components. Energy conservation is not a requirement that every measurable quantity remain constant.

On the Bloch map, this state moves around the equator. Its coordinates are $x=\cos\left(\omega_0 t\right)$, $y=-\sin\left(\omega_0 t\right)$, and $z=0$. The minus sign in y matters: at the first quarter-cycle the state is $|-_y\rangle$, not $|+_y\rangle$. A Y measurement can distinguish the direction of the phase change, whereas the cosine in X is unchanged when the sign of the angle is reversed.

The fair mixture of the two energy states behaves differently. It remains at the center of the Bloch ball and gives one-half for both X outcomes at every time. Waiting does not convert an unread energy lottery into the moving pure-state preparation. The two cases can agree at one time without agreeing throughout the experiment.

An oscillating signal alone does not distinguish quantum behavior from every classical model. A classical oscillator can also change phase while keeping its energy fixed. Our comparison is more specific: the coherent preparation and the fair energy mixture have the same energy-outcome probabilities, but only the coherent preparation gives the changing X probabilities calculated here.

A change that keeps the information

The evolution we have described is called *unitary*. Such a transformation preserves the normalization of every state and the overlaps between states. A known unitary operation has an inverse: mathematically, the original state can be recovered without first learning its amplitudes.[9]

Our example makes this concrete. Free evolution has multiplied the one component, relative to the zero component, by $e^{-i\omega_0 t}$. An accurately controlled operation multiplying it by $e^{+i\omega_0 t}$ cancels that change for any input amplitudes, not just the plus state. Producing that inverse in a laboratory requires the appropriate control. Waiting for negative time is not the instruction.

No information about which energy was present has been written into an external marker. This is different from making a measurement and ignoring its outcome. For example, an ideal Z measurement of the plus state, with its outcome left unread, leaves the qubit, considered by itself, in the fair energy mixture. Reversing a phase alone will not turn that mixture back into the original pure state.

There is a useful constraint here. A unitary cannot turn every possible unknown input into the same reset state. If it mapped both orthogonal inputs $|0\rangle$ and $|1\rangle$ to $|0\rangle$, their initially zero overlap would become one. Resetting a physical device therefore requires something beyond a closed, reversible operation on that device alone—such as transferring the unwanted distinction elsewhere. We will return to that cost when we examine computation.

For one isolated qubit, a unitary rotates the Bloch ball without moving an interior point to its surface. A pure state remains pure; the central mixture remains central. The way a system is changing is therefore as important as whether its displayed probabilities happen to change.

Turning a transition on

Free evolution under H_0 cannot transfer population from one energy level to the other. The matrix in Eq. (5.2) has no mixing terms. To prepare superpositions or deliberately exchange the levels, change the Hamiltonian by applying a field that couples them.

Drive the atom with an oscillating field near the transition frequency ω_0. With suitable coupling, small coherent contributions can build up into a substantial transfer. The strength of the interaction sets a second frequency, the *Rabi angular frequency*, Ω. It is not the transition frequency: ω_0 characterizes the level separation; Ω characterizes how strongly we drive the transition.[10]

A simple calculation becomes possible on resonance, when the drive frequency matches ω_0. Describe the amplitudes relative to a reference turning with the drive. This rotating description removes the rapid carrier-phase motion, much as describing two rotating hands by the angle between them removes their shared rotation. We must transform the

controls and measurement references consistently; changing the description is not itself a physical operation.

For a suitably chosen drive phase, the effective Hamiltonian is

$$H_{rot} = \frac{\hbar \Omega}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{5.6}$$

Now the off-diagonal entries mix the amplitudes. Acting on a zero-state input produces a term along the one state, and vice versa. The entries are still energies, not probabilities. The subscript “rot” reminds us that this is the generator in a rotating description, not the bare atomic energy operator with the level gap somehow removed from the laboratory.

For a sinusoidal drive, this form uses the *rotating-wave approximation*: rapidly oscillating terms are neglected. It is useful when the drive is near resonance and the coupling is sufficiently small compared with the transition frequency. We also neglect decay and excitation of other levels. Making a pulse arbitrarily strong does not preserve these assumptions.[11]

We already know how to solve this model. Its plus state is an eigenstate with eigenvalue $+\hbar \Omega / 2$, while its minus state has eigenvalue $-\hbar \Omega / 2$. Write the initial zero state as $(|+\rangle + |-\rangle)/\sqrt{2}$, give these two components their phase factors, and rewrite the result in the zero/one basis. The two output amplitudes make that recombination explicit.

The zero-state amplitude is $\left(e^{-i\Omega t/2} + e^{i\Omega t/2}\right)/2 = \cos\left(\Omega t/2\right)$, whereas the one-state amplitude is $\left(e^{-i\Omega t/2} - e^{i\Omega t/2}\right)/2 = -i\sin\left(\Omega t/2\right)$. The sum and difference of the two phase factors therefore give

$$|\psi_{rot}(t)\rangle = \cos\left(\frac{\Omega t}{2}\right)|0\rangle - i\sin\left(\frac{\Omega t}{2}\right)|1\rangle. \tag{5.7}$$

The label “rot” belongs to our representation of the state. Returning to the laboratory description restores the rapid phase evolution but does not change the probabilities for the two energy levels. Squaring the one component gives

$$P_Z(1;t) = \sin^2\left(\frac{\Omega t}{2}\right), t_\pi = \frac{\pi}{\Omega}. \tag{5.8}$$

The probability now rises from zero to one and returns. A pulse lasting t_π transfers the initial zero state to the one state, up to a global phase. It is called a pi pulse because the Bloch vector turns through π radians. Half that duration is a pi-over-two pulse. With the particular drive phase used in Eq. (5.6), it prepares $(|0\rangle - i|1\rangle)/\sqrt{2}$. Changing the drive phase chooses a different equatorial direction, including the plus state.[12]

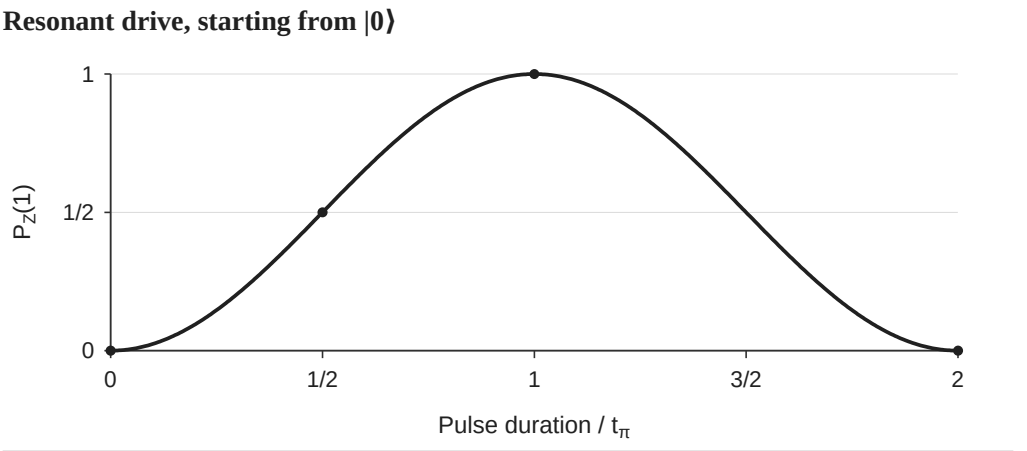


Figure 5.2. **Controlling the population with pulse duration.** The curve is the ideal resonant prediction in Eq. (5.8), starting from the zero state with constant Ω. The pi-over-two pulse gives equal energy probabilities; the pi pulse transfers the population; the two-pi pulse returns the initial state up to a global phase. The displayed halfway superposition uses the drive phase of Eq. (5.6). Each point represents a separate preparation, pulse, and energy readout. No continuous measurement, damping, or experimental data is shown.

For example, stipulate $\Omega/(2\pi) = 100,000$ cycles per second. A pi pulse then lasts five microseconds; a pi-over-two pulse lasts two and a half. The factor of two is worth checking: Ω measures the rate of the Bloch rotation, while the amplitudes contain half that angle.

The atom's bare average energy changes during this driven process. That is not a free source of energy: the control field can exchange energy with the atom. Conservation applies to the complete accounting, including the source and interactions, not to the bare two-level Hamiltonian while we are driving it.

Nor does the smooth curve mean that a particular atom continuously reports a fraction of an excitation. Between measurements, the model evolves amplitudes coherently. An energy measurement returns one of the two allowed outcomes. A curve assembled from many separately terminated trials is not a movie of repeated jumps by one atom.

A clock made from a comparison

We can now build an experiment in which waiting, rather than continuous driving, does most of the useful work.

Start in the zero state. Apply a short pi-over-two pulse with a phase chosen to prepare the plus state. Turn off the drive for an interval T. Then apply a second pi-over-two pulse and measure Z. Choose the second pulse's phase so that, with no accumulated relative phase error, it reverses the first preparation and returns the atom to zero.

The first pulse creates two coherent internal alternatives. They acquire different phases while the drive is off. The second pulse brings their contributions together in the measured populations. The structure resembles our optical interferometer: split, accumulate a phase difference, recombine. Here the alternatives occupy the same atom, not separate optical paths. This is Ramsey interferometry, named for Norman Ramsey's separated-field method.[13]

An actual pulse generator or laser supplies a phase reference. During the waiting interval, the atom evolves at its transition frequency while that reference advances at its own frequency ω_r. What the second pulse reads is their mismatch, not an absolute phase. Define the detuning $\delta = \omega_0 - \omega_r$. In the ideal short-pulse limit, our chosen readout convention gives

$$P_{return}(0;T) = \frac{1+\cos(\delta T)}{2}, \delta = \omega_0 - \omega_r . \tag{5.9}$$

This is the earlier interference calculation applied to the accumulated phase relative to the reference. On resonance, the relative phase stays aligned and the return is certain, even though the laboratory-frame amplitudes have continued to turn. At other frequencies, changing T or changing the reference frequency moves us through bright and dark settings.

Take a frequency mismatch of one cycle per second. In a quarter of a second, the mismatch accumulates a quarter-cycle: the return probability becomes one-half. At half a second it becomes zero; at one second it returns to one. The atom's actual transition frequency can be vastly greater than that mismatch. Measuring the slowly accumulating difference is the point.[14]

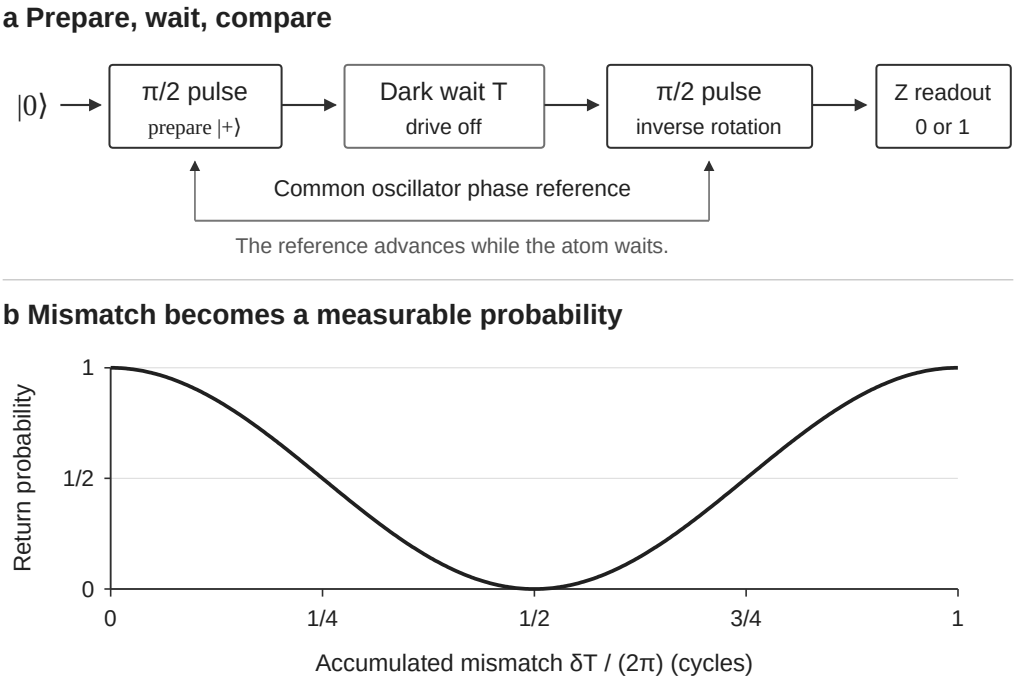


Figure 5.3. **Two pulses turn elapsed time into a frequency comparison.** The first pulse prepares the plus state; the second has the phase convention that returns it to zero when the detuning is zero. The common oscillator establishes both pulse phases while advancing during the dark interval. The lower curve shows Eq. (5.9) versus the accumulated mismatch $\delta T / (2\pi)$ in cycles. Pulses are idealized as short enough to neglect detuning during them, and loss, decay, and phase noise are absent. Blocks indicate operations, not apparatus dimensions. Probabilities are theoretical; no outcomes are discarded.

A clock can use repeated comparisons to adjust an oscillator toward an atomic transition and then count the oscillator's cycles. It does not obtain a continuous stream of perfectly timed ticks by watching an atom jump between levels. The reference, the interrogation sequence, the statistics, and the corrections are all part of the clock.[15]

Equation (5.9) is also a warning about inference. A cosine alone does not distinguish a positive frequency error from a negative one. Shifting the second pulse's phase by a known quarter-cycle supplies a complementary response. Longer waiting makes a small frequency difference accumulate more phase, but also allows more opportunity for phase fluctuations, decay, and ambiguity about how many cycles have elapsed. Sensitivity, stability, and usable interrogation time must be considered together.

The model meets its controls

These operations are not merely instructional diagrams. In a preprint posted in September 2026, Chris Caron and colleagues reported driving strontium-ion qubits with an integrated photonic laser. They used Rabi oscillations to examine population transfer and Ramsey measurements to examine phase coherence. The work illustrates why both tests matter: a device must control the amplitude transfer and maintain a sufficiently stable phase reference. It remains identified here as a preprint, rather than as a peer-reviewed result.[16]

The researchers also distinguished mechanisms limiting different measurements. Their analysis associated the decay of the Rabi oscillations with residual ion motion rather than simply assigning every fading signal to laser phase noise. The practical lesson is narrower and more useful than “the qubit lost its quantumness”: a particular experiment can reveal a particular imperfection, and the remedy depends on its cause.

Approximations must be tested as well. In a peer-reviewed study published in December 2025, Martijn Zwanenburg and colleagues treated fast control of a low-frequency superconducting qubit beyond the rotating-wave approximation. They included terms that the simpler description drops and tested corrections to the control pulses. This is not a failure of the Schrödinger equation. It is a case in which a simplified Hamiltonian no longer describes the desired operation accurately enough.[17]

The distinction will recur throughout this book. The evolution law, the Hamiltonian chosen to represent a device, the approximations used to solve it, and the quality of the implemented controls are separate parts of a prediction. A disagreement with data does not identify, by itself, which part needs revision.

What waiting does not explain away

Our isolated two-level model has no spontaneous decay. An atom prepared in the upper energy eigenstate remains there under H_0. A real excited atom can emit radiation because it interacts with electromagnetic modes we have left out. To describe that process, enlarge the system to include the field or use an effective open-system description of the atom. The two-level unitary model is useful over suitable intervals, not a promise of an infinitely stable excitation.[18]

Similarly, uncertainty in the phase from trial to trial can make an ensemble’s interference fade even when each trial is described by a coherent rotation. An interaction can also correlate the qubit with its surroundings, requiring a mixed state when we retain only the qubit’s description. Neither process is the same as the known, reversible phase accumulation in Eq. (5.4).

We have therefore found several distinct meanings of change. A global phase changes a vector representative but not the state it represents. Relative phases can change predictions while preserving every energy probability. A controlled interaction can

transfer population. A measurement or an uncontrolled environment can leave a different kind of record and a different description of the remaining system.

The Schrödinger equation supplies a common law for coherent evolution. The Hamiltonian tells us which evolution occurs. Knowing a state's coordinates without knowing the Hamiltonian is like knowing where a journey starts without knowing what will govern the motion.

So far we have stipulated two energy levels. In a molecule, we must learn where the levels come from. Bring atomic constituents together, and their interactions enter the Hamiltonian. The resulting energy states need not be states of either separated atom. A stable bond does not require electrons to stop; it requires a quantum description of the combined system and its energies.

We now have the language for that next step. The same equation that turns a qubit's phase can help explain why atoms form molecules.

Notes and sources

1. John Preskill, *Lecture Notes for Physics 219/Computer Science 219*, Chapter 2, “Foundations I: States and Ensembles,” §2.2, on physical realizations of a qubit. The atomic model here retains two internal states and treats other levels, motion, and environmental interactions as negligible during the specified intervals. The zero/one notation is reused from Chapter 4, but no spatial-route interpretation is carried over. Qubit realizations.

2. NIST, Handbook of Basic Atomic Spectroscopic Data, “Strong Lines of Hydrogen.” The listed optical values are air wavelengths and contain resolved components. The text rounds them to whole nanometres; the frequency equation separately specifies vacuum wavelength. This is an observed spectral pattern, not a measurement of a spatial orbital. Strong Lines of Hydrogen.

3. J. J. Balmer, “Notiz über die Spectrallinien des Wasserstoffs,” Annalen der Physik 261, 80–87 (1885); N. Bohr, “On the Constitution of Atoms and Molecules,” Philosophical Magazine 26, 1–25 (1913). Balmer’s empirical wavelength relation and Bohr’s atomic interpretation are distinct achievements. Balmer 1885. Bohr 1913.

4. NIST, “Atomic Spectroscopy: Spectral Lines,” distinguishes level differences, line wavelengths and transition rules. The bridge describes inference within a specified atomic and radiation model, without claiming that line positions alone determine states or interaction strengths. NIST Atomic Spectroscopy Spectral Lines.

5. Bureau International des Poids et Mesures, “The International System of Units (SI): Defining constants.” The exact SI value of h is $6.62607015 \times 10^{-34}$ joule seconds; $\hbar = h / (2\pi)$. The chapter’s numerical frequencies and durations are stipulated examples, not measurements of those constants or a specific atom. Defining constants.

6. Richard P. Feynman, Robert B. Leighton, and Matthew Sands, *The Feynman Lectures on Physics*, Vol. III, Chapter 8, §§8–3 to 8–4. The Hamiltonian generates time evolution and is Hermitian in this finite-dimensional closed-system model. In a fixed orthonormal basis its diagonal entries are real and opposite off-diagonal entries are complex conjugates. Equation (5.2) is the chapter’s diagonal example, not the most general qubit Hamiltonian. Hamiltonian and time evolution.

7. Feynman, Leighton, and Sands, Vol. III, Chapter 7, §7–1, on definite-energy states and phase evolution. Stationarity here means time-independent probabilities for fixed, time-independent measurements under a time-independent Hamiltonian. A mixture diagonal in the energy basis is also stationary. The shift of energy origin concerns adding a scalar multiple of the identity to the complete model, not altering one subsystem’s energy relative to an external reference. Energy and phase.

8. Equations (5.4)–(5.5), Figure 5.1, and the megahertz example are original calculations using Eq. (5.1) and the X, Y, and Z conventions of Chapter 4. With $E_1 > E_0$, the relative factor is $e^{-i\Delta E t/\hbar}$ and the equatorial Y coordinate is negative for small positive times. The chosen gap is not identified with a named atomic transition. For the underlying phase and probability rules, see John Watrous, IBM Quantum Learning, Basics of Quantum Information, “Single systems: Quantum information.” States and unitary operations.

9. Watrous, “Single systems: Quantum information,” section “Unitary operations”; Preskill, Chapter 3, “Foundations II: Measurement and Evolution,” §3.1. The inverse-phase example is an operation on the qubit, not a reversal of laboratory time. A unitary preserves inner products and the eigenvalues of a density matrix; a qubit-only reset to a fixed pure state is not such an operation. Unitary operations; measurement and evolution.

10. I. I. Rabi, “Space Quantization in a Gyrating Magnetic Field,” *Physical Review* **51**, 652–654 (1937), doi:10.1103/PhysRev.51.652. For the rotating-frame treatment of driven two-state systems, see also Feynman, Leighton, and Sands, Vol. III, Chapter 11. The chapter uses angular frequencies throughout formulas containing Ω or ω; dividing by 2π gives cycles per second. Rabi’s paper; two-state dynamics.

11. M. F. S. Zwanenburg and colleagues, “Single-qubit gates beyond the rotating-wave approximation for strongly anharmonic low-frequency qubits,” *Physical Review Research* **7**, 043290 (2025), doi:10.1103/z62h-kcnh. Published 12 December 2025. Equation (5.6) is the ideal resonant rotating-wave limit, with constant real coupling and a stated drive phase. Its eigenvalues are those of an effective rotating-frame generator, not the undriven atom’s two measured energies. Published article; author manuscript.

12. Equations (5.7)–(5.8), Figure 5.2, and the pulse-duration example are original calculations for Eq. (5.6). The initial state is $|0\rangle$; the pi-over-two result is $|-_y\rangle$ for this drive phase. A two-pi pulse gives $-|0\rangle$, the same pure state up to global phase. The resonance and constant-coupling assumptions are essential. The word “population” means the corresponding level-measurement probability, or the fraction estimated across repeated preparations.

13. N. F. Ramsey, “A Molecular Beam Resonance Method with Separated Oscillating Fields,” *Physical Review* **78**, 695–699 (1950), doi:10.1103/PhysRev.78.695. Figure 5.3 is an original functional diagram of a time-separated two-pulse protocol, not a reproduction of Ramsey’s molecular-beam apparatus. The second pulse is phased to invert the first on resonance; other phase conventions interchange bright and dark outputs. Original paper.

14. Equation (5.9) and the one-hertz example are original ideal calculations. Let the first pulse be a positive pi-over-two rotation about the Bloch Y axis and the second its inverse in the drive reference. The intervening phase operation is $diag\left(1, e^{-i\delta T}\right)$. Their product gives the stated zero-return probability. Detuning during the pulses is neglected, requiring pulses short compared with the relevant phase-mismatch timescale, while remaining within the chosen drive model’s validity.

15. N. F. Ramsey, “Experiments with Separated Oscillatory Fields and Hydrogen Masers,” *Reviews of Modern Physics* **62**, 541–552 (1990), doi:10.1103/RevModPhys.62.541; N. F. Ramsey and H. B. Silsbee, “Phase Shifts in the Molecular Beam Method of Separated Oscillating Fields,” *Physical Review* **84**, 506–507 (1951), doi:10.1103/PhysRev.84.506. The second paper explains how controlled phase shifts change the resonance response. Longer interrogation is not an unconditional improvement when noise, decay, cycle ambiguity, and oscillator stability are included. Clocks and separated fields; phase-sensitive readout.

16. C. Caron and colleagues, “Single- and Two-Qubit Gates Driven by an Integrated Photonic Laser,” arXiv:2609.07708v1, posted 7 September 2026. Preprint; no journal publication is asserted here. See §§II.4–II.5 for phase and population diagnostics using strontium-88 ions. This chapter’s figures are original ideal calculations, not those experimental data. Versioned preprint.

17. Zwanenburg and colleagues, *Physical Review Research* **7**, 043290 (2025), as in note 11. The experiment uses a superconducting fluxonium qubit and tests pulse corrections associated with counterrotating terms and noncomputational levels. Its relevance here is the domain of validity of an approximate Hamiltonian, not a claim that one simple formula describes every control platform. Published article; research group record.

18. Preskill, “Foundations II: Measurement and Evolution,” §§3.4.2–3.4.3, on dephasing and amplitude damping. Feynman, Leighton, and Sands, Vol. III, Chapter 7, §7–1, explicitly qualifies treating an excited atom as a stable definite-energy state when coupling to radiation is neglected. The full atom–field dynamics and the atom’s reduced description are different levels of modeling. Open-system models; excited-state qualification.

CHAPTER 6

Why Atoms Form Molecules

Two protons brought close together repel. Add one electron, and a bound molecule becomes possible. What has changed?

The charges have not changed their rules. The protons still repel each other, and each attracts the electron. Nor has the electron become a tiny tether stretched between them. The change is in the quantum states available to the combined system—and in their energies.

This is a useful place to begin chemistry because there is almost nothing to hide behind. The hydrogen molecular ion, H_2^+, contains two protons and one electron. The subscript counts hydrogen nuclei; the superscript records the net positive charge. It is not the neutral hydrogen molecule, which has a second electron. A single electron is enough for our first bond.[1]

In Chapter 5 we supplied an atom with two energy levels and calculated what happened next. Here we reverse the task. We supply the particles and their interactions, then ask which energies and states follow. The answer must explain not just why bringing the particles together can lower the energy, but why they do not simply collapse into one place.

Lower than what?

An energy is useful only with its reference stated. For this molecule, the relevant separated fragments are a hydrogen atom in its lowest electronic state and a bare proton, far apart. The electron is already bound to one proton in that comparison. We must improve on that arrangement, not merely on three particles with all their attractions removed.

The question is therefore precise: can the combined system have an energy below the energy of those separated fragments? If it can, separating it again requires energy. That is the energetic content of a bond in the example we are studying.

Begin by holding the two nuclei at a chosen separation, R. Calculate the electron's state and the total energy at that separation. Repeat at another separation. The resulting energy curve will tell us whether there is a favorable distance between the extremes of being far apart and being pressed too close together.

Holding the nuclei fixed is a calculational step, not a claim that nuclei are immovable. They are much heavier than electrons, and separating the electronic calculation from nuclear motion is often useful. This is the starting point of the Born–Oppenheimer treatment. We will let the nuclei move later. Near closely spaced electronic states, or at

sufficiently high precision, coupling between the motions can require a more complete calculation.[2]

What goes into the energy

Place nucleus A at one end of the separation and nucleus B at the other. Let r_A and r_B be the electron's distances from them. For fixed nuclei, our nonrelativistic Hamiltonian is

$$\hat{H}_R = \hat{T}_e - \frac{C}{r_A} - \frac{C}{r_B} + \frac{C}{R} \tag{6.1}$$

Here $C = e^2/(4\pi\varepsilon_0)$ collects the electrical constants, with e the magnitude of the electron's charge. The constant ε_0 is vacuum permittivity, which sets the electrostatic coupling in SI units. The two negative terms are attractions. The positive term is the repulsion between the protons. The remaining term, $\hat{T}_e$, is the electron's kinetic-energy operator. It plays the same energy-accounting role as momentum squared divided by twice the mass, but acts on a quantum state rather than on a specified classical trajectory.[3]

All four terms matter. Counting the attractions and forgetting the nuclear repulsion would not explain a molecule. It would describe a different problem. Counting only electrostatic energies would also be insufficient: changing the spatial form of the electron's state changes its kinetic energy.

We neglect relativity, magnetic interactions, and radiation here, and keep the electron's spin fixed. These choices are adequate for our first calculation of bonding, not for precision spectroscopy. The task is to find the lowest energy permitted by this Hamiltonian at each R.

The electron is no longer described by just two amplitudes. Its spatial wavefunction, $\psi(r)$, assigns an amplitude density throughout three-dimensional space. The squared magnitude, $|\psi(r)|^2$, is a probability density. Add it over a small volume to obtain the probability of finding the electron there; add it over all space and the result must be one. A height on a density plot is not itself the probability of an exactly specified point.

We will not solve for every possible spatial shape at once. We will choose two known shapes and see how much a carefully restricted calculation can explain. The restriction will stay visible, so that we can improve it rather than mistake it for an exact description.

Where the atomic orbital comes from

The two shapes we need are not freely drawn electron clouds. Each comes from a simpler physical problem: one electron attracted to one fixed proton. With the same electrical constant C as above, the stationary equation is

$$\left[-\frac{\hbar^2}{2m_e}\nabla^2-\frac{C}{r}\right]\phi(r)=E\,\phi(r).$$

The electron mass is m_e, and r is its distance from the proton. The symbol ∇^2 adds the second spatial derivatives: it measures how the amplitude bends in the three spatial directions. Through the kinetic-energy term, concentrating or sharply varying the state has an energy consequence. This is the spatial version of the Hamiltonian equation we used for two amplitudes in Chapter 5.[4]

Solving the differential equation is only part of the task. A bound orbital must have a finite integral of its squared magnitude, which we set to one. It must satisfy the regular condition at the proton and decay at large distance. There is no hard wall at an atomic radius. The equation and these conditions together select acceptable spatial states and energies. For this Coulomb problem, the negative bound-state energies form a discrete series; unbound electron states also exist and have a continuous range of energies. "Quantum" does not mean that every energy in every situation is discrete.

We can check the simplest bound solution without solving the entire family. Try a spherically symmetric exponential with adjustable decay length a. Away from the proton, applying the Hamiltonian gives

$$\frac{\widehat{H}_H e^{-r/a}}{e^{-r/a}}=-\frac{\hbar^2}{2m_e a^2}+\frac{1}{r}\left(\frac{\hbar^2}{m_e a}-C\right).$$

For this to be one constant energy at every r, the coefficient of 1/r must vanish. That fixes a to the Bohr radius a_0. Normalizing then gives the hydrogen 1s orbital and its energy:

$$a_0=\frac{\hbar^2}{m_e C},\ \phi_{1s}(r)=\frac{e^{-r/a_0}}{\sqrt{\pi a_0^3}},\ E_{1s}=-\frac{C}{2a_0}.$$

The letter s identifies zero orbital angular momentum; the 1 labels the lowest principal level. This calculation checks a candidate solution and its normalization. Establishing that it is the ground state, and finding the complete family, requires the fuller eigenvalue problem. Its bound energies are $E_n=-C/\left(2a_0n^2\right)$, with n = 1, 2, 3, and so on. Their differences recover the leading hydrogen spectral pattern discussed in Chapter 5. Schrödinger's 1926 work made this link through a wave equation and its acceptable solutions.[5]

The resulting energy scale is about 13.6 electronvolts below a separated electron and proton. It is not obtained by fitting a new orbital radius to each line. Nevertheless, agreement at this scale is not precision agreement: finite proton mass, relativistic effects,

radiative effects, and nuclear structure refine the comparison. A computed orbital is a model state whose consequences are tested through spectra and other measurements; a spectral line is not a direct image of its shape.

We can now translate this same normalized 1s function so that it is centered on A or on B. Each is an exact orbital of the corresponding one-proton approximation. Their combination will be a restricted trial description of the two-proton molecule, where neither function alone solves the new Hamiltonian.

Two centers, overlapping possibilities

Take the lowest-energy hydrogen orbital centered on A and call it ϕ_A. Take an identical orbital centered on B and call it ϕ_B. Each is a spherically symmetric, smoothly decaying amplitude: large near its own nucleus, smaller farther away. An orbital is a one-electron spatial state, not an orbit drawn through space.[6]

There is an important difference from the two optical routes. These orbitals overlap. The A-centered orbital extends into the region around B; the B-centered orbital extends into the region around A. They are not two perfectly distinguishable alternatives with disjoint support.

Chapter 4 gave us the mathematical test: their inner product. Write $S = \langle \phi_A | \phi_B \rangle$. For the real, positive hydrogen orbitals chosen here, S is positive and less than one at any nonzero finite separation. It approaches zero as the nuclei move far apart. This overlap must appear in the normalization.

The two simplest molecular trial states add or subtract the atomic shapes:

$$\psi_+ = \frac{\phi_A + \phi_B}{\sqrt{2(1+S)}}, \psi_- = \frac{\phi_A - \phi_B}{\sqrt{2(1-S)}} \tag{6.2}$$

The denominators are not decoration. Squaring and adding the numerator of the plus state over space gives $1 + 1 + 2S$: one from each normalized orbital and two overlap contributions. Dividing by $2(1+S)$ restores total probability one. For the minus state, the cross contributions have the opposite sign. This calculation is the continuous-space version of the inner products we already know.

In particular, we cannot take the coefficient of an A-centered orbital, square it, and announce the probability that the electron “belongs to A.” Those basis functions overlap. A probability needs a specified measurement; the names of the functions do not supply one.

Although the atomic functions overlap, the normalized plus and minus combinations are orthogonal to each other. Their symmetry makes their inner product vanish. They provide two useful reference states for this restricted calculation, not proof that the entire molecule is a qubit.

At the midpoint, the atomic amplitudes are equal. Addition reinforces them; subtraction cancels them. In fact, the minus state has a whole plane of zero amplitude halfway between the nuclei. Such a zero is called a node. On either side the amplitude can have opposite signs while its probability density remains nonnegative.

$$\left|\psi_{\pm}\right|^2 = \frac{\phi_A^2 + \phi_B^2 \pm 2\,\phi_A \phi_B}{2\left(1 \pm S\right)} \tag{6.3}$$

Equation (6.3) uses the real orbitals selected above. Its cross term is the interference contribution. Figure 6.1 shows how changing the relative sign changes both the amplitude and the density. The plus state is not an ordinary lottery that places the electron in an A-centered orbital on half the trials and a B-centered orbital on the rest.[7]

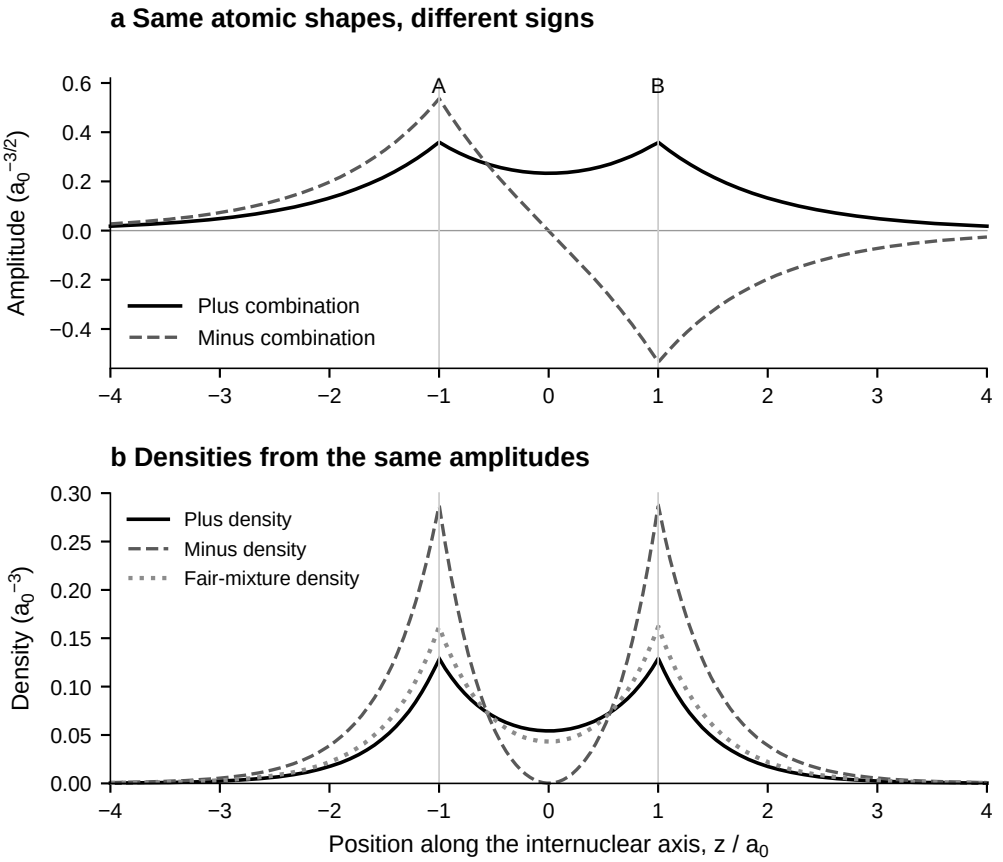


Figure 6.1. Adding amplitudes changes the spatial state. The upper panel shows the normalized plus and minus trial amplitudes along the line through the nuclei. The lower panel shows their squared magnitudes and, for comparison, the density of a fair classical mixture of the two atomic orbitals. Nuclei are at $z / a_0 = \pm 1$; the separation is $2\,a_0$. These are axial slices of three-dimensional functions, not one-dimensional probability distributions. The unit a_0 is the bohr. The curves use unmodified hydrogen 1s orbitals and are calculations, not measured electron tracks or images.

A bond requires more than a bright region

The extra density between the nuclei suggests why the plus combination might help. It gives the electron access to a region where both nuclei attract it. But the picture is not a complete energy calculation. The change in shape also changes kinetic energy, and the protons still repel. We need to evaluate the entire Hamiltonian.

Separate off the nuclear repulsion for a moment. Call the electron-only Hamiltonian $\widehat{H}_e$. Let d be its average energy in either one of the atomic trial orbitals, and let k be the off-diagonal matrix element $\langle \phi_A | \widehat{H}_e | \phi_B \rangle$. Symmetry makes the two diagonal entries equal and, for these real functions, the two off-diagonal entries equal.

The average electronic energies of the normalized combinations, followed by the total energies, are

$$\epsilon_{\pm} = \frac{d \pm k}{1 \pm S}, U_{\pm}(R) = \epsilon_{\pm} + \frac{C}{R} \tag{6.4}$$

For example, the plus numerator before normalization is $d + d + k + k$. Divide by $2(1+S)$ and the first formula follows. We have used precisely the overlap that would be lost by treating the atomic orbitals as orthogonal qubit labels.

Let us put numbers into this calculation. A convenient length unit is the bohr, a_0, about 0.0529 nanometers. A convenient energy unit is the hartree, E_h, about 27.2 electron volts. An electron volt is the energy gained by an electron accelerated through a potential difference of one volt. It equals one elementary charge multiplied by one volt, about 1.602×10^{-19} joules: an energy, not a voltage. In these units, the fixed-nucleus hydrogen ground-state energy is $-1/2$, and the proton repulsion at separation R bohr is $1/R$. The separated hydrogen atom and proton therefore have total energy -0.5 hartree.

At $R = 2a_0$, integrating the chosen hydrogen orbitals gives $S \approx 0.5865$, $d \approx -0.9725\, E_h$, and $k \approx -0.6992\, E_h$. Equation (6.4), including the $0.5\, E_h$ proton repulsion, then gives

$$U_+(2a_0) \approx -0.5538\, E_h, U_-(2a_0) \approx -0.1609\, E_h \tag{6.5}$$

The plus trial state is below the separated-fragment energy. The minus trial state is above it. This is already a quantitative reason to distinguish a bonding combination from an antibonding combination in this model. The labels summarize calculated behavior; they do not make it happen.[8]

Nor is the printed plus sign a universal instruction for making a bond. We chose both atomic functions positive. Reversing the sign convention for one would change which formula looks like a sum, without changing any physical state or energy. What matters is the spatial relationship and the Hamiltonian, not the typography.

We can even check which energy terms changed. For this fixed-shape plus state at $R = 2a_0$, the electronic kinetic energy is about $0.3863\, E_h$ and the total electrostatic

energy, including proton repulsion, is about $-0.9400\,E_h$. For the separated hydrogen atom and proton they are $0.5\,E_h$ and $-1\,E_h$. In this comparison the kinetic energy decreases more than the electrostatic energy increases, leaving the lower total in equation (6.5). The extra density between the protons therefore does not justify attributing the gain solely to more favorable electrostatic energy. This conclusion concerns the stated uncontracted trial functions; allowing their shapes to relax changes the division between kinetic and potential contributions.[9]

Finding the distance—and testing the approximation

Now repeat the calculation as the protons approach. Figure 6.2 plots both total trial energies relative to the separated fragments. The plus curve falls below zero, reaches a minimum, and rises steeply at short separation. In this limited model, its minimum occurs near $2.49\,a_0$, or 0.132 nanometers, and lies about 1.76 electron volts below the fragments.

This minimum expresses a balance. On its outer side, a small inward displacement lowers the energy; sufficiently far inward, a further inward displacement raises it. The energy curve therefore supplies an equilibrium distance for the clamped-nucleus calculation. It does not place an electron at rest between two protons.

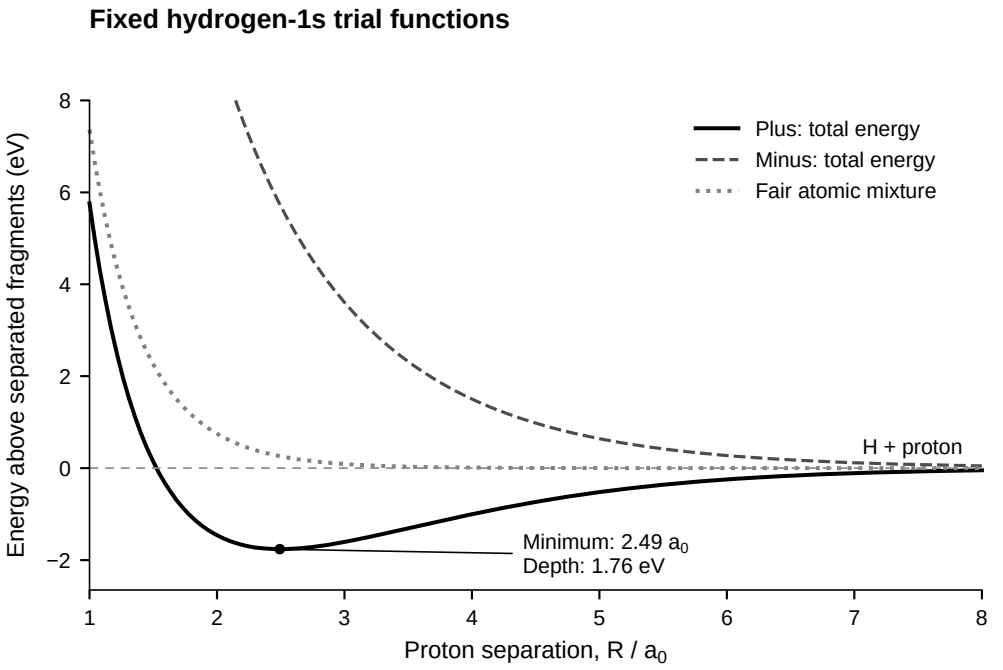


Figure 6.2. The energy comparison that tests binding. Total trial energies include electron kinetic energy, both electron–proton attractions, and proton–proton repulsion. Zero is the energy of a separated ground-state hydrogen atom and proton. The solid curve uses the plus combination; the dashed curve uses the minus combination. The dotted curve is the energy of the fair atomic-orbital mixture. The indicated minimum and depth belong to the fixed hydrogen-1s trial space, not to an exact calculation or an experiment. The restricted functions also omit long-range distortion of the atomic electron distribution.

The accuracy of the numbers is a different question from whether the calculation is intelligible. Our two atomic orbitals keep the size and shape of isolated hydrogen orbitals even when a second proton is nearby. There is no reason to expect that restriction to give the best molecular shape.

The variational principle gives us a disciplined way forward. For a normalized trial state, the average energy cannot lie below the true ground-state energy of the same Hamiltonian. To see why, imagine expressing the trial state in exact energy states: its average is a probability-weighted average of their energies. Such an average cannot be less than the lowest entry.

Consequently, an energy below the separated-fragment threshold is meaningful even from a limited trial state. But the particular minimum distance and well depth need improvement. The principle bounds the energy at fixed separation; it does not turn every approximate geometric prediction into a guaranteed bound.

As one improvement, allow the two orbitals to contract or expand together while retaining their simple exponential form. Optimizing this extra parameter and the separation lowers the trial minimum to about -0.5865 hartree, near $2.00\,a_0$. The depth

is now about 2.35 electron volts. These are two calculations with different permitted shapes—not two conflicting measurements.[10]

How much improvement have we bought? To answer fairly, compare calculations of the same quantity with the same physical assumptions. Each row below describes the lowest electronic state of H_2^+ with two point protons held fixed: the nonrelativistic Coulomb Hamiltonian of equation (6.1), including proton–proton repulsion. Each curve is minimized over separation. The reference is the same infinitely separated, fixed-nucleus ground-state hydrogen atom and proton, with energy $-0.5\,E_h$. Nuclear motion is excluded.

Table 6.1. How the permitted electronic shape affects a bond prediction.[11]

Electronic calculation	**Minimum separation R_e / a_0**	**Well depth D_e (eV)**
Fixed hydrogen 1s functions	2.49283	1.764
Common orbital scale optimized	2.0033	2.354
Accurate numerical reference	1.9972	2.793

The fixed orbitals predict a separation about 25 percent too large and a well about 37 percent too shallow relative to this reference. Allowing one scale parameter nearly repairs the distance, yet leaves the well about 16 percent too shallow. Agreement for one property does not certify another: the location of a minimum and its depth test different features of a curve.

This comparison isolates a particular limitation. We have changed the allowed electronic shapes while retaining the physical Hamiltonian. The remaining discrepancy therefore cannot be blamed on nuclear motion or on relativity, which all three rows omit. Enlarging the trial space can lower the variational energy further. Because the separated-atom reference is exact for this Hamiltonian, the restricted minima also give well depths no greater than the exact clamped-nucleus depth. There is no corresponding one-sided guarantee for the predicted separation.

The third row is itself a calculation, not a measurement. It tests how well we have solved the selected electronic problem. Testing the physical description against nature requires a further connection to a measured quantity. The nuclei move, and an experiment prepares particular states and detects particular transitions or fragments. We begin that connection next; Chapter 22 will carry it through to an experimental dissociation-energy inference.

A molecular energy eigenstate is stationary under the Hamiltonian of which it is an eigenstate, in the sense established in Chapter 5. Its probabilities do not show the electron repeatedly switching allegiance from one nucleus to the other. Superpositions of different energy states can evolve; a bonding eigenstate does not need a shuttle motion to hold the nuclei together. Our two-orbital combinations are eigenstates of the Hamiltonian restricted to that trial space; they are not exact eigenstates of the full Coulomb operator.

The nuclei also obey quantum rules

We have held the protons fixed long enough. Their separation is now a quantum variable too. The electronic energy curve becomes a potential energy in a second calculation: the motion of the nuclei. Even at the lowest allowed energy, they cannot be described as two classical objects permanently pinned at the bottom of the well.

For zero rotation, a simple radial model is

$$\left[-\frac{\hbar^2}{2\mu}\frac{d^2}{dR^2}+U_+(R)\right]\chi_v(R)=E_v\chi_v(R) \tag{6.6}$$

The reduced mass, μ, is half the proton mass for two equal nuclei. It accounts for their relative motion. The function χ_v describes the separation; $\left|\chi_v(R)\right|^2 dR$ gives the probability for a small interval of separations in this reduced radial description. The second derivative measures how the wavefunction bends as the separation changes. The integer v labels successive vibrational energy levels, starting at zero. This equation uses one electronic curve and omits coupling to other electronic states.[12]

The separation is positive. The regular reduced radial solution obeys $\chi_v(0)=0$, and a bound-state solution decays at large separation. Its normalization adds $\left|\chi_v(R)\right|^2 dR$ from zero to infinity to give one. There is no extra squared-radius factor: that factor has already been absorbed into this reduced radial function.

Its structure should now look familiar: kinetic energy plus a potential, acting on a state, gives an energy times that state. A molecule has not acquired a separate set of nonquantum rules merely because its nuclei are heavier.

Using the fixed-orbital plus curve from Figure 6.2, and a reduced mass of 918 electron masses, a numerical solution puts the lowest level about 0.111 electron volts above the curve's minimum. It remains about 1.65 electron volts below the separated-fragment threshold. Figure 6.3 shows this level and three higher ones. These are predictions of our explicitly restricted model, not precision spectral data.[13]

This second calculation answers a question that the drawing of a well alone leaves open: does the well actually support a bound nuclear state? Here it does. Confinement costs kinetic energy, and in a shallower or narrower well that cost could prevent a bound level. The minimum of an electronic curve is not, by itself, the whole molecular energy.

$$D_e=U(\infty)-U_{min},\ D_0=U(\infty)-E_{v=0} \tag{6.7}$$

The first quantity is the well depth. The second is the energy required to dissociate from the lowest vibrational level, with the same reference fragments and rotational specification. Our illustrative values are about 1.76 and 1.65 electron volts. Their difference is the zero-point vibrational energy in this model.

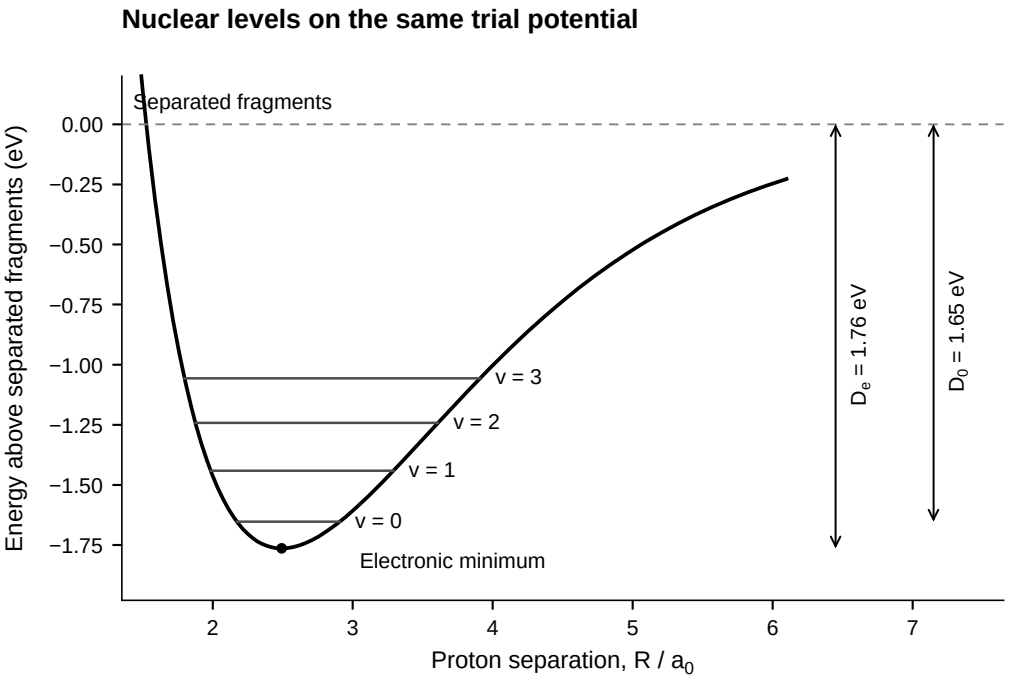


Figure 6.3. The well bottom is not the lowest molecular level. The curve is the same fixed-orbital plus potential as in Figure 6.2. Horizontal segments mark the four lowest eigenvalues of Eq. (6.6), drawn between their classical turning points for orientation; the nuclear wavefunctions do not stop abruptly at those points. The calculation uses zero rotation and $\mu = 918\, m_e$. The arrows distinguish D_e from D_0. All energies are relative to the model's separated fragments. Numerical levels are illustrative, not measured spectroscopy.

Calling the lowest state "vibrational" does not mean that its probability density must visibly slosh back and forth. An energy eigenstate has a stationary distribution of separations. A suitable superposition of levels can instead form a moving wavepacket. The same distinction between a stationary state and evolving relative phases applies to nuclear motion as to the atom in Chapter 5.

Having a bound state is not the same as forming it

There is a trap in the phrase "the system lowers its energy." A time-independent, isolated Hamiltonian does not make an arbitrary state relax into its ground state. Chapter 5 showed why: coherent evolution preserves the probabilities of its energy outcomes. Finding a lower-energy state is not a mechanism for reaching it.

If a hydrogen atom and a proton approach from far apart in a collision, their total energy does not vanish when they reach a favorable separation. To remain bound after an ordinary capture process, energy must be transferred elsewhere—for example to emitted radiation, another collision partner, or an external device. Which processes are allowed and how quickly they occur require a dynamical calculation beyond our curve.

There is no contradiction between a bond releasing energy when formed and requiring energy to break. These describe opposite directions between specified initial and final states, with the rest of the energy account included. The energy released is not destroyed; it has become available to some other part of the physical situation.

The existence of a bound molecule, the rate at which it forms, and its abundance in a hot or dilute environment are therefore different questions. An energy curve answers part of the first. It does not, alone, answer the other two.[14]

What changes when we add the second electron?

Add another electron and we obtain neutral molecular hydrogen, H_2. The new particle is attracted to both protons, but it also repels the first electron. We cannot predict the new bond by doubling the one-electron answer. There is an additional interaction and a new requirement on the state.

This extra structure also has an evidential history. Closely spaced spectral components and their changes in magnetic fields resisted an account based only on an electron's spatial motion. Pauli used an additional two-valued electron property in developing the exclusion principle published in 1925. Uhlenbeck and Goudsmit connected an intrinsic angular momentum and magnetic moment to these spectroscopic patterns in work of 1925–1926. The modern spin description preserves the successful relations without treating the electron as a mechanically spinning sphere.[15]

The earlier Stern–Gerlach experiment supplied a different observation: an inhomogeneous magnetic field split a beam of neutral silver atoms into two branches. The measured object was a deposited atomic-beam pattern, not a visible electron turning. In the modern description, the relevant silver ground-state magnetic response is dominated by its unpaired electron's spin. But the 1922 experiment originally tested space quantization; it did not begin with the later electron-spin theory. Observation and its eventual explanation have different dates.[16]

Exclusion adds a further requirement. Pauli used atomic spectroscopy and shell structure to constrain which electron configurations occur. In our nonrelativistic calculation, the antisymmetric electron state is an input supported by that wider body of evidence; it is not a consequence of Coulomb repulsion or of merely erasing particle names. Relativistic quantum field theory connects half-integer spin with fermionic statistics under additional physical assumptions. That spin–statistics result goes beyond the Hamiltonian written here. It explains why the spin label and the exchange rule belong together without pretending that this elementary bonding calculation has derived their connection.[17]

The requirement comes from the electrons being identical fermions. Exchanging their full labels—including position and spin—changes the sign of the total two-electron wavefunction:

$$\Psi\left(q_1, q_2\right) = -\Psi\left(q_2, q_1\right) \tag{6.8}$$

Here q_1 and q_2 each stand for spatial and spin variables. The minus sign does not describe a new electrostatic force. It restricts which joint states are allowed. In particular, two electrons cannot occupy the same complete one-electron state. This is the Pauli exclusion principle.[18]

Spin is an intrinsic quantum degree of freedom, not a claim that the electron is a little rotating ball. Two electrons can share the same spatial bonding orbital when their joint spin state is antisymmetric. The symmetric spatial part and antisymmetric spin part together supply the required minus sign. Different spin values do not turn the electrons into distinguishable species.

Putting both electrons into a bonding orbital is a useful first approximation near the molecule's equilibrium separation. Yet even this simple molecule exposes a limitation of describing the electrons independently. Their mutual repulsion makes the probability of one position depend on the other. A many-electron state must describe joint alternatives, not merely attach one isolated cloud to each particle.[19]

The difficulty becomes especially clear when we pull the molecule apart. At large separation the simple bonding orbital is approximately an equal sum of an A-centered and a B-centered orbital. Put both electrons independently into that sum and expand the product. Four spatial terms appear: both near A, one near each nucleus in either labeling, and both near B.

The separated ground-state fragments, however, are two neutral hydrogen atoms. A proposed dissociated state should not predict a substantial probability of leaving one atom negatively charged and the other bare. For the doubly occupied bonding orbital, the combined weight of those two charged arrangements tends to one-half as the atomic overlap vanishes. The approximation gives the wrong dissociation picture.

A more flexible joint state can repair it. In this minimal model, combine the configuration with two electrons in the bonding orbital with the configuration having two in the antibonding orbital. Their same-center terms can cancel while their one-electron-on-each-center terms remain. This is an interference calculation involving whole two-electron configurations, not a decision by two individually labeled particles to avoid one another.[20]

Here is the cancellation at vanishing atomic overlap. Write $A_j = \phi_A(r_j)$ and $B_j = \phi_B(r_j)$ for the two orbital values at coordinate r_j. The bonding and antibonding values are $g_j = (A_j + B_j)/\sqrt{2}$ and $u_j = (A_j - B_j)/\sqrt{2}$. Expanding the products gives $g_1 g_2 - u_1 u_2 = A_1 B_2 + B_1 A_2$. Both same-center terms have disappeared. Divide this spatial expression by √2 to normalize it, then multiply by the antisymmetric spin singlet. The coordinate labels do not make the electrons distinguishable.

The example explains why electron correlation matters without requiring us to solve a large molecule. A useful one-orbital picture in one range of separations can fail as the bond stretches. Improving the description means allowing the joint amplitudes needed for the question at hand, not simply drawing more elaborate electron paths.

A molecule is also an information system

Our small ion is not only a teaching example. In work published in 2025, David Holzapfel and colleagues prepared and coherently controlled the internal state of a single H_2^+ ion, using cooling together with quantum-logic operations involving a co-trapped atomic helper ion. They also demonstrated nondestructive readout and precise microwave spectroscopy. Their selected rotational manifold had rotational quantum number one, rather than the zero-rotation specification of our illustrative nuclear calculation.[21]

That experiment did not measure the well depth of our two-orbital model. It controlled finer internal structure that our first Hamiltonian omitted. The connection is nevertheless direct: the object whose binding we have been calculating can also carry a carefully prepared quantum state and participate in controlled information transfer.

Precision work keeps separating these levels of description. An August 2026 study by Ioana Doran and colleagues uses spectra of highly excited hydrogen and deuterium molecules to infer fine and hyperfine properties of their molecular-ion cores. Such comparisons demand more detailed interactions than the electrostatic model used here. A successful elementary explanation is a starting point for those refinements, not an excuse to ignore them.[22]

Computational quantum chemistry already makes predictions from quantum laws using classical computers. A quantum processor would be a different way to carry out selected parts of such calculations, not the first appearance of quantum mechanics in chemistry. The task, the required accuracy, and the resources still need to be specified.

We began with repelling protons. We now have a model in which a shared electronic state lowers the total energy, a separation at which the restricted energy is smallest, and bound levels for nuclear motion. We also know what the model has not established: a formation rate, a precision spectrum, or a universal picture of every chemical bond.

A wavefunction and a set of detector records provide different kinds of detail. How much information do they give us? To connect matter with communication and computation, we must learn to measure information as carefully as we compare energies.

Notes and sources

1. Richard P. Feynman, Robert B. Leighton, and Matthew Sands, The Feynman Lectures on Physics, Vol. III, Chapter 10, §10–1. The one-electron ion supplies a minimal example of covalent binding. This chapter uses explicit overlapping spatial orbitals rather than treating the two atomic functions as orthogonal. The reference fragments throughout are ground-state H plus a proton, unless neutral H_2 is explicitly introduced. Hydrogen molecular ion.

2. The Born–Oppenheimer construction used here first omits nuclear kinetic energy and solves an electronic problem at fixed nuclear coordinates. Equation (6.6) subsequently treats relative nuclear motion on one potential curve, omitting electronic-state couplings and diagonal corrections. For a modern account of precision modeling of this particular molecular ion, see V. I. Korobov, "Relativistic recoil corrections of order $m(Z\alpha)^6(m/M)$: Hydrogen-like atom and hydrogen molecular ion," arXiv:2506.03644v2 (2025). The corrections in that work are not included in the present numerical model. Versioned author manuscript.

3. Equation (6.1) is the fixed-nuclei Coulomb Hamiltonian, with $\hat{T}_e = -\left(\hbar^2/2m_e\right)\nabla^2$. For the position-space kinetic operator, probability density, and fixed-proton hydrogen approximation, see Feynman, Leighton, and Sands, Vol. III, Chapter 19, §§19–1 to 19–2. Proton–proton repulsion is included once in $\widehat{H}_R$, or added once to the electronic energy when $\widehat{H}_e$ is used instead. Position-space equation.

4. Feynman, Leighton and Sands, The Feynman Lectures on Physics, Vol. III, Chapter 19, §§19–1–19–2. The equation uses a fixed point proton and nonrelativistic Coulomb interaction. Regularity does not require a zero radial slope at the Coulomb singularity: the 1s orbital has the appropriate cusp. The Hydrogen Atom and The Periodic Table.

5. E. Schrödinger, "Quantisierung als Eigenwertproblem," Annalen der Physik 384, 361–376 (1926). The exponential substitution is a worked check in the present notation, not a derivation of every hydrogen state or an experimental fit. The integral $4\pi\int_0^\infty r^2 \exp(-2r/a_0)dr = \pi a_0^3$ fixes the stated normalization. Quantisierung als Eigenwertproblem.

6. The unmodified trial orbitals are $\phi_A = \left(\pi a_0^3\right)^{-1/2}\exp\left(-r_A/a_0\right)$ and the corresponding function at B. These are normalized hydrogen 1s functions for the fixed-proton, nonrelativistic problem. The length and energy units are $a_0 = \hbar^2/\left(m_e C\right)$ and $E_h = C/a_0$. Rounded conversions, 0.0529 nm and 27.2 eV, are sufficient for every reported physical scale here. See Feynman, Leighton, and Sands, Vol. III, Chapter 19, for the hydrogen solution. Hydrogen ground state.

7. Equations (6.2)–(6.3) and Figure 6.1 are original worked constructions from these trial orbitals. For separation R expressed in bohr, $S = e^{-R}\left(1 + R + R^2/3\right)$. The fair-mixture density is $\left(\phi_A^2 + \phi_B^2\right)/2$. The plus/minus density cross term is not present in that mixture. Figure 6.1 uses $R = 2$ and equal fixed orbital exponents; all three densities integrate to one in three dimensions.

8. All numerical electronic results are evaluations of the stated restricted trial space. In atomic units, $d = -1/2 - 1/R + \left(1 + 1/R\right)e^{-2R}$ and $k = -S/2 - \left(1 + R\right)e^{-R}$. Inserting them in Eq. (6.4) reproduces Eq. (6.5) and Figure 6.2. Generalized diagonalization of the electronic matrix with the overlap matrix gives the same two restricted eigenvalues. The elements can also be obtained by spatially integrating the normalized orbital products against the electron-only kinetic and attraction terms of Eq. (6.1). Nuclear repulsion is then added once, as in Eq. (6.4).

9. Original expectation-value calculation with the normalized plus state of equation (6.2). In atomic units, the kinetic off-diagonal element is $T_{AB} = -S/2 + \left(1 + R\right)e^{-R}$, giving $T_+ = \left(1/2 + T_{AB}\right)/\left(1 + S\right)$. Subtracting T_+ from U_+ gives the total electrostatic contribution. The comparison holds the atomic orbital exponent fixed and does not identify the energy components of a fully relaxed molecule.

10. The additional scale parameter uses normalized orbitals $\left(\zeta^3/\pi\right)^{1/2}\exp\left(-\zeta r\right)$ in atomic units. For positive ζ, use the atomic-unit numerical values to define $x=\zeta R$. The overlap and electronic matrix elements are $S=e^{-x}\left(1+x+\frac{x^2}{3}\right)$, $d=\frac{\zeta^2}{2}-\zeta-\frac{1-\left(1+x\right)e^{-2x}}{R}$, and $k=-\frac{\zeta^2 S}{2}+\left(\zeta^2-2\zeta\right)\left(1+x\right)e^{-x}$. Insert these in Eq. (6.4), including the nuclear repulsion $\frac{1}{R}$, and minimize the plus-state total energy over positive ζ and R. The choice $\zeta=1$ recovers note 8. Joint minimization gives $\zeta\approx 1.238$, $R\approx 2.003$ bohr, and $U_{+}\approx -0.586507$ hartree. Figure 6.2 deliberately retains $\zeta=1$, with $R_{min}\approx 2.49283$ bohr and $U_{min}\approx -0.564831$ hartree. Variational statements apply to the same specified Hamiltonian, not to omitted physical terms.

11. Klaus Ruedenberg and Michael W. Schmidt, “Why does electron sharing lead to covalent bonding? A variational analysis,” Journal of Computational Chemistry 28, 391–410 (2007), doi:10.1002/jcc.20553, section 3.1. Their accurate clamped-nucleus calculation gives a minimum of −0.602634066 hartree at 1.9972 bohr. The table’s reference depth is calculated relative to −0.5 hartree. The first two rows are original calculations with the trial functions specified here; the second optimizes both the common exponent and separation. None of these rows contains experimental data. Source.

12. Equation (6.6) is the reduced radial nuclear equation for zero rotational angular momentum. Its normalization is $\int_0^\infty \left|\chi\left(R\right)\right|^2 d\,R=1$; it is not the normalization of a three-dimensional unreduced radial function. Nuclear spin restrictions and hyperfine structure are outside this illustrative scalar model. The electronic curve already contains nuclear repulsion; adding it again here would double-count it.

13. The nuclear example uses the fixed $\zeta=1$ potential and $\mu=918\,m_e$. A Dirichlet finite-difference calculation on 0.1–20 bohr gives the four lowest energies as approximately −0.5607473, −0.5529579, −0.5456543, and −0.5388413 hartree at 0.0025-bohr spacing. Halving the spacing from 0.005 bohr changes these four energies by less than 0.000018 eV. Extending the interval to 0–30 bohr changes these four energies by less than 0.00000001 eV at either spacing. The quoted zero-point energy and D_0 are properties of this approximate potential; these stability checks do not give a rigorous numerical error bound or establish accuracy for the real molecule. Figure 6.3 displays the same calculation.

14. Binding and formation are distinguished by energy conservation and the spectral decomposition of a time-independent Hamiltonian. A pure incoming scattering state does not acquire a bound-state component merely by evolving under that unchanged closed-system Hamiltonian. Radiation, additional collision partners, or time-dependent controls enlarge the energy account. No particular radiative association rate or allowed transition is calculated here. The distinction follows from the unitary evolution developed in Chapter 5.

15. W. Pauli, “Über den Zusammenhang des Abschlusses der Elektronengruppen im Atom mit der Komplexstruktur der Spektren,” Zeitschrift für Physik 31, 765–783 (1925); G. E. Uhlenbeck and S. Goudsmit, “Spinning Electrons and the Structure of Spectra,” Nature 117, 264–265 (1926). The latter explicitly refers to their 1925 proposal and to magnetic-field spectral structure. Pauli exclusion paper. Spinning Electrons and the Structure of Spectra.

16. W. Gerlach and O. Stern, “Der experimentelle Nachweis der Richtungsquantelung im Magnetfeld,” Zeitschrift für Physik 9, 349–352 (1922). An English translation by M. Bauer is available as arXiv:2301.11343v1. The original record concerns silver-beam deflection and space quantization, preceding the electron-spin proposal. NIST lists neutral silver’s ground configuration as [Kr]4d^{10}5s and its term as $^2S_{1/2}$, supporting the modern spin-based account. Gerlach and Stern 1922. English translation of the original report. NIST Atomic Data for Silver.

17. W. Pauli, “The Connection Between Spin and Statistics,” Physical Review 58, 716–722 (1940). The relativistic result invokes physical conditions beyond nonrelativistic indistinguishability; it is not proved by the two-electron exchange equation in this chapter. The Connection Between Spin and Statistics.

18. Feynman, Leighton, and Sands, Vol. III, Chapter 4, on identical fermions and exchange amplitudes. Antisymmetry applies to the full spatial-and-spin state. A doubly occupied spatial orbital requires the antisymmetric two-electron spin state, not two distinguishable particles with permanent spin-name tags. Identical particles.

19. MIT OpenCourseWare, 5.61 Physical Chemistry (Fall 2007), lecture notes on helium and many-electron atoms, for electron–electron repulsion, independent-electron approximations, and antisymmetric states. Correlation here means that the joint electron description is not adequately represented by the stipulated independent occupation model; a density alone does not supply the complete joint state. Helium model; Many-electron states.

20. The dissociation example is an original expansion of two trial configurations. At vanishing atomic overlap, write $g=(A+B)/\sqrt{2}$ and $u=(A-B)/\sqrt{2}$. Then $g(1)g(2)-u(1)u(2)$ equals $A(1)B(2)+B(1)A(2)$. After normalization and multiplication by the antisymmetric spin singlet, only the neutral one-electron-per-center spatial terms remain. This demonstrates a minimal configuration correction, not a full calculation of molecular hydrogen at finite separation.

21. D. Holzapfel, F. Schmid, N. Schwegler, O. Stadler, M. Stadler, A. Ferk, J. P. Home, and D. Kienzler, “Quantum control of a single H_2^+ molecular ion,” Physical Review X 15, 031009 (2025), doi:10.1103/PhysRevX.15.031009. The work combines buffer-gas cooling with a co-trapped helper ion for quantum logic, preparation, coherent control, readout, and hyperfine spectroscopy. It is not a measurement of the restricted electronic curves plotted in this chapter. Published article; Author version, revised July 2025.

22. I. Doran, L. Jeckel, M. Beyer, Ch. Jungen, and F. Merkt, “Precision spectroscopy of the fine and hyperfine structures of high molecular Rydberg-Stark states: Metrology of molecular hydrogen ions,” Physical Review Letters 137, 063001 (7 August 2026), doi:10.1103/5rsv-mb6x. The corresponding full author manuscript is arXiv:2607.07636v1. The experiment studies highly excited neutral molecules whose spectra reveal properties of their ionic cores; it should not be confused with direct imaging of a bond or with the single-ion control experiment in note 21. Full author manuscript.

CHAPTER 7

How Much Information Is There?

Two records arrive from an experiment. One contains a thousand binary digits. The other contains ten. Which brings more information?

Before either arrives, you are told how it is made. The long record is guaranteed to contain a thousand zeros. The short record contains ten independent outcomes, each equally likely to be zero or one. The lengths and the production rules are already known. We are asking about the recorded outcomes, not whether the experiment ran or whether its promise was trustworthy.

You can write the first record before opening it. For the second, there are 1,024 equally likely possibilities. The shorter record resolves more uncertainty.

That conclusion does not make the long record useless. A predictable calibration result can be important. It means that importance, physical length, and information in the sense we are about to measure are different quantities. A thousand marks on a page do not necessarily represent a thousand new distinctions.

The last chapter connected a molecular state to its energies. Now we need a way to compare records and preparations without confusing the size of a description with what it lets us learn. The comparison will lead to two related measures: one for a classical probability distribution, and one for a quantum state. Their difference is as instructive as their similarity.

The alternatives behind a message

Imagine a source that produces one of eight symbols. They could label detector positions, molecular outcomes, or commands sent to a machine. If all eight are equally likely, three binary digits suffice to label them: there are eight strings from 000 through 111. Two digits would provide only four labels.

This is the counting argument from Chapter 1. It establishes a storage capacity. It does not yet exploit unequal probabilities. If one symbol arrives much more often than another, perhaps it should receive a shorter label. We could spend fewer digits on common messages and more on rare ones.

Claude Shannon made this distinction central to his 1948 theory of communication. His question concerned recovering a selected message, with its alternatives and their probabilities specified. It was not a proposal to rank the truth, beauty, or urgency of messages. A warning and a harmless status report can have the same information measure under the same probability model.[1]

We will use a *source* to mean a specified procedure for producing symbols, together with their probabilities. Let M label its next output, and let $p(m)$ be the probability of the

particular symbol m. The probabilities may come from a model or be estimated experimentally. Our first examples stipulate them exactly, so we can examine the reasoning without also fitting the model.

A number for surprise

An outcome with probability one brings no surprise relative to that model. Choosing between two equally likely alternatives supplies one bit. Choosing among eight equally likely alternatives supplies three.

A logarithm puts these cases on the same scale. The expression $\log_2 8 = 3$ asks what power of two equals eight. It counts how many doublings take us from one alternative to eight. For an outcome m with positive probability, define its *surprisal* by

$$s(m) = \log_2 \frac{1}{p(m)} = -\log_2 p(m). \tag{7.1}$$

Thus probabilities of one-half, one-quarter, and one-eighth give surprisals of one, two, and three bits. The logarithm also works between these convenient values. A bit is the unit of the measure; a surprisal need not be an integer.[2]

Why use this operation rather than the probability itself? Independent events multiply their probabilities. Two independent outcomes, each with probability one-half, have joint probability one-quarter. Their surprisals add: one bit plus one bit gives two. The logarithm turns the multiplication appropriate to independent alternatives into an additive information account.

The independence condition matters. Hearing an answer twice does not automatically supply twice the information. If the second answer is a guaranteed copy of the first, it introduces no new uncertainty once the first is known.

Surprisal concerns one possible outcome. To characterize the source, average over all its outcomes, weighting each by its probability. This is the Shannon entropy:

$$H(M) = \sum_m p(m)\, s(m) = -\sum_m p(m) \log_2 p(m). \tag{7.2}$$

The summation sign says to add one term for each symbol. We use the convention $0 \log_2 0 = 0$: a zero-probability alternative contributes nothing to the average. Observing something assigned probability zero would instead call the model into question. It would not give an ordinary finite surprisal under that model.

Entropy measures the average uncertainty of the source, not the surprise of every individual result. Rare events can carry large surprisal while contributing little to the average because they are rare. For d possible symbols, the greatest entropy is $\log_2 d$, attained when they are equally likely. Certainty gives zero.

A code we can actually read

Use Shannon's example: the source has four outputs, A, B, C, and D, with probabilities one-half, one-quarter, one-eighth, and one-eighth. A fixed-length code needs two binary digits per output. Try a variable-length alternative: give A the code 0, B the code 10, C the code 110, and D the code 111.

How could a receiver tell where one codeword ends? None is the beginning of another complete codeword. Read a 0 and A is finished. Read a 1 and continue: 10 means B, while 11 requires one more digit to distinguish C from D. Such a code is called *prefix-free*. Figure 7.1 draws its decoding tree.[3]

For example, ABAD becomes 0|10|0|111. The separators are shown only to help us check the encoding. The transmitted string is 0100111, seven digits long. Following the tree recovers the original four symbols without transmitting separators.

The mean number of digits per symbol is

$$\overline{L} = \frac{1}{2}(1) + \frac{1}{4}(2) + \frac{1}{8}(3) + \frac{1}{8}(3) = \frac{7}{4} = 1.75\,. \tag{7.3}$$

In this example the entropy is also 1.75 bits per symbol: each codeword length happens to equal its symbol's surprisal. The match is exact because the probabilities are the particular powers of one-half we chose. More general probabilities need not permit an exact match with whole-digit codewords.

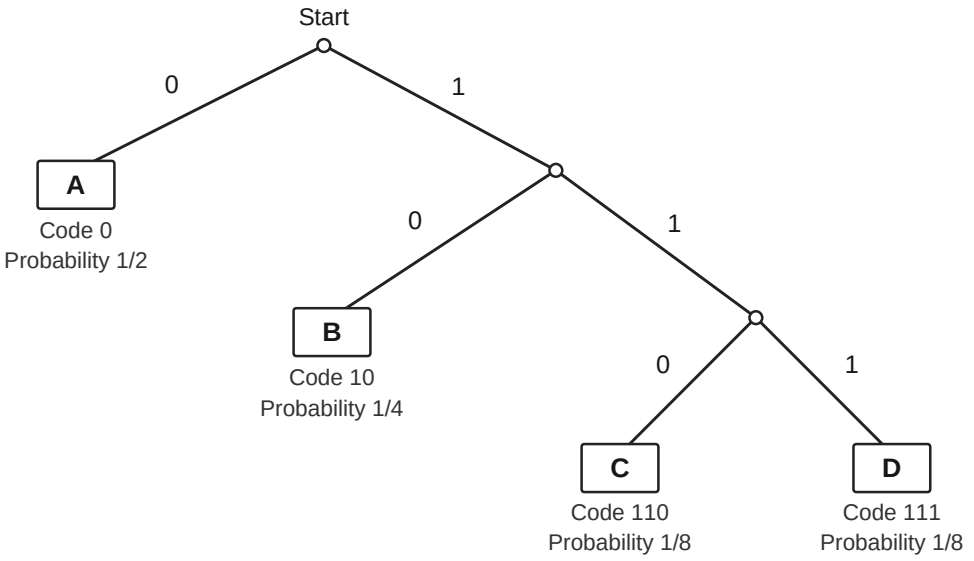


Figure 7.1. Short words for common outcomes. Each branch supplies the next binary digit. A symbol is complete only at a leaf; no leaf is an intermediate stop on the way to another symbol. The stipulated probabilities are A: 1/2, B: 1/4, C: 1/8, and D: 1/8. Codewords are 0, 10, 110, and 111. The codebook follows Shannon's example; this tree is an original drawing of its decoding rule, not a physical circuit or a set of measured frequencies.

The receiver needs the codebook and a way to identify the transmission's boundaries. We have agreed on them in advance. A practical file that must carry its own codebook or length information pays for those extras. And the saving is an average: DDDD takes twelve digits with our code, compared with eight in the fixed-length representation. Compression has not promised to shorten every message.

Nothing has been lost when this code is used correctly. The long and short representations identify the same original strings. We have removed a cost of the representation, not thrown away the rare symbols.

Less than one bit per binary result

Return to a source with only two outputs. Suppose the probability of one is p and the probability of zero is $1 - p$. Its entropy is the binary entropy function:

$$h_2(p) = -p \log_2 p - (1-p) \log_2 (1-p). \tag{7.4}$$

The subscript identifies a two-outcome function. At $p = 1/2$, it equals one bit. At either endpoint it equals zero: the source always produces the same result. Exchanging the names zero and one leaves the entropy unchanged, so the curve is symmetric. Figure 7.2 shows this behavior.

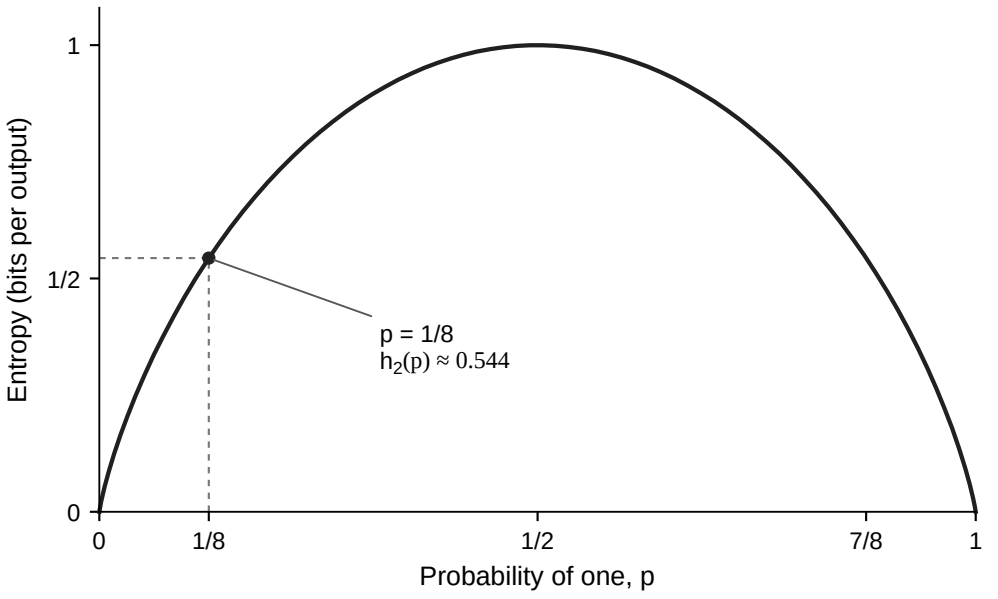


Figure 7.2. A binary alphabet need not supply one bit per use. The curve is the entropy of a stipulated two-outcome distribution, measured in bits. It reaches one only at equal probabilities and vanishes at certainty. The marked example has $p = 1/8$ and $h_2(p) \approx 0.544$. These are calculated distribution properties, not error bars, physical energies, or measured compression rates.

For $p = 1/8$, the result is approximately 0.544 bits per output. Yet a single output still needs a distinguishable zero or one. How could it use a fraction of a digit?

By encoding blocks rather than isolated outputs. Try just two at a time. The pairs 00, 01, 10, and 11 have probabilities 49/64, 7/64, 7/64, and 1/64. Assign them, in that order, the four codewords from Figure 7.1. Their average encoded length is 87/64 digits per pair, or $87/128 \approx 0.680$ per original output. The receiver can recover every pair. A fractional rate has emerged from whole-digit codewords.

This small code has not reached the entropy of 0.544. Longer blocks let us approach it. For independent repetitions of this source, most sufficiently long blocks contain a fraction of ones close to one-eighth. There are many such blocks, but far fewer than all the 2^n possible strings of length n. The probability is concentrated in a subset that can be indexed more economically.

The source-coding theorem makes the statement precise. For a known, independently repeated finite-alphabet source, zero-error variable-length block codes can approach $H(M)$ bits per symbol in average length from above. Alternatively, fixed-length block codes can approach that rate with an error probability that tends to zero as the block grows. These are two different guarantees; neither says that every finite block fits into exactly $nH(M)$ digits.[4]

For our biased binary source, the asymptotic rate is therefore about 0.544, not one, bits per output. The original ordering can be recovered, not merely the number of ones. Finite blocks require an allowance for overhead and for the chosen error guarantee.

There is no universal lossless compressor that shortens every n-digit binary string. There are 2^n inputs but only $2^n - 1$ binary strings of all shorter lengths combined, even counting the empty string. Distinct inputs cannot all be assigned distinct shorter outputs. A compressor succeeds by exploiting the source, not by evading that count.

The order can carry the difference

The assumption of independence deserves its own test. Consider two procedures for producing a thousand-digit record.

The first makes a fresh fair choice at every position. The second makes one fair choice for the first digit, then alternates deterministically: zero, one, zero, one, or the reverse. In the second procedure, each fixed position is still equally likely to be zero or one across repeated records. Every complete record also contains exactly five hundred of each.

Nevertheless, only two complete records are possible. Once the first digit is known, all the others follow. The whole record has one bit of entropy, not a thousand. The independent source has a thousand. Equal single-position probabilities do not establish equal joint distributions.[5]

We therefore need to distinguish the entropy of one output from the entropy of a sequence. The latter accounts for dependencies. A compressor that exploits the

alternating rule can outperform one that sees only the counts of zeros and ones. That does not beat information theory; it uses the information theory of the correct source.

Likewise, a frequency table calculated from one observed string is not, by itself, a complete source model. An apparently irregular record may have a known generating procedure, and an unusual record may come from an otherwise simple random source. Statistical assumptions must be tested, not read off from how a page looks.

Learning from a second record

Suppose a sender chooses a fair binary message M, and a receiver obtains a record R. The receiver's own output can look perfectly random in two very different situations: it might reproduce the sender exactly, or it might be an unrelated fair coin toss. In either case the record's entropy is one bit.

What distinguishes successful communication is the relationship between them. Let $H(M \mid R)$ denote the entropy of the message *conditional on* the receiver's record. For each possible record value, calculate the remaining message uncertainty, then average over those records. The vertical bar means "given"; it is not the magnitude sign used for amplitudes.

The joint entropy counts uncertainty in the pair. The mutual information measures the average reduction in message uncertainty supplied by the record:

$$\begin{aligned} H(M,R) &= H(R) + H(M \mid R), \\ I(M;R) &= H(M) - H(M \mid R). \end{aligned} \tag{7.5}$$

For a perfect copy, the conditional entropy is zero and the mutual information is one bit. For an independent fair output, the conditional entropy remains one and the mutual information is zero. A noisy receiver can generate unpredictable records without communicating anything about the intended message.[6]

Now stipulate that the receiver flips the message with probability one-eighth, independently on each use. Its output is still fair. But after seeing that output, the receiver assigns probabilities seven-eighths and one-eighth to the two possible messages. The remaining uncertainty is $h_2(1/8) \approx 0.544$, and the mutual information is about 0.456 bits per use.

The proportion of correct raw answers, 0.875, is not the amount of information in bits. They measure different things. A receiver that always flips the message would be completely reliable after relabeling its outputs; its mutual information is one bit. A flip probability of one-half makes the output independent and the mutual information zero.

Conditional entropy here is an average. Some particular records can leave greater uncertainty than there was before that record was seen, but conditioning cannot increase classical entropy on average. Mutual information is nonnegative and symmetric: it quantifies a statistical relationship, not a direction of causation.

The randomness of a measurement

We can apply Shannon entropy to quantum measurement records because the recorded outcomes have ordinary probabilities. Prepare the plus state from Chapter 4 on every trial. An ideal Z measurement gives zero or one with equal probabilities, so independently repeated Z outcomes have one bit of entropy per trial. An X measurement instead gives plus with certainty, so its outcome entropy is zero.

The preparation has not changed. The question has. It would be misleading to call either number, without qualification, "the entropy of the qubit." Each is the Shannon entropy of a specified measurement's outcomes.

This distinction is useful beyond a single qubit. In a 2025 preprint, Yu-Hsueh Chen and Tarun Grover investigated lossless compression of sampled configurations from several many-body quantum models as a way to estimate the entropy density of their chosen-basis outcome distributions. Their tests were numerical studies of specified models, not a demonstration that an ordinary file compressor stores an unknown quantum state. The object being compressed was a classical measurement record.[7]

A quantum state needs a measure that does not depend on which basis we happen to read. The fair mixture of zero and one differs from the pure plus state even though Z gives the same probabilities for both. Any complete, two-outcome basis measurement gives equal probabilities for that mixture; for the pure plus state, the X basis removes the outcome uncertainty entirely.

For a qubit, the smallest outcome entropy among these complete projective measurements is a basis-independent quantity: the von Neumann entropy. We are minimizing over genuine two-state basis measurements, not allowing an instrument that simply prints the same label whatever enters it. That trivial device would tell us nothing about the state.[8]

A measure for the state

Chapter 4 represented a qubit by three probability differences, x, y, and z, inside the Bloch ball. A small matrix packages exactly the same information, including mixtures:

$$\rho = \frac{1}{2}\begin{pmatrix} 1+z & x-i\,y \\ x+i\,y & 1-z \end{pmatrix}. \tag{7.6}$$

The Greek letter ρ, pronounced "rho," names the density matrix. Its diagonal entries are the Z probabilities. The off-diagonal entries retain the relationships needed to predict X and Y. It is not another list of amplitudes, and its entries are not all probabilities. For a pure state with amplitudes α and β, these entries are $|\alpha|^2$, $\alpha\,\beta^*$, $\beta\,\alpha^*$, and $|\beta|^2$ in the corresponding positions.[9]

For the pure plus state, all four entries are one-half. For an equal lottery over zero and one, form each pure-state matrix first and then average its entries: only the two diagonal

halves remain. Thus equal Z probabilities can accompany different off-diagonal entries. We are averaging state matrices, not the amplitudes of independently selected preparations.

There is always a basis in which this matrix is diagonal. Its diagonal entries in that basis are its eigenvalues: nonnegative numbers that add to one. The word is familiar from the Hamiltonian, but these eigenvalues are probabilities, not energies. For a qubit, let $r = \sqrt{x^2 + y^2 + z^2}$ be the distance from the Bloch ball's center. Choose the reference axis along that point, so that its coordinates in the new basis are zero, zero, and r. Substitution in Eq. (7.6) removes the off-diagonal entries and leaves

$$\lambda_{+} = \frac{1+r}{2}, \lambda_{-} = \frac{1-r}{2}. \tag{7.7}$$

These are also the outcome probabilities when the measurement direction follows the line through the state's Bloch point. That choice makes the difference between the probabilities as large as possible. Any other direction brings them closer to one-half and increases their binary entropy. At the center, every direction ties. This explains why the eigenbasis supplies the minimum we were looking for.

To calculate the quantum entropy, apply the Shannon formula to these eigenvalues:

$$S(\rho) = -\sum_{j} \lambda_j \log_2 \lambda_j = h_2\left(\frac{1+r}{2}\right). \tag{7.8}$$

The last equality is for a qubit; the eigenvalue sum defines the entropy in any finite dimension. As before, a zero eigenvalue contributes zero. A pure qubit lies on the surface, so its eigenvalues are one and zero and its entropy is zero. At the center they are one-half and one-half, giving one bit. Other mixed states lie between those limits.

Figure 7.3 follows states along the positive x radius. Their Z probabilities never change, but their quantum entropy does. The density matrix makes clear why: the off-diagonal entries grow as the point moves toward the pure plus state.

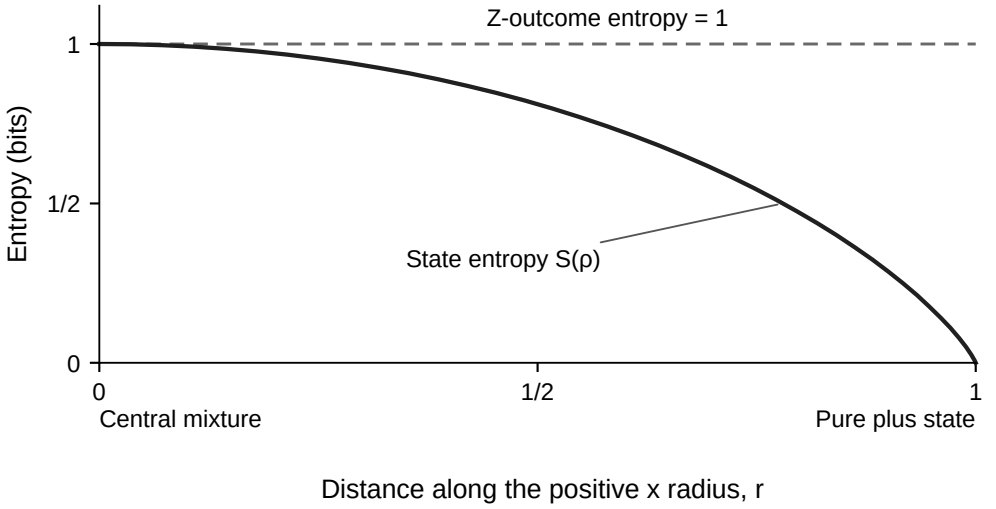


Figure 7.3. The same Z counts can conceal different state entropies. The family has Bloch coordinates $(r, 0, 0)$, with $0 \le r \le 1$. It can be prepared by choosing the plus state with probability $(1+r)/2$ and the minus state with probability $(1-r)/2$. Every member gives equal Z probabilities, hence Z-outcome entropy one bit. The solid curve is $S(\rho) = h_2((1+r)/2)$, also the X-outcome entropy for this family. Its endpoints are the central mixture and the pure plus state. Curves are ideal predictions, not measured data or the time evolution of one isolated qubit.

The entropy does not recover a unique preparation history. The fair route lottery and the fair plus/minus lottery both produce the same central density matrix when their selection records are unavailable. They therefore have the same quantum entropy. The number of entries in a laboratory's preparation menu is not a property of the delivered qubit alone.

Nor does zero entropy mean that a pure state is easy to calculate, that every measurement is predictable, or that the carrier is unable to convey a message. It identifies a property of the specified state. To ask how much a receiver can learn about a sender's choice, we must describe that choice as well.

A label is not automatically a readable bit

Consider a sender who tosses a fair coin. On one result, prepare $|0\rangle$; on the other, prepare $|+\rangle$. Send the qubit without the coin record. The classical selection has one bit of entropy. The two candidate quantum states, however, are not orthogonal.

The delivered mixture has Bloch coordinates $(1/2, 0, 1/2)$: the average of the zero and plus coordinates. Its radius is $1/\sqrt{2}$, giving eigenvalues approximately 0.854 and 0.146. Equation (7.8) gives $S(\rho) \approx 0.601$ bits. Averaging the two pure states' zero entropies would have missed the uncertainty introduced by the selection; we must first form the mixture, then calculate its entropy.

What would a Z measurement reveal? The zero preparation always gives zero. The plus preparation gives zero or one equally. Table 7.1 records these conditional probabilities.

Table 7.1. Two candidate inputs, one chosen measurement.

Sender's preparation	Selection probability	$P(R=0 \mid M)$	$P(R=1 \mid M)$
$\lvert 0\rangle$	$1/2$	1	0
$\lvert +\rangle$	$1/2$	$1/2$	$1/2$

The entries are ideal probabilities, not sample counts. The sender selects independently on each trial; the receiver has only the qubit and measures Z. No additional message-bearing record or shared entangled resource is available.

Combining the rows, the receiver sees zero with probability three-quarters and one with probability one-quarter. The record entropy is $h_2(1/4) \approx 0.811$ bits. Given the sender's label, the record entropy is zero on half the trials and one on half: an average of 0.5. Using the symmetric form of mutual information gives

$$I(M;R) = H(R) - H(R \mid M) = h_2(1/4) - \frac{1}{2} \approx 0.311. \tag{7.9}$$

The same example therefore gives four different information quantities. Each is measured in bits, but each concerns a different object or relationship. The comparison below keeps those questions separate.[10]

Table 7.2. Four numbers, four questions about the same source.

Quantity	Question it answers	Bits
Sender's message entropy H(M)	How uncertain is the sender's fair choice of preparation?	1
Received qubit's state entropy S(ρ)	What entropy follows from the delivered mixture's eigenvalues?	≈0.601
Z-record entropy H(R)	How uncertain is the outcome of this Z measurement?	≈0.811
Mutual information I(M:R)	How much message uncertainty does the Z record remove on average?	≈0.311

Values refer to the fair zero/plus source and the Z readout already specified in Eqs. (7.8)–(7.9). Rounded values describe the model, not measured frequencies. The two labels denote nonorthogonal states, so one readout does not reveal the full message bit. Changing the measurement changes the record quantities without changing the source's average-state entropy.

Could another measurement recover the entire bit? Not from this single delivered qubit. For an ensemble of pure signal states, the Holevo bound limits the mutual information obtainable by any measurement to $S(\rho)$ of the average state. Here that ceiling is approximately 0.601 bits, below one. It is an upper bound, not a promise that a single-use

measurement attains it. More outcome labels on an instrument do not remove the restriction.[11]

Choosing orthogonal zero and one states instead would give a central mixture with entropy one bit. Z would then recover the sender's bit perfectly. The classical selection probabilities are unchanged; the distinguishability of the physical encodings is different.

What quantum compression must preserve

Entropy also has a storage meaning for quantum signals, but the task is not to compress the detector's output file. The task is to encode quantum systems into a smaller quantum memory and later recover their states well enough that subsequent tests still behave correctly.

For an independently repeated source of pure signal states with average density matrix ρ, Schumacher's quantum source-coding theorem makes $S(\rho)$ the asymptotic compression rate in qubits per signal. Rates just above it permit reconstruction fidelity approaching one for sufficiently long blocks. The encoder need not learn which pure state was sent. This is a theorem about preserving quantum information, not a procedure for measuring amplitudes and writing them in a smaller notebook.[12]

For sufficiently long blocks, almost all of the source's probability weight lies in a typical subspace. Its dimension counts independent basis directions. For a block of n qubits from the stated independent source, that dimension grows approximately as $2^{nS(\rho)}$, compared with 2^n dimensions for the full block space. This is an asymptotic exponential scale, not an exact finite-block dimension. Coherent encoding preserves superpositions within the typical subspace. The small probability weight outside it accounts for an error that can vanish as the block grows. The encoding does not first measure which unknown signal label occurred; such a measurement could destroy the quantum information being compressed.

What does fidelity mean here? For a pure input $|\psi\rangle$ and recovered density matrix σ, use $F = \langle\psi|\sigma|\psi\rangle$: the probability of passing the ideal projection test for that input. It ranges from zero to one. This defines the recovery criterion; it does not give the encoder knowledge of an unknown state. The theorem requires the source-averaged fidelity of whole blocks to approach one, with their quantum memory counted and no free message-bearing side channel.

The limiting examples help. If every signal is the same known plus state, the receiver can prepare that state locally; there is no varying signal to specify. If the source independently chooses fair orthogonal states, the rate is one qubit per signal. For our nonorthogonal zero/plus source it is approximately 0.601, although the sender's coin labels still require one classical bit each if those exact labels must be preserved.

A successful quantum compressor preserves more than the average density matrix. Discarding every input and preparing an unrelated copy of the average mixture would reproduce some pooled statistics while losing the relationship to the individual inputs.

The fidelity requirement prevents that substitution. For the zero/plus source, measuring every input in Z and later rebuilding the recorded reference states would lose the plus state's coherence. Orthogonal zero/one signals are different: that same measurement can preserve them. The recovery requirement, not merely a smaller classical file, defines the task.

The asymptotic theorem does not specify the engineering cost of the encoder, the size of a useful finite block, or the quality of its controls. In a paper published in 2025, Jake Xuereb and colleagues examined how finite-temperature resources and imperfect clocks constrain a quantum compression protocol. Their analysis connects the information limit with physical requirements that the ideal coding theorem leaves separate.[13]

Counting is not yet computing

We can now distinguish the width of a register from the uncertainty of the source it stores, the entropy of a record from its information about another record, and the entropy of a quantum state from the randomness of one measurement.

The accounting also tells us what has not changed. A reversible relabeling of classical messages preserves their entropy. A closed unitary evolution preserves a density matrix's eigenvalues and hence its von Neumann entropy. The plus state can evolve into the minus state as in Chapter 5, changing a detector's prediction from certainty to impossibility, while its quantum entropy remains zero.[14]

The quantities measured here in bits are not energies or temperatures. Connections to thermodynamics require a physical model of the memory, its surroundings, and the operation performed. They cannot be inferred by calling a string "disordered." Likewise, a molecular wavefunction can be difficult to compute while describing a pure state of zero von Neumann entropy. Difficulty of calculation is not what this entropy measures.

Return to the two records. We would now ask how their outputs are generated, which dependencies are known, what must be recovered, and what other records are available. Only then would we assign an information measure.

A machine must do more than assign one. It must transform a physical representation, preserve the distinctions needed for an answer, and produce a result that can be read. Some transformations are reversible. Others discard distinctions from the working register and require us to ask where those distinctions went. We are ready to examine what it means to compute.

Notes and sources

1. Claude E. Shannon, “A Mathematical Theory of Communication,” *Bell System Technical Journal* **27**, 379–423 and 623–656 (1948). The introduction separates reconstruction of a selected message from its semantic importance. The notebook and eight-symbol examples in this chapter are stipulated source models; lengths, generation rules, and successful delivery are assumed known. Corrected reprint.

2. Shannon, §§6–7, on choice, uncertainty, and entropy. Base-two logarithms fix bits as the unit. Surprisal is assigned to a positive-probability outcome; Shannon entropy is its expectation. Additivity of surprisals for independent events follows from the logarithm product rule. The chapter uses discrete finite alphabets, not differential entropy for continuous variables. Original formulation.

3. David A. Huffman, “A Method for the Construction of Minimum-Redundancy Codes,” *Proceedings of the IRE* **40**, 1098–1101 (1952), doi:10.1109/JRPROC.1952.273898. The four probabilities and codewords in Figure 7.1 already appear in Shannon (note 1), §9; the drawing and the ABAD decoding trace are this book’s presentation. Huffman’s paper supplies a general minimum-redundancy construction, not the first appearance of this particular example. Its mean length and entropy both equal 7/4; no claim of improvement for every individual string is made. Prefix-free codewords ensure unique decoding; weighting their lengths by the stated probabilities gives the expected rate. Reprinted paper.

4. Shannon, §9, on the fundamental theorem for a noiseless channel; Yury Polyanskiy and Yihong Wu, MIT 6.441 lecture notes, Chapters 6–7, on variable-length and fixed-length lossless compression. The average-length zero-error and fixed-length vanishing-error formulations have different finite-block guarantees. Model, codebook, framing, and allowed side information must be included in an implemented resource account. Shannon; MIT notes.

5. The independent and alternating sources are original counterexamples to inferring joint entropy from one-position marginals. For fixed block length n, their joint entropies are n and one bit respectively. The alternating construction has one random starting digit and no subsequent random choices. Shannon’s treatment of sources with memory is not restricted to independent outputs. Shannon, §§2 and 6–7.

6. Polyanskiy and Wu, MIT 6.441, Chapters 1–2, on entropy, conditional entropy, and mutual information. The noisy binary example assumes a fair source and independent symmetric flips. Its mutual information is $1 - h_2(\varepsilon)$, not $1 - \varepsilon$. The claim about conditioning concerns the average over the conditioning variable; no causal direction is inferred. Entropy and information lecture notes.

7. Yu-Hsueh Chen and Tarun Grover, “Zipping many-body quantum states: a scalable approach to diagonal entropy,” arXiv:2502.18898v2, revised 10 August 2025. The version cited is a preprint. It studies Lempel–Ziv compression as an estimator of chosen-basis entropy density in specified many-body models, with numerical finite-size and sampling analyses. This is not Schumacher compression and not a general estimator of an unknown state’s von Neumann entropy from one fixed measurement basis. Versioned preprint.

8. John Preskill, *Quantum Information*, Chapter 10, “Quantum Shannon Theory,” updated June 2025, §§10.2–10.2.2. For a finite-dimensional state, the minimum Shannon entropy over complete rank-one projective measurements equals its von Neumann entropy; an eigenbasis attains the minimum. An arbitrary measurement with a single compulsory output is excluded from this minimization. Updated lecture chapter.

9. Preskill, Chapter 2, “Foundations I: States and Ensembles,” §2.3.2, and Chapter 10, §10.2. The density matrix uses the Bloch and Y-sign conventions established in this book’s Chapter 4. Its eigenvalues are $(1 \pm r)/2$. Figure 7.3 varies a family of preparation mixtures, not unitary time evolution from the center to the surface. The spectral entropy is basis-independent and the label of a particular ensemble decomposition is not inferred from the state alone. Bloch representation; Spectral entropy.

10. The zero/plus example and Table 7.1 are original calculations. The average state is $\rho = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 1/4 \end{pmatrix}$.

Its entropy is approximately 0.600876; the Z-record entropy is 0.811278; the Z mutual information is 0.311278 bits. The state entropy follows from the matrix eigenvalues. The record entropy and mutual

information follow from the joint distribution obtained by multiplying each row of Table 7.1 by its selection probability. Stated probabilities and rates are theoretical, not measured data.

11. Preskill, Chapter 10, §§10.6.1–10.6.2, on accessible information and the Holevo bound. For the stipulated pure-state encoding, $I(M;R) \leq S(\rho)$ for every receiver measurement. With mixed signal states the usual stronger expression subtracts their average entropy. The setup supplies no additional message-bearing system or preshared entangled resource. An upper bound is not identified with the best attainable single-use mutual information. Accessible information.

12. Benjamin Schumacher, "Quantum coding," *Physical Review A* **51**, 2738–2747 (1995), doi:10.1103/PhysRevA.51.2738. The discussion concerns an independently repeated pure-state source, ideal quantum encoding/decoding, and reconstruction fidelity approaching one as block length increases. The asymptotic rate is the entropy of the average source state, not the average of its pure constituents' entropies. Preserving the average density matrix alone is insufficient. The criterion here is source-averaged whole-block fidelity, with no free message-bearing side channel. See also Preskill, Chapter 10, §10.3.2, for the full typical-subspace argument and recovery criterion. Original paper.

13. Jake Xuereb, Tiago Debarba, Marcus Huber, and Paul Erker, "Quantum Coding with Finite Thermodynamic Resources," *Quantum Science and Technology* **10**, 025030 (2025), doi:10.1088/2058-9565/adb0e9. The author manuscript studies compression with imperfect timekeeping and finite-temperature resources; it is theoretical work, not an experimental compression benchmark. Journal year 2025 and manuscript revision year 2024 are distinct. Author manuscript and publication record.

14. Preskill, Chapter 10, §10.2.1, on unitary invariance of von Neumann entropy. A reversible classical relabeling is a permutation of probabilities; a quantum unitary preserves the spectrum of ρ. These statements concern the complete specified state or distribution. Discarding a correlated subsystem or conditioning on an outcome is a different operation. Thermodynamic costs are not calculated in this chapter. Entropy properties.

CHAPTER 8

What It Means to Compute

A box receives the numbers thirteen and eleven. A moment later, it displays twenty-four. What would make this a calculation rather than a coincidence?

The answer cannot be the display alone. A box permanently labeled twenty-four would pass this one test. A calculator must respond correctly when the inputs change. We need a rule connecting the inputs to the output, a procedure that implements the rule, and a physical device whose behavior follows that procedure.

There is another puzzle. Once the inputs and the addition rule are specified, the answer is determined. The machine has not received an extra message containing the sum. Yet before the calculation we may not know the answer, and afterward we do. Computation can make a consequence accessible without receiving new evidence about the outside world.

The previous chapter measured uncertainty. This chapter asks how a physical system can transform a representation into a usable answer. We will begin with an addition small enough to check by hand. Then we will ask a less familiar question: can the machine keep its answer while undoing the work that produced it?

A rule small enough to build

Write the inputs in binary, using place values one, two, four, eight, and so on. Thirteen is 1101: eight plus four plus one. Eleven is 1011: eight plus two plus one. The rightmost digit represents ones, just as it does in ordinary decimal notation. We will add from that end.

First consider a column with only two input digits, a and b, and no incoming carry. Each is zero or one. Two output digits suffice: a sum digit s for this column and a carry c for the next. Table 8.1 gives every possibility.

Table 8.1. Two input digits, a sum, and a carry.

a	b	Sum s	Carry c
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

The carry has weight two in this column's total: $a + b = s + 2c$. Entries are exact logical values, not probabilities or measured frequencies. No incoming carry is included in this half-adder table.

The sum digit is one exactly when the inputs differ. This is exclusive OR, written XOR or $\oplus$. The carry is one exactly when both inputs are one. This is AND, which for binary values is ordinary multiplication. Thus

$$a + b = s + 2c, s = a \oplus b, c = a\,b. \tag{8.1}$$

The first plus sign means ordinary addition; the circled plus keeps only whether the binary inputs differ. In particular, $1 + 1 = 2$, but $1 \oplus 1 = 0$. The missing two has not vanished: the carry records it in the next position.

A device implementing one of these elementary rules is called a logic gate. A gate is defined by its input–output behavior, not by its size or the material from which it is made. A physical implementation might use electrical switches; the same logical rule can be represented in other ways. Shannon's early work on switching circuits made the connection between such logical relations and networks of relays explicit.[1]

Our two-output device is a half adder. A full adder also accepts an incoming carry, r. One construction first computes $a \oplus b$, then combines that result with r. A carry is produced if both original inputs are one, or if the incoming carry meets exactly one original one:

$$s = a \oplus b \oplus r, c = a\,b \vee \left[r \left(a \oplus b \right) \right]. \tag{8.2}$$

The symbol $\vee$ means ordinary inclusive OR: the result is one if either argument is one. These rules give $a + b + r = s + 2c$ for all eight possible inputs. Figure 8.1 follows four such columns through our example.[2]

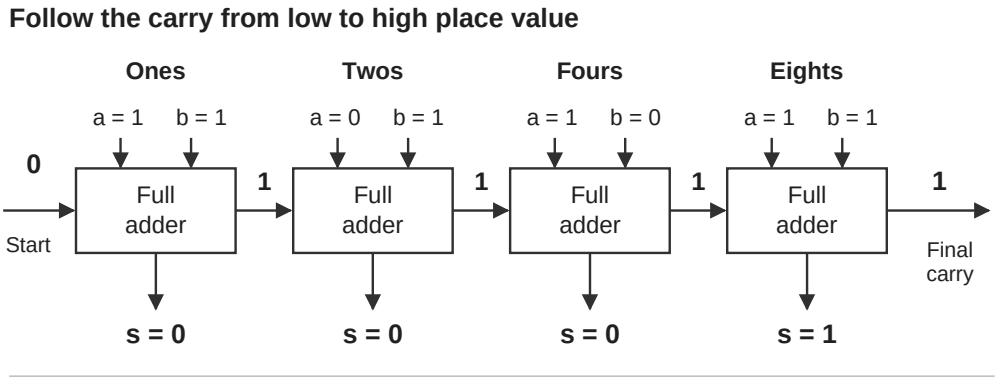


Figure 8.1. Following the carry through thirteen plus eleven. The inputs are 1101 and 1011 in conventional most-significant-first notation. The blocks run in the opposite order: ones, twos, fours, eights. Each is a full adder implementing Eq. (8.2). Carry starts at zero; the last carry supplies the fifth output digit. Read that carry, then the sum digits from right to left, to obtain 11000. This is an exact logical trace, not an electrical timing simulation or measured device.

A one-bit signal now does a recognizable job. It passes a carry from one column to the next. Nothing in the circuit needs to understand the number twenty-four. Each component follows a local rule; the arrangement makes their combined action perform addition.

In a real circuit, the signals need time to settle. A register—a collection of physical storage positions—can hold the inputs or capture a stable output. The logical drawing omits voltages, switching delays, and power supplies. Those omissions make the rule easier to see; they do not make the hardware unnecessary.

From a circuit to an algorithm

For longer inputs, repeat the same column operation. Start with carry zero, move from the least significant position toward the most significant, and retain the final carry. This is an algorithm: a specified procedure applicable to a family of inputs, not a memorized answer to one example.

For n-digit inputs, this ripple-carry construction uses n full-adder stages and produces up to $n+1$ output digits. We can build all the stages, or reuse a smaller arithmetic unit while storing the intermediate carry and our place in the input. The procedure and the physical arrangement are related, but they are not identical.[3]

The same small vocabulary reaches beyond addition. Take a finite truth table for one output bit. For each row that should produce one, build an AND condition that recognizes exactly that input: use an input directly where the row requires one, and NOT of that input where it requires zero. OR those conditions together. The circuit now produces one on precisely the required rows. An all-zero table needs only a fixed zero output.

This construction explains a limited but powerful kind of universality. AND, OR, and NOT, with wiring and fixed values, can implement any finite binary truth table. It does not promise a small circuit: listing all rows can itself require exponentially many entries. For a growing family of inputs, we also need a procedure for constructing or executing the corresponding circuits, not an unexplained new table at every size.

The contrast matters beyond arithmetic. A mathematical problem specifies what would count as a correct answer. An algorithm specifies how to obtain one. An implementation supplies the physical operations. A resource account states what those operations require. A single box labeled “solve” supplies none of the missing steps.

For addition, every input pair has one exact answer. Other tasks ask for an approximation, a sample from a distribution, or a result with a specified probability of error. A calculation of molecular energy, for example, needs an accuracy target and a specified Hamiltonian. Producing more printed digits does not establish that the underlying model is more accurate.

An algorithm may also use random choices. Then its guarantee concerns the distribution of its outputs or running times. Randomness is a resource to be specified, not permission to declare a favorable result after ignoring failed trials.

How the work grows

Our adder gives a simple way to count work without guessing the speed of a future computer. In the logical construction of Eq. (8.2), each column uses two XOR gates, two AND gates, and one OR gate. Counting each of those as one elementary gate gives $5n$ gates. Doubling the input length doubles this count.

This is a deliberately stated counting model. An XOR gate might require several components in a particular technology. Parallel operations can reduce elapsed time without reducing their total number. Other addition circuits arrange the carries differently. Gate count, circuit depth—the number of successive layers—and physical running time are different resources.

Compare a procedure that explicitly visits all 2^n binary strings of length n. Each extra digit doubles its number of visits. At $n = 64$, it must visit about 1.84×10^{19} strings. Even at a stipulated billion visits per second, with all other costs neglected, a complete pass takes about 585 years. That is arithmetic for an imagined rate, not a benchmark of an actual machine.[4]

The comparison illustrates linear versus exponential growth. A polynomial cost, such as n^2 or n^3, is often a useful theoretical dividing line from an exponential one. It is not a guarantee of practical usefulness: large powers, large prefactors, and demanding precision can still make a calculation inaccessible.

Nor does a slow exhaustive procedure prove that the problem itself requires exhaustive work. A better algorithm may exploit structure that the enumeration ignores. To claim an advantage, we must compare ways of solving the same task, with comparable accuracy and a consistent account of inputs and outputs. A lower count for one unnamed operation is not enough.

The length of an answer is no guide to the effort required to find it. A yes-or-no result occupies one bit whether it is trivial or extraordinarily demanding to calculate. Conversely, a million prescribed zeros require a long output but little uncertainty about what to print. Chapter 7's information measures and the present resource counts answer different questions.

Can the answer lead us back?

Return to the half adder. Its two inputs and two outputs occupy the same number of binary positions. Has it preserved everything?

No. Inputs 01 and 10 both give sum one and carry zero. The output tells us their total but not which input supplied the one. The map merges two alternatives. Even without shrinking the number of positions, it loses a distinction if we retain only the sum and carry.

This is logical irreversibility. An operation is logically reversible when its complete output uniquely determines its input. NOT, which exchanges zero and one, is reversible:

apply it again and the original value returns. AND, when its inputs are discarded and only its single result remains, is not. Three input pairs give output zero. Figure 8.2 makes that merger visible.

The word “complete” matters. If another register still holds the original inputs, the calculation has not removed their distinction from the machine as a whole. We must say which records remain, not infer information loss from the name of a gate.

Chapter 7 lets us quantify the half-adder example. Supply independent fair input bits. The carry–sum pairs 00, 01, and 10 occur with probabilities one-quarter, one-half, and one-quarter. Their entropy is 1.5 bits, compared with two bits for the input pair. Given output 01, one bit of uncertainty remains about the inputs; for the other outputs, none remains. Averaging gives

$$H(A,B \mid C,S) = 1/2\,\text{bit}. \tag{8.3}$$

Capital letters here denote the input and output random variables. The lost distinction amounts to half a bit on average, not one fixed binary position on every trial. This is an information calculation under a specified input distribution, not yet a prediction of heat.[5]

Keep the inputs, change the target

We can compute AND without merging the complete logical states. Retain the two input bits and add a third bit, t, called the target. Flip the target if both inputs are one; otherwise leave it alone:

$$(a,b,t) \mapsto (a,b,t \oplus ab). \tag{8.4}$$

This is the Toffoli operation. With the target initially zero, its final value is ab, so it contains the AND result. The inputs are still present. With any initial target, repeating the operation flips it twice or not at all and recovers the original triple. All eight possible triples have distinct successors.[6]

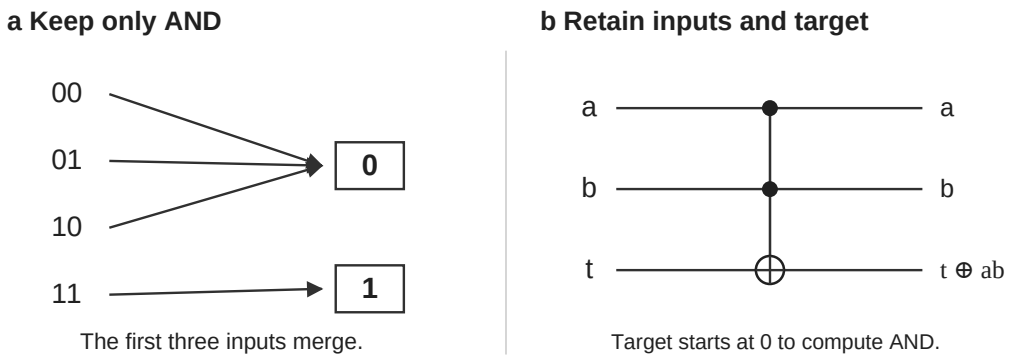


Figure 8.2. Retaining the inputs makes AND reversible. (a) Retaining only AND merges 00, 01, and 10 into output zero. (b) Toffoli retains a and b and flips t only when both controls are one. A filled dot denotes a control and a circled plus the target flip; the vertical line joins the participants in one operation. The reversible map is defined on all eight inputs, although the target is initialized to zero when used to calculate AND. Lines indicate logical registers, not particle paths.

A simpler reversible operation, controlled NOT, maps (a, t) to $(a, t \oplus a)$. A control value of one flips the target. A control value of zero does nothing. With a blank target, this copies a classical bit while retaining the original; with an identical copy as target, it returns the target to zero.

"Blank" means prepared in the agreed standard state. It does not mean that no physical system is there. The extra memory, its preparation, and its later reuse belong in the resource account.

More generally, for a function f with a fixed-length binary output, consider

$$(x, y) \mapsto (x, y \oplus f(x)). \tag{8.5}$$

Here x and y can be strings; XOR acts separately at each output position. Applying the map twice restores y, since a bit XORed with itself is zero. Starting from a zero string writes the answer into y. Keeping x makes the full map reversible even when f itself is many-to-one.

This notation describes a desired operation, not a free implementation of an arbitrary function. Its circuit must still compute f, and its cost must still be counted. Reversibility does not make a hard problem easy. It changes how the computation handles its records.

Keeping the answer, clearing the work

Saving every intermediate result would eventually fill a memory. Must we then erase the whole history, paying whatever physical cost we hoped to avoid?

A useful alternative is to run the work backward after preserving the answer. This is uncomputation. Charles Bennett's reversible-computation construction established the general strategy: retain what is needed to reverse the computation, copy a classical result, and reverse the working steps.[7]

Try a small example. The three input bits are a, b, and c. Our desired answer is $f(a,b,c) = (a \oplus b)c$: one exactly when the first two differ and the third is one. Give the machine two zeroed work bits, u and v, and a zeroed answer bit, q.

First apply controlled NOT from a to u, then from b to u. Now $u = a \oplus b$. Apply Toffoli with u and c controlling v, so $v = (a \oplus b)c$. Copy v into q with controlled NOT. The answer has reached its own register. Figure 8.3 shows the complete seven-gate procedure.

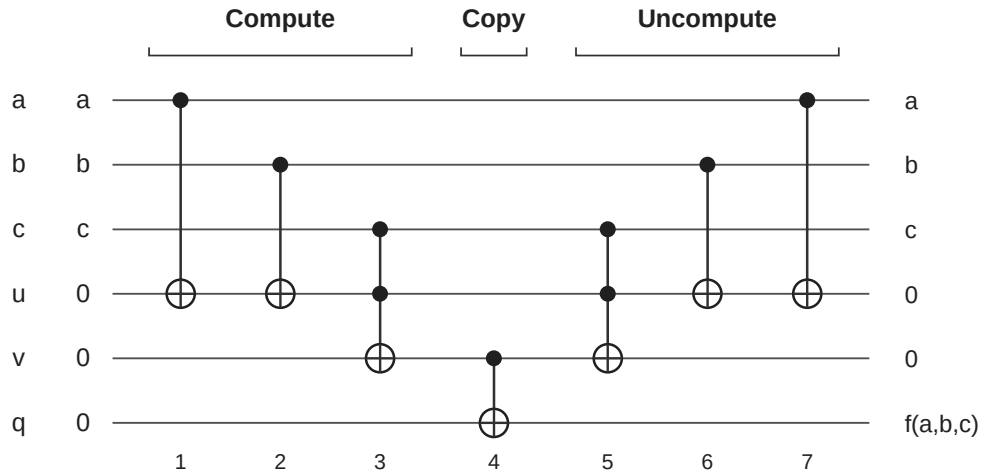


Figure 8.3. Compute, copy, uncompute. Registers are ordered a, b, c, u, v, q. Gates 1–3 compute $u = a \oplus b$ and $v = (a \oplus b)c$. Gate 4 writes the classical result into the initially zero answer register. Gates 5–7 undo the working steps, restoring $u = v = 0$ while leaving the inputs and answer intact. Only filled dots and circled targets participate; bare crossings are not controls. All operations proceed forward in time. The diagram shows an ideal logical circuit, not a claim about physical gate duration or heat.

Now reverse the working steps, not the copying step. Apply the same Toffoli again: v returns to zero because u and c are unchanged. Apply controlled NOT from b to u, then from a to u: u also returns to zero. The original inputs remain, the work bits are blank again, and q still holds the answer.

For inputs 101, the work bits pass through 00, 10, and 11. After the answer is copied, the reverse sequence restores 00. For inputs 111, the first two controls cancel in u, and the answer stays zero. The same circuit handles all eight input triples; it is not tailored to either example.[8]

No arbitrary unknown value has been forced into zero regardless of everything else. The controls retain the information needed to undo each change. This is why restoring the work bits is different from a blind reset.

There is a price in operations and storage. Our example uses seven gates instead of the three that first produce the answer in v, and retains the inputs. Larger constructions trade extra memory against recomputation. Reversibility supplies a way to organize those costs, not an unlimited supply of free space or time.[9]

The physical price of forgetting

So far “reversible” has described a logical map. A physically wasteful device could implement a perfectly reversible map. Conversely, a logical erasure can approach a thermodynamically reversible transfer of entropy. Logical reversibility and thermodynamic reversibility are not synonyms.

To connect them properly, specify a memory. Imagine a particle held in a controllable two-well potential. Left represents zero and right represents one. A sufficiently high barrier helps keep the stored value from changing over the storage interval. Initially, the two wells have the same shape and energy, with the particle equally likely to occupy either. Within each well it has the same thermal distribution.

The task is to reset the memory to the left well, regardless of its initial logical value. The potential must end in its original symmetric form, ready for reuse, but the final particle distribution is confined to the left. No other usable record tells the controller which well was initially occupied. The memory exchanges heat with an initially uncorrelated, large thermal reservoir at fixed absolute temperature T. Any auxiliary controller must be restored or included in the account; we cannot hide another expendable memory inside it. Under these assumptions, the two equally weighted alternatives contribute $k_B \ln 2$ to the memory’s thermodynamic entropy. Perfect reset removes that contribution without changing the within-well entropy. Boltzmann’s constant, k_B, converts the dimensionless information account into thermodynamic entropy measured in joules per kelvin. The second law requires a compensating entropy increase outside the memory. For our ideal fixed-temperature reservoir receiving mean heat $\langle Q_{bath} \rangle$, its entropy gain is that heat divided by T. Consequently,

$$\Delta S_{th,mem} = -k_B \ln 2, \langle Q_{bath} \rangle \geq k_B T \ln 2. \tag{8.6}$$

This is Landauer’s bound for the stipulated reset. The brackets denote an average over repetitions, not a minimum heat release on every fluctuating microscopic trajectory. The logarithm is natural; $\ln 2$ converts one bit to the corresponding natural-log measure.[10]

Here thermodynamic entropy has units of joules per kelvin, and heat has units of joules. For the memory probabilities in this thermodynamic model, Chapter 7’s entropy in bits becomes a thermodynamic entropy through the factor $k_B \ln 2$. Multiplication by absolute temperature supplies the corresponding energy scale.

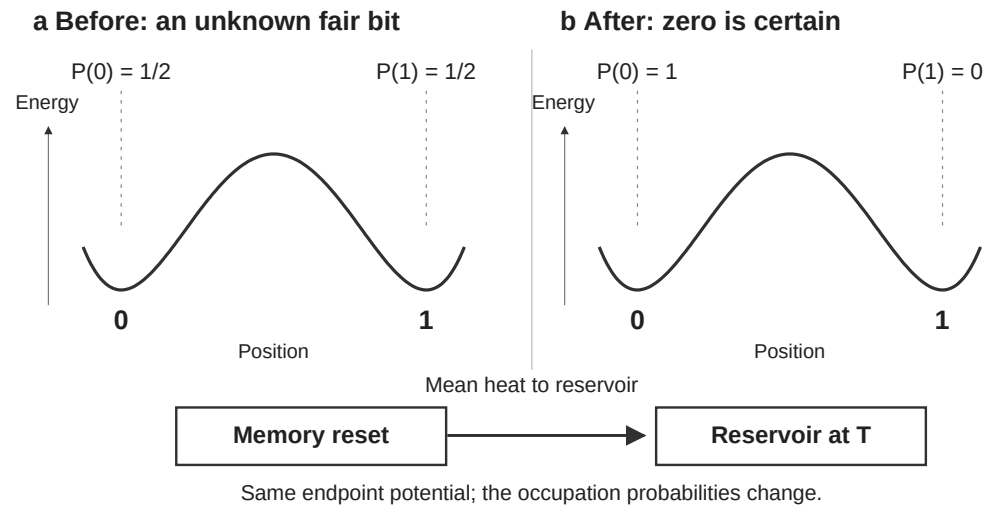


Figure 8.4. What a reset changes. The endpoint potential is the same symmetric two-well landscape before and after reset. The labels above the wells are logical occupation probabilities, not wavefunction amplitudes or density heights. Only the occupied alternatives change: a fair initial bit becomes a certain zero. The arrow identifies the reservoir receiving the exported heat in the stated model. The intervening control sequence is not drawn; the curve is schematic, with no physical energy or distance scale. Perfect reset is an ideal limiting task, not a finite-time experimental result.

At 300 kelvins, the bound is about 2.87×10^{-21} joules per reset. That is a calculated scale, not the electricity consumed by a practical computer instruction. In our symmetric memory model, the mean internal energy is the same at the beginning and end, so the supplied mean work equals the heat delivered to the bath. Different endpoint energies require a separate work–energy account.[11]

A high barrier is useful for retention, but its height is not the universal Landauer cost. A controller can reshape the potential while performing the reset. Neither "overcome the barrier" nor "flip a bit" is a substitute for calculating the full process.

At the ideal bound, the memory's entropy decrease and the reservoir's increase balance. Entropy has been transferred; no positive total entropy production is required in the limiting reversible process. Approaching that limit while also demanding reliable, rapid operation is a separate physical problem. Our formula does not promise exact perfect erasure in a finite time with bounded controls.

Which information is actually removed?

The assumptions change the answer. A memory already known to be zero needs no reset. A memory guaranteed to be one can be changed to zero by NOT, a one-to-one operation. Neither case removes an initially fair unknown distinction. That does not make a real NOT gate free of friction, electrical resistance, or control costs.

The relevant knowledge must be available through the preparation or physical controls. A person silently learning the value does not alter the thermodynamic behavior of an unchanged reset device.

If the initial one probability is p, the same symmetric-memory argument gives the lower bound $k_B T \ln 2 \, h_2(p)$ for perfect reset. The binary entropy h_2 is the function from Chapter 7. Its smaller value for a biased source can be useful only when the physical protocol can exploit that specified distribution.

An imperfect reset leaves another distinction. Suppose the initial bit is fair and the final probability of incorrectly remaining in one is ε, between zero and one-half. With the same within-well conditions, the remaining entropy is $h_2(\varepsilon)$ bits, giving

$$\langle Q_{bath} \rangle \geq k_B T \ln 2 \left[1 - h_2(\varepsilon) \right]. \tag{8.7}$$

At error probability one-half, no entropy reduction is required by this account. That is not a successful reset. Allowing error and reducing a heat bound are not the same as achieving equally reliable computation at lower cost. The mean must include unsuccessful trials, not only the favorable subset.[12]

A retained copy also changes the task. If another register contains the same classical bit, controlled NOT can clear the first while leaving the copy intact. The distinction still exists somewhere. Resetting the remaining copy for reuse brings back the entropy-reduction task. Throwing it away merely moves the stored distinction outside the chosen device; it does not perform the reset.[13]

Nor does a measurement universally require $k_B T \ln 2$ merely because it has two outcomes. Recording an outcome can correlate a system with a blank memory. Reusing that memory, stabilizing it, and operating the apparatus have their own costs. The reset bound belongs to a specified entropy-reducing task, not to every event we call “information processing.”

The bound meets a device

Recent experiments show why the physical qualifications matter. In a study published in 2025, Shingo Oikawa and colleagues used optically trapped microscopic particles and controlled potential shapes to test optimized finite-time transport and erasure. Their results linked the extra cost of rapid operation to the movement of a probability distribution, rather than treating the Landauer limit as a complete prediction for a finite-time protocol.[14]

A different limitation appeared in a 2026 study by Takase Shimizu and colleagues. They investigated a silicon dynamic random-access memory cell whose charge could be monitored at the single-electron level. Heat was inferred from electron-hop records and a calibrated electrostatic model. For their preparation and controls, heat release remained above the Landauer bound even when the charge-transfer stages were made very slow. Their analysis identified an initial nonequilibrium distribution that the available controls could not maintain through a reversible path.[15]

The two experiments are not contradictory. One tests carefully shaped transport; the other exposes a constraint of a particular memory architecture. Neither measures the wall-plug energy of an entire computer. A universal lower bound and an attainable device-specific cost are different statements.

The lesson reaches back to our adder. Knowing its truth table does not determine its heat production. We also need its physical states, input statistics, controls, speed, accuracy, and what happens to the records. The logical and thermodynamic descriptions must meet at a specified implementation.

What changes when the registers are quantum?

We have approached a quantum computer from an unexpected direction: a classical reversible circuit.

A reversible permutation of binary strings can act on the corresponding orthogonal quantum basis states. The ideal coherent version of Eq. (8.5) is

$$|x\rangle|y\rangle \mapsto |x\rangle|y \oplus f(x)\rangle. \tag{8.8}$$

It extends linearly to superpositions and preserves inner products. But a classical truth table alone does not guarantee this coherent implementation. A device might produce the correct answer for every basis input while leaking distinguishing records to its surroundings and destroying the interference needed for quantum work. The intended transformation must preserve the relevant phases as well.[16]

Nor does the word “quantum” remove the cost of computing f. The circuit still needs a construction. Its result must eventually be extracted through allowed measurements. Putting a computation into a superposition does not give the reader a separate printed answer for every component.

Even the familiar copying operation changes character. Controlled NOT takes $|0\rangle|0\rangle$ to $|0\rangle|0\rangle$ and $|1\rangle|0\rangle$ to $|1\rangle|1\rangle$. Apply the same operation to a plus state and a blank target. Linearity gives

$$|+\rangle|0\rangle \mapsto \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \tag{8.9}$$

Here $|00\rangle$ means that the two systems are in the zero reference states; $|11\rangle$ means both are in their one reference states. The right side is one joint state. It is not two independently prepared plus states, which would also allow 01 and 10 in a joint Z measurement. What copied the classical labels has produced a different relationship when its input is a superposition.[17]

We should not try to settle that relationship by looking at either system alone. That will be the next part of the book’s subject.

Our calculator has now done more than print twenty-four. We can explain how local rules produce the sum, how their cost scales, which distinctions the output omits, and how extra registers let us keep those distinctions long enough to undo the work. We can also say what must be specified before an information loss becomes a heat bound.

Matter has supplied states and energies. Information theory has supplied measures of uncertainty. Computation has supplied controlled procedures that connect an input to an answer. To take the next step, we must understand what happens when the information is carried not by an isolated system, but by the state of systems together.

Notes and sources

1. Claude E. Shannon, *A Symbolic Analysis of Relay and Switching Circuits*, MIT thesis, repository record. The historical claim here is the connection between Boolean relations and switching networks. Our arithmetic example and figures are original constructions, not reproductions of Shannon's apparatus. MIT repository.
2. Chris Terman, MIT 6.004, *Computation Structures* (Spring 2017), "Combinational Logic," especially functional specifications, sum-of-products construction, and universal building blocks. Equations (8.1)–(8.2), Table 8.1, and Figure 8.1 are original worked truth-table calculations. "Gate" denotes a chosen logical primitive, not a fixed transistor count. Annotated lecture.
3. Terman, "Sequential Logic," on storage, state, and timing; "Combinational Logic," on constructing a circuit from a finite truth table. The ripple-carry algorithm repeats one rule uniformly as input width increases. Finite-table universality does not make an arbitrary table small or provide a uniform efficient solution to every problem. Sequential logic; finite-table construction.
4. The resource examples are original counts under declared assumptions. The adder retains the full five-gate construction even in its first column, where the known zero carry permits simplification. At 10^9 visits per second, 2^{64} visits require approximately 584.542 years using 365.25 days per year. Loading, checking, and output costs are omitted from that illustrative rate. This is neither a hardware forecast nor a lower bound on every algorithm for a search problem.
5. For independent fair inputs, the deterministic half-adder map has input entropy two bits and output entropy 1.5 bits. The conditional entropy is their difference, 0.5 bit. The complete carry–sum distribution is $(1/4, 1/2, 1/4, 0)$ in the order 00, 01, 10, 11. This original example uses Chapter 7's classical entropy definitions. A physical dissipation statement would also require a model of which registers are retained or reset.
6. Adriano Barenco and colleagues, "Elementary gates for quantum computation," *Physical Review A* **52**, 3457–3467 (1995), doi:10.1103/PhysRevA.52.3457. The Toffoli and controlled-NOT maps in the text are permutations of all their basis inputs, including nonzero targets. Figure 8.2 compares a many-to-one map with a reversible extension; it does not equate logical universality with a fixed hardware cost. Author version.
7. Charles H. Bennett, "Logical Reversibility of Computation," *IBM Journal of Research and Development* **17**, 525–532 (1973), doi:10.1147/rd.176.0525. See also Bennett, "Notes on Landauer's principle, Reversible Computation and Maxwell's Demon," *Studies in History and Philosophy of Modern Physics* **34**, 501–510 (2003). The historical construction separates saving a classical output from reversing the working computation. Original publication; Bennett's retrospective.
8. Figure 8.3 is an original seven-gate circuit for $f(a, b, c) = (a \oplus b)c$. Register order is (a, b, c, u, v, q). Starting with $u = v = q = 0$, the output is $(a, b, c, 0, 0, f)$. Each gate is its own inverse; the inverse of the complete circuit uses them in reversed order. The complete map is a permutation of the 64 basis inputs. With clean work bits, the answer bit is flipped exactly when the first two inputs differ and the third is one, whether that answer bit starts at zero or one. These properties do not establish that the construction is optimal.
9. Ming Li and Paul Vitányi, "Reversibility and Adiabatic Computation: Trading Time and Space for Energy," *Proceedings of the Royal Society of London A* **452**, 769–789 (1996), doi:10.1098/rspa.1996.0039. The author version was deposited in 1997. The paper analyzes reversible simulation and resource tradeoffs. We use only the qualitative time–space–erasure distinction, not a particular quantitative optimality bound from its pebble-game analysis. Author version.
10. Rolf Landauer, "Irreversibility and Heat Generation in the Computing Process," *IBM Journal of Research and Development* **5**, 183–191 (1961), doi:10.1147/rd.53.0183. David Reeb and Michael M. Wolf, "An improved Landauer Principle with finite-size corrections," *New Journal of Physics* **16**, 103011 (2014), doi:10.1088/1367-2630/16/10/103011, gives explicit quantum assumptions and correction terms.

Equation (8.6) uses a large initially thermal reservoir, no initial memory–reservoir correlations, and no unaccounted auxiliary entropy sink. The two-well endpoints have identical conditional thermal distributions. Landauer publication; rigorous treatment.

11. The SI value is exactly $k_B = 1.380649 \times 10^{-23}$ joules per kelvin. At 300 K, $k_B T \ln 2 = 2.870978885 \times 10^{-21}$ joules. Positive heat denotes energy delivered to the reservoir; positive work denotes energy supplied to the memory. The stated equality of mean work and heat uses zero endpoint change in the memory's mean internal energy and a returned control potential. BIPM defining constants.

12. Equation (8.7) applies the entropy-reduction bound to a final symmetric-well population $(1-\varepsilon, \varepsilon)$ with unchanged conditional within-well states. Figure 8.4 depicts the perfect-reset limit, not a finite-duration protocol. For an initially biased source the removed binary entropy is $h_2(p)$; for the fair-input imperfect reset it is $1 - h_2(\varepsilon)$. Correlations, finite reservoirs, unequal endpoint energies, or nonthermal auxiliary resources require a correspondingly enlarged account. Reeb and Wolf.

13. Bennett's retrospective (note 7) distinguishes logical operations, correlations, and the thermodynamic cost of erasure. The controlled-NOT example maps the correlated classical pairs 00 and 11 to 00 and 10 when the second bit is the cleared target. Both initial alternatives remain distinguishable in the first bit. A measurement's record creation, storage, and cyclic reuse are different physical tasks. No universal heat cost per measurement outcome is asserted. Retrospective discussion.

14. Shingo Oikawa, Yohei Nakayama, Sosuke Ito, Takahiro Sagawa, and Shoichi Toyabe, "Experimentally achieving minimal dissipation via thermodynamically optimal transport," *Nature Communications* **16**, 10424 (2025), doi:10.1038/s41467-025-66519-9. The experiment uses optically trapped microparticles and controlled potentials, including optimized finite-time erasure. It tests additional finite-time constraints, not only the quasistatic Landauer bound. None of this chapter's diagrams reproduces its experimental data. Published article; author version.

15. Takase Shimizu, Kensaku Chida, Gento Yamahata, and Katsuhiko Nishiguchi, "Thermodynamic Constraints in Dynamic Random-Access Memory Cells: Experimental Verification of Energy Efficiency Limits in Information Erasure," *Physical Review Letters* **136**, 117103 (2026), doi:10.1103/1sgm-dhys. See Eqs. (2)–(4) and the Discussion in the published article; the corresponding author manuscript is arXiv:2505.23087v2, revised 31 March 2026. Charge-transition records and the calibrated electrostatic model determine the inferred heat; it is not direct calorimetry of the entire apparatus. Its restriction arises from the specified charge-memory architecture, preparation, and controls; it is not a theorem about every possible memory. The charge-transfer heat account does not include the entire readout apparatus's energy. Publication; versioned author manuscript.

16. Barenco and colleagues (note 6), on coherent gates acting on quantum states. Equation (8.8) is the linear extension of a complete reversible permutation on fixed-width registers, with no extra basis-dependent phases or retained distinguishing environment records. A classical truth-table test alone would not establish those conditions. The cost of constructing the circuit for f remains part of the computation. Quantum-gate framework.

17. Equation (8.9) follows by applying controlled NOT to each input component. In basis order 00, 01, 10, 11, its output amplitudes are $(1, 0, 0, 1)/\sqrt{2}$, unlike $(1, 1, 1, 1)/2$ for two independent plus states. Squaring the displayed amplitudes gives probabilities one-half for 00 and 11 and zero for 01 and 10; two independent plus states give one-quarter for each outcome. The text establishes the joint-state distinction without claiming that Z correlations alone certify entanglement; the next part develops that stronger question.

PART III

When Information Belongs to the Whole

Two records can agree without either observer knowing what the next answer will be. We have already seen how one ordinary random choice, copied and distributed, produces that situation. Agreement is a relationship; it is not yet a uniquely quantum one.

The last chapter ended with a different preparation. A controlled interaction acted on a superposition and a blank register. The result could not be replaced by two independent copies of the input. To understand what the interaction produced, we must describe the systems together.

In this part, the division between a whole and its parts becomes the central question. Which predictions can each observer make alone? Which require comparing their records? When does a shared description contain something that no mixture of individual preparations can reproduce?

We will build the answer from two qubits. Their joint amplitudes will lead to entanglement, and their measurement records will show how to distinguish it from ordinary correlation. Then we will ask a more demanding question: what could shared instructions explain if we did not assume that the devices contained qubits at all? Bell’s argument will turn that question into an experimental test.

The same distinctions will govern what can be copied, what can be transferred, and why a record left in the surroundings can make interference inaccessible. We will need to track not only a system’s state, but also which other systems and records remain available. Ignoring a partner, reading its result, and reversing an interaction are different operations.

There is no need to begin with an invisible influence racing between distant objects. We can begin with a preparation, a pair of measurements, and a comparison that ordinary local descriptions fail to explain. The question is not merely whether the two records match. It is what changes when the observers choose a different way to read them.

CHAPTER 9

Two Systems, One Quantum State

Two laboratories receive one qubit each. Each observer measures Z and records a zero or a one. Viewed separately, the two records look like fair coin tosses. Put them side by side, aligning the trials, and every result agrees.

That is a relationship worth explaining, but it is not yet evidence of entanglement. A source could toss one ordinary coin, prepare two zeros or two ones according to the result, and send one to each laboratory. The records would agree perfectly. Two matching envelopes need no remarkable physics.

Chapter 8 ended with a different procedure. A coherent controlled-NOT operation acted on a plus state and a zero state. It produced a joint superposition of zero–zero and one–one. The Z records look just like those of the coin-selected source. Does the difference survive anywhere that an experiment can reveal?

To find out, we must keep track of the pair, not just the probability table for either member. We will call the observers Alice and Bob, and their systems A and B. These are names for two separately addressable qubits, not new kinds of particles. Each observer can choose the X, Y, or Z measurements developed in Chapter 4. Their settings are calibrated against agreed references, and each trial begins with a fresh pair.

Four alternatives, one list of amplitudes

Two qubits have four reference combinations: $|00\rangle$, $|01\rangle$, $|10\rangle$, and $|11\rangle$. The first position always belongs to A, the second to B. The symbol $|01\rangle$ means zero for A and one for B; it is not the binary number one in an unlabeled register.

A general pure state of the pair assigns an amplitude to each combination:

$$|\Psi\rangle = c_{00}|00\rangle + c_{01}|01\rangle + c_{10}|10\rangle + c_{11}|11\rangle,$$
$$\sum_{a,b=0}^{1} |c_{ab}|^2 = 1. \tag{9.1}$$

Capital psi names the joint state. The two subscripts on each coefficient identify the corresponding joint alternative. Squaring its magnitude gives the probability for that pair of Z outcomes. The normalization adds probabilities for four distinguishable records, not amplitudes for four ways of reaching one record.

Now prepare the systems independently in pure states $\alpha|0\rangle + \beta|1\rangle$ and $\gamma|0\rangle + \delta|1\rangle$. Their joint description is the tensor product, written with the symbol $\otimes$. For this example its entire meaning is visible in the expansion:

$$\begin{aligned}&\left(\alpha|0\rangle + \beta|1\rangle\right) \otimes \left(\gamma|0\rangle + \delta|1\rangle\right) \\ &= \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle.\end{aligned} \tag{9.2}$$

We multiply one amplitude from each system to obtain a joint amplitude. If both inputs are plus states, every coefficient is one-half, so each joint Z result has probability one-quarter. The local coin tosses are independent. The tensor symbol is a way of assembling the description, not an interaction acting between the particles.[1]

This distinction in size will matter later. Two two-dimensional state spaces make a four-dimensional joint space. Three qubits have eight reference combinations; n qubits have 2^n. An arbitrary pure joint state needs a coefficient for each. But an independently prepared product state has a much shorter description: specify each factor. The large space does not mean that every state is equally difficult to describe, or that a detector can print all its amplitudes.

When the product fails

Apply controlled NOT to $|+\rangle_A|0\rangle_B$, retaining coherence as in Chapter 8. The zero-control contribution remains zero–zero; the one-control contribution becomes one–one. Therefore

$$|+\rangle_A|0\rangle_B \rightarrow |\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \tag{9.3}$$

The right side is one of the four standard Bell states. Its name will become more significant in the next chapter. For now, ask whether it can be written as a product like Eq. (9.2).

The zero–zero coefficient would require $\alpha\gamma \neq 0$, and the one–one coefficient would require $\beta\delta \neq 0$. All four factors would consequently be nonzero. But then $\alpha\delta$ and $\beta\gamma$ could not vanish. The unwanted zero–one and one–zero terms would appear. No pair of individual pure states produces this joint vector.

A pure state that cannot be factored across the chosen systems is entangled. That is a mathematical property we can check, rather than an adjective for a mysterious connection. For a normalized two-qubit pure vector, the compact factorization test is $c_{00}c_{11} - c_{01}c_{10} = 0$. A product passes it; our Bell state gives one-half instead of zero.[2]

Here is why the zero condition is sufficient. Arrange the coefficients in a square array, with Alice's label choosing the row and Bob's the column. Relabel the references, if necessary, so c_{00} is nonzero. The second row's first entry is $\frac{c_{10}}{c_{00}}$ times the first row's first entry. The zero condition forces $c_{11} = \frac{c_{10}}{c_{00}}c_{01}$, so the same ratio also gives the second entry. Factor out the common row: it describes Bob, while the two row weights describe Alice. Normalizing those two factors gives the product in Eq. (9.2). A row of zeros simply means that one Alice reference has zero weight.

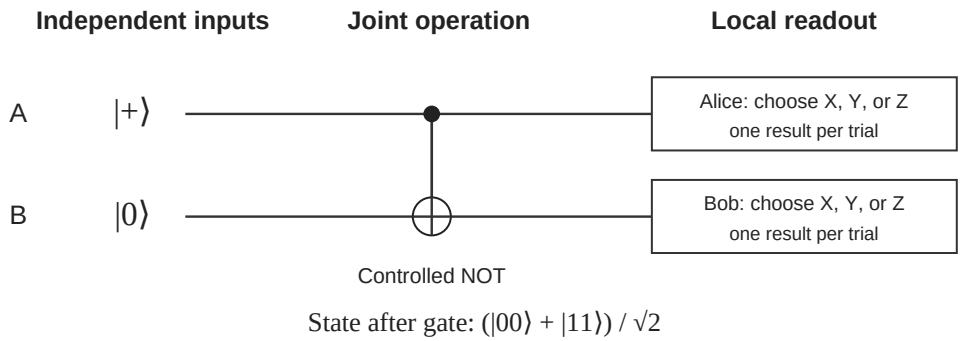


Figure 9.1. An interaction prepares the joint state. The two input registers are independently prepared in plus and zero. The filled control dot and circled target implement the same coherent controlled-NOT operation as in Chapter 8. Afterward, each register can be measured locally. For Z on both, only 00 and 11 occur, with equal probabilities. Wires identify registers and operation order, not paths through space; no line between the final observers represents a communication signal. This is an ideal circuit, not a report of measured counts.

Two qubits have not become one qubit. Their joint space still has four dimensions. Nor does the failure of a product say that neither part has a state. Each has a valid local description; it will be mixed rather than pure. The part of the description we are missing lies in how their outcomes are related.

Creating entanglement also requires more than placing two qubits near one another. The controlled interaction must preserve the amplitudes and their relative phase. In our ideal circuit, applying the same controlled NOT again returns the pair to $|+\rangle_A|0\rangle_B$. This reversal acts jointly. Separate unitary operations on A and B cannot turn an entangled pure state into a product, because their inverses would then create entanglement from a product without any joint operation.[3]

The comparison that the Z records miss

Return to the ordinary alternative. The source makes an independent fair choice on each trial, then sends $|00\rangle$ or $|11\rangle$. Its choice record is unavailable to the observers. Call this the correlated lottery. It is not the same as choosing A and B independently: the lottery deliberately makes their Z values agree.

Now let both observers measure X instead, on fresh trials. For the Bell state, substitute $|0\rangle = (|+\rangle + |-\rangle)/\sqrt{2}$ and $|1\rangle = (|+\rangle - |-\rangle)/\sqrt{2}$ into Eq. (9.3). The crossed terms cancel, leaving

$$|\Phi^+\rangle = \frac{|++\rangle + |--\rangle}{\sqrt{2}}. \tag{9.4}$$

The X results agree every time too. Neither individual result is certain; the relationship is.

The correlated lottery behaves differently. A zero–zero input gives each of the four joint X results with probability one-quarter. A one–one input does the same. Averaging these two probability tables leaves one-quarter for each result. The probability of agreement is

only one-half. Here a change of measurement distinguishes preparations that Z could not.[4]

Y supplies a useful warning against an overly simple slogan. With the $|+_y\rangle$ and $|-_y\rangle$ conventions of Chapter 4, the Bell state gives opposite Y signs with certainty. "Entangled" does not mean "the same answer to every identical question." Table 9.1 collects the predictions.

Table 9.1. Agreement depends on both the state and the setting.

Preparation	Agreement ZZ	Agreement XX	Agreement YY
Bell state $\|\Phi^+\rangle$	1	1	0
Correlated Z lottery	1	$\frac{1}{2}$	$\frac{1}{2}$
Independent fair mixture	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$

Each column gives the probability that the two outcome labels agree when both observers use the named setting. Columns refer to separate fresh trials. All three preparations give fair local outcomes in every setting. The last row is the product of two independent, maximally mixed qubits. Entries are ideal probabilities, not measured frequencies.

Each setting uses a different collection of identically prepared pairs. Alice and Bob do not measure Z, then X, then Y on the same undisturbed pair. They also need trial labels when comparing their records. Shuffling Bob's results independently would erase the visible pairings without changing either laboratory's totals.

There is a larger class of ordinary alternatives than this one lottery. A source could coordinate many different local preparations, not just zeros and ones. Excluding our particular lottery is therefore not yet a general test of entanglement. We will build that stronger test shortly.

Keeping the pair, keeping only one part

The density matrix from Chapter 7 lets us represent both preparations in one language. In the reference order 00, 01, 10, 11, they are

$$\rho_{\Phi}=\frac{1}{2}\begin{pmatrix}1&0&0&1\\0&0&0&0\\0&0&0&0\\1&0&0&1\end{pmatrix},\ \rho_{lot}=\frac{1}{2}\begin{pmatrix}1&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&1\end{pmatrix}. \tag{9.5}$$

For a pure vector, a matrix entry is one amplitude multiplied by the conjugate of another. Thus the Bell state has corner entries linking 00 and 11. For the lottery, form each pure-

state matrix first, then average with the selection probabilities. Its corner entries are zero. The diagonals agree, explaining the identical Z records; the off-diagonal terms distinguish the coherent pair from the lottery.

What description predicts every measurement Alice can make on A alone? In a joint Z measurement, sum over Bob's two alternatives: $P_A(0) = P(00) + P(01)$ and $P_A(1) = P(10) + P(11)$. The density-matrix version does the same job while retaining the terms Alice needs for other settings. For a pure pair, its entries are $(\rho_A)_{aa'} = \sum_{b=0}^{1} c_{ab} c^*_{a'b}$.

For example, Alice's off-diagonal entry is $(\rho_A)_{01} = c_{00} c_{10}{}^* + c_{01} c_{11}{}^*$. The first product pairs coefficients with Bob's label zero; the second pairs coefficients with his label one. We do not include $c_{00} c_{11}{}^*$: it connects different Bob labels. The sum retains coherence relevant to Alice's measurements without combining distinguishable Bob alternatives as one coherent outcome.

Keep Alice's row and column labels; match Bob's label and sum it out. This operation is called the partial trace. In the Bell example, Alice's zero diagonal receives one-half from 00, her one diagonal receives one-half from 11, and her off-diagonal entry is zero. The corner coherence of the joint matrix connects different Bob labels, so it does not survive this sum. Both preparations give

$$\rho_A = \rho_B = \frac{I_2}{2} = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix}. \tag{9.6}$$

Here I_2 is the two-dimensional identity matrix. Each local state sits at the center of the Bloch ball. Every complete qubit basis measurement gives equal local probabilities. No measurement restricted to one delivered member distinguishes these two sources, although comparing their X records does.[5]

Taking a partial trace is not an instruction to measure, destroy, or physically discard Bob's qubit. It is a restriction on which predictions we ask the description to provide. Bob can leave his system untouched while Alice uses the reduced state. Conditioning on a particular result that Bob reports is a different task.

The two local matrices do not reconstruct the joint matrix. Their tensor product would be $I_4/4$: four independent, equally probable Z pairs. It would lose even the ordinary correlation of the lottery. Local descriptions are complete for local questions, not for every question about the combined system.

A pure whole with uncertain parts

The Bell density matrix has eigenvalues one, zero, zero, zero. Its von Neumann entropy is zero. Each reduced matrix has eigenvalues one-half and one-half and entropy one bit. The correlated lottery instead has two eigenvalues one-half, giving joint entropy one bit while retaining the same local entropies.

These results do not assign incompatible uncertainties to the same object. They describe different objects: A, B, and AB. Nor does the Bell pair's zero state entropy make both individual outcomes predictable. It means that the complete preparation is pure, just as a single plus state is pure despite random Z outcomes.

Chapter 7 defined classical mutual information from the entropies of two records and their joint distribution. The corresponding quantum quantity uses density matrices:

$$I_q(A:B) = S(\rho_A) + S(\rho_B) - S(\rho_{AB}). \tag{9.7}$$

It equals two bits for the Bell pair and one for the correlated lottery. In contrast, measuring Z on both Bell qubits produces two classical records with mutual information one bit: knowing one answer identifies the other. The quantum quantity concerns the state before that measurement; it is not a promise that one local readout supplies two classical bits.[6]

Table 9.2. The state and its readout have different entropy accounts.

Quantity (bits)	Bell pair	Correlated Z lottery
Joint-state entropy	0	1
Entropy of either local state	1	1
Quantum mutual information before readout	2	1
Classical mutual information between the two Z records	1	1

Both preparations and the local Z readouts are those already defined in this chapter. The first three rows concern density matrices before readout; the last concerns classical outcome records. Identical local matrices or Z-record information do not specify the full joint state. No row is an extra message capacity available from a single local measurement.

Something else has become possible. A system can have a mixed state without our having tossed an unrecorded preparation coin for that system. It can be part of a pure entangled whole. Calling every mixed state mere ignorance of a privately assigned pure state would miss this distinction. The joint predictions determine whether that story works.

Check the inference

Alice and Bob each describe their own qubit by the maximally mixed state, with entropy one bit. Does that determine the entropy of the pair? Has taking a partial trace physically destroyed the correlations?

The local descriptions do not settle the joint question. The Bell pair has joint entropy zero; the correlated zero–zero/one–one lottery has joint entropy one bit. Two independently prepared maximally mixed qubits have joint entropy two bits. All three cases give the same local matrix to Alice and to Bob. Substituting into the definition of quantum mutual information gives two, one, and zero bits, respectively.

The difference becomes testable through joint records. Independent maximally mixed qubits give all four Z pairs equally often; the other two preparations give only equal Z labels. Comparing X records then distinguishes the Bell preparation from the correlated Z lottery. Taking the partial trace merely specifies predictions for one side alone. It neither turns the Bell pair into either lottery nor measures Bob's qubit. A complete local description can therefore leave a physically important joint distinction unresolved.

What a remote result changes

Suppose Alice measures Z on her member of the Bell pair and obtains zero. Conditional on that result, Bob's state is zero. Conditional on Alice obtaining one, his state is one. If she measures X instead, her plus and minus outcomes identify the corresponding plus and minus states for Bob.

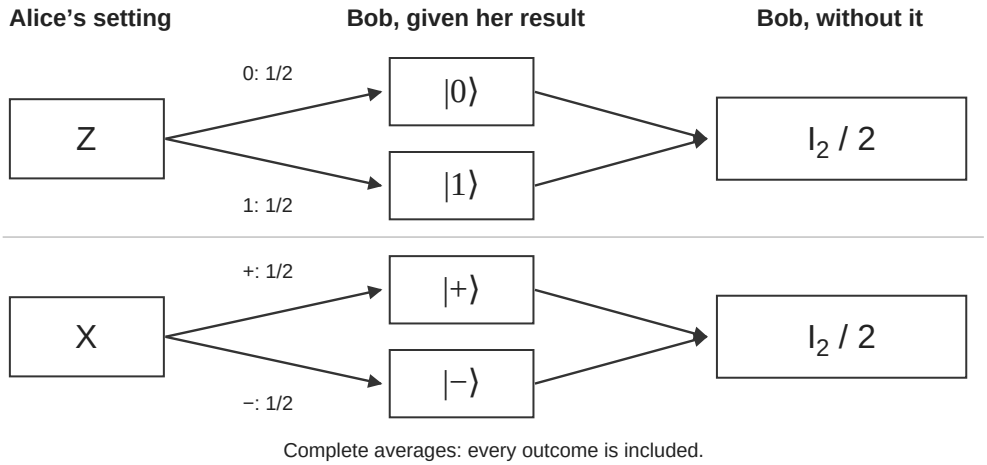


Figure 9.2. A labeled subset is not the unsorted ensemble. Fresh Bell pairs are used in the two rows. Alice chooses Z or X, with equiprobable outcomes. The boxes in the middle show Bob's conditional state when the corresponding Alice outcome is known. Averaging over the unavailable label gives the same reduced state in both rows. Arrows organize a probability calculation, not the propagation of a physical influence. Bob cannot sort by Alice's outcome until he receives a record identifying it; no outcomes are omitted from the averages.

Does this let her choose Bob's state and send an instant message? The missing step is control over her result. She can choose the setting, not which random outcome it produces. Without her record, Bob must average over both possibilities. In either case,

$$\frac{|0\rangle\langle 0|+|1\rangle\langle 1|}{2}=\frac{|+\rangle\langle +|+|-\rangle\langle -|}{2}=\frac{I_2}{2}. \tag{9.8}$$

The outer product $|0\rangle\langle 0|$ denotes the density matrix of the zero state, and similarly for the other terms. Both sums describe the same ensemble available to Bob without Alice's outcome. Figure 9.2 separates that unconditional description from the descriptions of the outcome-labeled groups.

If Alice later sends her setting and results through an ordinary communication channel, Bob can sort his corresponding records and verify the predicted correlations. A group

selected using her zero outcomes is not the same statistical question as his full unsorted record. The extra information is the selection label, and it has to reach him.

Even a deliberate local phase change makes the point. Alice can turn the Bell plus state into $|\Phi^-\rangle = (|00\rangle - |11\rangle)/\sqrt{2}$ by reversing the sign of her one component. The joint state changes: X agreement becomes X disagreement. But both local matrices remain $I_2/2$. The difference appears in their comparison, not in Bob's isolated data.

Quantum theory extends this conclusion beyond these examples. An operation confined to A cannot change B's reduced state when every outcome of the operation is included and no system or message is sent to B. Selecting a successful branch can change a conditional description; it does not give Bob knowledge of which trials belong to that branch. This is the no-signalling distinction between local operations and communicated outcomes.[7]

Distance does not create the entanglement, and no continuing exchange between the qubits is required to retain it in the isolated model. The preparation established the joint state. Unwanted interactions during separation can spoil it; whether the intended state survives is an experimental question, not something guaranteed by the word "entangled."

A test against every separable preparation

Our lottery was one way to distribute correlated but unentangled qubits. The general version chooses a classical label k with probability p_k, prepares a local state for A and another for B, and forgets the label. Its density matrix is

$$\rho_{sep} = \sum_k p_k \rho_A^{(k)} \otimes \rho_B^{(k)}, \; p_k \geq 0, \; \sum_k p_k = 1. \tag{9.9}$$

Such a state is separable. The label can coordinate the preparations, so separable does not mean independent. A mixed state is entangled precisely when no decomposition of this form exists. Finding a decomposition that uses entangled states is not enough: even an unentangled mixed state may admit one. It is the impossibility of any product-state mixture that matters.[8]

We can test that entire class using just two settings, provided X and Z really are the calibrated, perpendicular qubit measurements of Chapter 4. Give each outcome the numerical label +1 or −1; for Z, zero is +1 and one is −1. Let C_X be the average product of Alice's and Bob's labels when both measure X. Equivalently, it is the probability of agreement minus that of disagreement. Define C_Z in the same way.

For these binary tests, agreement contributes +1 and disagreement −1. The correlation is therefore twice the agreement probability minus one. The bound below also uses Chapter 4's restriction on one qubit: its X and Z coordinates cannot both have magnitude one. The witness combines that local restriction with the assumed product or mixture structure.

For one product preparation, the X correlation is $x_A x_B$, where each x is its local Bloch coordinate. The Z correlation is $z_A z_B$. The two-coordinate arrows (x_A, z_A) and (x_B, z_B) each have length at most one, because their full Bloch vectors fit inside the unit ball. Their dot product cannot exceed the product of their lengths. Consequently,

$$\begin{aligned} C_X + C_Z &= x_A x_B + z_A z_B \\ &\le \sqrt{x_A^2 + z_A^2}\sqrt{x_B^2 + z_B^2} \le 1. \end{aligned} \tag{9.10}$$

Every product preparation obeys the bound. Averaging over classical labels preserves it, so every separable state obeys it too. The Bell pair has $C_X = C_Z = 1$, giving two. It violates the separable bound. We have now excluded every preparation in Eq. (9.9), not just the zero–one lottery.[9]

This is an entanglement witness: a measured quantity with a bound for unentangled states. Suppose, as a stipulated example rather than experimental data, each setting gives 90 percent agreement. Each correlation is then $0.9 - 0.1 = 0.8$, and their sum is 1.6. Imperfect correlations can still establish entanglement. In an experiment, the uncertainty and calibration bounds must leave the sum reliably above one; a favorable fluctuation alone is not a certificate.

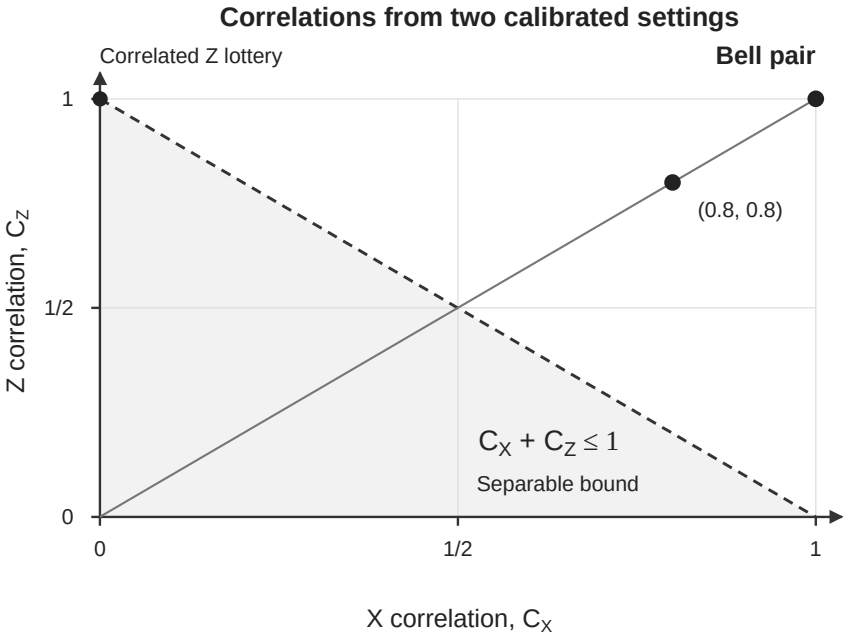


Figure 9.3. Two correlations can rule out separability. In the displayed nonnegative quadrant, all separable two-qubit states lie on or below the diagonal $C_X + C_Z = 1$. The Bell pair is at (1,1); the correlated Z lottery is at (0,1). The marked noisy example is at (0.8,0.8). Its line from the origin corresponds to mixing a Bell pair with the maximally mixed two-qubit state. Passing above the boundary certifies entanglement under calibrated local X/Z measurements. Points below it are inconclusive for this witness, not automatically separable. Coordinates are theoretical correlations, not measured data.

The witness need not identify every entangled state. For instance, the Bell minus state has $C_X = -1$ and $C_Z = 1$ and would fail this particular test, although changing the sign in the witness would detect it. Nor can mismatched source preparations be combined indiscriminately: testing Z on one source and X on a different source can fake the desired sum. Settings must sample the same preparation procedure, with selection effects and losses accounted for.

There is a crucial limit to our conclusion. We used quantum mechanics for each local qubit, including the constraint that no qubit state makes both X and Z outcomes certain. Two uncharacterized classical boxes could simply share two random instructions: agree on one whenever asked X and agree on the other whenever asked Z. They would reproduce these same-axis statistics without obeying the local qubit model. Our witness is not a Bell test. The next chapter will examine what can be ruled out without assuming those internal quantum descriptions.[10]

A smaller share of entanglement

The equal amplitudes of the Bell pair are not essential. Consider the family

$$|\Psi_\lambda\rangle = \sqrt{\lambda}\,|00\rangle + \sqrt{1-\lambda}\,|11\rangle,\ 0 \le \lambda \le 1. \tag{9.11}$$

At either endpoint it is a product. At any interior value, the same factorization test as before fails. The reduced states have eigenvalues λ and $1-\lambda$, so each has entropy $h_2(\lambda)$, the binary entropy of Chapter 7. For a pure bipartite state this reduced entropy measures its entanglement; the two parts have the same nonzero spectrum. A general theorem, the Schmidt decomposition, guarantees that suitable choices of local bases put every two-qubit pure state into this form. The displayed family illustrates that theorem; the factorization calculation alone does not prove it.[11]

Take $\lambda = 4/5$. Both Z measurements still agree with certainty, but zero–zero occurs four times as often as one–one. Local entropy is about 0.722 bits. Expanding in the X basis gives $C_X = 2\sqrt{\lambda(1-\lambda)} = 0.8$, so X agreement is 0.9 and the witness sum is 1.8. Perfect Z agreement alone therefore does not measure how much entanglement the pure pair contains.

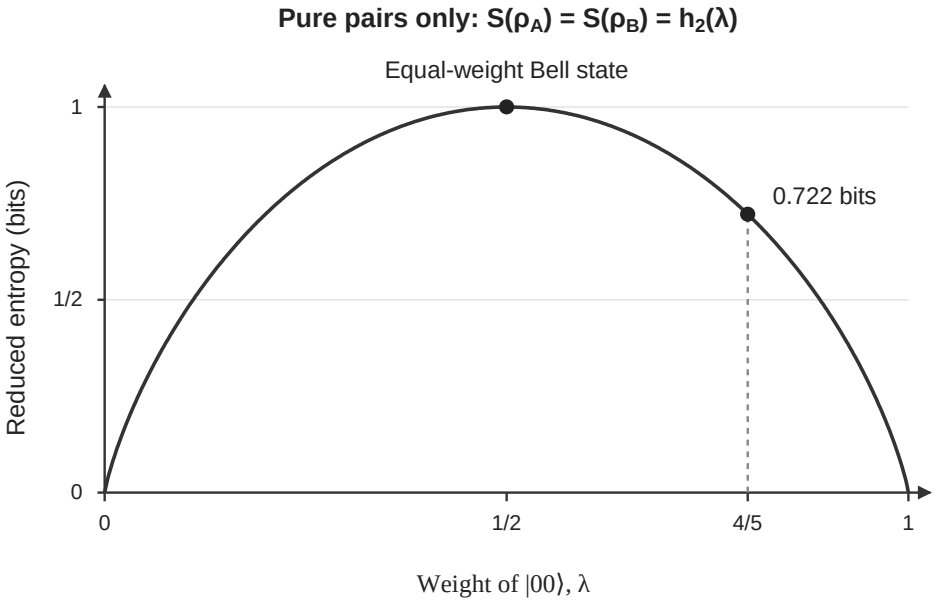


Figure 9.4. Entanglement varies even when Z agreement stays perfect. For the pure family in Eq. (9.11), either reduced entropy is $h_2(\lambda)$ bits. The product endpoints give zero, the equal-weight Bell state gives one, and $\lambda = 4/5$ gives approximately 0.722. All members give perfect Z agreement. The curve is a property of stipulated pure states, not a universal entanglement measure for mixed states and not measured data.

The restriction to a pure pair matters. The correlated lottery also has local entropy one bit, yet it is not entangled. We cannot use a subsystem's mixedness alone as an entanglement meter when the complete state may itself be mixed. The operational importance of reduced entropy for pure pairs was established through protocols that concentrate many partly entangled pairs into fewer maximally entangled pairs using local operations and classical communication.[12]

What an experiment must establish

No laboratory receives a density matrix printed on a particle. It receives preparations, controls, and outcomes. A proposed circuit predicts a state; measurements test whether the device actually prepared it. A witness may establish entanglement without determining every matrix entry. Full state reconstruction asks for more information and requires additional measurement settings.

The difference remains useful with more than two systems. In a paper published in August 2026, Dian Wu and colleagues used photonic experiments to certify several three- and four-qubit entangled states from suitably chosen correlations involving only subsets of the particles. Different target states required different subset sizes. The study concerns designed entanglement witnesses, not an assertion that every many-particle state can be certified from the same few local measurements.[13]

Entangled pairs also do work. In a 2025 experiment, D. Main and colleagues connected two trapped-ion processing modules, separated by about two metres, through an optical

link. Shared entanglement, local operations, and exchanged classical measurement records let them implement a gate between circuit qubits in different modules. The classical link was part of the procedure, not an optional confirmation of a gate completed instantaneously at a distance. Generating the remote resource and using it were distinct stages.[14]

Neither experiment is reproduced by our ideal figures. Their relevance is the same distinction we have followed throughout the chapter: an intended joint state, evidence for its properties, and a protocol that uses those properties are separate achievements.

The whole is not a hidden third particle

Entanglement is always relative to a stated division into systems. Here A and B are distinguishable registers with separately defined operations. A single carrier can also have several degrees of freedom, but that does not make every mathematical decomposition a pair of distant laboratories. Likewise, the electron correlations of Chapter 6 require careful choices about modes, spin, and the indistinguishability of electrons. Our two named qubits should not be imported into that setting without specifying the division.

Changing a local basis cannot remove pure-state entanglement: the factorization question is independent of how each part is labeled. Changing which degrees of freedom count as the parts asks a different question. The extra structure belongs to the joint state relative to that division; it is not another object lodged between the carriers.[15]

Return to the matching records. At first they could have come from one ordinary coin copied twice. We now know how a second measurement can expose the difference, how to calculate the state available to either observer, and how to test a whole class of unentangled preparations. We also know why a predictable relationship is not a controllable message.

One question remains. Perhaps the devices have more elaborate shared instructions than our quantum-state picture allows them. Can an experiment rule out those instructions without trusting the inside of each device? To find that limit, we must let Alice and Bob choose their questions independently and compare more than identical settings. That is the argument Bell made possible.

Notes and sources

1. John Watrous, IBM Quantum Learning, Basics of Quantum Information, “Multiple systems: Quantum information,” sections on tensor products and entangled states. Equations (9.1)–(9.2) fix the register order A then B. Dimension counting concerns a state space, not the number of independently readable classical messages. The product expansion is reproduced here as a worked algebraic construction. Tensor products and joint states.

2. John Preskill, Quantum Information, Chapter 2, “Foundations I: States and Ensembles,” §§2.3.1–2.4.1. The determinant test applies to a normalized pure two-qubit vector: its 2-by-2 coefficient matrix has rank one exactly when it factors. It does not test entanglement of a general mixed density matrix. Equation (9.3) is this book’s ideal controlled-NOT construction, not an inferred experimental state. Composite systems and entanglement.

3. Watrous, “Multiple systems: Quantum information,” on independent operations and controlled NOT. A product of invertible local operations preserves pure-state factorability. The disentangling operation discussed here is a joint controlled NOT. Outcome-conditioned local measurements are a different class of operation and can leave a product state together with a classical record. Operations on multiple systems.

4. Equation (9.4), Table 9.1, and all displayed probabilities are original calculations using the established X, Y, and Z conventions. For the Bell-plus state the correlations XX, YY, ZZ are +1, −1, +1. For the fair 00/11 lottery they are 0, 0, +1. The table’s independent mixture is $I_4/4$. All three preparations have maximally mixed marginals. The full outcome tables follow by applying the corresponding local measurement projectors to each density matrix; this includes mismatched settings.

5. Preskill, Chapter 2, §2.3.1, on reduced density operators and the partial trace. The general pure-state formula in the text matches Bob’s index before summing. It does not sum coherent amplitudes over distinguishable Bob outcomes. Equation (9.6) is valid for both specified preparations, not for arbitrary entangled states. No measurement of B is required to define its reduced partner state. Reduced states.

6. Preskill, Quantum Information, Chapter 10, “Quantum Shannon Theory,” June 2025 revision, §10.2.1, Eq. (10.69). Equation (9.7) defines quantum mutual information in bits. The values 2 and 1 are calculated from the full and reduced density matrices. The classical mutual information after local Z readout is one bit in either correlated example. Quantum mutual information is not a per-readout classical communication capacity. Entropy and correlations.

7. Preskill, Chapter 2, §§2.5.2–2.5.3; Chapter 3, “Foundations II: Measurement and Evolution,” §§3.1–3.2. A local trace-preserving quantum channel on A leaves the reduced state of B unchanged. Selective branches can have different normalized conditional states; their probability-weighted sum is unchanged. Figure 9.2 uses ideal projective measurements and fresh Bell pairs. There is no postselection in its unsorted averages. Ensembles and signalling; Local channels and measurement.

8. Reinhard F. Werner, “Quantum states with Einstein-Podolsky-Rosen correlations admitting a hidden-variable model,” Physical Review A 40, 4277–4281 (1989), doi:10.1103/PhysRevA.40.4277. Equation (9.9) is the finite-dimensional definition of separability as a mixture of product states. The discussion uses contemporary terminology rather than adopting every historical label in that paper. Original research article.

9. The two-setting witness is derived explicitly in Eq. (9.10) from the Bloch-ball constraint and the Cauchy–Schwarz inequality; convex averaging extends it to Eq. (9.9). The same witness is the two-qubit case of Géza Tóth and Otfried Gühne, “Entanglement detection in the stabilizer formalism,” Physical Review A 72, 022340 (2005), doi:10.1103/PhysRevA.72.022340, Theorem 1, Eqs. (11)–(12). See also Otfried Gühne and Géza Tóth, “Entanglement detection,” Physics Reports 474, 1–75 (2009), doi:10.1016/j.physrep.2009.02.004, for witness theory. Figure 9.3 uses $\rho(v)=v\rho\Phi+(1-v)I_4/4$, so $CX=CZ=v$; this witness detects $v>1/2$. It is sufficient, not necessary. The 0.8 example is stipulated, not measured. Witness theory, author version. Original witness paper.

10. The witness assumes trusted qubit dimensions and calibrated local noncommuting X/Z observables, setting choices independent of the preparation, and comparable sampling. These are stronger device assumptions than a Bell-local model requires. A pair of classical devices can share independent random X,

Y, and Z instructions, use equal signs for X/Z and opposite signs for Y, and reproduce all nine Pauli-setting distributions of this Bell state. Other directions expose stronger constraints; that argument is deferred to Chapter 10. Werner's paper distinguishes entanglement from the failure of local hidden-variable models. Entanglement and hidden-variable distinction.

11. Preskill, Chapter 2, §§2.4–2.4.1, on the Schmidt decomposition. For pure bipartite states, the reduced density matrices have the same nonzero eigenvalues. The two-qubit family in Eq. (9.11) has spectrum λ, $1-\lambda$ locally. The $\lambda=4/5$ entropy is 0.7219280949 bits, XX correlation 0.8, and X agreement probability 0.9. Figure 9.4 and these values are original evaluations of the stated model. Schmidt form and reduced spectrum.

12. Charles H. Bennett, Herbert J. Bernstein, Sandu Popescu, and Benjamin Schumacher, "Concentrating partial entanglement by local operations," Physical Review A 53, 2046–2052 (1996), doi:10.1103/PhysRevA.53.2046. The resource interpretation concerns many identically prepared pure pairs with suitable local protocols and classical communication. Local entropy alone is not an entanglement measure for an arbitrary mixed pair. Published paper and author manuscript.

13. Dian Wu, Fei Shi, Jia-Cheng Sun, Bo-Wen Wang, Xue-Mei Gu, Giulio Chiribella, Qi Zhao, and Jian Wu, "Experimental Multipartite Entanglement Detection with Minimal-Size Correlations," Physical Review Letters 137, 100201 (2026), doi:10.1103/w6fl-nx59. Published 31 August 2026. The reported photonic tests use two-body correlations for specified three-/four-qubit W and four-qubit Dicke states, and three-body correlations for a four-qubit cluster state. Only these documented certification claims are used here; the book's two-qubit witness is not their experimental protocol. The published abstract supports the certification claims retained here. The earlier author manuscript, arXiv:2510.22918v1 (27 October 2025), provides additional methodological context; no additional numerical result from that version is attributed here to the final article. Published article and abstract. Author manuscript.

14. D. Main, P. Drmota, D. P. Nadlinger, E. M. Ainley, A. Agrawal, B. C. Nichol, R. Srinivas, G. Araneda, and D. M. Lucas, "Distributed quantum computing across an optical network link," Nature 638, 383–388 (2025), doi:10.1038/s41586-024-08404-x. Published 5 February 2025. The experiment separates heralded remote-entanglement generation from its use with local gates, classical measurement exchange, and conditional corrections. No instantaneous communication or deterministic success of each resource-generation attempt is implied. Peer-reviewed open-access article.

15. Subsystems and their accessible operations are specified throughout this chapter. The elementary tensor-product argument concerns separately addressable qubit registers, not arbitrary labeling of identical particles. For the bipartition dependence and local-basis invariance, see Preskill, Chapter 2, §§2.3–2.4; for identical-particle amplitudes, Richard P. Feynman, Robert B. Leighton, and Matthew Sands, The Feynman Lectures on Physics, Vol. III, Chapter 4. No claim that exchange antisymmetry alone supplies two independently addressable qubits is made. Subsystem structure; Identical particles.

CHAPTER 10

The Limit of Shared Instructions

Imagine receiving two sealed instruments. Each has two buttons and a small display. Their supplier claims that the instruments contain an entangled pair. You are allowed to choose the buttons and record the answers, but not to inspect the machinery. What could establish that the claim is more than clever programming?

Matching answers will not suffice. The instruments could share a list made before delivery. Even agreement for two different kinds of question is easy to arrange: put two answers on the list, one for each button. Chapter 9's entanglement witness excluded that possibility only because we trusted the instruments to perform particular qubit measurements.

Now withdraw that trust. The boxes may contain large classical computers, shared random numbers, and elaborate memories. Give them every such advantage. We will impose restrictions on which information can reach them during a trial, not on how simply they must be built.

John Bell's 1964 argument made this change of perspective possible. A family of explanations could be constrained by its causal structure, without specifying the particles' hidden machinery. We will develop a later, experimentally useful version associated with John Clauser, Michael Horne, Abner Shimony, and Richard Holt: the CHSH test. Its decisive arithmetic fits on a small table.[1]

A game with the questions kept apart

Keep Alice and Bob in separate laboratories. On each trial, a referee supplies Alice with a binary setting x and Bob with a binary setting y. Each is chosen fairly and independently, after the shared resource has been distributed. Alice returns one bit, a; Bob returns one bit, b. Neither learns the other setting before both answers are fixed.

They may agree on any strategy beforehand. They may share common causes and records. What they cannot do is exchange information about the current questions during the response interval, or arrange for their shared instructions to predict those questions. Later, the records can be brought together and scored.

The rule is to give matching answers unless both settings are one. In that last case, give different answers. Using the binary operations from Chapter 8, a win means

$$a \oplus b = x\,y\,. \tag{10.1}$$

The circled plus is XOR: zero for agreement, one for disagreement. The product $x\,y$ is one only when both settings are one. Table 10.1 lists every setting pair. The referee asks only one pair on a trial; the four rows are four possible trials, not four measurements on the same pair of particles.

Table 10.1. The four questions and the required relationship.

x	y	To win	Both always answer 0
0	0	Same answers	Win
0	1	Same answers	Win
1	0	Same answers	Win
1	1	Different answers	Lose

The settings are independent fair bits, so each row has probability 1/4. The last column is one fixed local strategy. A trial includes an answer on both sides; rules for missing detections must be specified before scoring.

Always answering zero wins on three rows. Could a more ingenious strategy win on all four? The obstacle is that each observer sees only half of the question. Alice cannot tell whether her setting one was paired with Bob's zero or his one.[2]

Why the fourth instruction cannot fit

First consider a fixed instruction sheet. Let a_0 and a_1 be Alice's prescribed answers to her two settings, and b_0 and b_1 Bob's. To win on every possible row, the sheet would have to satisfy

$$\begin{aligned} a_0 \oplus b_0 &= 0 \,, a_0 \oplus b_1 = 0 \,, \\ a_1 \oplus b_0 &= 0 \,, a_1 \oplus b_1 = 1 \,. \end{aligned} \tag{10.2}$$

The first three relations say that a_0, b_0, a_1, and b_1 must all agree. The fourth says that two of them must disagree. Equivalently, XOR all four equations: every answer appears twice and cancels, leaving zero on the left and one on the right. No instruction sheet can do that.

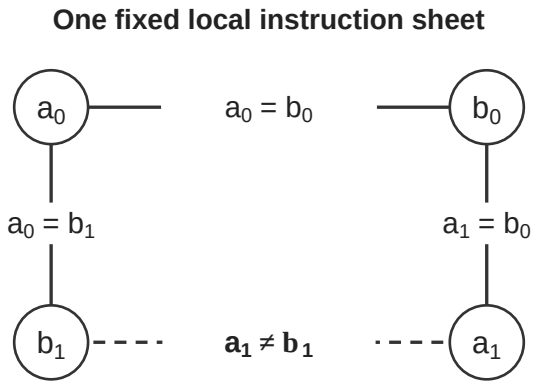


Figure 10.1. Three agreements force the fourth. Each circle is an answer on a hypothetical fixed local instruction sheet. The three solid edges require equality, which makes all four entries equal. The dashed edge requires inequality and cannot then be satisfied. The diagram compares four possible setting pairs; it does not depict four simultaneous measurements or a signal between laboratories.

At least one row fails. Since the rows are equally likely, the winning probability of every fixed local sheet is at most three-quarters. Our always-zero strategy attains that limit.

Randomness does not repair the contradiction. The devices may draw a different sheet on every trial, using shared randomness. But an average of scores no greater than three-quarters cannot exceed three-quarters. Private random choices can also be resolved into local response sheets before averaging. The bound does not require predetermined outcomes; it covers probabilistic strategies as well.

A compact expression states the full class of models. Let λ collect the shared information and any relevant past history. Then a Bell-local model has the form

$$P(a,b \mid x,y)=\sum_{\lambda} q(\lambda)\,P_A(a \mid x,\lambda)\,P_B(b \mid y,\lambda). \tag{10.3}$$

Once the shared information is specified, Alice's response depends on her setting, and Bob's on his. The weights $q(\lambda)$ do not change with the current settings. That last condition is measurement independence: the setting choices do not select different hidden-instruction ensembles. A continuous hidden variable would replace the sum by an integral without changing the argument.[3]

The four entries on a sheet are a representation of this model, not observations we claim to make together. The contradiction constrains possible local response rules. No assumption about the Hilbert-space dimension, the orientation of a polarizer, or the material inside a box entered the bound.

$$w_{local}\le\frac{3}{4}. \tag{10.4}$$

Here w is the probability of winning, or the expected fraction of wins over repetitions. A finite local record can exceed three-quarters by chance. The bound is not an instruction to declare a discovery after the first four wins. We will return to the statistics.

The score that laboratories usually report

Bell experiments often express the same test using correlations. For each setting pair, define E_{xy} as the probability that the outputs agree minus the probability that they disagree. It runs from minus one to plus one. Equivalently, relabel bit zero as $+1$ and bit one as -1, and average the product of the labels.

Agreement has probability $(1+E_{xy})/2$. Disagreement has probability $(1-E_{xy})/2$. Average the first expression for the first three rows and the second for the last row:

$$\begin{aligned} S_{CHSH} &= E_{00}+E_{01}+E_{10}-E_{11}, \\ w &= \frac{1}{2}+\frac{S_{CHSH}}{8}. \end{aligned} \tag{10.5}$$

Thus the upper local bound is $S_{CHSH} \le 2$. The subscript distinguishes this score from the entropy called S earlier in the book. A relabeled version gives the familiar two-sided bound $\left|S_{CHSH}\right| \le 2$. We will keep the scoring convention of Table 10.1 throughout.[4]

Equation (10.5) weights the four setting pairs equally. A finite record need not contain equal numbers of them. Suppose, as an invented example, both boxes always answer zero and the setting pairs 00, 01, 10, and 11 occur 2, 2, 2, and 4 times. The boxes win 6 of 10 trials. Each separately estimated correlation is nevertheless one, giving $S_{CHSH} = 2$. Inserting that estimate into Eq. (10.5) gives three-quarters, not six-tenths. Nothing is inconsistent: the raw win fraction uses the observed setting frequencies, while the four-correlation estimate gives each setting pair equal weight. Our later statistical test counts raw wins under fresh, uniform setting choices.

Notice the change from Chapter 9. The witness there compared identical, calibrated X and Z settings. Here we compare all four combinations of two settings per laboratory. We are no longer constraining how a qubit can answer two questions; we are constraining how any local instruction model can coordinate answers to independently selected questions.

A quantum strategy that exceeds the bound

Now give the boxes a specific quantum implementation. On every trial distribute the Bell pair $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. Alice measures Z when $x = 0$ and X when $x = 1$. Bob uses two directions halfway between those axes: one tilted toward positive X from positive Z, the other tilted toward negative X.

These are directions on the Bloch map, not necessarily angles between pieces of apparatus. Label the outcome along the chosen direction zero and the opposite outcome one. If θ is the angle from positive Z toward positive X, the zero reference state is

$$|0_\theta\rangle = \cos(\theta/2)|0\rangle + \sin(\theta/2)|1\rangle. \tag{10.6}$$

Its opposite reference is orthogonal. The half-angle is essential: a quarter-turn on the Bloch sphere carries the zero state to the plus state. Alice's two angles are 0 and $\pi/2$; Bob's are $\pi/4$ and $-\pi/4$. Figure 10.2 shows all four on the same map.[5]

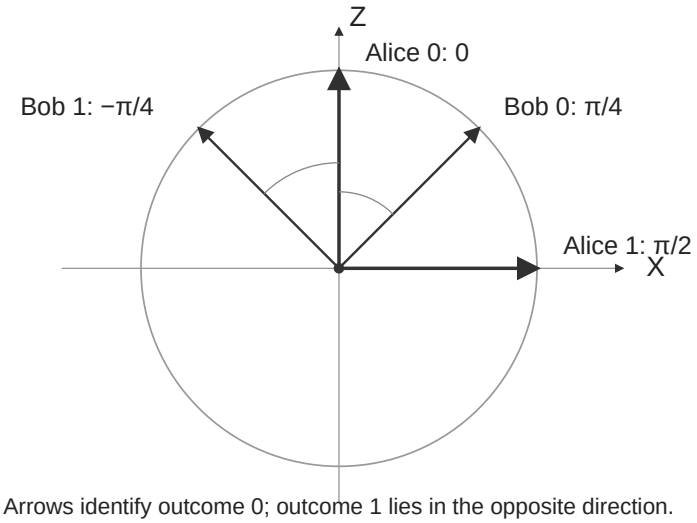


Figure 10.2. The four measurement directions of the quantum strategy. Arrows indicate the outcome-zero reference directions on the X–Z great circle of the Bloch sphere. Alice chooses 0 or π/2; Bob chooses π/4 or −π/4. The unshown outcome-one reference is opposite each arrow. Three Alice–Bob angle differences have magnitude π/4; the fourth is 3π/4. The angles describe qubit states, not optical polarizer settings or particle trajectories.

We can calculate the result without appealing to a picture of instantaneous influence. Suppose Alice's angle is α and Bob's is β. Their joint zero–zero amplitude is the overlap of the two reference states with the Bell pair:

$$\begin{aligned} \langle 0_\alpha 0_\beta | \Phi^+ \rangle &= \frac{\cos(\alpha/2)\cos(\beta/2) + \sin(\alpha/2)\sin(\beta/2)}{\sqrt{2}} \\ \langle 0_\alpha 0_\beta | \Phi^+ \rangle &= \frac{\cos[(\alpha-\beta)/2]}{\sqrt{2}}. \end{aligned} \tag{10.7}$$

The second line uses the cosine subtraction identity. Squaring gives half of $\cos^2[(\alpha-\beta)/2]$. The one–one probability is the same. Adding those two distinguishable records gives the total agreement probability. Consequently,

$$\begin{aligned} P(a=b \mid \alpha,\beta) &= \cos^2[(\alpha-\beta)/2], \\ E(\alpha,\beta) &= \cos(\alpha-\beta). \end{aligned} \tag{10.8}$$

The first three setting pairs have correlation $1/\sqrt{2}$. The fourth has correlation $-1/\sqrt{2}$—exactly the row on which the scoring rule asks for disagreement. Table 10.2 follows the calculation. Every row has the same winning probability, approximately 0.8536.

Table 10.2. The same scoring rule, a quantum implementation.

(x, y)	$\alpha - \beta$	E_{xy}	**Winning relation**
(0, 0)	$-\pi/4$	$1/\sqrt{2}$	Agreement
(0, 1)	$\pi/4$	$1/\sqrt{2}$	Agreement
(1, 0)	$\pi/4$	$1/\sqrt{2}$	Agreement
(1, 1)	$3\pi/4$	$-1/\sqrt{2}$	Disagreement

Each required relation has probability (1 + 1/√2)/2. Entries are exact predictions for the Bell-plus state and ideal local measurements. The setting pairs refer to separate trials. No measured data or detection losses are included.

$$S_{CHSH} = 2\sqrt{2}, w_{quantum} = \frac{2+\sqrt{2}}{4} \approx 0.8536. \tag{10.9}$$

This is greater than three-quarters. The mathematical bound has not been broken by a bad sum. The quantum probability table cannot be represented by Eq. (10.3) under its stated conditions. Entanglement supplies a coordination resource that those local response sheets do not contain.[6]

Quantum theory does not allow perfect success in this game. The largest quantum CHSH value is $2\sqrt{2}$, known as Tsirelson's bound. Our calculation shows that this value is attainable. A separate general theorem establishes that no quantum strategy for this game can exceed it, even with higher-dimensional devices. A sufficiently convincing score above that limit would challenge the quantum model or the experimental assumptions, not demonstrate an extra-strong version of ordinary entanglement.[7]

What appears in either notebook

For any one of the four setting pairs, the two winning output pairs each have probability about 0.4268, and the two losing pairs each have probability about 0.0732. Alice's zero probability is their appropriate sum: one-half. Bob's is also one-half. Changing either distant setting changes no local marginal probability.

Alice cannot use the arrangement to select a bit in Bob's notebook. The distinction from shared local instructions appears only in the joint table assembled after the settings and results are compared. Communication is needed for that comparison; the Bell pair does not supply a way around it.

Bell locality and no signalling are therefore different conditions. Equation (10.3) implies that a distant setting cannot alter a local distribution. Its converse is false: the quantum table we have just calculated has unchanged local distributions but exceeds the Bell-local bound. No-signalling alone does not mean that correlations must have a shared-instruction explanation.[8]

A failure to exceed two proves less. The state may be noisy, the measurements poorly chosen, or the sample too small. It does not by itself show that the source is unentangled. Certification has a direction: a justified violation excludes the local class; an inconclusive test does not establish membership in it.

Giving the separation a clock

The arithmetic is simple enough. Arranging a physical experiment that actually tests its assumptions is harder. If Alice's current setting reaches Bob before he commits his answer, a classical device can win every time: let Alice answer zero, send x, and let Bob answer $x\,y$. The scoring rule has not changed; the information available to the devices has.

To exclude communication at or below the speed of light during the relevant interval, locate the laboratories far enough apart and generate settings and answers quickly enough. The setting-generation events and the distant outcome-registration events must be spacelike separated, allowing for the full event intervals and their timing and position uncertainties. In ordinary language, even a light-speed signal carrying that setting would arrive too late.

As an illustrative layout, take laboratories 300 metres apart. Light in vacuum takes about one microsecond to cross that distance. If both settings are generated near the same time and both answers are irrevocably recorded 0.40 microseconds later, the cross-laboratory events can be spacelike separated. Figure 10.3 shows these stipulated values, not a particular experiment.[9]

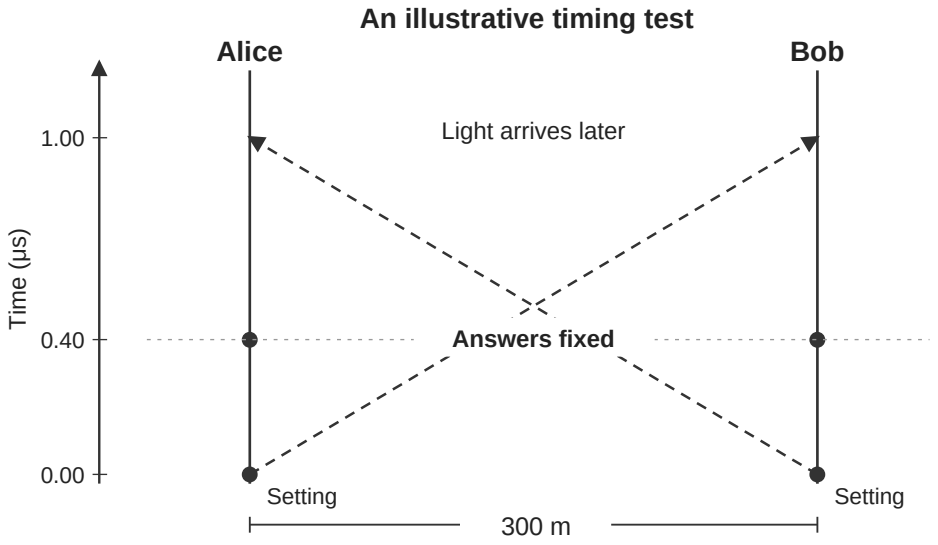


Figure 10.3. The distant question would arrive too late. This idealized spacetime diagram places Alice and Bob 300 m apart. Time increases upward. Both settings are generated at time zero and local answers are completed at 0.40 μs. Dashed diagonals show light-speed signals from either setting event, arriving at the opposite laboratory only after about 1.00 μs. Vertical lines mark laboratory locations. A real analysis must bound the full setting-generation and outcome-recording intervals and their uncertainties; simultaneous point events are only the drawing's simplification.

Distance alone is not the test. The complete timing chain matters: when the setting becomes available, when the measurement is influenced by it, and when the answer becomes a fixed record. A late timestamp attached to an answer that was chosen earlier would not establish the required sequence.

Setting independence needs attention too. A pseudorandom schedule already known to the boxes lets them predict both questions and coordinate accordingly. The ideal fair-input bound assumes fresh choices independent of the shared information, including the relevant history. Real analyses can allow quantified imperfections, but they must use the corresponding bound rather than silently retaining three-quarters.

This assumption is not a claim about a person's metaphysical freedom. It specifies a statistical and causal relationship between the setting sources and the tested systems. No finite experiment can exclude every proposed hidden correlation connecting its entire past. The operational task is to identify the assumption and exclude or bound the mechanisms that the experiment can test.

The missing answers can manufacture a victory

Now consider a less obvious shortcut. Suppose we score only trials on which both instruments produce a detection. If the probability of producing an answer depends on the local setting and shared instructions, the surviving sample need not obey the original bound.

Here is a completely local construction. Before each trial, share a fair random bit g, independent of the questions. Bob always answers $b = g\ y$. Alice answers zero only when $x = g$; otherwise her instrument reports no result. On every retained trial, $x = g$, so $a \oplus b = g\ y = x\ y$. The retained sample wins every time.

Only half the trials survive. Importantly, that same half-survival rate holds for every setting pair. Even apparently balanced detection totals would not reveal the selection problem by themselves. The accepted ensemble contains a hidden correlation between the setting and the shared bit.

Specify instead, before collecting data, that Alice's missing result is assigned the bit zero. All trials can now be scored. The four equally weighted retained cases win. Of the four previously omitted cases, those with $y = 0$ win and those with $y = 1$ lose. The complete score is six wins out of eight: three-quarters, just as required. Figure 10.4 keeps both accounts visible.[10]

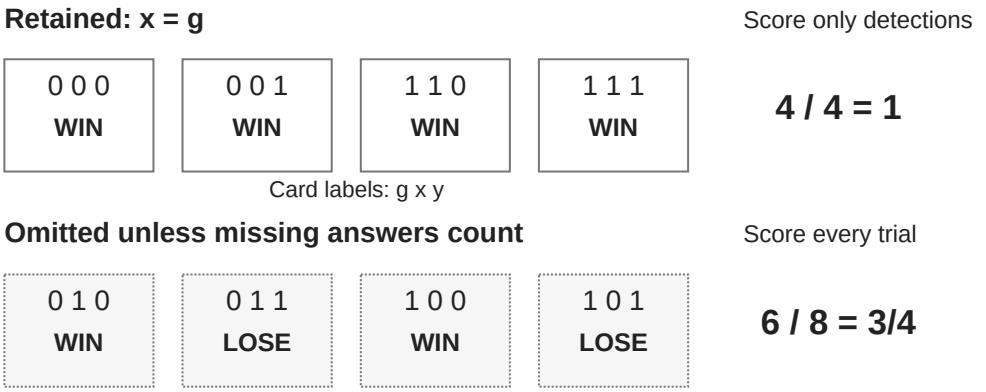


Figure 10.4. The same local device, two different denominators. The construction uses independent fair bits g, x, and y, giving eight equally likely cases. Alice reports only when x = g; Bob reports b = gy. All four retained cases win. Assigning Alice's missing result the preset bit zero restores the four omitted cases: two wins and two losses. Postselected scoring gives 4/4; complete scoring gives 6/8. The counts enumerate a stipulated local model, not experimental observations. Acceptance is one-half for each setting pair.

This is the detection loophole in a form we can check. A claim that the detected pairs are representative requires justification; it cannot be supplied by the size of the observed violation. Other selection procedures, including how detection times are paired into trials, require the same care.

Selection before the questions can have a different status. An event-ready experiment first obtains a herald announcing that a shared resource is available, then generates the independent settings and retains every resulting trial. Preparation attempts that failed before that announcement are not retroactively rejected losing answers. The timing and independence of the herald are what make this different from our selective detector.[11]

A finite record, not an infinite average

Even with sound timing and trial accounting, there remains chance. Suppose we choose in advance to run N trials, with fresh independent fair settings on each. Under the local model, the conditional chance of winning the next trial is at most three-quarters even if the devices remember and adapt to everything that happened earlier.

Here is why memory does not defeat the bound. For any particular history, let q_i be the conditional winning probability on the next trial; it is at most three-quarters. We can represent that trial mathematically by drawing a fresh uniform number U_i between zero and one and declaring a win when $U_i \le q_i$. Beside it, use the same number for a comparison trial that wins when $U_i \le 3/4$. Every win of the modeled device is then a comparison win. The comparison trials are independent and each wins with probability three-quarters, even though the device's thresholds can depend on its past. Its total wins are therefore bounded by their total. This is a mathematical representation of the probabilities, not a claim that the apparatus draws these extra numbers.

For example, if the next-trial win chance after a particular history is 0.60, uniform draws up to 0.60 win in both accounts. Draws above 0.60 and up to 0.75 add comparison wins only. Draws above 0.75 win in neither account. This nesting holds for every permitted threshold, even when the device adapts to its history.

That observation gives a rigorous upper bound on the probability of obtaining at least k wins. It is the upper tail of a binomial distribution:

$$p_{tail} \le \sum_{j=k}^{N} \binom{N}{j} \left(\frac{3}{4}\right)^{j} \left(\frac{1}{4}\right)^{N-j}. \tag{10.10}$$

The binomial coefficient counts placements of j wins among N positions. The two powers weight wins and losses. Here the expression bounds adaptive local models too; it does not require identical, independent behaviour inside the devices. The fresh-setting condition and fixed trial count do the work.[12]

For a stipulated record of 850 wins in 1,000 trials, the right side is about 8.8×10^{-15}. That is an illustrative calculation, not data from a Bell experiment. It is a bound on the probability that an allowed local model produces a score at least this large. It is not the probability that local models are true, and it is not a guarantee that the apparatus was correctly characterized at its external boundaries.

The protocol must also specify when to stop. Repeatedly checking the same fixed-sample formula and ending at the most favorable moment changes the statistical question. Planned stopping rules or methods valid under such monitoring are needed. Likewise, imperfect setting generators and outcome-dependent exclusions require their own treatment. A tiny number cannot correct a wrong sampling model.

What the experiments established

In 2015, Bas Hensen and colleagues performed an event-ready experiment using electron spins in diamond separated by 1.3 kilometres. The heralded preparation, rapid independent setting choices, and efficient spin readout addressed the locality and detection loopholes together. They reported 245 trials, a CHSH value of 2.42 ± 0.20, and a local-model test with $p \leq 0.039$ that allowed memory in the devices. The quoted uncertainty comes from the authors' conventional independent-trial, Gaussian analysis; the separate local-model test allows memory and accounts for measured setting-source imperfections. The modest trial count matters to the strength of that particular record.[13]

Table 10.3. A Bell claim uses both a comparison bound and a statistical test.

Quantity	Value	What it establishes
Bell-local CHSH bound	2	An expectation bound for Eq. (10.3), with its stated setting and locality assumptions.
Ideal quantum maximum	$2\sqrt{2} \approx 2.828$	A theoretical maximum achieved by the ideal strategy above; not an experimental result.
Hensen et al. CHSH estimate	2.42 ± 0.20	245 event-ready trials; the quoted uncertainty uses the authors' conventional independent-trial, Gaussian analysis.
Hensen et al. local-model test	$p \leq 0.039$	Reported bound from a separate finite-data analysis allowing device memory and measured setting imperfections; not a probability that locality is true.

Reported results are from the experiment described in note 13, not from our invented 850-of-1,000 example. The ±0.20 is one standard deviation from the authors' conventional independent-trial Gaussian analysis; it is not the memory-allowing test. The 196 observed wins give a linear score of 2.40, while separately normalized setting correlations give about 2.42. Those estimators need not coincide in a finite record. The p-value is not the probability that local models are true.

In the same year, separate experiments led by Marissa Giustina and Lynden Shalm used entangled photons and efficient detectors to test Bell inequalities while addressing the principal detection and locality loopholes. Their protocols and statistical analyses were not simply repetitions of our idealized game. The common achievement was to bring the theoretical assumptions into a complete experimental design.[14]

"Loophole-free" in this context does not mean free of every conceivable assumption. It identifies major experimental escape routes that were closed within stated causal and statistical conditions. The result is stronger than demonstrating entanglement with calibrated measurements, and more precise than declaring that experiments have disproved reality.

Within the quantum description, a valid Bell violation also certifies entanglement: separable states produce Bell-local probabilities for local measurements. But a Bell test does not select one interpretation of quantum mechanics. A theory that permits nonlocal

hidden dynamics, or changes the setting-independence assumption, lies outside the class bounded by Eq. (10.3). Its further claims would need their own tests.

From a limit on explanations to a useful resource

The same logic can establish properties of devices without trusting a detailed model of their interiors. This is the point of device-independent certification. The boundary is still real: the settings, trial records, allowed communication, and statistical procedure must satisfy the protocol. What is removed is the assumption that the boxes faithfully contain the advertised state and calibrated measurements.

In a 2025 experiment, Simon Storz and colleagues used remote superconducting circuits to bound both the quality of an entangled state and the measurements from Bell-test records. This approach is called self-testing. Its guarantees concern identifiable quantum degrees of freedom up to permitted changes of local representation; they are not a microscopic photograph of everything inside a cryostat.[15]

Randomness is another application. A file may pass ordinary frequency tests yet have been fully determined by a seed known to someone else. Bell-based certification instead constrains what an allowed predictor could know under a specified physical model. Finite-data analysis and a randomness extractor turn that constraint into an output with a quantified closeness to ideal randomness. An extractor is an algorithm that concentrates suitably bounded raw randomness into a shorter, nearly uniform output; its additional inputs and assumptions must also be accounted for. A large CHSH score alone is not the complete protocol.

Gautam Kavuri and colleagues reported a public randomness beacon in 2025 that combined Bell-based generation with cryptographic records making the extraction process auditable. Its successful rounds released 512-bit public outputs. Unpredictability before generation and secrecy after publication are different properties; a public beacon is deliberately not a supply of private keys.[16]

A May 2026 paper by Anatoly Kulikov and colleagues demonstrated randomness amplification with superconducting circuits. It relaxed the idealization of perfect input randomness by using a protocol with a quantified model of imperfect setting sources. That is not the generation of unpredictability from a completely prearranged input: residual unpredictability and restrictions on correlations remain part of the conditions. The advance is in what can be certified with imperfect resources, not in eliminating assumptions by decree.[17]

The instruction sheet we can no longer supply

We began with sealed boxes, allowing arbitrary shared preparation, complicated local machinery, and stochastic responses. The restriction was not simplicity. Each side had to answer without the other current question, and the questions had to be independent of the shared instructions.

Quantum theory predicts a score that this class cannot reach on average. Experiments connect that prediction to tests of timing, losses, and finite statistics. The conclusion does not follow from matching records alone. It emerges from comparing the joint relationships across independently chosen questions.

Those relationships are still not a channel for choosing a distant outcome or reading every amplitude. Next we will ask what can be transferred: why an unknown quantum state cannot be copied universally, yet can have its role moved from one system to another.

Notes and sources

1. John S. Bell, “On the Einstein Podolsky Rosen paradox,” Physics Physique Fizika 1, 195–200 (1964), doi:10.1103/PhysicsPhysiqueFizika.1.195; John F. Clauser, Michael A. Horne, Abner Shimony, and Richard A. Holt, “Proposed Experiment to Test Local Hidden-Variable Theories,” Physical Review Letters 23, 880–884 (1969), doi:10.1103/PhysRevLett.23.880. The game is a later presentation of CHSH, not a claim that Bell’s 1964 paper used this game. Bell’s paper; CHSH paper.

2. John Watrous, IBM Quantum Learning, Basics of Quantum Information, “Entanglement in action: CHSH game.” The present scoring convention is $a \oplus b = x\,y$, with independent uniform settings. Tables and figures are original constructions. No same-pair simultaneous measurement of all four settings is assumed. Game formulation.

3. John F. Clauser and Michael A. Horne, “Experimental consequences of objective local theories,” Physical Review D 10, 526–535 (1974), doi:10.1103/PhysRevD.10.526. Equation (10.3) explicitly separates local factorization from independence of the current settings and the hidden information. Local randomness can be absorbed into the mixture of deterministic response functions for this finite scenario; determinism is not an additional physical premise. Stochastic local models.

4. Equation (10.5) is the elementary conversion between CHSH correlations and the game’s expected score under uniform setting probabilities. In a finite sample, the raw win fraction and a score assembled from four separately normalized conditional-frequency estimates need not be exactly related by this formula. The statistical example uses raw wins, fresh uniform draws, and a fixed total number of trials. For the distinction between separately normalized and linear trial statistics in memory-bearing experiments, see Jonathan Barrett, Daniel Collins, Lucien Hardy, Adrian Kent, and Sandu Popescu, “Quantum nonlocality, Bell inequalities, and the memory loophole,” Physical Review A 66, 042111 (2002), doi:10.1103/PhysRevA.66.042111, §§II–III; author version quant-ph/0205016v3. The 2, 2, 2, 4 setting-count example is a constructed illustration, not data from that paper. CHSH formulation. Memory analysis.

5. John Preskill, Lecture Notes for Physics 219/Computer Science 219, Chapter 4, “Quantum Entanglement,” §§4.3.1–4.3.3; Watrous, “CHSH game.” The reference state in Eq. (10.6) uses Bloch coordinates $(\sin\theta, 0, \cos\theta)$. Its orthogonal partner may be chosen as $-\sin(\theta/2)|0\rangle + \cos(\theta/2)|1\rangle$. Optical linear-polarization angles generally differ by a factor of two from these Bloch angles. Entanglement lecture; Worked quantum game.

6. Equations (10.7)–(10.9) and Table 10.2 are exact calculations for the stipulated Bell-plus state. The complete table is $P(a,b|x,y) = \left[1 + (-1)^{a \oplus b} \cos(\alpha_x - \beta_y)\right]/4$. The table can be obtained directly using tensor-product measurement projectors. Summing its four outcomes gives one; summing over either party’s outcomes gives one-half for each outcome of the other party. These values are not experimental data.

7. B. S. Cirel’son, “Quantum generalizations of Bell’s inequality,” Letters in Mathematical Physics 4, 93–100 (1980), doi:10.1007/BF00417500. Cirel’son is another transliteration of Tsirelson. The bound concerns bipartite quantum correlations with two binary settings per party, without communication during the responses. It is not a bound on arbitrary postselected tables. An accessible derivation is Preskill, Chapter 4, §4.3.2, Eqs. (4.39)–(4.49). Quantum bound.

8. The local marginal calculation follows directly from note 6. For the general quantum no-signalling statement, see Preskill, Chapter 2, “Foundations I: States and Ensembles,” §2.5, and Chapter 3, “Foundations II: Measurement and Evolution.” Outcomes of a local instrument must be included, not selected using a remote record. Bell-locality is stronger than operational no-signalling. Ensembles and signalling; Local operations.

9. Figure 10.3 stipulates a 300-m separation, simultaneous idealized setting events, and 0.40-μs local response intervals. Using the exact SI vacuum speed $c = 299792458$ m/s gives a crossing time about 1.00069 μs. This is a timing example, not a reconstruction of a published layout. Setting independence is a separate condition from spacelike separation. BIPM defining constants.

10. The selective-answer example and Figure 10.4 are original enumerations of all eight equally likely triples (g, x, y). Retention has probability one-half for every setting pair, but conditioning on it correlates g with x. With a preset zero for Alice’s missing answer the full score is 3/4. For the general detector-efficiency

issue see Anupam Garg and N. David Mermin, “Detector inefficiencies in the Einstein–Podolsky–Rosen experiment,” Physical Review D 35, 3831–3835 (1987), doi:10.1103/PhysRevD.35.3831. Detector-efficiency analysis.

11. David Elkouss and Stephanie Wehner, “(Nearly) optimal P values for all Bell inequalities,” npj Quantum Information 2, 16026 (2016), doi:10.1038/npjqi.2016.26, treats event-ready selection, memory, imperfect setting generators, and statistical protocol design. Valid heralding fixes trial inclusion before the independent current questions; it is not outcome-dependent filtering after those questions. Event-ready analysis.

12. Elkouss and Wehner, especially Eq. (18) of their published article, provide the binomial-tail bound for win/lose games with a bound on conditional winning probability. Equation (10.10) specializes it to ideal uniform-input CHSH. For N = 1,000 and k = 850 the exact tail is approximately $8.78792265 \times 10^{-15}$. Fixed N and the conditioning on past history are essential. The earlier author version, arXiv:1510.07233v3 (15 March 2016), gives the bound in Eq. (20) and an event-ready proof in Appendix B, Lemma 3. The text’s uniform-number coupling is an alternative elementary explanation for the stated fixed-trial-count setting. It does not validate arbitrary optional stopping or source bias. Finite-data bound.

13. B. Hensen and colleagues, “Loophole-free Bell inequality violation using electron spins separated by 1.3 kilometres,” Nature 526, 682–686 (2015), doi:10.1038/nature15759. The 245 trials, quoted CHSH estimate, and p-value bound are the authors’ reported results. The relevant author-manuscript text and Fig. 4 distinguish the conventional uncertainty estimate from the complete memory-allowing analysis. Its 196 wins give the linear statistic 2.40; separately normalized correlations give approximately 2.42. These are different estimators of the same probability-level quantity, not conflicting experimental results. These are the authors’ reported aggregate results, not a new analysis of their raw time-tagged records. Their event-ready spin experiment is distinct from the hypothetical spacetime and missing-detection examples in this chapter. Publication; Author manuscript.

14. M. Giustina and colleagues, “Significant-Loophole-Free Test of Bell’s Theorem with Entangled Photons,” Physical Review Letters 115, 250401 (2015), doi:10.1103/PhysRevLett.115.250401; L. K. Shalm and colleagues, “Strong Loophole-Free Test of Local Realism,” Physical Review Letters 115, 250402 (2015), doi:10.1103/PhysRevLett.115.250402. Each reports a photonic experiment and its own trial, timing, and statistical analysis. Giustina and colleagues; Shalm and colleagues.

15. Simon Storz and colleagues, “Complete Self-Testing of a System of Remote Superconducting Qubits,” Physical Review Letters 135, 030801 (2025), doi:10.1103/nv7d-k3wr. Published 15 July 2025. The work bounds state and measurement fidelities using Bell data. Self-testing is understood up to appropriate local representations and unused degrees of freedom, with the protocol’s external assumptions retained. The published abstract supports the stated certification result. Experimental and finite-sample details are described in the earlier author version arXiv:2408.01299v1 (2 August 2024); no performance number specific to that version is attributed here to the final article. Published research; Author version.

16. Gautam A. Kavuri and colleagues, “Traceable random numbers from a non-local quantum advantage,” Nature 642, 916–921 (2025), doi:10.1038/s41586-025-09054-3. Published 11 June 2025. Bell-based certification and cryptographic traceability address different parts of the process. The 512-bit output is per successful beacon round, not per photon pair. The text reports the published demonstration, not the service’s current operating status. The published abstract supports the 512-bit statement. Protocol details and assumptions are described in the earlier author manuscript arXiv:2411.05247v1; the output claim here is the published result. Published research; Author manuscript.

17. Anatoly Kulikov and colleagues, “Experimental randomness amplification,” Nature 653, 1033–1038 (2026), doi:10.1038/s41586-026-10521-8. Published 27 May 2026. The protocol uses a specified weak-source model, finite-data security analysis, and extraction; device independence does not remove its source, isolation, or record-processing conditions. The earlier author manuscript is titled “Device-Independent Randomness Amplification.” No preprint-only numerical result is attributed here to the final article. The published paper specifies the weak-source model and the extraction procedure. The earlier technical formulation is arXiv:2412.17931v1 (23 December 2024); its assumptions should not be read as a guarantee for arbitrary input sources. Published research; Earlier technical manuscript.

CHAPTER 11

What Cannot Be Copied—and What Can Be Transferred

A laboratory has prepared one qubit. Its state is not supplied as a formula, and the preparation cannot simply be repeated. Before trying a delicate operation, the experimenter wants a backup. Why not copy the qubit, keep one version safe, and work on the other?

For a classical record, this is an ordinary precaution. For an arbitrary quantum state, it asks for a machine that the quantum rules do not permit. Yet those same rules allow the state to be transferred to another laboratory without sending its original carrier there.

The distinction is not between two names for the same trick. A copier must leave two systems that each reproduce the input state. A transfer must leave one usable successor. We will first show why the copying requirement fails, then follow a transfer through every operation, message, and possible result.

Chapter 10 tested explanations without assuming what was inside the devices. Here we return to a specified quantum model: trusted gates, a prepared entangled pair, and defined measurements. We are designing a protocol and deriving its predictions. Whether a particular apparatus implements it is a further question.

The copier that works on two labels

Controlled NOT already copies a classical bit into a blank target. With the second qubit initially in $|0\rangle$, it takes $|00\rangle$ to $|00\rangle$ and $|10\rangle$ to $|11\rangle$. Both possible Z labels are copied perfectly.

Now supply the plus state instead. A coherent gate acts linearly, so it acts on each input component and adds the results. The output is

$$CNOT\left(|+\rangle|0\rangle\right)=\frac{|00\rangle+|11\rangle}{\sqrt{2}}\neq|+\rangle|+\rangle. \tag{11.1}$$

Two independent plus states would have four equal amplitudes, including $|01\rangle$ and $|10\rangle$. The actual output has neither of those terms. In a joint Z measurement, its answers always agree; independently prepared plus states disagree half the time. Figure 11.1 compares the complete tables.

a The same gate, three inputs

Input + zero target		Actual coherent output
$\|0\rangle\,\|0\rangle$	⟶	$\|0\rangle\,\|0\rangle$
$\|1\rangle\,\|0\rangle$	⟶	$\|1\rangle\,\|1\rangle$
$\|+\rangle\,\|0\rangle$	⟶	$(\|00\rangle + \|11\rangle)\,/\,\sqrt{2}$

b Joint Z probabilities for a plus input

	00	01	10	11
Actual: Bell pair	1/2	0	0	1/2
Requested: two plus states	1/4	1/4	1/4	1/4

Figure 11.1. Copying the reference labels is not copying every state. A coherent controlled-NOT gate with a zero target copies either Z-basis input. On a plus input it produces a Bell pair, not two plus states. The lower table gives all four joint Z probabilities for the actual and requested outputs. These are ideal calculations, not experimental counts. The failed comparison rules out this candidate copier; the following proof rules out a universal perfect copier.

This is not a defect in controlled NOT. It is a conflict between what the device is required to do on different inputs. The no-cloning result, established in influential 1982 papers by Wootters and Zurek and by Dieks, makes that conflict general.[1]

A proof that leaves room for the machinery

Allow the copier a blank target, a power supply, auxiliary systems, and a memory of its operations. Collect everything except the source and target into a machine state $|M_0\rangle$. The initial target and machine must be fixed independently of the incoming state, with no pre-existing correlations to it. Otherwise we might have hidden its preparation recipe inside the copier.

An initially mixed machine does not evade the argument. We can represent it as one part of a larger pure preparation, leaving the extra reference system untouched. Include that reference in the machine description. The enlarged preparation is still fixed independently of the input; it supplies no hidden recipe for the state to be copied.

Take two possible pure inputs, $|\psi\rangle$ and $|\phi\rangle$. A perfect device would perform both transformations

$$\begin{aligned} U\,|\psi\rangle|0\rangle|M_0\rangle &= |\psi\rangle|\psi\rangle|M_\psi\rangle, \\ U\,|\phi\rangle|0\rangle|M_0\rangle &= |\phi\rangle|\phi\rangle|M_\phi\rangle. \end{aligned} \tag{11.2}$$

We even allow its final machine states to differ. Each perfect copy must have the stipulated pure local state. A subsystem in a pure state cannot remain correlated with the

rest. Applying that fact to both copies gives the product form on the right, even though the final machine state may depend on the input. Keeping the full surroundings permits a unitary description of a deterministic quantum operation, including a procedure that uses measurements and records.[2]

Chapter 5 supplied the decisive property: a unitary preserves inner products. Call the magnitude of the input overlap $s = |\langle \psi | \phi \rangle|$. Initially the blank and machine contribute factors of one. After copying, the two copies each contribute s. Consequently,

$$s = s^2 |\langle M_\psi | M_\phi \rangle| \leq s^2. \tag{11.3}$$

The inequality follows because normalized states have overlap magnitude at most one. But for distinct, nonorthogonal inputs, $0 < s < 1$, and squaring makes the number smaller. The required equality is impossible.

For example, zero and plus have $s = 1/\sqrt{2}$, about 0.707. Their proposed two-copy outputs could have overlap no larger than 0.5. The same physical operation would have made the alternatives more distinguishable than unitarity permits. No extra notebook or hidden auxiliary system repairs that mismatch.

Orthogonal inputs have $s = 0$ and create no contradiction; identical states have $s = 1$ and create none either. These are precisely the cases our everyday copying intuition overlooks. The prohibition concerns one device that works perfectly on every unknown input, not the preparation of several systems from the same known recipe.

Unknown means that the recipe is not supplied as an additional usable input. It does not mean that the equipment behaves differently according to what an observer privately knows. A written description can be copied; possessing one physical specimen does not supply that description.

What the restriction does not forbid

Approximate copying is possible. Bužek and Hillery constructed a universal two-output qubit copier whose individual outputs each have fidelity $5/6$ with a pure input. Here fidelity means the probability that an ideal test for that input state accepts the output. The value is the same for every pure input, but it is not one; the two outputs are also not independent perfect replicas.[3]

A restricted, heralded procedure can also make exact copies on some trials. Promise, for example, that the input is either zero or plus, with the two candidate states specified in advance but the chosen one unannounced. A suitable device can flag success and then leave two exact copies; it must also permit an announced failure. Its guarantee does not extend to arbitrary other inputs. Keeping the failure branch in the full description preserves the overlap relation; selecting only successes changes the task. Likewise,

preparing many fresh specimens and estimating their state, as in Chapter 4, is not extracting an exact recipe from one specimen.

There is no comparable obstacle to moving a state without retaining a copy. Exchanging the states of two systems is a reversible operation. Sending a carrier through a suitable physical channel can also transfer its state. Teleportation will accomplish the transfer using a different resource account: a previously shared entangled pair and a classical message.[4]

Three qubits and a prior agreement

Alice receives the input qubit S. She also holds A, one half of a Bell pair shared with Bob, who holds B. Throughout the calculation, register order is S, A, B. The input is independent of the resource pair. Begin with

$$|\psi\rangle_S = \alpha|0\rangle + \beta|1\rangle, |\Phi^+\rangle_{AB} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \tag{11.4}$$

The input amplitudes satisfy $|\alpha|^2 + |\beta|^2 = 1$. Neither participant needs to know their values. Bob's qubit is not a blank zero waiting to be inscribed: considered alone, it initially has density matrix $I_2/2$. Its useful feature is the shared state with A.

Someone must establish that pair physically. It might be prepared in one place and distributed, or generated through a network with heralded attempts. That earlier step costs resources and obeys ordinary transmission constraints. We assume it has succeeded before Alice starts operating on the supplied input.

Alice and Bob must also agree on the meaning of their basis states, the phases of their controls, and how the two outcome bits are labeled. A mismatched reference is an implementation error, not a new teleportation rule. For now, all gates, measurements, storage, and classical transmission are ideal.

The protocol proposed by Bennett and colleagues in 1993 uses a joint measurement by Alice, followed by a correction selected from its communicated result. We will implement that measurement with two familiar gates, so its four branches can be calculated directly.[5]

Turn the joint question into two readable bits

First Alice applies controlled NOT from S to A. Both qubits are in her laboratory. She then applies the Hadamard operation to S: the balanced transformation $|0\rangle \mapsto |+\rangle$ and $|1\rangle \mapsto |-\rangle$ used by our earlier recombiner. Here its symbol H names a gate, not the energy operator called the Hamiltonian in Chapter 5.

Finally she measures S and A in Z. Call the S result m and the A result n. Each is zero or one. Figure 11.2 shows the complete circuit, including the classical message to Bob.

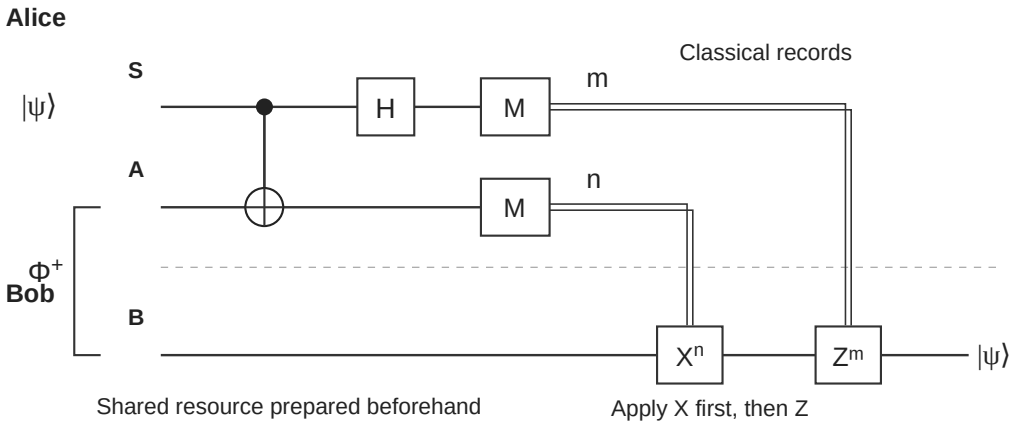


Figure 11.2. A complete teleportation circuit. Register order is S, A, B. A and B begin in the shared Bell-plus state; the bracket marks that resource, not a signal. Alice applies controlled NOT S→A, then H on S, then Z readout of both. Double lines carry the classical outcomes m and n. Bob applies X if n=1, then Z if m=1. Both gates before Alice's readout are local to her laboratory; no gate joins Alice and Bob after the input arrives. Wires show register history and operation order, not spatial trajectories or equal durations. M denotes measurement in the Z basis.

Why these two gates? They turn the four orthogonal Bell states of S and A into the four ordinary labels 00, 01, 10, and 11. Together with the Z readouts, they therefore implement a Bell-basis measurement. Measuring S and A in Z without the preceding gates would be a different measurement and would not implement this protocol.[6]

We can check the algebra without guessing the unknown input. Initially the three-qubit numerator is $\alpha|000\rangle+\alpha|011\rangle+\beta|100\rangle+\beta|111\rangle$, with denominator $\sqrt{2}$. Controlled NOT changes the last two terms to $\beta|110\rangle+\beta|101\rangle$. Apply H to the first position and collect terms by Alice's two labels:

$$
\begin{aligned}
|\Omega\rangle = &\frac{1}{2}|00\rangle_{SA}\left(\alpha|0\rangle+\beta|1\rangle\right)_B \\
&+\frac{1}{2}|01\rangle_{SA}\left(\beta|0\rangle+\alpha|1\rangle\right)_B \\
&+\frac{1}{2}|10\rangle_{SA}\left(\alpha|0\rangle-\beta|1\rangle\right)_B \\
&+\frac{1}{2}|11\rangle_{SA}\left(-\beta|0\rangle+\alpha|1\rangle\right)_B .
\end{aligned}
\tag{11.5}
$$

Each parenthesis on Bob's side is normalized. The factor one-half therefore gives probability one-quarter for each of Alice's four outcomes, whatever α and β are. The branches are alternatives for one pair of readouts, not four copies of the input.

Four outcomes, one recovered state

Bob needs only two elementary corrections. Here X and Z name gates, not the measurement procedures that shared those letters earlier. The gate X exchanges zero and one. The gate Z leaves zero unchanged and reverses the sign of the one component. They act without first learning the amplitudes. Table 11.1 gives every case.

Table 11.1. Every outcome has a prescribed correction.

Alice: $m\,n$	Bob before correction	Operations at Bob
00	$\alpha\|0\rangle+\beta\|1\rangle$	None
01	$\beta\|0\rangle+\alpha\|1\rangle$	X
10	$\alpha\|0\rangle-\beta\|1\rangle$	Z
11	$-\beta\|0\rangle+\alpha\|1\rangle$	X, then Z

Every row occurs with probability 1/4 under the ideal assumptions. The state column is conditioned on the indicated record; the table does not assign four simultaneous local states to B. X is applied first when both corrections are required. Every corrected row gives the original input state; no outcomes are rejected.

In compact notation, Bob's conditional state is $X^n Z^m|\psi\rangle$. An exponent zero means do nothing, and an exponent one means apply the gate. Operators act from right to left, so his correction is $Z^m X^n$: first X when needed, then Z. Thus

$$P(m,n)=\frac{1}{4},\ Z^m X^n\left(X^n Z^m|\psi\rangle\right)=|\psi\rangle. \tag{11.6}$$

Both gates square to the identity. We have kept their order explicit; silently exchanging their order can introduce a minus sign on a branch. Such an overall sign does not change that branch's state, but a consistent convention makes the calculation easier to check.

Take a numerical input with a genuinely complex component: $|\psi\rangle=\left(\sqrt{3}|0\rangle+i|1\rangle\right)/2$. Suppose Alice's record is 11. Bob's conditional state is $\left(-i|0\rangle+\sqrt{3}|1\rangle\right)/2$. X exchanges its components, giving $\left(\sqrt{3}|0\rangle-i|1\rangle\right)/2$; Z reverses the remaining minus sign and restores the input. The same two corrections work without Bob knowing which numerical example was supplied.[7]

After ideal qubit-preserving readout, Alice's S and A registers are in $|m\rangle$ and $|n\rangle$. They no longer retain the unknown input state. An absorbing optical implementation may destroy the measured carriers instead. Either way, teleportation has not left Alice a second usable copy. The shared A–B Bell pair has also been consumed, rather than returned intact for another transfer.

The message does not contain the amplitudes

Two ordinary bits cannot be an exact written description of arbitrary complex amplitudes. Here they do not even reveal which input was supplied: all four messages have probability one-quarter for every input. The message identifies a correction, not a point on the Bloch sphere.

But before it arrives, does Bob already have something readable? Let $\rho = |\psi\rangle\langle\psi|$. Without Alice's outcome, include every branch:

$$\rho_B^{unread} = \frac{1}{4}\sum_{m,n=0}^{1} X^n Z^m \rho Z^m X^n = \frac{I_2}{2}. \tag{11.7}$$

One way to verify this is to add the four density matrices. Sign changes cancel the off-diagonal entries; the exchanged diagonal entries average to one-half. In Bloch language, their four directions average to the center. No measurement on B alone can distinguish two choices of input before the message is used.[8]

The two bits have different jobs. Suppose, just to separate those jobs, that n arrives first and Bob applies X^n. Averaging over the still-missing m leaves $(\rho + Z\rho Z)/2$. Its Z probabilities are correct, but its relative-phase coherence is gone from that conditional description. The second bit supplies the remaining correction. Figure 11.3 uses the numerical input to show the difference.

One input: (√3|0⟩ + i|1⟩) / 2

Record available and action at Bob	PZ(0)	PY(+)
No message Average all four branches	1/2	1/2
Only n; apply X^n Average over the unknown m	3/4	1/2
Both m and n; apply X^n, then Z^m No unknown correction remains	3/4	≈ 0.933

Columns are alternative tests, not repeated readings of one qubit.

Figure 11.3. A correct population is not yet the recovered state. The stipulated input is (√3|0⟩ + i|1⟩)/2. With no outcome record, B is maximally mixed. Given n and after X^n, averaging over the unknown m gives diagonal probabilities 3/4 and 1/4 but no off-diagonal coherence. With both bits and both corrections, the input is recovered. Z and Y columns are theoretical predictions for alternative measurements on fresh trials, not consecutive readouts of one qubit. The final Y-plus probability is (2+√3)/4 ≈ 0.933. This staged delivery is an explanatory variant; the protocol may send both bits together.

Classical communication remains indispensable. Copying the two-bit message to a third observer does not provide that observer with Bob's resource qubit and its correlations.

And the message cannot outrun a physical signal. A conditional state written beside a branch is not evidence that a usable message has arrived instantaneously.

Nor need Bob always apply a separate hardware correction immediately. A processor can sometimes track the required correction and adapt subsequent operations and readout consistently. That is an implementation of the same accounting. Sorting already measured data afterward is not, by itself, a corrected quantum system available for an arbitrary later computation.

Would matching envelopes do?

Chapter 9 compared the Bell pair with an ordinary source that sends zero–zero on half the trials and one–one on the rest. Their Z records match perfectly. Could that less demanding resource run the same transfer?

Replace only A and B's shared state by that lottery and keep every operation and correction unchanged. The output becomes the input's diagonal part: its zero probability is $|\alpha|^2$, its one probability is $|\beta|^2$, and its off-diagonal entries vanish. It is the same channel as measuring the input in Z and preparing the recorded state at the receiver. To see the cancellation, write the lottery as an equal mixture of the Bell-plus and Bell-minus states. With the same protocol, Bell plus gives the input state and Bell minus gives that state with Z applied. Averaging the two outputs preserves the diagonal entries and cancels the off-diagonal entries.

For a plus input, the desired output is pure plus. The lottery instead produces $I_2/2$. It passes a test that asks only for equal Z counts, but its X-plus probability is one-half rather than one. Alice still has four possible messages, and sending them all does not repair the missing relationship.

This locates what the entangled resource contributes. Its coherent relationship between zero–zero and one–one supports the phase transfer that the matching classical alternatives cannot supply. We need not say that two random bits somehow contain a continuous state description. We can compare two fully specified preparations and calculate which part of the output each preserves.

A test stronger than reproducing one local state

There is a more demanding transfer test. The input S may be entangled with another system R that neither Alice nor Bob touches. A channel that merely recreates S's local density matrix could destroy that relationship. Successful teleportation must move S's role in the joint state, not just reproduce a few local totals.

Use a concrete input: let R and S begin in $\frac{|00\rangle+|11\rangle}{\sqrt{2}}$. Considered alone, S is maximally mixed. A device could output a fresh maximally mixed qubit and reproduce every local test on S, yet that output would have no correlation with R. A real transfer must instead preserve the joint Bell relationship between R and the receiver's output. The untouched reference is a test of what the channel preserves, not an extra message supplied to Bob. Write a general pure state of R and S as $|r_0\rangle|0\rangle_S+|r_1\rangle|1\rangle_S$. The R vectors include the weights; they need not be normalized or orthogonal. Repeat Eq. (11.5) with those vectors in place of the scalar coefficients. The gates still act only on S, A, and B. After Bob's correction, every branch contains

$$|r_0\rangle|0\rangle_S+|r_1\rangle|1\rangle_S \rightarrow |r_0\rangle|0\rangle_B+|r_1\rangle|1\rangle_B. \tag{11.8}$$

The relationship with R is preserved. Mixtures follow by averaging. In this precise sense, the ideal protocol acts as an identity channel from the input qubit to the receiver, including its external correlations.[9]

Start with a Bell pair between R and S, and another between A and B. The corrected result is a Bell pair between R and B, although those two systems need never interact directly. This is entanglement swapping. The result still depends on prior physical distribution, the local joint measurement, and its communicated outcome; the drawing of a new pairing is not a new route for superluminal transmission.

What would count as a successful experiment?

The ideal calculation accepts every one of the four Bell outcomes. A real apparatus may recognize only some of them, lose photons, or damage a stored qubit while waiting. Successful-branch quality, probability of success, waiting time, and rate of delivered outputs are separate numbers.

For a pure test input and an output density matrix σ, define

$$F=\langle\psi|\sigma|\psi\rangle. \tag{11.9}$$

This fidelity is the probability of passing a test aligned with the intended state. Perfect transfer gives one. Fidelity 0.9 does not necessarily mean that nine out of ten trials are exact and the tenth is wholly wrong; many noise processes yield that same number.

In an experiment, fidelity is estimated from repeated test preparations and calibrated measurements, not read from a single output. The estimate must distinguish statistical uncertainty from preparation and readout errors. Pooling tests of different inputs also requires a justified model of the device's stability. An output that remains available for later use and a correlation verified only after that output has been detected are different experimental achievements.

A device that measures Z and prepares its result at the receiver transfers zero and one perfectly. It fails on superpositions. Test uniformly chosen pure qubit states, require an output on every trial, supply only one input copy, and allow no shared entanglement or quantum transmission: the optimal measure-and-prepare average fidelity is $2/3$. That is a theorem for a specified task, not a universal threshold for every advertised teleportation experiment.[10]

The simple Z strategy shows that this average is attainable; the general optimality bound also rules out better strategies using other measurements. For an input at Bloch coordinate z it gives $F=\left(1+z^2\right)/2$. Uniformly distributed directions have average $z^2=1/3$. Changing the input ensemble, giving extra copies or labels, or allowing selective reporting requires the appropriate new benchmark. Known verification recipes must not leak to a supposedly blind transfer device.

For an explicit noise example, retain perfect local operations but replace the resource by a Bell pair mixed with a completely mixed pair:

$$\omega_{AB}(v)=v\,|\Phi^+\rangle\langle\Phi^+|+(1-v)\frac{I_4}{4},$$
$$\sigma(v)=v\,\rho+(1-v)\frac{I_2}{2},F(v)=\frac{1+v}{2}. \tag{11.10}$$

Here $0\le v\le 1$ is the stipulated mixing weight, not a measured visibility. The pure Bell component transfers the state perfectly; the independent maximally mixed component supplies a maximally mixed output. Linearity gives their weighted average. At $v=0.8$, every pure input has fidelity 0.9. Figure 11.4 follows the whole family.[11]

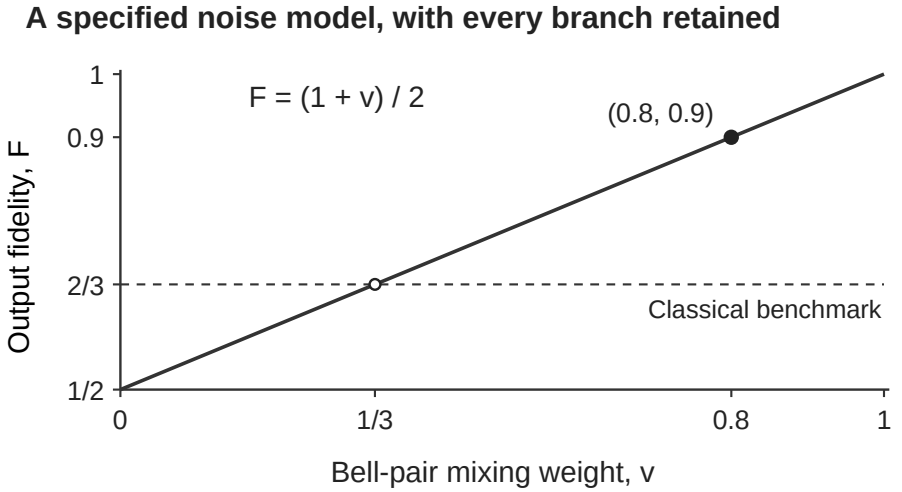


Figure 11.4. Resource noise and output quality are different quantities. With the resource family and ideal local operations in Eq. (11.10), every pure input has output fidelity (1+v)/2. The marked v=0.8 example gives 0.9. The dashed line is the 2/3 deterministic single-copy measure-and-prepare benchmark for uniformly distributed pure qubit inputs; the curve crosses it at v=1/3. No unsuccessful trials are removed. This is an original channel calculation, not experimental data, a rate estimate, or a general formula for arbitrary noise.

This calculation does not need a new Bell violation on each transferred qubit. The certification task is different from Chapter 10's uncharacterized-box test. Ordinary teleportation verification relies on stated input, control, measurement, and sampling assumptions. A device-independent guarantee requires an additional suitable protocol.

The same noisy resource also connects the three chapters. At mixing weight 0.6, Eq. (11.10) gives teleportation fidelity 0.8. The completely mixed component contributes zero to each correlation, so the Chapter 9 witness sum is 1.2, above its separable bound, while the Chapter 10 CHSH strategy gives about 1.70, below its Bell-local bound. These are predictions for one specified state family and fixed tests. They show why useful transfer and entanglement certification need not coincide with a violation in a chosen Bell test; failure of that test does not establish locality for every possible measurement.

From a circuit to a network

The direction from theory to experiment is visible here. Bennett and colleagues proposed the protocol in 1993; Bouwmeester and colleagues reported an optical teleportation experiment in 1997. The proposal specified a transformation and the resources it would need. An experiment then had to realize and test those relationships with actual sources, detectors, and selected detection events. The ideal all-branch protocol and a particular optical demonstration are not interchangeable performance claims.

In 2022, Hermans and colleagues demonstrated qubit teleportation between non-neighbouring nodes of a three-node network based on spins in diamond. They first established the end-to-end resource through an intermediate node, heralded its readiness, and then prepared and teleported the input. Real-time outcome-dependent correction and protection of stored states were essential parts of the experiment. Unconditional teleportation after a ready signal did not mean that every earlier attempt to establish entanglement succeeded.[12]

A different engineering question is whether quantum signals can share fibres with ordinary data traffic. Thomas and colleagues reported in 2024 a teleportation experiment over 30.2 kilometres of fibre carrying a 400-gigabit-per-second classical channel. They used spectral and temporal filtering and coincidence detection to suppress unwanted light. The quantum-sharing span used laboratory fibre spools; deployed fibre formed an additional part of the classical link. Their verification was conditioned on successful detection events, not a lossless deterministic delivery on every attempt. Bob’s target photon was detected before the Bell-state measurement occurred. The reported transfer was therefore verified retrospectively through correlations; it did not leave a stored output waiting for the result and available for a later operation.[13]

The physical implementation also need not reproduce our three-qubit circuit literally. In August 2026, Lou and colleagues reported reconfigurable teleportation across 100 spatial optical channels, combining programmable holographic encoding with all-optical feedforward. The published abstract reports optical-mode array and image transfer with fidelities above the corresponding classical benchmarks. These are optical-mode signals with a different encoding and control architecture, not 100 perfect copies of one unknown qubit. The relevant benchmark depends on that optical encoding and input ensemble; the single-qubit 2/3 limit derived for a different task cannot be imported unchanged. A physical forward link remains part of the protocol.[14]

In photonic hardware, realizing the required joint measurement is itself a resource problem. Restrictions on optical elements, auxiliary photons, and detection constrain which Bell outcomes can be identified. Results about those restrictions are not a prohibition on Bell measurement by every possible interaction. Adding a gate box to our diagram cannot make the required hardware free.[15]

A transfer without a surviving duplicate

Return to the request for a backup. Its failure is now a calculation: a universal perfect copier would violate the overlap relations preserved by quantum dynamics. Knowing how to prepare a state, approximating it, and transferring its role are different tasks.

Teleportation succeeds because it does not perform that forbidden task. The input is consumed as a usable original, a shared Bell pair is spent, and two recorded outcomes select the receiver's correction. The message is not a list of amplitudes, and the resource is not an instantaneous communication channel. Together they implement a transfer that neither supplies alone.

Bob needs a carrier and local controls even though Alice's original carrier does not travel to him. Creating the resource and operating the protocol still require physical transmission and energy.

Throughout this protocol, keeping track of records made the difference. Alice's record permits a deliberate correction. A record unintentionally left in the surroundings can instead hide interference from the system we are watching. To understand why the quantum world so often looks ordinary, we must next follow those uncontrolled records.

Notes and sources

1. W. K. Wootters and W. H. Zurek, “A single quantum cannot be cloned,” Nature 299, 802–803 (1982), doi:10.1038/299802a0; D. Dieks, “Communication by EPR devices,” Physics Letters A 92, 271–272 (1982), doi:10.1016/0375-9601(82)90084-6. The restriction concerns exact, deterministic copying of an arbitrary state supplied without a usable preparation description. Equation (11.1) and Figure 11.1 are original applications of controlled NOT, not reproductions of either paper's figures. Wootters and Zurek; Dieks.

2. John Preskill, Lecture Notes for Physics 219/Computer Science 219, Chapter 4, “Quantum Entanglement,” §4.5.2, and Chapter 3, “Foundations II: Measurement and Evolution,” §3.3.2, Eq. (3.80), on unitary representations of quantum operations. The initial auxiliary state is fixed independently of the input. A mixed auxiliary state can be purified by enlarging the machine. Pure exact copies factor from the remaining degrees of freedom; normalized final machine states have overlap magnitude at most one. This gives Eq. (11.3) even when measurements, records, and discarded systems are included. The proof does not assume an unchanging final machine. No-cloning proof; Quantum operations.

3. Vladimír Bužek and Mark Hillery, “Quantum copying: Beyond the no-cloning theorem,” Physical Review A 54, 1844–1852 (1996), doi:10.1103/PhysRevA.54.1844. The symmetric universal $1 \to 2$ qubit copier gives each reduced output fidelity $5/6$ for every pure input. Fidelity here is the squared-overlap convention in Eq. (11.9). The author paper’s Eq. (3.16) gives the corresponding density matrix; its root-fidelity convention reports the square root of 5/6. The isometry is given in Eq. (3.29). These are different conventions for the same reduced output, not conflicting performance values. Applying that isometry and tracing out the other output and the machine gives each copy’s reduced density matrix. The two outputs are correlated, not independent perfect copies. Author manuscript; Published article.

4. Lu-Ming Duan and Guang-Can Guo, “Probabilistic Cloning and Identification of Linearly Independent Quantum States,” Physical Review Letters 80, 4999–5002 (1998), doi:10.1103/PhysRevLett.80.4999. Exact heralded cloning with nonzero success probabilities is possible for a specified finite linearly independent set of pure states. It includes an announced failure branch and is not universal deterministic copying. Equation (12) of the author manuscript quant-ph/9804064v1 gives the two-state efficiency bound. For the zero/plus example, an ideal equal-success construction attains 2 minus the square root of 2, approximately 0.586, for either promised input. This is an ideal promised-input model, not an experimental success rate; the failure branch is part of the allowed operation. Direct transmission and a SWAP operation, mentioned separately in the text, transfer a state rather than leave a second copy; teleportation changes the resources used for that transfer. Restricted probabilistic cloning.

5. Charles H. Bennett, Gilles Brassard, Claude Crépeau, Richard Jozsa, Asher Peres, and William K. Wootters, “Teleporting an unknown quantum state via dual classical and Einstein-Podolsky-Rosen channels,” Physical Review Letters 70, 1895–1899 (1993), doi:10.1103/PhysRevLett.70.1895. The resource count in this chapter is one initially independent Bell pair, Alice's complete local Bell measurement, a two-bit classical message, and conditional corrections on Bob's existing qubit. Entanglement distribution and storage precede the input operation; their physical costs are not removed. Original paper; IBM research record. For the later optical demonstration, see D. Bouwmeester, J.-W. Pan, K. Mattle, M. Eibl, H. Weinfurter, and A. Zeilinger, “Experimental quantum teleportation,” Nature 390, 575–579 (1997), doi:10.1038/37539. The history comparison distinguishes the 1993 proposal from this specific optical implementation.

6. John Watrous, IBM Quantum Learning, Basics of Quantum Information, “Entanglement in action: Quantum teleportation”; Preskill, Chapter 4, §4.4.2. The present circuit fixes order S, A, B, with m the S readout and n the A readout. These labels must not be imported unchanged from a diagram using a different register order. Controlled NOT S→A followed by H on S maps $\Phi^+, \Psi^+, \Phi^-, \Psi^-$ to 00, 01, 10, 11 respectively. All gates before measurement are in Alice's laboratory. Protocol and reference systems; Lecture treatment.

7. Equations (11.5)–(11.6), Table 11.1, and the complex-amplitude example are original calculations. The pre-correction branch is $X^n Z^m |\psi\rangle$; the inverse is $Z^m X^n$. The four unnormalized branch vectors follow by applying controlled NOT and Hadamard to the three-qubit input. Their squared norms are one-quarter, and the stated corrections restore the input density matrix in every branch. Phase conventions are fixed

throughout. Postmeasurement descriptions of Alice's carriers assume ideal repeatable projective readout when those carriers are retained.

8. Equation (11.7) is the Pauli average of the input state. Without either bit the output is $I_2/2$; given n and after X^n, the remaining average is $(\rho + Z\rho Z)/2$. For Figure 11.3's input, Bloch coordinates are $(0, \sqrt{3}/2, 1/2)$, so the recovered Y-plus probability is $(2+\sqrt{3})/4$. A conditional description requires access to the relevant record. Tracking a correction through subsequent operations is possible only with a consistent gate and readout account. For the general no-signalling argument, see Preskill, Chapter 3. Local operations and records.

9. Watrous, “Quantum teleportation,” discusses inputs correlated with external systems. Equation (11.8) supplies the linear reference-system argument; mixed inputs follow by linearity. Replacing the resource by the fair 00/11 lottery gives the Z-dephasing channel, not the identity channel, by the Bell-plus/Bell-minus mixture argument in the text. For entanglement swapping, see M. Żukowski, A. Zeilinger, M. A. Horne, and A. K. Ekert, “ ‘Event-ready-detectors’ Bell experiment via entanglement swapping,” Physical Review Letters 71, 4287–4290 (1993), doi:10.1103/PhysRevLett.71.4287. A corrected R–B pairing still requires prior resources and communicated outcomes. Reference-system preservation; Entanglement swapping.

10. S. Massar and S. Popescu, “Optimal Extraction of Information from Finite Quantum Ensembles,” Physical Review Letters 74, 1259–1263 (1995), doi:10.1103/PhysRevLett.74.1259. For one copy of a uniformly distributed pure qubit, optimal deterministic measurement-and-repreparation has mean fidelity 2/3. The text's Z strategy attains that value by the spherical average of z^2; that example alone is not a proof of optimality. The benchmark assumes no message-bearing preparation label, shared entanglement, or quantum transmission, and does not silently discard difficult inputs. Preskill, Chapter 4, §4.4.3, also states this classical limit. Optimal estimation benchmark.

11. Equation (11.10) and Figure 11.4 are original evaluations of the standard protocol with the stated resource family. The maximally mixed part produces a maximally mixed output; all four Bell outcomes are retained. This is a channel formula for one noise model, not an optimality claim for arbitrary noisy resources. For the broader relation between entangled-resource quality and teleportation channels, see Michał Horodecki, Paweł Horodecki, and Ryszard Horodecki, “General teleportation channel, singlet fraction, and quasidistillation,” Physical Review A 60, 1888–1898 (1999), doi:10.1103/PhysRevA.60.1888. General channel analysis.

12. S. L. N. Hermans, M. Pompili, H. K. C. Beukers, S. Baier, J. Borregaard, and R. Hanson, “Qubit teleportation between non-neighbouring nodes in a quantum network,” Nature 605, 663–668 (2022), doi:10.1038/s41586-022-04697-y. Their three optically linked diamond-spin nodes are called Alice, Bob, and Charlie; the middle node establishes the outer-node resource by entanglement swapping. The ready signal precedes preparation of the test input. Unconditional input transfer after heralded readiness is distinct from deterministic success of every entanglement-generation attempt. The book's diagrams are not their apparatus. The published protocol in Figs. 1 and 4 separates conditional readout filtering from the unconditional mode, which accepts every Bell-state readout pattern after readiness. The account here summarizes the authors’ protocol and reported observations; it does not present a new reconstruction of their raw or calibration data. Open-access experiment.

13. Jordan M. Thomas, Fei I. Yeh, Jim Hao Chen, Joe J. Mambretti, Scott J. Kohlert, Gregory S. Kanter, and Prem Kumar, “Quantum teleportation coexisting with classical communications in optical fiber,” Optica 11, 1700–1707 (2024), doi:10.1364/OPTICA.540362. Published 20 December 2024. The quantum-coexistence span is 30.2 km of laboratory fibre spools carrying 400-Gbps classical traffic. An additional 48-km deployed segment belongs to the classical communication link. Fourfold-coincidence verification conditions the reported teleportation quality on detected events; these results are not attributed to unconditional transmission on a fully deployed 30.2-km quantum link. Section 4 identifies the demonstration as a posteriori teleportation: Bob’s target photon is detected before the Bell-state measurement. Storage and later use of the target would require an additional implementation; the reported retrospective verification is not such a stored output. Published open-access article.

14. Yanbo Lou, Jiabin Wang, Yuyan Zou, Lingyue Hou, Shengshuai Liu, and Jietai Jing, “Hundred-Channel Reconfigurable Quantum Teleportation,” Physical Review Letters 137, 080801 (2026), doi:10.1103/rfz9-

3prw. Published 20 August 2026. The publisher's abstract reports programmable holographic encoding of 100 addressable spatial modes, measurement-free all-optical feedforward, and transfer of an optical array and image above the corresponding classical fidelity limits. This summary is limited to the published abstract: it makes no additional claim about detailed loss, rate, numerical fidelity, or a universal qubit benchmark. The encoding differs from the chapter's discrete three-qubit example. Published abstract and record.

15. Norbert Lütkenhaus, John Calsamiglia, and Kalle-Antti Suominen, “Bell measurements for teleportation,” Physical Review A 59, 3295–3300 (1999), doi:10.1103/PhysRevA.59.3295. The author manuscript is titled “On Bell measurements for teleportation.” Its limitations concern specified linear-optical and photodetection resources, not arbitrary two-qubit interactions. A complete Bell measurement is allowed by quantum theory even where a particular hardware model cannot implement it deterministically. The abstract-only 2026 example in note 14 is not treated as a counterexample to a theorem with different encoding and resource assumptions. Author manuscript; Published article.

CHAPTER 12

How the Quantum World Becomes Familiar

The light that lets us see an object also carries information away from it. A photon scattered toward one observer and a different photon scattered toward another can supply two records of the same position. What makes an object easy to see may also make some of its quantum interference difficult to observe.

Until now, we have mostly protected the systems we wanted to study. We matched the interferometer's routes, controlled the qubits' phases, and kept track of measurement records. Chapter 11 used a deliberately transmitted record to complete a transfer. Now consider records that the apparatus never intended to make: in scattered light, a passing gas molecule, or an uncontrolled degree of freedom inside a device.

We will follow one such interaction from beginning to end. The probabilities of its two Z outcomes will stay fixed. Yet an interference pattern will fade. The explanation will require neither a person watching nor a new rule that suddenly applies to large objects. It will require a larger quantum state.

The process is called environmental decoherence. Its immediate subject is the loss of accessible interference between specified alternatives when the system becomes correlated with its surroundings. That is a physical prediction we can calculate and test. Whether the calculation also selects one definite outcome is a separate question, and we will keep it separate.[1]

One record, without an observer

Return to a qubit S with reference states $|0\rangle$ and $|1\rangle$. They may label the two optical routes, but the calculation also applies to other encodings. Let E be an additional system, initially in a fixed pure state $|e_r\rangle$. The subscript means ready. For now, S and E are initially independent and together form an isolated model.

Suppose their interaction leaves either reference state of S unchanged but makes a corresponding change in E. By linearity, a superposition then evolves as follows:

$$\begin{aligned} &|0\rangle|e_r\rangle \mapsto |0\rangle|e_0\rangle, \\ &|1\rangle|e_r\rangle \mapsto |1\rangle|e_1\rangle, \\ &\big(\alpha|0\rangle+\beta|1\rangle\big)|e_r\rangle \mapsto \alpha|0\rangle|e_0\rangle+\beta|1\rangle|e_1\rangle. \end{aligned} \tag{12.1}$$

All environmental states shown are normalized, and $|\alpha|^2+|\beta|^2=1$. The two possible final environmental states need not be orthogonal. If they are identical, they contain no distinction between the two alternatives. If they are orthogonal, an appropriate measurement of E can distinguish the associated labels perfectly. The magnitude of their overlap controls this distinguishability. States that differ only by a phase represent the same environmental preparation; that phase can shift the system's interference pattern without reducing its contrast.

The symbol E does not require a hot bath or a vast room. One controlled auxiliary qubit can play this role. Calling it an environment tells us that we are interested in predictions for S without retaining E's full description. The surroundings in a real experiment may, of course, contain an enormous number of such degrees of freedom.[2]

Nobody needs to read the record. What matters is whether the interaction has produced distinguishable physical states, not whether someone has learned their names.

Where the interference goes

Use the partial trace from Chapter 9 to obtain the state of S alone. Define the complex number γ to be the overlap of the two environmental states, in the order written here:

$$\gamma=\langle e_1|e_0\rangle,\ \rho_S=\begin{pmatrix}|\alpha|^2 & \alpha\beta^*\gamma\\ \alpha^*\beta\gamma^* & |\beta|^2\end{pmatrix}. \tag{12.2}$$

The diagonal entries are unchanged. A Z measurement still gives the original zero and one probabilities. The off-diagonal entries, which help predict other measurements, have acquired the environmental overlap. The star denotes complex conjugation. This is the mathematical place where the unwanted record enters.

We can see why without treating the partial trace as a mysterious deletion. In the joint density matrix, the cross term contains $|e_0\rangle\langle e_1|$. Summing over a complete environmental basis turns that operator into $\langle e_1|e_0\rangle$. If the accompanying records are orthogonal, the contributions cannot interfere in a measurement that examines only S.

At zero overlap, S has the same density matrix as the ordinary lottery that prepares zero with probability $|\alpha|^2$ and one with probability $|\beta|^2$. Every measurement on S alone then agrees between those descriptions. But the complete state in Eq. (12.1) is still a coherent joint state, not that lottery applied to the whole pair.

For example, a plus input and perfectly distinguishing qubit records give the Bell state $(|00\rangle+|11\rangle)/\sqrt{2}$. Chapter 9 showed how joint X measurements distinguish it from a lottery over 00 and 11. The relative-phase information has become a property of joint correlations. It has not turned into a phase value that a local detector can print.[3]

A record we can make stronger

Give the environmental marker an explicit two-state design. Leave it in zero when S is zero; when S is one, turn it toward one:

$$|e_0\rangle = |0\rangle, |e_1\rangle = \cos\theta\,|0\rangle + \sin\theta\,|1\rangle, \gamma = \cos\theta. \tag{12.3}$$

Here θ runs from zero to $\pi/2$. It sets the interaction strength in this illustrative model, not a measured detector angle. A controlled rotation implements the map unitarily. At zero angle it makes no record; at $\theta = \pi/2$ the records are perfectly distinguishable.

Prepare S with equal amplitudes and relative phase ϕ, namely $\left(|0\rangle + e^{i\phi}|1\rangle\right)/\sqrt{2}$. Apply the marking interaction, then the fixed recombiner from the early chapters. Call its plus output A and its minus output B. Equation (12.2) gives

$$P(A) = \frac{1+\gamma\cos\phi}{2}, P(B) = 1 - P(A), V = \frac{P_{max} - P_{min}}{P_{max} + P_{min}} = \gamma. \tag{12.4}$$

The last expression defines the fringe visibility for this phase scan. At full visibility, A ranges from certainty to impossibility. At $\theta = \pi/3$, the overlap is one-half: its maximum is three-quarters and its minimum one-quarter. At $\theta = \pi/2$, A stays at one-half whatever phase we choose. Figure 12.1 shows the three cases.[4]

This fading pattern does not mean that some photons have been removed. In our model each trial still produces one output, and the probabilities add to one. Nor does a fifty-fifty split at one setting diagnose complete decoherence: a fully coherent preparation gives that split at a quarter-cycle. The scan tests a relationship that one setting leaves hidden.

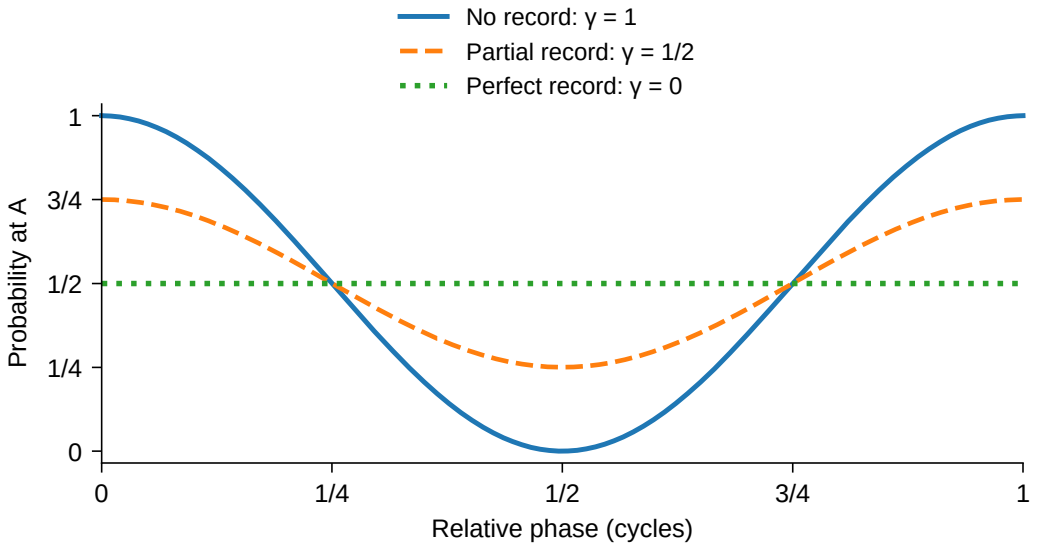


Figure 12.1**. A record reduces the fringe, not the arrival rate.** The curves give Eq. (12.4) for environmental overlaps $\gamma = 1, 1/2$, and 0. Phase is measured in cycles; $\phi = 2\pi$ times the horizontal coordinate. The marker is not read and every trial is retained. These are theoretical probabilities for an ideal lossless, balanced interferometer, not measured counts. At each point $P(B) = 1 - P(A)$. A complex overlap would also shift the phase of the fringe; its magnitude would set the visibility under the same balanced assumptions.

Many small imprints

Now let S interact in turn with N initially independent environmental units. Assume that the units start in a product state, do not interact with one another, and each undergoes the same kind of label-preserving interaction. The environmental state associated with zero is a product of N records, as is the one associated with one. Their overlap is

$$\gamma_N = \prod_{k=1}^{N} \langle e_1^{(k)} | e_0^{(k)} \rangle , V_N = |\gamma_N| . \tag{12.5}$$

The multiplication sign says to multiply one overlap for each unit. If each overlap is the real number 0.9, the visibility after ten interactions is about 0.349. After fifty it is about 0.00515. After a hundred it is about 0.0000266. None of the individual records perfectly distinguishes the alternatives. Together, their joint states become very nearly orthogonal. Figure 12.2 follows this accumulation.

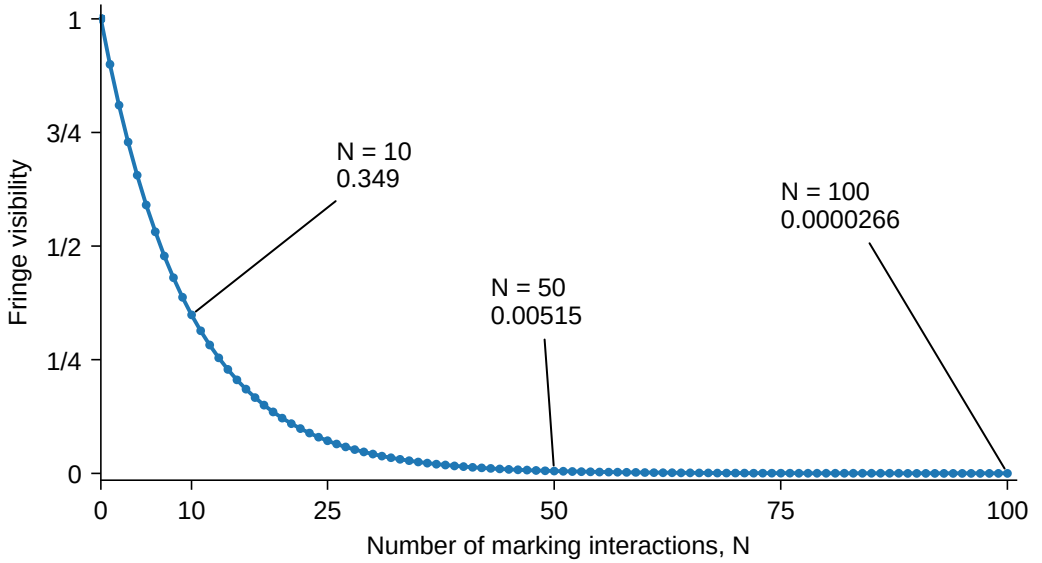


Figure 12.2**. Weak imprints accumulate.** Each fresh environmental unit has the stipulated conditional overlap 0.9, giving $V_N = 0.9^N$. Points are shown at integer N; the joining line guides the eye. The horizontal axis counts successive marking interactions, not seconds or particle mass. Initial product records, label-preserving couplings, and the absence of subsequent record interactions are assumptions of this model. The curve is not a measured decoherence law.

Small visibility also limits what a system-only experiment can detect. For an initially balanced superposition, the largest difference in the probability of any one S-only measurement event between the reduced state and its fully dephased counterpart is $|\gamma_N|/2$. At a hundred of these interactions it is only about thirteen parts in a million. That statement concerns access to S alone, not a joint experiment on S and every record.[5]

This is a bound for one event on one supplied system. Many independent repetitions can resolve a small probability difference. Almost indistinguishable on one trial does not mean indistinguishable with unlimited data.

The product formula is not a universal timetable. Correlated initial environments and repeated system–environment encounters can change it. In a small controlled environment, a later joint interaction can restore coherence; the reversal below is one example. But once S is no longer coupled to E, interactions confined to E can redistribute its records, but cannot change S's reduced state when all outcomes are included. Record redistribution and a revival of system-only interference are different effects.

The physical mechanism is nevertheless testable. In 2003, Hornberger and colleagues varied the background gas in a fullerene interferometer and measured suppression of interference consistent with collisional decoherence. In 2004, Hackermüller and colleagues heated C_{70} molecules before interferometry and compared the fading contrast with a model of thermal photon emission. The outgoing radiation could carry information about spatial alternatives without anyone collecting it.[6]

The comparison concerns fringe contrast, not merely how many molecules reach the detector. The gas experiment accounted for velocity selection and the detector's finite acceptance. In the thermal experiment, a heating-and-cooling model constrained by ion-count measurements linked laser heating to molecular temperature. Neither experiment simply read a decoherence factor from one detected molecule.

The chemistry of the molecule had not stopped being quantum. What was being tested was coherence in its center-of-mass motion. A statement about decoherence must name the degree of freedom, the alternatives, and the timescale. Size and temperature matter through interactions; they are not switches that turn off quantum mechanics.

A fading pattern is not a diagnosis

Our marking interaction leaves the Z populations fixed. If zero and one are energy eigenstates of the bare system Hamiltonian, their mean energy therefore stays fixed too. Decoherence can occur without relaxation from the higher energy level. This does not say that all control hardware or the environment exchanges no energy; it says that losing phase coherence and losing the system's excitation are different questions.

An ordinary phase fluctuation can produce the same reduced map. On each trial, choose a known kind of phase operation, but let its actual angle vary in an uncontrolled way. Pooling the trials averages their density matrices. A fair choice between doing nothing and applying Z removes the off-diagonal entries just as perfect marking does.

These are operationally identical channels on S at that stage, but their physical implementations need not be identical. The pure, initially independent environment in Eq. (12.1) is a useful model, not a theorem that every noisy laboratory process creates precisely that entangled pair. A mixed environment can produce different patterns of correlations.

A slowly varying phase error may be refocused by an echo sequence: exchange the two states halfway through the waiting interval so that a nearly constant frequency offset acts with opposite signs in the two halves. Hahn's spin-echo work introduced this kind of recovery. Rapidly changing noise need not cancel, and an echo is not a general means of recalling information carried away by arbitrary scattered particles.[7]

For a constant offset, each original component spends one half-interval in each reference state. Both therefore acquire the same total phase. The halfway exchange still leaves their labels swapped; a final X pulse restores them for an arbitrary input. This reasoning fails when the two intervals accumulate different uncontrolled phases.

One contrast curve is therefore not enough to tell us which remedy to build. We may need to stabilize a reference, suppress unwanted coupling, or reconstruct how records leave the system. The word decoherence should sharpen that investigation, not replace it.

Three ways to handle the record

For a perfectly controlled marker, the distinction between losing access and destroying the joint state can be tested directly. Use controlled NOT with E initially zero. An arbitrary input $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ becomes $\alpha|00\rangle + \beta|11\rangle$. Applying the same controlled NOT again restores $|\psi\rangle|0\rangle$ exactly. This is a joint operation, and it works without knowing the amplitudes.

Doing something to E alone is different. Ignoring its record does nothing to the state of S. Resetting E into a standard state cannot restore the unknown superposition either: the old distinction must be transferred to other degrees of freedom. An operation confined to E, with all its outcomes included and no subsequent action on S, leaves S's reduced state unchanged.

There is a third procedure. Rewrite the joint state in E's plus/minus basis:

$$\alpha|00\rangle + \beta|11\rangle = \frac{|\psi\rangle_S|+\rangle_E + Z|\psi\rangle_S|-\rangle_E}{\sqrt{2}}. \tag{12.6}$$

An X measurement of E gives plus or minus with equal probability. After plus, S is the original state; after minus, applying Z restores it. The outcome must guide the correction. Figure 12.3 compares this recovery with ignoring the marker or undoing the joint interaction.[8]

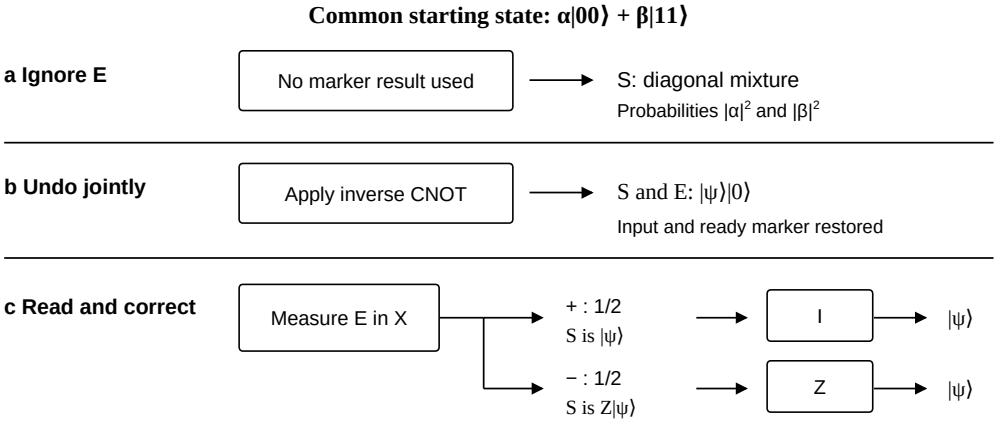


Figure 12.3**. Ignoring, reversing, and reading a record are different operations.** All three rows start from the same marked state $\alpha|00\rangle + \beta|11\rangle$. In the first, no marker outcome is available and S is described by its diagonal mixture. In the second, a joint inverse controlled NOT restores the input and the ready marker. In the third, an X measurement of E gives two conditional states, each with probability one-half; feedforward applies Z only after minus. Both branches are retained. Rows describe alternative protocols, not successive stages. The diagram assumes ideal controls and no inaccessible additional record.

This is the logic of a quantum eraser. The complementary marker measurement does not reveal which reference label was recorded. It sorts the data into phase-related groups, or supplies a record for active correction. It does not change detector results already registered. If the marker result arrives after S has been measured, sorting reveals correlations among existing records; it does not create a recovered qubit for a later calculation.

With more environmental units, reversing only the first marking operation is not enough if the remaining units still distinguish the alternatives. One must control the relevant larger state or use a valid recovery protocol. A missing laboratory notebook is not the same physical act as undoing every correlation it describes.

Why some records are stable

The interaction that hides phase relationships from measurements on S alone also explains why some recorded alternatives remain reproducible. It treats the zero and one states differently from their superpositions. Either reference state survives on S while the marker changes. When the marker states distinguish the alternatives, a superposition with both amplitudes nonzero instead becomes correlated with the marker. The surviving reference alternatives are the pointer states of this model, named after the distinguishable positions of a measuring instrument's pointer.[9]

This preference comes from the interaction, not from how we decide to write the density matrix. Every density matrix can be diagonalized, but that mathematical convenience does not identify which states an actual environment repeatedly records. Changing the coupling can change the preferred alternatives.

For spatial records, scattered light can distinguish sufficiently separated positions. It does not thereby make an object a mathematical point with exact position and momentum. A useful classical description also needs its own dynamics, appropriate resolution, and states that remain sufficiently localized over the interval of interest. Calculations of robust wavepackets show how such conditions can arise; suppression of interference alone is not a derivation of every classical equation of motion.[10]

Nor can the system's own motion always be neglected while its surroundings monitor it. A 2026 study by Chisholm, Palma, and Innocenti examined a qubit model in which the system's evolution competes with the interaction that records it. They analyzed how readable records arise and vary over time in that setting, and how the relevant pointer states should be identified. The simple rule 'the environment always measures one fixed basis' is not general enough.[11]

Many observers, one distinction

The light from a page usually reaches many observers in different photons. To explain their agreement we need more than decoherence of the page considered alone. We need records that different observers can actually obtain.

Build a deliberately simple model. Prepare S in plus and N environmental qubits in zero, with $N \geq 2$. Apply controlled NOT from S to each environmental target. The result is

$$|\Psi_{SE}\rangle = \frac{|0\rangle|0\cdots0\rangle + |1\rangle|1\cdots1\rangle}{\sqrt{2}}. \tag{12.7}$$

Each environmental qubit is a perfect Z record. One observer can read the first, another the second, and their Z results agree. Neither needs access to the rest. These are not independent random answers: they are multiple records of the same distinction. The total string of environmental Z outcomes has only one bit of entropy.

For two environmental qubits, keeping only the first leaves S and that record in a mixture of zero–zero and one–one. The omitted second record distinguishes those alternatives, so their coherence is absent from this reduced description. Keeping both records together with S retains the pure joint state. This is the difference between a proper fragment and the whole environment.

For any nonempty fragment F containing fewer than all N environmental qubits, trace out the remainder. At least one omitted record is orthogonal between the two alternatives, so the coherence term vanishes. If the fragment contains m qubits, the retained state is

$$\rho_{SF} = \frac{1}{2}|0,0^m\rangle\langle 0,0^m| + \frac{1}{2}|1,1^m\rangle\langle 1,1^m|, 1 \leq m < N. \tag{12.8}$$

The notation 0^m means m zero labels, not zero raised to a numerical power. Every such fragment carries the Z distinction. None reveals the relative phase of the entire superposition. That phase remains detectable in suitable measurements of the full joint system, not in a single observer's fragment.

The entropy account is particularly revealing. S has entropy one bit. A nonempty proper fragment has entropy one, and its joint state with S also has entropy one. Their quantum mutual information is therefore $1 + 1 - 1 = 1$. For the entire environment together with S, the state is pure and the mutual information is two. Thus

$$\begin{gathered} I(S:F_0) = 0, \\ I(S:F_m) = 1\,\text{bit}\,(1 \leq m < N), \\ I(S:F_N) = 2\,\text{bits}. \end{gathered} \tag{12.9}$$

For two environmental qubits, the three cases can be read directly. With no fragment, there is no shared record and the mutual information is zero. With either one-qubit fragment, the three entropies are one, one, and one, giving one bit. With both environmental qubits, the system entropy and environment entropy are each one, but the joint entropy is zero, giving two bits. The full-environment value concerns the pure joint state; reading the two environmental Z labels still reveals only one shared classical bit.

Figure 12.4 shows how the same pattern extends to six environmental qubits: many small fragments each supply the same one-bit distinction.[12]

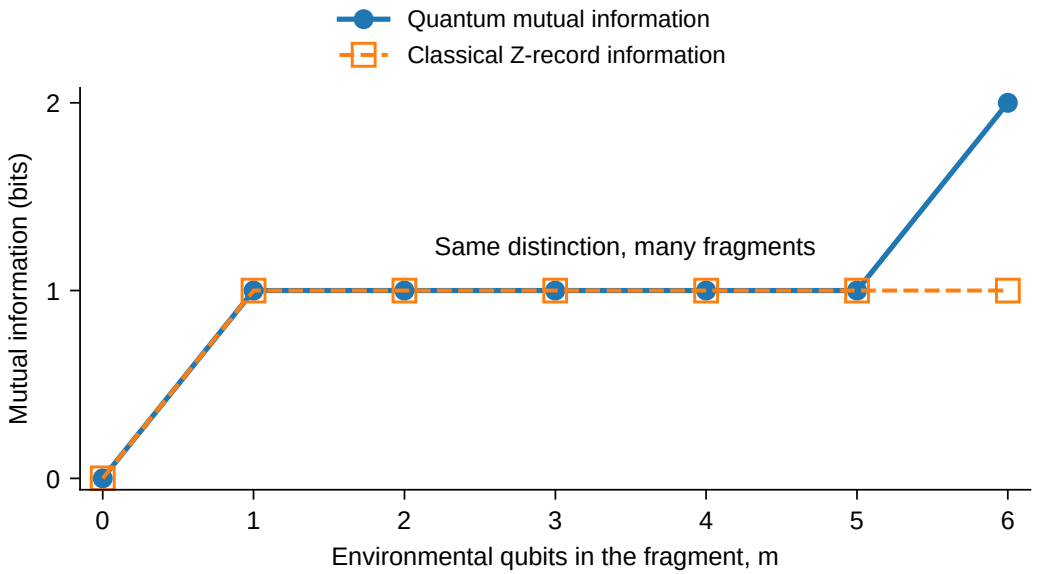


Figure 12.4**. Repeated access is not repeated independent information.** For Eq. (12.7) with six environmental qubits, circles give the quantum mutual information between S and a fragment of m qubits. Squares give the classical mutual information between a Z measurement on S and the fragment's complete Z record. Both equal one for nonempty proper fragments; at m=6 only the quantum quantity rises to two. Every outcome is included. The straight segments connect discrete fragment sizes and do not describe a continuous physical process. Values are exact calculations for the ideal branching state, not experimental observations.

There is no conflict with no-cloning. The interaction replicates the distinguishable zero/one labels, not an arbitrary input state. In particular, a plus input has not made N extra plus states. It has produced the entangled state in Eq. (12.7).

Quantum Darwinism names the research program studying how such repeated environmental records make selected properties accessible to many observers. The selection concerns which information survives and spreads under the dynamics. It does not mean that a quantum state reproduces biologically or that the environment decides which outcome should happen.[13]

Our perfect-record example establishes the plateau and the readable Z records separately. In more general states, a plateau in total quantum mutual information alone need not

establish that independent observers can recover the same classical information without disturbing what others can learn. Accessibility and the structure of the records require additional tests.[14]

A stronger claim would say that the fragments share no correlations beyond the recorded system label. That requires the whole collection to factor once the label is fixed, not merely each pair. Two independent fair bits and their XOR illustrate the gap: every pair is independent, yet any two determine the third. Readable records need not satisfy that stronger independence requirement.

In a 2025 superconducting-circuit experiment, Zhu and colleagues engineered system–environment interactions and examined information in different environmental fragments. Measurements in several bases were used to reconstruct density matrices and infer information measures, including distinctions between classical and quantum correlations. The study also probed branching structures through conditioned system states; that geometric-state reconstruction retained only outcomes within the chosen logical subspace. This was a controlled investigation of the record-forming mechanism, not the observation of a unique macroscopic outcome emerging from unitary evolution alone.[15]

What the calculation has not chosen

We can now put the measuring instrument itself into the model. Let its ready state be $|A_r\rangle$, and let its two pointer records be $|A_0\rangle$ and $|A_1\rangle$. Including the environment, linear evolution gives

$$\big(\alpha|0\rangle+\beta|1\rangle\big)|A_r\rangle|e_r\rangle \mapsto \alpha|0\rangle|A_0\rangle|e_0\rangle+\beta|1\rangle|A_1\rangle|e_1\rangle. \tag{12.10}$$

When the environmental records are effectively orthogonal, the reduced system–apparatus state is effectively a mixture of zero with pointer zero and one with pointer one. Interference between the two records is inaccessible through that reduced description. The pointer alternatives can be stable, and further parts of the environment can hold consistent copies.

But no step has replaced the right-hand side of Eq. (12.10) by just its first term or just its second. Taking a partial trace is not an outcome-selection mechanism. It calculates probabilities for a restricted set of measurements. Once a particular pointer record is supplied, the usual conditional-state rule tells us how to update predictions; that rule is not derived merely by making an environmental overlap small.

We have also used the Born rule throughout: squared amplitudes and traces gave our probabilities. The calculation does not establish that rule by using it. Decoherence explains why particular interference terms become negligible and why certain records are robust. By itself, this unitary argument does not provide a complete account of why a run

has the one definite outcome we experience. That is the remaining measurement problem, not a defect in the arithmetic.[16]

Different interpretations and modified dynamical theories address that remaining question in different ways. The distinction between suppressing interference and selecting an outcome helps us assess their claims. Testing an environmental-decoherence model does not automatically choose among those accounts, and observing a fading fringe does not by itself establish a new collapse law.

Familiarity without a change of rules

We began with the light by which an object becomes visible. We now have a model of how a physical interaction can preserve a selected distinction while dispersing the phase relationships needed for other questions. One local state can lose purity while the larger state remains pure. A record can be unread yet matter. Many records can agree without becoming many copies of an arbitrary quantum state.

This is not a line drawn through the world with quantum objects on one side and classical objects on the other. It is an account of particular systems, interactions, records, and accessible measurements. A molecule can retain quantum electronic structure while its center-of-mass interference is suppressed. A carefully controlled device can expose coherence in a degree of freedom that a less controlled experiment would describe classically.

Part III has followed information into correlations, across laboratories, through a transfer, and into the surroundings. Next we will control those relationships deliberately, constructing quantum programs that make a chosen answer accessible while preserving the interference they need.

Notes and sources

1. Erich Joos and H. Dieter Zeh, ‘The emergence of classical properties through interaction with the environment,’ Zeitschrift für Physik B 59, 223–243 (1985), doi:10.1007/BF01725541. Environmental decoherence concerns specified alternatives and a system–environment division. The opening's scattered-light example is explanatory, not a reconstruction of one experiment. Original analysis.

2. Wojciech H. Zurek, ‘Environment-induced superselection rules,’ Physical Review D 26, 1862–1880 (1982), doi:10.1103/PhysRevD.26.1862. Equations (12.1)–(12.2) are this book's worked pure-environment, label-preserving model. The input environment is fixed independently of S; initial correlations, other system transitions, and environmental mixedness require a more general calculation. Original theory.

3. John Preskill, Lecture Notes for Physics 219/Computer Science 219, Chapter 3, ‘Foundations II: Measurement and Evolution,’ §§3.1–3.2 and 3.4.2. The reduced state is obtained by tracing, not by conditioning on an outcome. With $\gamma = \langle e_1 | e_0 \rangle$, the upper-right entry is $\alpha \beta^* \gamma$. Complete-system unitarity and reduced-system nonunitarity refer to different objects. The Bell comparison uses the earlier book chapters. Quantum operations and dephasing.

4. Berthold-Georg Englert, ‘Fringe Visibility and Which-Way Information: An Inequality,’ Physical Review Letters 77, 2154–2157 (1996), doi:10.1103/PhysRevLett.77.2154. Equation (12.4) is derived for this chapter's balanced, lossless model, with a real nonnegative overlap. For arbitrary complex γ the probability is $\frac{1 + \mathrm{Re}\left(\gamma e^{-i\phi}\right)}{2}$ and the scan visibility is $|\gamma|$. Background counts, unequal weights, and imperfect recombination change the experimental relation. Visibility and distinguishability.

5. Equation (12.5), Figure 12.2, and all numerical values are original model calculations. Initial environmental states and the two conditional final states factor across units. For a balanced input, the reduced density matrix minus its dephased version has eigenvalues $\pm |\gamma_N| / 2$. Its trace distance is $|\gamma_N| / 2$, the maximum difference for a single measurement-event probability. At N=100 this is approximately 1.32807×10^{-5}. Numerical checks do not establish these assumptions for an experimental bath. An all-outcome channel acting only on the environment leaves the reduced system state unchanged, even when environmental fragments interact with one another. This is the same local-channel identity used in Chapters 9 and 11; subsequent joint system–environment evolution is a different operation. The stated trace distance bounds one measurement-event probability on one copy, not the power of arbitrarily many repetitions.

6. Klaus Hornberger and colleagues, ‘Collisional Decoherence Observed in Matter Wave Interferometry,’ Physical Review Letters 90, 160401 (2003), doi:10.1103/PhysRevLett.90.160401; Lucia Hackermüller, Klaus Hornberger, Björn Brezger, Anton Zeilinger, and Markus Arndt, ‘Decoherence of matter waves by thermal emission of radiation,’ Nature 427, 711–714 (2004), doi:10.1038/nature02276. These experiments probe spatial coherence of fullerene center-of-mass motion. Their gas-scattering and photon-emission mechanisms are different from the ideal qubit markers in our figures. Gas-collision experiment; Thermal-radiation experiment. In the author manuscripts, Hornberger et al., Eq. (5), Fig. 2 and its discussion distinguish visibility from uniform loss and include finite detector acceptance and velocity selection. Hackermüller et al., Figs. 2–4 and Methods, constrain a heating-and-cooling model with ion-count data rather than measuring temperature on each interfering molecule. These are interpretations of the reported experiments and models, not new analyses of their raw records.

7. Erwin L. Hahn, ‘Spin Echoes,’ Physical Review 80, 580–594 (1950), doi:10.1103/PhysRev.80.580. For a constant phase per half interval, $U_\phi = diag\left(1, e^{i\phi}\right)$ gives $X U_\phi X U_\phi = e^{i\phi} I$, including a final restoring X pulse. More generally, refocusing depends on the noise and control model. For dephasing without mandatory system–environment entanglement, see G. García-Pérez and colleagues, ‘Decoherence without entanglement and quantum Darwinism,’ Physical Review Research 2, 012061(R) (2020), doi:10.1103/PhysRevResearch.2.012061. Spin echoes; Different correlation mechanisms.

8. Marlan O. Scully and Kai Drühl, ‘Quantum eraser: A proposed photon correlation experiment concerning observation and “delayed choice” in quantum mechanics,’ Physical Review A 25, 2208–2213 (1982), doi:10.1103/PhysRevA.25.2208. Equation (12.6) is an original two-qubit version with explicit active feedforward. Ignoring, outcome-conditioning, and coherently reversing the marker are distinct operations. The no-signalling restriction on an unconditioned E-only channel is the quantum-channel result in Preskill, Chapter 3, not a claim about postselected states. Original proposal; Local quantum operations.

9. Wojciech H. Zurek, ‘Pointer basis of quantum apparatus: Into what mixture does the wave packet collapse?,’ Physical Review D 24, 1516–1525 (1981), doi:10.1103/PhysRevD.24.1516. The chapter's exact pointer states are the two labels preserved by its controlled interaction, with intervening system dynamics neglected. The historical title is not used as evidence that the present unitary calculation selects one outcome. Pointer-state construction.

10. Wojciech H. Zurek, Salman Habib, and Juan Pablo Paz, ‘Coherent states via decoherence,’ Physical Review Letters 70, 1187–1190 (1993), doi:10.1103/PhysRevLett.70.1187. Robust wavepacket states depend on both system dynamics and environmental coupling. Our qualitative discussion does not claim exact point trajectories, universally fixed pointer states, or a complete classical limit for every Hamiltonian. Dynamical stability of states.

11. Diana A. Chisholm, G. Massimo Palma, and Luca Innocenti, ‘Emergence of quantum Darwinism and pointer states for noncommuting evolutions,’ APS Open Science 1, 000024 (2026), doi:10.1103/75jr-ltct; published 22 May 2026. This is a theoretical qubit-model study, not an experiment or a general proof for arbitrary environments. The published statement concerns the extent of redundant, accessible records under competing evolution and a pointer-state definition informed by spectrum broadcast structure. The earlier author version has a slightly different title. Published article; Published full text.

12. Robin Blume-Kohout and Wojciech H. Zurek, ‘Quantum Darwinism: Entanglement, branches, and the emergent classicality of redundantly stored quantum information,’ Physical Review A 73, 062310 (2006), doi:10.1103/PhysRevA.73.062310. Equations (12.7)–(12.9) specialize the record argument to an ideal equal-weight branching state. Taking partial traces of the displayed branching state gives the reduced matrices used in the entropy calculation. The mutual-information jump at the full environment is not two independently readable classical bits. Branching-state theory.

13. Harold Ollivier, David Poulin, and Wojciech H. Zurek, ‘Objective Properties from Subjective Quantum States: Environment as a Witness,’ Physical Review Letters 93, 220401 (2004), doi:10.1103/PhysRevLett.93.220401. Redundant records concern repeated access to selected information across environmental fragments. The book's example copies orthogonal reference labels and does not clone arbitrary superpositions. Environment-as-witness proposal.

14. Thao P. Le and Alexandra Olaya-Castro, ‘Strong Quantum Darwinism and Strong Independence are Equivalent to Spectrum Broadcast Structure,’ Physical Review Letters 122, 010403 (2019), doi:10.1103/PhysRevLett.122.010403. Total quantum mutual information is not automatically accessible classical information. Conditions on accessibility and interfragment structure are distinct from observing a plateau alone. No claim that every decohering environment produces redundant records is made. Accessibility and record structure. The formal pairwise condition in that paper's Eq. (9) is insufficient for more than two environmental fragments. A. Feller, B. Roussel, I. Frérot, and P. Degiovanni, Comment, Physical Review Letters 126, 188901 (2021), doi:10.1103/PhysRevLett.126.188901, supplies a readable-record counterexample with conditional three-fragment correlations. Le and Olaya-Castro, Reply, Physical Review Letters 126, 188902 (2021), doi:10.1103/PhysRevLett.126.188902, accept replacement by vanishing conditional multipartite mutual information and retain their full-independence formulation. That stronger factorization is not necessary merely for each observer to read the system label. Published Comment. Published Reply.

15. Zitian Zhu, Kiera Salice, Akram Touil, Zehang Bao, Zixuan Song, Pengfei Zhang, Hekang Li, Zhen Wang, Chao Song, Qiujiang Guo, H. Wang, and Rubem Mondaini, ‘Observation of quantum Darwinism and the origin of classicality with superconducting circuits,’ Science Advances 11, eadx6857 (2025), doi:10.1126/sciadv.adx6857; published 1 August 2025. The published main text and quantum-state-tomography Methods are available in the PubMed Central full-text record PMC12315987. Information measures are inferred from reconstructed density matrices; a single outcome does not supply an entropy or

discord value. The geometric-state measurement excludes outcomes outside the chosen logical subspace; that postselection is distinct from the all-outcome ideal model in this chapter. The earlier author manuscript is arXiv:2504.00781v2, revised 3 April 2025. The chapter uses the documented circuit-based investigation of branching and fragment information, not a claim that it resolves the definite-outcome problem. None of the book's plotted curves is experimental data from this study. Publication; Full author manuscript; Publication record.

16. Maximilian Schlosshauer, ‘Quantum Decoherence,’ Physics Reports 831, 1–57 (2019), doi:10.1016/j.physrep.2019.10.001, §§7.1–7.2. The author version states that it matches the published article. The interpretive distinction is between reduced-state interference suppression and the occurrence of a definite outcome. Equation (12.10) is a linear measurement-chain model using the Born rule, not a derivation of a stochastic collapse or of the Born rule itself. The review is used for this foundational analysis; experimental claims are sourced to original research. Author analysis, full text.

PART IV

Computing with Quantum Rules

A useful interference pattern is not just one we can observe. It is one we can arrange to answer a question.

The earlier parts gave us the necessary pieces. A pure state is described by amplitudes, controlled interactions transform the state, and a measurement produces a record. With several systems, the relevant relationships may belong to the whole. Unwanted records can make those relationships inaccessible; a deliberately chosen record can complete a transfer or guide a correction.

We will now organize these pieces into programs. The task is not to put a machine into a complicated state and hope that an answer appears. We must specify the problem, construct a sequence of operations, and show why its final records reveal the quantity we asked for. Every preparation, auxiliary register, and repetition has a place in that account.

Our first program will be small enough to follow completely. It will learn a relationship between two possible function values without returning those values separately. The mechanism will be the same interference that first made a detector go dark, but the phase will now be supplied by a computation rather than a known adjustment.

Then we will ask how interference can improve a search and how repeated structure in arithmetic can become measurable. These examples will let us compare different kinds of computational advantage. A reduction in the number of calls to a specified operation is not automatically a reduction in elapsed time, and a successful algorithm for one problem is not a solution to every difficult problem.

The question throughout this part is practical and precise: what must we do to make a particular answer accessible? We begin with two input alternatives, one controlled operation, and a measurement whose meaning we can derive.

The rules we can now use

Before building the programs, gather the rules that made the earlier examples calculable. They concern a state, its transformations, the systems described together, and the records of measurements. This is a guide to the formulation already used in the book, not a claim that quantum theory has only one possible set of starting axioms.

A pure state is represented by a normalized vector of complex amplitudes. A common phase multiplying the whole vector changes no prediction. A density matrix also describes mixtures and a subsystem considered without its partners. It is Hermitian: interchanging rows and columns and conjugating every entry leaves it unchanged. Its eigenvalues are nonnegative and sum to one. For a qubit, the Bloch ball and the density

matrix package the same state information; neither is a table of answers to every possible question. Chapters 3, 4, and 7 supplied these descriptions.[1]

The probability rule must be paired with a specified measurement. For the ideal projective measurements used in our examples, let P_j project onto the subspace assigned to outcome j. The projectors are mutually orthogonal and sum to the identity. The Born rule gives

$$p_j = tr\left(P_j \rho\right). \tag{IV.1}$$

The trace adds a matrix's diagonal entries. For a pure input and a one-dimensional outcome, this becomes the squared magnitude of the corresponding overlap. Coherent contributions to that one overlap add before squaring. Probabilities of distinct recorded outcomes add afterward. An ordinary lottery over preparations averages probabilities, not amplitudes. These are the distinctions that changed the dark detector's response.

For a closed system, a specified Hamiltonian generates unitary evolution. Its action on a density matrix is

$$\rho \mapsto U \rho U^{\dagger}, U^{\dagger} U = I. \tag{IV.2}$$

The dagger denotes the adjoint, and here also the inverse. This evolution preserves inner products and does not first measure the input amplitudes. The rule does not determine the Hamiltonian of a particular molecule or device: its interactions, controls, and approximations must still be specified. An open subsystem need not evolve unitarily when its partners are left out of the description, as Chapter 12 showed.

Systems together require a joint state. For separately specified registers, their state spaces combine by a tensor product. Independently prepared states give a product state; a general joint state need not. Predictions restricted to one register use the partial trace over the others. That operation changes the description available for a question, not the physical preparation. Chapter 9 used it to explain how a pure entangled pair can have mixed local states.

A recorded outcome and an unsorted collection of outcomes require different predictions. For the ideal projective instrument that retains only the selected projection, a result with $p_j > 0$ leaves

$$\rho_j = \frac{P_j \rho P_j}{p_j}. \tag{IV.3}$$

If the outcome is not available, average these conditional states with their probabilities. Instruments that measure and then rotate, absorb, or replace a carrier can leave different output states, even with the same outcome probabilities. Their operations must be specified separately. The conditional rule describes what follows a given result; using it does not derive the Born rule or explain away the interpretive question of individual outcomes.[2]

These starting rules are not interchangeable with their consequences. The no-cloning theorem follows from requirements on one copying device and the preservation of overlaps. A Bell bound follows from the stated local-response assumptions; quantum predictions can exceed that bound. Teleportation is a constructed protocol. An experimental fidelity is a measured or inferred performance quantity. Each has its own argument and evidence.

We can now put the rules to work deliberately. A coherent pure-state preparation supplies amplitudes. A sequence of allowed transformations makes a chosen relationship affect the final probabilities. A record supplies an outcome, and an inference connects it to the original task. The next chapters build that connection rather than asking a detector to reveal an entire state at once.

Notes and sources

1. The state and probability statements summarize the finite-dimensional framework developed in Chapters 3, 4, 7, and 9. John Preskill, Quantum Information, Chapter 2, “Foundations I: States and Ensembles,” §§2.1–2.3 and 2.5; Richard P. Feynman, Robert B. Leighton, and Matthew Sands, The Feynman Lectures on Physics, Vol. III, Chapter 3, “Probability Amplitudes.” The listed rules are a working formulation, not a unique axiomatization. Equation (IV.1) specializes the Born rule to a complete projective measurement. State framework Amplitude rules.

2. Preskill, Quantum Information, Chapter 3, “Foundations II: Measurement and Evolution,” updated October 2018, §§3.1–3.3. Equation (IV.3) is the projection instrument, not the most general state update for a given observable. In general a quantum instrument must specify both outcome probabilities and conditional operations. The pure-state and density-matrix calculations in this recap do not constitute an empirical derivation of the probability postulate or an independent solution of the measurement problem. Measurement and evolution.

CHAPTER 13

A Program Made of Interference

So far, we have adjusted the phase in an interferometer and predicted which detector will respond. Now reverse the purpose of the experiment. Let an operation we do not know supply the phase, and use the detector to learn something about that operation.

This is the beginning of a quantum program. The apparatus must do more than display interference: the pattern must answer a specified question. We will build one complete example, including the preparation, the unknown operation, the final measurement, and the cost being compared.

The example is small on purpose. There are only two possible arguments to a function, zero and one. We want to know whether its two answers agree. The program will reveal that relationship without returning both answers separately. Its most important component will be a minus sign.

The interferometer becomes a circuit

Begin with the balanced transformation already used to recombine our optical routes. As a qubit gate it is called Hadamard, written H. This letter names an operation here, not the energy operator of Chapter 5. In the zero/one basis,

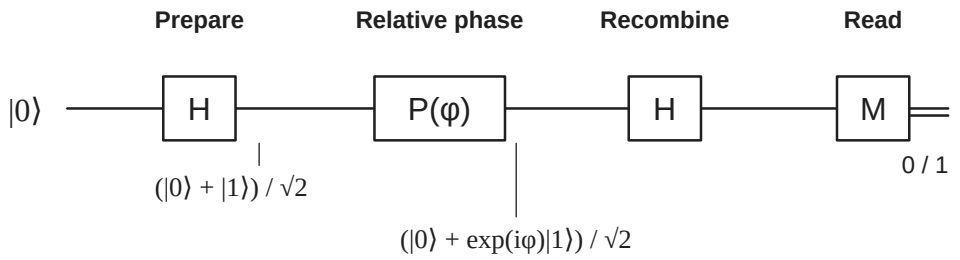
$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, H\begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} \alpha+\beta \\ \alpha-\beta \end{pmatrix}. \tag{13.1}$$

Each output amplitude is a normalized sum or difference of the inputs. Applying H twice returns every input state: $H^2 = I$. A fair random choice would not behave this way. For a zero input, inserting an ideal Z measurement between the gates and ignoring its result changes the certain return to a fifty-fifty output. An ideal repeatable X measurement at the same point leaves the plus state unchanged. The gate changes amplitudes coherently; it does not merely make the next recorded bit random.[1]

Put a phase gate $P(\phi)$ between the two Hadamards. It leaves the zero component unchanged and multiplies the one component by $e^{i\phi}$. Starting from zero gives

$$H\,P(\phi)\,H\,|0\rangle = \frac{1+e^{i\phi}}{2}|0\rangle + \frac{1-e^{i\phi}}{2}|1\rangle. \tag{13.2}$$

This is the calculation from Chapter 3 in circuit form. The zero probability is $(1+\cos\phi)/2$; the one probability is its complement. Figure 13.1 shows which operation produces each change.



φ = 0: output 0 is certain. φ = π: output 1 is certain.

Figure 13.1. A circuit representation of the familiar interference experiment. Read operations from left to right. The first H creates equal amplitudes; P(φ) changes their relative phase; the final H combines them before Z readout. The two output amplitudes are those in Eq. (13.2). Wires identify a register through successive operations, not measured particle tracks. There is no intermediate measurement, route marker, or loss in this ideal model. The measurement label M means read the Z basis, not apply a Z phase gate.

The ordering deserves attention. Circuit diagrams run left to right, but in a product of operators the rightmost acts first. Thus $H\,P(\phi)\,H$ means exactly the sequence drawn. Order is physical: $H\,Z\,|0\rangle = |+\rangle$, whereas $Z\,H\,|0\rangle = |-\rangle$. The same gates in a different order can give different later measurement results. Here Z as a gate reverses the sign of the one component.[2]

For two Hadamards with no phase gate, the final zero amplitude receives two contributions of one-half. The final one amplitude receives one-half and minus one-half. The first pair adds to one; the second cancels. If we replace the intermediate state by an unread lottery over zero and one, either selected state gives equal final probabilities instead. A description that tracks only the intermediate probabilities misses the distinction, even though the same two intermediate alternatives remain available.

One question about two possible answers

Let a device evaluate a fixed but unknown binary function f. Supply zero and it returns $f(0)$; supply one and it returns $f(1)$. There are four possible functions, listed in Table 13.1. We want one bit, $d = f(0) \oplus f(1)$: zero if the answers agree, one if they differ. The circled plus is XOR.

Table 13.1. Four functions, one question.

$f(0)$	$f(1)$	Relationship	$d = f(0) \oplus f(1)$
0	0	Same (constant)	0
0	1	Different (balanced)	1
1	0	Different (balanced)	1
1	1	Same (constant)	0

Each row is one possible fixed function, not a measured frequency. “Balanced” means that its two outputs are different. The task asks for the last column, not the two individual function values. No probability distribution over rows is required for the exact algorithm.

Classically, evaluate both inputs and compare. Could one evaluation guarantee the answer? Suppose the one queried input is zero and its answer is zero. The function could still be the first or second row. Those rows require different answers to our question. The same ambiguity occurs for either queried input and either reply.

Randomizing which input to query does not supply the missing value. No one-query classical procedure is correct with certainty for all four functions. Two evaluations suffice. If the four functions are chosen with equal probability, one classical reply leaves the two classes equally likely; no one-query guessing rule then beats one-half average success.

The comparison concerns access through a specified interface. We cannot inspect the device's code or receive its function label through another channel. The quantum procedure will receive a stronger kind of access: one coherent use of the same reversible function-evaluation rule. The distinction between these interfaces is part of the result, not something to conceal.[3]

An operation, not a printed answer

Use two qubits, a query register Q and a target T. In every state vector below, Q is written first. The allowed query is the reversible extension from Chapter 8:

$$U_f |x\rangle_Q |t\rangle_T = |x\rangle_Q |t \oplus f(x)\rangle_T . \tag{13.3}$$

The target is flipped when $f(x) = 1$ and left alone when $f(x) = 0$. Keeping x makes the full map reversible. Applying it twice restores both registers, even when the initial target is one. A quantum query applies this rule linearly to superpositions, with the phases shown and without leaving a distinguishing record elsewhere.

Such a device is often called an oracle. The name does not grant it magical powers. It specifies an operation whose uses we count separately. Building that operation can itself require substantial computation. An ordinary service that measures its input, calculates a

value, and returns a classical bit does not automatically offer the coherent interface in Eq. (13.3). Getting the four reference-state answers right is not enough to establish it.[4]

We could initialize T in zero and use it to hold a function value. For this program, prepare it instead in $|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$. The reason is a one-line calculation:

$$X|-\rangle = \frac{|1\rangle - |0\rangle}{\sqrt{2}} = -|-\rangle. \tag{13.4}$$

A flip leaves the target in the same pure state, apart from a minus sign. When the decision to flip is controlled by Q, that sign can become relative between Q's components:

$$U_f(|x\rangle|-\rangle) = (-1)^{f(x)}|x\rangle|-\rangle. \tag{13.5}$$

For a definite x, the sign multiplies the entire vector and changes no prediction. For a superposition of different x values, the signs need not agree. Then they alter the relative phase. This is phase kickback: a controlled operation supplies a phase to the control amplitudes while the target returns to its original state. Nothing travels backward along a circuit wire. The name describes the algebra.[5]

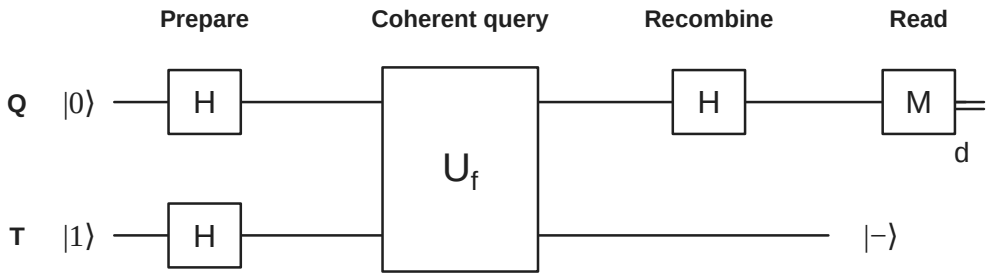
The complete program

Prepare Q in zero and T in one. Apply H to each. Q becomes plus, and T becomes minus. Apply the unknown U_f once. Equation (13.5) gives

$$U_f(|+\rangle|-\rangle) = \frac{(-1)^{f(0)}|0\rangle + (-1)^{f(1)}|1\rangle}{\sqrt{2}} \otimes |-\rangle. \tag{13.6}$$

Q's zero and one probabilities are still one-half. Measuring Q in Z now would miss the result. The function has changed the relationship between its amplitudes, not their magnitudes. T contains no separate readable copy of either function value; it has factored out in the same minus state.

Finally apply H to Q, leave T alone, and measure Q in Z. Figure 13.2 contains the complete procedure.



One fixed program; no measurement of the target is required.

Figure 13.2. One coherent query, followed by one answer. Register order is Q then T. Begin in $|0\rangle|1\rangle$, apply H to both, use the same unknown U_f once, and apply H only to Q before measuring it in Z. The recorded bit is $d = f(0) \oplus f(1)$ with certainty in the ideal model. T ends in $|-\rangle$ and need not be measured. The U_f block spans both registers and counts as one query, not necessarily one native hardware gate. No outcome is rejected. The measurement label M means read the Z basis, not apply a Z phase gate.

The two final Q amplitudes are

$$\alpha_0 = \frac{(-1)^{f(0)} + (-1)^{f(1)}}{2}, \alpha_1 = \frac{(-1)^{f(0)} - (-1)^{f(1)}}{2}. \tag{13.7}$$

If the function values agree, the two signs in the first numerator reinforce, while those in the second cancel. If they differ, the first numerator cancels and the second reinforces. In either case the surviving amplitude has magnitude one. Equivalently, the final joint state is $(-1)^{f(0)}|d\rangle|-\rangle$. Its overall sign does not affect the answer.

Check the less visually convenient case, $f(0) = 1$ and $f(1) = 0$. Immediately after the query, Q has state $(-|0\rangle + |1\rangle)/\sqrt{2} = -|-\rangle$. The last H gives $-|1\rangle$. Its amplitude is negative, but the probability of recording one is positive and equals one. The signs matter before the squaring, exactly as they did at the dark detector.

All four rows are handled by the same preparation, query location, final gate, and readout. This is the exact version of Deutsch's algorithm. It uses one coherent query where a classical zero-error procedure requires two evaluations. The original 1985 proposal and the later exact formulation are distinguished in the source notes.[6]

The answer is not a record of both values. The constant-zero and constant-one functions leave final states differing only by a global sign; this procedure cannot tell them apart. The two differing-value functions likewise give the same recorded answer. The output retains precisely the relation the task asked for.

We can describe the result in information terms too. Under an equal lottery over the four possible functions, the supplied choice has two bits of uncertainty. Our final answer resolves only the one-bit division into agreeing and differing pairs. Its certainty on each

instance does not mean that we have reconstructed the full choice. The two alternatives left within each class are deliberately invisible to this program.

What the extra qubit contributes

The target's preparation is essential. Prepare it in plus instead, and $X|+\rangle=|+\rangle$. Flipping it introduces no minus sign. The query then leaves the query superposition unchanged for every function, and the final H always produces zero. Merely sending a superposition through the box is not an algorithm.

This small example also keeps the two registers unentangled at the labeled boundaries of the ideal gates. The target remains a common factor in Eq. (13.6). That does not make the query restriction irrelevant, or establish that entanglement is unimportant in larger algorithms. It shows why neither 'many states at once' nor 'entanglement' can replace a calculation of what this particular program does.[7]

The same gate vocabulary also gives the entangling construction from Chapter 9. Prepare two zeros, apply Hadamard to the first register, and then controlled NOT from the first to the second. The intermediate state is $|+\rangle|0\rangle$; the output is $(|00\rangle+|11\rangle)/\sqrt{2}$. Only equal Z labels occur, although neither local label is certain. Changing the target preparation changes this conclusion: the minus target in the program just analyzed remains a common factor and supplies phase kickback instead. An entangling operation need not produce entanglement on every input. Here the preparation, not a slogan about the gate, determines which relationship the circuit creates.

The same mechanism works beyond a sign. Suppose the unitary V has a known target eigenstate $|u\rangle$ with eigenvalue $e^{i\vartheta}$. A controlled V applies V only on the one branch. Then

$$C(V)(|+\rangle|u\rangle)=\frac{|0\rangle+e^{i\vartheta}|1\rangle}{\sqrt{2}}\otimes|u\rangle. \tag{13.8}$$

An operation on the target has supplied the phase needed for the familiar interference experiment. But the resources remain specific: we require the eigenstate and a controlled implementation. Access to an unspecified device that performs V does not by itself give us every desired controlled version. Here we assume that controlled operation is supplied. Later algorithms will use this connection between eigenvalues, phase, and measurement much more extensively.[8]

Why the working memory must be clean

The interface also puts a requirement on any scratch memory inside the query device. A reversible calculation can retain intermediate results. That is useful while computing, but

dangerous when different input components must later interfere. If the working memory ends in distinguishable states for zero and one, it acts like the environmental marker in the previous chapter.

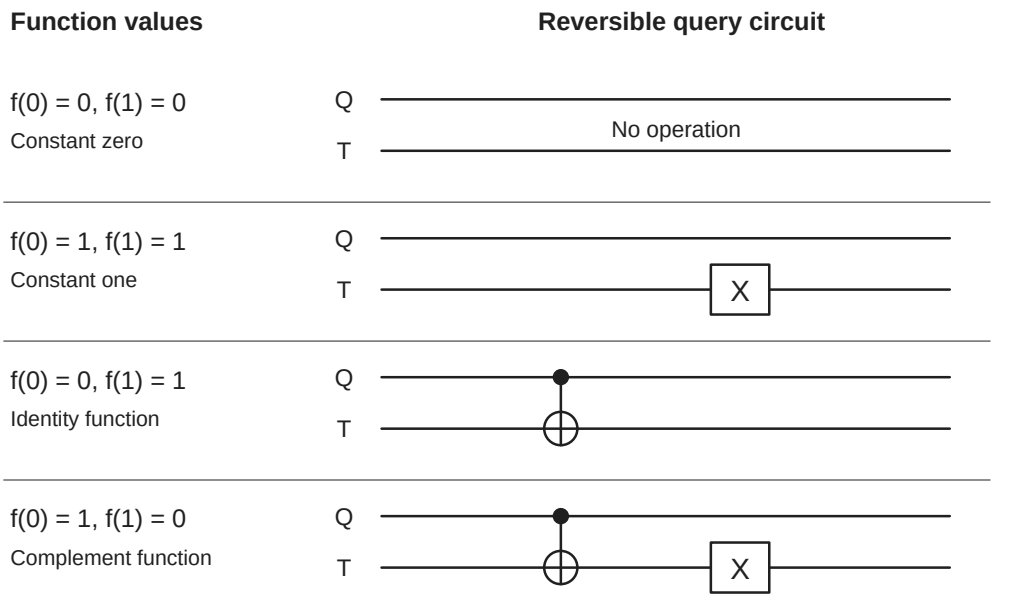
Consider a direct way to manufacture a phase from a classical function. Add a blank work bit W. Compute f into W while retaining the input. Apply Z to W, so that a computed one contributes a minus sign and a computed zero does not. Now reverse the function evaluation. The work bit returns to zero while the input component retains the factor $(-1)^{f(x)}$. The reversal removes the output label from W; it does not remove the phase already attached to the joint amplitude.

For a superposed query, the common final work state can be factored out. Without that reversal, the work states would distinguish the two components whenever the function values differ. Combining the query amplitudes alone would then be the wrong calculation. Uncomputation, first introduced as a way to clear working storage, also protects the interference a program needs.

This compute–phase–uncompute construction invokes the reversible evaluation twice. It is a useful general construction, not the one-query algorithm above disguised as a cheaper circuit. Our minus target obtains the sign in one supplied query because it is already an eigenstate of the target flip. In a larger implementation, auxiliary memory must end without a record distinguishing the query inputs. Returning it to its common ready state is one way to meet that requirement. The cost of doing so belongs inside its implementation account.[9]

Open the box for an accounting check

For our tiny problem, we can display every possible query implementation. The constant-zero function uses the identity. The constant-one function flips T. The function $f(x)=x$ uses controlled NOT from Q to T. The function $f(x)=1 \oplus x$ uses that controlled NOT followed by a flip of T. Figure 13.3 makes these alternatives explicit.



The row is hidden in the query task, but shown here for verification.

Figure 13.3. The four function-evaluation boxes. Each row implements Eq. (13.3) on all four input basis states, including a nonzero target. An empty wire is the identity, X flips its register, and a filled dot with a circled target is controlled NOT. Operations run left to right. A supplier chooses one row; the querying program is not given that row's label or permission to inspect its gates. The constructions show that the interface is physically specified, not that identifying an exposed circuit requires a quantum algorithm.

These implementations make the calculation easy to verify. They also expose a common mistake in interpreting demonstrations. Someone who sees which row was built already knows whether the function is constant. Choosing the row oneself, executing the quantum circuit, and recovering that known answer demonstrates the transformation; it does not establish a useful speed advantage over reading the chosen circuit description.

In the query comparison, the supplier holds that description and the algorithm has only the stipulated access. Both the restriction and the coherent resource are explicit. A classical simulation given the gate description can reproduce the calculation without contradicting the one-query lower bound for a different access model.[9]

Outside the query box the program uses three Hadamard gates, two initialized registers, and one final qubit measurement. The first two H gates can occupy one logical layer. Preparing T from a hardware zero state additionally requires a flip or an equivalent initialization. Repetitions, resource preparation, and whatever operations realize the query must be counted when asking about elapsed time or energy.

A pair of classical processors could evaluate zero and one simultaneously. That still makes two value queries, although it might reduce their elapsed time. Our comparison counts queries rather than processors or seconds. A coherent call could take longer or require more control than either classical call. The exact separation remains a statement

about permitted operations, not a claim about the fastest way to answer this two-entry question in a laboratory.

The same care applies to larger circuits. Arbitrary single-qubit gates together with controlled NOT suffice to construct general finite-qubit unitary transformations. This universality result supplies building blocks, not a guarantee of an economical construction for every desired transformation. A box labeled with a difficult calculation has not paid for its contents.[10]

When the right truth table gives the wrong program

Now let the oracle leave a record of Q's zero/one label in an unwanted auxiliary system. It may still pass every test made with basis inputs. But after a superposed input, the record reduces the off-diagonal coherence of Q. If it distinguishes the two labels perfectly, the final Hadamard produces a fair answer instead of the required certainty.

Chapter 12 provides a quantitative test. Let the two normalized environmental record states have real overlap γ, between zero and one. Suppose an additional coherent phase error δ is imposed on Q's one component before the last H. Populations are unchanged. Averaging over the inaccessible record gives

$$P(\text{correct}) = \frac{1 + \gamma \cos \delta}{2}. \tag{13.9}$$

This probability is the same for each of the four functions, with its own correct label. With no phase error, a partial record of overlap 0.8 reduces it to 0.9. With no record but a half-cycle phase error, the program is certainly wrong. A systematic coherent error can reverse an answer; it need not look like extra random noise. Figure 13.4 separates these effects.[11]

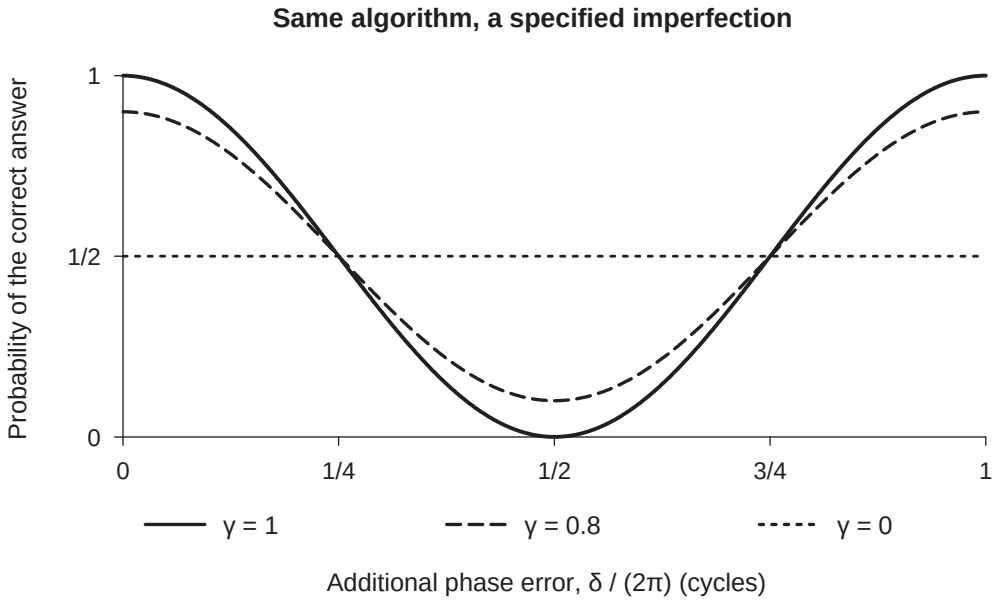


Figure 13.4. The answer depends on both coherence and calibration. Curves show Eq. (13.9) for environmental overlaps $\gamma = 1$, 0.8, and 0. The horizontal axis is the additional relative-phase error in cycles, $\delta/(2\pi)$. A perfect retained label record gives success $1/2$ at every phase. With no record, a half-cycle error reverses the answer completely. All trials count, readout is ideal, and the populations before the final H are balanced. These are calculations for a stated noise model, not measured performance or a universal gate-error law.

Suppose the final classical readout also flips the recorded answer with independent probability ϵ, equally for zero and one. Averaging correct and incorrect reports gives

$$P(\text{reported correct}) = \frac{1 + (1 - 2\epsilon)\gamma \cos\delta}{2}. \tag{13.10}$$

For the stipulated values $\gamma = 0.8$, $\delta = \pi/3$, and $\epsilon = 0.02$, this is 0.692. The values describe an invented test case, not a device benchmark. With a stable preparation and independent repetitions of this error model, repeated runs estimate this imperfect probability more accurately. They do not make the underlying channel ideal. Majority voting can help when the correct answer remains more likely and the errors satisfy the assumed model; it cannot rescue an unrecognized systematic reversal.[12]

Each repetition starts from a new preparation and uses the oracle again. In the ideal exact algorithm one trial suffices. In an actual implementation, repetitions used to verify a device or estimate an answer are additional work. These are different questions about the same circuit.

From a diagram to a running device

A drawn H is not a universal piece of hardware. Different carriers implement the same logical operation through different controls. A compiler must map the circuit's registers to available qubits, translate gates into supported operations, and respect which qubits can interact. Routing may introduce extra swaps; scheduling must account for waiting intervals. The logical circuit and the resulting pulse sequence therefore have different resource counts.[13]

Readout has its own physics. Eoin Kelly and colleagues converted spin information into a charge record and held that record long enough for reliable detection. Fits to repeated records gave an average single-qubit, single-shot state-preparation-and-measurement fidelity of 97.0 percent, with a reported one-sigma uncertainty of 0.5 percentage points. This combined figure is neither a gate fidelity nor our program’s success probability; an average also does not establish equal error rates for the two labels. The final measurement box therefore needs its own preparation, conversion, and detection account.[14]

Classical comparisons must be equally concrete. Our explicit four query implementations, combined with H, belong to a family called stabilizer circuits. Chapter 16 will show how constraints describe their states compactly and make the simulation possible. With the basis-state inputs and readout used here, this family admits efficient classical simulation even when its circuits produce entanglement. The arbitrary phase gate in our earlier interferometer is not generally in this restricted family. Interference is real, but its presence alone does not establish computational intractability for a classical machine.[15]

Classical simulations of more general circuits also continue to improve. Manuel Rudolph and Joseph Tindall used two-dimensional tensor networks—linked arrays that represent a state without storing every amplitude separately—to simulate and sample selected circuit families. Their comparisons show that circuit geometry and required accuracy can strongly affect the classical workload. Two errors must be kept separate: how closely the stored representation matches the circuit state, and how faithfully the sampling procedure draws from that representation. This is not a theorem of efficient simulation for every circuit; it is a reason to test a claimed advantage against methods suited to the actual problem.[16]

An answer made accessible

We began with a known phase and a predicted detector response. We now have a program in which an unknown operation supplies the signs, coherent recombination turns

their relationship into a measurement outcome, and the outcome answers a precise question.

Nothing has read out a separate result from every term of a superposition. The program has arranged one useful distinction. Nor have we obtained that distinction from an unrestricted classical interface: coherent access was a stated resource, and the one-query advantage belongs to that access model.

This is a productive way to think about quantum programming. Choose a task, identify a transformation that encodes the needed relationship, preserve it long enough to use it, and calculate how the final measurement reveals it. Then count the full procedure and compare it with an appropriate alternative.

Our two-answer problem allowed complete cancellation in one step. Searching among many candidates will require a different pattern. The next program will repeatedly change signs and recombine amplitudes so that one desired result becomes progressively more likely. The question is no longer whether interference occurs, but how much useful work it can do.

Notes and sources

1. John Watrous, IBM Quantum Learning, Basics of Quantum Information, ‘Single systems: Quantum information,’ on Hadamard and unitary operations. Eq. (13.1) fixes the same normalized-sum/difference convention used throughout this book. The interrupted two-Hadamard example starts in zero and uses a nonselective Z measurement. Its comparison is an ideal repeatable X measurement of the intermediate plus state. The examples have different output channels, not different applications of the Born rule. States and gates.

2. Richard P. Feynman, Robert B. Leighton, and Matthew Sands, The Feynman Lectures on Physics, Vol. III, Chapter 3, ‘Probability Amplitudes.’ Eq. (13.2) is an original evaluation of the chapter's circuit, using addition across coherent alternatives and multiplication along successive operations. $P(\phi) = diag(1, e^{i\phi})$; no uncontrolled path record or loss is included. Fig. 13.1 uses register notation, not a reconstruction of an optical layout. Amplitude rules.

3. John Preskill, Quantum Information, Chapter 6, ‘Quantum Algorithms,’ November 2020 revision, §6.1. The classical lower bound is for zero-error computation with ordinary value queries, whereas the quantum interface supplies the coherent reversible U_f. The uniform-function average-success statement is an original enumeration of the eight deterministic one-query decision rules. Private randomization only forms mixtures of them. Query model and Deutsch's problem.

4. Preskill, Chapter 5, ‘Classical and quantum circuits,’ July 2015 revision, §§5.2–5.3. Eq. (13.3) is a reversible basis permutation with its coherent linear extension fixed. Basis-dependent extra phases or retained distinguishing records define a different channel, even if basis truth-table tests succeed. Work registers inside an implementation must be restored or remain independent of the input; see the uncomputation in this book's Chapter 8. Reversible and quantum circuits.

5. Richard Cleve, Artur Ekert, Chiara Macchiavello, and Michele Mosca, ‘Quantum Algorithms Revisited,’ Proceedings of the Royal Society A 454, 339–354 (1998), doi:10.1098/rspa.1998.0164, §§1–2. The author version is arXiv:quant-ph/9708016v1 (8 August 1997). Eqs. (13.4)–(13.5) develop its controlled-eigenvalue mechanism with an explicit register order, normalized states, and an X eigenstate as target. Full author text; Publication record.

6. David Deutsch, ‘Quantum theory, the Church–Turing principle and the universal quantum computer,’ Proceedings of the Royal Society A 400, 97–117 (1985), doi:10.1098/rspa.1985.0070. The exact deterministic one-query formulation used here is the later improvement explained by Cleve and colleagues (§2), not a claim that Deutsch's original protocol succeeded conclusively on every trial. Eqs. (13.6)–(13.7), Table 13.1, and Fig. 13.2 are worked constructions for all four functions. Substituting each function's two values into Eqs. (13.6)–(13.7) checks its ideal output directly. Original proposal; Exact formulation.

7. Watrous, Fundamentals of Quantum Algorithms, ‘Quantum query algorithms: Deutsch's algorithm.’ The ideal states at the labeled gate boundaries factor as a query state times the unchanged minus target. This observation is not a claim about the intermediate dynamics of every physical gate implementation, or about arbitrary larger algorithms. Our tensor order Q,T is fixed explicitly and should not be inferred from a software package's bit-order convention. Worked algorithm.

8. Cleve and colleagues, §1, on controlled unitaries and phase kickback. Eq. (13.8) assumes $V|u\rangle = e^{i\vartheta}|u\rangle$ and a supplied coherent controlled-V operation. It does not assert a generic way to add control to an unknown physical black box. Preparing the target eigenstate, providing the control interface, and realizing the final readout are separate resources. Controlled-eigenvalue construction. See also Mateus Araújo, Adrien Feix, Fabio Costa, and Časlav Brukner, ‘Quantum circuits cannot control unknown operations,’ New Journal of Physics 16, 093026 (2014), doi:10.1088/1367-2630/16/9/093026. Its circuit-model restriction does not exclude implementations with extra known subspaces or paths; those supply additional structure. Original no-control analysis.

9. The compute–phase–uncompute example uses two reversible evaluations and a work-bit Z operation; it is not counted as one query. Fig. 13.3 gives original two-register gate constructions of all four U_f maps. The

last row uses $f(x) = 1 \oplus x$; its target X and CNOT commute. Gate descriptions are shown for verification, but are unavailable in the stated oracle-identification task. An exposed classical description and a restricted function-query interface are different computational inputs. Applying each circuit to all four basis inputs, including their signs, establishes the reversible action. Linearity fixes its action on superpositions, and Eqs. (13.6)–(13.7) give the resulting program output.

10. Adriano Barenco and colleagues, 'Elementary gates for quantum computation,' Physical Review A 52, 3457–3467 (1995), doi:10.1103/PhysRevA.52.3457. Arbitrary one-qubit gates and CNOT form an exactly universal family. This is not a claim of efficient implementation of every unitary, nor a claim that the restricted H/X/CNOT example by itself is universal. Preparation and readout are additional operations. Author manuscript.

11. Eq. (13.9) is an original noise calculation using the label-preserving environment model in Chapter 12. After the ideal query, apply a relative phase δ and a dephasing channel of real multiplier γ to Q, then H and Z readout. In Q's density matrix the upper-right entry is $(-1)^d \gamma e^{-i\delta} / 2$. The reduced map alone does not identify whether physical marking or classical phase noise produced it. Fig. 13.4 plots this stipulated model, not experimental data. Quantum channels and dephasing.

12. Eq. (13.10) adds an independent, symmetric flip of the final recorded bit. It follows by weighting the correct report with $1 - \epsilon$ and the incorrect report with ε. The worked 0.692 probability is exact for the stated parameters. Tracing out the environmental record in the overlap model gives Eq. (13.9); averaging the two possible classical readout flips gives Eq. (13.10). No general noise independence across repeated experimental trials is established by this calculation.

13. IBM Quantum Documentation, 'Introduction to transpilation,' accessed 30 September 2026. Layout, routing, basis translation, optimization, and scheduling connect a logical circuit to a specified device. These are implementation stages, not extra queries hidden inside an oracle comparison. The chapter does not depend on a particular SDK version or promise a gate count for an unspecified machine. Official implementation documentation.

14. Eoin G. Kelly and colleagues, 'Identification and mitigation of errors in hole spin qubit readout,' Nature Electronics (published 29 September 2026), doi:10.1038/s41928-026-01705-1. The reported 97.0(5)% is an average single-qubit, single-shot SPAM fidelity. Figure 4c–d and Methods, "State probability determination," describe thresholded records, fitted Rabi oscillations and inferred confusion matrices. The average is over their diagonal entries and both qubits; parentheses denote the authors' one-sigma interval. The quoted fidelity and uncertainty are the authors' reported estimates from those fits. The experiment investigates germanium hole spins, conversion and relaxation errors, and double-latched charge detection; it does not implement the circuit or the symmetric error model in this chapter. Published full text.

15. Scott Aaronson and Daniel Gottesman, 'Improved Simulation of Stabilizer Circuits,' Physical Review A 70, 052328 (2004), doi:10.1103/PhysRevA.70.052328. The cited author version, arXiv:quant-ph/0406196v5, was revised in 2008. Efficient simulation applies to the stated stabilizer preparations, gates, and measurements. The unrestricted continuous phase gate used earlier to illustrate interferometry is not generally a stabilizer gate; the four discrete oracle implementations used for Deutsch's problem are. Full author text.

16. Manuel S. Rudolph and Joseph Tindall, 'Simulating and sampling from quantum circuits with 2D tensor networks,' npj Quantum Information, doi:10.1038/s41534-026-01372-2. The publisher released the peer-reviewed accepted article on 24 September 2026 and identified it as not yet the final Version of Record. The cited Article in Press version's Discussion and Methods, "Sampling from tensor network states," distinguish finite-representation error from the sampler's approximation to that representation. The numerical circuit-family comparisons support the geometry-and-accuracy distinction; they are not a theorem of efficient general quantum-circuit simulation. No performance curve in this chapter reproduces their data. Publisher record; Accepted full text.

CHAPTER 14

Search by Changing the Odds

A checker can recognize an answer without telling us where to find one. Give it a proposed arrangement, and it reports whether the arrangement satisfies the rules. Ask it to suggest a proposal, and it has nothing to say.

Imagine four candidates, labeled 00, 01, 10, and 11. Exactly one passes a fixed test. We may use the checker, but we are not given the successful label or permission to inspect the checker's internal construction. A blind first guess succeeds with probability one-quarter. What could a quantum program do differently?

It cannot begin by knowing which answer deserves a larger probability. Nor can it obtain four separately readable answers from one measurement. Instead, it can make the checker change a relative sign, then use an operation that treats all four labels alike. Together those two changes can direct the entire final probability toward the successful candidate.

That is the remarkable claim we will calculate. The four-candidate case makes every contribution visible. A larger search will reveal both the square-root improvement discovered by Lov Grover and the work hidden inside the word ‘check’.[1]

A sign that records success

Write the checking rule as $f(x)$: it equals one when candidate x is acceptable and zero otherwise. For now, assume exactly one acceptable candidate and ideal coherent control. The function stays fixed throughout a run. A successful search returns its label, not just the statement that an answer exists.

Chapter 13 supplied the required interface. A reversible evaluation acts on a candidate register and a target bit. With the target prepared in the minus state, its conditional flip becomes a conditional sign:

$$\begin{aligned} U_f|x\rangle|t\rangle &= |x\rangle|t \oplus f(x)\rangle, \\ U_f|x\rangle|-\rangle &= (-1)^{f(x)}|x\rangle|-\rangle. \end{aligned} \tag{14.1}$$

The unchanged target factors out. We will therefore draw just the operation on the candidate register, call it O_f, and remember that it can be realized through one coherent use of this supplied evaluation. The letter O denotes the oracle, not the number zero. The oracle's physical implementation may contain many gates.[2]

Prepare the two candidate qubits in zero and apply a Hadamard to each. Their joint state is an equal superposition of the four labels. Choose 10 as the successful label for our worked calculation; the program is not told that choice:

$$|s\rangle = \frac{|00\rangle + |01\rangle + |10\rangle + |11\rangle}{2},$$
$$O_f|s\rangle = \frac{|00\rangle + |01\rangle - |10\rangle + |11\rangle}{2}. \tag{14.2}$$

The amplitudes have magnitude one-half, not one-quarter. Squaring their magnitudes gives the four probabilities. After the oracle, those probabilities are still one-quarter each. Stopping to measure here would throw away the opportunity: the useful difference is a sign that this measurement cannot reveal.

Nothing has certified to the experimenter that 10 is the answer. The checker has affected the state through its rule. The next operation must turn that effect into a readable distinction without being designed around the unknown successful label.

The operation that makes the difference

After the sign change, the four amplitudes are one-half, one-half, minus one-half, and one-half. Their arithmetic mean is one-quarter. Now reflect each amplitude about that mean: an amplitude the same distance above the mean ends the same distance below it, and conversely.

$$\bar{a} = \frac{1}{N}\sum_{x=0}^{N-1} a_x, a'_x = 2\bar{a} - a_x. \tag{14.3}$$

Here N is the number of candidate labels; it is four in the current example. The bar over a means their arithmetic mean, not complex conjugation. We are transforming amplitudes before calculating probabilities.

Each unsuccessful amplitude was one-half. Its new value is twice one-quarter minus one-half: zero. The successful amplitude was minus one-half. Its new value is twice one-quarter plus one-half: one. A final measurement therefore returns 10 with certainty. Figure 14.1 shows all four labels, including those whose contributions cancel.

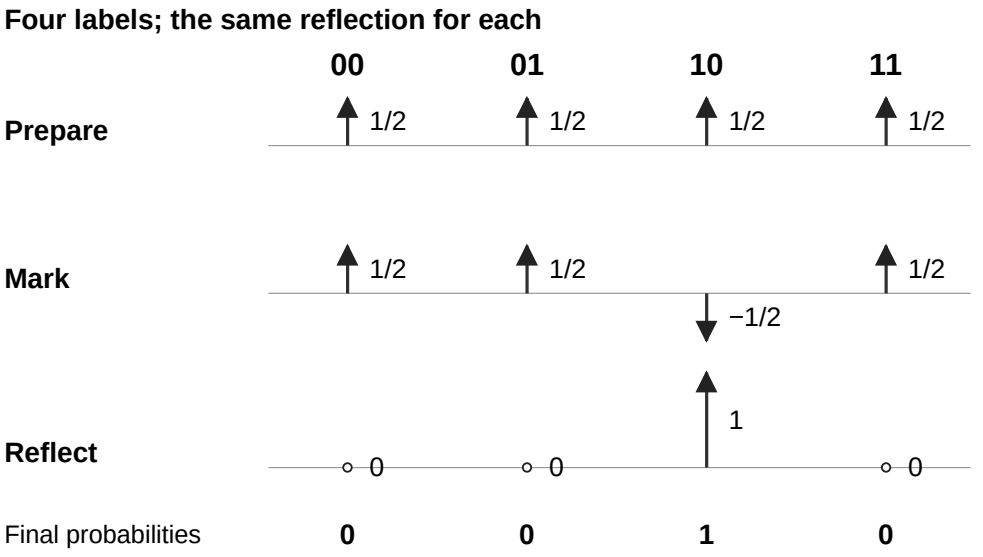


Figure 14.1. A relative sign becomes a certain answer. Columns correspond to the fixed reference order 00, 01, 10, 11. The first two rows show amplitudes before and after the oracle for the stipulated successful label 10; their squared magnitudes are unchanged. The final row applies the same reflection to every component, $a'_x = 2\bar{a} - a_x$, with $\bar{a} = 1/4$ after the oracle. Only the successful label remains. These are exact state-vector values, not measured frequencies, independent parallel trials, or particle paths.

Had the checker accepted another label, the minus sign would have appeared in another position. The same mean and the same reflection would then have selected that position. Neither the initial Hadamards nor the final reflection has changed. The unknown instance enters only through the checker.

But how could a machine reflect about a mean it has not measured? The question exposes a helpful distinction between describing an operation and implementing it. Equation (14.3) looks like a calculation that first estimates four unknown numbers. It is actually one fixed linear transformation of the whole vector.

For N equally weighted labels, write $|s\rangle = N^{-1/2}\sum_x |x\rangle$. The transformation is

$$D = 2|s\rangle\langle s| - I\,,\; D|v\rangle = 2|s\rangle\langle s|v\rangle - |v\rangle\,. \tag{14.4}$$

The identity I leaves a vector unchanged. The outer product $|s\rangle\langle s|$ keeps its component along $|s\rangle$. Thus D leaves that component unchanged and reverses every perpendicular component. Applying it twice restores the original vector. It preserves lengths and overlaps: it is a unitary reflection, not a measurement or a process that increases total probability.[3]

For any amplitude list, $\langle s|v\rangle$ is its sum divided by $\sqrt{N}$. Substitution gives Eq. (14.3) directly. The rule works for complex amplitudes too, although our first search stays

within real numbers. No detector reads the mean, and no classical processor receives the individual amplitudes.

When $N = 2^n$, the reflection has a useful construction. Undo the initial Hadamards, retain the sign of the all-zero component while reversing every other sign, then apply the Hadamards again. This implements D. For two qubits, the middle sign operation is Z on each qubit followed by controlled Z, which changes only the 11 sign once more. Checking the four labels gives signs $+, -, -, -$. For larger registers, the corresponding controlled operation must be built and counted; its box is not free.[4]

Two reflections, one turn

Why does the trick not simply deliver certainty after one check for any number of candidates? The mean answers part of the question. For one acceptable label among N, each initial amplitude is $1/\sqrt{N}$. After its sign is reversed, the mean is $(N-2)/(N\sqrt{N})$. The reflection makes the acceptable amplitude $(3-4/N)/\sqrt{N}$. For large N, that is about three times its original value—but still small.

The next move is to repeat the pair of operations: mark, then reflect. Call one complete pair a search round, $G = D O_f$. The order matters: the rightmost operator acts first. To understand repeated rounds, we can replace the long amplitude list by a two-dimensional drawing without replacing the N-candidate problem by a two-candidate one.

Allow M acceptable labels, with $0 < M < N$. Let $|g\rangle$ be their normalized equal superposition, and let $|b\rangle$ be the normalized equal superposition of the unacceptable labels. These states are orthogonal because their label sets do not overlap. All acceptable components remain equal throughout this ideal algorithm, as do all unacceptable components. The state therefore stays in the plane they span:

$$|s\rangle = \sqrt{\frac{N-M}{N}}\,|b\rangle + \sqrt{\frac{M}{N}}\,|g\rangle$$
$$|s\rangle = \cos\theta\,|b\rangle + \sin\theta\,|g\rangle,\ \sin^2\theta = \frac{M}{N}. \tag{14.5}$$

Measure the angle from the bad direction toward the good direction. The initial angle θ has a sine that is an amplitude for the whole acceptable subspace; its squared sine is the total success probability. A measurement still returns an individual label within that subspace.

The oracle reverses the good component and leaves the bad component unchanged. Geometrically, it reflects the arrow across the bad axis. D then reflects across the initial direction $|s\rangle$. Two reflections about axes separated by θ produce a rotation through 2θ. Figure 14.2 follows the first round. The same pair of reflections adds another 2θ on every round.[5]

$$G^j|s\rangle = \cos\big((2j+1)\theta\big)|b\rangle + \sin\big((2j+1)\theta\big)|g\rangle,$$
$$P_j = \sin^2\big((2j+1)\theta\big). \tag{14.6}$$

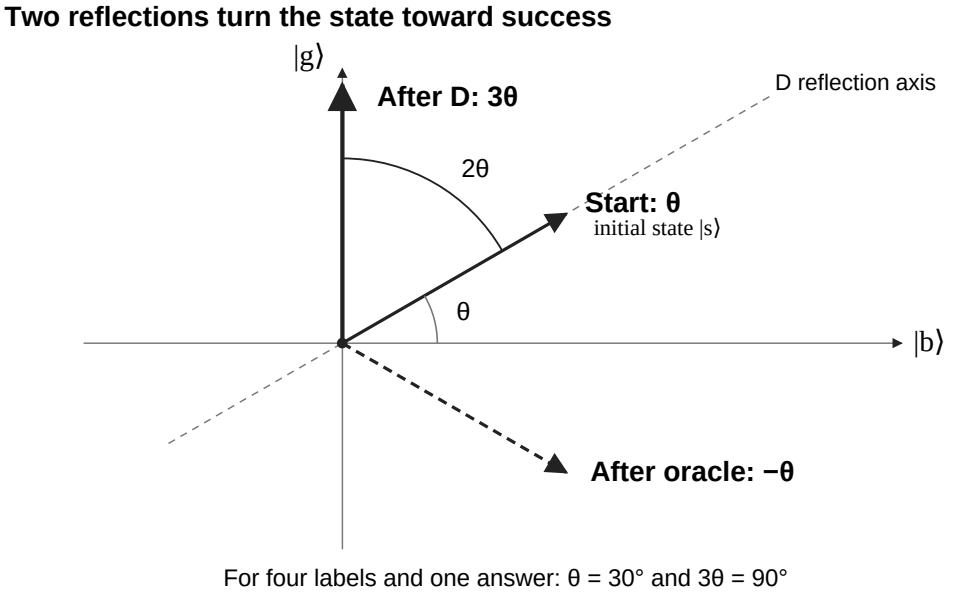


Figure 14.2. A search round is a rotation made from two reflections. The axes denote the normalized bad and good superpositions, not spatial directions or the real and imaginary parts of one amplitude. The incoming arrow lies at θ; the oracle reflects it to $-\theta$; D reflects that arrow about the initial direction to 3θ. The angle shown is $\theta = \pi/6$, the four-candidate, one-answer case. The final arrow then lies exactly on the good axis. Dashed extensions identify reflection axes; no measurement occurs between reflections.

For four candidates and one answer, $\sin\theta = 1/2$, so $\theta = \pi/6$: thirty degrees. One round takes the arrow from thirty to ninety degrees, giving certainty. That unusually neat landing explains the small example. It was not evidence that one query suffices for every search.

The two-direction construction assumes that acceptable and unacceptable labels both exist. If no label is acceptable, the success probability is zero and no rotation can create an answer. If every label is acceptable, the initial preparation already succeeds: a measurement returns an acceptable label without a search round.

In a large search with a small acceptable fraction, θ is small and approximately $\sqrt{M/N}$ in radians. Reaching the vicinity of ninety degrees takes roughly $(\pi/4)\sqrt{N/M}$ rounds. Repeated independent uniform guesses require roughly N/M checks to find an acceptable result. The square root enters because the coherent rotation starts with an amplitude of size $\sqrt{M/N}$, while independent trials succeed with its square.[6]

When to stop

Take sixty-four candidates with one acceptable label. The initial success probability is $1/64$. After one round it is about 13.48 percent; after three, 59.14 percent; after six, 99.66 percent. But after twelve rounds it is only about 0.0071 percent. Twice as many rounds have almost undone the achievement.

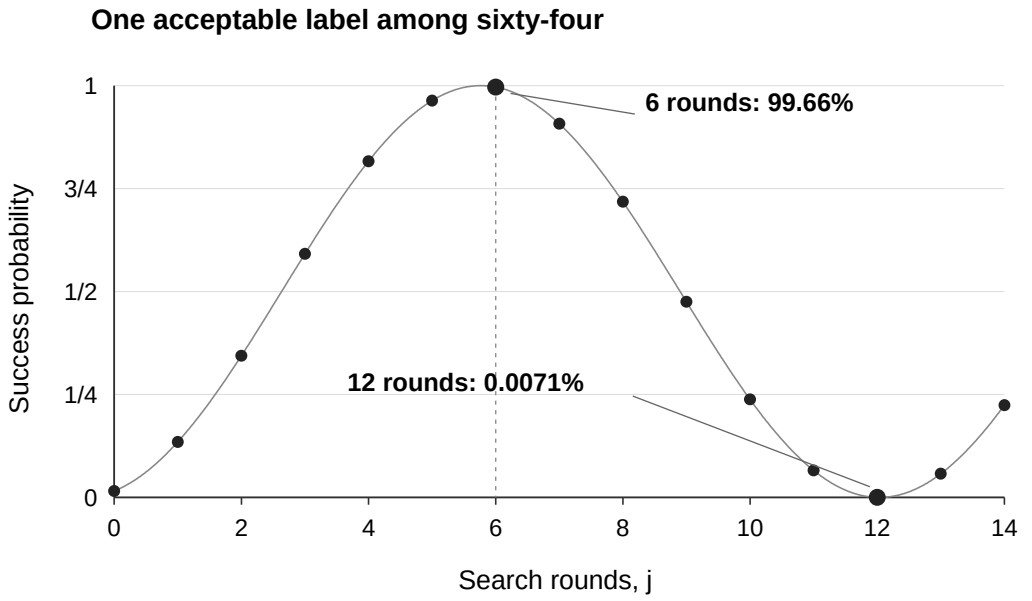


Figure 14.3. More rounds do not always mean more success. Circles show the exact probabilities P_j for $N = 64$, $M = 1$, and integer round counts. Six rounds give approximately 0.996586; twelve give approximately 0.000070506. The connecting curve displays the same mathematical expression between integer arguments only as a visual guide. Each point describes a fresh preparation followed by the chosen number of uninterrupted rounds and one final measurement. No running measurement, gate noise, or experimental data is included.

There is no trapdoor at the answer that holds the state once it arrives. The evolution is reversible. Unless the program stops, the next round keeps turning. For known M, choose a nearest nonnegative integer j to the first peak's continuous location, $\pi/(4\theta) - 1/2$. Integer rounding means that the basic algorithm need not reach probability exactly one. Rounding moves the final angle away from ninety degrees by at most the initial angle. The resulting failure probability is therefore no greater than the original acceptable fraction. This bound is useful when that fraction is small. When more than half the labels are acceptable, measuring the initial state already succeeds with probability greater than one-half.

We also cannot watch a candidate measurement after every oracle call and expect the same trajectory. In the four-label example, a complete unread record of the label destroys the relative-sign coherence and leaves the maximally mixed candidate state, $I_4/4$. The reflection leaves that state unchanged. The success probability stays one-quarter.

Observing progress through fresh repeated experiments is different from interrupting the same coherent run.

What if M is unknown? One can vary the run length randomly within an expanding range, measure and verify a candidate, and restart after failure. Boyer and colleagues proved a square-root expected query bound for a suitable schedule when acceptable candidates exist. The schedule avoids repeatedly stopping at an unfortunate angle. If none exist, verification never succeeds; a finite ‘not found’ conclusion needs a specified error guarantee and stopping rule, not endless hopeful repetition.[7]

Verification is an additional use of the checking rule, not a reading of the sign left by the phase oracle. The supplied reversible evaluation can provide the value by using a zero target and reading that target after a candidate has been measured. A sign change on one definite candidate, by itself, is only a global phase and supplies no acceptance record. Those verification calls and all failed attempts belong in the query account.

There are also modified phase sequences with fixed-point guarantees, designed to avoid the ordinary overshoot over a specified range of initial success probabilities. A positive lower bound on the initial success probability and a chosen error tolerance determine the sequence and its cost. The guarantee concerns the output of a complete designed sequence; it does not say that every internal operation monotonically improves success. Nor does it make indefinitely repeating the original pair of reflections safe.[8]

A checker need not contain the answer

A common objection now becomes precise: to reverse the successful amplitude, must we already know the successful label? No. The oracle needs a procedure for testing a proposed label. Recognition and discovery are different computational tasks.

Consider three binary choices, a, b, and c, subject to two rules: a must equal b, and b must differ from c. This tiny problem is deliberately easy to solve by inspection. It lets us examine a recognizer without hiding arithmetic in an unnamed subroutine.

Use two zeroed working qubits, u and v. Write the first rule's truth value into u and the second into v:

$$u = 1 \oplus a \oplus b,\ v = b \oplus c,\ f(a,b,c) = u\,v. \tag{14.7}$$

A flip of u followed by controlled NOT from a and b calculates the first value. Controlled NOT from b and c calculates the second. Both checks are true exactly when u and v are one. A controlled-Z gate between them supplies a minus sign to that case. Reverse the five calculation gates afterward. The work qubits return to zero, but the desired phase remains.

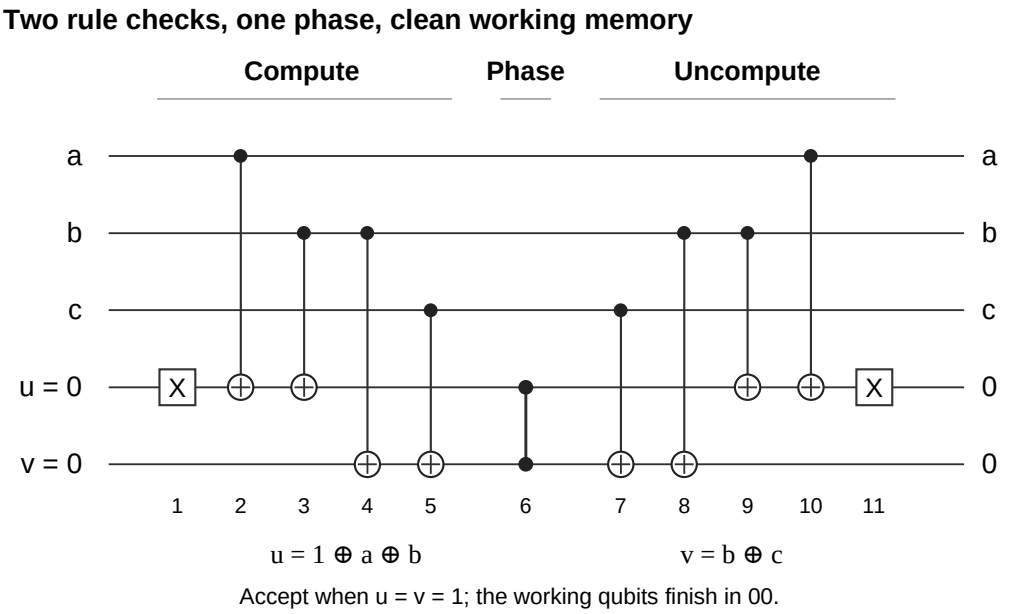


Figure 14.4. Build the checks, attach a phase, remove the working record. Registers are ordered a, b, c, u, v; only u and v must begin at zero. Gates 1–5 calculate Eq. (14.7); gate 6 is controlled Z, indicated by two filled dots joined vertically; gates 7–11 reverse the computation. A circled plus is an X target and a filled dot on its connecting line is a CNOT control. Bare crossings do not participate. The exact circuit uses two X gates, eight CNOT gates, and one controlled-Z gate. It supplies the phase $(-1)^{f(a,b,c)}$ on clean-work inputs, without measuring the checks. Wires denote registers and operation order, not particle paths or equal gate durations.

The accepted labels are 001 and 110. Starting with equal amplitudes on all eight labels, one search round leaves their equal superposition, $(|001\rangle + |110\rangle)/\sqrt{2}$. The success probability is one, but the program returns one acceptable assignment on measurement, not both as separate answers. The same geometry worked because the acceptable fraction was again one-quarter.

This phase checker contains eleven gates in the stated logical vocabulary, not one elementary switch. A platform that implements controlled Z as a Hadamard, a controlled NOT, and another Hadamard would count thirteen instead. Its actual pulses and error-protection operations need a further accounting.[9]

Reversal matters for more than tidiness. Without it, different candidates can leave different values in the working qubits. Those records participate in the full state and obstruct the candidate-only interference we calculated. The inverse computation removes the records coherently. It does not erase the unknown input or first learn which candidate will win.

The simple rules above also reveal their answers classically, so this circuit is not a demonstration of practical speedup. In a larger problem, checking may be a long arithmetic calculation. Grover's algorithm changes how often the checker is invoked; it

does not excuse us from constructing it or from asking whether its internal structure permits a better classical search.

Amplifying a better starting point

Uniform preparation is convenient, but not sacred. Suppose a controllable unitary preparation A prepares a state $|\psi\rangle = A|0\rangle$ that already produces an acceptable result with probability p. Perhaps it builds in a constraint instead of assigning weight to impossible candidates. The good and bad components can be normalized just as before, with $\sin^2\theta = p$.

The required reflection now follows that prepared state rather than the uniform direction:

$$D_A = 2|\psi\rangle\langle\psi| - I = A\left(2|0\rangle\langle 0| - I\right)A^{\dagger}. \tag{14.8}$$

The dagger denotes the inverse of this unitary preparation. To implement the reflection, apply the inverse preparation, reflect about the all-zero state of the full preparation register, then prepare forward again. The two-reflection argument is unchanged. Starting probability p can be amplified using a number of suitable operations proportional to $1/\sqrt{p}$, rather than the $1/p$ independent repetitions needed on average.[10]

But every reflection now pays for A and its inverse. A complicated starting distribution can reduce the query count while increasing the time for each round. An ordinary program that measures or discards working records cannot simply be inserted unchanged: its coherent implementation must preserve or restore what the inverse needs.

This is why amplitude amplification is more general than searching a directory. It amplifies the useful component of a known, controllable preparation. The output can be an acceptable classical label or a desired quantum subspace. What makes the improvement possible is access to both a test and the reflections, not a special privilege granted to anything called a database.

Counting calls—and counting the machine

The square-root query advantage is not merely a comparison with an especially foolish classical program. For a uniformly hidden single acceptable label among N, a classical method that makes q distinct value queries and then, after all failures, guesses an unqueried label succeeds with probability at most $(q+1)/N$, for $q < N$. The queried labels account for q possible locations; the last guess adds one. Randomizing the strategy cannot improve this average bound.

Thus one classical check in our four-label task can succeed at most half the time, even with a free final guess. Three checks can guarantee the label by elimination. A policy that requires an actual positive check of the returned label may use a fourth. These are slightly

different tasks; neither changes the linear growth of classical query requirements at fixed, nontrivial success probability.[11]

Quantum lower-bound results establish square-root growth as optimal for unstructured search in this oracle model. A different quantum circuit cannot reduce a general black-box search to a logarithmic number of queries while retaining the same guarantee. That statement does not prove that every problem encoded as a search is hard: an explicit problem description may reveal structure absent from the black-box interface.[12]

For a physical comparison, include the whole run. Let t_{init} be its preparation time, t_O the time for one phase marking including cleared workspace, t_D the reflection time, and t_{out} the time for readout and reliable classical verification. With independent fresh retries of the same run and success probability $P_j > 0$,

$$T_Q(j) = \frac{t_{init} + j(t_O + t_D) + t_{out}}{P_j}. \tag{14.9}$$

This is an expected time to a verified answer, not a worst-case deadline. The denominator includes failed runs. It assumes that verification rejects incorrect candidates without rejecting correct ones; an imperfect verifier requires a different calculation. Peak single-run success need not minimize this quotient. Shorter, cheaper attempts may be worth repeating.[13]

The division has a simple reason. Every attempt pays the same run cost. After a failure, the fresh restart faces the same expected remaining time as at the beginning. Thus the expected total equals one run cost plus the failure probability times that expected total. Solving this relation divides the run cost by the success probability.

Use a numerical illustration, not a hardware forecast. There is one acceptable label among 65,536, so the candidate register needs sixteen qubits before workspace or error protection. A run of 201 rounds succeeds with probability about 0.999988. Measure time in units of one classical candidate check. Stipulate that preparation, readout, and verification together cost four units. Table 14.1 changes only the duration of a quantum round.

Table 14.1**. The round count does not determine the elapsed time.**

Method and stipulated timing	Mean time to a verified answer
Classical: distinct candidates checked until success	32,768.5 units
Quantum: 201 rounds at 100 units per round	20,104.2 units
Quantum: 201 rounds at 1,000 units per round	201,006.4 units

One acceptable label is uniformly hidden among 65,536. The classical method samples without replacement and confirms the positive result. Quantum runs use Eq. (14.9), ideal P_{201}, four additional time units per attempt, and fresh independent retries. A unit is the stipulated duration of one classical check; no processor speed is asserted. Both methods are judged by a verified output. These are illustrative expectations, not hardware measurements or a comparison with every possible structured classical solver.

Both quantum rows make the same number of oracle calls per run. One wins this timing comparison; the other loses it. For this chosen 201-round policy, the crossover is about 163 classical-check time units per complete quantum round. Optimizing the run length before restarting can change that crossover. The relevant question is not whether 201 is smaller than 32,768.5. It is whether the implemented work corresponding to those numbers is smaller.

Stored data introduce another cost. A coherent lookup must correlate an address with the requested data while preserving superpositions, then remove temporary data when the phase is formed. Preparing or maintaining that access is not accomplished by putting the label ‘memory’ on a wire. Conversely, a rule that computes its answer directly need not use a large stored database at all. Classical indexing, parallel workers, reusable preprocessing, and the quantum memory arrangement belong in the same comparison.[14]

Where the idea meets current research

A laboratory can test the state transformation without claiming an end-to-end search advantage. In 2017, Figgatt and colleagues used trapped-ion qubits to run searches for every one-answer and every two-answer choice in an eight-label space: eight instances of the first kind and twenty-eight of the second. Testing all those instances probed more than a single favorable marking. The reported output populations were estimated from repeated fluorescence measurements and corrected for preparation and readout errors. Their success statistic concerns the returned label, not every phase of the final quantum state. It did not establish that the apparatus located an unknown database entry faster than the best classical machine.[15]

Recent work asks harder implementation questions. In a paper published in 2026, Martijn Brehm and Jordi Weggemans compared fault-tolerant quantum search estimates with a strong classical solver on Boolean constraint problems whose structure could be varied. Their conclusions depended strongly on that structure and on whether the resource model counted total costly gates or permitted extensive parallel execution. Many projected advantages disappeared in the studied regimes. The comparison combined measured classical runtimes with quantum circuit-cost bounds and fits across sampled instance sizes. These were model-based resource comparisons, not timings from an operational large quantum search engine.[16]

A different use of the same reflections appeared in a 2025 proposal by Omar Nagib, Mark Saffman, and Klaus Mølmer. They used Grover-style amplification to design preparation protocols for selected collective entangled atomic states, with conditional phase changes supplied by photon scattering from an optical cavity. This was a theoretical protocol and error analysis, not a demonstration of database search. Its nonideal model tracks the

fidelity of the prepared state separately from the probability that a heralded preparation succeeds. Its point here is constructive: an algorithm first introduced to find a label can also help prepare matter in a chosen quantum state.[17]

The distinction between a paper's proposal, its calculation, and its experimental result matters just as much as the distinction between a gate and an oracle. A resource estimate can identify a promising regime or a prohibitive cost. An experiment can test an implementation. Neither should be silently promoted into the other.

What has actually been amplified?

No probability has been created. Normalization has held at every step. No new information has been printed beside every candidate. The coherent checker supplied signs; a fixed reflection made them interfere; repeated reflections redirected amplitude toward the acceptable subspace. Measurement turned that prepared distribution into one label.

The entire pure state can have zero von Neumann entropy throughout this calculation, while the entropy of its eventual label distribution changes. The amplification is therefore neither an injection of uncertainty nor a thermodynamic cooling law. It is a controlled change in which outcomes a specified measurement is likely to reveal.

It also does not reduce generic exhaustive search to an easy task. With $N = 2^n$, square-root search still takes order $2^{n/2}$ queries. Halving an exponent can be enormously useful, but an exponential remains an exponential. More powerful improvements must exploit more specific structure.

Return to the checker that offered judgments but no suggestions. We have made its judgments useful in a new way, without supplying the answer beforehand. The achievement is exact in the four-candidate model, scalable as a query improvement, and conditional as a practical advantage. Its mechanism is a pattern of signs and reflections that we can calculate.

The next algorithm will look for a different kind of structure: a repetition hidden in arithmetic. Instead of steering probability toward a marked label, we will arrange interference so that a period becomes visible. The same rules will make a different question answerable.

Notes and sources

1. Lov K. Grover, ‘A fast quantum mechanical algorithm for database search,’ Proceedings of the 28th Annual ACM Symposium on Theory of Computing, 212–219 (1996), doi:10.1145/237814.237866. The sign inversion and inversion-about-average mechanism follows this construction. The four-candidate instance and all figures are original explanatory calculations, not experimental data. Original paper.

2. Richard Cleve, Artur Ekert, Chiara Macchiavello, and Michele Mosca, ‘Quantum Algorithms Revisited,’ Proceedings of the Royal Society A 454, 339–354 (1998), doi:10.1098/rspa.1998.0164, on controlled eigenvalues and phase kickback. Equation (14.1) assumes a fixed coherent reversible evaluation, no extra input-dependent phases, and no retained distinguishing workspace. One query is not one physical gate. Author text.

3. The reflection argument is a direct derivation: the projector $P=|s\rangle\langle s|$ satisfies $P^2=P$, so $D^2=(2P-I)^2=I$ and D is Hermitian. Its action preserves the uniform component and reverses the orthogonal subspace. Grover's original construction supplies the historical context; no nonlinear mean-estimation operation is assumed. Grover's construction.

4. Adriano Barenco and colleagues, ‘Elementary gates for quantum computation,’ Physical Review A 52, 3457–3467 (1995), doi:10.1103/PhysRevA.52.3457, on constructing controlled gates. For n qubits, conjugate the sign reflection about the all-zero state by a Hadamard on each qubit. In the two-qubit case the middle operation is exactly a Z on each qubit followed by controlled Z. The stated seven-gate implementation of D counts four H gates, two Z gates and one CZ. It is not a general native-hardware count. Author manuscript.

5. Michel Boyer, Gilles Brassard, Peter Høyer, and Alain Tapp, ‘Tight bounds on quantum searching,’ Fortschritte der Physik 46, 493–505 (1998), arXiv:quant-ph/9605034, §§2–3. The good/bad plane is invariant for uniform initialization and an exact binary phase oracle. Figure 14.2 fixes the first-round example at N=4, M=1. The plane is not an extra two-level physical register. Full author manuscript.

6. Gilles Brassard, Peter Høyer, Michele Mosca, and Alain Tapp, ‘Quantum Amplitude Amplification and Estimation,’ Contemporary Mathematics 305, 53–74 (2002), doi:10.1090/conm/305/05215. Equation (14.6) also follows by multiplying two 2-by-2 reflection matrices. The square-root approximation assumes M/N is small; zero and unit success fractions are separate endpoints. The plotted probabilities use Eq. (14.6) directly, without the small-angle approximation. Full author manuscript.

7. Boyer and colleagues, §4, analyze unknown solution counts and a randomized schedule. The expected square-root finding bound presumes at least one solution; deciding that none exist requires a bounded-error stopping protocol. The N=64 values and Figure 14.3 use the exact formula, not empirical frequencies. An unread complete candidate-label measurement produces dephasing; for the uniform four-label preparation the resulting state is $I_4/4$. Unknown-count analysis. In the author version quant-ph/9605034v1, §4, Lemma 2 and Theorem 3, the randomized-length average is bounded below after a sufficient window size. Its table-lookup/uncomputation convention is not silently equated with this chapter's one-call phase kickback from a supplied Boolean evaluation. Verification adds a value call. A finite stopping rule must specify an attempt limit and bound the probability of missing an existing solution. When no solution exists, every verification fails; a failure to find one within that limit must be reported with the declared error guarantee.

8. Theodore J. Yoder, Guang Hao Low, and Isaac L. Chuang, ‘Fixed-Point Quantum Search with an Optimal Number of Queries,’ Physical Review Letters 113, 210501 (2014), doi:10.1103/PhysRevLett.113.210501. Its error-controlled fixed-point guarantee uses different phase sequences and assumptions about initial overlap. The text does not identify it with unlimited repetition of the basic π-phase Grover round. Author manuscript. In arXiv:1409.3305v2, Eqs. (1)–(3) give the error envelope and initial-overlap range; Eqs. (6)–(11) define the phase-dependent sequence. The guarantee is not a claim of monotonic improvement after each gate. Its query convention counts two Boolean-oracle calls per generalized iterate, unlike the basic pi-phase round supplied here.

9. Figure 14.4 and Eq. (14.7) are an original reversible checker. The five forward gates are X(u), CNOT(a,u), CNOT(b,u), CNOT(b,v), CNOT(c,v); then CZ(u,v); then the five forward gates in reverse order. Clean inputs acquire phase −1 precisely for 001 and 110. For a superposition with both work qubits initially zero, linearity combines these basis-input actions. The inverse gates return both work qubits to zero while retaining the

conditional phase. Charles H. Bennett, ‘Logical Reversibility of Computation,’ IBM Journal of Research and Development 17, 525–532 (1973), provides the general uncomputation framework. Bennett's paper.

10. Brassard and colleagues, ‘Quantum Amplitude Amplification and Estimation,’ especially §2. Equation (14.8) assumes access to a unitary preparation and its inverse, a good-subspace phase operation, and clean or fully included work registers. The amplification is relative to that preparation's success probability. It does not assume that an arbitrary irreversible heuristic can be inverted at its original classical cost. Amplitude amplification.

11. The classical (q+1)/N bound is an elementary counting argument for one uniformly hidden solution, ordinary value queries, and an allowed final unqueried guess. Repeated queries reveal nothing new. An adversarial hidden label gives the corresponding worst-case obstruction. The positive-confirmation policy in Table 14.1 checks until it encounters the successful label; its expected number of checks is (N+1)/2. These are declared access and output conventions, not universal costs for every structured search.

12. Charles H. Bennett, Ethan Bernstein, Gilles Brassard, and Umesh Vazirani, ‘Strengths and Weaknesses of Quantum Computing,’ SIAM Journal on Computing 26, 1510–1523 (1997), doi:10.1137/S0097539796300933; Christof Zalka, ‘Grover's quantum searching algorithm is optimal,’ Physical Review A 60, 2746–2751 (1999), doi:10.1103/PhysRevA.60.2746. Their lower bounds concern the specified black-box access model. They are not lower bounds for every explicitly represented optimization or satisfiability problem. BBBV author manuscript. Zalka author manuscript.

13. Equation (14.9) follows from a geometric number of independent attempts, each with fixed cost and positive success probability. Chris Cade, Marten Folkertsma, Ido Niesen, and Jordi Weggemans, ‘Quantifying Grover speed-ups beyond asymptotic analysis,’ Quantum 7, 1133 (2023), doi:10.22331/q-2023-10-10-1133, develops explicit instance-sensitive cost comparisons. Table 14.1 is our stipulated timing model, not a result quoted from that paper. Reliability, repeated-run independence, setup reuse, and verification costs are explicit assumptions. Published paper.

14. Vittorio Giovannetti, Seth Lloyd, and Lorenzo Maccone, ‘Quantum Random Access Memory,’ Physical Review Letters 100, 160501 (2008), doi:10.1103/PhysRevLett.100.160501. Coherent address access is a physical resource with an architecture, not an automatic property of a classical data file. No universal loading-time bound or universal impossibility of useful database search is asserted here. Parallelization and the availability of an explicit problem description also change the comparison; see Zalka and the resource studies cited above. Author manuscript.

15. C. Figgatt, D. Maslov, K. A. Landsman, N. M. Linke, S. Debnath, and C. Monroe, ‘Complete 3-Qubit Grover search on a programmable quantum computer,’ Nature Communications 8, 1918 (2017), doi:10.1038/s41467-017-01904-7. The trapped-ion study runs all eight single-result and all twenty-eight two-result oracles with one search iteration. Its small-instance experimental tests are distinct from an end-to-end elapsed-time advantage. The reported results are described in the open-access article. Published full text. Published Results, “Data,” and Figs. 3–4 distinguish algorithm success probability from squared statistical overlap of measured and ideal populations. All displayed data are corrected for state-preparation/readout errors, including multi-ion crosstalk; the stated uncertainties are statistical. A population overlap is not a fidelity test of all quantum phases. The population statistics quoted here are the authors' corrected experimental estimates.

16. Martijn Brehm and Jordi Weggemans, ‘Assessing fault-tolerant quantum advantage for k-SAT with structure,’ Quantum 10, 1975 (2026), doi:10.22331/q-2026-01-20-1975; published 20 January 2026. The study uses classical benchmarks and fault-tolerant resource estimates, comparing gate-count and gate-depth models on specified structured random instances. Its regime-dependent findings are not a theorem that no quantum search can be useful. The relevant resource model and comparisons appear in §§2.6, 3 and 4.3. The benchmark uses generated, tunably structured instances, measured CaDiCaL runtimes, quantum resource bounds, and fits to medians; it does not certify precise future runtime on every industrial instance. Error-reduction repetitions are serialized in this study, even though the gate-depth model permits parallel gates within a run. Published article. Full author text.

17. Omar Nagib, M. Saffman, and K. Mølmer, ‘Efficient Preparation of Entangled States in Cavity QED with Grover's Algorithm,’ Physical Review Letters 135, 050601 (2025), doi:10.1103/3fzf-wsr2; published 28 July

2025. The work proposes atom–cavity protocols and analyzes their imperfections; it is not a new experimental search benchmark. The discussion uses author version arXiv:2501.18881v3 and the publisher's bibliographic record. The publisher records an 11 March 2026 correction to a reference; none of the present chapter's numerical results is taken from that reference. Published record. Full author text. The author-version discussion around Fig. 4 treats heralded fidelity and success probability separately. Its illustrative parameters omit internal and mirror losses, so the plotted performance is conditional on those omissions. The present chapter uses the protocol to illustrate state preparation and does not adopt that performance as a device prediction.

CHAPTER 15

The Hidden Rhythms of Arithmetic

A number can conceal a repetition. Multiply by the same amount again and again, keeping only the remainder after division, and an apparently wandering sequence can return to its beginning. That return can reveal factors of the number used as the divisor.

The difficult part is finding the repetition without waiting through the entire sequence. The period may be enormous even when the rule generating it is short. Peter Shor showed how a quantum computer could turn this arithmetic structure into an interference pattern, then use an ordinary calculation to recover factors.[1]

We will follow the whole chain for twenty-one. Its factors are easy enough to know in advance, which lets us check the result. But the procedure will be built from twenty-one and a chosen multiplier—not from the factors or the period we hope to discover. The example must explain a method, not conceal an answer inside a circuit.

The search in Chapter 14 amplified an acceptable candidate. Here there is no marker attached to the correct factor. Instead, many coherent contributions will reinforce at outputs related to a repetition. A measurement will supply one such output. It will not print the period or the factors directly.

Arithmetic that comes back

To calculate modulo twenty-one, divide by twenty-one and keep the remainder. Thus thirty-two becomes eleven, because $32 = 21 + 11$. We write $32 \equiv 11 \pmod{21}$; the three-line equality means that the two numbers have the same remainder. It does not mean they are equal as ordinary integers.

Now start with one and repeatedly multiply by two, reducing after every multiplication. The values stay small even though the unreduced powers double. Table 15.1 shows two complete cycles.

Table 15.1. **Powers can repeat even while their exponents increase.**

Exponent x	Remainder $2^x \bmod 21$
$0, 6, 12$	1
$1, 7$	2
$2, 8$	4
$3, 9$	8
$4, 10$	16
$5, 11$	11

Read exponents in numerical order to obtain $1, 2, 4, 8, 16, 11, 1, \ldots$. Grouping equal remainders exposes their spacing. These are exact arithmetic values, not measurement outcomes or a table supplied to the quantum circuit.

The first positive exponent that returns one is six: $2^6 = 64$, whose remainder is one. The sequence then repeats because multiplying by two follows the same rule from the same starting value. This smallest return exponent is called the order of two modulo twenty-one. We will call it r.

More generally, choose an integer a between two and $N - 1$, where N is the number we want to factor. First calculate their greatest common divisor, written $\gcd(a, N)$. If it is greater than one, it already supplies a factor. If it is one, a and N are coprime: they share no divisor greater than one. Multiplication by a is then reversible among the remainders coprime to N. A finite reversible sequence must return, and its order satisfies

$$a^r \equiv 1 \pmod{N}, r = \text{smallest positive return exponent}. \tag{15.1}$$

Calculating the sequence until it returns would find r. That is not the promised quantum method. A large order could require a prohibitive number of such steps. First we should understand why the order would help at all.

How a period becomes a factor

Suppose the order is even. Calculate $u = a^{r/2} \bmod N$. Its square has remainder one, so $u^2 - 1$ is divisible by N. Ordinary algebra separates that difference of squares:

$$(u - 1)(u + 1) \equiv 0 \pmod{N}. \tag{15.2}$$

When u is neither one nor minus one modulo N, the two neighboring numbers can expose nontrivial divisors through $\gcd(u - 1, N)$ and $\gcd(u + 1, N)$. When N is composite, a product can be divisible by N without either factor being divisible by all of N.

In fact, under these conditions both greatest common divisors are nontrivial. If one neighbor were coprime to the modulus, we could multiply by its modular inverse in Eq. (15.2); the other neighbor would then be divisible by the whole modulus. That would

make the halfway value one or minus one, contrary to our condition. Neither neighbor is itself divisible by the whole modulus either.

For our example, $r = 6$ gives $u = 2^3 = 8$. The two greatest common divisors are $\gcd(7, 21) = 7$ and $\gcd(9, 21) = 3$. The latter needs only successive divisions: twenty-one leaves remainder three when divided by nine, and nine divides exactly by three. We have recovered $21 = 3 \times 7$.[2]

Not every choice of a works this way. For twenty-one, choosing four gives order three, which is odd. Choosing twenty gives order two, but $u = 20$ is minus one modulo twenty-one; the greatest common divisors are then one and twenty-one, which teach us nothing. The procedure must recognize these outcomes and choose again. Standard classical checks also handle even numbers, primes, and powers of a single prime before the general quantum step.

After that preprocessing, at least half of uniformly selected coprime bases have an even order with a useful halfway value. This is a number-theoretic guarantee for an odd composite with at least two distinct prime factors, not the probability of recovering the order from one quantum measurement. A nontrivial divisor may itself be composite; apply the procedure again when a complete prime factorization is required.

The reduction has a clear boundary. Finding the order is the unfamiliar task; checking a proposed return and extracting greatest common divisors are ordinary arithmetic. A correct final factor can be verified by division without trusting the device that suggested it.

Build the repetition without knowing its length

Use one register for an exponent x and another for its modular power. Give the exponent register nine qubits, so it has $Q = 512$ labels, from zero through 511. This choice is made using N alone: $N^2 \le Q < 2N^2$. This exponent range provides the precision needed to recover a small-denominator fraction from the readout, as we will see below. Five qubits suffice for the value register, since the possible remainders are zero through twenty. Working qubits needed by the arithmetic are additional.

Prepare the exponent register in all zeros and apply a Hadamard to each qubit. Prepare the value register in the integer one. Then perform a coherent modular exponentiation. With clean working registers left implicit, the result is

$$|\Psi\rangle = \frac{1}{\sqrt{Q}} \sum_{x=0}^{Q-1} |x\rangle_X |a^x \bmod N\rangle_V . \tag{15.3}$$

Each exponent is now correlated with the corresponding remainder. This does not create a readable table of all those values. Measuring both registers now would produce one pair of labels. The purpose of the preparation is the interference we have not yet performed.

The arithmetic block also needs a construction. Write the exponent in binary as $x = \sum_j 2^j x_j$, where x_j is its digit of weight 2^j. Precompute the constants $a^{2^j} \bmod N$ by repeated squaring. Then, for every exponent bit equal to one, multiply the value register by its corresponding constant, modulo N. These are controlled reversible operations, acting coherently when the control is in a superposition.[3]

For $a = 2$ and $N = 21$, the first constants are $2, 4, 16, 4$. To evaluate exponent thirteen, whose binary form is 1101, the active weights are one, four, and eight. Their constants give $2 \times 16 \times 4 = 128$, leaving remainder two. We never needed to write out the unreduced power 2^{13}, and the same method works for much longer exponents.

Each multiplication is a permutation of valid remainder labels because its constant is coprime to N. It can be extended reversibly to the unused register labels as well. Carries and other temporary records must be cleared coherently. Nine controlled multiplication blocks produce Eq. (15.3) in our example; each block may contain many gates. Neither the factors nor r were needed to choose those blocks.

The circuit is specified by a and N—not by the period

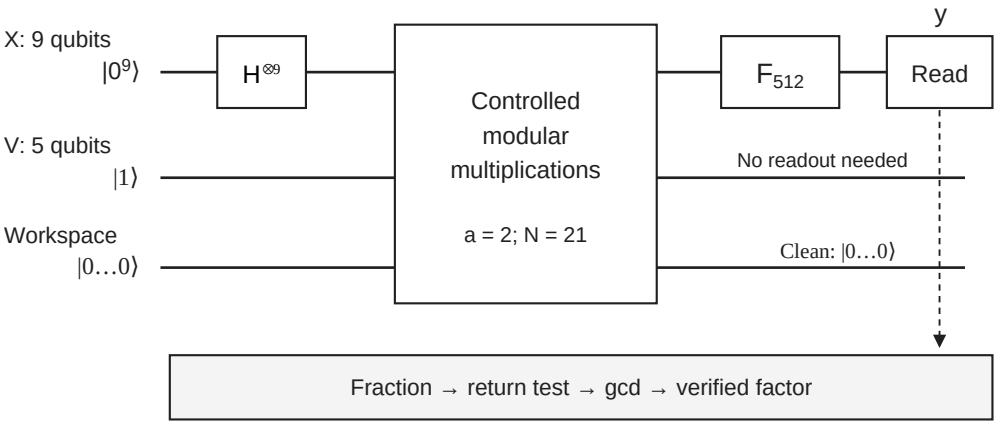


Figure 15.1. Two data registers connect modular arithmetic to a classical answer. The exponent register starts in $|0^9\rangle$ and receives nine Hadamards. The value register starts in the integer state $|1\rangle$, not five ones. Modular exponentiation is compiled from $a = 2$ and $N = 21$ by repeated squaring, with arithmetic workspace restored to zero. The Fourier transform acts only on the exponent register; its nine-bit readout is processed classically. The value register need not be measured. Bundled wires show registers and operation order, not single-qubit gates or equal durations.

A record that preserves the right alternatives

To see the structure of Eq. (15.3), imagine measuring the value register. Suppose its result is one. Only exponents that produced that remainder remain in the conditional exponent state:

$$|\psi_0\rangle = \frac{|0\rangle + |6\rangle + |12\rangle + \cdots + |510\rangle}{\sqrt{86}}. \tag{15.4}$$

The allowed labels form a comb with spacing six. Another remainder gives an offset comb: the result two selects $1, 7, 13, \ldots, 511$, for example. We do not retain only the result one and throw the others away. Every remainder supplies a comb with the same spacing. Their finite lengths differ slightly because 512 is not divisible by six.

Why exactly this spacing? Equal remainders mean that the larger power divided by the smaller has remainder one; division is permitted because the base is coprime to the modulus. The exponent difference must therefore be a multiple of the order. Otherwise dividing that difference by the order would leave a smaller positive return, contradicting its definition. Conversely, every multiple of the order returns the same remainder.

The value measurement is an explanatory option, not a required step. Leaving that register unread produces the same exponent statistics as averaging all its conditional outcomes with their probabilities. What remains coherent within each group is exactly what we need: different exponents that produced the same remainder. A record of the remainder does not distinguish them.[4]

This is subtler than saying that every record destroys the algorithm. An extra uncontrolled record of the exact exponent would distinguish all the teeth and destroy their interference. Conversely, reversing the entire exponentiation before using its correlations would restore the initial uniform state and remove the useful pattern. We clear the temporary arithmetic workspace; we keep the specified input–output correlation.

Ask the amplitudes for a rhythm

The quantum Fourier transform combines an input pattern with a family of phase patterns. Each output label y assigns a different phase increment as the input label x increases. The contributions to that one output are added before their magnitude is squared.

For Q labels, choose the convention

$$F_Q|x\rangle = \frac{1}{\sqrt{Q}} \sum_{y=0}^{Q-1} e^{2\pi i x y/Q} |y\rangle. \tag{15.5}$$

The exponential is the turning unit arrow from Chapter 3. For a fixed y, increasing x by one advances its angle by $2\pi y/Q$. We may call y a frequency label, but no particle must oscillate in space at that frequency. The pattern is indexed by integers in a register.[5] The phase patterns for different input labels have zero overlap: their relative arrows step evenly around a complete circle and sum to zero. Each pattern is also normalized. Thus the transform maps one orthonormal basis to another, preserving lengths and overlaps. It redistributes probability without reading the incoming amplitude list or merging distinguishable inputs.

A smaller calculation makes the rule visible. Temporarily use eight labels and the state $(|0\rangle+|2\rangle+|4\rangle+|6\rangle)/2$. This is a Fourier example with spacing two, not a smaller implementation of the twenty-one calculation. Four equal contributions reach each output. At $y=0$, their phase factors are all one. At $y=1$, they are $1, i, -1, -i$, which cancel. At $y=4$, they are all one again, because every two-label step advances a full phase cycle.

$$F_8\frac{|0\rangle+|2\rangle+|4\rangle+|6\rangle}{2}=\frac{|0\rangle+|4\rangle}{\sqrt{2}}. \tag{15.6}$$

The remaining outputs cancel too. Figure 15.2 shows the addition for three representative labels. We did not measure the four incoming amplitudes. One fixed unitary operation has redirected where they reinforce.

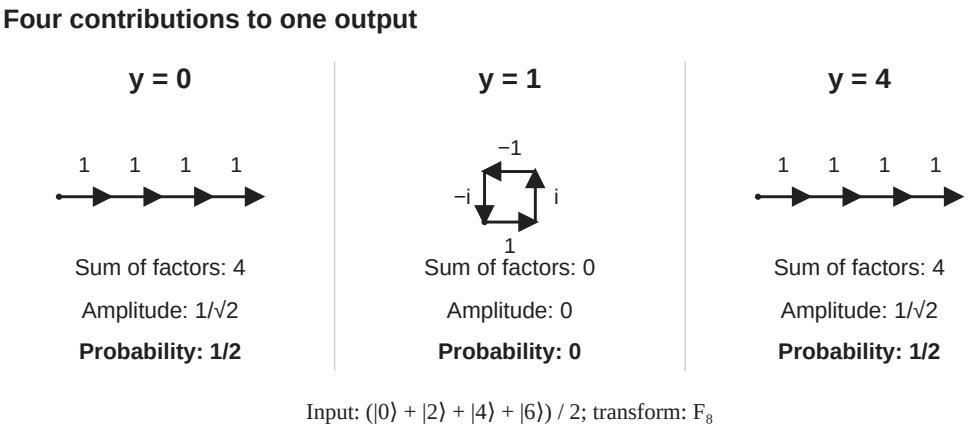


Figure 15.2. Repeated spacing selects reinforcing phase patterns. For the eight-label input in Eq. (15.6), the contributions to any one output have equal magnitude $1/(2\sqrt{8})$. The diagrams show their phase factors at $y=0,1,4$, drawn head to tail at a common scale. Four aligned factors give output amplitude $1/\sqrt{2}$; the quarter-turn sequence closes and gives zero. All unshown output amplitudes are also zero. Arrow directions represent complex phases, not particle paths. This separate toy calculation is not the $N=21$, $Q=512$ factorization.

For a general comb beginning at x_0, with L teeth separated by r, Eq. (15.5) gives the output amplitude

$$c_y = \frac{e^{2\pi i x_0 y/Q}}{\sqrt{LQ}} \sum_{j=0}^{L-1} e^{2\pi i j r y/Q}. \tag{15.7}$$

The leading factor turns all the contributions to this output together and does not affect its probability. Successive terms in the sum differ in phase by $2\pi r\, y/Q$. They reinforce when $r\, y/Q$ is close to an integer. Thus the likely outcomes lie near

$$\frac{y}{Q} \approx \frac{k}{r}, k = 0, 1, \ldots, r-1. \tag{15.8}$$

The unknown spacing has become a denominator in a fraction. The transform did not need that spacing as an input: F_Q is fixed once we choose the register size. The symbol r enters our analysis of the result, not the instructions used to build the transform.

When the peaks do not land on integers

If r divides Q exactly, a complete comb gives r equally probable outputs separated by Q/r, with zero probability elsewhere. Our example deliberately avoids that convenience. Six does not divide 512. Its six main peaks lie near $0, 85, 171, 256, 341, 427$, and probability also spreads into neighboring labels. This is a finite-register effect present even with perfect gates, not necessarily experimental noise.[6]

A plot like Figure 15.3 could be estimated by repeating the experiment many times. But the factoring procedure need not first reconstruct a smooth spectrum or identify all six peaks. It processes each measured integer. One favorable output can already be useful; some very probable outputs are not.

Suppose the nine-bit readout is $y = 85$. The ratio $85/512$ is about 0.166016, close to one-sixth. Its binary precision was fixed by the size of the register. We must now recover a plausible small-denominator fraction without trying every possible period.

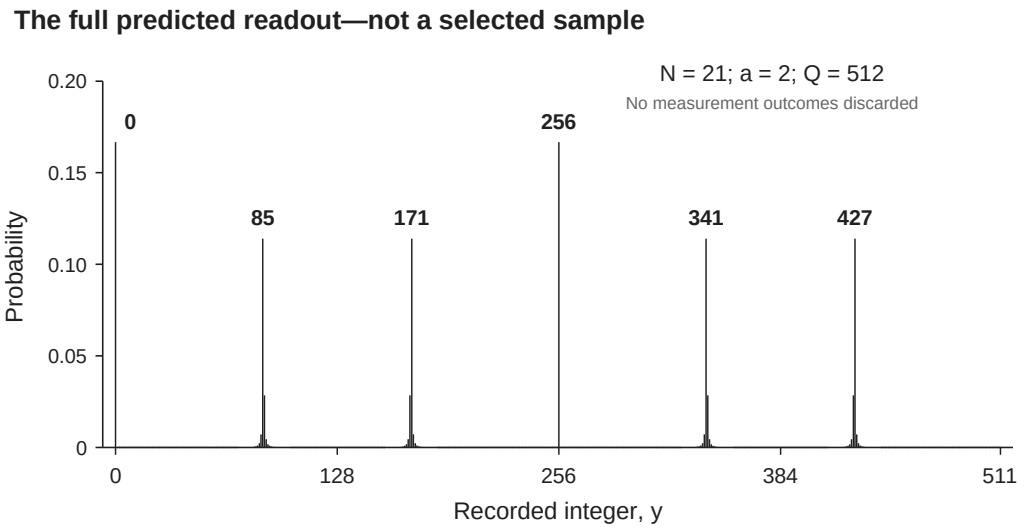


Figure 15.3. A finite register produces peaks with neighboring weight. Every vertical stem is the theoretical probability for one of the 512 exponent outcomes after F_{512}, with $a = 2$ and $N = 21$. The calculation includes all value-register outcomes with their probabilities; there is no selection of a favorable remainder. The six labeled outputs together carry about 78.93 percent of the total probability. Labels zero and 256 each have probability about 0.166672; the other four labeled peaks each have about 0.113989. The remaining probability is retained in the smaller stems. One run yields one label, not this entire plot.

A fraction becomes a tested answer

Continued fractions do this using repeated division. Divide 512 by 85: the quotient is six and the remainder two. Divide 85 by two: the quotient is forty-two and the remainder one. One more division finishes. These steps give

$$\frac{85}{512} = \frac{1}{6 + \dfrac{1}{42 + \dfrac{1}{2}}}. \tag{15.9}$$

Stopping after the first six suggests $1/6$. Taking the next stage gives $42/253$; the complete expression returns $85/512$. These successive approximations are called convergents. We seek denominators below N, since the order is smaller than N. Six is the nontrivial candidate in this example; 253 and 512 are too large.[7]

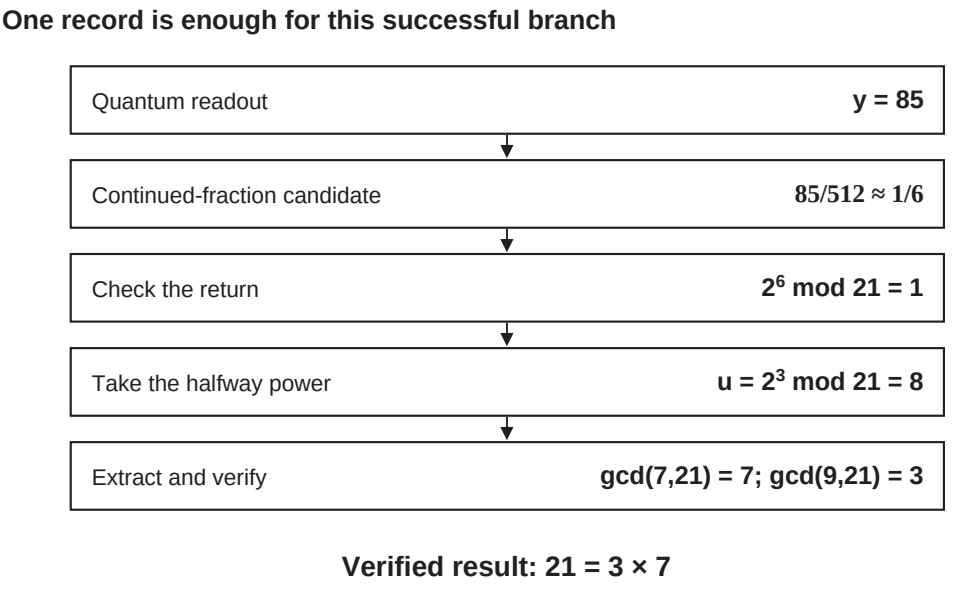


Figure 15.4. One usable record, followed by ordinary arithmetic. The stipulated output $y = 85$ leads to the convergent $1/6$. The return test accepts six, its halfway modular power is eight, and the two greatest common divisors yield seven and three. The result is verified by multiplication. A different output may fail the return or factor tests and require another sample or additional classical processing. No period or factor is supplied to the quantum circuit. Boxes show a worked inference, not a claim that every run follows this successful branch.

The choice of register size makes this more than a lucky guess. For an outcome nearest to Qk/r, the error in y/Q is at most $1/(2Q)$. Because $Q \geq N^2$ and $r < N$, that is sufficiently small for the reduced fraction k/r to appear among the continued-fraction convergents. This statement applies to those favorable outcomes—not to every label in the distribution. Adding one exponent qubit doubles the label range and halves the spacing between neighboring readout fractions. The arithmetic order stays fixed, and some outcomes remain unhelpful; the larger register's implementation costs must also be counted.

Test the candidate six by calculating $2^6 \bmod 21 = 1$. This establishes a return. In general, that test alone does not prove the return is the smallest one: a multiple of the true order also passes. The factor step still has to be checked. Here six is even, the halfway value is eight, and the greatest common divisors give seven and three. Their product confirms the answer. Figure 15.4 keeps the quantum sample and the classical conclusion separate.

Why not use the strongest peak? The output zero is compatible with $k = 0$ for every order and reveals no denominator. The output 256 gives exactly one-half. It corresponds here to $k = 3$ and $r = 6$, but reducing the fraction has removed a factor of three from its denominator. Table 15.2 shows the difference between a strong spectral peak and an immediately useful period candidate.

Table 15.2**. Likely records do not all reveal the full order.**

Readout y	Nearby reduced fraction	Denominator d	$2^d \bmod 21$
0	0 / 1	1	2
85	1 / 6	6	1
171	1 / 3	3	8
256	1 / 2	2	4
341	2 / 3	3	8
427	5 / 6	6	1

The six rows describe prominent possible outcomes of the same ideal run, not six consecutive required measurements. A remainder of one passes the displayed return test. Failed reduced denominators may still support more extensive classical post-processing. These rows omit the smaller spectral peaks only for this comparison; Figure 15.3 includes all outcomes.

For a deliberately basic rule, examine every positive even continued-fraction denominator below twenty-one. Require its modular power to return one, reject a halfway value of one or minus one, and retain a greatest common divisor only after verifying that it is a nontrivial divisor. Across all 512 outputs, this succeeds with probability about 32.84 percent in our ideal example. That is a calculation using the complete distribution, including unhelpful records—not a selected set of successful runs.[7]

An unhelpful readout can be followed by a fresh run at the same suitable base. For this ideal base-two example, twelve independent repetitions of the stated basic rule give more than 99 percent probability of a verified factor. But repeatedly using a base whose true order is odd, or whose halfway value is minus one, does not repair that arithmetic failure; choose another base. More capable classical processing can combine partial denominators or test a controlled range of alternatives. Ekerå's analysis, published in 2024, proves high single-run order-recovery probabilities when two limited classical searches are included. Its success guarantee concerns that enlarged procedure, not simply reading the denominator of the first fraction we notice.[8]

The quantum part is therefore a sampler feeding a classical inference. Failure to extract a factor from one sample does not refute the predicted distribution. Nor should a plausible-looking denominator be accepted without the arithmetic tests.

Check a less helpful outcome

Suppose the twenty-one experiment returns 256 instead of 85. Has the circuit failed? Would measuring the value register, or keeping a record of the exact exponent, change that conclusion?

The record 256 is a prominent outcome of the ideal distribution. Its fraction is one-half, whose reduced denominator is two. Yet two squared has remainder four modulo twenty-one, so this denominator fails the return test. The underlying order is six: reduction of the

fraction has hidden a factor of three. The basic procedure may start a fresh run; more extensive classical processing can use such partial information. An unhelpful record is therefore compatible with a correctly functioning circuit.

Measuring the value register and averaging every result leaves the same exponent statistics. Exponents with the same remainder remain coherent within each group. An unread perfect record of the exact exponent would instead distinguish every term, remove those coherences, and make the subsequent Fourier readout uniform. The useful question is which alternatives the record distinguishes. Assessing the implementation requires statistics from fresh trials, not rejecting it because one allowed outcome failed to yield factors.

The general guarantee also accounts for the chance of a useful record. Sufficiently accurate Fourier outcomes occur often enough, and a sufficient fraction preserve the order when their fractions are reduced, that independent attempts with suitable random bases need grow only polynomially with the input length. One can set a finite attempt budget for a chosen failure probability. Every attempt pays for a new preparation, quantum operations, readout, and classical checks; reaching the budget without a factor is a reported failure, not a proof of primality.

A short circuit for a long list of labels

Equation (15.5) contains a sum over Q outputs for every input. That is a large mathematical description. It need not be a circuit with Q successive steps. For $Q = 2^m$, the usual exact Fourier construction uses m Hadamards, $m(m-1)/2$ controlled phase rotations, and a final reversal of bit order. The rotations progressively assemble the binary fractions that specify the phases.[9]

For an eight-label transform, write an output as $y = 4\,y_2 + 2\,y_1 + y_0$. Its phase factor separates into $e^{\pi i x\, y_2} e^{\frac{\pi i x}{2} y_1} e^{\frac{\pi i x}{4} y_0}$. Each output bit contributes its factor when it is one, and contributes one otherwise. Hadamards and controlled phase rotations assemble these binary fractions rather than visiting every output in sequence. This factorization describes a basis input; a superposition of inputs need not leave the output qubits in a product state.

For nine qubits, this is nine Hadamards and thirty-six controlled rotations. Four swaps reverse the output order if we implement that reversal physically; when reading immediately, we can instead relabel the recorded bits. These are logical operations in a chosen vocabulary, not a count of native pulses. Small-angle rotations must be synthesized with a suitable accuracy, or omitted under a justified approximate-transform error bound.[10]

If the Fourier transform is followed immediately by readout, another implementation can replace its controlled rotations by measurements and outcome-dependent one-qubit rotations. Griffiths and Niu showed that this reproduces the required output statistics. It is

not permission to measure the exponent register at any earlier point: the measurement order and subsequent corrections are part of the construction.[11]

The modular arithmetic is the heavier part of straightforward factoring circuits. Let n be the bit length of N. Our exponent register has about $2n$ qubits. Schoolbook reversible constructions use order n controlled modular multiplications, each assembled from order n^2 elementary logical operations, giving an order-n^3 baseline. This is a baseline for one coherent modular-exponentiation attempt, before retries, rotation synthesis, and error protection. Faster arithmetic and different memory tradeoffs change that count. Neither a multiplication block nor its cleanup is free.[12]

The important scale is the input length n, not the value of N, which grows exponentially with n. Shor's algorithm performs factoring with polynomial quantum and classical resources under its computational assumptions. It does not follow that a polynomial classical factoring algorithm is impossible. No such general classical algorithm is known. The usual large-input comparison is with general-purpose number-field-sieve methods and their subexponential heuristic cost estimates—not a proved lower bound ruling out every possible classical discovery.[13]

There is also no conflict with the search lower bound in Chapter 14. The new program has exploitable arithmetic structure, including efficient controlled powers. It is not a black-box search over prospective factors. And a compact Fourier circuit does not hand us an exponentially long classical list of Fourier coefficients: preparation and extraction of the requested output remain separate tasks.

What a demonstration—and an estimate—can claim

A small example must preserve the distinction between knowing an instance and using its solution. Constants calculated from a and N are legitimate inputs to a compiler. Replacing the modular arithmetic by a short circuit that already assumes its period is a different demonstration. Smolin, Smith, and Vargo made this problem explicit in their analysis of oversimplified factoring experiments: the size of a displayed integer alone does not establish the scale of the computation performed.[14]

The calculations in this chapter simulate specified small quantum transformations and states. They are not a hardware factoring record. At larger scales, error protection, arithmetic workspace, gate synthesis, and the speed of control decisions separate logical qubits from a complete physical machine.

Craig Gidney and Martin Ekerå's 2021 analysis gives a concrete example: about twenty million noisy physical qubits and roughly eight hours on average to factor a 2,048-bit RSA integer. The model assumes a planar grid with nearest-neighbor connections, a 0.1 percent physical gate-error rate, one-microsecond error-correction cycles, and ten-

microsecond control reactions. The estimate accounts for error protection, routing, auxiliary states, and repeated attempts. These are estimates for a specified design, not a performed computation, a minimum requirement for every design, or a forecast of a machine's arrival. Gidney's 2025 preprint develops different tradeoffs among arithmetic, storage, and auxiliary-state production.[15]

The logical constructions continue to change too. In a preprint revised in September 2026, Kahanamoku-Meyer and colleagues develop shallow Fourier circuits using a carefully qualified approximation guarantee. Their construction without extra working qubits does not bound the error equally on every possible input; a related construction supplies an all-input guarantee with different resource requirements. The distinction matters inside a larger algorithm: a cheap subroutine is useful only if its error guarantee covers the states that algorithm actually supplies.[16]

These advances refine an account that our small example can already make explicit: which arithmetic is executed, which phase relationships are preserved, which record is obtained, and how much further work is needed to turn it into a verified result.

The rhythm, not a list of answers

We began with an arithmetic sequence returning to one. Reversible modular powers correlated exponents that shared a remainder. Fourier interference favored fractions related to their spacing. One measured integer led, through continued fractions and greatest common divisors, to two verified factors.

No operation read every exponent's answer. No detector announced the factors without interpretation. The program arranged a probability distribution whose samples expose a specific structure, and classical arithmetic completed the task. The role of measurement was limited, but sufficient.

The three programs in this part now ask different things of interference: compare two function values, amplify an acceptable subspace, and reveal a period. Their successes are not interchangeable. To understand the power of quantum computation, we must ask which structure a problem offers and which resources its algorithm requires. That is the next question.

Notes and sources

1. Peter W. Shor, “Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer,” SIAM Journal on Computing 26, 1484–1509 (1997), doi:10.1137/S0097539795293172. The preliminary factoring and discrete-logarithm algorithms were presented in 1994. This chapter develops a complete small order-finding example rather than using search over candidate factors. Its parameters and numerical tables are original calculations. Original author manuscript. Full author text.

2. The factoring reduction is classical: for a coprime base and an even order, a nontrivial square root of one modulo an odd composite yields a nontrivial greatest common divisor. The chapter’s $N = 21$ examples are directly verified. Primality, perfect-power tests, and initial gcd checks are classical preprocessing; a failed base is replaced. The guarantee for general factoring is probabilistic, not that every base produces a useful square root. See Shor, §5. Reduction and success analysis. For an odd composite with at least two distinct prime factors, a uniform unit has a useful even order with probability at least one-half. This arithmetic event is separate from quantum sampling and denominator recovery. The useful-base guarantee and the probability of a usable Fourier record both enter the retry analysis; neither is an experimental success rate.

3. Vlatko Vedral, Adriano Barenco, and Artur Ekert, “Quantum networks for elementary arithmetic operations,” Physical Review A 54, 147–153 (1996), doi:10.1103/PhysRevA.54.147. The author version arXiv:quant-ph/9511018v1, dated 16 November 1995, bears the title “Quantum Networks for Elementary Arithmetic Operations.” Modular exponentiation is assembled from controlled modular multiplication with classically precomputed constants. Our definition acts as multiplication on $0 \le v < N$ and as the identity on unused value labels, an admissible reversible extension. Register order is exponent, value, workspace. Arithmetic construction. Publication record.

4. Equation (15.3) and Figure 15.1 stipulate ideal coherent arithmetic, a uniform exponent register, a value register initially one, and clean scratch. For $N = 21$, $a = 2$, $Q = 512$, the six residue classes have lengths $86, 86, 85, 85, 85, 85$, with probabilities $L_s / 512$. An explanatory value measurement is averaged over every result. The partial trace retains exponent coherences exactly when the two exponents give the same value. This follows from the orthogonality of different value labels; no value-register postselection is used.

5. Richard Cleve, Artur Ekert, Chiara Macchiavello, and Michele Mosca, “Quantum Algorithms Revisited,” Proceedings of the Royal Society A 454, 339–354 (1998), doi:10.1098/rspa.1998.0164, on the Fourier transform, phase estimation, and order finding. Equation (15.5) uses the positive-exponent transform; some equivalent presentations use the inverse and correspondingly reflected output labels. The unitary acts on amplitudes, not on a list that is first measured. Equation (15.6) is a separate eight-label calculation. Full author text.

6. For the exact finite-register distribution in Figure 15.3, let L_s count integers $s + jr < Q$. The unconditioned probability is $p(y) = Q^{-2} \sum_{s=0}^{r-1} \left| \sum_{j=0}^{L_s - 1} e^{2\pi i j r y / Q} \right|^2$. This includes the different class lengths and every Fourier output. The six indicated peaks carry total probability 0.7893015002. The same distribution follows by applying Eq. (15.5) to the exponent register of Eq. (15.3), then summing probabilities over every value-register label. No plotted ordinate is measured data. See Ekerå, §1.3, for a general distribution analysis. Finite-register analysis.

7. The continued-fraction convergents of $85 / 512$ are $0 / 1, 1 / 6, 42 / 253, 85 / 512$. The approximation error for $1 / 6$ is $1 / 1536$. A reduced fraction within $1 / (2r^2)$ of a real number appears as a continued-fraction convergent; our choice $Q \ge N^2$ supplies this accuracy for nearest-peak labels because $r < N$. If k and r have a common divisor, the recovered reduced denominator is smaller than r. Return verification establishes a multiple of the order, not necessarily the minimal order. Figure 15.4 and Table 15.2 are original arithmetic checks. Testing all convergents with even denominator below 21, then applying return and factor tests, succeeds on 54 possible outputs whose total probability is 0.3284172755; this is not an

optimized post-processing bound. See Cleve and colleagues, §6, and Shor, §5. Fraction recovery. Order-finding reduction.

8. Martin Ekerå, "On the Success Probability of Quantum Order Finding," ACM Transactions on Quantum Computing 5(2), Article 11 (2024), doi:10.1145/3655026. The theorem is cited from author version arXiv:2201.07791v2, revised 28 November 2022. The theorem adds two limited classical searches and provides high single-run recovery guarantees with its stated parameters. It is not a guarantee that every measured label reveals the order by its first reduced denominator. Our elementary processing table is not an implementation of that optimized algorithm. Published record. Full author analysis. The cited bound is Theorem 3 in §4.2 of that author version: a bounded search over nearby frequency labels is combined with bounded classical recovery of missing factors in a reduced denominator. The result's register, search-range, and smoothness parameters matter. The theorem and section numbers given here refer to that author version.

9. The conventional exact m-qubit Fourier decomposition has m Hadamards and $m(m-1)/2$ two-qubit controlled phases. A reversal requires $\lfloor m/2 \rfloor$ swaps when performed physically. Including the output reversal matches the bit order of the positive-exponent Fourier matrix in Eq. (15.5). Our nine-qubit counts exclude routing, synthesis, scratch arithmetic, and error correction. See Cleve and colleagues, §4, for the transform construction. Fourier circuit.

10. Don Coppersmith, "An approximate Fourier transform useful in quantum factoring," IBM Research Report RC19642 (1994); archived as arXiv:quant-ph/0201067 in 2002. Dropping sufficiently small rotations allows an efficiently controlled approximation; the upload date is not the date of the original result. Finite-precision gate synthesis and the total algorithmic error budget are additional implementation questions, not a requirement to execute infinitely precise real numbers. Original report, archived author version.

11. Robert B. Griffiths and Chi-Sheng Niu, "Semiclassical Fourier Transform for Quantum Computation," Physical Review Letters 76, 3228–3231 (1996), doi:10.1103/PhysRevLett.76.3228. Their construction applies when the Fourier transform is immediately followed by measurement, using successive measurement outcomes to control later rotations. It preserves the required classical output statistics, not an unmeasured quantum Fourier output available for arbitrary later coherent use. Full author text.

12. Vedral, Barenco, and Ekert's arithmetic construction (note 3) gives the elementary cubic logical-operation baseline for modular exponentiation. The estimate here counts polynomial-size reversible arithmetic, with $O(n)$ controlled multiplications and $O(n^2)$ operations per schoolbook multiplier. It is not a claim that cubic scaling is optimal, a clock-time estimate, or a physical-qubit total. Our worked register uses nine exponent and five value qubits, plus unspecified implementation scratch. Reversible resource account.

13. Shor's polynomial-time result concerns the quantum computational model and classical post-processing. The absence of a known polynomial-time classical algorithm is not a proof that none exists. Resource estimates distinguish this asymptotic comparison from hardware feasibility; see Gidney and Ekerå's 2021 analysis (note 15). The black-box search lower bound from Chapter 14 has different access assumptions and does not constrain this structured arithmetic algorithm. Algorithmic result. Physical resource analysis. For the standard number-field-sieve comparison, see Fabrice Boudot and colleagues, "Comparing the difficulty of factorization and discrete logarithm: a 240-digit experiment," CRYPTO 2020, LNCS 12171, 62–91, author paper arXiv:2006.06197, §2. Jonathan Lee and Ramarathnam Venkatesan, "Rigorous Analysis of a Randomised Number Field Sieve," arXiv:1805.08873, distinguish heuristic estimates from what their randomized variant proves. These are not proofs that classical polynomial-time factoring is impossible. Classical cost comparison. Scope of rigorous sieve analysis.

14. John A. Smolin, Graeme Smith, and Alexander Vargo, "Oversimplifying quantum factoring," Nature 499, 163–165 (2013), doi:10.1038/nature12290. Their analysis warns that compiled demonstrations can exploit known answers, making the displayed integer's size a misleading measure of the computation. The published abstract states this compilation concern. The author manuscript, arXiv:1301.7007, titled "Pretending to factor large numbers on a quantum computer," develops the argument. This methodological warning does not assess every later factoring experiment. The present simulations use modular multiplication generated from a, N without an order-dependent circuit. Published analysis and abstract.

15. Craig Gidney and Martin Ekerå, “How to factor 2048 bit RSA integers in 8 hours using 20 million noisy qubits,” Quantum 5, 433 (2021), doi:10.22331/q-2021-04-15-433; author manuscript arXiv:1905.09749v3, revised 13 April 2021, §§2–3 and Tables 2–3. The model accounts for error correction, layout, auxiliary-state production, and repeated attempts. The authors describe its resource figures as approximate, with substantial systematic uncertainty. Their optimized factoring construction differs from our elementary circuit; its resources are not obtained by scaling the fourteen displayed register qubits. Published analysis. Author paper and code. For later arithmetic and storage tradeoffs, see Craig Gidney, “How to factor 2048 bit RSA integers with less than a million noisy qubits,” arXiv:2505.15917v1, submitted 21 May 2025, §§2–3. Its residue-system construction and approximate period finding use stated assumptions. This citation supports the discussion of design choices; the numerical example above uses the 2021 model. Version record. Full technical text.

16. Gregory D. Kahanamoku-Meyer, John Blue, Thiago Bergamaschi, Craig Gidney, and Isaac L. Chuang, “A log-depth in-place quantum Fourier transform that rarely needs ancillas,” arXiv:2505.00701v3, revised 14 September 2026. The full preprint develops “optimistic” circuits with an averaged, basis-independent error guarantee, not a uniform worst-case guarantee for every input. Its all-input reduction changes ancillary-space and connectivity requirements. No experimental implementation or journal acceptance is asserted here. The elementary Fourier transform in this chapter is exact and does not use that approximation. Version record. Full preprint. Definition 1 uses a basis-independent average squared vector error. Its factoring application uses the multiplier construction and randomized value-register input analyzed in Appendix E; it does not follow merely by substituting an average-error transform into an arbitrary fixed-input circuit.

CHAPTER 16

Power Has a Shape

The same list can pose very different questions. Does it contain an acceptable answer? Do an odd number of its entries pass the test? Does a rule generating the entries repeat? Each question asks for a small result. Their computational demands need not be alike.

We have now built three quantum programs. One compared two function values, one amplified acceptable candidates, and one exposed a period in arithmetic. Their circuits all used interference. Yet their improvements had different sizes and different explanations. There is no single multiplier called “quantum speed” that we can attach to an arbitrary calculation.

To understand that difference, we will examine two revealing cases: a question that provably resists a large quantum query advantage, and an entangled quantum computation whose results a classical computer can reproduce efficiently. Neither requires abandoning quantum mechanics. Both require looking more closely at what the task asks and what the circuit actually does.

Three successes, three different claims

Deutsch’s problem gave us an unknown two-entry function through a restricted interface. A classical procedure needed both values for an exact comparison. A coherent query could imprint their relative sign, and a final Hadamard turned that sign into the answer. One query replaced two; the program did not recover the entire function.

Grover’s search repeatedly reflected the state toward an acceptable subspace. With one acceptable label among N candidates, the required number of coherent checks grows as the square root of N, rather than proportionally to N for classical black-box search with a fixed, substantial success probability. Matching lower bounds establish that this scaling cannot be improved arbitrarily within that access model.[1]

Shor’s algorithm used something a generic checker did not supply: efficiently computable modular powers with a repeating structure. Its combined quantum computation and classical processing grows polynomially with the number of bits in the integer. Compared with the best known general classical factoring methods, this is a superpolynomial improvement in how the cost grows. Its comparison with classical factoring is not a matching lower-bound theorem. No polynomial-time classical algorithm for general factoring is known, but its impossibility has not been proved.[2]

Table 16.1**. The three programs do not make the same claim.**

Program	Relationship exposed	Established improvement
Deutsch	Whether two values agree	One coherent query instead of two classical value queries for an exact answer.
Grover	Membership in an acceptable set	Square-root query scaling for unstructured search; an optimal scaling in that model.
Shor	A period in modular powers	Polynomial-time quantum factoring in input bit length; no comparable classical impossibility theorem.

Query statements hold under the stated coherent or classical interfaces. Factoring also counts the arithmetic and classical processing. None of these entries is a hardware running time.

Counting an oracle call as one step leaves the work inside that call unspecified. An implementation must supply the coherent operation and count its cost, including any work needed to restore temporary registers for reuse. A stored classical list does not by itself supply that coherent interface. Preparation, repeated calls, and readout must all fit within the cost of the complete task.

The distinction matters even at the level of exponents. If there are $N = 2^n$ candidates, square-root search uses a number of checks proportional to $2^{n/2}$. That can be an enormous improvement, but it is still exponential in the address length n. By contrast, a polynomial such as n^3 has a different growth law. “Faster” and “polynomial” are not interchangeable.

One query instead of two is a fixed-factor improvement. Square-root search changes a growth law within a specified access model. Polynomial quantum factoring is a stronger scaling result relative to the best classical methods known. An “exponential speedup” needs both a stated input-size convention and an identified classical comparison. Comparing only with exhaustive enumeration can exaggerate the improvement over better classical algorithms.

A question that cannot be amplified away

Keep a list of N unknown binary entries, accessible only by queries. Instead of looking for a one, ask whether the total number of ones is odd. This is parity: XOR all the entries. Changing any one entry reverses the answer.

For a classical procedure restricted to fewer than N value queries on every run, some entry remains unread. Under a fair independent distribution of the entries, that missing bit makes the final parity equally likely to be zero or one, even after all replies are known. Such a procedure cannot succeed with probability at least two-thirds on every possible list. It needs N queries under this fixed worst-case budget.

A quantum procedure can do better, but not dramatically. For even N, split the entries into pairs. Deutsch’s exact comparison finds each pair’s XOR using one coherent query. XOR the pair results classically. For the eight-entry list 10110100, the pair results are 1,

0, 1, and 0, giving total parity zero. Four coherent queries have replaced eight ordinary ones.

Could a more elaborate quantum program achieve the square-root improvement of search? Here the answer is no: at least $N/2$ queries are required even when an error probability up to one-third is allowed on each input. Beals and colleagues' polynomial method gives a particularly instructive reason.[3]

A polynomial is a sum of terms built from powers and products of variables; its degree is the largest total power in a term. In the usual coherent bit-query model, after q queries each amplitude is a polynomial of degree at most q in the unknown entries. Squaring magnitudes and summing over all outcomes counted as output one gives a polynomial of degree at most $2q$. Gates that do not depend on the unknown list cannot increase that degree.

Check the smallest case first. What polynomial reports odd parity for two binary entries? If the entries are b_1 and b_2, the expression $b_1 + b_2 - 2b_1b_2$ gives zero for 00 and 11 and one for 01 and 10. It has degree two, consistent with the exact one-query Deutsch procedure.

Now group these inputs by their number k of ones. For k equal to zero, one, or two, the exact acceptance probabilities are zero, one, and zero. The same values are supplied by $g(k) = k(2-k)$. The intermediate real values of k are only a mathematical extension, not fractional entries in a physical list. Even with the allowed error, an acceptance polynomial must be below one-half at both endpoints and above one-half in the middle. It therefore crosses one-half twice and needs degree at least two. The argument below extends this alternating pattern to a list of any length.

The query itself explains the degree count. At a fixed address, the two possible reply amplitudes are combined with weights given by the addressed bit and by one minus that bit. One query adds at most one power. Because every entry is zero or one, any positive power of an entry equals that entry. Removing repeated powers leaves products of distinct entries without increasing the degree.

Average the probability of reporting odd parity over all lists containing exactly k ones. The result, $g(k)$, is still a polynomial of degree at most $2q$. To see why averaging does not introduce higher powers, consider a product of two different entries. It equals one exactly when both positions contain ones. Among weight-k lists this happens with probability $k(k-1)/[N(N-1)]$, a quadratic in k. Products of more distinct entries follow the same counting rule; their average degree does not grow.

Correctness requires $g(k)$ to be at most one-third at every even k, and at least two-thirds at every odd k. The variable between integer values is a mathematical extension, not a claim that a list can contain half an entry.

Consequently, $g(k)$ must cross one-half between every two neighboring integers from zero to N. The polynomial $g(k) - 1/2$ has at least N distinct zeros. A nonzero polynomial of degree d cannot have more than d distinct zeros. Therefore

$$\deg(g) \le 2q, \deg(g) \ge N, q \ge N/2. \tag{16.1}$$

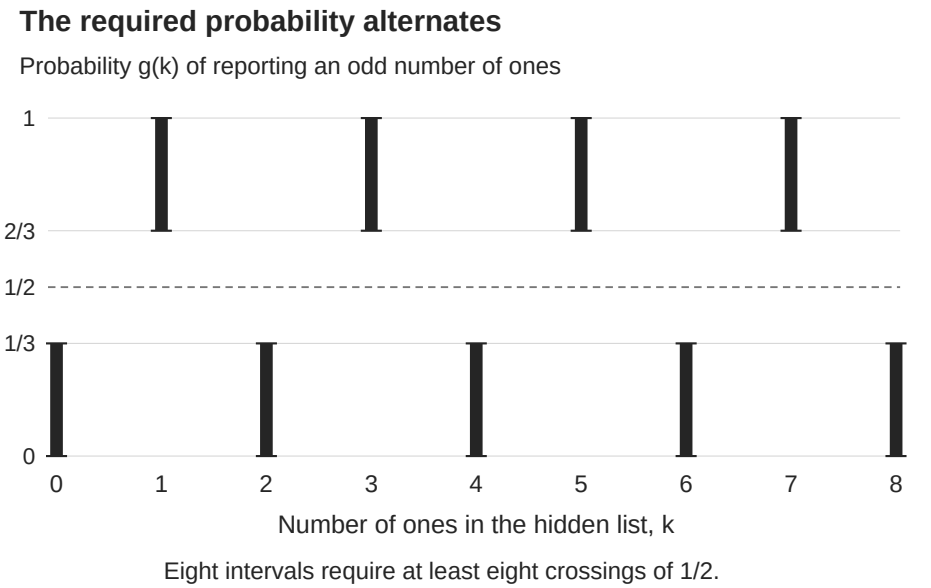


Figure 16.1. Parity demands repeated changes of answer. For an eight-entry list, the vertical segments mark allowed values of the symmetrized probability $g(k)$: at most $1/3$ for even k, at least $2/3$ for odd k. A continuous polynomial meeting all nine conditions must cross $1/2$ in each of the eight intervening intervals. The drawing shows constraints, not a fitted polynomial, a physical trajectory, or measured data. The degree bound then gives at least four coherent bit queries.

This is a limit on all quantum algorithms with that query access and guarantee, not merely on a strategy we happened to try. Search asks whether some entry is acceptable. Parity asks for a relationship sensitive to every entry. The same oracle and the same quantum laws give different limits because the questions differ.

Query counts are whole numbers, so the lower bound rounds upward for an odd-length list. Pair all but the last entry, perform the exact pair comparisons, and query the remaining bit once. This attains the rounded bound as well.

It is equally important not to extend the theorem beyond its premise. If a short formula generates the list and its form immediately reveals the parity, inspecting that formula is a different input model. A query lower bound does not prohibit using information that the query problem deliberately withheld.

What a complexity class promises

Computer science groups problems by the resources needed as their inputs grow. For classical yes-or-no problems, P contains those solvable deterministically in polynomial time. BPP allows classical randomness and a probability of correctness of at least two-thirds on every input. BQP makes the corresponding allowance for quantum computation, with efficiently constructible circuits and polynomial resources. The letters are names for families of tasks, not grades assigned to individual machines.[4]

Efficient construction requires one classical algorithm that, given an input length, produces the circuit for that length in time polynomial in that length. This condition is called uniformity. A short circuit chosen after someone has solved each instance would hide the work instead of accounting for it. The class names here organize yes-or-no questions; returning factors or drawing samples requires specifying those different outputs too.

The success guarantee is not an average over easy instances. Every permitted input must meet it. The particular fraction two-thirds is conventional: independent repetitions and majority voting can reduce the error with modest extra work when this bounded-error guarantee already holds. Repeating a calculation that is systematically wrong is something else.

A classical deterministic calculation is a special case of a randomized one. A quantum computer can implement a classical randomized calculation with polynomial overhead. Thus

$$\mathrm{P} \subseteq \mathrm{BPP} \subseteq \mathrm{BQP}. \tag{16.2}$$

Containment is known; strict separation between BPP and BQP in the ordinary, unrestricted computational model remains unproved. The oracle separations we have discussed are rigorous results with additional access rules. They do not, by themselves, settle that larger question.

NP asks a different question: can a yes answer be supported by a polynomial-length classical certificate that a deterministic computer checks in polynomial time? For a set of Boolean constraints, a proposed satisfying assignment is such a certificate. Checking its entries against the constraints is different from finding that assignment. NP does not mean “non-polynomial”; it includes every problem in P.[5]

An NP-complete problem is one in NP to which every NP problem can be translated with polynomial overhead. An efficient general quantum algorithm for one would therefore have consequences throughout NP. No such general algorithm is known. Shor’s factoring result does not supply one, and the black-box search bound does not prove that one is impossible. To settle the latter would require controlling every way of exploiting the explicit problem structure, not only querying a checker.

The unresolved boundary is important: quantum algorithms have established substantial new capabilities without giving us a complete map of efficient computation.

An enormous state with a short description

An arbitrary pure state of n qubits has 2^n amplitudes in the computational basis. That is a large possible description, not a requirement that every state needs that much storage. The state made by applying a Hadamard to each zero qubit is simply $|+\rangle^{\otimes n}$. All its amplitudes are equal. A short recipe specifies the entire list.

Length of description and ease of prediction are separate. A short circuit can specify a state without offering an efficient classical way to calculate its measurement probabilities. The next example supplies both a short description and a rule for using it.

Entanglement alone does not remove this possibility. Begin with three zeros. Apply H to the first qubit, controlled NOT from the first to the second, and controlled NOT from the second to the third. The result is

$$|G_3\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}}. \tag{16.3}$$

This is a three-qubit Greenberger–Horne–Zeilinger state. No product of individual pure states gives its amplitudes. Yet we have described it in one line. There is a more systematic description that survives an entire family of subsequent gates.[6]

Write $Z\,Z\,I$ for Z on the first two registers and the identity on the third. Here these letters describe a product operator, not three successive questions asked of one qubit. Acting on $|G_3\rangle$, both $Z\,Z\,I$ and $I\,Z\,Z$ return the same state: the neighboring bits agree. The operation $X\,X\,X$ also returns it, exchanging 000 and 111. These three properties can be written

$$Z\,Z\,I\,|G_3\rangle = |G_3\rangle,\; I\,Z\,Z\,|G_3\rangle = |G_3\rangle,\; X\,X\,X\,|G_3\rangle = |G_3\rangle. \tag{16.4}$$

The first two conditions restrict a pure state to $\alpha|000\rangle + \beta|111\rangle$. The third requires $\alpha = \beta$. Normalization then fixes the state up to global phase. We have specified it by three constraints, called stabilizers, rather than storing eight coefficients.

The description can follow an operation without reconstructing those coefficients. If $\mathrm{P}|\psi\rangle = |\psi\rangle$ and the state is changed by a gate U, the updated constraint is $U\,\mathrm{P}\,U^{\dagger}$. The dagger denotes the inverse of this unitary gate. Acting with the updated constraint on $U|\psi\rangle$ cancels the neighboring inverse and forward operations, leaving $U\,\mathrm{P}|\psi\rangle = U|\psi\rangle$. The property follows the state.

Now apply H to every register. The identities $H\,Z\,H = X$ and $H\,X\,H = Z$ exchange the corresponding letters in each constraint. The new state is stabilized by $X\,X\,I$, $I\,X\,X$, and $Z\,Z\,Z$. Expanding its amplitudes gives

$$H^{\otimes 3}|G_3\rangle = \frac{|000\rangle + |011\rangle + |101\rangle + |110\rangle}{2}. \tag{16.5}$$

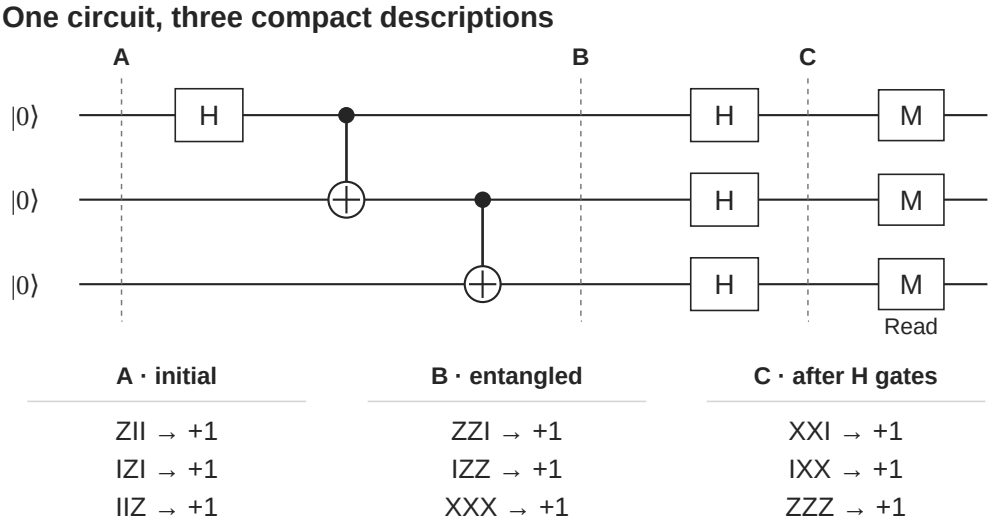


Figure 16.2. Follow constraints instead of an amplitude array. The circuit first prepares $|G_3\rangle$, then applies H to every qubit. At each dashed boundary, the three listed product operators have eigenvalue $+1$ on the state and specify it uniquely up to global phase. The lists are an alternative classical description, not intermediate measurements. The M boxes mean read the Z basis, not apply Z phase gates. Final Z readout is uniform over 000, 011, 101, and 110. All gates and outputs are ideal; wires identify registers, not spatial paths.

The four possible final strings all contain an even number of ones. To reproduce this Z-output distribution classically, choose the first two bits fairly and independently, then set the third to their XOR. Each permitted string appears with probability one-quarter. No eight-component state vector was needed.

The construction extends to n qubits. A chain of controlled NOTs makes $\left(|0^n\rangle + |1^n\rangle\right)/\sqrt{2}$. Hadamards on all registers then produce equal amplitudes on all even-parity strings and zero on the rest. There are 2^{n-1} nonzero amplitudes. A classical sampler still needs only $n-1$ independent fair bits; their XOR supplies the last. At a hundred qubits, 2^{99} nonzero amplitudes have not defeated a hundred-bit sampling procedure.

This particular sampler reproduces only the specified Z measurement. The stabilizer method is more general. Starting from computational-basis states, circuits built from H, controlled NOT, and the phase gate that multiplies $|1\rangle$ by i can be simulated efficiently,

including appropriate Pauli measurements. These are Clifford gates. Each gate updates a compact set of product-operator constraints. This is the Gottesman–Knill framework, with efficient implementations developed by Aaronson and Gottesman.[7]

For a pure stabilizer state of n qubits, n independent, mutually commuting generators, each with n Pauli letters and a sign, specify the state. Commuting means that applying any two in either order has the same effect. That is order n^2 symbols, rather than order 2^n amplitudes. The simulation works because these gates transform the generators into the same kind of object. It is a rule for updating a compact description, not a lucky recognition of one final pattern.

The selected generators commute, so their constraints can hold together. Independent means that none is obtained by multiplying a selection of the others; a redundant constraint would supply no new information. For instance, knowing both $Z_1 Z_2 = +1$ and $Z_2 Z_3 = +1$ already fixes $Z_1 Z_3 = +1$. A pure n-qubit stabilizer state needs n independent compatible constraints, not merely n listed equations.

There is no contradiction with Bell's theorem. The classical simulator receives a circuit and measurement specification and samples the joint outcome distribution. It is not separated devices required to answer independent questions without communication. A simulation algorithm and a Bell-local causal model have different rules.

Beyond one classical shortcut

More general gates can make that compact update rule fail. For example, the T gate multiplies the one component by $e^{i\pi/4}$. Conjugating X by T gives $(X+Y)/\sqrt{2}$, rather than one product of Pauli letters. A representation that followed only one such product must now accommodate a sum.

But applying T separately to a thousand independent qubits still leaves a product of a thousand one-qubit states. That is easy to describe. Escaping one classical simulation method does not establish that all methods fail. Nor does counting unfamiliar gates tell us whether their arrangement computes anything useful.

Other methods exploit other structure. Vidal's sufficient condition concerns a pure-state computation. For each division of the qubits into two groups, the Schmidt rank counts the nonzero paired terms in the Schmidt decomposition introduced in Chapter 9. A product state has rank one. The GHZ state above has rank two across every division into two nonempty groups: all-zero on one side is paired with all-zero on the other, and likewise for all-one. If the largest such rank across all divisions grows at most polynomially with the number of qubits at every step, his method gives an efficient classical simulation. A small rank only at the end, or a small entropy by itself, is a different promise. Circuit structure, state preparation, and the requested measurement all matter.[8]

Evolution is not a search for the ground state

A useful distinction appears in the simulation of interacting systems. A quantum computer can approximate broad classes of evolution with efficiently specified local interactions, using resources polynomial in the system size, simulated time, relevant interaction strengths, and inverse allowed error. Lloyd's universal-simulation construction established an early general route. This statement presupposes access to the initial state. Preparing that state and extracting the required answer have to be counted separately; a theorem about evolution does not guarantee that either is efficient.[9]

Consider an assignment of binary variables to some constraints. Give every failed constraint an energy penalty $\Delta > 0$. The resulting diagonal Hamiltonian has $H_C|x\rangle = E(x)|x\rangle$, where $E(x)$ is the total penalty. A zero-energy assignment, if one exists, satisfies every constraint. For a polynomial-size list of local constraints, the terms are straightforward to specify.

Prepare a superposition and let this Hamiltonian act. Chapter 5 immediately gives

$$e^{-iH_C t/\hbar} \sum_x \alpha_x |x\rangle = \sum_x \alpha_x e^{-iE(x)t/\hbar} |x\rangle . \tag{16.6}$$

Each component changes phase. None changes magnitude. The probability of obtaining a zero-penalty assignment in a computational-basis measurement has not increased at all. Figure 16.3 gives an eight-label example.

Adding mixing operations can turn those phases into population changes; Grover's reflections showed one way. Designing a sequence that reliably favors low energy is then an additional algorithmic problem. Simply writing an optimization task as a Hamiltonian does not solve it.

The same distinction protects our understanding of chemistry. Simulating the evolution of a prepared molecular state, estimating its mean energy, and preparing the ground state are different tasks. A stationary bonding state does not have to be difficult to maintain merely because finding an accurate description of it is difficult. Conversely, reproducing a short interval of dynamics does not automatically reveal an equilibrium property or a reaction rate.

Waiting changes phases, not the chance of low energy

Uniform input; phase interval $\Delta t/\hbar = \pi/2$

Candidate	000	001	010	011	100	101	110	111
Energy E/Δ	0	1	1	2	1	2	2	3
Phase factor	1	−i	−i	−1	−i	−1	−1	i
P before	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8
P after	1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/8

The unique zero-cost label is 000. Its probability is still 1/8.

Figure 16.3. An energy-dependent phase does not by itself favor low energy. The illustrative cost is $E(x) = \Delta(x_1 + x_2 + x_3)$ and the initial state is a uniform superposition of eight labels. At $\Delta t / \hbar = \pi / 2$, each component has acquired the displayed phase factor. Every label still has probability $1/8$, including the unique zero-cost label 000. This deliberately easy cost function illustrates the distinction between evolution and optimization; it is not a hard benchmark. The figure is an exact ideal-model calculation.

The answer must fit through the measurement

A quantum state, a sample from a measurement, and a classical table of all outcome probabilities are three different outputs. Our even-parity sampler produces one string cheaply. Printing its full table already requires exponentially many entries. Its compactness is useful precisely because we do not insist on expanding it into that table.

For an experiment we may need only a mean. Suppose a chosen binary outcome has probability p, and m fresh, independently prepared trials produce a fraction $\hat{p}$ of ones. A useful bound is

$$\Pr\left(|\hat{p} - p| \geq \epsilon\right) \leq 2e^{-2m\epsilon^2}. \tag{16.7}$$

The symbol ϵ sets the allowed absolute error. This version of Hoeffding's inequality bounds fluctuations of independent outcomes; it does not correct a detector bias, a drifting source, or a wrong circuit. To make its upper bound at most a chosen failure probability δ, it suffices to take[10]

$$m \geq \frac{\ln(2/\delta)}{2\epsilon^2}. \tag{16.8}$$

Since the trial count must be a whole number, round the right-hand side upward. With $\delta = 0.01$ and $\epsilon = 0.01$, the prescription gives 26,492 trials. The tolerance is one percentage point of probability: an estimate within 0.01 of the true probability, rather

than one percent of its value. Tightening the tolerance to 0.001 raises it to 2,649,159. A short circuit can require a long experiment when its output must be estimated precisely.

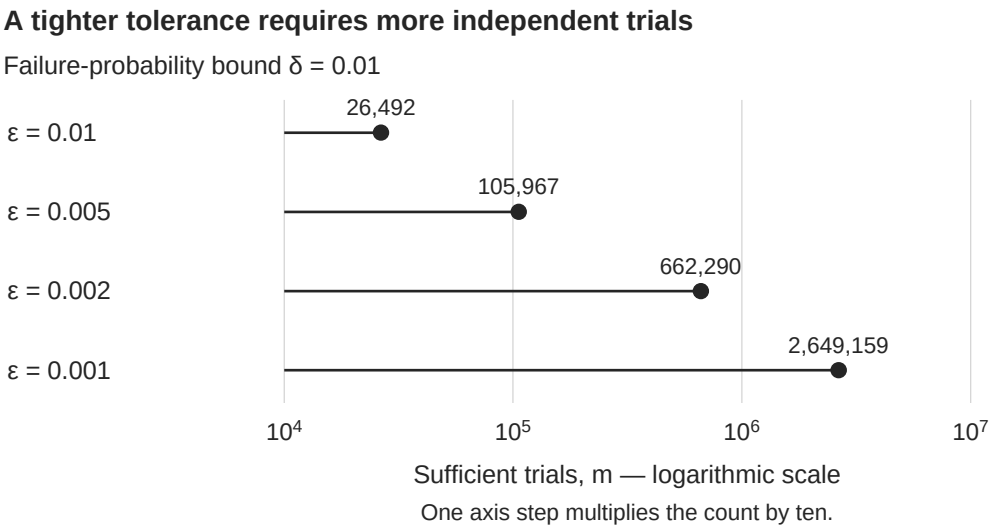


Figure 16.4. Precision changes the number of preparations. The four rows apply Eq. (16.8) at fixed $\delta = 0.01$. Each smaller absolute tolerance requires more independent repetitions under this sufficient sampling prescription. The final row tightens the first tolerance tenfold and requires about a hundred times as many trials. These are calculated sufficient counts, not a universal lower bound, experimental sample sizes, or an error estimate for biased measurements.

These are sufficient counts for this particular readout method, not a theorem that every quantum estimation protocol must use them. At fixed success probability, coherent amplitude-estimation procedures can reduce inverse-square precision dependence to an inverse-linear number of coherent calls, when the required preparation operations, controls, and inverses are available. That stronger access has its own implementation cost.[11]

The bounded-error definitions above likewise require a usable separation between the output probabilities for yes and no. Distinguishing a probability of at least two-thirds from one of at most one-third is not the same promise as deciding on which side of one-half an exponentially tiny deviation lies. Encoding a hard answer in such a deviation has not made it cheaply readable.

Which kind of evidence is being offered?

A theorem can establish a scaling advantage under explicit access rules. A resource estimate can ask where an implementation might overtake a classical competitor. An experiment can test an executed task under a specified benchmark. These statements support one another only when their tasks, accuracies, and resources match. None should silently substitute for the others.

A September 2026 preprint by Chang and colleagues makes this distinction even within classically tractable circuit families. They compare the cost of estimating expectation values through quantum execution and through classical simulation. Their analysis separates measurements, quantum processing, and classical processing, including whether a preprocessing cost can be reused. When an input is supplied as unknown quantum states, acquiring information about it is a resource; a known classical description is a different starting point. Their classical-simulation route can therefore still require quantum measurements to acquire the input information its simulator needs. Their hardware-level comparisons are estimates with stated limitations, not an experimental race. Polynomial classical simulability need not settle the better finite-size implementation.[12]

An experiment can also have a purpose quite different from factoring or predicting a molecule. In 2025, Liu and colleagues reported certified randomness using a trapped-ion quantum processor. The demonstration used 56-qubit challenge circuits and certified 71,313 bits of entropy under its protocol and a restricted adversary model. It connected sampled quantum outputs with substantial classical verification. This is not the separated-box Bell certification of Chapter 10: computational assumptions are part of the guarantee. Its specific achievement should neither be dismissed for being a sampling task nor promoted into proof that arbitrary difficult calculations are now faster.[13]

Classical algorithms remain part of the comparison. A better simulator can change an earlier estimate of the cost of reproducing particular records without changing a single law of physics. Equally, a proof of polynomial simulation does not promise that a poorly implemented simulator will beat every quantum device at every size. The appropriate conclusion depends on whether we are comparing growth laws, finite workloads, or an experimental performance claim.

The shape of the advantage

We can now say more than that quantum computers use superposition and interference. A useful program must create a relationship that the task needs, transform it coherently, and make enough of it readable at an acceptable cost. Classical shortcuts must be compared against that complete task, not against the most cumbersome possible representation of its state.

Deutsch’s relative sign answered one comparison. Grover’s repeated reflections increased the chance of an acceptable result. Shor’s Fourier pattern exposed arithmetic repetition. Parity limited what the same query access could achieve, while the stabilizer example showed why even a large entangled state can admit a small classical description. These are different shapes of structure, not exceptions to one universal promise.

The detector in the prologue stayed dark because contributions canceled. We can now arrange such cancellations to answer questions, and we can distinguish a successful interference experiment from a successful computational comparison. The remaining challenge is physical: build the carriers, realize the operations, and protect the relationships long enough for the answer to survive. That is where the next part begins.

Notes and sources

1. Lov K. Grover, “A fast quantum mechanical algorithm for database search,” Proceedings of STOC 1996, 212–219, doi:10.1145/237814.237866; Charles H. Bennett, Ethan Bernstein, Gilles Brassard, and Umesh Vazirani, “Strengths and Weaknesses of Quantum Computing,” SIAM Journal on Computing 26, 1510–1523 (1997), doi:10.1137/S0097539796300933. The search statement is bounded-error black-box query complexity, with a marked item promised; constants and implementation costs are separate. Deutsch’s exact comparison was developed in Chapter 13, following Cleve and colleagues. Table 16.1 compares claims with different scopes, not commensurate clock times. Grover, author paper. Search lower bounds. Exact Deutsch formulation.

2. Peter W. Shor, “Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer,” SIAM Journal on Computing 26, 1484–1509 (1997), doi:10.1137/S0097539795293172. Polynomial scaling concerns input bit length, a uniform quantum construction, suitable accuracy, and classical post-processing. The result does not prove an unconditional superpolynomial classical lower bound for factoring. The present chapter does not claim that factoring is NP-complete. Full author paper. The general number-field-sieve benchmark is subexponential but superpolynomial in input bit length under its standard heuristic analysis. The comparison is with known methods, not a proof against all classical algorithms. See Alexander M. Dalzell and colleagues, “Quantum algorithms: A survey of applications and end-to-end complexities,” arXiv:2310.03011, §6.1, especially “Comparable classical complexity” and “Speedup” (author text, accessed 6 October 2026). Algorithm and cost comparison.

3. Robert Beals, Harry Buhrman, Richard Cleve, Michele Mosca, and Ronald de Wolf, “Quantum Lower Bounds by Polynomials,” Journal of the ACM 48, 778–797 (2001), doi:10.1145/502090.502097, especially the polynomial representation and the parity argument. In the bit-query model, amplitudes have multilinear degree at most q, and output probabilities degree at most $2q$; averaging a degree-d monomial over weight-k inputs gives a polynomial in k of degree at most d. Adaptive measurements can be deferred in this query model. The chapter uses a worst-case fixed query budget and correctness on every input; expected-query conventions can give different classical constants. For even N, pairwise exact Deutsch queries attain $N/2$. Figure 16.1 is an original constraint diagram, not the graph of an acceptance polynomial between integer inputs. Full author paper. Published record. Independent parity analysis. The cited author version is quant-ph/9802049v3 (30 September 1998): see its symmetrization lemma, polynomial representation and Proposition 6.4. Its randomized classical parity count uses expected queries; it is not the fixed maximum used in this chapter. For odd length, the quantum count is the ceiling of half the length, attained by pairing and one remaining single-bit query.

4. Ethan Bernstein and Umesh Vazirani, “Quantum Complexity Theory,” SIAM Journal on Computing 26, 1411–1473 (1997), doi:10.1137/S0097539796300921. The circuit description uses the standard uniform polynomial-time model, efficiently specified gates or finite-gate approximations, classical inputs, and bounded two-sided error. Equation (16.2) does not assert strict containments. Oracle separations are not unconditional separations in the ordinary input model. See also John Preskill, Chapter 6, “Quantum Algorithms,” for a pedagogical development of computational classes and the limitation of black-box arguments. Primary publication. Preskill, computational complexity. John Watrous, “Quantum Computational Complexity,” arXiv:0804.3401v1 (2008), Definition 2 and §§IV.1–IV.3, gives the explicit polynomial-time circuit-generation convention and bounded-error definition. Generation time is polynomial in input length, not in the number of bits needed merely to write that length. Factoring returns factors; its decision variants, rather than that multi-output task by definition, are examples in BQP. Finite gate sets or efficiently computable approximations exclude encoding inaccessible answers in gate parameters. Circuit model and output conventions.

5. Stephen A. Cook, “The Complexity of Theorem-Proving Procedures,” Proceedings of STOC 1971, 151–158, doi:10.1145/800157.805047. The manuscript uses contemporary certificate and polynomial-reduction terminology. NP concerns deterministic verification of classical certificates for yes instances, not efficient verification of arbitrary optimality claims. The relationship of NP to BQP is not resolved by the oracle parity

or search bounds; those bounds impose a restricted input interface. Author-hosted original paper. Publication record.

6. Equations (16.3)–(16.5) and Figure 16.2 are original worked constructions. Register order is 1, 2, 3, with the first bit written leftmost. The two controlled NOTs are $1 \rightarrow 2$ and $2 \rightarrow 3$. Each listed stabilizer has eigenvalue $+1$; the operators commute, and the three independent constraints select one normalized pure state up to global phase. Before the first H, the generators are ZII, IZI, IIZ; after it, XII, IZI, IIZ; after the first controlled NOT, XXI, ZZI, IIZ. No intermediate measurement is performed.

7. Scott Aaronson and Daniel Gottesman, "Improved Simulation of Stabilizer Circuits," Physical Review A 70, 052328 (2004), doi:10.1103/PhysRevA.70.052328. The cited author version is arXiv:quant-ph/0406196v5 (2008 revision). With stabilizer inputs, Clifford gates, and Pauli measurements, a tableau stores a polynomial-size description and permits efficient measurement sampling. The phase gate in the narrative is conventionally called S; it is not the T gate. For the original even-parity family, the displayed sampler is exact for the specified joint Z readout. The broader guarantee covers Clifford processing and Pauli measurements; arbitrary non-Pauli measurements are outside the stated guarantee. Availability of a full circuit description distinguishes this task from oracle and Bell-local access models. Full author treatment.

8. Guifré Vidal, "Efficient classical simulation of slightly entangled quantum computations," Physical Review Letters 91, 147902 (2003), doi:10.1103/PhysRevLett.91.147902. The cited author version is quant-ph/0301063v2 (26 February 2003). Equation (3) maximizes Schmidt rank over all bipartitions; Theorem 1 requires that maximum to be polynomially bounded throughout a pure-state computation of polynomially many local gates. This is a sufficient condition, not a necessary condition for every efficient simulator. A small final entropy alone does not supply it. The T-gate calculation is an original one-qubit conjugation: $T\,X\,T^{\dagger} = (X + Y)/\sqrt{2}$. Its failure to preserve a single Pauli operator is not a theorem that every circuit containing T is classically hard. Full author analysis.

9. Seth Lloyd, "Universal Quantum Simulators," Science 273, 1073–1078 (1996), doi:10.1126/science.273.5278.1073. The simulation claim concerns suitably specified local dynamics, with simulated time and approximation accuracy included, not arbitrary state preparation or automatic ground-state optimization. Equation (16.6) is an exact original consequence of diagonal time-independent evolution. Figure 16.3 uses $H_C = \Delta \sum_{j=1}^{3} (I - Z_j)/2$, uniform input, and $\Delta t/\hbar = \pi/2$. It is deliberately an easy noninteracting example of what evolution alone does not change, not a complexity lower bound for optimization. Original article copy. Publication record. The basic product-formula route assumes finitely many local degrees of freedom with efficiently implementable, bounded local terms. Its cost is polynomial in the stated simulation parameters; it does not promise polynomial cost in the bit length of an exponentially long requested time or an exponentially small tolerance. The constraint example establishes a distinction between tasks, not a lower bound for every ground-state algorithm.

10. Wassily Hoeffding, "Probability Inequalities for Sums of Bounded Random Variables," Journal of the American Statistical Association 58, 13–30 (1963), doi:10.1080/01621459.1963.10500830. Equations (16.7)–(16.8) apply the two-sided bounded-variable inequality to independent, identically distributed binary outcomes. The stated probability is over the records of fresh trials, not a posterior probability assigned to an unknown constant. Figure 16.4 rounds the sufficient count upward at each tolerance; it does not plot a lower bound or incorporate systematic errors. Original paper copy. Publication record.

11. Gilles Brassard, Peter Høyer, Michele Mosca, and Alain Tapp, "Quantum Amplitude Amplification and Estimation," Contemporary Mathematics 305, 53–74 (2002), doi:10.1090/conm/305/05215. Amplitude estimation can improve precision dependence given suitable coherent preparation, its inverse, and controlled amplification operations. It is not the independent-readout procedure of Eq. (16.8). No claim of unlimited precision from a fixed number of samples is made. Full author paper. Theorem 12 of author version quant-ph/0005055v1 supplies a worst-case absolute error proportional to the reciprocal of the coherent iteration budget at fixed success probability. Further confidence amplification has additional cost. This does not replace detector calibration or provide inverses for an unknown state source.

12. Su Yeon Chang, Supanut Thanasilp, Zoë Holmes, and M. Cerezo, “The cost of simulating classically tractable quantum circuits and dynamics,” arXiv:2609.11847v1, submitted 10 September 2026. The citation is to this preprint version; no journal publication is asserted here. Its central task is expectation-value estimation for specified input states, circuit families, and observables, not general output-distribution sampling. The comparison distinguishes state acquisition, reusable preprocessing, and repeated evaluation. The hardware-level estimates neglect or simplify some prefactors and architectural effects, as stated in §V.3; they are not measured end-to-end timings. Our figures reproduce none of its numerical results. Version record. Full technical text. Section V.3 also omits noise and protection costs, some gate and control overheads, memory effects, and constant prefactors. Its serial busy-time expression is not a parallel time-to-solution benchmark. Acquiring reusable information about the input may consume quantum samples; the relevant reuse assumptions belong to the comparison. These distinctions are retained without endorsing any particular numerical crossover.

13. Minzhao Liu and colleagues, “Certified randomness using a trapped-ion quantum processor,” Nature 640, 343–348 (2025), doi:10.1038/s41586-025-08737-1, published 26 March 2025. The 56-qubit challenge-circuit demonstration reports certification of 71,313 bits of entropy under the protocol and its restricted adversary assumptions. Entropy certified is not silently identified with a final uniform-string length. The computational certification and classical verification costs differ from Bell-based device-independent randomness. The result is not an unconditional proof of BPP differing from BQP or a claim of generic application speedup. The reported entropy bound is stated in the published article. Open-access published article. More precisely, the reported bound is conditional smooth min-entropy relative to the initial classical side information. The stated case uses soundness one part in a million, smoothing one part in four million, and a classical adversary budget four times that of Frontier within the paper’s restricted strategy family. The strategy fixes in advance which rounds use an honest single quantum sample and which return deterministic classical results, and excludes postselection. The extracted string contains 71,273 bits, distinct from the 71,313-bit entropy bound.

PART V

Making Quantum Information Useful

The programs in the previous part were precise enough to calculate. A Hadamard combined amplitudes, an interaction attached a phase, and a measurement produced a record. What do those instructions demand of a physical machine?

No device offers unlimited access without disturbance. To use a quantum state, we must reach it with controls and eventually connect it to a record. To preserve it, we must limit uncontrolled transitions and phase changes, and prevent other systems from acquiring distinctions that the calculation needs to keep coherent. The challenge is not simply to shut the world out. It is to choose when, how, and to what the device opens.

This part follows that challenge through four tasks: building a controllable carrier, protecting its information, moving that information between locations, and using its sensitivity to learn about the world. A long-lived state is not automatically easy to control. A successful transfer must preserve the relevant relationships, not merely one set of detector totals. A good sensor must distinguish the influence of interest from unwanted disturbances.

We will begin with a small circuit and ask what an experimentalist must supply in place of each symbol. Atoms, electrical circuits, spins, and light provide different answers. Their differences expose the work that the diagram leaves implicit: preparing states, arranging interactions, recognizing leakage and loss, and producing trustworthy records.

Then we will confront the apparent contradiction at the heart of protection. An unknown quantum state cannot be backed up by ordinary copying, and measuring it can alter what we hoped to save. Error correction must obtain information about damage without extracting the protected information itself.

The goal is not a ranking of technologies. It is an understanding of what useful quantum information requires from matter. We know what the ideal operations would do. Now we must ask how a device can keep those promises.

CHAPTER 17

A Qubit Is Something You Build

In the diagram, the gate is a small square. In the laboratory, it may require a shaped microwave pulse, a pair of laser beams, or an optical path held steady while the room vibrates. The drawing says what must happen to the state. It does not yet say what physical event will make it happen.

Let us give the engineer a modest assignment. Prepare two zero states, turn the first into plus, entangle the pair, wait, and read both qubits. Repeat with a different measurement setting. We already know the ideal predictions. Now we want a device whose records justify using those predictions.

This assignment contains most of the difficulty in miniature. The device must isolate a useful quantum degree of freedom, but also let us reach it. It must couple two qubits deliberately without recording their private alternatives elsewhere. Finally, it must make a record strong enough to read, at the moment when making that record is wanted.

A qubit is therefore not just a small object with two labels. For an experiment, it is a physical encoding together with the means to prepare, control, preserve, and read it. DiVincenzo's implementation criteria organized these requirements into a common language for very different technologies. We will follow the requirements through one task rather than declare a winner among materials.[1]

Choose the two states—and account for the others

An isolated atom has a characteristic set of energy levels; the experimentalist does not create them. But choosing a pair, holding the atom, controlling its surroundings, and connecting the pair to reliable operations are engineering tasks. Two internal states can serve as zero and one while the atom's position is used only to hold or move the carrier.

A superconducting circuit takes a different route. Its qubit states belong to a collective electrical degree of freedom, not to the spin of one electron. A circuit containing a Josephson junction has a nonlinear energy spectrum. In a transmon, the design balances reduced sensitivity to unwanted charge fluctuations against the need to distinguish neighboring transitions.[2]

That last requirement is easy to miss. A quantum harmonic oscillator has equally spaced energy levels. For a drive that couples neighboring levels, resonance with the lowest transition also gives resonance with the next. Merely pointing to its two lowest levels does not make all higher ones irrelevant. Unequal spacing helps us address the intended pair selectively.

$$\omega_{01} = \frac{E_1 - E_0}{\hbar}, \omega_{12} = \frac{E_2 - E_1}{\hbar}, \Delta = \omega_{12} - \omega_{01} \tag{17.1}$$

The frequencies are angular frequencies, measured in radians per second. The quantity Δ measures the departure from equal spacing; it is called the anharmonicity. Its sign can be negative, as in a transmon whose second step is smaller than its first. Figure 17.1 shows why the difference, rather than the sheer size of either energy, matters for selectivity.

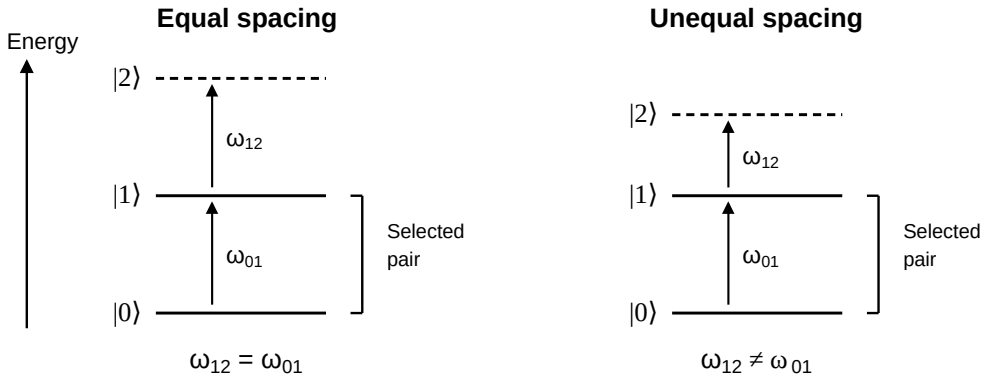


Figure 17.1. Two selected levels do not remove the rest. Both schematic spectra contain three levels. Equal spacing on the left makes the two neighboring transition frequencies coincide. Unequal spacing on the right separates them. The bracket identifies the intended zero/one subspace; occupation of level two is leakage from that encoding. Vertical position denotes energy, not a particle's height or orbit. Level spacings are schematic and are not measured data.

If a pulse leaves some probability in $|2\rangle$, calling that a bit flip is inaccurate. A bit flip stays inside the zero/one space. Leakage leaves it. For a normalized state of the larger device, the accounting is

$$\mathrm{P}(0) + \mathrm{P}(1) = 1 - \mathrm{P}(\mathrm{leak}) \tag{17.2}$$

The missing probability has not disappeared from quantum mechanics. It lies outside the chosen encoding. A readout that cannot recognize it may misleadingly assign it to zero or one. The two-dimensional model and the detector's labels then conceal different problems.

Keeping only trials that remain inside the encoding answers a different question. The fraction in one among those retained trials is its original population divided by the total population still in zero or one. Reporting that fraction together with the probability of retention keeps both the quality of the retained events and the cost of leakage visible.

Our original photon has the same issue in another form. One photon in either of two optical modes represents one qubit. No photon in either mode is a loss event, not its

logical zero. Extra photons also lie outside that encoding. Deciding which physical possibilities count as data, leakage, or loss comes before counting usable qubits.[3]

What a pulse must avoid

Chapter 5 made a resonant pulse exchange the populations of two levels. A real drive may also reach the levels we excluded. We can expose that omission with a small calculation instead of merely asking for “better control.”

Keep three levels and drive at ω_{01}. Use a reference frame that follows the drive, removing that known phase rotation from the amplitudes. Let Ω denote the on-resonance Rabi angular frequency of the intended transition, in radians per second. In this frame, stipulate the constant Hamiltonian

$$\frac{H_{\mathrm{rot}}}{\hbar} = \begin{pmatrix} 0 & \Omega/2 & 0 \\ \Omega/2 & 0 & \sqrt{2}\,\Omega/2 \\ 0 & \sqrt{2}\,\Omega/2 & \Delta \end{pmatrix} \tag{17.3}$$

The rows and columns refer to zero, one, and two, in that order. The first off-diagonal pair drives the intended transition. The second couples one to two. We have chosen its strength to be $\sqrt{2}$ times larger, as in an oscillator-like three-level approximation. The last diagonal entry gives the unwanted level’s detuning in this rotating description. This is a specified approximate model, not a universal Hamiltonian for every qubit.[4]

Begin in zero. As in Chapter 5, evolve this initial state with the constant Hamiltonian for the pulse duration; the squared magnitudes of its three final amplitudes give the three populations. With the third level absent, a rectangular pulse of duration $t_\pi = \pi/\Omega$ would give one with certainty. Now retain the third level, choose $\Delta/(2\pi) = -200$ megahertz, and calculate the same nominal pulse. At ten nanoseconds, the model gives $\mathrm{P}(1) \approx 0.9590$ and $\mathrm{P}(2) \approx 0.0263$. The remaining probability, about 0.0147, is still in zero. At twenty nanoseconds, those probabilities become approximately 0.9887, 0.00748, and 0.00384.

These are original calculations with stipulated parameters, not device measurements. Doubling the duration halves the drive strength in this comparison. It improves the intended population transfer, but neither result is a full gate-fidelity measurement: we have tested only one input and one kind of output. A gate must also handle superpositions correctly.

For these pulses, the angular drive strength divided by two pi is fifty megahertz at ten nanoseconds and twenty-five megahertz at twenty nanoseconds. Both are small beside the nominal carrier at five gigahertz assumed in the note. That supports discarding terms that oscillate rapidly with the carrier, the rotating-wave approximation. Yet those strengths are appreciable fractions of the anharmonicity of two hundred megahertz. A useful rotating-wave model can therefore still predict substantial leakage from the chosen two-level encoding.

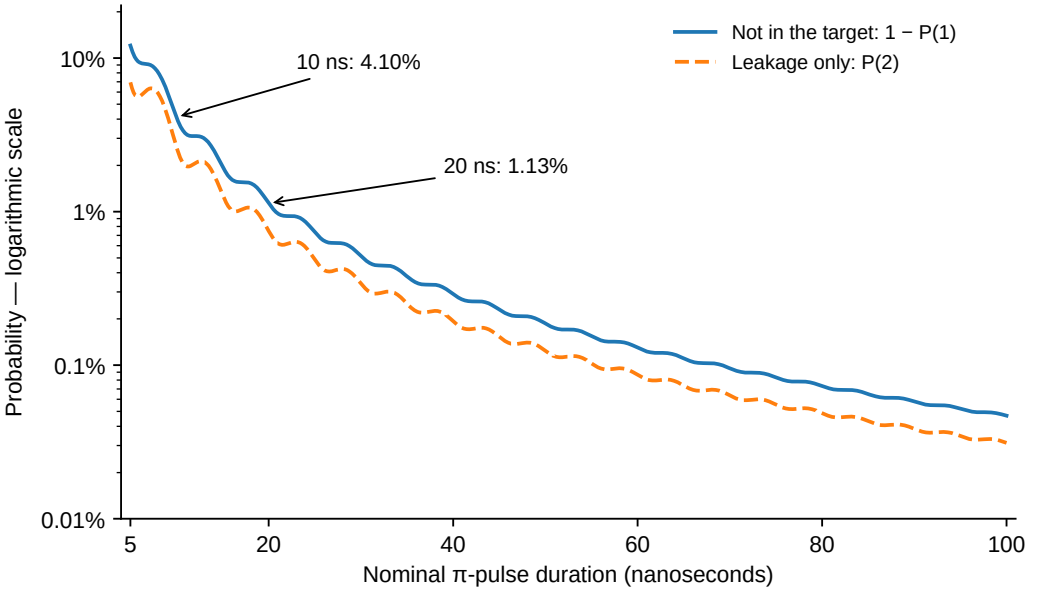


Figure 17.2. A shorter nominal flip can leave the intended space. For Eq. (17.3), each point starts in zero and uses a separate rectangular pulse with $\Omega t_{\pi} = \pi$ and $\Delta/(2\pi) = -200$ MHz. The solid curve gives $1 - \mathrm{P}(1)$; the dashed curve gives only leakage $\mathrm{P}(2)$. Their difference is the residual zero population. The vertical scale is logarithmic: each labeled step multiplies the probability by ten. No relaxation, dephasing, pulse-shape optimization, or measurement error is included. The curves are calculated, not experimental benchmarks or universal speed limits.

Pulse shape gives us more freedom than simply choosing long or short. Motzoi and colleagues' DRAG construction adds a second, phase-shifted control related to how the first envelope changes in time. It suppresses unwanted excitation while correcting effects that a two-level treatment misses. Our rectangular-pulse example does not implement that improvement.[5]

Making the pulse slower also gives uncontrolled interactions more time to act. The useful duration depends on the actual spectrum, available controls, and noise. There is no universally best pulse length hidden in the word qubit. The engineering task is to produce the intended transformation before competing processes significantly change it.

Different carriers, the same obligations

The physical encodings in Table 17.1 solve different parts of that task in different ways. The table identifies examples, not exhaustive definitions. An ion is an electrically charged atom held by trapping fields. A neutral atom can be held by forces produced by a tightly focused light field. A semiconductor quantum dot uses electrical confinement to isolate a carrier whose spin can encode the state. A photonic circuit guides and mixes optical modes. Their control mechanisms are not interchangeable merely because their state vectors share a notation.[6]

Table 17.1**. The label does not identify the physical encoding.**

Example carrier	Where zero and one live	A route to a classical record
Trapped ion	Two selected internal atomic states	State-dependent fluorescence
Neutral atom	Two selected internal atomic states	Fluorescence or state-selective detection
Transmon circuit	Two levels of a nonlinear electrical mode	Microwave resonator response
Semiconductor spin	Two spin states of a confined carrier	Spin-to-charge conversion and sensing
Single photon	One photon in either of two modes	Detection at the corresponding outputs

These are representative encodings and readouts, not rankings. Each row omits auxiliary levels, controls, detectors, and preparation hardware. In the photonic row a missing photon is outside the logical pair; in the atomic rows position is not the internal-state label. Different implementations within a row can use different readout procedures.

The distinction between carrying and controlling information is especially important for atoms. Cooling the atom's motion and initializing its internal state are different jobs. A well-localized atom need not be in the desired internal state; an internally prepared atom need not have motion suitable for the next gate. Each unwanted degree of freedom must either be controlled or shown not to matter for the operation.

An interaction that belongs to the pair

Independent rotations cannot turn two independently prepared pure states into an entangled pair. Something must act jointly: a direct interaction, a shared mediator, or a measurement protocol supplied with the necessary resources. The line joining two gates must have a physical explanation.

A simple mathematical example adds energy J only to the joint state $|11\rangle$. Over an interval τ, this gives

$$U_{\text{pair}} = \text{diag}\left(1, 1, 1, e^{-i\chi}\right), \chi = J\tau/\hbar \tag{17.4}$$

The diagonal lists the factors for 00, 01, 10, and 11 in that order. At $\chi = \pi$, the last factor is minus one. This is controlled Z, or CZ. It leaves every computational-basis probability unchanged at that instant, but adds a relative phase that subsequent gates can use. One-qubit rotations convert it into the controlled NOT already familiar:

$$\text{CNOT} = (I \otimes H)\,\text{CZ}\,(I \otimes H) \tag{17.5}$$

Here H is Hadamard, not the Hamiltonian. The rightmost operation acts first. This identity specifies what a conditional-phase interaction must accomplish; it does not assert that every device literally has only the energy term we wrote. Real gates may use auxiliary states and must return the relevant workspace without unwanted records.

An auxiliary system can change during a gate. What matters at completion is whether its final state distinguishes alternatives that the qubits must keep coherent. A common change independent of those alternatives need not damage the operation. A residual

record that depends on them can reduce the qubits' coherence when the auxiliary system is left out of the description.

For trapped ions, collective motion can mediate an interaction between internal states. State-dependent forces make the motion participate during the gate. The control sequence must finish with the required internal-state transformation and without residual information in the motion that distinguishes its components. Sørensen and Mølmer developed gates that can achieve this without requiring an initially pure motional ground state under their specified conditions.[7]

A 2025 experimental preprint by Hughes and colleagues advances that idea through smooth changes of gate detuning. Their electronically controlled gates suppress residual entanglement between spin and motion, reducing dependence on ground-state cooling. This is not permission to ignore motion: it is a more effective way of arranging how motion enters and leaves the operation.[8]

Neutral-atom gates can instead temporarily excite atoms to Rydberg states, in which an electron occupies a highly excited, spatially extended state. Strong interactions shift the energy of joint excitation. A pulse resonant for one atom may then be off resonance when its neighbor is already excited. This blockade can be used in a controlled phase sequence. The Rydberg excitation is a temporary control resource; it need not be the state in which the qubit spends its storage time.[9]

Superconducting circuits can exchange interactions through electrical coupling, a resonator, or a controllable coupling element. Opening a path for a desired interaction also opens opportunities for unwanted coupling and energy loss. Circuit quantum electrodynamics puts both coherent interactions and microwave readout into this engineered electromagnetic setting.[10]

Photons offer another route. Passive optical elements mix modes readily, but do not automatically supply a deterministic interaction between two independent photonic qubits. Ancillary photons, joint detection, and outcome-dependent operations can supply effective entangling procedures. Such a procedure must count successful events, consumed photons, and failed attempts, rather than calling a detector-conditioned operation an ordinary always-successful gate.[11]

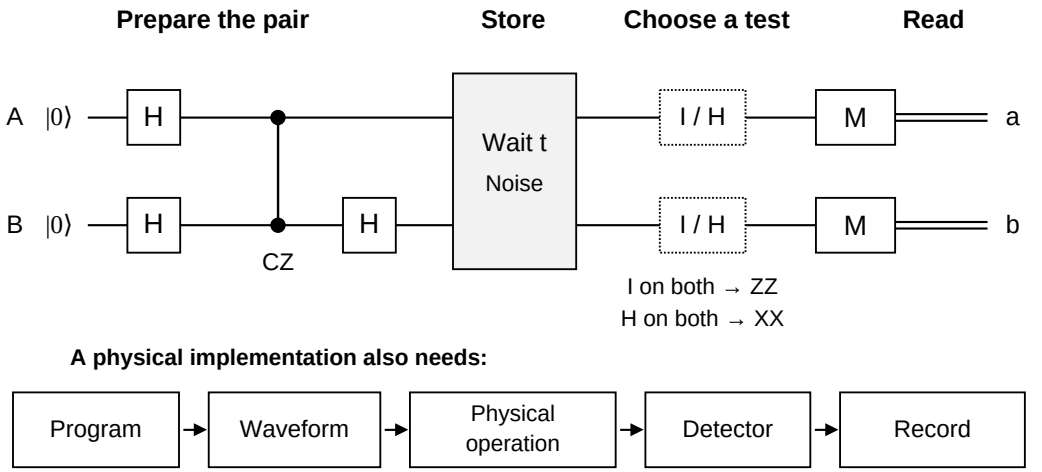


Figure 17.3. One logical task has several physical stages. The preparation uses H on both initially zero qubits, CZ, then H on the lower qubit; Eq. (17.5) makes this equivalent to preparing a Bell-plus pair. The shaded interval is storage. Each trial then uses either no analyzer rotation for ZZ or H on both for XX, followed by Z readout. The lower line identifies the classical control and recording chain needed in a device. It is a functional diagram, not a layout or a timing scale. All output pairs are retained. M denotes measurement in the Z basis.

A record is an engineered conversion

At the end of Figure 17.3, a state has to become an electrical or optical record. In fluorescence readout, a selected internal state repeatedly scatters light while another is comparatively dark. In a circuit, a probe can acquire a state-dependent response from a microwave resonator. In a quantum dot, an allowed tunneling event can turn a spin distinction into a charge distinction; Elzerman and colleagues demonstrated electrical single-spin readout using that conversion.[12]

The detector does not necessarily print the state label. It may supply a photon count, a voltage trace, or a pair of signal quadratures. A calibrated decision rule assigns an outcome. Overlapping distributions, missed events, and state changes during the acquisition can make that assignment wrong.

Model the readout as an ideal measurement of the chosen label followed by classical assignment errors. Suppose it reports one incorrectly with probability ϵ_0 when the input label is zero, and reports zero incorrectly with probability ϵ_1 when it is one. If the true one probability is p_1, the reported probability is

$$r_1 = \epsilon_0 + \left(1 - \epsilon_0 - \epsilon_1\right) p_1 \tag{17.6}$$

The two terms follow by considering both possible true labels. For $p_1 = 0.8$, $\epsilon_0 = 0.03$, and $\epsilon_1 = 0.07$, the report is $r_1 = 0.75$. More repetitions converge toward 0.75, not 0.8. A correction requires calibrated error probabilities and propagates their uncertainty. This is a stipulated assignment model, not a general description of every measurement.[13]

When the two true labels produce identical reporting probabilities, even perfectly known calibration cannot recover the lost population information. If those probabilities are

merely close, inversion magnifies uncertainty. Calibration must therefore establish how well the detector distinguishes the labels, as well as how often it reports either one.

A combined state-preparation-and-measurement score also includes mistakes made before the test began. It cannot be assigned entirely to the detector or entirely to a gate. A missing atom must be distinguished from a dark internal state where the encoding requires it. Likewise, a high-quality result conditional on a photon reaching a detector does not establish that the photon usually reaches it.

The controlled record-making of measurement is not opposed to Chapter 12's explanation of decoherence. It uses that mechanism deliberately. The difficulty is to make the intended record without exposing neighboring stored information to the same light, fields, or electronics. Reset and cooling must also be scheduled while other qubits may still contain useful states.

Two clocks for a waiting qubit

How long can the Bell pair survive that schedule? A single coherence time cannot answer every question. We need a model of the relevant errors, and experiments that test it.

For an illustrative model, let an excited qubit relax toward zero with survival factor $q = e^{-t/T_1}$. Let its off-diagonal coherence decay by $c = e^{-t/T_2}$ after accounting for the known free phase. Assume constant rates, negligible thermal excitation, no leakage, and independent relaxation and pure dephasing. The density matrix evolves as

$$\rho(t) = \begin{pmatrix} 1 - q\,\rho_{11}(0) & c\,\rho_{01}(0) \\ c\,\rho_{10}(0) & q\,\rho_{11}(0) \end{pmatrix} \tag{17.7}$$

The diagonal change transfers excitation probability to zero. Even without pure dephasing, relaxation multiplies coherence by the square root of the survival factor. Independent pure dephasing contributes a further factor. Their exponential decay rates therefore add: $1/T_2 = 1/(2T_1) + 1/T_\phi$, where T_ϕ describes pure dephasing. Thus $T_2 \le 2T_1$: relaxation itself limits coherence, even without the additional dephasing. These are model parameters for specified experiments, not a universal conversion between any two published lifetime numbers.[14]

A Ramsey contrast measurement and an echo sequence can yield different decay times when slowly varying frequency errors are present. The echo cancels some errors that undisturbed waiting does not. Quoting the longer result without naming the sequence changes the meaning of the claim.

Now prepare a perfect Bell-plus pair and let each qubit undergo Eq. (17.7) independently with the same parameters. Let P_{ZZ} mean the probability that the two Z outcomes agree; define P_{XX} similarly. Direct calculation gives

$$\mathrm{P}_{ZZ}(\text{agree}) = 1 - q + q^2,$$
$$\mathrm{P}_{XX}(\text{agree}) = \frac{1 + c^2}{2}. \tag{17.8}$$

We can understand the first expression without a large matrix. The initial zero–zero component continues to give agreement. From the one–one component, agreement occurs when both excitations survive or both decay; disagreement occurs when only one survives. The X result instead depends on the surviving off-diagonal relationship between zero–zero and one–one. Each local channel contributes one factor of c.

Check the prediction. As relaxation continues, the two qubits' Z-measurement outcomes begin to agree more often again. Does that establish that their entanglement is recovering?

The first prediction contains a trap for an unwary tester. Z agreement initially gets worse, then recovers toward certainty as both qubits relax to zero. The pair has not healed. In the long-time limit it is an unentangled pair of zeros, with X agreement only one-half. Figure 17.4 shows why a favorable score for one question is not a complete memory test.[15]

The second setting resolves this particular ambiguity. Agreement in two chosen settings still does not fully characterize an arbitrary state or establish how a gate acts on every input. Additional preparations and measurements are needed for those different claims.

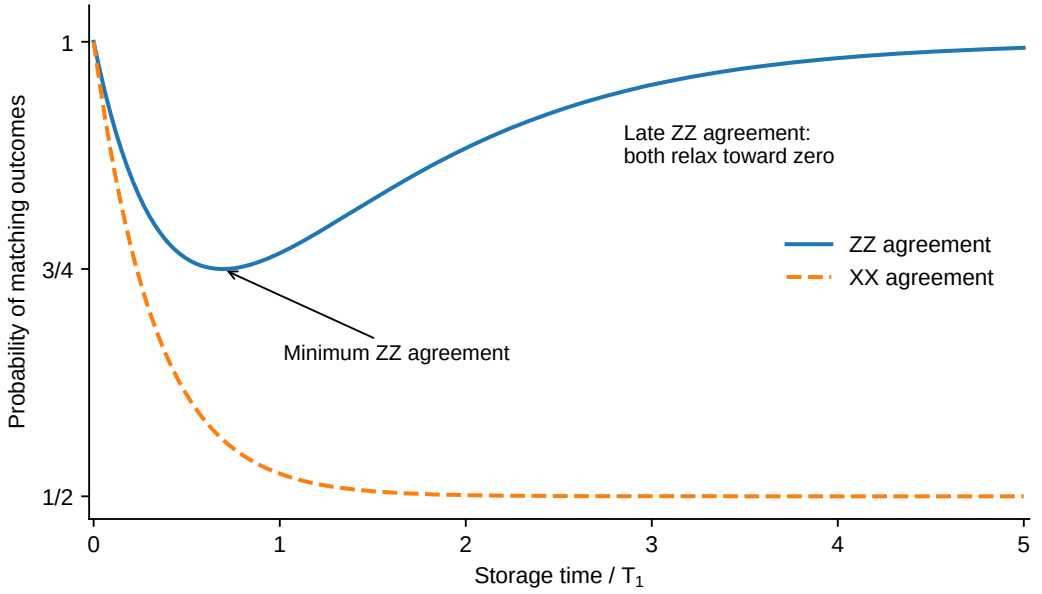


Figure 17.4. Agreement can return without recovery of the Bell pair. The curves use a perfect initial Bell-plus state, identical independent channels from Eq. (17.7), and $T_\phi = T_1$, hence $T_2 = 2T_1/3$. The horizontal axis is storage time in units of T_1. ZZ agreement reaches a minimum of 3/4 at $t/T_1 = \ln 2$, then approaches one as both qubits relax to zero. XX agreement approaches 1/2. Measurements and preparation are ideal; each point uses fresh trials. These are model probabilities, not experimental data or an entanglement measure.

Choose $T_1 = T_\phi = 100$ microseconds and wait twenty microseconds. The model predicts $P_{ZZ} \approx 0.8516$ and $P_{XX} \approx 0.7744$. These figures include storage error only. Suppose each final recorded bit is additionally flipped with independent symmetric probability ϵ. Agreement survives if both bits flip or neither does; disagreement becomes agreement if exactly one flips. For either setting,

$$P_{\text{reported}} = \frac{1}{2} + (1-2\epsilon)^2 \left(P_{\text{true}} - \frac{1}{2} \right) \tag{17.9}$$

At a two-percent flip probability, the reported agreements become about 0.8240 and 0.7529. The first loss occurred in the stored state; the second in the readout model. The same final number could have several causes. Our calculation separates them because we specified the whole sequence. In a real device, correlations between the errors would require a different account.

From good components to a working machine

A qubit that performs well while its neighbors are idle may behave differently while they are driven or measured. An available interaction between every pair does not mean every pair can interact simultaneously. Transport, shared lasers, coupling elements, cooling, and classical decisions all constrain the schedule.

Ransford and colleagues' 2026 trapped-ion processor illustrates this distinction concretely. Its all-to-all connectivity among the physical qubits is realized by moving ions between storage and operation regions. The study separately characterizes transport-related memory errors and disturbance from measurement and reset. Connectivity is a capability of the complete architecture, not a claim that every ion has a permanent direct gate connection to every other.[16]

Characterizing gates also takes more than preparing one state and asking whether it returned. The randomized-benchmarking protocol considered here executes randomly selected Clifford gate sequences, each followed by an inverse calculated from the intended ideal sequence. It then studies how the return probability changes with length. Under its noise assumptions, the decay separates an average gate-error parameter from preparation and readout contributions. Magesan, Gambetta, and Emerson developed a rigorous framework for that inference; it does not make a single fitted number a complete noise model.[17]

Evered and colleagues' 2026 study, first reported in April and published in October, reports neutral-atom entangling gates with estimated raw fidelity 99.854(4) percent and loss-postselected fidelity 99.941(3) percent. The difference is not merely a better estimate of the same quantity: the latter excludes trials with atom loss. The work combines stronger, smoothly shaped controls with state-selective readout and rapid calibration. Its two reported metrics expose part of the implementation account rather than erase it.[18]

A 2025 Nature paper from the PsiQuantum team demonstrates an integrated photonic stack containing sources, interferometers, detectors, and network interfaces. Its quantum benchmarks are explicitly conditioned on detection. The experiment also uses cryogenic superconducting detectors: transmitting information in light does not mean that every part of the apparatus operates at room temperature. Component quality, probability of delivery, and system operation remain distinct achievements.[19]

Keeping a device running adds another timescale. Chiu and colleagues maintained an array of more than three thousand neutral atoms for over two hours by replenishing atoms while protecting nearby stored qubits. That 2025 result concerns sustained operation with replacement, not two hours of undisturbed coherence in the same entangled state. The apparatus must preserve active information while preparing the resources needed next.[20]

Materials remain part of the same story. Bland and colleagues' 2025 transmon study connects longer lifetimes with reduced dielectric loss and improved junction fabrication. A cleaner material stack can improve the interval available for operations. It does not, by itself, specify the fidelity of an entire multi-qubit program.[21]

The device must keep its promises together

Our modest assignment now contains a complete experimental chain. Select physical states. Prepare them. Apply controls whose action is known beyond one truth table. Produce a joint interaction without leaving unwanted records. Preserve the state during everything that follows. Convert selected properties into calibrated records, and replenish what the next run requires.

The requirements pull in different directions. Isolation protects a stored phase; access permits a gate and a readout. A strong interaction can shorten a pulse while reaching unwanted levels. A detector can improve one record while disturbing a neighboring qubit. The achievement is not to maximize each item separately, but to make them work together in the experiment we intend to run.

We have still used individual physical qubits as though their imperfections could only be reduced by improving the hardware. The next step is more surprising: arrange several imperfect physical systems so that the information they jointly carry can outlast any unprotected member. Such an encoded unit is a logical qubit. Its quality depends on the encoding, repeated checks, controls, and noise—not on a universal conversion factor from a count of atoms or circuits.

But how can those checks discover damage without reading the unknown state we want to protect? No-cloning rules out making ordinary backup copies. Measurement can itself remove the needed relationships. The next chapter will show how error correction works within those restrictions, rather than pretending that a sufficiently well-built qubit makes them disappear.

Notes and sources

1. David P. DiVincenzo, “The Physical Implementation of Quantum Computation,” Fortschritte der Physik 48, 771–783 (2000). The five implementation criteria concern characterized qubits, initialization, relevant coherence times, gates, and measurement; two further criteria concern communication. Only this organizing framework is used, not the paper’s historical forecasts or numerical error thresholds. The chapter’s Bell-pair task is an original construction. Full author paper.

2. Jens Koch and colleagues, “Charge-insensitive qubit design derived from the Cooper pair box,” Physical Review A 76, 042319 (2007), doi:10.1103/PhysRevA.76.042319. The transmon is a nonlinear circuit mode with a larger spectrum, not an intrinsically isolated two-level object. Figure 17.1 is an original schematic. The frequency difference in Eq. (17.1) fixes our sign convention for anharmonicity. Author manuscript. Published article.

3. Equation (17.2) defines leakage relative to a selected subspace and a complete normalized state. For the photonic example the subspace contains exactly one photon across two modes. Vacuum and multiphoton sectors are excluded. This is the same single-photon encoding developed in Chapters 1–4. See Emanuel Knill, Raymond Laflamme, and Gerald J. Milburn, “A scheme for efficient quantum computation with linear optics,” Nature 409, 46–52 (2001), doi:10.1038/35051009. Author manuscript. Published article.

4. Equation (17.3) is an original three-level numerical example based on a resonantly driven weakly anharmonic ladder. It uses a rotating-wave approximation, relative coupling $\sqrt{2}$, and a rectangular envelope. With a nominal $\omega_{01}/(2\pi) = 5$ GHz, the stipulated drives are small compared with the carrier frequency. The simulation omits higher levels and dissipative processes; pulse parameters are not fitted to a device. Figure 17.2 uses t_π from 5 to 100 ns. For each pulse duration, diagonalize the Hermitian matrix in Eq. (17.3), resolve the initial zero state in its eigenvectors, apply the energy-dependent phases for that duration, and transform back before squaring the amplitudes. This is equivalent to solving the constant-Hamiltonian Schrödinger equation. The rotating-wave approximation and restriction to two levels are separate approximations: small drive strength relative to the carrier does not by itself imply small unwanted excitation relative to the anharmonicity.

5. Felix Motzoi, Jay M. Gambetta, Patrick Rebentrost, and Frank K. Wilhelm, “Simple Pulses for Elimination of Leakage in Weakly Nonlinear Qubits,” Physical Review Letters 103, 110501 (2009), doi:10.1103/PhysRevLett.103.110501. The DRAG control construction uses additional quadrature and phase/frequency corrections; it is not implemented in Figure 17.2. The full author version’s three-level Hamiltonian supplies the physical motivation for the illustrative model, not its chosen numerical results. Full author text. Published article.

6. Table 17.1 is a representative mechanism map, not a survey of comparative performance. For internal-state and photonic encodings see DiVincenzo (note 1); for spin encoding, Daniel Loss and David P. DiVincenzo, “Quantum computation with quantum dots,” Physical Review A 57, 120–126 (1998), doi:10.1103/PhysRevA.57.120. The readout examples are supported by notes 10, 12, 19, and 20. An implementation may encode a qubit in more than one physical carrier. Spin-qubit proposal.

7. Anders Sørensen and Klaus Mølmer, “Entanglement and quantum computation with ions in thermal motion,” Physical Review A 62, 022311 (2000), doi:10.1103/PhysRevA.62.022311. Shared motion mediates an internal-state operation while suitable dynamics remove residual internal–motional entanglement. Thermal robustness is conditional on the model and control regime, not independence from arbitrary motional noise. Equations (17.4)–(17.5) are a separate illustrative conditional-phase construction, not their full gate Hamiltonian. Author manuscript. Published article.

8. A. C. Hughes and colleagues, “Trapped-ion two-qubit gates with >99.99% fidelity without ground-state cooling,” arXiv:2510.17286v1, submitted 20 October 2025. The cited version is a preprint. It reports electronically controlled gates with smooth detuning ramps and tests their dependence on initial motion. The title’s fidelity is not used as a cross-platform ranking or the success probability of this chapter’s Bell experiment. Versioned record. Full author text. The title’s figure is a protocol-specific benchmark estimate. A fit using sequences through 200 Cliffords differs from a fit using all 500; the authors discuss sequence-length dependence and the assumed state-preparation-and-measurement contribution.

9. Dieter Jaksch, J. I. Cirac, Peter Zoller, S. L. Rolston, Robin Côté, and M. D. Lukin, “Fast Quantum Gates for Neutral Atoms,” Physical Review Letters 85, 2208–2211 (2000), doi:10.1103/PhysRevLett.85.2208. Rydberg interactions provide state-dependent shifts used in entangling protocols. The narrative gives the blockade mechanism, not a claim that every Rydberg pulse realizes the ideal diagonal operator in Eq. (17.4). Contemporary implementation is discussed in note 18. Author manuscript. Published article.

10. Alexandre Blais, Ren-Shou Huang, Andreas Wallraff, S. M. Girvin, and R. J. Schoelkopf, “Cavity quantum electrodynamics for superconducting electrical circuits: An architecture for quantum computation,” Physical Review A 69, 062320 (2004), doi:10.1103/PhysRevA.69.062320. Circuit–resonator coupling supports controlled interactions and state-dependent microwave readout. This early architecture paper is used for the mechanism, not contemporary measured performance. Author manuscript. Published article.

11. Knill, Laflamme, and Milburn (note 3) establish a route to computation with linear optics, ancillary photons, detection, and feedforward. Measurement-assisted photonic operations and deterministic unitary gates on two persistent matter qubits have different success and resource accounting. Figure 17.3 is an original logical circuit; it does not assert that its CZ block has the same direct physical realization in every platform. Original author paper.

12. J. M. Elzerman, R. Hanson, L. H. Willems van Beveren, B. Witkamp, L. M. K. Vandersypen, and L. P. Kouwenhoven, “Single-shot read-out of an individual electron spin in a quantum dot,” Nature 430, 431–435 (2004), doi:10.1038/nature02693. The experiment converts spin information to a charge signal sensed electrically. Fluorescence and circuit-resonator readout are separate mechanisms; see DiVincenzo (note 1) and Blais and colleagues (note 10). Author manuscript. Published article.

13. Equation (17.6) is an original application of total probability to an assumed binary assignment channel. It assumes calibrated, stable conditional error rates and no extra leakage/loss outcome. It does not describe arbitrary measurement back action or identify errors in state preparation. The numerical example is stipulated. Equation (17.9) additionally assumes independent, equal symmetric flips on the two reported outcomes.

14. John Preskill, Lecture Notes for Physics 219/Computer Science 219, Chapter 3, “Foundations II: Measurement and Evolution,” updated October 2018, §§3.4.2–3.4.3 and §3.5. Equation (17.7) composes zero-temperature amplitude damping with independent Markovian dephasing. The pure-dephasing factor is e^{-t/T_ϕ}, so $c=\sqrt{q}\,e^{-t/T_\phi}$. Known phase evolution is compensated. Non-Markovian noise, thermal excitation, echo protocols, leakage, and correlated errors require appropriate alternatives; the stated T_2 relation is not imposed on arbitrary empirical fits. Authored lecture notes.

15. Equations (17.8)–(17.9), Figure 17.4, and the 20-microsecond example are original channel calculations. The initial pair is $|\Phi^+\rangle$, each local map is Eq. (17.7), and preparation and analyzer gates are ideal. Apply the two independent local maps in Eq. (17.7) to the initial Bell-pair density matrix, then sum the matching-outcome probabilities in each chosen measurement basis. ZZ agreement has minimum 3/4 at $q=1/2$, independently of the chosen pure-dephasing rate. Neither plotted quantity is claimed to be an entanglement measure.

16. Anthony Ransford and colleagues, “A 98-qubit trapped-ion quantum computer with all-to-all connectivity,” Nature 655, 81–86 (2026), doi:10.1038/s41586-026-10676-4, published 17 June 2026. The 7 July 2026 publisher correction changes an author’s name, not the device results. The discussion uses its storage/operation architecture, transport, and measurement-reset characterization, not its broader classical-simulation claims. Updated published full text.

17. Easwar Magesan, Jay M. Gambetta, and Joseph Emerson, “Scalable and Robust Randomized Benchmarking of Quantum Processes,” Physical Review Letters 106, 180504 (2011), doi:10.1103/PhysRevLett.106.180504. The full author version is titled “Robust randomized benchmarking of quantum processes.” Average decay parameters require the protocol’s noise assumptions and account for preparation/readout coefficients; they are not automatically worst-case errors or independent failure probabilities that can be multiplied for an arbitrary circuit. Full author text. Published article. In the Markovian description, gate-independent and time-independent errors give the basic exponential decay. The extension for sufficiently weak variation across gates and time includes first-order corrections.

18. Simon J. Evered and colleagues, “High-fidelity entangling gates and nonlocal circuits with neutral atoms,” Science 394, 197–201 (8 October 2026), doi:10.1126/science.aei1312; initial author version arXiv:2604.25987v1, submitted 28 April 2026. The principal fidelities quoted here also appear in the published abstract. The detailed benchmarking and selection account follows arXiv:2604.25987v1; agreement with the final journal methods and supplement has not been established for this edition. Reported fidelities are 99.854(4)% raw and 99.941(3)% after loss postselection; parentheses are the authors’ reported uncertainty in the final digits. Those values are not identified with the success probability of a complete circuit, and their ratio is not used to infer a loss rate. Versioned record. Full author text. These estimates come from echo randomized-benchmarking fits, with results averaged across eight CZ gate sites; loss postselection retains trials in which neither atom is lost at the end of the sequence. Published article.

19. The PsiQuantum team, “A manufacturable platform for photonic quantum computing,” Nature 641, 876–883 (2025), doi:10.1038/s41586-025-08820-7, published 26 February 2025. The published full text explicitly conditions its quantum benchmarks on photon detection and describes heralded, nondeterministic generation and superconducting detectors. Integrated building blocks are not presented here as a complete fault-tolerant machine. No conditional fidelity is substituted for unconditional delivery probability. Published full text.

20. Neng-Chun Chiu and colleagues, “Continuous operation of a coherent 3,000-qubit system,” Nature 646, 1075–1080 (2025), doi:10.1038/s41586-025-09596-6, published 15 September 2025. The reported multi-hour array operation uses replenishment, separated preparation/storage zones, shielding, and decoupling. It is not a claim that one unchanged many-qubit state retained coherence for the full operating duration. Only this operational distinction is used here. Published full text.

21. Matthew P. Bland and colleagues, “Millisecond lifetimes and coherence times in 2D transmon qubits,” Nature 647, 343–348 (2025), doi:10.1038/s41586-025-09687-4, published 5 November 2025. The published article connects tantalum-on-silicon devices and junction processing with reduced loss and improved coherence. The manuscript uses that materials mechanism, not a universal lifetime for every transmon, an array-wide simultaneous benchmark, or a demonstrated full-program accuracy. Versioned record. Full author text. The coherence improvement highlighted here is measured with Hahn echo, and the article distinguishes the results for its best individual qubit from the averages over the device sample. Published article.

CHAPTER 18

Repairing Information Without Reading It

A repairer receives a register containing an unknown quantum state. Something may have gone wrong, but asking the register what it contains would risk destroying precisely what must be saved. The repairer needs a different question: not “What is the message?” but “Which relationships have been disturbed?”

That distinction makes quantum error correction possible. A carefully chosen measurement can reveal damage while leaving the encoded alternatives unresolved. Its result supplies instructions for recovery, not a description of the amplitudes.

The last chapter improved the physical carrier. Here we improve the organization of the information it carries. We will first build a code that handles one restricted kind of error. Its failures will show why a complete quantum code must protect more than a classical pattern. Finally, we will ask what happens when the checking apparatus is imperfect too. Shor’s 1995 construction showed, under ideal recovery, how an unknown qubit could be protected against arbitrary errors on any one of nine constituent qubits without cloning it.[1]

One state spread over three carriers

For a classical bit, repetition provides an elementary safeguard: store 000 or 111. If exactly one position flips, majority voting recovers the original bit. But we cannot protect a qubit by measuring it in Z, making three copies of that result, and later guessing the lost phase.

Instead, take $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ and two auxiliary qubits prepared in zero. Apply controlled NOT from the input to each auxiliary. Linearity gives

$$|\psi\rangle|00\rangle \rightarrow \alpha|000\rangle + \beta|111\rangle = |\psi_L\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1. \tag{18.1}$$

The subscript L means logical: the information we want to preserve is now encoded in a joint state. Three physical qubits provide eight computational-basis labels, but this code uses only the two-dimensional subspace spanned by $|000\rangle$ and $|111\rangle$. There is still one unknown qubit, not three independent copies of it.

The controls need not know α or β. They implement the same encoding for every input. We will initially suppose that encoding, checks, and recovery are perfect, and that the storage interval introduces at most one X error—a flip of one physical zero/one label. This three-qubit repetition code will not yet protect against phase errors.[2]

Reading all three physical qubits and taking a majority would recover a classical label, not the unknown state. It would distinguish 000 from 111 and remove their relative-phase coherence from the retained register. We need to read less.

Ask whether the entries agree

Both allowed codewords satisfy two relations: the first and second bits agree, and the second and third agree. Those statements remain certain in any superposition of the two codewords. A check that reports only disagreement can therefore leave their amplitudes untouched.

In the operator language of Chapter 16, these are the stabilizers $Z_1 Z_2$ and $Z_2 Z_3$. Each returns the same encoded state:

$$Z_1 Z_2 |\psi_L\rangle = |\psi_L\rangle , Z_2 Z_3 |\psi_L\rangle = |\psi_L\rangle . \tag{18.2}$$

The product of two Z signs is plus when the labels agree and minus when they differ. We record plus as syndrome bit zero and minus as syndrome bit one. A syndrome is the collection of check results. It identifies a pattern of violated relations; it is not a readout of the logical zero or one.[3]

To measure the first relation, prepare a fresh auxiliary in zero. Apply controlled NOT from data qubit 1 to that auxiliary, then from data qubit 2 to the same auxiliary. Its final bit is their XOR. Only now measure the auxiliary. A second auxiliary similarly checks qubits 2 and 3. Figure 18.1 shows all four interactions.

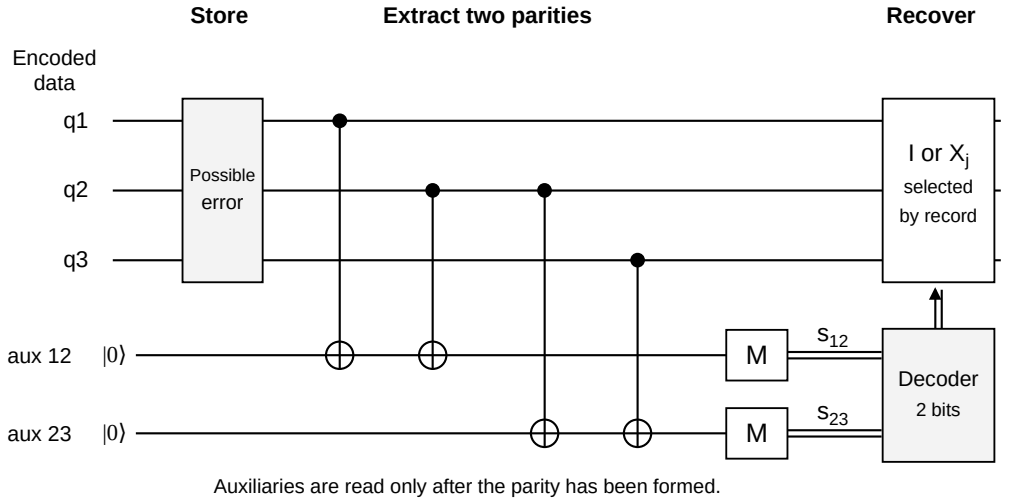


Figure 18.1. The auxiliaries read parity, not the individual labels. The data register is already encoded before the displayed interval. After a possible data error, four controlled NOTs write $s_{12} = q_1 \oplus q_2$ and $s_{23} = q_2 \oplus q_3$ into fresh zero auxiliaries. Only the auxiliaries are measured in Z. Filled dots are controls; circled crosses are target flips. Bare wire crossings are not controls. Double lines carry classical bits to a decoder that selects the recovery in Table 18.1. The diagram is an ideal logical protocol, not a physical layout or a fault-tolerant implementation. M denotes measurement in the Z basis.

The auxiliary cannot tell whether agreement came from two zeros or two ones: either pair flips it an even number of times. For an undamaged codeword, it finishes in zero in both cases. That shared final auxiliary state is what permits the two data amplitudes to remain coherent. Merely hiding a more detailed measurement result from the repairer would not provide the same protection.

The timing matters. Measuring the first auxiliary after only its first controlled NOT would reveal the first data label. Leaving an uncontrolled record of that intermediate value would be harmful too. Completing both interactions coherently leaves only the parity. The apparatus must implement the less revealing question, not merely give its output a less revealing name.

Table 18.1. **A syndrome selects a recovery under the one-flip assumption.**

Error during storage	s_{12}	s_{23}	Recovery
None	0	0	I
X_1	1	0	X_1
X_2	1	1	X_2
X_3	0	1	X_3

The table assumes either no error or one physical X error, with perfect check extraction and readout. I means no operation. The same row applies to every logical input. Multiple errors and phase errors are not covered by this table's decoding assumption.

For example, a flip on qubit 2 changes the encoded state to $\alpha|010\rangle+\beta|101\rangle$. Both components disagree on both checked pairs. Including the two auxiliaries after extraction gives

$$(\alpha|010\rangle+\beta|101\rangle)|11\rangle_{\text{aux}}. \tag{18.3}$$

The measured syndrome is certainly 11, regardless of the amplitudes. Apply X_2 and the original encoded state returns. If the input was $(\sqrt{3}|0\rangle+i|1\rangle)/2$, the factor i survives along with the unequal weights. Recovery has not estimated either one. Reversing the original encoding, when desired, returns that input to the first carrier and the other two to zero.[4]

A small rotation is not a prewritten error label

Real disturbances need not be complete flips. Suppose qubit 2 undergoes a coherent rotation, by an angle that the repairer does not know. The rotation is a sum of an unchanged contribution and a flipped contribution. After extracting the syndrome, the full state is

$$\begin{aligned}|\Psi\rangle &= \cos(\epsilon/2)|\psi_L\rangle|00\rangle_{\text{aux}} \\ &\quad -i\sin(\epsilon/2)X_2|\psi_L\rangle|11\rangle_{\text{aux}}.\end{aligned} \tag{18.4}$$

Here ϵ is the rotation angle, and the underlying error operation is $\cos(\epsilon/2)I - i\sin(\epsilon/2)X_2$. A syndrome measurement gives 00 with probability $\cos^2(\epsilon/2)$ and 11 with probability $\sin^2(\epsilon/2)$. Neither probability depends on the logical state. In the second branch apply X_2; in the first do nothing. Both leave the same $|\psi_L\rangle$ intact.

This does not mean the rotation secretly selected a complete flip beforehand. The error began as a coherent combination. The check correlated its two correctable contributions with distinguishable records. The recovery works for both records, without learning the angle or throwing either branch away.[5]

We can express the same recovery coherently, before measuring the auxiliaries: condition X_2 on their 11 label. The resulting state factors as

$$|\psi_L\rangle[\cos(\epsilon/2)|00\rangle - i\sin(\epsilon/2)|11\rangle]_{\text{aux}}. \tag{18.5}$$

The unknown logical state and the error information are now in different factors. This is the central separation. The auxiliaries may be measured and recycled without reading the logical amplitudes. Recycling is a physical task, with the entropy and resource accounting discussed in Chapter 8; it is not free deletion of the record.

The argument also preserves correlations with an untouched reference. Replace the two scalar amplitudes by two reference-system vectors and repeat the calculation: the recovery still factors out the same joint logical–reference state. Protecting a qubit means preserving those relationships, not merely obtaining a plausible local probability table.[6]

When the syndrome points to the wrong repair

Now let qubits 1 and 2 both flip. Their pair still agrees; qubits 2 and 3 disagree. The syndrome is 01—the same as a lone X_3. The table prescribes X_3, producing the combined operation $X_1X_2X_3$. It exchanges $|000\rangle$ with $|111\rangle$: a logical X error. The code returns to its allowed subspace, but with the wrong logical transformation.

A syndrome therefore need not identify a unique physical history. Recovery succeeds only when the histories it confuses have an equivalent effect on the stored information. An all-zero syndrome is not proof that nothing happened either: three flips leave both parity checks satisfied.

For a simple quantitative test, let each data qubit flip independently with probability p during one storage interval. Keep all checks and recoveries perfect. Recovery fails logically when two or three flips occur, so

$$p_L = 3p^2(1-p) + p^3 = 3p^2 - 2p^3. \tag{18.6}$$

At p = 0.01, this gives 0.000298: about three logical flips per ten thousand intervals instead of one unprotected flip per hundred. Figure 18.2 follows the whole comparison. This is a logical-flip probability, not the infidelity of every possible input; a logical plus state, for instance, is unchanged by a logical X.[7]

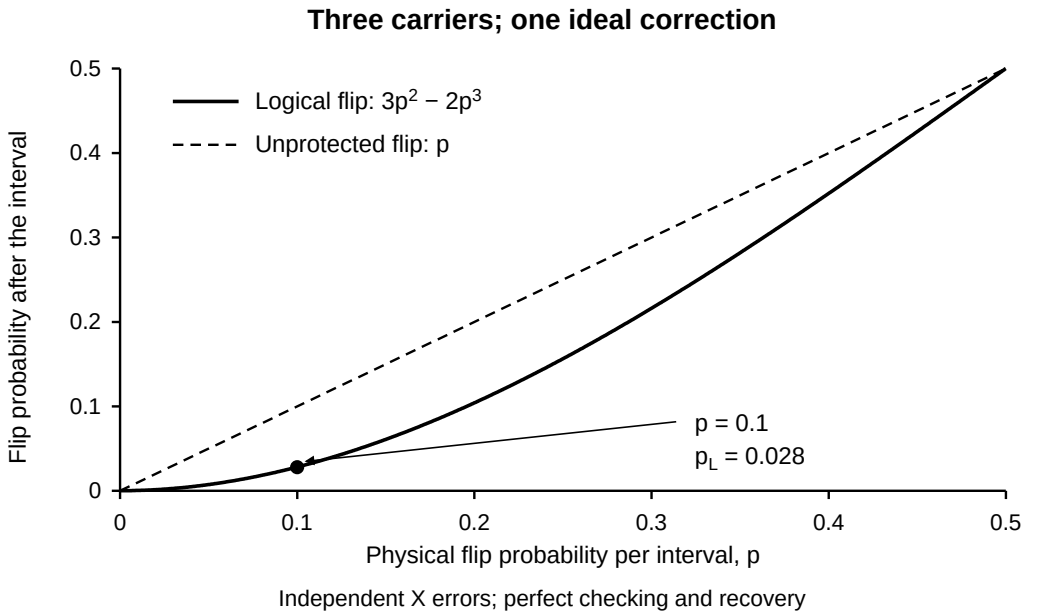


Figure 18.2. An improvement with its assumptions attached. The solid curve is the logical-X probability $p_L = 3p^2 - 2p^3$ after one ideal recovery; the dashed line is the unprotected physical-X probability p. Independent, identical data flips and perfect encoding, checks, and recovery are assumed. All error patterns are included. At p = 0.1 the two probabilities are 0.028 and 0.1. The crossing at one-half is a property of this restricted model, not a threshold for a noisy universal quantum computer. These are calculated probabilities, not experimental data.

Independence is doing real work in this estimate. Suppose instead that one rare common disturbance flips all three carriers together. Every such event passes the parity checks and acts logically, even when its probability is very small. Tripling the number of carriers does not turn that probability into its cube. The physical pattern of errors matters as much as their average frequency.

There is a more immediate limitation. A single phase flip gives

$$Z_1|\psi_L\rangle = \alpha|000\rangle - \beta|111\rangle. \tag{18.7}$$

Both parity checks still report agreement. The logical phase has changed without announcing itself. Repetition has protected one distinction while leaving another exposed. Rotating the encoding with Hadamards exchanges the roles of X and Z errors, but does not by itself protect both at once.[8]

Protect the signs as well as the labels

Shor's nine-qubit code combines two layers of protection. Group nine physical qubits into three blocks of three. Within each block use one of two fixed states, which differ by a relative sign:

$$
\begin{aligned}
|B_{\pm}\rangle &= \big(|000\rangle \pm |111\rangle\big)/\sqrt{2}, \\
|0_L^{(9)}\rangle &= |B_+\rangle \otimes |B_+\rangle \otimes |B_+\rangle, \\
|1_L^{(9)}\rangle &= |B_-\rangle \otimes |B_-\rangle \otimes |B_-\rangle.
\end{aligned}
\tag{18.8}
$$

Encode the unknown state as $\alpha|0_L^{(9)}\rangle + \beta|1_L^{(9)}\rangle$. These are two known codewords, not three copies of an unknown block state. The amplitudes still appear only once in the superposition between the logical alternatives.[9]

Six Z-pair checks test adjacent labels inside the three blocks. They locate a single physical X error within its block, just as before. But now each block also has a sign that can be checked without reading which logical state was encoded.

The product $X_1X_2X_3$ has eigenvalue plus one on $|B_+\rangle$ and minus one on $|B_-\rangle$. Measuring that block sign alone would reveal the logical alternative. Instead, compare signs between blocks: measure $X_1X_2X_3X_4X_5X_6$, and then $X_4X_5X_6X_7X_8X_9$. Their outcomes say whether neighboring block signs agree. Figure 18.3 collects all eight checks.

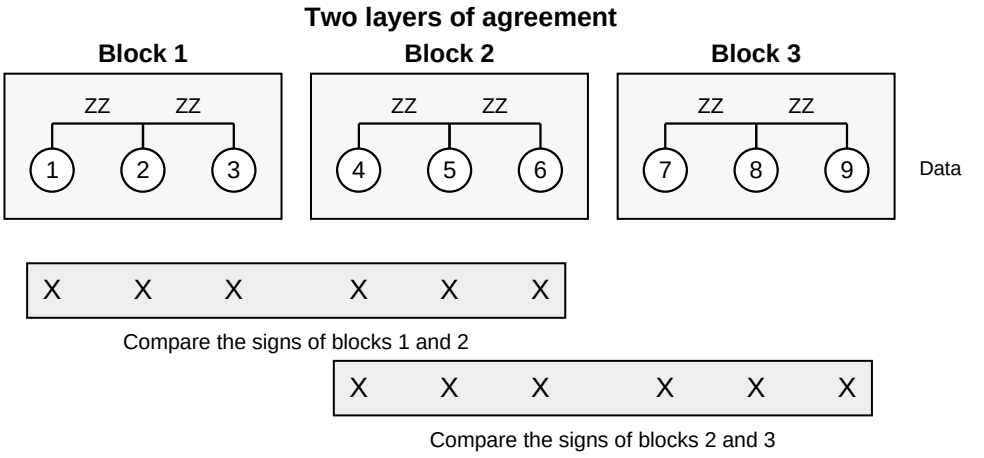


Figure 18.3. Eight checks leave one logical qubit. Three outlined boxes group the nine data qubits. The six short links denote Z-pair parity checks within blocks. The two horizontal bands denote products of X on the first six and last six qubits, comparing block signs rather than reading them separately. All eight operators stabilize both codewords in Eq. (18.8). Any X-type and Z-type check overlap on zero or two data qubits, so they commute. Lines and bands show operator support, not gates already supplied by the hardware. Auxiliary qubits and a fault-tolerant measurement circuit would be additional resources.

A phase flip on qubit 2 changes the sign of the first block in either logical codeword. The first six-X check becomes negative and the second stays positive. A Z applied anywhere in the first block restores the state. The check need not distinguish Z_1 from Z_2 or Z_3:

they act identically on the encoded subspace. This useful ambiguity is called degeneracy.[10]

The X-type checks and Z-type checks commute because their overlaps contain an even number of anticommuting one-qubit factors. Thus their relationships can be measured together without requiring simultaneous definite values of individual X and Z labels. The six within-block constraints restrict each block to the span of 000 and 111. Within that space, the two sign comparisons allow only all three blocks in the plus state or all three in the minus state, and their superpositions. Those are exactly the two logical codewords in Eq. (18.8).

A Y error combines an X and a Z, with $Y = i\,X\,Z$. It is handled by the corresponding two kinds of syndrome. More generally, any operator acting on one qubit can be expanded as

$$E = a\,I + b\,X + c\,Y + d\,Z. \tag{18.9}$$

The coefficients can be complex. Correcting this operator span with one consistent recovery handles continuous rotations and the one-qubit operators describing environmental noise, not just a classical lottery over four error names. Knill and Laflamme's correction conditions formalize when the syndrome can be independent of the logical information. The nine-qubit code satisfies them for every error confined to one physical qubit, including an unknown location. Errors affecting several qubits need not be correctable.[11]

The demand is stronger than showing that each named error has some inverse. All correctable contributions must use one fixed recovery procedure, which may choose different corrections for different syndromes. If two error contributions are coherent, their overlap matters too. A check that leaves different records for the logical alternatives would damage a superposition even if it restored each alternative separately. The correction conditions require those overlaps to depend on the error contributions without carrying information about the logical state. Equation (18.5) shows that separation explicitly in the smaller code.

Environmental noise can require several operators, each describing a possible environmental branch, whether or not the surroundings are measured. Each branch operator can have the form in Eq. (18.9), but need not be unitary; the full noisy evolution need not be one reversible gate. Relaxation illustrates the distinction. An excitation may leave a qubit and enter its surroundings; the branch with no emitted excitation can also change relative amplitudes. Neither branch is simply an ordinary bit flip. Each nevertheless acts on the affected two-dimensional system through an operator of the form in Eq. (18.9). A recovery that corrects the entire span handles both branches and their external correlations. It need not retrieve the emitted excitation or identify the original logical state.

The repairer can make mistakes

So far the checking apparatus has been perfect. A real auxiliary may start in the wrong state; a gate may disturb the data; a detector may report the wrong syndrome. Repeating our ideal recovery box does not make those faults disappear.

Worse, an interaction can spread a fault. For controlled NOT, an X on the control before the gate becomes X on both outputs. A Z on the target becomes Z on both. Written as propagation identities, with C denoting controlled NOT,

$$\begin{aligned} C\left(X \otimes I\right)C^{\dagger} &= X \otimes X\,, \\ C\left(I \otimes Z\right)C^{\dagger} &= Z \otimes Z\,. \end{aligned} \tag{18.10}$$

An auxiliary that interacts successively with several data qubits can therefore carry a single fault into several positions. A naïve measurement of a six-qubit check may destroy the protection its result was meant to support. Fault tolerance means designing preparations, measurements, gates, and recovery so that faults remain within the correction capability, under a specified noise model. Verified auxiliary states, additional flags, and carefully ordered interactions are ways to constrain this propagation.[12]

Consider a bare way to measure the first six-X check. Prepare one auxiliary in plus, use it as the control of six controlled NOTs targeting data qubits 1 through 6 in order, and read the auxiliary in X. Ideally this extracts the required parity. But an X fault on the auxiliary just after the third interaction propagates X onto data qubits 4, 5, and 6. On this code their product is a logical Z: it changes a protected relative phase while leaving every stabilizer satisfied. The auxiliary's X readout can still report plus. A correct ideal measurement circuit can therefore fail as a fault-tolerant one.

Repeated checks supply another tool: time. In the three-qubit example, a data flip on qubit 2 changes both check values and, if not yet corrected, keeps them changed. A single erroneous report from one check can instead create a one-round blip. Figure 18.4 compares those two particular histories.

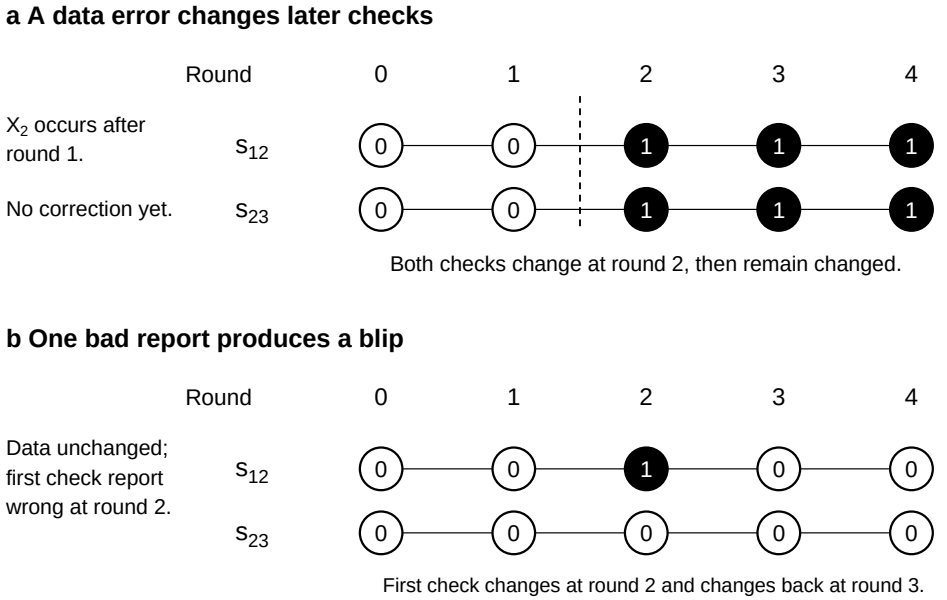


Figure 18.4. A history can distinguish a data change from a bad report. The register begins with known syndrome 00. In the upper example, X_2 occurs between rounds 1 and 2; no correction is applied during the displayed window. Both parities remain changed. In the lower example, the data remain untouched but the first check is misreported only in round 2. The indicated events are changes relative to the preceding round, not additional measurements of the logical state. These stipulated histories illustrate inference from time; longer or correlated fault sequences can make different physical histories ambiguous.

A decoder uses the pattern of check changes, the circuit, and an error model to infer a recovery. The inferred recovery need not identify the error that actually occurred. Surface-code memories extend this idea across space and time: local parity checks constrain possible error chains, while certain chains crossing the encoded region act as logical errors without leaving a remaining endpoint syndrome. Noisy measurements make repeated rounds part of the decoding problem.[13]

This is why demanding an all-zero syndrome after every cycle is the wrong standard. A long experiment can contain many detected physical faults while the logical information survives. Conversely, a decoder can choose the wrong recovery and return every check to its expected value. The final checks then look satisfactory, but the logical state has changed. Testing the memory requires logical measurements in complementary settings, not just a count of how often the checks agree.

The selected correction can sometimes be tracked in a classical record instead of immediately applied as another physical gate. Subsequent operations and interpretations must then respect that record. Offline decoding can establish a memory's performance; a running computation also needs sufficient decoding throughput and timely answers whenever a later operation depends on them.

More hardware must buy more protection

For the stabilizer codes considered here, distance is the smallest number of physical positions on which an undetectable, nontrivial logical Pauli operation can act. Stabilizers themselves do not count: they leave every encoded state unchanged. Under ideal syndrome measurements, distance d permits correction of any t unknown-position errors when $2t < d$. The nine-qubit code has distance three. Our earlier three-qubit repetition encoding has quantum distance one: a single Z already acts logically without a syndrome.

In a planar surface-code patch, data errors can form a chain whose interior leaves no unmatched check events. An appropriate chain joining opposite boundaries can act on logical information without a detectable endpoint. Enlarging the patch can force the shortest harmful chain to cross more data qubits. Distances three, five, and seven therefore require at least three, five, and seven affected positions, respectively, for an undetectable logical Pauli operation under ideal checks. Noisy checks also require the time history used by the decoder; distance alone is not an experimentally measured failure probability.

Distance does not count every resource. The nine data qubits still need auxiliary systems and checks; protecting gates adds more work. A single logical-qubit count conceals the chosen error model, target reliability, operation, and available time. There is no fixed exchange rate between physical and logical qubits.

Computation adds a further demand: change the logical state deliberately without losing this protection. An operation can preserve the code space and still spread one hardware fault into several data errors. Logical gates therefore need fault-tolerant implementations of their own; a good idle memory is not automatically a reliable processor. Auxiliary preparations and classical decisions used during those gates belong to the same resource account.

Fault-tolerance theorems establish that sufficiently accurate components can support longer computations to a chosen accuracy, under specified assumptions about faults, fresh auxiliaries, and control. At a fixed target accuracy, the concatenated-code theorem cited here bounds the number of noisy circuit locations by the ideal circuit size multiplied by a fixed power of its logarithm. The locations include preparations, gates, storage steps, and measurements. This is an asymptotic guarantee within the theorem’s model, not a resource estimate for every machine. The threshold belongs to a code, circuit, decoder, and noise model—not to the word quantum. The simple formula in Eq. (18.6) assumed a perfect repairer and is not such a theorem.[14]

From repeated records to an error rate

Experiments must test whether enlarging the protection improves the stored information. The Google Quantum AI study in Nature’s 2025 volume prepared known logical X or Z eigenstates, repeated parity-extraction cycles, and finally read the data qubits. A decoder combined those final readings with the syndrome history. The measured question was

whether its corrected logical result agreed with the preparation. The reported duration sweep extended to 250 cycles; each basis-averaged point comprised 100,000 repetitions. The distance-seven memory used 49 data qubits, 48 check qubits, and four additional qubits for leakage removal.[15]

For each chosen duration, repeated preparations supply a count of correct and incorrect logical results. Their disagreement fraction estimates $p_L(t)$, where t counts correction cycles. This is already an inference: a detector supplies physical records, while the encoding and decoder determine which records count as a logical disagreement. The result tests this memory-and-decoder combination.

To understand the next inference, consider a simple model with a constant independent probability ε of reversing the tested logical label in each cycle. An odd number of reversals gives disagreement; an even number restores the label. Let D_t be the probability of agreement minus the probability of disagreement. One more cycle multiplies this difference by 1−2ε. With ideal boundaries, $D_0 = 1$, so

$$p_L(t) = \frac{1 - (1 - 2\varepsilon)^t}{2}$$

This elementary derivation explains why the accumulated disagreement is not simply tε at all durations. The probability approaches one-half as the initial label becomes uninformative. The linear approximation $p_L(t) \approx t\,\varepsilon$ is useful only while the accumulated error is small. It also explains why a per-cycle estimate is not itself an observed fraction of damaged hardware components.

For an exponential fit allowing a boundary factor A, write $1 - 2\,p_L(t) = A(1 - 2\varepsilon)^t$. Taking logarithms gives a straight line. Its intercept is the logarithm of A, where A is the extrapolated probability of agreement minus that of disagreement at zero cycles; the slope b gives $\varepsilon = (1 - \exp(b))/2$. A duration-dependent drift or correlated fault process can invalidate this simple description. Residuals—the differences between observations and the fitted curve—therefore carry physical information rather than being inconvenient points to hide.

The study fitted each code and logical basis separately, then averaged the inferred rates. Its author supplement also enlarged the fit uncertainty to account for residual variation beyond counting noise. Figure 18.5 replots those tabulated estimates; it does not recreate the underlying shot records. The lower rates at larger distance are visible across the displayed preparations and code locations, including variation among small patches.

A further fit compares distances. If each increase of two in d reduces the rate by a factor Λ, then $\log \varepsilon_d$ is linear in $(d+1)/2$ with slope $-\log \Lambda$. The published neural-network result was $\Lambda = 2.14 \pm 0.02$ and $\varepsilon_7 = (0.143 \pm 0.003)$ percent per cycle. These uncertainties concern specified fitted quantities; they are neither guarantees for every shot nor a complete allowance for future changes in the device.

That distance-seven result used offline decoding. Real-time distance-five operation was a separate demonstration. The evidence supports improved encoded memory over the tested sizes and conditions. It leaves further tasks: maintaining the error model as hardware grows, obtaining timely decoding decisions, and performing protected logical gates.

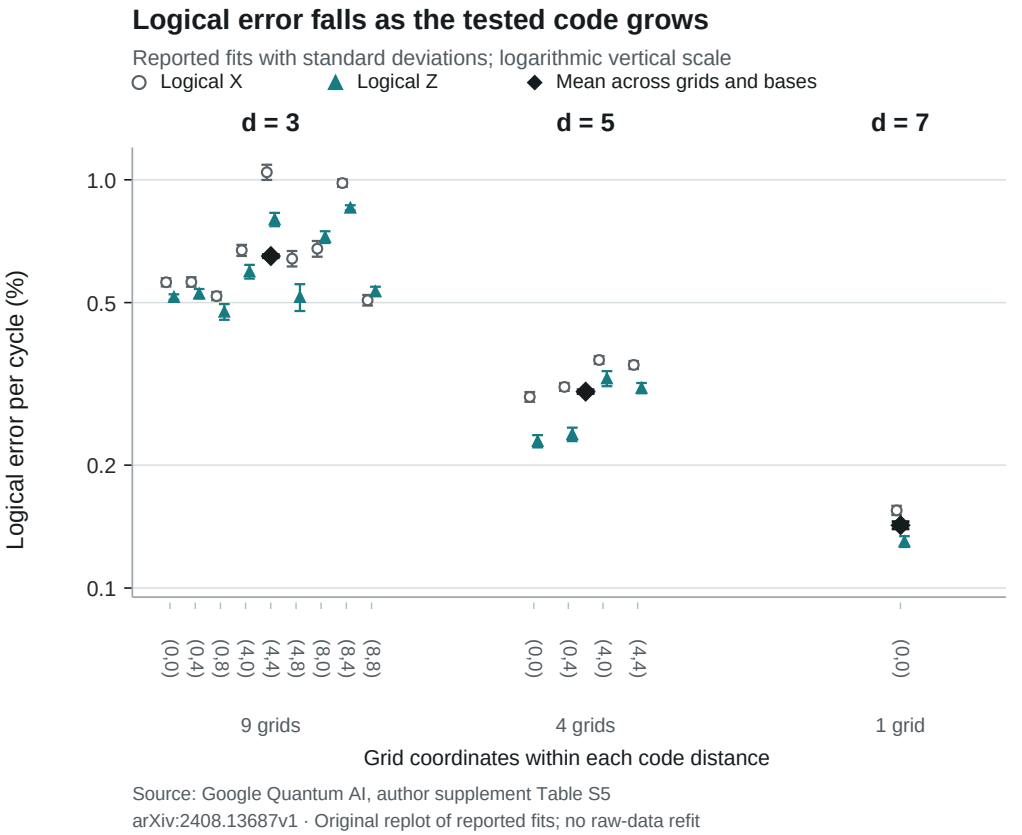


Figure 18.5. Logical performance improves as the tested code grows. Small markers show reported neural-network-decoder estimates for each grid and logical basis; bars show their reported standard deviations. Diamonds are this book's arithmetic means of the rounded table entries, with uncertainties propagated as specified in note 15. Rates are inferred from experimental memory records, not direct counts of individual physical faults. Source: author supplement Table S5 to Google Quantum AI and Collaborators, arXiv:2408.13687v1. The figure is an original replot of tabulated numbers, not a new raw-data analysis.[15]

Keeping the checks trustworthy is itself an ongoing control task. Sivak and colleagues' July 2026 study used reinforcement learning driven by detection events in repeated memory experiments; each shot prepared the state afresh. Against injected drift, the controls reduced fluctuations in the logical error rate. The feedback was the check-event pattern—not unknown logical amplitudes. Uninterrupted long computations and larger codes were simulated, not demonstrated in those memory experiments.[16]

A flagged loss is a different problem

Chapter 17 distinguished leakage and particle loss from an ordinary bit flip. A reliably flagged failure also identifies where protection is needed. Such an erasure is easier to locate than an unannounced error: an ideal distance-d code can correct up to d−1 erased positions, rather than only the smaller number of arbitrary unknown-position errors. But the flag must be accurate, and effects on interacting neighbors still count.[17]

Recovery must handle every error compatible with the record. When locations are unknown, two candidates can act on different sets; their comparison can therefore involve twice the assumed number of affected positions. For a known erased set, both candidates stay within that same set, so the comparison does not double its size. The code must exclude a nontrivial logical difference in either case.

A location flag does not supply the erased state. Recovery must use the quantum relationships retained in the other carriers. For the nine-qubit code, each individual data qubit is maximally mixed for every logical input: an isolated carrier does not reveal which logical state was encoded. This is different from our restricted three-qubit code, whose individual Z probabilities still depend on the input. Protecting against general local damage requires a stronger distribution of the information than protecting against flips alone.

The condition concerns the entire erased set, including correlations inside it. Looking separately at two maximally mixed carriers would not establish that their joint state is independent of the logical input. The Shor code has that stronger independence for any two lost positions, consistent with its distance-three guarantee.

An August 2026 Nature experiment by the D-Wave Quantum Inc. team demonstrates why this distinction reaches inside a gate. Each dual-rail qubit used one excitation distributed between two microwave cavities; losing that excitation left the encoding. The researchers implemented an entangling gate while retaining a strong hierarchy between detectable losses and residual unflagged errors. They also examined how a loss could dephase the other qubit. Their gate measurements and accompanying surface-code simulations support different claims: an experimentally characterized building block, and predictions for a larger correction scheme. Removing flagged trials from a fidelity estimate does not make those losses disappear from a working computer’s resource account.[18]

Keep the state, give the damage a record

The repairer can now answer the opening question. Encoding makes particular relationships certain for every logical input. The checks distinguish the error effects that require different repairs while leaving the protected superposition unresolved. Recovery

then returns its role to the encoded register, including correlations with systems the repairer never touched.

Nothing has cloned an unknown state. Nothing has reconstructed lost amplitudes from a majority of measured labels. Extra physical systems have made the error information separable from the information being protected. The mechanism succeeds only while its error assumptions, controls, and decoding remain adequate.

Storage and transmission now become closely related tasks. A qubit travelling through a noisy link also needs its relationships preserved, while failures, resources, and classical records must be accounted for. The next chapter takes the protected message out of the memory and asks how it can reach another place.

Notes and sources

1. Peter W. Shor, “Scheme for reducing decoherence in quantum computer memory,” Physical Review A 52, R2493–R2496 (1995), doi:10.1103/PhysRevA.52.R2493. This original nine-qubit construction corrects arbitrary errors on one constituent qubit. The opening’s repairer is an explanatory situation, not a historical reconstruction. Original publication.

2. John Preskill, Quantum Information, Chapter 7, “Quantum Error Correction,” updated March 2026, §§7.1–7.2. Equation (18.1) is the three-qubit repetition isometry, not a universal cloning map. Initial blank carriers, ideal encoding, and the restricted X-error model are specified separately. Logical information occupies a subspace; the physical carriers are not three independent instances of the input. Updated draft lecture chapter.

3. Daniel Gottesman, Stabilizer Codes and Quantum Error Correction, Caltech PhD thesis (1997), arXiv:quant-ph/9705052v1 (28 May 1997), Chapters 3 and 5. The two commuting observables here are $Z_1 Z_2$ and $Z_2 Z_3$, with identity on the unlisted position. Their eigenvalues are converted to bits by $+1 \mapsto 0$ and $-1 \mapsto 1$. Measuring a parity coherently is different from reading and forgetting its constituents. Full author thesis.

4. Figure 18.1, Table 18.1, and Eq. (18.3) are original explicit constructions. Register order is data 1,2,3, then auxiliaries 12,23. The extraction sequence is CNOT $1 \rightarrow$ aux12, $2 \rightarrow$ aux12, $2 \rightarrow$ aux23, $3 \rightarrow$ aux23. For each data basis word, these four controlled NOTs leave the data unchanged and write its two adjacent parities into the auxiliaries. Linearity then gives their action on every superposition; conditioning the recovery on the auxiliary record gives the branches in Table 18.1. Auxiliary readout occurs only after both contributions to a given parity have been combined.

5. Equations (18.4)–(18.5) are exact original evaluations for $\exp\left(-i\epsilon X_2/2\right)$ and perfect parity extraction. A scalar global phase on a conditional branch does not alter that branch’s state. Coherently controlling the recovery before readout gives the factorization in Eq. (18.5). No pre-existing stochastic Pauli selection is assumed. For correction of a linear span of errors, see Emanuel Knill and Raymond Laflamme, “Theory of quantum error-correcting codes,” Physical Review A 55, 900–911 (1997), doi:10.1103/PhysRevA.55.900. Published article. Author version v1, 26 April 1996.

6. Knill and Laflamme, as in note 5. For code projector P and a set of correctable error operators, the conditions are $\mathrm{P} E_a^\dagger E_b \mathrm{P} = c_{ab} \mathrm{P}$. They apply to products of errors, not merely separate tests of basis codewords. They permit degeneracy and ensure that one recovery corrects the whole operator span. The reference-system argument follows by linearity: replacing the two logical amplitudes by reference-system vectors leaves the same recovery factorization, including entangled inputs. Mixtures follow by averaging those recovered states. Resetting syndrome memory is a separate thermodynamic operation, as developed in Chapter 8. Correction conditions. The product appears because an inner product between two errored code states contains the adjoint of the first error followed by the second. Within the code, that comparison must reduce to a scalar depending only on the two errors, multiplied by the original logical inner product. This is what prevents the error record from distinguishing logical alternatives.

7. Equation (18.6) and Figure 18.2 enumerate all eight independent X-error patterns during one interval, with identical flip probability p and ideal recovery. The logical channel is $\left(1-p_L\right)\rho + p_L X \rho X$ after decoding. Values are $p_L = 0.000298$ at $p = 0.01$ and $p_L = 0.028$ at $p = 0.1$. Correlated faults, errors in the check circuit, phase errors, leakage, and finite-speed control are excluded. The formula is not a universal threshold or a state-independent fidelity.

8. Preskill, Chapter 7, §§7.1–7.2; Gottesman, Chapter 3. The three-qubit code’s basis words differ in three bit positions, but its distance against general quantum errors is one. Any single Z acts as logical Z. Applying Hadamard to every data qubit exchanges X and Z protection; it does not correct an arbitrary one-qubit error. Equation (18.7) is the explicit undetected-error example. Repetition and phase protection. Quantum distance.

9. Shor's original paper, note 1, and Gottesman, §3.1. Equation (18.8) fixes one standard convention for the nine-qubit code. Its factors $|B_+\rangle$ and $|B_-\rangle$ are individually normalized, so both nine-qubit codewords are normalized and orthogonal. The illustrative encoding is not claimed to be a fault-tolerant preparation circuit. The two normalized codewords in Eq. (18.8) are the columns of a 512-by-2 encoding matrix in the nine-qubit computational basis. Shor construction. Stabilizer formulation.

10. Figure 18.3 uses generators Z_1Z_2, Z_2Z_3, Z_4Z_5, Z_5Z_6, Z_7Z_8, Z_8Z_9, $X_1X_2X_3X_4X_5X_6$, and $X_4X_5X_6X_7X_8X_9$. These generate the same code as equivalent choices with different within-block or block-to-block pairings. They are independent and commute. Operators Z within one block differ by stabilizers and therefore act identically on the code. The error set contains identity and the three Pauli operators at each of nine positions. For each pair, form the adjoint of the first followed by the second. Its restriction to the code is either zero or a scalar times identity, as required by the correction conditions in note 6. Generators and degeneracy.

11. Knill and Laflamme, note 5. Identity and the three Pauli matrices span the space of operators on a qubit. For a physical channel, its Kraus operators—not just a guessed classical error distribution—can be expanded in this span. The nine-qubit code corrects the span of identity and all single-position Pauli operators with a common recovery. A channel involving products on several positions can exceed that span. Correction assumes no further faults in the ideal recovery being analyzed. General noise and correctability.

12. Gottesman, Chapter 5; Panos Aliferis, Daniel Gottesman, and John Preskill, "Quantum accuracy threshold for concatenated distance-3 codes," Quantum Information and Computation 6, 97–165 (2006), arXiv:quant-ph/0504218v3 (21 October 2005). Equation (18.10) is a direct two-qubit operator calculation; the dagger denotes the adjoint/inverse. The displayed parity circuit is deliberately ideal, not a verified fault-tolerant gadget. No claim that a single bare auxiliary can safely measure every high-weight check is made. Fault propagation. Fault-tolerant construction and assumptions. In the bare six-X measurement, the auxiliary controls data positions 1 through 6 in that order and is measured in X. Its X fault after the third coupling induces X on positions 4, 5, and 6, a logical Z in the stated code convention, while allowing the auxiliary to report plus.

13. Eric Dennis, Alexei Kitaev, Andrew Landahl, and John Preskill, "Topological quantum memory," Journal of Mathematical Physics 43, 4452–4505 (2002), doi:10.1063/1.1499754. Noisy syndrome information requires a spacetime history rather than an assumption that every nonzero syndrome is a new data error. Figure 18.4 is a separate three-qubit illustration, with no correction or further data changes during the displayed window. Its two histories do not exhaust possible faults. Logical boundary conditions depend on the surface-code geometry. Author version, 24 October 2001.

14. Aliferis, Gottesman, and Preskill, note 12, establish a threshold theorem for specified fault models and fault-tolerant constructions. The distance definition and $2t < d$ correction criterion concern general Pauli support and ideal checks; circuit faults require the stronger fault-tolerance account. For the Shor code, $X_1X_2X_3$ is a weight-three logical Z and $Z_1Z_4Z_7$ a weight-three logical X in Eq. (18.8)'s convention. No nontrivial weight-one or weight-two logical Pauli exists. A universal numerical threshold is not asserted. Threshold theorem. Distance and logical operators. Theorem 1 treats independent stochastic faults and a fixed allowed error in the final output distribution. The latter is measured by the sum of absolute probability differences, not directly by a logical-failure probability. Its count includes all circuit locations. Theorem 6 treats a local non-Markovian model with a sufficiently small fault-amplitude norm bound, allowing specified shared-bath correlations rather than arbitrary correlated noise. The control assumptions include parallel operations and fresh or resettable auxiliaries.

15. Google Quantum AI and Collaborators, "Quantum error correction below the surface code threshold," Nature 638, 920–926 (2025), doi:10.1038/s41586-024-08449-y; online 9 December 2024. The 28 April 2026 correction, doi:10.1038/s41586-026-10559-8, repairs Fig. 3a's repetition-code axis and legend, not the surface-code results used here. Figure 18.5 uses the individually tabulated experimental fit estimates in the author version arXiv:2408.13687v1, supplementary Table S5, rather than extracting pixels from a figure. For m listed estimates, this book computes their mean and $\sqrt{\sum_i \sigma_i^2}/m$, following that supplement's fixed-

code-ensemble convention. This uncertainty does not describe variation from randomly selecting a new patch. The rounded table gives means 0.65033%, 0.302625%, and 0.1425% for distances 3, 5, and 7; the last agrees with the published 0.143% after rounding. The unnumbered reversal equation is the book's elementary independent-reversal derivation, not a claim that all device faults are independent. Underlying records: doi:10.5281/zenodo.13273331. No raw records were re-decoded for this replot. Published article. Tabulated author supplement. The same 28 entries appear in supplementary Table S7 of the published article. The displayed aggregate uncertainties reproduce the source's propagation of the individual fit uncertainties; no additional covariance or shared-systematic-error estimate is inferred from the rounded table. Published supplementary table.

16. Volodymyr Sivak and colleagues, "Reinforcement learning control of quantum error correction," Nature 655, 879–884 (2026), doi:10.1038/s41586-026-10759-2; published 8 July 2026. Under injected drift, controller steering reduced the logical-error-rate distribution's standard deviation 2.4-fold; that factor describes the spread, not the mean. The 3.5-fold figure also includes decoder steering using logical-error estimates, unlike the detection-event controller. Long computations and larger codes were simulated. Neither procedure is tomography of an unknown logical state. Published full text.

17. Preskill, Chapter 7, §§7.3.4, 7.4.1–7.4.2, and 7.4.4, and Gottesman, §2.3, on quantum erasure, distance, and information held by an erased set. Known erased locations permit up to d−1 erasures to be corrected ideally; unlocated arbitrary errors require $2t < d$. An erasure flag supplies a location, not the missing amplitudes. Inaccurate flags, leakage propagation, correlated faults, and noisy restoration affect an implemented guarantee. "More tolerant" does not justify omitting flagged failures from a throughput or resource count. Erasure conditions. Primary code theory.

18. D-Wave Quantum Inc., "An entangling gate for dual-rail erasure qubits," Nature 656, 47–53 (2026), doi:10.1038/s41586-026-10822-y; published 5 August 2026. The published article uses a corporate author with a consortium list. Its experiment implements a controlled-Z operation between two cavity-encoded qubits and separately analyzes erasure rates, residual Pauli errors, and conditional dephasing associated with loss. Surface-code scaling predictions are simulations. Conditional gate-quality estimates are not unconditional logical-memory performance or a demonstration of a complete error-corrected processor. Published article. The displayed surface-code simulations assume noisy controlled-Z gates and perfect erasure checks delayed until the end of each syndrome round. Additional gate, idle, and measurement errors would change those predictions.

CHAPTER 19

Sending States and Protecting Messages

The photon has left Alice's laboratory. Bob's apparatus records nothing. Has a zero arrived, has a one arrived, or has the carrier failed to arrive at all?

A communication system must distinguish those possibilities before it can claim to have delivered anything. It must also say what it intended to deliver: an unknown quantum state, a shared secret key, or an ordinary message protected by that key. Those are different tasks. A device can succeed at one without performing the others.

Chapter 18 protected information inside a register. Now the register's relationships must survive a journey. We will begin with a deliberately simple channel, then build a method for sharing a key across a channel that an adversary may control. Along the way, a public conversation will become as important as the photons. Its contents need not all be secret, but its origin cannot be left in doubt.

A missing carrier is a third possibility

Use the photonic encoding from our earlier chapters: exactly one photon in either of two modes represents $|0\rangle$ or $|1\rangle$. A superposition uses both mode amplitudes. The vacuum, with neither mode occupied, is outside this two-dimensional space. It is not the zero state.

For an ideal loss-only model, suppose the photon survives with probability η, independently of its encoded state. If it survives, its state is unchanged. Otherwise Bob receives vacuum. For an input density matrix ρ, the channel gives

$$\mathrm{E}_{\eta}(\rho) = \eta\,\rho + (1-\eta)\,|\,\mathrm{vac}\,\rangle\langle\,\mathrm{vac}\,|. \tag{19.1}$$

The original matrix is understood to occupy the single-photon subspace of this larger output space. The two terms describe distinguishable possibilities, not coherent alternatives to be added before squaring. This is a quantum erasure channel. Its erasure flag says that the state is missing, not what the missing amplitudes were.[1]

At $\eta = 0.01$, the surviving outputs can have perfect fidelity while ninety-nine percent of the inputs are lost. Figure 19.1 keeps both facts visible. Retaining only detections would answer a question about successful arrivals, not about delivery on every attempt.

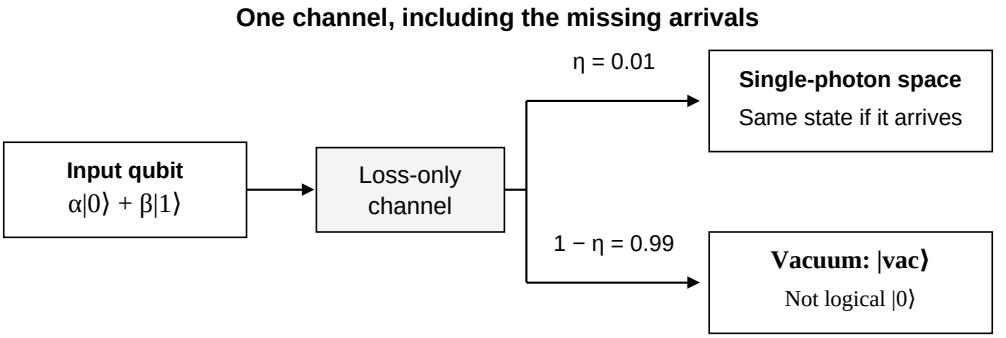


Figure 19.1. Loss changes the outcome space. The input occupies the single-photon subspace spanned by zero and one. The ideal state-independent channel either preserves that state, with probability η, or supplies an orthogonal vacuum flag, with probability $1-\eta$. The worked value is $\eta = 0.01$. Branches organize the complete channel, not two copies or measured photon trajectories. An ordinary absorbing detector does not both identify an arrival and retain its unknown quantum state; a usable stored output requires an appropriate interface.

A conventional photodetector usually absorbs the photon it detects. Its click is useful for a measurement-based task, but does not leave the unknown qubit waiting in memory. Transferring a state into a memory and heralding that transfer without reading its logical value requires additional equipment. Even interpreting silence requires an agreed trial window and a calibrated account of missed detections and background counts.

Loss accumulates rapidly with distance. If a fibre has a constant attenuation a, measured in decibels per kilometre, its transmission over length L is

$$\eta(L) = 10^{-aL/10}. \tag{19.2}$$

A stipulated $a = 0.20$ gives transmission 0.1 over fifty kilometres, 0.01 over a hundred, and 0.0001 over two hundred. Each additional fifty-kilometre segment keeps one-tenth of what entered it. These are chosen model values, not specifications for every fibre. Source efficiency, connectors, conversion, detection, and storage add their own losses or errors.[2]

For a classical file we can retain the original and resend a missing packet. An arbitrary unknown qubit does not come with that backup. If its only carrier is erased, simply asking Alice to try again does not reconstruct the state. Sending a brighter signal is not a universal remedy either: producing many perfect replicas was precisely the task excluded in Chapter 11.

Prepare the connection before sending the state

Teleportation changes which part of the procedure must risk the link. First try to establish a shared entangled pair. A failed attempt loses a resource still being prepared, not the unique input waiting to be transferred. Only after readiness is reliably announced does Alice couple that input to her half of the pair and send the two classical correction bits.

The receiver still needs a carrier, storage, and controls. A false herald or a damaged pair causes trouble; classical messages still take time. But the distinction between preparing a connection and using it is valuable. Repetition can improve the chance of establishing the former without cloning the latter.

A repeater can divide the resource preparation into shorter links. Put a station between Alice and Bob, with a separate memory for each link. Establish one pair between Alice and the first memory, another between the second memory and Bob. A Bell measurement on the middle memories, followed by its communicated correction record, joins the outer memories into an entangled pair. It is entanglement swapping, not a quantum signal being measured, copied, and retransmitted.[3]

In this ideal example, the two-bit record identifies which of four equally likely Bell states the outer memories share. Discarding the record leaves a maximally mixed, unentangled state. Keeping it and applying the indicated correction supplies a known shared pair.

Why bother with memory? Suppose each elementary link succeeds independently with probability s per time slot. Without storage, requiring both successes in the same slot takes $1/s^2$ slots on average. With perfect storage, stop retrying a link once it succeeds and wait for the other. Let T_1 and T_2 be their individual waiting times. The wait for both is their maximum:

$$\mathrm{E}\left[\max\left(T_1, T_2\right)\right] = \frac{2}{s} - \frac{1}{s\left(2-s\right)}. \tag{19.3}$$

The reasoning is ordinary probability. Each link alone takes $1/s$ slots on average. Add the two waiting times and subtract the time to the first success. That first success occurs with probability $1-\left(1-s\right)^2 = s\left(2-s\right)$ in a slot. At $s = 0.1$, storing successes reduces the mean wait from 100 slots to about 14.74.[4]

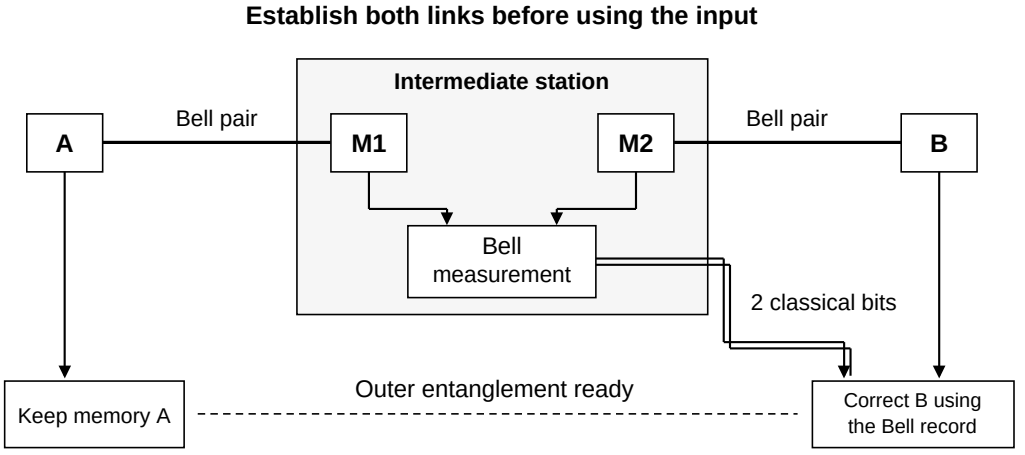


Figure 19.2. Store successful links before joining them. In the upper row, two independently established Bell pairs connect A to M1 and M2 to B. M1 and M2 are separate memories at one station. The lower row performs a joint Bell measurement there; its two classical outcome bits select the outer-pair correction. The unknown input is not yet being teleported. For the timing example, each elementary link succeeds independently with probability s per slot, perfect memories retain early successes, and joining is deterministic. A real repeater must also count storage noise, heralding latency, local failures, and consumed resources.

These ideal slots include enough time to know which attempts succeeded. They omit memory decay and a probabilistic joining step. Real protocols may consume several pairs to improve their quality, or use quantum codes to protect transmission. The calculation explains one advantage of storage; it is not a universal rate formula for a repeater.

Knaut and colleagues' 2024 experiment established entanglement between diamond-based quantum memories through a thirty-five-kilometre loop of deployed telecom fibre. Wavelength conversion and control of the stored states were part of the link. The fibre length was not the straight-line separation of laboratories, and generating that link was not a demonstration of a complete multi-hop repeater chain.[5]

A key is not the message

Now change the task. Alice and Bob want matching random classical bits that an outsider cannot predict. They do not need to deliver an arbitrary unknown qubit for later computation. They can measure every accepted signal, provided their final records have the required secrecy.

Why is a shared key useful? Let M be a binary message and K an equally long, uniformly random, independent secret key. A one-time pad sends

$$C = M \oplus K, C \oplus K = M. \tag{19.4}$$

For each possible message and observed ciphertext C, exactly one key explains that ciphertext. Uniform secret keys therefore leave every message with its original probability. The secret is not in the XOR operation, which everyone may know. It is in the unavailable key.[6]

Reuse the same key on two messages and the protection changes: $C_1 \oplus C_2 = M_1 \oplus M_2$. A relation between the messages is exposed. The pad also does not authenticate a message. Flipping a ciphertext bit flips the corresponding decoded bit without announcing the alteration. Confidentiality and integrity are separate jobs.

Quantum key distribution, or QKD, addresses the establishment of fresh key material. It does not send the eventual message by putting every letter into an unknown quantum state. After key establishment, the protected message can travel over an ordinary classical connection. The encryption method still determines how much key is needed and what additional assumptions protect the data.

A public conversation must have an origin

Before sending the quantum signals, Alice and Bob need an authenticated classical channel. “Public” means an adversary may read its messages. “Authenticated” means an accepted message can be attributed to the intended participant, except with a specified failure probability. The adversary may still block delivery.

Without authentication, Eve could establish one apparently legitimate session with Alice and another with Bob, pretending to be each person to the other. Good quantum statistics within both sessions would not reveal that substitution. A photon’s arrival does not establish who sent it.

Information-theoretic authentication can begin with a shorter pre-shared secret and spend fresh authentication material as required. A successful QKD protocol can produce enough new key to cover that cost and leave a surplus. Alternatively, authenticated public-key credentials can establish identity under computational assumptions. In either case, the starting trust and its cost belong in the account. QKD expands or uses an authenticated relationship; it does not create identity from nothing.[7]

Four preparations, two questions

The Bennett–Brassard protocol introduced in 1984, called BB84, uses states we already know. Alice chooses a fresh fair bit and independently chooses Z or X. Table 19.1 specifies the state she sends. Bob independently chooses a basis and measures before learning hers.[8]

Table 19.1**. The bit determines a state only after the basis is specified.**

Alice’s basis	Bit 0	Bit 1
Z	$\lvert 0\rangle$	$\lvert 1\rangle$
X	$\lvert +\rangle = (\lvert 0\rangle + \lvert 1\rangle)/\sqrt{2}$	$\lvert -\rangle = (\lvert 0\rangle - \lvert 1\rangle)/\sqrt{2}$

These are ideal single-qubit signal states with agreed phase and measurement conventions. Matching bases give the encoded bit with certainty in the noiseless model. Opposite bases give either outcome with probability one-half. A preparation’s other physical properties must not reveal its bit or basis through an unmodeled side channel.

In our version, Z–Z trials will supply prospective key bits; X–X trials will test the complementary relationship. With fair choices, each class occupies one-quarter of the trials on average. The remaining half have mismatched bases. Differences between the recorded bits on mismatched-basis trials are expected from the measurement rule and are not counted as key errors.

Specify a fixed block of transmissions in advance. Only after the quantum phase ends and Bob's detection positions are fixed do the parties announce their bases. They use agreed random selection rules if more matching positions than needed are available, and abort if too few remain. The X test outcomes are revealed; the Z key values are not. This selection is called sifting. Its timing matters: revealing previous basis choices during an adaptive stopping procedure can invalidate a security analysis that assumed an unbiased test sample.[9]

For this first model, assume exact single-photon states, no unintended source leakage, private random choices, and secure endpoints. The receiver has no memory between signals. Its acceptance of an incoming signal is described by one common accept-or-discard operation before either basis measurement. This requires more than equal detection efficiencies averaged over calibration signals. The adversary may control the intervening channel. These are premises to check against an implementation, not benefits supplied automatically by the acronym BB84.

A disturbance we can calculate

Try one concrete attack. Eve intercepts each photon, guesses Z or X fairly, measures, and sends a replacement in the state corresponding to her result. Consider only a trial on which Alice and Bob chose the same basis. Half the time Eve guessed that basis and introduces no error. Half the time she guessed the other, leaving Bob a random answer. Thus

$$Q_{\text{intercept}} = \frac{1}{2} \times 0 + \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}. \tag{19.5}$$

The quarter is the expected error fraction in the matched data for this attack. It is not a guarantee that every disturbed photon is recognized. Even under independent attacks, a sample of m test bits has probability $(3/4)^m$ of showing no errors. Real optical noise can also cause disagreement, so an error does not identify an eavesdropper by name.[10]

More importantly, Eve is not obliged to use our simple strategy. She can keep quantum systems, interact coherently with many signals, and wait for the public discussion. A security proof must bound what she could retain under the allowed attacks and the observed statistics. The intercept example explains one mechanism; it is not that proof.

The connection to Chapter 18 is useful. In an entanglement-based description of BB84, label errors and phase errors require different safeguards. The X tests constrain the

unobserved phase errors of the prospective Z key. They do not measure X and Z on each key photon. Shor and Preskill's proof connects this complementary checking to quantum error correction, while the implemented key-processing steps can remain classical.[11]

Suppose Eve always measures in Z and resends her result. She learns every prospective Z-key bit without causing any Z disagreement in this ideal model. The X tests disagree half the time. Agreement between the key strings would miss the evidence those complementary tests supply.

Agreement is not yet secrecy

Alice and Bob must first reconcile their noisy raw strings. Public parity checks can help Bob locate discrepancies without Alice announcing every bit. But Eve hears those checks too. A parity may reveal a relationship even when none of its constituent bits is revealed. Reconciliation leakage must be deducted from the security account. They then compare a short verification tag produced from each string by the same hash function, selected using fresh randomness. If the tags disagree, they abort.

Here is a small example of the remaining problem. Suppose three independent fair bits are a, b, c, and Eve learns $t = a \oplus b$. The raw string had eight equally likely possibilities; the public parity leaves four. Keeping a and b as a supposed two-bit secret would fail: Eve still knows their XOR. Keeping a and c instead gives four equally likely outputs for either value of t.

One public parity; two candidate keys

Eve knows a ⊕ b = 0

Raw abc	Keep ab	Keep ac
000	00	00
001	00	01
110	11	10
111	11	11
	Only 2 values	All 4 values

Eve knows a ⊕ b = 1

Raw abc	Keep ab	Keep ac
010	01	00
011	01	01
100	10	10
101	10	11
	Only 2 values	All 4 values

Within either parity group, each of the four raw strings is equally likely.

Figure 19.3. Removing a digit is not automatically removing the leaked information. Three independent fair bits form the raw string. Eve knows only the public parity $t = a \oplus b$. For either parity, the proposed key ab has just two possible values, while ac has all four values equally often. The eight rows exhaust this stipulated classical example. Choosing ac is justified only by its specified side information; it is not a general privacy-amplification algorithm for an adversary holding arbitrary quantum systems.

Privacy amplification generalizes the goal: turn partially secret material into a shorter key that is nearly uniform and nearly independent of the adversary's information. Suitable

randomly selected hash functions accomplish this even against quantum side information when the required entropy bound holds. Fresh randomness, independent of the raw string and Eve's prior information, selects the hash function. The selection seed may then be public. The secrecy comes from the remaining uncertainty in the raw string, not from hiding the rule used to shorten it.[12]

Security therefore asks for more than a string that looks random. The two keys must agree. Any allowed test involving the key, Eve's quantum systems and the public transcript should have little ability to distinguish the actual output from an ideal independent key. That comparison covers the whole protocol, including abort, rather than every accepted record separately. Authentication failures also enter the protocol's guarantee. A procedure that always aborts can avoid releasing an insecure key, but is not a useful key service.

Follow a block all the way to the usable key

A familiar asymptotic BB84 calculation, with ideal single-photon preparations and ideal one-way reconciliation, provides an achievable secret fraction

$$r_\infty = 1 - h_2(e_Z) - h_2(e_X). \tag{19.6}$$

Here h_2 is Chapter 7's binary entropy. The errors in the two bases have different jobs: reconciliation of the Z string and limitation of Eve's knowledge through the X test. A negative expression supplies no positive key rate from this prescription. The fraction is per retained Z bit, not per emitted photon. Equal two-percent errors give approximately 0.717 in the asymptotic model; that is not yet a finite experiment's key yield.[13]

For a finite block, an error rate estimated from a sample needs a safety allowance. Let ℓ count the secret output bits before any separate authentication reserve. One rigorous finite-key budget for the ideal source has the form

$$\ell \le n\left[1 - h_2\left(Q_{\text{tol}} + \mu\right)\right] - L_{\text{EC}} - \Delta_{\text{sec}}. \tag{19.7}$$

The symbol n counts raw key bits; Q_{tol} is the pre-agreed tolerated X-test error. The positive allowance μ accounts for limited test data. L_{EC} bounds reconciliation leakage, while Δ_{sec} covers verification and the chosen secrecy guarantee. These quantities come from a matching security analysis. The first term supplies a conservative allowance for secret content under the phase-error bound; the subtractions account for public disclosures and the required security guarantees. We choose an integer output length no greater than the resulting budget. Satisfying this bound certifies a chosen key length under that model; it does not establish the longest key every possible protocol could produce. If the allowed phase-error fraction reaches one-half, this prescription supplies no positive key.[14]

For the same raw-key length, a larger test sample reduces the statistical allowance in this model. A smaller tolerated error or less public leakage leaves more usable output, provided the protocol passes its tests. Demanding smaller secrecy and correctness errors

increases the security deductions. These are changes to a specified proof budget, not guarantees that a different apparatus will satisfy its premises.

Take a constructed record of ten million transmissions, eighty thousand detections, and twenty thousand matches in each basis. Forty thousand detections have mismatched bases. Reveal the twenty thousand X bits for testing and keep twenty thousand Z bits for reconciliation. Suppose the test shows exactly two-percent disagreement, passing the pre-agreed tolerance, and reconciliation succeeds while disclosing at most four thousand bits.

The correctness parameter bounds the probability of accepting unequal keys. The secrecy parameter bounds the protocol's deviation from an ideal independent key, with abort included; it is not the probability that Eve guesses the whole key. Using the formula in note 14, with secrecy parameter 10^{-9} and correctness parameter 10^{-12}, gives $\mu \approx 0.04702$ and $\Delta_{\text{sec}} \approx 100.66$. The effective phase-error allowance is therefore about 6.70 percent, not the observed two percent. Its forty-bit verification tag is covered by the security deduction, outside the reconciliation budget. Substitution permits $\ell = 8,805$ key bits. Figure 19.4 follows every stage.

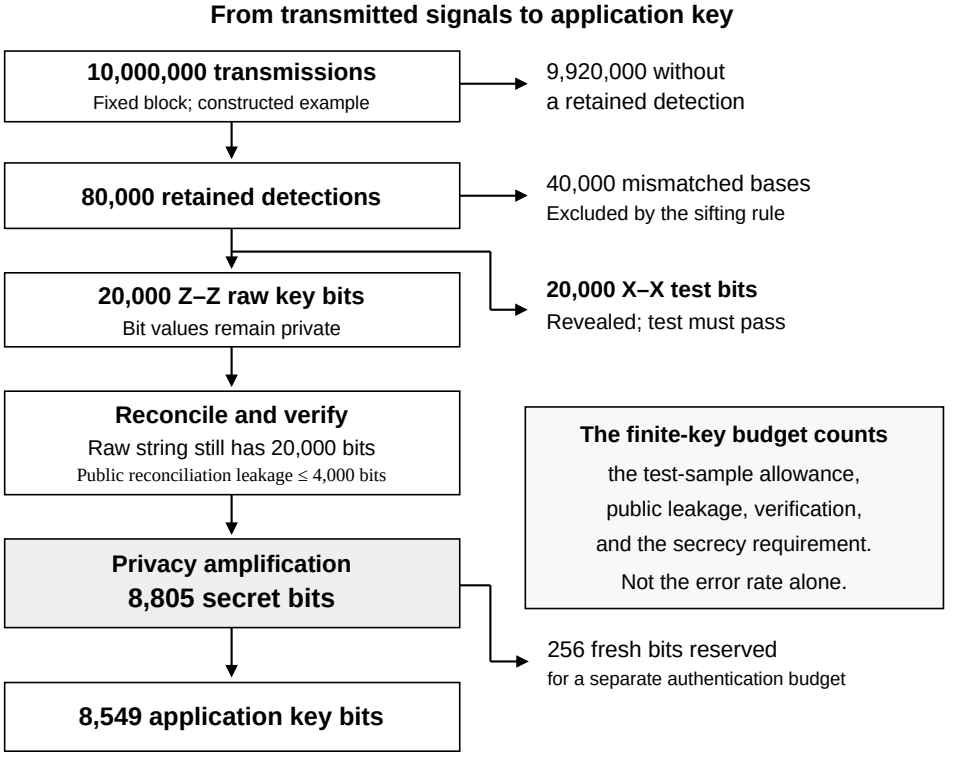


Figure 19.4. A complete key budget uses several denominators. The diagram is a constructed finite-block example, not observed network data. Detection positions and basis selection obey the stated fixed-block model; mismatched and test positions do not enter the raw key. The 8,805-bit output is conditional on passing the test and verification and on the assumptions of note 14. The independent authentication account reserves 256 fresh bits in this example; its security must be justified separately. Remaining application key is 8,549 bits from ten million transmissions. Control time, aborts, and unused attempts belong in a measured rate.

If a separately justified authentication budget reserves 256 fresh bits, 8,549 remain for applications. Dividing by ten million gives about 0.000855 net bits per transmission. A rate per second additionally needs the complete elapsed time, including preparation, public processing, and unsuccessful blocks. The photon count, the raw key length, and the usable secret length are not interchangeable.

This is arithmetic under a stipulated protocol and device model, not a certificate for an unexamined apparatus. A changed detection rule, leaked basis label, or differently selected sample can make the calculation inapplicable. The finite-key proof is not replaced by inserting a small observed error into its formula.

The device can disclose what the model concealed

An attenuated laser pulse is not automatically one photon. Some pulses contain several photons in the same prepared state. An adversary can retain one while allowing another to continue, then use the announced basis later. No unknown state has been cloned: the source supplied the extra copies. Decoy-state protocols vary the emitted intensity and use the resulting statistics to constrain the single-photon contribution. Their model must account for the source's photon-number and phase properties.[15]

Detectors can also respond differently to signals the model treated identically. A 2026 paper by Currás-Lorenzo and colleagues develops finite-key security analysis that jointly accounts for source imperfections and detector-efficiency mismatch. Its importance here is methodological: separately reassuring specifications do not automatically combine into a security proof. The constraints have to enter the same analysis.[16]

Different architectures remove different trust requirements. Measurement-device-independent QKD sends signals from both users to an untrusted joint-measurement station, while retaining assumptions about their sources and local security. Device-independent QKD instead uses Bell-test evidence to constrain otherwise uncharacterized quantum devices. It still needs suitable private setting choices, isolated laboratories, authenticated discussion, and finite-data analysis. Neither name means assumption-free security.[17]

Lu and colleagues' 2026 single-atom experiment makes those distinctions concrete. Nodes in one laboratory were connected by spooled fibre. Results through a hundred kilometres supported positive asymptotic device-independent key rates; the finite-data demonstration used eleven kilometres and 624 hours of records. Its analysis estimated an extractable finite-size rate of 0.112 bits per heralded event. The measurements were not spacelike separated, so laboratory isolation remained an assumption rather than the timing guarantee used in Chapter 10. These are distinct qualifications of a substantial result, not interchangeable distance records.[18]

An intermediate station is not necessarily a quantum repeater either. A trusted key relay can establish separate keys on successive links and help relay a final key, but its access must be included in the trust model. An untrusted measurement station, a trusted classical relay, and a memory that supports entanglement swapping perform different jobs.

Protection without a quantum channel

Post-quantum cryptography takes another route. It uses classical computations designed to resist attacks by quantum as well as classical computers. No photon-level quantum link is required. Its security rests on the assumed difficulty of specified mathematical problems and on correct implementation, rather than on a measured disturbance of transmitted qubits.

NIST's 2024 standards separate these jobs. ML-KEM establishes shared key material using a key-encapsulation mechanism; ML-DSA and SLH-DSA provide digital signatures for authentication and integrity. A signature is not encryption, and key establishment is not a complete messaging system. The standard records, including their correction notices, are the appropriate sources for an implementation—not a book's informal description.[19]

QKD and post-quantum cryptography therefore are not two names for the same technology. A post-quantum signature may authenticate a QKD conversation, in which case that computational premise must remain explicit. A successful quantum exchange does not protect a key later exposed by a compromised endpoint. Both routes need a complete account of identity, stored secrets, implementation, and the task being secured.

What has arrived?

Return to Bob's silent detector. We now know the questions to ask. Was the carrier erased or merely missed? Was the intended output a preserved state or a classical outcome? Was an announced success legitimate? Which public records were released, and what claim does the final key calculation actually support?

A useful channel is more than a line between two boxes. It includes encodings, controls, records, and a test of the promised result. Preserving a state, distributing entanglement, creating a key, and protecting a message can support one another without becoming the same task.

For communication we have tried to prevent uncontrolled changes from corrupting the signal. In the next chapter, some changes will become the signal: a field, a force, or an elapsed time will shift a quantum state in a way we deliberately measure. The same sensitivity that threatens a transmitted phase can help us learn something about the world.

Notes and sources

1. Charles H. Bennett, David P. DiVincenzo, and John A. Smolin, “Capacities of Quantum Erasure Channels,” Physical Review Letters 78, 3217–3220 (1997), doi:10.1103/PhysRevLett.78.3217. Equation (19.1) embeds the qubit input in a three-dimensional output with an orthogonal erasure flag. Loss is state-independent and survival is otherwise ideal. For a reference-correlated input, the full channel is $\eta\,\rho_{RB}+(1-\eta)\,\rho_R\otimes|\mathrm{vac}\rangle\langle\mathrm{vac}|$, with the surviving space relabeled B. No nondestructive heralding apparatus follows merely from writing the flag. Author version v2, 12 March 1997.

2. Equation (19.2) follows from the definition of decibel attenuation. Values $a=0.20$ dB/km and $\eta=0.01$ in Figure 19.1 are stipulated, not measured. Bob’s subsequent detector efficiency must not be counted as fibre transmission unless the combined quantity is explicitly defined. For a concrete experimental separation of fibre, conversion, and detection losses, see Knaut and colleagues in note 5. Experimental loss accounting.

3. H.-J. Briegel, W. Dür, J. I. Cirac, and P. Zoller, “Quantum Repeaters: The Role of Imperfect Local Operations in Quantum Communication,” Physical Review Letters 81, 5932–5935 (1998), doi:10.1103/PhysRevLett.81.5932. The paper’s nested purification architecture is one repeater construction, not the complete class. Figure 19.2 is this book’s ideal two-link swapping example. M. Żukowski and colleagues, “ ‘Event-ready-detectors’ Bell experiment via entanglement swapping,” Physical Review Letters 71, 4287–4290 (1993), doi:10.1103/PhysRevLett.71.4287, provides the underlying swapping proposal. Repeater proposal. Swapping proposal. In the ideal two-Bell-pair convention used here, the four middle outcomes have equal probability. Discarding their record leaves the outer pair maximally mixed; the appropriate conditional correction returns a known Bell pair.

4. Equation (19.3) is an original waiting-time calculation for independent geometric random variables on slots 1,2,… . It follows from $\max(T_1,T_2)=T_1+T_2-\min(T_1,T_2)$ and per-slot probability $s(2-s)$ for at least one success. At $s=0.1$ the stored-link expectation is $280/19$ slots, versus 100 for simultaneous success without retention. Equivalently, sum the probability that at least one link is still unfinished after each nonnegative integer number of slots. With independent links this is twice the single-link survival tail minus its square, giving the same expectation. Perfect persistent memories, equal parallel link attempts, reliable heralds, and deterministic joining are assumed. Slot duration includes the elementary communication latency; no physical rate or general repeater scaling is claimed.

5. C. M. Knaut and colleagues, “Entanglement of nanophotonic quantum memory nodes in a telecom network,” Nature 629, 573–578 (2024), doi:10.1038/s41586-024-07252-z, published 15 May 2024. The full published article reports memory entanglement through a 35-km deployed Boston-area fibre loop, with frequency conversion and error-detecting memory operations. This is not a 35-km direct geographical separation, a deterministic success on every attempt, or a demonstrated chain of functioning repeaters. None of the chapter’s figures reproduces its data. Published PDF at author laboratory.

6. Christopher Portmann and Renato Renner, “Cryptographic security of quantum key distribution,” arXiv:1409.3525v1 (11 September 2014), especially its discussion of composing key distribution with a one-time pad. Equation (19.4) and the ciphertext-reuse identity are direct binary algebra; ideal confidentiality assumes a uniform independent key, secrecy from the adversary, matching message length, and no reuse. The ciphertext can remain malleable without authentication. The chapter does not recommend deploying a home-built encryption protocol. Full author analysis.

7. Mark N. Wegman and J. Lawrence Carter, “New hash functions and their use in authentication and set equality,” Journal of Computer and System Sciences 22, 265–279 (1981), doi:10.1016/0022-0000(81)90033-7; Portmann and Renner, Appendix D, for composition of authentication and QKD. A public transcript is available to Eve, but undetected forgery must be bounded. A pre-shared authentication resource is not silently counted as new output; reuse and fresh-key requirements depend on the chosen authentication scheme. Authentication need not guarantee delivery or protect a compromised endpoint. Original research record. Authentication and composition.

8. Charles H. Bennett and Gilles Brassard, “Quantum cryptography: Public key distribution and coin tossing,” Proceedings of the IEEE International Conference on Computers, Systems and Signal Processing (Bangalore, 1984), 175–179. The archived author copy was posted in 2020, not authored then. Table 19.1 uses this book’s Z/X conventions. The following fixed-block, Z-key/X-test variant is not asserted to

reproduce every procedural detail of the original paper. Basis information and detection positions, but not raw Z values, are publicly announced after quantum transmission. Archived original paper.

9. Corsin Pfister, Norbert Lütkenhaus, Stephanie Wehner, and Patrick J. Coles, "Sifting attacks in finite-size quantum key distribution," New Journal of Physics 18, 053001 (2016). Their LCA-sifting and single-basis parameter-estimation combination supplies uniform sampling with no basis-information leakage during transmission. Our version fixes the block, announces afterwards, chooses uniformly within matching sets, and aborts below the pre-agreed quotas. Basis-independent detection is an additional device premise. Iterative sifting is not universally forbidden; it needs a proof compatible with its stopping and disclosure rules. The original paper's X-key/Z-test letters are interchanged here. Author version v3, 11 May 2016.

10. Equation (19.5) is the explicitly enumerated intercept–resend example with independent, uniformly chosen BB84 bases and ideal replacement states. The test is conditioned on Alice and Bob using the same basis; no basis-dependent filtering is allowed. $(3/4)^m$ assumes independent repetitions of this particular attack and checks for no errors at all. It is neither a general finite-key bound nor the probability that Eve is absent. Enumerate Alice's four preparations and Eve's two basis choices. In matched Alice–Bob trials, Eve's matching basis gives no error; her other basis gives equal probabilities for Bob's two outcomes. Averaging these cases gives Eq. (19.5). For why sampling premises change the result, see Pfister and colleagues. Sampling and intercept–resend analysis. The separate always-Z example has zero Z-key disagreement and full Eve knowledge of those bits, but one-half disagreement in matched X tests. It does not use the random-basis quarter-error model of Eq. (19.5).

11. Peter W. Shor and John Preskill, "Simple Proof of Security of the BB84 Quantum Key Distribution Protocol," Physical Review Letters 85, 441–444 (2000), doi:10.1103/PhysRevLett.85.441. The proof connects BB84 to entanglement purification with complementary error correction. The phase-error description is a virtual test used in the proof, not an instruction to measure both complementary observables on each retained key bit. Reconciliation is not sufficient to establish secrecy. Author version v2, 12 May 2000.

12. Marco Tomamichel, Christian Schaffner, Adam Smith, and Renato Renner, "Leftover Hashing Against Quantum Side Information," IEEE Transactions on Information Theory 57, 5524–5535 (2011), doi:10.1109/TIT.2011.2158473. Quantum-proof hashing requires an appropriate entropy bound and a suitably independent seed. Figure 19.3 is an original exact classical example only: $t = a \oplus b$, so ab reveals a relation while ac is independent of t. Its selected fixed map is not a universal extractor for arbitrary side information. Public reconciliation, verification, and sampling records enter the adversary's information account. Author version v1, 12 February 2010.

13. Equation (19.6) gives the usual ideal single-photon, asymptotic one-way BB84 achievable fraction per raw key bit. It uses complementary phase-error estimation and ideal reconciliation cost $h_2(e_Z)$. Symmetric errors specialize it to $1 - 2h_2(e)$. The approximately eleven-percent zero of that expression is not a universal tolerable-error threshold for every QKD protocol. Source flaws, finite samples, different processing, and side channels alter the guarantee. The 0.7171189 value at two percent is an original numerical evaluation. Complementary-error security proof. Finite and asymptotic analysis.

14. Marco Tomamichel, Charles Ci Wen Lim, Nicolas Gisin, and Renato Renner, "Tight finite-key analysis for quantum cryptography," Nature Communications 3, 634 (2012), doi:10.1038/ncomms1631; full author version arXiv:1103.4130v2. Equation (19.7) uses preparation quality one and their Eq. (2): $\mu = \sqrt{[(n+k)/(nk)][(k+1)/k]\ln(4/\epsilon_{\text{sec}})}$ and $\Delta_{\text{sec}} = \log_2[2/(\epsilon_{\text{sec}}^2 \epsilon_{\text{cor}})]$. Here $n = k = 20{,}000$, $Q_{\text{tol}} = 0.02$, $L_{\text{EC}} = 4{,}000$, $\epsilon_{\text{sec}} = 10^{-9}$, and $\epsilon_{\text{cor}} = 10^{-12}$. The resulting upper budget is 8,805.1286 bits; choose 8,805. The entropy term is capped at one for a phase-error allowance above one-half, as in the source theorem. The key/test basis names are exchanged. The calculation uses fixed-block uniform sampling as in note 9, not the problematic iterative sifting of the early presentation. Ideal qubit preparations, basis-independent detection, secure endpoints, private randomness, and the specified receiver model are premises. Reconciliation leakage excludes the separately covered verification tag; failed verification or an exceeded budget aborts. Security includes abort weighting; authentication adds its own error bound and cost. Figure 19.4's counts and 256-bit fresh authentication reserve are stipulated bookkeeping, not measured data or a proof that that reserve suffices for an unspecified scheme. Full finite-

key analysis. Secure fixed-block sampling. A sufficient receiver model applies the same acceptance operation before either basis measurement. Together with the ideal basis-independent source ensemble and delayed announcements, this permits conditioning first on the detection pattern and then applying uniform fixed-block sampling. This is an explicit combination of the cited models; Pfister's formal sampling proposition does not itself specify optical loss. Equal averaged efficiencies alone do not establish that premise. Verification selects a universal$_2$ hash uniformly with fresh independent randomness and uses a forty-bit tag for the stated correctness parameter. Privacy amplification selects a separate, independent universal$_2$ hash. The four-thousand-bit budget bounds key-dependent reconciliation communication and excludes verification. The security deduction covers the verification ceiling once, as in the cited bound.

15. Hoi-Kwong Lo, Xiongfeng Ma, and Kai Chen, "Decoy State Quantum Key Distribution," Physical Review Letters 94, 230504 (2005), doi:10.1103/PhysRevLett.94.230504. Decoy intensities constrain the contributions of photon-number components under a characterized source model. A phase-randomized coherent pulse has a Poisson photon-number distribution, not exactly one photon. Source side channels, intensity correlations, and incomplete phase randomization require appropriate extensions; observing a low total error is not alone a security proof. Author version v4, 12 May 2005.

16. Guillermo Currás-Lorenzo, Margarida Pereira, Shlok Nahar, and Devashish Tupkary, "Security of quantum key distribution with source and detector imperfections through phase-error estimation," npj Quantum Information 12, 129 (2026), doi:10.1038/s41534-026-01264-5. The publisher lists 18 May 2026 online publication and 6 August 2026 Version of Record. The accepted author manuscript is arXiv:2507.03549v2, dated 31 August 2026. Its lifting theorem incorporates bounded detector-efficiency mismatch in active-basis-choice BB84-type phase-error proofs, including a combined finite-key treatment of stated source and detector flaws. It is a theoretical security result, not an experiment or protection against arbitrary unspecified device defects. Published article. The theorem assumes classical announcements follow the completed quantum communication stage. Accepted author version v2.

17. Hoi-Kwong Lo, Marcos Curty, and Bing Qi, "Measurement-device-independent quantum key distribution," Physical Review Letters 108, 130503 (2012), doi:10.1103/PhysRevLett.108.130503. MDI QKD removes the need to trust the measurement station while retaining assumptions about sources and local laboratories. For finite-data device-independent security, see Rotem Arnon-Friedman, Frédéric Dupuis, Omar Fawzi, Renato Renner, and Thomas Vidick, "Practical device-independent quantum cryptography via entropy accumulation," Nature Communications 9, 459 (2018), doi:10.1038/s41467-017-02307-4. Bell-based security and MDI security are distinct. No claim that arbitrary Bell violation alone yields a positive finite key is made. MDI author version v2, 28 May 2012. Device-independent finite-data proof.

18. Bo-Wei Lu and colleagues, "Device-independent quantum key distribution over 100 km with single atoms," Science 391, 592–597 (2026), doi:10.1126/science.aec6243. The author manuscript is arXiv:2602.09596v1, posted 10 February 2026, following journal publication on 5 February. Its finite-rate analysis uses an achievable reconciliation-leakage bound. It reports nodes housed in one laboratory, spooled fibre, and no spacelike separation of readouts. At 100 km the reported key-rate assessment is asymptotic; at 11 km, 1.208 million heralded events collected over 624 hours support an estimated extractable finite-size rate of 0.112 bits/event with soundness error 10^{-5}. This is not asserted to be an already delivered final key at 100 km or a loophole-free spacetime test. Laboratory isolation, authenticated public communication, and finite-data assumptions remain. The chapter's finite-key example is a different, trusted-device BB84 protocol. Full author article and supplement. Journal record.

19. National Institute of Standards and Technology, FIPS 203, Module-Lattice-Based Key-Encapsulation Mechanism Standard; FIPS 204, Module-Lattice-Based Digital Signature Standard; FIPS 205, Stateless Hash-Based Digital Signature Standard, all issued 13 August 2024. Standards status is stated as of 6 October 2026. FIPS 203 and 204 display planning notices dated 17 November 2025 and 31 July 2026, respectively; this chapter does not reproduce implementation instructions or claim to implement corrected standards. ML-KEM is key establishment; ML-DSA and SLH-DSA are signatures. Their computational security goals do not require quantum transmission and are not proofs that no future attack can exist. Authentication through a computational signature retains that assumption during the QKD exchange. FIPS 203 and notices. FIPS 204 and notices. FIPS 205.

CHAPTER 20

Measuring What We Could Not Measure Before

A magnetic field changes the phase of a stored qubit. In the last chapter, that change could spoil a message. Here it can become the message: evidence that a field is present, and a way to estimate its strength.

The distinction comes from how we prepare and read the system. A memory should preserve an unknown input despite disturbances. A sensor begins with a controlled preparation and lets a particular disturbance alter it. We then compare the result with a reference.

Sensitivity alone is not enough. An instrument that responds strongly to everything cannot tell us which influence produced its record. We need a known response, a way to distinguish the desired signal from other changes, and an uncertainty that includes the resources used to obtain it.

We will build that account around one small experiment. Then we will add more probes, entangle them, and ask when the promised improvement survives. The surprise will not be that quantum rules abolish measurement limits. It will be that the limit changes when we change which physical resources and disturbances the experiment must accommodate.

Let the field write a phase

Take two selected states of a spin sensor, labeled $|0\rangle$ and $|1\rangle$. Work near a known background field. Let B denote the additional, approximately constant magnetic-field component along the sensor's sensitive direction. Over a small enough range, suppose the transition's angular frequency shifts by κB. The calibrated coefficient κ tells us how strongly this particular transition responds.

Prepare the plus state. Let the field act for a known interval τ, pronounced "tau." Describe the phase relative to an oscillator tracking the unshifted transition. Under the two-level, coherent model of Chapter 5,

$$\delta = \kappa B, \phi = \delta \tau = \kappa B \tau,$$
$$|\psi_B\rangle = \frac{|0\rangle + e^{-i\phi}|1\rangle}{\sqrt{2}}. \tag{20.1}$$

Here δ is the frequency difference from the reference, in radians per second. The corresponding effective Hamiltonian is $\hbar\delta|1\rangle\langle 1|$, apart from a common energy. The field changes a relative phase; it need not move probability between the two energy levels. Detecting those populations directly would miss the change. A second controlled pulse makes it visible, as in Ramsey's separated-interaction method.[1]

Choose the readout phase so that the probability of one named result rises through one-half as B passes through zero. In our conventions, this result tests $|-_y\rangle = (|0\rangle - i|1\rangle)/\sqrt{2}$. Equivalently, apply a positive quarter-cycle phase to the one component, then Hadamard and record output zero. Allow a known loss of contrast:

$$p(B) = \frac{1 + C\sin(\kappa B \tau)}{2}, 0 < C \le 1. \tag{20.2}$$

The dimensionless number C is the fringe contrast. This simplified model assumes a centered fringe, a contrast independent of the small field being estimated, no rejected trials, and calibrated readout. More general instruments can have an offset and unequal detection responses. Figure 20.1 follows one trial from preparation to record.

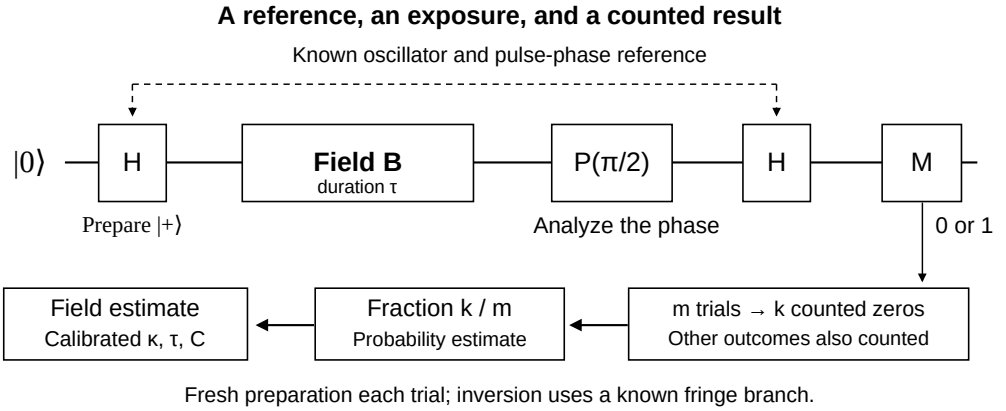


Figure 20.1. A controlled change becomes a statistical estimate. Each trial prepares plus, exposes the sensor to the field for τ, and measures in a fixed quadrature using $\mathrm{P}(\pi/2)$, H, and Z readout. The phase gate multiplies the one amplitude by i; output zero is the counted result. All m trials enter the fraction k/m. The oscillator, pulse phase, response coefficient, and timing supply the reference and calibration. Blocks are functional stages, not a spatial layout or equal durations. No field value is returned by an individual click. M denotes measurement in the Z basis.

A nitrogen–vacancy centre in diamond provides one physical route to such a sensor. It is a crystal defect involving a substitutional nitrogen and a neighboring missing carbon site; selected electron-spin states can be controlled with microwave pulses and read out optically. Maze and colleagues demonstrated magnetic sensing with an individual centre in 2008. A single localized probe already does useful quantum sensing without entanglement between several probes. Calling a sensor quantum identifies its physical operation; claiming a quantum enhancement additionally requires a comparison with a specified alternative.[2]

A click is not a field value

Repeat the preparation, exposure, and measurement m times. Count the named result k times. The fraction $\hat{p} = k/m$ estimates a probability, not the field directly. To infer the field we must invert the calibrated response on a specified portion of its fringe.

For independent trials with fixed probability p, the fraction has standard deviation $\sqrt{p(1-p)/m}$. Even ideal preparation and detection do not remove this spread. It comes from the quantum probabilities of the chosen measurement, often called projection noise. Itano and colleagues investigated these population fluctuations in two-level systems.[3]

A small change in field changes the probability by the local slope times that field change. Dividing the probability uncertainty by the slope gives an approximate local field uncertainty:

$$\sigma_B \simeq \frac{\sqrt{p(1-p)}}{\sqrt{m}\,|\partial p/\partial B|}, \sigma_B|_{B=0} \simeq \frac{1}{C\kappa\tau\sqrt{m}}. \tag{20.3}$$

We take $\kappa > 0$ in this example. The derivative is the slope, not a new physical observable. This is a local, large-sample error estimate on a known fringe branch, not a guarantee for every finite record. It can also be obtained from the classical Fisher information of the two possible outcomes. That quantity measures how much their probabilities change with the parameter; its role is statistical, not an extra detector.[4]

For a worked scale, stipulate $\kappa/(2\pi) = 28$ gigahertz per tesla, $\tau = 10$ microseconds, and $C = 0.8$. These chosen values are not a measured device specification. Out of 10,000 trials, suppose 5,400 give the counted result. Then $\sin\phi = 0.1$. On the branch near zero, $\phi \simeq 0.10017$ radians and $\hat{B} \simeq 56.94$ nanoteslas. Equation (20.3), evaluated at that estimate, gives about 7.12 nanoteslas of statistical standard uncertainty. A nanotesla is one-billionth of a tesla.[5]

The same record does not uniquely determine a field over an unlimited range. Sine repeats, and different phases can have the same value even within one cycle. A shorter exposure, a second readout phase, or prior coarse measurements can identify the appropriate branch. Those additional measurements belong in the resource account. Measuring the same probability more precisely does not, by itself, resolve an ambiguity in the response.

Precision is not calibration

Suppose the oscillator has an unknown frequency offset. The accumulated phase now depends on κB plus that offset. One fringe cannot determine both separately. Likewise, an unnoticed offset in the detector's reported probability can imitate a field near our working point. Smaller counting fluctuations do not distinguish those explanations.

Precision describes the statistical spread of a specified estimator. Calibration connects its scale and reference to the quantity being measured. A complete uncertainty account must also address calibration errors, drift, unwanted field components, and assumptions about the signal's time dependence. Reversing the signal, interleaving reference measurements, or using a second transition can provide additional comparisons. None is supplied merely by adding trials.

There is also a time resolution to the answer. For a changing field, the phase depends on its weighted history during the exposure. A long integration can improve sensitivity to a

steady component while averaging away a rapidly changing one. Sensitivity, spatial resolution, bandwidth, and unambiguous range are different properties of a sensor.

Time counts between the measurements too

Longer exposure makes a given field accumulate more phase. Why not wait indefinitely? Because contrast can decay, and because fewer complete trials fit into a fixed observing time.

Let n independent probes work in parallel in each trial. Suppose each has contrast $C = e^{-\tau/T_2}$, and let t_d be the time per trial spent preparing, resetting, and reading rather than accumulating the signal. For many trials, total laboratory time is approximately $\mathrm{T} = m\left(\tau + t_d\right)$. Near the working point, define the sensitivity coefficient η_B by

$$\eta_B \equiv \sigma_B \sqrt{\mathrm{T}} \simeq \frac{e^{\tau/T_2}\sqrt{\tau + t_d}}{\kappa\,\tau\sqrt{n}}. \tag{20.4}$$

Its units are tesla times the square root of seconds, commonly written tesla per square root of hertz. This convention assumes the uncertainty averages down as $1/\sqrt{\mathrm{T}}$; it is not a universal noise-spectrum normalization. Our T_2 is the decay constant of the stated exponential model, not whichever lifetime happens to be longest on a specification sheet.

With negligible dead time, this expression is smallest at $\tau = T_2/2$. Earlier readout wastes phase accumulation; later readout loses too much contrast. Dead time changes the optimum because every restart now has an additional cost. Other noise laws give other optima. A useful exposure time is an answer to a specified optimization problem, not simply “as long as possible.”[6]

Let several probes accumulate the same phase

Now spend the same n sensing probes on one entangled preparation. Use the state that places equal amplitudes on all zeros and all ones. Expose every probe to the same field for the same interval. Each one component acquires its own phase, so the joint state becomes

$$|\Psi_B\rangle = \frac{|0^n\rangle + e^{-in\phi}|1^n\rangle}{\sqrt{2}}, \phi = \kappa B \tau. \tag{20.5}$$

This is a Greenberger–Horne–Zeilinger, or GHZ, state. The notation 0^n means a string of n zeros. The relative phase has grown n times as fast without increasing the exposure duration. Read it by applying the known quarter-cycle phase to one probe, Hadamards to all probes, and then measuring all in Z. Count whether the number of recorded ones is even. Its probability is $\left[1 + \sin\left(n\phi\right)\right]/2$ ideally, or $\left[1 + C_n \sin\left(n\phi\right)\right]/2$ with a stipulated joint contrast C_n.[7]

One parity result comes from the whole block. The n results inside it are not n independent samples of that parity. For m fresh blocks, ideal controls, and negligible phase uncertainty about the working branch, the two strategies give

$$\sigma_{B,\mathrm{ind}} \simeq \frac{1}{\kappa\,\tau\,\sqrt{n\,m}}, \sigma_{B,\mathrm{ent}} \simeq \frac{1}{\kappa\,\tau\,n\,\sqrt{m}}. \tag{20.6}$$

The entangled strategy improves the standard uncertainty by $\sqrt{n}$ in this noiseless comparison. The independent $1/\sqrt{n}$ behavior is the standard quantum limit for these equally exposed, uncorrelated probes. The $1/n$ behavior is often called Heisenberg scaling. These are local precision scalings under a declared resource model, not permission to measure anything with unlimited accuracy. Extra coherent passes through a signal can also change the account; counting only physical carriers would then miss the repeated interactions.[8]

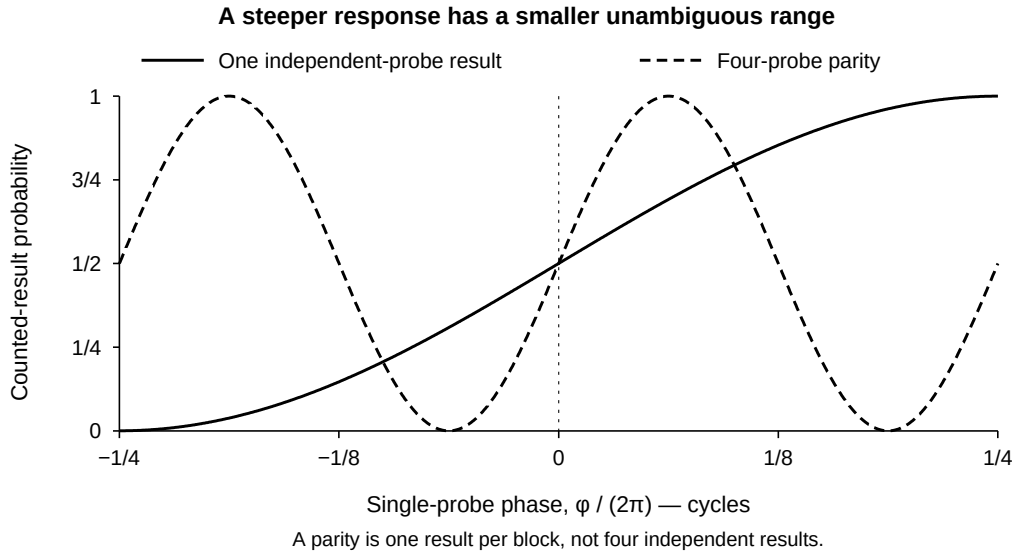


Figure 20.2. A steeper fringe also repeats more often. The solid curve is the counted-result probability of each independent probe; the dashed curve is even-parity probability for a four-probe entangled block. Both use ideal contrast and the same single-probe phase ϕ. An independent four-probe trial yields four individual records; an entangled trial yields one block parity. The curves alone are not a complete resource comparison. Near zero, the parity slope is four times larger, but its phase pattern repeats four times within one single-probe cycle. Points on either curve are theoretical probabilities, not measured data.

The stronger slope carries a price visible in Figure 20.2: more repetitions of the fringe within the same field range. An initial estimate must be good enough to distinguish the intended branch, or a larger protocol must obtain that information. Entangling gates, failed preparations, and parity-readout errors add further costs. Keeping only successfully detected blocks can yield an attractive conditional curve without delivering the same precision per attempted trial.

Loss makes that last warning quantitative. Suppose each probe survives independently with a field-independent probability s, known losses are flagged, and the survivors are otherwise ideal. Independent probes still supply, on average, $n\,m\,s$ useful individual results from m attempted trials. A maximally entangled block needs every member:

losing one and tracing it out removes the phase between all zeros and all ones. Only a fraction s^n of blocks retains that phase.

At the working point, a surviving block has the n-fold phase slope, but there are only about $m s^n$ usable block results. For four probes with survival one-half, the ideal independent phase uncertainty is $1/\sqrt{2m}$; the entangled one is $1/\sqrt{m}$. The latter is worse. The flags reveal which attempts failed, not the lost phase. These statements concern this particular GHZ protocol without a loss-recovery code, not every possible entangled sensor. They follow by including failures in the resource count rather than comparing only survivors.[8]

An advantage that noise can remove

There is a particularly clean test of the ideal comparison. Suppose each probe independently loses off-diagonal coherence by the factor $e^{-\gamma\tau}$, where $\gamma = 1/T_2$ is constant. The noise preserves populations and is Markovian in this model: the decay rate does not depend on earlier history. The all-zero/all-one coherence of the entangled block then receives n such factors, giving $C_n = e^{-n\gamma\tau}$.

Set dead time and control errors to zero so we can isolate this effect. For the same total laboratory time and the same number of probes, the squared sensitivity coefficients are

$$\eta_{\mathrm{ind}}^2 = \frac{e^{2\gamma\tau}}{n\kappa^2\tau}, \eta_{\mathrm{ent}}^2 = \frac{e^{2n\gamma\tau}}{n^2\kappa^2\tau},$$
$$\min_{\tau}\eta_{\mathrm{ind}}^2 = \min_{\tau}\eta_{\mathrm{ent}}^2 = \frac{2e\gamma}{n\kappa^2}. \tag{20.7}$$

The independent probes prefer $\tau = 1/(2\gamma)$. The maximally entangled block prefers the shorter time $1/(2n\gamma)$. At their separate optima they tie. The extra phase gain has been exactly offset by faster loss of coherence and the number of trials available. Here e is the exponential constant from Chapter 3, approximately 2.718; the minimum is still proportional to the dephasing rate. Figure 20.3 shows the comparison for four probes.

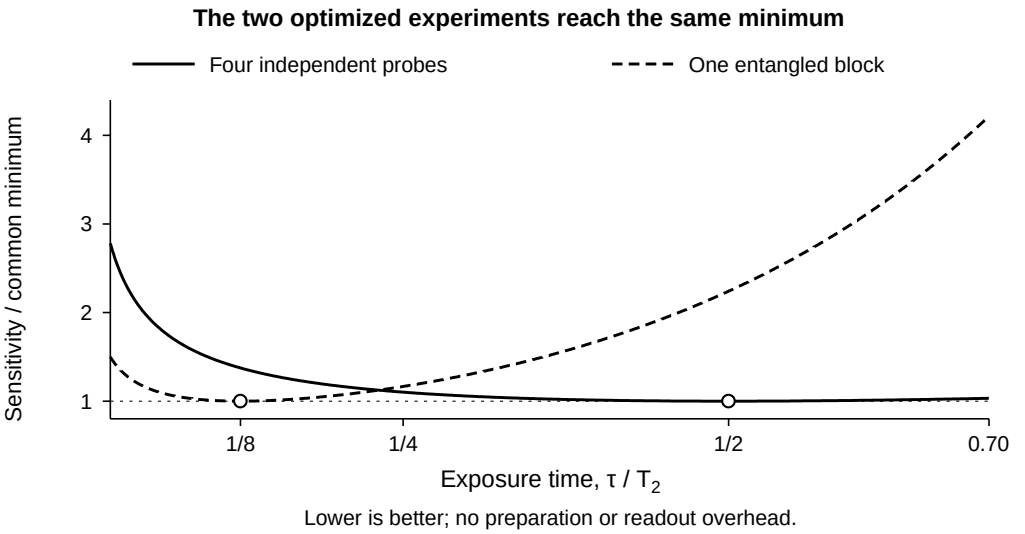


Figure 20.3. Optimize both experiments before comparing them. Curves show the square root of Eq. (20.7), divided by their common minimum, for four probes. Lower is better. Independent probes reach one at $\tau / T_2 = 1/2$; the maximally entangled block reaches one at $1/8$. The same number of sensing probes and total observing time are used, with independent exponential dephasing, negligible dead time, perfect preparation, and ideal readout. This is an exact comparison of two specified strategies, not an optimality theorem over all entangled states or an experimental benchmark.

Huelga and colleagues established this cancellation for maximally entangled probes in the independent-dephasing model. Their analysis also examined partially entangled preparations, so the lesson is not that entanglement never helps in noise. Changing the state, noise correlations, available controls, or signal can change the answer. It is that the noiseless scaling alone cannot decide the useful experiment.[9] Giving both strategies the same positive restart cost also breaks the tie in favor of independent probes.

Squeeze the noise without flattening the signal

Maximal entanglement is not the only design. Spin squeezing redistributes fluctuations so that the measured quantity varies less while retaining enough response to the signal. Use axes in which the mean collective spin points along X and a small signal rotation about Y changes the Z population difference. A measurement of the collective operator J_z returns half the number of zeros minus half the number of ones. Write $\langle J_x \rangle$ for the corresponding mean in the X setting, estimated on separate preparations. We use dimensionless spin units. Independent aligned probes have variance $(\Delta J_z)^2 = n/4$ and mean $n/2$. For the small rotation specified here, the change in mean collective Z spin per radian has magnitude equal to the magnitude of the mean collective X spin. The phase uncertainty is noise divided by this response slope, just as in Eq. (20.3). The metrological squeezing parameter packages both:

$$\xi_R^2 = \frac{n(\Delta J_z)^2}{|\langle J_x \rangle|^2} = \frac{R}{C_s^2}, \sigma_\phi \simeq \frac{\xi_R}{\sqrt{nm}}. \tag{20.8}$$

Here $R = 4\left(\Delta J_z\right)^2 / n$ is the variance relative to independent probes, and $C_s = 2|\langle J_x \rangle| / n$ is the retained spin contrast. A value $\xi_R^2 < 1$ improves this local phase measurement at fixed resources. Wineland and colleagues developed this connection between squeezing and spectroscopic precision.[10]

We can make the comparison with actual allowed states. For two probes, take

$$|\chi\rangle = \sqrt{0.9}|{+}{+}\rangle - \sqrt{0.1}|{-}{-}\rangle .$$

This normalized state has $\langle J_x \rangle = 0.8$ and variance $\left(\Delta J_z\right)^2 = 0.2$. The independent reference $|{+}{+}\rangle$ has mean 1 and variance 0.5. Now introduce a controlled loss of directional information: on one-quarter of preparations apply Z to both probes, and discard the classical record of that choice. This changes the mean response to 0.4 while leaving the J_z variance at 0.2. Both rows therefore correspond to valid quantum preparations.

Table 20.1. Equal noise reduction can accompany unequal response.

Preparation	Variance ratio R	Contrast C_s	Precision factor ξ_R
Independent pair $\|{+}{+}\rangle$	1	1	1
Pure pair $\|\chi\rangle$	0.40	0.80	0.791
Mixture with a forgotten joint phase flip	0.40	0.40	1.581

All rows use two probes and the same local phase readout. These are exact constructed states, not experimental measurements. The last column is $\sqrt{R}/C_s$, the standard-uncertainty ratio to the independent pair at equal repetition count and with no extra overhead. The mixture is

$$\rho_{mix} = \frac{3}{4}|\chi\rangle\langle\chi| + \frac{1}{4}(Z \otimes Z)|\chi\rangle\langle\chi|(Z \otimes Z).$$

Preparation and randomization costs are excluded from this local comparison. Equal variance reductions need not give equal metrological gains.

The last two rows have identical readout variance, but the mixture is less useful than the independent reference. The forgotten choice reverses the mean spin on some preparations without reversing the measured noise reduction. Dividing by the weakened response reverses the apparent advantage. In 2025, Wu and colleagues demonstrated spin squeezing in a room-temperature ensemble of nitrogen–vacancy centres, using interactions between the defects. Their inferred squeezing parameter includes both fluctuations and the retained collective spin response. A practical field-sensitivity gain also needs an account of the field interaction, readout and time per trial.[11]

Choose which disturbance survives

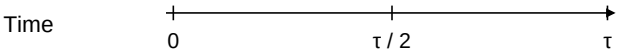
Protection and sensing can help each other only when the unwanted disturbance can be distinguished from the wanted one. Consider the echo operation from Chapter 12. Swap zero and one halfway through the exposure, and restore the labels at the end. In the ideal instantaneous-pulse limit, the phase becomes

$$\phi = \kappa \int_0^{\tau} y(t) B(t)\, dt\,, \; y(t) = +1 \text{ then } -1\,. \tag{20.9}$$

The integral adds small phase contributions across the interval. The control sign $y(t)$ changes at the halfway swap. A constant field contributes equal and opposite halves and disappears. A field that changes from $+B_s$ to $-B_s$ at the same instant contributes with the same sign in both halves. Its amplitude remains measurable while a constant background is rejected. Figure 20.4 gives both cases.

The echo rejects one waveform and retains another

What is multiplied	First half	Second half	Accumulated phase
Control y(t)	+1	−1	X at the midpoint; restore at the end.
Constant field	+Bc	+Bc	
Weighted y(t)B(t)	+Bc	−Bc	**0** Equal halves cancel.
Synchronized field	+Bs	−Bs	
Weighted y(t)B(t)	+Bs	+Bs	**κBsτ** Equal halves add.



Prescribed signals, instantaneous pulses, every outcome retained.

Figure 20.4. The pulse sequence selects a signal, not just a long lifetime. The control sign is positive in the first half and negative in the second. A constant field B_c therefore gives effective contributions $+B_c$ and $-B_c$, with zero net phase. A synchronized field of amplitude B_s that reverses halfway gives $+B_s$ in both weighted halves and phase $\kappa B_s \tau$. Time profiles are prescribed ideal examples with instantaneous pulses; a final restoring swap or its equivalent in the analyzer is included. Other waveforms and unknown signal timing require their own response calculation.

Thus an echo-protected coherence time cannot simply be inserted into the unpulsed static-field formula. The sequence that extends the lifetime may cancel the signal. A September 2026 article by Yamamoto and colleagues illustrates the importance of naming the sequence: a diamond electron spin at 4 kelvin reached 11.2(8) seconds under a Carr–Purcell–Meiboom–Gill pulse sequence with 24,000 pulses, while the reported Hahn-echo time was 6.8(1) milliseconds. The parentheses state the reported uncertainty

in the last digits: 11.2(8) means 11.2 ± 0.8. These are different controlled experiments, not an eleven-second unpulsed measurement of a static field.[12]

Error correction raises the same question more sharply. A code that exactly corrects the span of I and Z on a sensing qubit also corrects the small phase rotation generated by Z. The corrected encoded state then loses the phase response we intended to read. Any information in the correction record needs its own sensing analysis. Useful sensing codes must preserve the signal generator while suppressing the relevant noise. Zhou and colleagues derived precise conditions for doing this with Markovian noise, noiseless auxiliaries, and arbitrarily fast, ideal controls. Those are substantial physical assumptions, not automatic benefits of encoding.[13]

Fields, forces, and clocks

A frequency comparison need not seek a magnetic field at all. In an optical clock, a magnetic shift may instead be a nuisance. Dietze and colleagues' 2026 trapped-ion experiment used entangled states designed to cancel first-order magnetic sensitivity while retaining sensitivity to an optical frequency. They compared the clock with a strontium reference. Their results distinguish statistical frequency stability at a fixed probe time from the best operating point after changing that time—the same distinction that mattered in our noise calculation.[14]

A force can write a phase through position rather than an internal transition. For a weak, constant force along x, its potential energy is $-F\,x$. Along two prescribed alternatives, its contribution to their relative phase is

$$\Delta\phi_F = \frac{F}{\hbar}\int_0^{\tau}\left[x_1(t) - x_0(t)\right]dt\,. \tag{20.10}$$

For a nearly constant separation d, this contribution is $F\,d\,\tau/\hbar$. A larger separation or longer exposure gives a larger response. This is not the complete phase of every force interferometer: trajectories, laser phases, other potentials, and changes caused by the force must also be included. Light-pulse atom interferometers turn such comparisons into acceleration and gravity measurements without requiring an entangled atomic input.[15]

Reading a mechanical displacement introduces another issue: the probe can push what it measures. In an optical interferometer, quantum fluctuations contribute both to the readout and to radiation-pressure forces on the mirrors. The optical field has two complementary wave components, called quadratures: relative to a reference, one is aligned with the wave and the other is shifted by a quarter-cycle. Their quantum fluctuations cannot both be made arbitrarily small. Squeezing redistributes those fluctuations rather than eliminating them. The useful choice depends on the frequency band and on the mechanical response.

Ganapathy and the LIGO detector collaboration demonstrated frequency-dependent squeezing in full-scale gravitational-wave detectors in 2023. A filter cavity rotated the squeezed optical quadrature as a function of frequency, reducing both high-frequency

shot noise and low-frequency quantum radiation-pressure noise over the relevant bands. That is an improvement in an operating instrument, not a claim that every noise source or calibration uncertainty vanished. Optical loss and technical noise remained in the performance account.[16]

Sensitivity made interpretable

Return to the unwanted phase with which we began. We have turned it into an estimate by specifying a preparation, an interaction, a reference, a readout, and a statistical model. We have also found several ways that an impressive number can mislead: a steep but ambiguous fringe, narrow fluctuations with little response, a long protected lifetime that rejects the intended signal, or a comparison that omits preparation time.

Quantum enhancement becomes meaningful only after that account is made. Entanglement can steepen a response or redistribute fluctuations. It can also make a state more vulnerable to the very noise that limits its exposure. A useful sensor selects a relationship that the environment can change in an informative way, then preserves enough control to interpret the change.

This completes the path from building a qubit to protecting it, transmitting it, and deliberately using its sensitivity. The next part uses these capabilities to study other systems. Can one controlled quantum system reproduce the behavior of another, and what evidence would make its answer a trustworthy scientific prediction?

Notes and sources

1. Norman F. Ramsey, “A Molecular Beam Resonance Method with Separated Oscillating Fields,” Physical Review 78, 695–699 (1950), doi:10.1103/PhysRev.78.695. Equations (20.1)–(20.2) are this book’s two-level, single-parameter Ramsey model, not a reconstruction of Ramsey’s apparatus. The reference removes the known background transition frequency; the signal is a small additional field component with positive calibrated coefficient κ. The input is plus. The counted outcome is $|-_y\rangle$, consistent with the earlier chapters’ Y convention. Population transfer, uncontrolled parameter-dependent contrast, and other levels are omitted. Original paper.

2. J. R. Maze and colleagues, “Nanoscale magnetic sensing with an individual electronic spin in diamond,” Nature 455, 644–647 (2008), doi:10.1038/nature07279. The cited experiment uses coherent control and optical readout of individual nitrogen–vacancy centres. Its actual magnetic-field sensing protocol and frequency band are not identified with the unpulsed, static-field example here. No contemporary record claim or spatial-resolution number is inferred from our ideal two-level model. Published experiment. The demonstrated sensor used microwave coherent control, optical preparation and fluorescence readout. Its spin-echo signal band and photon-shot-noise limitation differ from our ideal projection-noise calculation. Published copy at MIT.

3. W. M. Itano and colleagues, “Quantum projection noise: Population fluctuations in two-level systems,” Physical Review A 47, 3554–3570 (1993), doi:10.1103/PhysRevA.47.3554; erratum, Physical Review A 51, 1717 (1995). The elementary binomial variance in our example follows from independent binary outcomes with fixed probability. Common-mode fluctuations, correlations between probes, drift across trials, and readout bias are not included. This noise is not removed simply by making the detector ideal. Publication and erratum record. Published PDF at NIST. The erratum, doi:10.1103/PhysRevA.51.1717, corrects an introductory characterization of another author’s interpretation and states that the article’s conclusions are unaffected. 1995 erratum.

4. Samuel L. Braunstein and Carlton M. Caves, “Statistical distance and the geometry of quantum states,” Physical Review Letters 72, 3439–3443 (1994), doi:10.1103/PhysRevLett.72.3439. For this fixed binary measurement the classical Fisher information is $\mathrm{I}_B = \left(\partial_B p\right)^2 / \left[p\left(1-p\right)\right]$. Equation (20.3) is the delta-method, local asymptotic standard uncertainty and agrees with the inverse square root of $m\,\mathrm{I}_B$. It is not asserted to be a finite-sample confidence interval, a globally unbiased estimator on a periodic response, or an optimization over all possible quantum measurements. Primary statistical analysis. Finite counts can fall outside the probability range allowed by the stipulated contrast. Fitting then requires a constrained likelihood and a stated interval procedure; applying an unrestricted inverse sine is not sufficient. Published copy at author institution.

5. The 10,000-trial record and all its parameters are constructed examples. $\kappa = 2\pi \times 28 \times 10^9$ radians per second per tesla, $C = 0.8$, and $\tau = 10^{-5}$ seconds give $\hat{B} = \arcsin\left(0.1\right) / \left(\kappa\,\tau\right) \simeq 56.936$ nanoteslas. Substitution in Eq. (20.3) gives approximately 7.118 nanoteslas. The value 28 GHz/T is used as a stipulated electron-spin-scale coefficient, not a precision constant or a measured response of the Maze apparatus. Calibration parameters are treated as known exactly only in this illustrative uncertainty calculation. All trials, including the uncounted outcome, enter the binomial likelihood.

6. Equation (20.4) is an original extension of the local binomial calculation to n independent probes, m repetitions, exponential contrast, and deterministic per-trial dead time. It assumes enough trials to use $m \simeq \mathrm{T} / \left(\tau + t_d\right)$. With $t_d = 0$, minimizing $e^{2\tau/T_2}/\tau$ gives $\tau = T_2/2$. Positive dead time gives $\tau_* = \left[T_2 - 2t_d + \sqrt{T_2^2 + 12\,T_2 t_d + 4\,t_d^2}\right] / 4$. The noise model, not an arbitrary measured echo lifetime, determines T_2. For a physical clock analysis including preparation time and oscillator limitations, see Kai Dietze and colleagues, cited in note 14. Clock timing and noise account. Our decay-time convention is defined by the contrast itself, and the stated optimum follows from that definition.

7. J. J. Bollinger, W. M. Itano, D. J. Wineland, and D. J. Heinzen, “Optimal frequency measurements with maximally correlated states,” Physical Review A 54, R4649–R4652 (1996), doi:10.1103/PhysRevA.54.R4649. Our GHZ calculation uses n identical, simultaneously exposed probes and no nonlinear enhancement of the signal Hamiltonian. A phase gate $\mathrm{P}(\pi/2)$ on one register followed by H on every register and Z readout gives even-parity probability $\left[1+\sin(n\phi)\right]/2$. Dephasing changes the contrast as explicitly specified later. Figure 20.2 is an original calculation of these probabilities. The even-parity probability is obtained by summing the output probabilities over all bit strings containing an even number of ones. Published proposal. Published PDF at NIST.

8. Vittorio Giovannetti, Seth Lloyd, and Lorenzo Maccone, “Quantum Metrology,” Physical Review Letters 96, 010401 (2006), doi:10.1103/PhysRevLett.96.010401. Precision scalings require a resource convention. B. L. Higgins, D. W. Berry, S. D. Bartlett, H. M. Wiseman, and G. J. Pryde, “Entanglement-free Heisenberg-limited phase estimation,” Nature 450, 393–396 (2007), doi:10.1038/nature06257, explicitly counts repeated passes. Equation (20.6) holds near a known operating branch, at fixed interrogation time, with equal probe exposure counts nm and ideal measurements; it does not assert a globally unambiguous single-shot estimate or that entanglement is necessary in every resource model. The independent-erasure example is an original calculation: for $n \geq 2$, any missing GHZ member leaves a phase-independent mixture on the surviving block; only the s^n all-survive branch contributes phase information. With heralded, phase-independent losses its per-attempt Fisher information is $n^2 s^n$, versus ns for n independent ideal probes. The example assumes no recovery code and negligible counting fluctuations in the retained sample sizes. Resource-based metrology. Repeated-pass experiment. The cited author versions are Giovannetti and colleagues, quant-ph/0509179v1 (26 September 2005), and Higgins and colleagues, arXiv:0709.2996v2 (12 June 2008). Metrology author version v1. Repeated-pass author version v2.

9. S. F. Huelga, C. Macchiavello, T. Pellizzari, A. K. Ekert, M. B. Plenio, and J. I. Cirac, “Improvement of Frequency Standards with Quantum Entanglement,” Physical Review Letters 79, 3865–3868 (1997), doi:10.1103/PhysRevLett.79.3865. The cited author version is arXiv:quant-ph/9707014v1 (4 July 1997). Equation (20.7) reproduces the independent-dephasing cancellation for independent Ramsey probes versus a maximally entangled block with each interrogation time separately optimized. It is not a claim that all entangled probes tie the independent optimum: the original work also analyzes partially entangled states. Figure 20.3 plots this restricted comparison, not a universal quantum limit. Setting a nonzero fixed gate or readout overhead changes the result. Published paper. Author version v1. With the same positive overhead per attempted trial, the rescaled GHZ objective has overhead multiplied by the block size. Its separately optimized minimum is therefore worse for blocks larger than one in this specific model.

10. D. J. Wineland, J. J. Bollinger, W. M. Itano, F. L. Moore, and D. J. Heinzen, “Spin squeezing and reduced quantum noise in spectroscopy,” Physical Review A 46, R6797–R6800 (1992), doi:10.1103/PhysRevA.46.R6797. Equation (20.8) uses dimensionless collective spins, an X-directed mean, a small rotation about Y, and J_z readout. Table 20.1 is an original two-qubit construction. With $J_x = (X \otimes I + I \otimes X)/2$ and $J_z = (Z \otimes I + I \otimes Z)/2$, direct evaluation gives $\langle J_z \rangle = 0$, $\langle J_z^2 \rangle = 0.2$, and $\langle J_x \rangle = 0.8$ for $|\chi\rangle$. Conjugation by Z⊗Z reverses J_x and preserves J_z^2. Mixing the unchanged and conjugated preparations in proportions 3:1 therefore halves the mean while preserving the variance. Both density operators are positive and trace one. The precision factors are $\sqrt{0.4}/0.8$ and $\sqrt{0.4}/0.4$. These exact moments establish state consistency, not an experimentally demonstrated preparation or a practical gain including preparation overhead. Published article.

11. Weijie Wu and colleagues, “Spin squeezing in an ensemble of nitrogen–vacancy centres in diamond,” Nature 646, 74–80 (2025), doi:10.1038/s41586-025-09524-8, published 1 October 2025. The cited author text is arXiv:2503.14585v1 (18 March 2025). The experiment characterizes sub-projection-noise fluctuations using interaction-enabled noise spectroscopy in an interacting room-temperature ensemble. The inferred metrological squeezing parameter includes both fluctuation variance and mean spin length. Practical field sensitivity further depends on the sensing interaction, implemented readout and time per trial. None of Table 20.1’s numbers comes from that experiment. Published abstract and record. Author version v1.

12. Takashi Yamamoto and colleagues, “Ten-second electron-spin coherence in isotopically engineered diamond,” npj Quantum Information, doi:10.1038/s41534-026-01368-y, published online 28 September 2026. The publisher’s Article in Press PDF is cited in that form, without final pagination. The reported 11.2(8)-second decoupling time uses 24,000 pulses. The 6.8(1)-millisecond Hahn-echo result includes mitigation of mains-frequency noise. Neither is unpulsed static-field interrogation. Equation (20.9) and Figure 20.4 are separate ideal filter calculations. The waveform is synchronized, pulses are instantaneous, and the final swap is restored or accounted for in the analyzer. Other control sequences have different transfer functions. Publisher record. Published Article in Press. CPMG denotes Carr–Purcell–Meiboom–Gill. The quoted single-centre result was obtained at 4 kelvin. The Hahn-echo comparison used mains synchronization and phase-estimate feedforward; the CPMG protocol was not mains-synchronized.

13. Sisi Zhou, Mengzhen Zhang, John Preskill, and Liang Jiang, “Achieving the Heisenberg limit in quantum metrology using quantum error correction,” Nature Communications 9, 78 (2018), doi:10.1038/s41467-017-02510-3. The published analysis gives conditions in terms of the signal Hamiltonian and the span generated by Markovian noise operators, with noiseless auxiliaries and arbitrarily fast, ideal controls. The Markovian description is assumed to remain valid in that control limit. Our narrower observation follows from the Knill–Laflamme conditions: exact correction of $\{ I , Z \}$ implies $\Pi Z \Pi = c \Pi$ on the code, removing that first-order logical signal when the signal generator is Z. It does not rule out sensing a differently structured signal or protocols using additional resources. Published PDF at author institution. Here the recovery statement concerns the encoded state’s coherent phase response. Finite-interval correction records can depend on the parameter; extracting information from them is a different statistical task.

14. Kai Dietze and colleagues, “Entanglement-Enhanced Optical Ion Clock,” Physical Review Letters 136, 073601 (2026), doi:10.1103/dyqm-k8p6, published 17 February 2026. The cited version is the published article. It uses two calcium-40 ions with selected opposite magnetic sensitivities, laser-referenced optical phases, and comparison to a strontium-87 lattice clock. Its Ramsey, entangled, and classically correlated reference protocols have distinct contrast and timing accounts. Fixed-probe-time improvements are not silently extended to an ideal zero-dead-time optimum over all probe durations. The experiment is not a realization of our simple common-field GHZ magnetometer; suppressing magnetic nuisance while preserving optical-frequency sensitivity is the relevant connection. Publication record. Published full text. The reported comparisons concern frequency instability. Evaluation of systematic clock shifts remained future work in that report.

15. Mark Kasevich and Steven Chu, “Atomic interferometry using stimulated Raman transitions,” Physical Review Letters 67, 181–184 (1991), doi:10.1103/PhysRevLett.67.181. Equation (20.10) is the force-potential contribution from $V = - F x$ on two prescribed paths, with branch-one phase minus branch-zero phase. It assumes the perturbation is weak enough to evaluate those paths as specified and does not replace the complete action, control-phase, and trajectory calculation. A stationary separation is an illustrative simplification, not a full light-pulse gravimeter. Cold-atom interferometry need not use interatomic entanglement to exploit single-particle coherence. Primary atom-interferometry experiment. Published copy at Duke.

16. D. Ganapathy and colleagues (LIGO O4 Detector Collaboration), “Broadband Quantum Enhancement of the LIGO Detectors with Frequency-Dependent Squeezing,” Physical Review X 13, 041021 (2023), doi:10.1103/PhysRevX.13.041021. The study reports filter-cavity-based frequency-dependent squeezing in full-scale detectors, reducing shot noise and quantum radiation-pressure noise over different frequency bands. These are optical continuous-variable resources, not the spin-squeezed state or GHZ scheme calculated here. The improvement is band-dependent and limited by optical loss and technical noise; no factor is assigned to every frequency or every type of uncertainty. The record cited describes the published demonstration, not a claim about the observatories’ current operating state. Published experiment. Published full text.

PART VI

Simulating Systems, Understanding the Claims

We have learned to prepare quantum states, change them, protect them, send them, and use their sensitivity. Now we turn those capabilities toward a different purpose: understanding something outside the machine.

A quantum simulator need not resemble the object it represents. An atomic spin can stand for an excitation moving between sites; a collection of controlled qubits can represent electronic configurations in a molecule. The connection is made by a dictionary between states, interactions, time scales, and measurements. A successful experiment must justify that dictionary, not merely produce complicated behavior.

This final part follows the journey from a scientific question to a defensible answer. We will first build a small simulation whose entire prediction can be checked. It will let us separate the model of the target from its encoding, the algorithm from the device, and the measured record from the quantity inferred. We will also see why a simulator can pass one convincing test while still answering another question incorrectly.

Then we will return to chemistry. The energies and states introduced through a small molecular ion will become ingredients in predictions about molecules and reactions. An electronic energy, an energy difference, and an experimentally measured rate are connected, but they are not interchangeable outputs. Each connection adds assumptions that must remain visible.

Those examples will prepare us to assess claims of advantage. What was calculated? At what accuracy? What did the comparison include? A new capability, a reliable scientific prediction, and a practical improvement over existing methods require different evidence. Recognizing their differences makes genuine progress easier to identify.

Finally, we will return to the larger questions that brought us here: what a state describes, what a measurement establishes, and what information contributes to our understanding of reality. We will approach those questions with the calculations and distinctions we have earned. First, we give one quantum system a precise assignment: imitate another.

CHAPTER 21

Letting One Quantum System Imitate Another

Suppose we want to know whether an excitation placed at one end of a tiny system will reach the other. We could study the system itself. We could calculate its behavior on an ordinary computer. Or we could build a different quantum device whose controlled evolution gives the same answer.

The third possibility sounds almost too easy. Every quantum system evolves according to quantum rules. What distinguishes a simulator from an unrelated experiment that also happens to be quantum?

A simulator needs a correspondence. Its preparations must represent the target's preparations, its changes must represent the target's changes, and its measurements must answer the target's questions. A resemblance between two graphs is not enough.

Feynman's proposal to use quantum systems for simulation made this correspondence a route to computation. Lloyd later established a general gate-based construction for suitably specified local dynamics. We will begin with a much smaller task, easy enough to solve independently. That will expose what must be carried over faithfully before a larger calculation can teach us something new.[1]

Two places for one excitation

Take two sites, left and right, each able to hold an excitation. Restrict the model to exactly one excitation in total. The reference states are $|L\rangle = |10\rangle$ and $|R\rangle = |01\rangle$: excited on the left or excited on the right. The sites remain where they are. It is the excitation's distribution that can change.

Let the two sites exchange excitation with coupling $g > 0$, and let their uncoupled excitation energies differ by $\hbar\Delta$. Both g and Δ are angular-frequency quantities. Subtracting a common energy leaves the Hamiltonian, in the ordered basis L, R,

$$H_{\mathrm{T}} = \hbar \begin{pmatrix} \Delta/2 & g \\ g & -\Delta/2 \end{pmatrix}. \tag{21.1}$$

The subscript T means target. The diagonal entries give the energy mismatch; the off-diagonal entries couple the alternatives. A negative diagonal entry does not mean that the apparatus contains a negative amount of energy. We have chosen a reference and retained only the difference relevant to this model.

Assume that no excitation is lost or added, that other states remain irrelevant, and that the parameters stay fixed. Under those conditions the two alternatives form an invariant

subspace: evolution does not leave it. A general pair of two-state sites would need four reference states. Our restriction leaves two, so one qubit can represent this particular problem. It cannot represent every state of the unrestricted pair.[2]

Prepare $|L\rangle$ at time zero. Our first requested output is $\mathrm{P}_R(t)$, the probability of finding the excitation on the right after a specified time. No detector is asked to supply its full journey or every amplitude.

A prediction we can check without the simulator

The matrix is small enough to solve exactly. Define $\omega = \sqrt{g^2 + \Delta^2/4}$. Squaring the Hamiltonian gives $H_{\mathrm{T}}^2 = \hbar^2 \omega^2 I$. In the power series for its exponential, even powers therefore collect into a cosine and odd powers into a sine:

$$U_{\mathrm{T}}(t) = e^{-iH_{\mathrm{T}}t/\hbar} = I\cos(\omega t) - i\frac{H_{\mathrm{T}}}{\hbar\omega}\sin(\omega t). \tag{21.2}$$

Apply this to the left input. The right amplitude is $-i(g/\omega)\sin(\omega t)$. Squaring its magnitude gives

$$\mathrm{P}_R(t) = \frac{g^2}{g^2 + \Delta^2/4}\sin^2\left(t\sqrt{g^2 + \Delta^2/4}\right). \tag{21.3}$$

At zero mismatch, the probability can reach one. At $\Delta = 2g$, its maximum is one-half. At $\Delta = 4g$, the maximum is one-fifth. A larger mismatch makes the oscillation faster here while reducing its greatest transfer. Frequency alone would therefore be a poor measure of transport. These are consequences of our Hamiltonian, not universal rules for every pair of coupled objects.[3]

For a physical scale, stipulate $g/(2\pi) = 1$ kilohertz and $\Delta = 2g$. The first transfer maximum occurs after about 177 microseconds. Those chosen values let us translate the equations into an experiment; they are not measurements of a particular material.

Simple two-amplitude oscillations also have classical coupled-wave analogues. The purpose of this small example is to expose a checkable state–operation–observable correspondence and the limits of its validation tests. The oscillation curve alone does not establish nonclassicality or computational advantage.

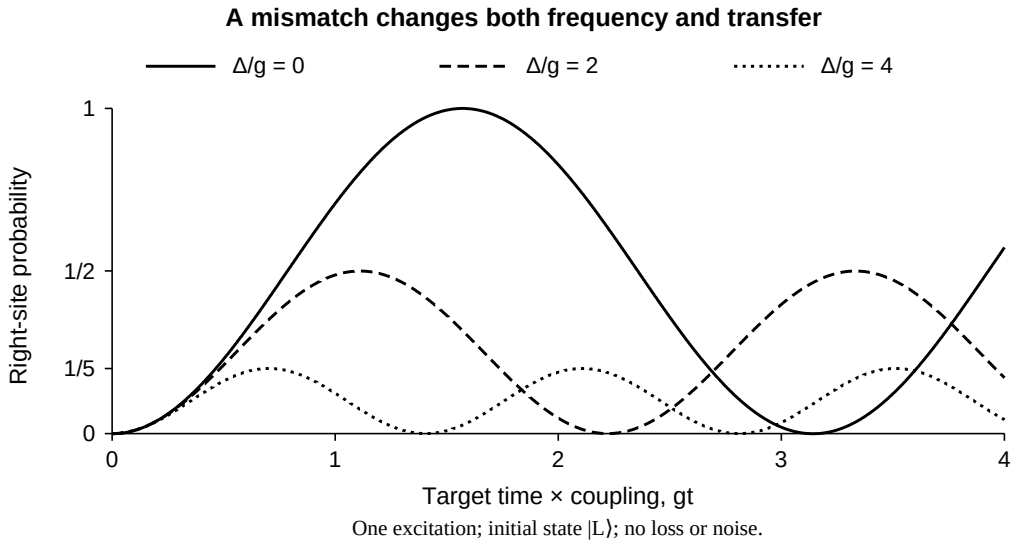


Figure 21.1. A mismatch changes both the rhythm and the reach. The curves give Eq. (21.3) for an initial left excitation and three fixed ratios Δ / g. The horizontal coordinate is target time multiplied by g, not laboratory seconds. All outcomes are retained: $\mathrm{P}_L = 1 - \mathrm{P}_R$. The model conserves one excitation and includes no loss or noise. Curves are exact theoretical predictions, not measured data or trajectories of a particle.

Give the device a dictionary

Now choose a different physical carrier: one controlled qubit. Map $|L\rangle$ to its $|0\rangle$ and $|R\rangle$ to its $|1\rangle$. Call this mapping V. A zero in the simulator now means an excitation on the target's left site, not an empty target. Labels acquire their meaning from the encoding.

The initial preparation and the final test must follow the same map. Preparing simulator zero represents a left excitation. Measuring simulator one asks whether the target excitation is on the right. Testing a coherent superposition of left and right instead requires a correspondingly different measurement on the simulator.

$$H_{\mathrm{S}} = s\, V\, H_{\mathrm{T}}\, V^{\dagger}, U_{\mathrm{S}}(\tau) V = V\, U_{\mathrm{T}}(s\tau), t = s\tau . \tag{21.4}$$

Here $s > 0$ scales the interaction strengths, and τ is laboratory running time. The equality follows by substituting the mapped Hamiltonian into the exponential. Both sides describe the same encoded evolution. If the simulator's strengths are a hundred times larger, its 1.77-microsecond interval represents our target's 177 microseconds. That ratio is a conversion between two physical evolutions, not a speedup over a classical calculation.[4]

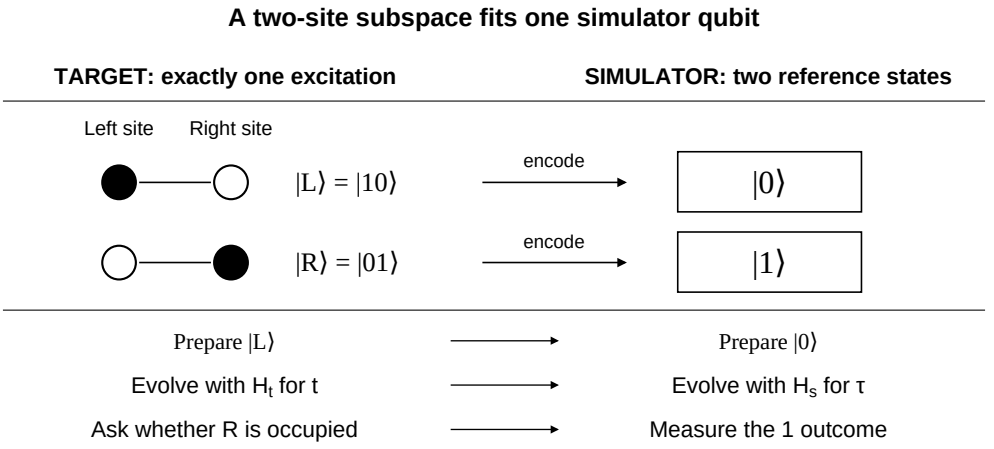


Figure 21.2. The device inherits a dictionary, not the material. The target has two sites and exactly one excitation. The simulator uses one qubit, with $|L\rangle \mapsto |0\rangle$ and $|R\rangle \mapsto |1\rangle$. Preparation, dynamics, and readout all use this correspondence. Filled circles mark excited sites; open circles mark unexcited sites. Rows are basis alternatives, not copies or trajectories. The simulator's zero is not the target vacuum. The time relation $t = s\tau$ applies when every retained Hamiltonian term is scaled by the same positive factor, as in Eq. (21.4).

The dictionary is not only a way to rename the final detector. Suppose the target begins in an equal superposition of left and right. The simulator must begin in plus, with the appropriate phase, rather than in an ordinary lottery over zero and one. Otherwise the wrong preparation could evolve under the right Hamiltonian. Likewise, an unknown relative phase between preparation and readout would change the question the apparatus actually asks.

This is the simplest kind of analogue simulation: engineer a Hamiltonian whose simultaneous action represents the target. In the present case, a detuned drive on the qubit supplies the same transverse coupling and longitudinal mismatch as the atomic model of Chapter 5, in an appropriate rotating description. The carrier need not contain two physical sites.

Analogue simulation becomes more ambitious when many controlled constituents interact. Jaksch and colleagues showed how ultracold atoms in an optical lattice could realize a model with tunable hopping and on-site interactions. Greiner and colleagues' 2002 experiment used that setting to study a transition from a superfluid state with coherence across the lattice toward a Mott-insulating state, in which interactions suppress fluctuations of the number of atoms at each site. Changing the laser intensity altered the balance between hopping and interaction energy. Images of the atoms after release tested changes in coherence, alongside measurements of the excitation response. The controlled atoms supplied an experimentally accessible realization of the model; they were not a miniature sample of every solid it might help us understand.[5]

Build the same evolution out of different steps

A programmable gate device need not hold both terms on at once. For this digital comparison, set the time-scale factor to one. In the simulator basis, write $H_x = \hbar g X$ and $H_z = \hbar \Delta Z / 2$. The target generator is their sum. We already know how to implement rotations produced by either term separately.

The difficulty is their order. A rotation about X followed by one about Z is generally not the same operation as the reverse order. Applying each for the entire requested interval does not reproduce their simultaneous action.

Divide target time into r pieces of duration $\delta = t / r$. For each piece, apply the Z evolution and then the X evolution. Operators act from right to left, giving

$$U_r^{(1)}(t) = \left[e^{-iH_x \delta / \hbar} e^{-iH_z \delta / \hbar} \right]^r, \delta = t / r. \tag{21.5}$$

The individual gates are $R_z(\Delta \delta)$ and $R_x(2 g \delta)$ in the usual half-angle rotation convention. No measurement separates them. Each gate acts on the amplitudes left by the preceding one.

For a sufficiently short interval, both the simultaneous and sequential operations begin with the same first-order change. Their first mismatch involves products of the two terms in different orders. Making the pieces shorter reduces that discrepancy. At fixed total time and bounded Hamiltonian strengths, this first-order product formula has an error bounded by a constant times one over the number of pieces. The constant depends on the problem; choosing a large integer alone is not an accuracy certificate.[6]

A symmetric sequence cancels the leading ordering error. Use half of a Z step, a full X step, and the remaining half Z step:

$$U_r^{(2)}(t) = \left[e^{-iH_z \delta / (2\hbar)} e^{-iH_x \delta / \hbar} e^{-iH_z \delta / (2\hbar)} \right]^r. \tag{21.6}$$

For fixed time and strengths, its error bound falls as $1 / r^2$. Figure 21.3 shows both constructions. Adjacent half Z steps can merge, leaving $2r + 1$ rotations rather than $3r$ for the symmetric sequence, compared with $2r$ for the first-order sequence. These are logical rotation counts; pulse synthesis, waiting, and imperfect gates have their own costs.[7]

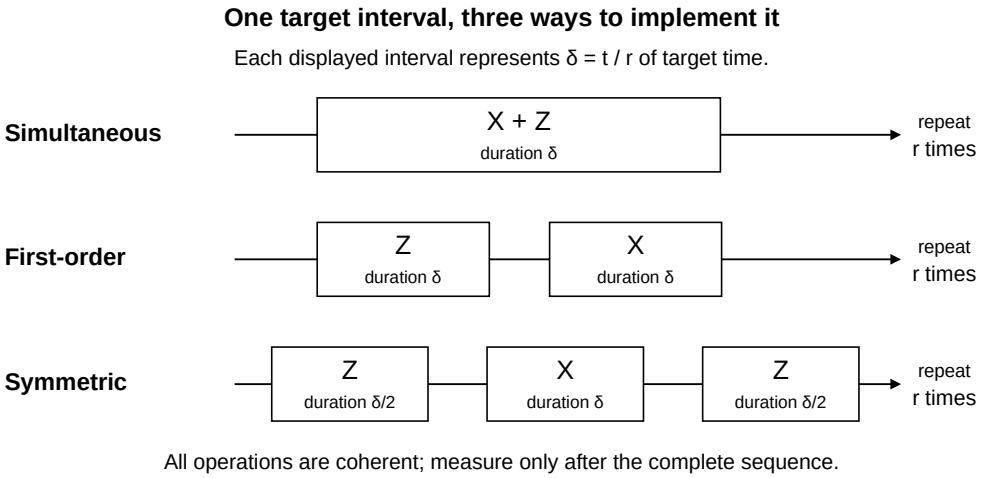


Figure 21.3. Sequential controls approximate a simultaneous interaction. Each row represents the same target-time interval, with $\delta = t / r$. The first row applies both terms simultaneously. The other rows repeat the displayed gate sequence r times, without intermediate measurement. Boxes labeled Z or X denote evolution under H_z or H_x for the stated target duration, not measurements of those observables. Adjacent half Z steps in the symmetric sequence can be merged. Box widths are diagrammatic, not a claim of equal total laboratory time.

There is no contradiction in using a succession of different controls to represent simultaneous forces. The target's small change is assembled from contributions, much as a diagonal displacement can be assembled from horizontal and vertical steps. Here the steps are rotations of a quantum state, so their noncommuting order leaves a calculable remainder. Refinement controls that remainder; it does not require a particle to follow the diagram's boxes as a spatial path.

Digitizing has made one approximation adjustable. It has not made the simulator exact at finite step size, and it has not removed hardware error. Taking more steps can reduce the mathematical ordering error while accumulating more imperfect operations. Analogue controls and digital gates can also be combined; the distinction concerns how the intended evolution is implemented, not two mutually exclusive kinds of matter.

The answer depends on what we ask

Set $\Delta = 2g$ and $gt = 1$, still beginning on the left. The exact right probability is about 0.487841. Table 21.1 compares the two digital constructions. Alongside the probability, it reports how far the complete output state lies from the exact state.

For normalized pure states $|\psi\rangle$ and $|\phi\rangle$, use their trace distance:

$$D(\psi, \phi) = \sqrt{1 - |\langle \psi | \phi \rangle|^2}. \tag{21.7}$$

Zero means the same physical pure state, allowing an irrelevant global phase. The distance is the largest possible difference in the probability of a single measurement event

when comparing those states. It therefore bounds the error for every one-qubit measurement, not just right-site detection.[8]

Table 21.1**. One probability can hide different state errors.**

Pieces r	Right probability Both formulas	State distance First-order	State distance Symmetric
1	0.708073	0.637197	0.279782
2	0.545324	0.281778	0.064884
4	0.502032	0.132288	0.015792
8	0.491374	0.064237	0.003920
16	0.488723	0.031671	0.000978

All entries are original calculations for $\Delta = 2g$, $gt = 1$, and initial $|0\rangle$. The exact right probability is 0.487841. Distances compare the full pure output with exact evolution. Equality of the two formulas' population predictions is exact for this model and input, not a general property of product formulas. There is no noise, fitted data, or postselection.

The two formulas predict exactly the same right probability at each listed step count, yet their state errors differ substantially. Here the symmetric sequence changes the first-order output by a final rotation about Z, apart from an initial global phase. That can change coherence without changing the Z populations.

A different measurement sees the improvement. At eight pieces, the exact probability of the X-plus outcome is about 0.987841. First-order evolution gives 0.969948; the symmetric sequence gives 0.987540. The more accurate state matters when that is the question. If our task genuinely asks only for the right probability at this time, the two constructions perform equally in that respect.[9]

A readout adds a separate uncertainty. Ten thousand independent ideal trials estimate a binary probability with standard deviation at most 0.005. The sixteen-piece population error here is about 0.000882, smaller than that statistical scale. More repetitions could make the bias visible; they would not remove it. Refining the formula and collecting more data address different parts of the prediction.

A perfect return can still be the wrong journey

Now give the analogue device the wrong sign of detuning: it implements $\widetilde{H} = \hbar(gX - \Delta Z/2)$ instead of the intended generator. Beginning from zero, it nevertheless reproduces $P_R(t)$ exactly at every time. Equation (21.3) contains the detuning only through its square.

At our worked time, the wrong-sign device gives X-plus probability about 0.012159 instead of 0.987841. The population record alone could not distinguish these very different predictions. Figure 21.4 retains both measurements.

An echo test does not automatically resolve the error. Let each device perform its own evolution and then its own exact inverse. Both return every input perfectly. That

establishes reversibility of those operations, not that either forward Hamiltonian matches the intended target. A shared calibration error can disappear from a self-consistency test.[10]

Two forward dynamics can pass the same return test

Initial $|0\rangle$; $g\,t = 1$; target $\Delta = 2g$

Test on a fresh preparation	Intended $H/\hbar$ $gX + gZ$	Wrong-sign $H/\hbar$ $gX - gZ$
Right-site probability	0.487841	0.487841
X-plus probability	**0.987841**	**0.012159**
Return after its own inverse	1	1

The inverse undoes what the device did—not necessarily what it should have done.

Figure 21.4. A perfect return is not complete validation. At $g\,t = 1$ and positive target detuning $\Delta = 2\,g$, the two devices give identical right-site probabilities but sharply different X-plus probabilities. Each also returns the initial state with certainty when followed by its own exact inverse. Every entry is a theoretical probability for a separate fresh experiment. The inverse is of the implemented generator, not an independently certified target inverse. The result distinguishes tests of one observable, tests of a state, and tests of reversibility.

A useful validation program therefore combines different kinds of evidence. Calibrate the terms through independent experiments. Compare with exact classical calculations where those are affordable. Change the initial state and the measurement, rather than fitting one curve ever more closely. Check how results respond to step size, system size, and known limiting cases. Agreement between devices with different error mechanisms can add evidence that repeating one device cannot.

A July 2026 preprint by Dalyac and colleagues tests this approach across three neutral-atom processors. In their cross-device benchmark, a noise model calibrated on one processor was used to predict measurements on the others without device-specific refitting. The 36-atom comparisons show useful agreement. Further tests examine residual calibration offsets and limits of the classical reference. These are experimentally testable predictions in a tractable regime, not a blanket certificate for larger calculations that cannot be checked the same way.[11]

When a requested quantity has no exact classical comparison at the desired scale, trust must come from the combination of justified approximation bounds and such independent checks. Conservation of one excitation, for example, would not catch our sign error: both Hamiltonians conserve it. Passing a necessary test does not make that test sufficient.

What changes when the system grows?

Our two-site problem is easy for a classical computer. Even a pure state with one excitation distributed over a thousand sites needs only a thousand amplitudes in the site basis. Quantum behavior alone does not make an intractable computation.

Allow many sites to be occupied, and the counting changes. Twenty two-state sites with exactly ten excitations have $\binom{20}{10} = 184,756$ basis configurations. Forty with twenty excitations have $137,846,528,820$. The binomial coefficient counts which sites are occupied. An interaction can make the energy of one occupation depend on neighboring occupations, coupling the description of the constituents.

The enlarged space still does not prove classical difficulty. Chapter 16 supplied large entangled states with compact descriptions. Symmetries, weak entanglement, free-particle structure, and a limited set of requested observables can keep a problem manageable. The case for a useful simulator concerns the actual family of states and questions, not the longest array we could choose to write down.

Digital simulation offers systematic constructions for broad classes of local Hamiltonians: assemble the evolution from interactions involving small subsets of the registers. The relevant cost includes the number and strengths of those terms, target evolution time, accuracy, and access to their description. Product formulas are one route. Low and Chuang's qubitization gives another with different, explicitly specified access requirements. Neither grants a free implementation of an arbitrary Hamiltonian.[12]

Some mappings are more surprising than one site label becoming one qubit label. In a study published in September 2026 in Nature Physics, De and colleagues used thirteen active ion spins to investigate how an energy-bearing connection between charges can break in a one-dimensional model. Boundaries between oppositely oriented neighboring spins represented charges; down-spin domains represented the connections, called strings. Laser controls set the model's fields and couplings. Spin measurements on separately prepared runs then located the encoded charges and showed how their distribution changed with evolution time. When the string's energy cost per unit length was larger, observed charge pairs formed near its edges and spread into the interior. The simulator investigated a specified lower-dimensional theory, not the complete theory of quarks and gluons or literal quarks appearing in the trap.[13]

Such a mapping must also preserve the question. Measuring an ion's spin can represent a field variable, while a charge may require comparing two spins. Reading every carrier in a convenient basis is useful only when the resulting records support the intended target observable.

Prepare the state—and the environment

An evolution algorithm takes an initial state as an input. It does not automatically supply the state a scientific question needs. Starting our device on the left gives transport from a localized excitation. Starting in an energy eigenstate gives stationary probabilities for fixed measurements. The same Hamiltonian does not make those experiments interchangeable.

Nor does letting the isolated device run under a fixed Hamiltonian make an arbitrary input relax into the ground state. Its energy-state weights remain fixed. Preparing a ground state, approximating thermal equilibrium, following a quench, and measuring a reaction rate are different tasks. Each needs its own preparation and readout account. A long coherent lifetime does not turn waiting into cooling.

Sometimes the target itself is open. Then we must reproduce the relevant interaction with a bath, not insist that every change be unitary on the retained registers. Barreiro and colleagues' 2011 trapped-ion experiment combined coherent gates with optical pumping of an auxiliary ion to engineer dissipative dynamics. The environment was a designed part of the simulation.[14]

Chapter 12 gives a small example: let a fresh auxiliary record part of the zero–one distinction, then set that auxiliary aside. Repeating with new auxiliaries produces the prescribed reduction of coherence. Reusing the same unreset auxiliary is a different model because it can retain memory. Uncontrolled hardware noise is not automatically the target bath, even when both make a fringe fade.

Some observables can nevertheless tolerate imperfect simulation. Kashyap and colleagues' 2025 theoretical analysis establishes accuracy guarantees for specified local observables of geometrically local open systems, with stated stability and noise assumptions. It supports asking how much error the scientific quantity suffers, rather than demanding closeness of every part of a huge state. It does not say that arbitrary hardware noise is harmless.[15] The bounds retain conditions on evolution time, spatial locality and the strength of the noise.

From a record to a prediction about the system

We can now separate the complete task. A physical question motivates a model. An encoding maps that model to available degrees of freedom. A procedure realizes its evolution approximately. A device executes the procedure imperfectly. Measurements produce samples from which the requested quantity is estimated. Each connection can be tested, and each can introduce a different error.

A simulator that follows its assigned Hamiltonian exactly can still be a poor model of a molecule if important interactions were omitted. Conversely, a faithful scientific model

can be implemented by a poorly calibrated device. More shots repair neither mistake. We should report uncertainty at the level of the final quantity, identifying which contributions are bounded, estimated, or not yet controlled.[16]

Our excitation has completed its journey. We know its exact transfer probability, how to encode it in another carrier, how sequential gates approximate simultaneous interactions, and how a plausible validation test can miss the wrong phase. The example is small enough to reveal the entire chain. The purpose of scaling is to retain that discipline when the calculation is no longer small.

One quantum system can imitate another because the same mathematical relations can be embodied in different matter. The scientific value lies in which relations we preserve and which question they answer. Next we will use that perspective on molecules, following the path from electronic states to chemical predictions.

Notes and sources

1. Richard P. Feynman, “Simulating physics with computers,” International Journal of Theoretical Physics 21, 467–488 (1982), doi:10.1007/BF02650179; Seth Lloyd, “Universal Quantum Simulators,” Science 273, 1073–1078 (1996), doi:10.1126/science.273.5278.1073. Feynman’s discussion originated in a 1981 lecture; the cited publication is from 1982. Lloyd’s construction concerns suitably specified local dynamics and counts time and accuracy, not arbitrary state preparation. The chapter’s two-site model is an original worked example. Feynman, publisher record. Feynman, original article copy. Lloyd, original article copy.

2. Equation (21.1) retains the one-excitation sector of $H/\hbar=(\Delta/2)(n_L-n_R)+(g/2)(X_L X_R+Y_L Y_R)$, where $n_j=(I-Z_j)/2$. In the full reference order 00,01,10,11, the retained ordered basis is 10,01. This Hamiltonian preserves total excitation number. The simulator qubit compresses that invariant sector only; vacuum, double excitation, other internal states, leakage, and loss are excluded. The reduction can be checked by selecting the rows and columns for 10 and 01 from the four-dimensional Hamiltonian, then comparing their evolution with Eq. (21.2). No particle-number symmetry is presumed for arbitrary physical hardware.

3. Equations (21.2)–(21.3) follow by squaring the specified two-by-two Hamiltonian and summing its exponential series. At $g=\Delta=0$, the continuous limit is the identity and zero transfer. Figure 21.1 uses $g>0$ and horizontal coordinate $g\,t$. The stipulated $g=2\pi\times 1{,}000$ radians per second and $\Delta=2g$ give the first maximum at $t=\pi/(2\sqrt{2}\,g)=176.776695$ microseconds. The formula can also be reproduced by diagonalizing the printed matrix or integrating its two coupled Schrödinger equations. None of these figures is experimental data.

4. Equation (21.4) uses an isometry between the target’s retained two-dimensional sector and the simulator qubit; within those spaces it is unitary. The inverse or adjoint is denoted by a dagger. The initial state and the observable must both be transformed by this map. A common energy shift adds only a global phase. The relation $t=s\,\tau$ assumes a constant scale multiplying every retained term, not just the coupling. Figure 21.2 does not assert that a simulator running faster than a slow target also outperforms classical numerical calculation.

5. D. Jaksch, C. Bruder, J. I. Cirac, C. W. Gardiner, and P. Zoller, “Cold Bosonic Atoms in Optical Lattices,” Physical Review Letters 81, 3108–3111 (1998), doi:10.1103/PhysRevLett.81.3108. M. Greiner, O. Mandel, T. Esslinger, T. W. Hänsch, and I. Bloch, “Quantum phase transition from a superfluid to a Mott insulator in a gas of ultracold atoms,” Nature 415, 39–44 (2002), doi:10.1038/415039a. The latter’s original article was archived on arXiv in June 2025; that date is not the experiment’s publication year. These sources support the model-to-atom mapping, not a speedup claim. Optical-lattice model. Greiner, published record. Full original article, later archive.

6. Lloyd, note 1; Andrew M. Childs, Yuan Su, Minh C. Tran, Nathan Wiebe, and Shuchen Zhu, “Theory of Trotter Error with Commutator Scaling,” Physical Review X 11, 011020 (2021), doi:10.1103/PhysRevX.11.011020. The first-order product uses chronological Z then X, so $[H_x,H_z]=-i\hbar^2 g\Delta Y$. The asymptotic operator error is bounded as $O(1/r)$ at fixed finite strengths and total time. It is not an error guarantee depending on step count alone. No intermediate measurements or stochastic alternation are used. Full author treatment. Published paper. The error norm here is the spectral operator norm, the largest change the operator difference can produce on a normalized input. For the table’s dimensionless parameters, a conservative first-order upper bound is one divided by the step count. The linked author version is A Theory of Trotter Error, arXiv:1912.08854v3 (3 February 2021); its Eqs. (143) and (150) give the two-term error bounds.

7. Equation (21.6) is the symmetric second-order product formula. For fixed parameters its operator error has an $O(1/r^2)$ bound; the statement is not a monotonicity theorem for every observable at every coarse step. Figure 21.3 shows an unoptimized sequence. Merging adjacent Z halves leaves r X rotations and $r+1$ Z rotations. If each separate generator can only be applied at the original strength, their durations sum to twice

the target time in this example; stronger controls can change that laboratory-time account. The logical step width is not a physical speed guarantee. Product-formula error analysis. For the table's dimensionless parameters, a conservative symmetric operator-error bound is one divided by twice the square of the step count. These are bounds on ideal controls, with no hardware noise included.

8. John Watrous, The Theory of Quantum Information (Cambridge University Press, 2018), Chapter 3, on distinguishability and the trace norm. The linked text is the author's 2018 prepublication copy. Equation (21.7) is the trace distance of two pure-state density matrices. It equals the maximum difference of probabilities for a single event over all quantum measurements. For these qubits the normalized trace distance ranges from zero to one; it is not the squared overlap convention called fidelity in Chapter 11. For numerical evaluation near identical states, taking half the trace norm of the difference of their density matrices avoids subtracting a nearly unit overlap from one. Author's prepublication text. The pure-state trace-norm identity is Eq. (1.186), and the operational distinction is developed in Theorem 3.4. Dividing the trace norm by two gives the distance used here.

9. Table 21.1 is an original exact matrix comparison, not empirical data. Set $g = t = 1$ and $\Delta = 2$. If $A = e^{-iH_z\delta/(2\hbar)}$, then the complete symmetric sequence equals $A\,U_r^{(1)}\,A^{-1}$. For initial zero, the rightmost factor is a global phase and the leftmost factor preserves Z populations. This explains the identical population predictions for all positive integer r in this particular comparison. The exact value is 0.4878407820. The independent-trial standard deviation is $\sqrt{\mathrm{P}(1-\mathrm{P})/m}$; at $m = 10,000$ it cannot exceed 0.005. This is a standard deviation, not a finite-sample confidence guarantee.

10. Figure 21.4 changes only the detuning's sign while retaining the initial zero and laboratory measurement references. At $g\,t = 1$ and target $\Delta = 2\,g$, the exact target X-plus probability is 0.9878407820 and the wrong-sign probability is 0.0121592180; both right probabilities are 0.4878407820. Each generator followed by its own exact inverse gives identity for every input. This original counterexample limits what the specified tests establish; it does not assert that every echo-based verification protocol is uninformative. If only the population is the scientific target, the sign difference is irrelevant to that restricted observable.

11. Constantin Dalyac, Sergi Julià-Farré, Lucas Leclerc, Vittorio Vitale, and colleagues, "Noise-aware emulation and cross-device validation of neutral atom analog quantum processing units," arXiv:2607.28364v1, submitted 30 July 2026. The full preprint reports comparisons on three same-generation neutral-atom processors using 36-atom arrays. Its single calibrated noise-parameter set is tested across devices and regimes, with limitations discussed. This is vendor-associated experimental research, identified as a preprint, not independent certification of all larger workloads. Versioned record. Full author manuscript. The no-refitting statement refers to the three-device benchmark in Sec. III. Section VI C separately infers a residual detuning offset, while Sec. VI D and Appendix E report that strong-interaction correlation simulations are not fully converged by about one microsecond. The latter comparison supports qualitative trends rather than an exact classical reference.

12. Guang Hao Low and Isaac L. Chuang, "Hamiltonian Simulation by Qubitization," Quantum 3, 163 (2019), doi:10.22331/q-2019-07-12-163. Its complexity statements assume particular coherent access to a Hamiltonian encoding and accompanying operations. They cannot be translated into free access to any requested many-body interaction. The original site-configuration counts in the narrative are binomial coefficients, not complexity lower bounds. Lloyd's local-interaction result and Low–Chuang's access model are different constructions, not two constant-cost replacement boxes. Published article. Accepted author version v3. The single-excitation amplitude count refers to a pure state. A general mixed state in that sector instead requires a density matrix, whose size is the square of the sector dimension. The linked accepted author manuscript is arXiv:1610.06546v3 (11 July 2019).

13. Arinjoy De, Alessio Lerose, De Luo, and colleagues, "String-breaking dynamics in a quantum simulator," Nature Physics, published 23 September 2026, doi:10.1038/s41567-026-03422-0. The cited version is the published journal article. Thirteen active spins in a fifteen-ion chain implement the mapped one-dimensional lattice gauge model; external static regions are represented through programmed fields. The narrow claim used here concerns the state/observable mapping and observed edge-facilitated dynamics. It is not a simulation of full quantum chromodynamics or a demonstrated computational speedup. Published full text. Charges are domain walls and strings are down-spin domains in this mapping. At weak or zero string tension,

created pairs can remain near the edges; propagation into the interior is reported at larger tension. September 2026 is the publication date. Publisher-supplied article text. Publication status.

14. Julio T. Barreiro, Markus Müller, and colleagues, “An open-system quantum simulator with trapped ions,” Nature 470, 486–491 (2011), doi:10.1038/nature09801. The experiment combines coherent controls with optical pumping of an auxiliary ion to implement specified dissipative maps. The linked arXiv:1104.1146v1 manuscript (6 April 2011) identifies itself as a pre-review submission; its system–environment construction uses one auxiliary ion reset by optical pumping. The fresh-marker example in this chapter instead reuses the ideal dephasing model of Chapter 12; it does not reproduce the experimental pumping protocol. State preparation by engineered dissipation is distinct from unchanged isolated evolution. Published article. Pre-review author manuscript.

15. Vikram Kashyap, Georgios Styliaris, Sara Mouradian, J. Ignacio Cirac, and Rahul Trivedi, “Accuracy Guarantees and Quantum Advantage in Analog Open Quantum Simulation with and without Noise,” Physical Review X 15, 021017 (2025), doi:10.1103/PhysRevX.15.021017. The published theory treats local observables of geometrically local Lindblad dynamics and selected stable steady-state quantities. Its guarantees have locality, time, accuracy, and stability conditions; its complexity comparisons additionally have computational premises. Only the scoped accuracy point is used here, not unconditional practical advantage or tolerance of arbitrary noise. Published full theory. The noisy bounds assume spatially local, bounded noise terms. The fixed-time dynamics result retains explicit time dependence; the uniform long-time and steady-state results additionally require rapid mixing of the selected local observable. The analysis assumes the specified simulation and calibrated noise model rather than certifying an uncharacterized device.

16. The closing error account separates physical-model choice, encoding or truncation, algorithmic approximation, executed controls, and statistical estimation. These categories are bookkeeping for the same specified final observable, not independently measured universal error bars. The chapter’s exact two-dimensional calculations omit target-model error by definition. Numerical agreement between implementations verifies those calculations; it does not establish the adequacy of a two-site Hamiltonian for an unspecified real material. The diagrams and comparison values are original calculations for the stated models; they contain no experimental data copied from the cited papers.

CHAPTER 22

From Electrons to Chemical Predictions

A chemist has two proposed routes from the same reactant. One leads to product P, the other to product Q. Which product will appear more often? A computer returns an electronic energy with eight digits after the decimal point. The question is still unanswered.

The number may describe one arrangement of nuclei, at no specified temperature, with no solvent and no account of how either product forms. Even a perfectly executed quantum calculation cannot supply physical conditions that were never put into its model.

Yet electronic energies can be essential to the answer. They help determine molecular structures, the relative cost of approaching different arrangements, and the statistical weights that enter a rate calculation. The task is to connect those pieces without mistaking any one of them for the final prediction.

First return to Chapter 6's molecular ion and follow one prediction all the way from its electronic curve to a dissociation threshold. Then add the second electron: a small neutral-hydrogen calculation will make electronic estimation explicit. A separate two-route example will show how thermal and kinetic assumptions can reverse a chemical conclusion.

One molecule, one completed prediction

How much energy is required to separate the hydrogen molecular ion, initially in its lowest vibrational level with zero rotation, into a ground-state hydrogen atom and a proton? This is a question about specified initial and final states of H_2^+, not about neutral H_2 and not about the rate at which a collision forms a bond.

Use exactly the restricted model from Chapter 6: nonrelativistic electrostatic interactions, two fixed hydrogen 1s trial functions for the electron, one electronic energy curve, and a nuclear reduced mass of 918 electron masses. The separated-fragment energy is $U_\infty = -0.5$ hartree. At each proton separation R, the electronic calculation provides a total curve $U(R)$ that already includes proton–proton repulsion. That curve is the input, not the answer, to the nuclear calculation.

Represent nuclear separation on a grid and solve the zero-rotation radial equation (6.6), with the stated boundary conditions. Its lowest eigenvalue E_0 includes nuclear kinetic energy. The energy needed for dissociation from that level is

$$D_0 = U_\infty - E_0 = D_e - \left(E_0 - U_{min}\right). \tag{22.A}$$

Here U_{min} is the curve's minimum and $D_e = U_\infty - U_{min}$ is the clamped-nucleus well depth. The subtraction removes the energy already present in the lowest nuclear level. For the printed trial curve, the minimum is about -0.564831 hartree and the nuclear ground energy is about -0.560747 hartree. Thus $D_e \simeq 1.764$ electron volts, the zero-point contribution is about 0.111 electron volts, and $D_0 \simeq 1.653$ electron volts.[1]

That is one complete model prediction: the molecule fixes a Hamiltonian and reference fragments; the trial functions supply an electronic curve; the grid represents nuclear motion; an eigenvalue supplies a level; and an energy difference supplies the energetic dissociation threshold. All those calculations can be performed on a classical computer using quantum theory. A quantum processor might replace an expensive electronic step in a larger problem, but it would not remove the nuclear step or the definition of the requested observable.

Check the prediction. If further grid refinement leaves the calculated energy unchanged within the numerical tolerance, what have we established—and what remains untested about the molecular model?

That is evidence of stability for the tested grids and domain, not proof that the algorithm is correct or the molecular model accurate. The grid refinement leaves the displayed 1.653-electron-volt result unchanged, while taking the well depth itself as the answer would miss about 0.111 electron volts. Solving the electronic problem more accurately at just one separation does not repair that omission or determine the shape of the curve sampled by nuclear motion. Nor should the deeper minimum obtained by optimizing an orbital scale in Chapter 6 be combined with nuclear levels computed on the older, fixed-orbital curve. That would mix two different models.

Now put the model beside an experimental inference. The target is the energy required to take the ion from its electronic ground state, lowest vibrational level and zero rotation to an infinitely separated ground-state hydrogen atom and proton, with no final relative kinetic energy. This is D_0, not the clamped-nucleus depth D_e. A useful historical determination lets us see how such a number is obtained, rather than presenting it as a fact without a measurement behind it.

In 1994, A. Balakrishnan, V. Smith and B. P. Stoicheff used fluorescence-excitation spectroscopy near 84 nanometers to locate a neutral-hydrogen dissociation threshold producing $H(1s) + H(2s)$. Delayed quenching of the excited $2s$ atoms separated the dissociation signal from overlapping molecular lines. The reported threshold, referred to the neutral molecule's ground level, was $118\,377.06 \pm 0.04\ cm^{-1}$. Spectroscopists often express energy as a wavenumber, $\tilde{\nu} = E/(hc)$; multiplying by hc restores energy units.

That threshold leaves one atom excited. Subtracting the independently known atomic $1s$–$2s$ excitation energy therefore gives the energy required to produce two ground-state atoms:

$$D_0(H_2) = E_{th}\left[H(1s) + H(2s)\right] - \left[E_{2s}(H) - E_{1s}(H)\right].$$

The experiment yielded $D_0(H_2)/(hc) = 36\,118.11 \pm 0.08\ cm^{-1}$. This is still the neutral molecule. To obtain the ion's dissociation energy, combine it with atomic and molecular ionization energies. Define each positive ionization energy as the cost of removing an electron to rest at infinity, with the initial and final species in the specified ground levels. There are two routes from neutral H_2 to the same endpoint, $H(1s) + p + e^-$. First dissociate the molecule and then ionize one atom: the cost is $D_0(H_2) + I(H)$. Or first ionize the molecule and then dissociate the molecular ion: the cost is $I(H_2) + D_0(H_2^+)$. Equality of these energy differences gives

$$D_0(H_2^+) = D_0(H_2) + I(H) - I(H_2).$$

Using the ionization energies then available, the authors inferred $D_0(H_2^+)/(hc) = 21\,379.37 \pm 0.08\ cm^{-1}$, or approximately 2.6507 electron volts. This is an experimentally inferred energy assembled from a dissociation threshold and separate spectroscopic inputs. It was not a direct measurement of our electronic energy curve.[2]

Our restricted prediction, 1.653 electron volts, is about 1.00 electron volt, or 38 percent, smaller. The discrepancy is enormously larger than the grid-convergence change. Chapter 6 already exposed substantial underbinding in the fixed-orbital electronic curve when compared with a more accurate solution of the same clamped-nucleus Hamiltonian. That restricted electronic shape is the principal weakness here. Relativistic, radiative, spin and electron–nuclear coupling corrections matter for precision work; listing them does not explain away this much larger discrepancy.

The comparison has also made the experimental assumptions visible. A threshold must be assigned to identified initial levels and final fragments; frequencies need calibration; other measured energy differences enter the subtraction. An energetic threshold is not automatically the position or strength of an absorption line. The reported uncertainties belong to that experimental inference, while the numerical convergence estimate belongs to our calculation. They measure different sources of uncertainty.

We have used a historical result to display the chain, not to claim the latest precision benchmark. At the accuracy needed to judge this elementary model, the conclusion is decisive: it demonstrates binding and supports an intelligible calculation of nuclear motion, but gives a poor quantitative dissociation energy. Improving it means repairing the physical approximation and calculating the same observable again. More digits from the original grid cannot do that.

Keep this completed chain in mind as the electronic problem grows. The one-electron ion made every stage visible. Neutral hydrogen will show what a quantum energy estimator measures, and the later reaction example will expose additional assumptions needed to turn electronic energies into competing rates. Those examples extend the reasoning; they do not share one set of numerical outputs.

First specify the molecule

A molecular formula is not yet a complete computational input. We need the nuclear positions, the number of electrons, the charge, and the electronic state or symmetry sector being investigated. For a reaction, we also need to identify which structures and pathways are being compared.

At fixed nuclear coordinates R, an electronic Hamiltonian includes electron kinetic energy, attraction to the nuclei, and electron–electron repulsion. In the Born–Oppenheimer construction, solve that electronic problem and include the repulsion between the fixed nuclei once:

$$\begin{aligned} H_{\mathrm{e}}(R)\Psi_k &= E_{\mathrm{e},k}(R)\Psi_k, \\ U_k(R) &= E_{\mathrm{e},k}(R) + V_{\mathrm{nn}}(R). \end{aligned} \tag{22.1}$$

The index k labels an electronic energy state. Moving the nuclei changes the operator and its solutions, producing an energy surface rather than one universal molecular number. A surface minimum describes a locally stable arrangement in this fixed-nuclei calculation. Nuclear motion, which Chapter 6 treated separately, still has to be included where the question requires it.[3]

A conventional transition structure is a saddle: the energy decreases along one displacement through it and rises along the other internal directions. Finding such a point does not by itself establish which reactant and product it connects. The path must be checked. Nor is a transition structure necessarily an electronically excited state. A reaction can pass over a nuclear-coordinate barrier while remaining on its lowest electronic surface.

Near strongly interacting electronic surfaces, following one surface independently can cease to be adequate. That is a limitation of the physical approximation, not a request for more decimal places in the same calculation.

Choose which electronic possibilities to retain

To make the spatial problem finite, expand the electronic description in a chosen collection of functions. Molecular orbitals are combinations of those functions. An orbital together with a spin label is a spin orbital; a list of occupied spin orbitals specifies an antisymmetric many-electron basis state, a Slater determinant. The antisymmetry enforces the exchange rule from Chapter 6.

Two electrons placed among four spin orbitals give six possible occupation patterns. This is a list of joint alternatives, not a list saying which named electron owns which orbital. In a minimal description of hydrogen, two important alternatives doubly occupy either the bonding spatial orbital or the antibonding one. Call these configurations Φ_0 and Φ_1. Their combination repaired the dissociation picture in Chapter 6.

The matrix elements themselves can be assembled on a classical computer from the chosen spatial functions and interactions. That task is not the same as finding the coefficients of the many-electron ground state. A compact list of interactions can couple an enormous number of allowed occupations. Encoding the list does not require first knowing the solution.

A larger calculation may retain many orbitals and configurations. An active-space calculation allows selected electrons to rearrange among selected orbitals, while treating the others more restrictively. Solving that active problem exactly does not recover correlations excluded by its construction. Full configuration interaction likewise means all permitted configurations within a specified finite orbital basis—not all possible physics.

Balanced choices along a reaction path matter. An orbital important near a stretched bond may look unimportant at the reactant geometry. Bensberg and Reiher developed an automated way to select corresponding active spaces along changing structures. Keeping the same number of orbitals is not, by itself, a guarantee of keeping the same quality of description.[4]

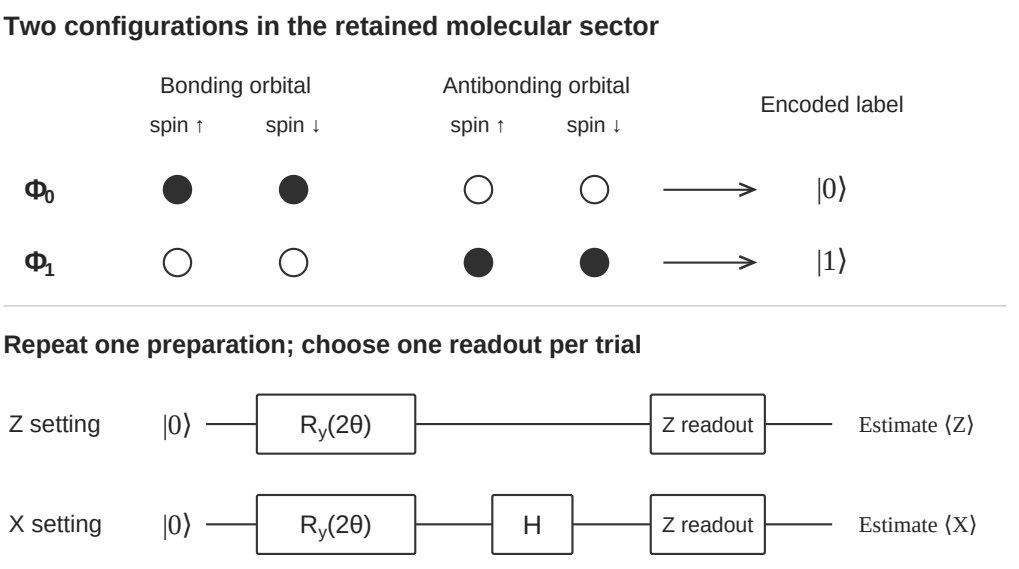


Figure 22.1. An occupation label can become a qubit label. The four columns are ordered spin orbitals: two spin states of the bonding orbital, followed by two of the antibonding orbital. Filled circles indicate occupation; open circles indicate vacancy. The paired configurations span the two-dimensional sector used in the example, with determinant phases chosen consistently. The two readout rows use separate, freshly prepared copies of the same trial state. A Z readout estimates the population difference; H followed by Z estimates X. One encoded qubit represents this restricted configuration space, not the complete states of two arbitrary electrons.

A small calculation with real molecular coefficients

Yamamoto and colleagues reported a one-qubit Hamiltonian for molecular hydrogen at its equilibrium geometry in a minimal basis. Use their rounded coefficients, with the two encoded labels interchanged so the lower diagonal entry comes first. In the ordered configuration basis Φ_0, Φ_1 it is[5]

$$\frac{H_{\mathrm{mol}}}{E_{\mathrm{h}}} = \begin{pmatrix} -1.11701 & -0.18092 \\ -0.18092 & 0.47509 \end{pmatrix}. \tag{22.2}$$

The unit E_{h} is the hartree. One hartree per molecule corresponds to about 627.51 kilocalories per mole. The constant part of this reported molecular operator is already included; adding nuclear repulsion again would double-count it. These coefficients come from an electronic-structure calculation, not a fit to our upcoming reaction example.

Try a normalized real superposition, with an adjustable angle θ:

$$|\psi(\theta)\rangle = \cos\theta\,|\Phi_0\rangle + \sin\theta\,|\Phi_1\rangle,$$

$$\frac{E(\theta)}{E_{\mathrm{h}}} = -1.11701 + 1.59210\sin^2\theta - 0.36184\sin\theta\cos\theta. \tag{22.3}$$

The first configuration alone gives -1.11701 hartree. Adding a little of the second costs diagonal energy, but the cross term can more than compensate. The minimum occurs near $\theta = 0.11174$ radians, about 6.40 degrees. Its energy is approximately -1.137310 hartree: lower by about 12.74 kilocalories per mole, even though the second configuration has only 1.24 percent of the occupation probability.[6]

Because this matrix is real and symmetric, its lowest eigenvector can be chosen real. Our one-angle family therefore contains the exact ground state of the printed model. In a larger calculation, a convenient circuit family may not contain the desired state at all. Finding its best angle would then solve a restricted optimization, not the full electronic problem.

A lottery with exactly those two occupation probabilities would omit the cross term. Its energy would rise above the first configuration’s value, not fall below it. The improvement comes from the joint amplitudes and their relative sign. Figure 22.2 compares the coherent calculation with that lottery.

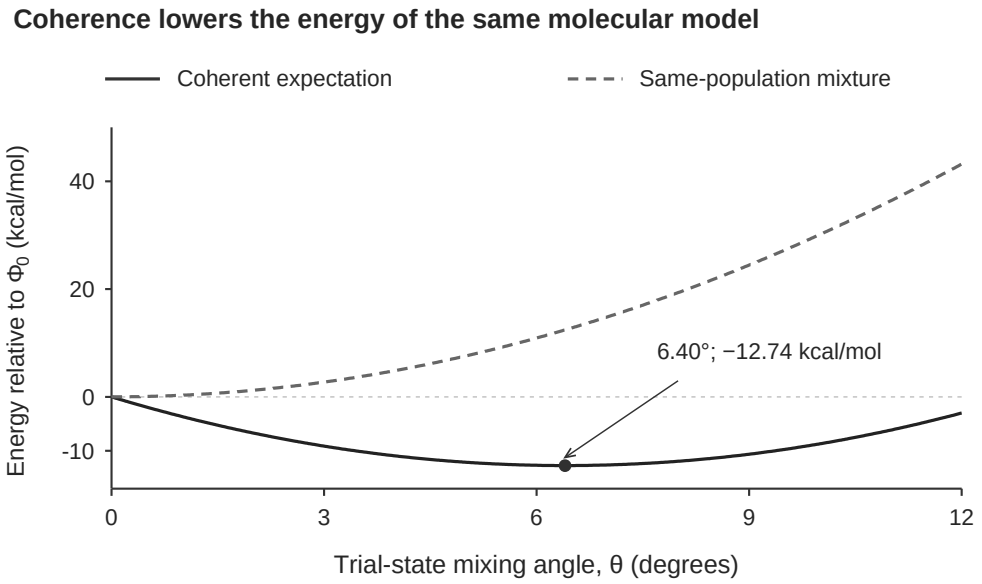


Figure 22.2. The same occupations do not give the same energy. The solid curve is the expectation of Eq. (22.2) in the pure state of Eq. (22.3). The dashed curve uses a classical mixture of the two configurations with the same diagonal probabilities. Energy is relative to the Φ_0 diagonal value, converted to kilocalories per mole. The minimum is approximately -12.74 at 6.40 degrees. Curves are calculations from the published, rounded Hamiltonian—not measured energies, a dissociation curve, or the separate reaction budget in Table 22.1.

This is a tiny matrix and an easy classical calculation. Its role is to reveal what an electronic solver must recover. The lowest eigenvalue is exact for the printed matrix, but that matrix remains a finite-basis molecular approximation at one geometry. Solving it more accurately cannot enlarge the basis or move the nuclei.

How a quantum processor would estimate the energy

Encode Φ_0 as qubit zero and Φ_1 as qubit one. The same operator becomes

$$\frac{H_{\mathrm{mol}}}{E_{\mathrm{h}}} = -0.32096\, I - 0.79605\, Z - 0.18092\, X\,. \tag{22.4}$$

A rotation through 2θ about the Bloch Y axis prepares the trial state from zero. Measure Z on one set of preparations and X on another, as Figure 22.1 shows. The average numerical signs, each between minus one and plus one, supply the two terms. A classical optimizer changes the angle and requests another estimate. This is the basic variational quantum eigensolver strategy demonstrated by Peruzzo and colleagues: quantum preparations and measurements coupled to classical optimization.[7]

The X measurement is essential. Measuring occupation strings alone supplies the diagonal probabilities, which our coherent state and its lottery share. It would miss the energy-lowering term. In larger encodings the Hamiltonian becomes a sum of products of qubit operators, and the measurement plan must recover the expectations that actually

enter that sum. The final energy is not obtained by averaging the binary integers printed by the register.

Let $\bar{z}$ and $\bar{x}$ denote the measured averages. For independent, ideal samples numbering m_Z and m_X, the estimator and its variance are

$$\frac{\hat{E}}{E_{\mathrm{h}}} = -0.32096 - 0.79605\,\bar{z} - 0.18092\,\bar{x},$$
$$\frac{\mathrm{Var}\left(\hat{E}\right)}{E_{\mathrm{h}}^2} = \frac{0.79605^2\left(1-\langle Z\rangle^2\right)}{m_Z} + \frac{0.18092^2\left(1-\langle X\rangle^2\right)}{m_X}. \tag{22.5}$$

Each variance is the familiar fluctuation of repeated binary outcomes. At the exact model ground state, ten thousand trials in each setting still give an energy standard uncertainty of about 1.57 kilocalories per mole. The state has definite energy, yet this particular readout estimates it through two separately fluctuating quantities. A direct energy measurement and this sum of sample averages are different procedures.[8]

For the specified Hamiltonian, the true expectation in any physical trial state cannot lie below its ground energy. That variational bound does not constrain every finite-sample estimate. Noise in the counts can push an estimate lower. Choosing the lowest among many noisy estimates can accentuate the problem. Once the trial state is selected, fresh evaluation data can assess it without reusing the favorable fluctuations that helped select it.

Another route reads energy from time evolution. An energy component accumulates phase $e^{-iEt/\hbar}$. Controlled evolution and phase estimation, related to the Fourier construction of Chapter 15, can infer that phase and hence the energy. The initial state must have enough overlap with the desired energy state; the evolution, phase range, accuracy, and repetitions all cost resources. The procedure does not automatically prepare the ground state.[9]

Yamamoto and colleagues' 2026 experiment used this second route with error-corrected trapped-ion registers for the hydrogen model. Their reported estimate, $-1.136(13)$ hartree, was consistent with the finite-basis reference but retained a statistical uncertainty of 0.013 hartree, about 8.16 kilocalories per mole. Error-detected rejected shots and calibrated noise-aware processing were part of the procedure. Its fault-tolerant protection was partial. The result was a molecular-energy computation, not a chemical rate prediction or a precisely measured complete-basis molecular energy.[10]

There is more than one way to choose a useful subspace

A quantum processor can also help select configurations rather than return the energy directly. In Robledo-Moreno and colleagues' 2025 study, measured occupation strings helped construct reduced electronic spaces whose Hamiltonians were then diagonalized classically. This sample-based quantum diagonalization used both quantum hardware and

substantial classical computation. The samples were not measurements of all the wavefunction's amplitudes.[11]

The classical alternatives keep changing too. Zhang and Otten's study, released as peer-reviewed accepted research in September 2026, introduced a classical method that starts from random determinants and repeatedly expands and trims the retained space. Its numerical successes do not prove that every molecule has an easily found compact description. They do show why an enormous full matrix is not the only classical competitor.[12]

A useful comparison therefore asks how accurately each complete procedure predicts the same quantity. A smaller circuit, a lower variational energy, or a larger represented orbital space can be progress without yet answering that question.

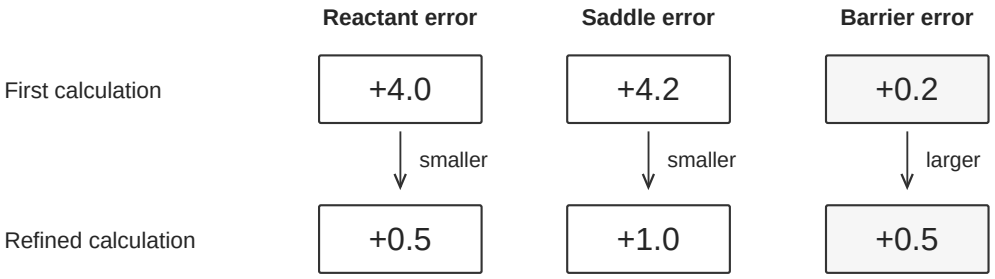
Better energies can give a worse barrier

Return to a reaction. Suppose an approximate calculation has errors ϵ_{A} at the reactant and $\epsilon_{\ddagger}$ at a fixed saddle geometry, relative to exact energies of the same model. Subtracting gives

$$\widetilde{\Delta U}^{\ddagger} - \Delta U^{\ddagger} = \epsilon_{\ddagger} - \epsilon_{\mathrm{A}}. \tag{22.6}$$

Imagine two positive variational errors: 4.0 kilocalories per mole at the reactant and 4.2 at the saddle. The barrier error is only 0.2. A refined calculation lowers those errors to 0.5 and 1.0. Both individual energies improve, but the barrier error grows to 0.5. Figure 22.3 keeps all four errors visible.[13]

A barrier is a difference, not either energy separately



All errors in kcal/mol, relative to one unchanged model at fixed geometries.

Figure 22.3. Improving both upper bounds can worsen their difference. The entries are stipulated positive variational errors at two fixed geometries, measured relative to the exact energies of one unchanged model. In the refinement, both errors decrease, while their difference increases from 0.2 to 0.5 kilocalories per mole. These are not measured errors or uncertainty intervals. A variational bound on each energy is not a bound on a barrier's error; error cancellation must be assessed rather than presumed.

Large energies often share contributions that cancel in a difference. That can help enormously, but the cancellation must survive the change in geometry and electronic

character. Using unrelated approximations on the two sides may remove it. Conversely, agreement between two calculations with the same omitted interaction may preserve a common bias. Matching printed digits is not an independent physical test.

The nuclei and the temperature re-enter

A potential-energy barrier is not yet a free-energy barrier. At a specified temperature, the reactant and the transition region have different accessible nuclear motions and statistical weights. Zero-point vibration, thermal excitation, and entropy alter their relative cost.

For a conventional molecular treatment, write the molar activation free energy schematically as

$$\Delta G_j^{\ddagger} = \Delta U_j^{\ddagger} + \Delta E_{\mathrm{ZP},j}^{\ddagger} + \Delta H_{\mathrm{th},j}^{\ddagger} - T\,\Delta S_j^{\ddagger}. \tag{22.7}$$

The label j identifies a route. Here the thermal enthalpy correction excludes zero-point energy, which is already written separately. Entropy counts the relevant accessible arrangements and motions; it is not an informal score for how disorderly a drawing looks. These molecular thermodynamic quantities require a temperature and consistent standard states.[14]

For example, one route may require a more constrained arrangement of the molecule than another. Even with a lower electronic barrier, fewer thermally available motions may lead into that transition region. The free-energy comparison counts both the energetic cost and the statistical restriction. The molecule does not read the table and choose a route; the competing fluxes emerge from its dynamics and preparation.

In a harmonic treatment, the unstable transition coordinate is not counted as another bound vibrational mode. Flexible molecules may also need several conformations and a better treatment of internal rotations. A room-temperature distribution need not be adequately represented by the single geometry with the lowest electronic energy.

Solvent introduces further interactions and statistical averaging. Replacing surrounding molecules by a dielectric medium is one useful model, not the removal of solvent from the problem. Explicit solvent configurations offer another route with their own sampling demands. Standard-state conversions and the actual protonation and charge states must remain consistent. We will omit solvent in the worked example rather than pretend its effects are universal constants.[15]

Now construct two gas-phase unimolecular routes from the same thermalized reactant A. The following budget is stipulated to test the inference; it is not derived from the hydrogen matrix or attributed to a measured reaction. All entries are molar differences from A at 298.15 kelvins.

Table 22.1**. A lower electronic barrier need not give the faster route.**

Contribution (kcal/mol)	Route to P	Route to Q
Fixed-nuclei barrier $\Delta U^{\ddagger}$	17.00	17.60
Zero-point correction $\Delta E_{\mathrm{ZP}}^{\ddagger}$	0.50	0.30
Thermal enthalpy correction $\Delta H_{\mathrm{th}}^{\ddagger}$	0.30	0.10
Entropy contribution $-T\,\Delta S^{\ddagger}$	2.20	1.00
Activation free energy $\Delta G^{\ddagger}$	20.00	19.00

The numbers form a constructed budget at 298.15 K, with a common reactant and consistent standard states. Thermal enthalpy excludes the separately listed zero-point term. No solvent correction is included. Electronic barriers favor P, but the completed free-energy budget favors Q. These are not experimental data or results for molecular hydrogen.

A difference becomes a rate

For our unimolecular, thermally equilibrated model, transition-state theory gives

$$k_j(T) = \kappa_j(T)\,\frac{k_{\mathrm{B}}T}{h}\exp\left[-\frac{\Delta G_j^{\ddagger}}{RT}\right]. \tag{22.8}$$

Here k_j has units of inverse seconds, k_{B} is Boltzmann's constant, h is Planck's constant, and R is the molar gas constant. The exponential penalizes reaching a high-free-energy transition region. The prefactor supplies a frequency scale. Eyring's formulation connects statistical populations with reactive flux; a suitable transmission factor κ_j accounts for specified departures from the simplest crossing model.[16]

The transition region is not a second stable reservoir that stores molecules indefinitely. The theory relates its statistical weight to the rate of crossing a chosen boundary. If a trajectory crosses and promptly returns, counting the first crossing as a permanent reaction can overestimate the rate. Tunnelling can require a different correction. Those dynamical questions remain even after the energies at the relevant geometries are known.

For the arithmetic, set both transmission factors to one and assume no competing pathways or return reactions. At 298.15 K, $RT \approx 0.592485$ kilocalories per mole. Table 22.1 gives $k_{\mathrm{P}} \approx 0.01359$ per second and $k_{\mathrm{Q}} \approx 0.07348$ per second.

The useful comparison depends on the difference of the two barriers:

$$d = \Delta G_{\mathrm{P}}^{\ddagger} - \Delta G_{\mathrm{Q}}^{\ddagger}, \quad \frac{k_{\mathrm{Q}}}{k_{\mathrm{P}}} = e^{d/(RT)}, \qquad f_{\mathrm{Q}} = \frac{k_{\mathrm{Q}}}{k_{\mathrm{P}} + k_{\mathrm{Q}}}. \tag{22.9}$$

The fraction f_{Q} is the share of product Q for this two-channel irreversible first-order model, starting from A alone. The common precursor decays with the sum of the rates; each route receives its corresponding share. With $d = 1.00$ kilocalorie per mole, Q forms about 5.41 times as fast as P and receives about 84.4 percent of the products.[17]

Using the electronic barriers alone would have favored P by about 2.75 to one under that same exponential comparison. The free-energy corrections reverse the ranking. An arbitrarily precise electronic calculation, followed by omitting those corrections, would confidently answer the wrong question.

Nor does a lower product energy necessarily identify the product that forms fastest. Equilibrium preferences depend on product free energies; branching under our assumptions depends on competing activation free energies. Returns, intermediate reservoirs, diffusion, or incomplete thermalization can require an entire kinetic network rather than this two-rate ratio.

Ask how accurate the conclusion must be

A one-kilocalorie-per-mole error is often used as a convenient chemical accuracy scale. Our example shows why it is not a universal permission slip. At this temperature, changing one activation free energy by that amount changes its predicted rate by a factor of about 5.41 when the prefactor is fixed. Keeping that factor below two requires an error smaller than $RT\ln 2$, about 0.411 kilocalories per mole.

Uncertainty in the barrier difference also depends on shared errors. For statistically characterized uncertainties,

$$u_d^2 = u_{\mathrm{P}}^2 + u_{\mathrm{Q}}^2 - 2\,\mathrm{Cov}\left(\Delta G_{\mathrm{P}}^{\ddagger}, \Delta G_{\mathrm{Q}}^{\ddagger}\right). \tag{22.10}$$

The covariance records how the two errors vary together. If both barriers subtract the very same computed reactant free energy, that common contribution cancels exactly from their difference. It does not cancel from either absolute rate. Independent shot noise, uncertain thermal corrections, and shared model errors should not all be combined as though they were independent random draws. Unquantified omissions are not converted into measured standard deviations by this formula.[18]

Suppose the decision requires Q to form at least three times as fast as P. Under the present equal-prefactor model, the condition is $d \geq RT\ln 3$, about 0.651 kilocalories per mole. A justified bound $d = 1.00 \pm 0.40$ establishes that Q is faster, but its lower endpoint permits a ratio of only 2.75. Tightening the bound to 1.00 ± 0.20 raises that minimum to 3.86 and supports the stronger conclusion. Figure 22.4 distinguishes those questions.[19]

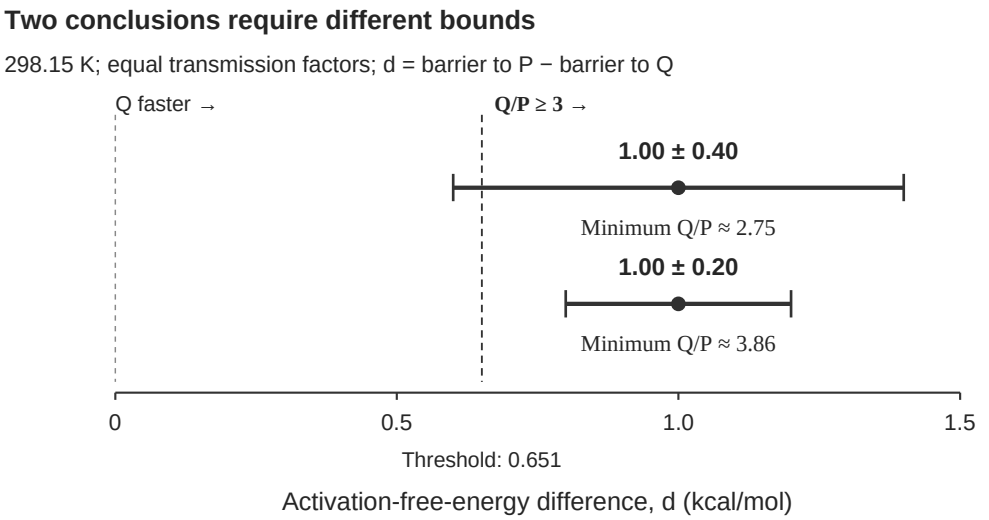


Figure 22.4. The required precision depends on the decision. The horizontal coordinate is the activation-free-energy difference d, at 298.15 K with equal transmission factors. Q is faster to the right of zero; a Q/P rate ratio of at least three requires $d \geq 0.651$ kcal/mol. The two displayed intervals are hypothetical justified bounds, not confidence intervals estimated from data. Both establish a preference for Q, but only the narrower interval establishes the three-to-one claim. Their worst-case rate ratios are approximately 2.75 and 3.86.

Where should the next calculation go? If sample noise dominates the relevant difference, more preparations may help. If a missing conformer or an uncertain solvent response determines the ranking, refining a fixed-geometry electronic energy may not. Sensitivity analysis in a reaction network can identify which uncertain inputs actually change the predicted outcome. Proppe and Reiher's kinetic-network work develops that connection between uncertain energies and mechanism inference.[18]

The chemical conclusion is the end of the calculation

The two routes have now supplied an answer, but only after we named the conditions under which the answer holds. We needed electronic structure, a nuclear and thermal model, a rate expression, and a rule for converting competing rates into products. Each step introduced a different possible error.

Agreement with one measured rate is therefore not the end of validation. Changing the temperature tests the predicted balance of energy and statistical weight. Comparing additional structures or pathways tests whether a favorable cancellation persists. Each new comparison should be predicted without silently refitting the quantity being tested. A model that explains several independent changes earns more trust than one adjusted to reproduce one favorable number.

A quantum processor can contribute a better electronic state, a better energy estimate, or useful configurations for a classical calculation. Its value need not be diminished because

other steps remain classical. But its output must be judged where those steps meet: the energy difference, rate, spectrum, or chemical decision the investigation actually needs.

That returns us to the number with eight decimal places. We can now ask what Hamiltonian it belongs to, which approximation limited it, how its uncertainty was assessed, and what changes when it is carried through the rest of the prediction. Precision becomes useful when it survives that journey.

The next question is how to compare the whole undertaking with its alternatives. A faster subroutine is one achievement. A trustworthy chemical conclusion obtained more effectively is another. Establishing when the first becomes the second is the subject of the next chapter.

Notes and sources

1. Equation (22.A) restates Chapter 6, Eq. (6.7), for the same restricted hydrogen molecular-ion curve, not the neutral-hydrogen Hamiltonian of Eq. (22.2). To reproduce the nuclear level, evaluate the electronic curve from Chapter 6 on a uniform grid and represent the second derivative in Eq. (6.6) by a centered three-point difference. Add the curve to the diagonal of the resulting tridiagonal Hamiltonian and solve for its lowest eigenpair. The accompanying script checks residuals and discrete normalization for the first four eigenpairs on the two grids and domains specified in Chapter 6, note 13. With R from 0.1 to 20 bohr, Dirichlet boundaries, step 0.0025 bohr, and reduced mass 918, it gives E_0 = −0.5607472658 hartree and D_0 = 1.653017314 eV; the computed well depth is 1.764141174 eV. These are finite-grid results for a restricted Born–Oppenheimer model, not measured energies or a rigorous total-error bound for the real molecule. These checks test the assembled finite matrices and sensitivity to the stated refinements; they do not bound omitted physical interactions.

2. A. Balakrishnan, V. Smith and B. P. Stoicheff, “Dissociation energies of the hydrogen and deuterium molecules,” Physical Review A 49, 2460–2469 (1994), doi:10.1103/PhysRevA.49.2460. The values and ± uncertainties are quoted as reported, without assigning a different confidence convention. Conversion uses 1 cm^{-1}=1.2398419843×10^{-4} eV. The comparison here concerns the ground electronic, vibrational and rotational dissociation threshold to H(1s)+p; it does not resolve hyperfine components. This historical result is sufficient to assess the large error of the stated trial model and is not presented as the latest determination. Source.

3. Equation (22.1) states the fixed-nuclei, nonrelativistic Born–Oppenheimer energy account used in Chapter 6. The electronic operator excludes nuclear repulsion when V_{nn} is added explicitly. For the kinetic and Coulomb terms see Richard P. Feynman, Robert B. Leighton, and Matthew Sands, The Feynman Lectures on Physics, Vol. III, Chapters 19 and 10. The saddle-point and rate construction is developed in the primary reaction dataset cited in note 14. Coupled electronic surfaces and nuclear nonadiabatic effects are not calculated here. Position-space Hamiltonian. Molecular-ion treatment.

4. Moritz Bensberg and Markus Reiher, “Corresponding Active Orbital Spaces along Chemical Reaction Paths,” Journal of Physical Chemistry Letters 14, 2112–2118 (2023), doi:10.1021/acs.jpclett.2c03905. The full author version is arXiv:2212.12883v2, revised 13 April 2023. Its orbital-mapping and selection construction addresses balanced descriptions across geometries, not a universal guarantee from keeping a fixed orbital count. Our two-paired-configuration illustration is not a reproduction of its 1-pentene calculations. Published record. Author paper.

5. Kentaro Yamamoto and colleagues, “Quantum Error-Corrected Computation of Molecular Energies,” PRX Quantum 7, 020319 (2026), doi:10.1103/m7j3-5sk6, published 30 April 2026, Sec. II and Eq. (1). The published coefficients are $H / E_{\mathrm{h}} = 0.79605\,Z - 0.18092\,X - 0.32096\,I$. Conjugation by X interchanges encoded labels and gives Eq. (22.2). We use the printed five-decimal coefficients; we do not recompute their molecular integrals or claim additional basis or geometric accuracy. The scalar offset is retained once. Figure 22.1 is original. Published full paper.

6. Equation (22.3) and Figure 22.2 are original evaluations of Eq. (22.2). The angle is one-half of $\arctan\left(0.18092 / 0.79605\right)$; the eigenvalue is $-0.32096 - \sqrt{0.79605^2 + 0.18092^2}$ in hartree. The coherent lowering is 0.0203001999 hartree. Diagonalizing Eq. (22.2) and minimizing Eq. (22.3) provide two ways to reproduce this value. Conversions use the 2022 CODATA hartree energy $4.3597447222060 \times 10^{-18}$ J, exact SI Avogadro constant, and the thermochemical calorie, 4.184 J. Constant uncertainty and coefficient rounding are distinct from sampling uncertainty; our displayed conversions are deliberately rounded. NIST hartree value. SI defining constants. The thermochemical-calorie conversion is specified in NIST’s Guide to the SI, Appendix B.8 and footnote 9. Exact calorie conversion.

7. Alberto Peruzzo and colleagues, “A variational eigenvalue solver on a photonic quantum processor,” Nature Communications 5, 4213 (2014), doi:10.1038/ncomms5213. The experiment introduced a hybrid variational energy-estimation workflow. Our real-angle trial family, one-qubit dictionary, and separate X/Z

readout are a worked construction for Eq. (22.2), not its experimental data or molecule. The single-qubit circuit is classically trivial; no speed advantage is asserted. Published full article.

8. Jarrod R. McClean, Jonathan Romero, Ryan Babbush, and Alán Aspuru-Guzik, "The theory of variational hybrid quantum-classical algorithms," New Journal of Physics 18, 023023 (2016), doi:10.1088/1367-2630/18/2/023023. Equation (22.5) is an original independent-sample calculation for two Pauli averages. At the model optimum, $\langle Z \rangle = 0.9751329761$ and $\langle X \rangle = 0.2216205741$. Ten thousand fresh trials in each basis give standard uncertainty 0.00249497053 hartree, or 1.56562 kcal/mol. The number is not a confidence interval, device error, or uncertainty of direct projective energy measurement. Fresh validation addresses selection of favorable statistical fluctuations, not systematic bias. Full author analysis. Publication. The linked author manuscript is arXiv:1509.04279v1, submitted 14 September 2015.

9. Alán Aspuru-Guzik, Anthony D. Dutoi, Peter J. Love, and Martin Head-Gordon, "Simulated Quantum Computation of Molecular Energies," Science 309, 1704–1707 (2005), doi:10.1126/science.1113479. This is the original molecular-energy phase-estimation proposal. The general phase account also follows the controlled-evolution and Fourier constructions developed earlier in this book. It requires appropriate input overlap and coherent evolution; it is not a ground-state preparation theorem. Different energy-estimation variants have different depth, sampling, and inference costs. The modern experiment is specified separately in note 10. Original publication. The available author manuscript is arXiv:quant-ph/0604193v1, deposited 26 April 2006, after the journal publication. Author manuscript.

10. Yamamoto and colleagues (note 5), Secs. III B and V. The reported $-1.136(13)$ hartree means statistical uncertainty 0.013 hartree in the reported convention; the finite-basis reference is -1.13731. Their error-detection gadgets reject 4.7 percent of total shots, excluded from the retained sample count of 2,186. Noise-aware inference uses calibrated parameters. These are experiment-specific qualifications, not values generated by our variational calculation. Published results. Section II A estimates a systematic truncation contribution of about 0.001 hartree in the finite evolution and rotation construction. The reported 0.013-hartree statistical uncertainty does not include every systematic contribution, and the implementation is only partially fault tolerant.

11. Javier Robledo-Moreno and colleagues, "Chemistry Beyond the Scale of Exact Diagonalization on a Quantum-Centric Supercomputer," Science Advances 11, eadu9991 (2025), doi:10.1126/sciadv.adu9991. The author version is arXiv:2405.05068v3, revised 13 July 2025. Sample-based quantum diagonalization combines quantum occupation samples, configuration recovery, and classical projected-Hamiltonian diagonalization. "Beyond exact diagonalization" refers to the full represented space, not every approximate classical competitor. No end-to-end speedup is inferred here from a qubit count or from the dimension of that space. Publication record. Full author paper.

12. Hao Zhang and Matthew Otten, "An auto-initializing algorithm to determine electronic ground states from random Slater determinants," Nature Communications, doi:10.1038/s41467-026-77968-1, published 29 September 2026 as an Article in Press. The cited source is the publisher's peer-reviewed accepted version, which may differ from the final Version of Record. TrimCI iteratively expands and trims a determinant space using the Hamiltonian's couplings. Its benchmark results do not establish universal efficiency, and the discussion notes circumstances in which good prior reference states favor other methods. The discussion makes no claim of a universal speedup or an experimental quantum advantage. Accepted article Publisher record.

13. Equation (22.6) and Figure 22.3 are original counterexamples using positive variational energy errors at fixed geometries. The errors are not statistical standard uncertainties. Each can be realized by a normalized mixture of two exact model eigenstates, so their ordering is consistent with the variational principle. The example does not assert a variational upper or lower bound on a difference, nor does it include the additional error from shifting an optimized geometry. The same chosen Hamiltonian and energy references must be retained throughout.

14. Kevin Spiekermann, Lagnajit Pattanaik, and William H. Green, "High accuracy barrier heights, enthalpies, and rate coefficients for chemical reactions," Scientific Data 9, 417 (2022), doi:10.1038/s41597-022-01529-6. The full primary computational study separates electronic energies, geometry/frequency calculations, and transition-state-theory rates. It limits its rate dataset to rigid reactions rather than extending

a harmonic treatment uncritically to flexible species. Equation (22.7) is our schematic molar thermodynamic decomposition; Table 22.1 is an entirely stipulated budget, not data from the paper. Standard states and temperature are common to its two routes. Published full study.

15. Aleksandr V. Marenich, Christopher J. Cramer, and Donald G. Truhlar, "Universal Solvation Model Based on Solute Electron Density and on a Continuum Model of the Solvent Defined by the Bulk Dielectric Constant and Atomic Surface Tensions," Journal of Physical Chemistry B 113, 6378–6396 (2009), doi:10.1021/jp810292n. This primary model is an example of solvent represented through a continuum rather than individually sampled solvent molecules. No solvent energy, conformer correction, or solution-phase prediction is computed in the present chapter. Such additions would require corresponding definitions, parameters, and uncertainty assessment. Primary model and abstract. Publication. A full copy of the published article is available from an institutional host. Published article copy.

16. Henry Eyring, "The Activated Complex in Chemical Reactions," Journal of Chemical Physics 3, 107–115 (1935), doi:10.1063/1.1749604. Equation (22.8) uses a thermally equilibrated unimolecular reactant, consistent standard-state activation free energies, and an explicitly supplied transmission factor. The worked example sets both factors to one. Barrier recrossing, tunnelling, pressure dependence, and nonadiabatic pathways may require further dynamics; they are not certified by a saddle energy. A bimolecular rate has a different standard-state concentration account and is not assigned units of inverse seconds by this formula without that account. Original publication.

17. Equation (22.9), Table 22.1, and the numerical rate predictions are original stipulated examples. At 298.15 K, $RT = 0.5924849497$ kcal/mol and $k_{\mathrm{B}}T/h = 6.21244 \times 10^{12}$ per second. The rates are 0.01358815005 and 0.07347936941 per second; their ratio is 5.407606565 and Q fraction 0.843935487. For A initially one and products initially zero, $A(t) = e^{-(k_{\mathrm{P}}+k_{\mathrm{Q}})t}$ and $Q(t) = f_{\mathrm{Q}}\left[1 - A(t)\right]$. The same populations follow by integrating the two first-order rate equations with the stated initial conditions. These probabilities describe only the specified irreversible parallel channels, not a general product-yield formula. SI constants.

18. Jonny Proppe and Markus Reiher, "Mechanism Deduction from Noisy Chemical Reaction Networks," Journal of Chemical Theory and Computation 15, 357–370 (2019), doi:10.1021/acs.jctc.8b00310. The full author version is arXiv:1803.09346v3, revised 19 December 2018. Its uncertainty propagation and kinetic sensitivity analysis connect uncertain inputs to mechanism deductions. Equation (22.10) is the exact variance identity for a difference, not a claim that truncation, solvent, and hardware errors follow known Gaussian distributions. A shared reactant term cancels algebraically in the two-route difference if it is literally the same quantity in both; residual saddle and correction errors do not thereby disappear. Full primary analysis. Publication.

19. Figure 22.4 and the tolerance calculations are original consequences of Eq. (22.9) at 298.15 K with fixed equal transmission factors. The three-to-one threshold is $RT\ln 3 = 0.6509112466$ kcal/mol. Lower endpoints 0.60 and 0.80 give ratios 2.752979973 and 3.858371752. The intervals are hypothetical bounds encompassing all errors relevant to this stipulated rate ratio, not empirical confidence intervals or permission to replace model uncertainty with shot uncertainty. Unknown unequal transmission factors would add $RT\ln\left(\kappa_{\mathrm{Q}}/\kappa_{\mathrm{P}}\right)$ to the effective barrier difference. Numerical consistency within the model does not establish the accuracy of a real reaction prediction.

CHAPTER 23

When Is an Advantage Really an Advantage?

Two teams receive the same chemical question: will one product form at least three times as fast as another? The first team finishes its electronic calculation in a minute. The second takes an hour and supplies a defensible answer to the question. Which team was faster?

The clocks stopped at different places. The minute measured a subroutine. The hour measured a completed inference. A fair comparison must follow both procedures to the same destination.

This need not diminish the faster electronic calculation. It may be the improvement that makes the complete prediction affordable. But to discover that, we must put it back into the procedure that needs it. The relevant quantity is not how quickly a machine produces a number. It is what that number allows us to conclude, with what reliability, and at what total cost.

We have reached a point where the earlier chapters can answer these questions together. Algorithms supply operations. Hardware supplies imperfect implementations. Statistics connects records to estimates. Models link those estimates to the system's behavior. A claim of advantage crosses all four boundaries.

Decide where the race ends

Begin by specifying the input and the required output. An integer to be factored, a molecular Hamiltonian, and a supply of unknown quantum states are different inputs. One candidate answer, a sample from a distribution, and a complete probability table are different outputs. Giving one competitor a state already prepared while charging the other for preparing it changes the race.

The required accuracy belongs in the question too. Finding a low-energy configuration is not the same as proving that no better one exists. Returning a rate within a factor of two is not the same as reporting an electronic energy within a fixed tolerance. A solver allowed occasional failure needs a stated success requirement.

Only then can we compare resources. Elapsed time, total processor time, memory, energy, and monetary cost need not rank methods in the same order. A faster result obtained with many more processors can be valuable; the extra processors should remain visible. Rønnow and colleagues emphasized how the kind of comparator and the performance measure determine what a quantum-speedup claim establishes.[1]

Table 23.1**. The evidence must match the kind of claim.**

Evidence	What it can establish	What it does not supply
Algorithmic theorem	A guarantee under specified input access, error, and resource rules.	The running time of an unbuilt physical machine.
Resource estimate	A projected complete cost under stated hardware and algorithm assumptions.	A measured advantage or a date when it will be available.
Executed benchmark	Observed performance on named tasks against tested alternatives.	A proof against every future classical method.

These are different forms of evidence, not a ranking of scientific worth. An experiment can test a theoretical prediction; an estimate can use measured component properties. Their assumptions and requested outputs must still match.

There is also useful progress that is not a speed comparison. A simulator may let us vary an interaction that is inaccessible in the original material. An encoded memory may preserve a state long enough for a new experiment. We should describe those capabilities precisely rather than make their value depend on an unrelated claim that all classical computers have been overtaken.

The answer has an accuracy requirement

Return to the two competing routes from Chapter 22. Keep its deliberately simple conditions: a common thermalized reactant, two irreversible unimolecular channels, equal transmission factors, and a temperature of 298.15 kelvins. Let d be the activation free energy for the route to P minus that for the route to Q. Then

$$\frac{k_Q}{k_P} = e^{d/[RT]}, d_* = RT\ln 3 \approx 0.651\,\text{kcal/mol}. \tag{23.1}$$

The requested three-to-one conclusion requires $d \geq d_*$. A confident ranking of which route is faster is a weaker answer. A calculation too uncertain to establish the stronger claim has not completed this particular task.[2]

Suppose a method returns $\hat{d} = 0.80$ kilocalories per mole. Its sampling analysis supplies a half-width $u_{\text{stat}} = 0.04$ at a stated coverage level. Separately, suppose we have justified a numerical-approximation bound $b_{\text{num}} = 0.03$ and a bound $b_{\text{model}} = 0.15$ on the remaining physical-model discrepancy relevant to this difference. These are constructed budgets, not experimentally established uncertainties.

On the event that the sampling interval covers its intended quantity, a conservative account gives

$$|\hat{d} - d| \leq b_{\text{model}} + b_{\text{num}} + u_{\text{stat}}. \tag{23.2}$$

We have added bounds, not independent variances. The triangle inequality needs no assumption that the model and numerical errors fluctuate independently. If those two bounds really hold and the statistical interval has 95-percent coverage, the enlarged

interval has at least that coverage too. An unknown omitted mechanism is not a justified bound merely because we assign it a small number.

The lower endpoint is 0.58, below the required 0.651. Now spend a hundred times as many independent measurement trials. Under the same regular sampling model, the statistical half-width becomes about 0.004. The lower endpoint improves only to 0.616. The electronic readout is more precise, but the requested conclusion is still unresolved.

Instead, suppose additional physical analysis reduces the model bound to 0.04 while leaving the original sampling budget unchanged. The lower endpoint becomes 0.69 and supports the claim. Figure 23.1 compares these three accounts. The valuable next calculation was the one that removed the limiting uncertainty, not necessarily the one performed on the most advanced processor.

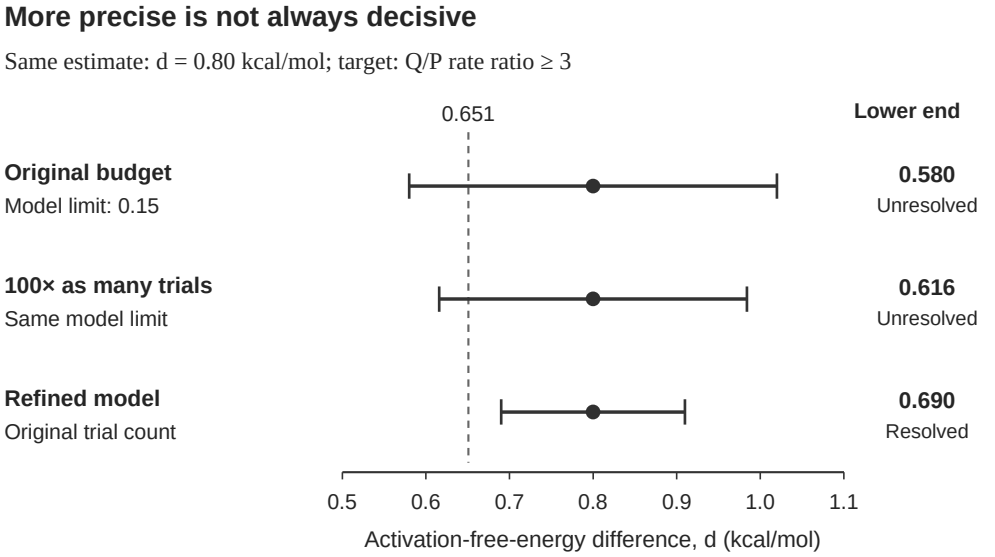


Figure 23.1. The useful improvement is the one that resolves the question. The center is $\hat{d} = 0.80$ kcal/mol in every row. The first interval uses model, numerical, and sampling allowances of 0.15, 0.03, and 0.04. The second changes only sampling to 0.004; the third instead changes only the model allowance to 0.04. The dashed line is $RT \ln 3$ at 298.15 K. Only the third lower endpoint exceeds it. These are hypothetical justified budgets under the stated reaction model, not measured confidence intervals or a universal error decomposition.

Check what more trials can establish

If a hundredfold increase in trials is insufficient, could indefinitely many trials establish the three-to-one rate claim under the original model and numerical bounds?

Even if the statistical half-width approached zero and the central estimate remained 0.80, the lower endpoint would approach only 0.80 − 0.15 − 0.03 = 0.62 kilocalories per mole. That remains below the required threshold of approximately 0.651. The hundredfold

increase reduces the statistical half-width by about tenfold under the stipulated independent-trial model; it does not reduce either of the other bounds.

This does not show that the three-to-one claim is false. It shows that the present evidence cannot establish it. The useful next step must address the limiting physical or numerical uncertainty, or obtain independent evidence that constrains the requested rate ratio. Reducing a model bound requires justification from analysis or observations; changing the number in an error budget is not an improvement. More precise readout and a better-supported physical prediction answer different parts of the problem.

A hundred times faster—inside the calculation

Now suppose both procedures can meet the same accuracy requirement. An all-classical workflow takes 100 minutes per case: 90 for its demanding core calculation and 10 for everything else. Replace the core by a quantum-assisted procedure that is a hundred times faster. Its core time becomes 0.9 minutes.

The full time is not 0.9 minutes. Even without any new overhead, it is 10.9. Suppose the replacement also requires 20 minutes for additional preparation, control, data transfer, and processing. The complete time becomes 30.9 minutes: an improvement of about 3.24, not a hundred. Figure 23.2 keeps the omitted work on the same scale.

The arithmetic generalizes. Let f be the fraction of the original time spent in the accelerated component, s its acceleration factor, and a the added time divided by the original total. For this sequential model, the complete speedup is

$$S = \frac{T_C}{T_H} = \frac{1}{(1-f) + f/s + a}. \tag{23.3}$$

Here C denotes classical and H denotes the hybrid procedure. With $f = 0.9$, even infinitely fast replacement cannot exceed a tenfold improvement when $a = 0$. Additional work lowers that limit. Conversely, an improvement that removes several expensive stages can do more than accelerating just one. The model must describe the actual dependency chain.[3]

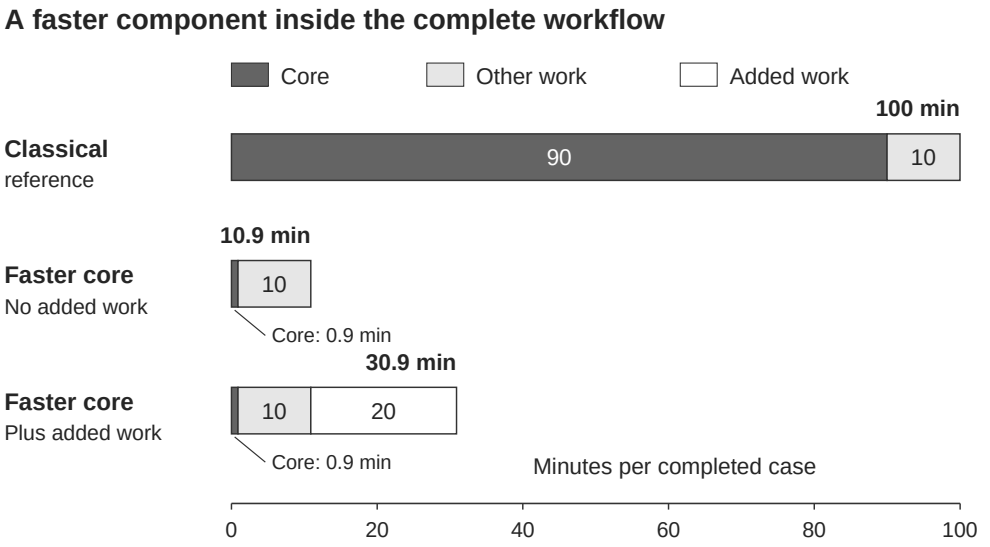


Figure 23.2. A fast core is not the complete running time. The three bars share one linear time scale. The reference spends 90 minutes in its core and 10 elsewhere. A hundredfold core acceleration gives 0.9 minutes but leaves the other ten; adding twenty minutes of replacement-specific work gives 30.9 overall. Segment styles identify the cost categories, and the narrow 0.9-minute segment is retained. Values are stipulated sequential costs for equally adequate outputs, not hardware measurements or projections for a particular machine.

These times already include the repetitions needed for the required accuracy. Calling one quantum shot the replacement for a full classical solve would compare different amounts of work. Overlapping stages or parallel execution can change elapsed time, but then the schedule, rather than a simple sum, must be analyzed. Faster gates do not automatically remove waiting for an adaptive classical decision.

Fault-tolerant estimates add another translation. A logical operation may require many physical operations, supporting qubits, and decoding. Babbush and colleagues examined how such overheads affect possible crossovers, especially for quadratic algorithmic improvements. Their analysis is a resource comparison under specified technologies, not a theorem that a quadratic improvement can never be useful.[4]

One prediction or a reusable service?

A calculation performed once and a service used repeatedly have different economics. Suppose the classical method additionally needs 30 minutes of reusable setup, while the hybrid method needs 600. Retain our per-case times of 100 and 30.9 minutes. For a batch of K cases,

$$T_C(K) = 30 + 100\,K,$$
$$T_H(K) = 600 + 30.9\,K. \tag{23.4}$$

For one case, the totals are 130 and 630.9 minutes; the hybrid loses. For ten cases, they are 1,030 and 909; it wins. The first integer batch size favoring the hybrid is nine. All

times in Eq. (23.4) are in minutes. None of the algorithms or hardware changed. The workload changed.

Reuse must be genuine and permitted on both sides. A classical solver may reuse an index, an orbital transformation, or a fitted approximation. A quantum device may reuse compilation and calibration only while they remain adequate. Setup that must be repeated for each new molecule or after drift is not a once-only investment. Report both the cold-start cost and the cost of an additional case when the distinction matters.[5]

Failures also belong inside the clock. Suppose an independently repeated attempt returns a verifiably acceptable answer with probability p. The number of attempts sufficient for at least one success with probability $1-\delta$ is

$$r = \mathrm{ceil}\left(\frac{\ln \delta}{\ln(1-p)}\right), 0 < p < 1. \tag{23.5}$$

The function ceil means round upward to a whole number. A one-second attempt with $p = 0.02$ needs 228 attempts for 99-percent success. A five-second attempt with $p = 0.5$ needs seven, or 35 seconds. The faster individual attempt is slower under this reliability requirement. Every attempt includes preparation and verification; p is stipulated exactly here, not estimated from a favorable handful of trials.[6]

This formula assumes we can recognize success. A noisy estimate with an unknown bias is not a flagged failed attempt. Repeating it may produce many equally misleading numbers. Recognizing an error, bounding it, and merely hoping it was absent are different procedures.

The price of recovering a small signal

Consider a simple version of error mitigation. The ideal quantity is a mean μ of a measurement with outcomes $+1$ and -1. Suppose a known readout error attenuates it: the actual outcome Y has mean $\lambda\mu$, with $0 < \lambda \le 1$. Independent symmetric sign errors of probability $(1-\lambda)/2$ produce exactly this model.

Dividing the observed average by λ recovers an unbiased estimate. It also enlarges the fluctuations. For m retained trials,

$$\hat{\mu} = \frac{\overline{Y}}{\lambda}, \qquad \mathrm{Var}(\hat{\mu}) = \frac{1-\lambda^2\mu^2}{m\lambda^2}. \tag{23.6}$$

At $\mu = 0$, a standard uncertainty of 0.02 requires 2,500 ideal trials. With $\lambda = 0.2$, it requires 62,500 retained trials. If only a state-independent quarter of attempts supplies a usable record, collecting that many records takes 250,000 attempts on average. Figure 23.3 shows how correcting the mean and retaining enough data are different achievements.[7]

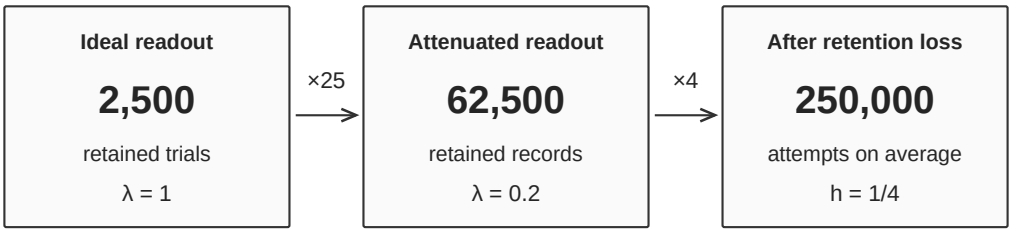


Figure 23.3. An unbiased mean can require many more attempts. At $\mu = 0$, Eq. (23.6) gives 2,500 trials for standard uncertainty 0.02 with ideal readout, or 62,500 retained records with known attenuation $\lambda = 0.2$. Independent retention probability $h = 1/4$ makes the latter require 250,000 attempts on average. The comparison is to the ideal readout of this same statistical task, not to the best classical algorithm. All numbers are stipulated; calibration time and processing add further costs. The final count is an expectation, not a high-confidence deadline.

We have assumed that retention does not favor one sign. Outcome-dependent rejection can alter the mean itself and requires another model. We have also treated λ as known. If its true value is 0.18 but the analysis divides by 0.20, the estimated mean approaches 0.9μ, however many trials we collect. Calibration uncertainty must travel through the correction too.

More general mitigation methods combine many noisy experiments, sometimes with signed weights, to estimate an ideal observable. Temme, Bravyi, and Gambetta developed foundational constructions of this kind. They do not leave behind an error-corrected quantum state available for arbitrary later operations. They repair an inference, under a noise account.[8]

That can be valuable. But the sampling and calibration costs must be included before a circuit becomes a performance claim. Takagi and colleagues established lower bounds on the largest possible spread of mitigation estimates, including exponential growth with depth under specified local depolarizing noise. A conservative sampling budget based on that range can be large, while low variance may permit fewer samples. Those theorems have defined noise and protocol premises; they are not a declaration that every useful observable must become inaccessible.[9]

The other team need not imitate the circuit

A classical competitor should solve the scientific task, not necessarily reproduce every internal amplitude of the proposed quantum calculation. If a few observables are enough,

an algorithm tailored to those observables is a legitimate opponent. If a useful approximation meets the requested tolerance, requiring the classical side to compute an exact answer changes the competition.

A revealing example followed Kim and colleagues' 2023 experiment with a 127-qubit processor. The experiment estimated observables of a kicked Ising model using error mitigation and compared them with available classical calculations. Tindall and colleagues subsequently developed a tensor-network simulation tailored to that model's geometry. It gave more accurate values in exactly checkable comparisons and supported revised estimates at greater depth, where some results relied on convergence and extrapolation. A cumbersome full-state calculation was not the only classical route.[10]

The later calculation did not make the earlier control of 127 qubits disappear. It changed the evidence for the particular computational comparison. Nor did it prove that every larger circuit is easy. The meaningful object was a specified experiment, its requested outputs, and the methods available to reproduce them.

Chemistry presents the same issue. A full configuration-interaction matrix may be impossibly large while a selected set of configurations, a tensor-network state, or another approximation gives the requested energy difference adequately. Lee and colleagues examined the evidence for exponential quantum advantage in ground-state chemistry with such competing considerations in view. Difficulty of one expanded representation is not an unconditional lower bound on every classical method.[11]

Hybrid methods need one more comparison. Suppose quantum samples select configurations that a classical solver then uses to estimate an energy. Replace the sampler by a serious classical alternative, retune fairly, and compare the completed outputs at matched total budgets. This component-replacement test, often called an ablation, asks whether the quantum contribution improves that workflow. Also compare an independent classical solver that need not use the same workflow at all. Figure 23.4 separates the two questions.

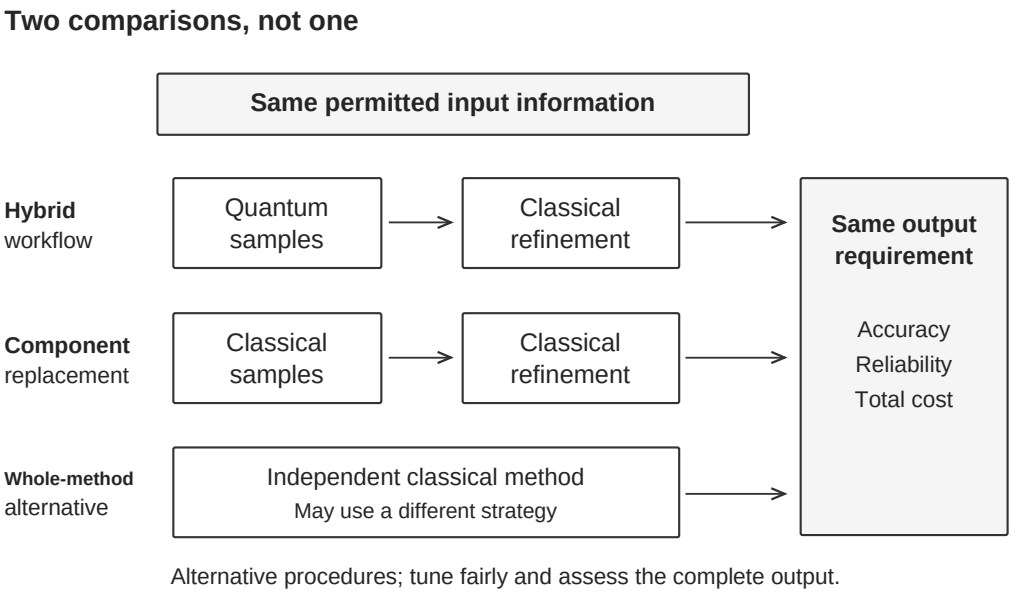


Figure 23.4. Test the quantum contribution and the complete alternative. The top two routes compare quantum and classical candidate generation followed by the same kind of classical refinement, with appropriate retuning and total-cost accounting. The bottom route lets a classical method solve the task by another strategy. All routes receive the same permitted input information and face the same final accuracy and reliability test. This is a design for a comparison, not a claim that the boxes have equal cost or that any pictured route wins.

What establishes that an answer is good enough?

Factoring offers a convenient check: multiply the proposed factors. A low molecular energy is harder to certify. The variational principle supplies an upper bound from a valid trial state, but says nothing by itself about how far below it the true ground energy lies. Agreement between two upper bounds does not close that gap.

If an independent argument also supplies a lower bound L, while the valid trial calculation gives U, then

$$L \le E_0 \le U, U - L \le \epsilon \Rightarrow 0 \le U - E_0 \le \epsilon. \tag{23.7}$$

This is a certificate at tolerance ϵ for the specified Hamiltonian. Obtaining a tight lower bound may itself be difficult. Uncertain measured expectations need valid bounds before being used as U. And even a rigorous interval for that Hamiltonian does not certify interactions omitted from it. The distinction between solving a model and predicting a molecule remains.[12]

We should not demand an equally expensive exact classical solution as the only possible evidence. That would rule out learning from a genuinely new computational capability by definition. Instead, ask which guarantee the output needs: a cheaply checked factor, a rigorous approximation bound, a statistically justified estimate, or converging independent physical evidence. These are different routes to trust, with different residual uncertainties.

Benchmarking adds safeguards against flattering ourselves. Choose the accuracy target before examining the result. Separate development instances from held-out tests. Include timeouts and failed instances. Allow comparable tuning effort, and state whether the claim concerns typical cases, difficult tails, or every member of a promised family. A benchmark chosen because one implementation already wins on it answers a narrower question.

Koch and colleagues' Quantum Optimization Benchmarking Library, published in 2026, makes that discussion concrete through ten model-independent problem classes, public instances, and explicit reporting of solutions and resources. Its classical baselines are starting points for comparison, not a decree that no better method exists. An acceptable feasible solution and a proven optimum remain separately reportable achievements.[13]

A learning task needs the same complete comparison

The same comparison now moves from a chemical conclusion to prediction on new records. The question is still whether the complete procedure reaches the required output quality, with all preparation, computation, measurement, and testing costs included. A different application does not relax that accounting.

Suppose a proposal promises to classify new records more accurately by encoding their numerical features into qubits. A high training score cannot establish that promise: the task concerns records not used to fit the model. Separate training data, validation data used for choosing settings, and a final held-out test before comparing procedures. Any normalization or feature selection learned from data belongs inside that separation. Give the competing methods the same permitted inputs and comparable tuning budgets.

A small construction makes the quantum component visible. Let a record contain one angle x, measured in radians. This is a deliberately simple classical input, not an unknown supplied quantum state. A Y rotation set by that angle prepares

$$|\psi(x)\rangle = \cos(x/2)|0\rangle + \sin(x/2)|1\rangle. \tag{23.8}$$

To compare records x and z, prepare with x, apply the inverse preparation for z, and count returns to zero. The return probability supplies a similarity called a kernel:

$$K(x,z) = \left|\langle \psi(z)|\psi(x)\rangle\right|^2 = \cos^2\left[(x-z)/2\right]. \tag{23.9}$$

Kernel methods use such similarities to fit a prediction rule to labelled examples. The quantum device can estimate kernel entries while a classical procedure performs the fitting. This division of work follows the feature-space approach developed by Schuld and Killoran; the one-qubit map here is chosen for an exact comparison, not as a claim of difficult classical computation.[14]

Indeed, expand the cosine using the identities from Chapter 3:

$$K(x,z)=\frac{1+\cos x\cos z+\sin x\sin z}{2}. \tag{23.10}$$

The right side is the ordinary dot product of the three-component vectors $(1,\cos x,\sin x)/\sqrt{2}$ and $(1,\cos z,\sin z)/\sqrt{2}$. A classical program given the same angles can calculate the identical kernel directly to the requested numerical tolerance. No amplitudes need to be recovered from an unknown state. Thus, for any fixed kernel-based fitting procedure, this replacement preserves its ideal input matrix and its prediction rule. For example, let training select coefficients a_i and a bias b in the score $f(x)=\sum_i a_i K(x_i,x)+b$, with positive score selecting one class. Equation (23.10) reduces the fitted score to $A+B\cos x+C\sin x$, where the three coefficients can be accumulated once from the training result. Future ideal predictions then require only that short expression. This is a complete classical replacement for this model, not an attempt to simulate every internal step of its circuit.

Now count one proposed implementation. With 100 training records and 1,000 test records, constructing every distinct off-diagonal training similarity and every test-to-training similarity requires 4,950 + 100,000 = 104,950 kernel estimates. The ideal diagonal entries are already one. At 10,000 independent shots per estimated entry, the count is 1,049,500,000 circuit executions, before tuning, calibration, fitting, or data movement. This is the cost of that specified entry-by-entry procedure, not a lower bound on every quantum learning method. A different batching or inference procedure needs its own count.[15]

Ten thousand ideal Bernoulli shots give standard error at most 0.005 for one entry; that is not a simultaneous guarantee for the entire matrix or a guarantee of classification accuracy. For fixed fitted coefficients, entry errors bounded by ϵ give score error at most $\epsilon\sum_i |a_i|$. Points whose ideal score has magnitude greater than that error bound keep their label. Training on noisy entries can also change the coefficients, so a full assessment must include that additional effect.

The final accuracy target needs equal care. Suppose it is at least 90 percent correct on a specified population, and the frozen classifier is tested once on 1,000 independent records from that population. A constructed result of 905 correct answers has point accuracy 90.5 percent, but does not meet that target under a one-sided 95-percent Hoeffding lower-bound rule. The lower bound is only about 86.63 percent. A constructed result of 940 correct answers gives about 90.13 percent and does meet that particular rule. These are arithmetic examples, not observed classifier results. They certify neither a distribution shift nor superiority to a competitor; a comparison of two models requires an appropriate paired analysis.[15]

The useful comparison therefore keeps predictive performance and complete cost together. Test the exact classical replacement as well as independently designed classical models. When evaluating an ablation, retune fairly rather than deliberately weakening the replacement. Bowles, Ahmed, and Schuld's 2024 simulation study illustrates how model rankings and the apparent contribution of entanglement depend on those choices in the small classification tasks they examined.[16]

Classical input files are not the only learning setting. In a photonic experiment published in 2025, Liu and colleagues used entangled probes and retained reference modes to learn a multimode displacement process. The resource comparison concerned the number of uses of that process against specified entanglement-free strategies, not the running time of a classifier given an ordinary table of features. The earlier experiment–memory distinction now becomes a constructive opportunity: preserving a quantum relationship until a joint measurement can make a different learning task more economical.[17]

Our little classifier and that experiment therefore answer different questions. One permits an explicit classical replacement; the other uses access to a quantum process and an entangled reference. The lesson is not a verdict for or against everything called quantum machine learning. It is how to locate the proposed improvement: in the available data, the representation, the learning rule, the measurement, or the cost of the complete prediction.

From a forecast to an observed advantage

The most useful forecasts expose enough of the full procedure to reveal what could prevent success. In a 2026 preprint, Eklund and colleagues analyze a quantum route through chemical dynamics: preparing a specified initial electronic-and-nuclear state, evolving a nonrelativistic Coulomb model, and extracting reaction outcomes. Logical resource estimates include stages that an evolution-only count leaves outside the calculation. They remain estimates, with state-description, grid, truncation, and algorithmic assumptions—not measured running times of such a machine.[18]

A forecast can guide an experiment before it wins a race. If preparation dominates, improve preparation. If decoding or measurement dominates, shortening a simulated evolution may contribute little. If the physical model cannot distinguish the chemical alternatives, no hardware speed will repair that omission. Identifying the limiting stage is an actionable result.

An observed advantage can then be stated without pretending to know every future algorithm: on a declared workload, with the required output quality, a specified quantum-assisted procedure used less of a named resource than the credible tested classical alternatives. Publish the task, code, settings, unsuccessful runs, and resource account so that the comparison can be challenged. New classical methods may move the boundary; better quantum methods may move it again.

This empirical statement is different from a theorem bounding an entire class of algorithms. Both matter. The theorem can identify structure worth exploiting; the benchmark can establish whether exploiting it pays on the machines and instances actually considered. Neither needs to borrow the other's certainty.

The answer that survives the comparison

Return to the two teams. We would now ask for more than their fastest time or their lowest energy. What did they receive? What had to be returned? Did the uncertainty resolve the chemical question? Which setup could be reused? How many attempts failed? Could another method reach the same conclusion without following their circuit?

When those questions are answered, a quantum contribution need not perform the entire calculation to be decisive. A better correlated state can enable a classical inference that was previously inadequate. A faster subroutine can shorten a complete workflow. A new measurement can make an otherwise inaccessible physical comparison possible. The value lies in the completed task, not in excluding classical work from it.

The detector with which this book began did not announce an advantage. It revealed a rule about amplitudes. We have learned how that rule can support programs, how their implementations can fail, and how their results can earn a scientific interpretation. One question remains broader than any benchmark: what has this information-centered account taught us about the physical world itself? The final chapter returns to that question without mistaking a successful technology for an answer to every question about reality.

Notes and sources

1. Troels F. Rønnow, Zhihui Wang, Joshua Job, Sergio Boixo, Sergei V. Isakov, David Wecker, John M. Martinis, Daniel A. Lidar, and Matthias Troyer, “Defining and detecting quantum speedup,” Science 345, 420–424 (2014), doi:10.1126/science.1252319. The full author manuscript, arXiv:1401.2910v1, submitted 13 January 2014, distinguishes scopes of comparison and performance measures. Table 23.1 is an original classification of evidence, not a reproduction of that paper’s taxonomy or an empirical claim about a current machine. Full author paper. Publication.

2. Equations (23.1)–(23.2) and Figure 23.1 are original continuations of Chapter 22’s stipulated two-route model. They use 298.15 K, equal transmission factors, and $d = \Delta G_{\mathrm{P}}^{\ddagger} - \Delta G_{Q}^{\ddagger}$. Exact SI k_B and N_A, with 4,184 J per thermochemical kilocalorie, give $RT \ln 3 \approx 0.650911$ kcal/mol. The three intervals are [0.58,1.02], [0.616,0.984], and [0.69,0.91]. Numerical and model terms are assumed valid deterministic bounds; only the sampling term has statistical coverage. A hundredfold increase in independent samples reduces a regular fixed-coverage, large-sample half-width tenfold at unchanged variance. It is not a guarantee for arbitrary adaptive intervals. For propagation from energies to reaction conclusions, see Jonny Proppe and Markus Reiher, “Mechanism Deduction from Noisy Chemical Reaction Networks,” Journal of Chemical Theory and Computation 15, 357–370 (2019), doi:10.1021/acs.jctc.8b00310. Full reaction-uncertainty analysis. SI constants. The reaction-network author manuscript is arXiv:1803.09346v3, revised 19 December 2018. The exact thermochemical-calorie conversion is specified in NIST’s Guide to the SI, Appendix B.8 and footnote 9. Exact calorie conversion.

3. Equation (23.3) and Figure 23.2 are direct cost arithmetic for a sequential workflow. $T_H = T_C\left[(1-f) + f/s + a\right]$, where $a \geq 0$ counts added replacement-specific work. The examples use $T_C = 100$ minutes, $f = 0.9$, $s = 100$, and $a = 0$ or 0.2. Whole-workflow speedups are approximately 9.174 and 3.236. Every per-case time is assumed to include the work needed for the same requested quality. They are neither measured data nor forecasts. Pipeline overlap, competing resource constraints, or a replacement that changes several stages requires a different schedule model.

4. Ryan Babbush, Jarrod R. McClean, Michael Newman, Craig Gidney, Sergio Boixo, and Hartmut Neven, “Focus beyond Quadratic Speedups for Error-Corrected Quantum Advantage,” PRX Quantum 2, 010103 (2021), doi:10.1103/PRXQuantum.2.010103. The resource analysis considers fault-tolerant overhead, operation times, and classical parallelism under stated assumptions. Only that implementation-versus-scaling distinction is used here, not a universal impossibility claim or a contemporary numerical crossover obtained from its 2021 assumptions. Author paper. Published article.

5. Equation (23.4) adds separately stipulated reusable setup to the per-case example. The crossover is $K > 570/69.1 \approx 8.249$, hence nine cases. At $K = 8$ the totals are 830 and 847.2 minutes; at $K = 9$ they are 930 and 878.1. Setup is genuinely reused across these identical-cost cases, and calibration renewal is excluded. This construction is not a measured batch benchmark or permission to omit classical preprocessing while charging quantum preprocessing.

6. Equation (23.5) follows from independent failure probability $(1-p)^r$. See Rønnow and colleagues, arXiv:1401.2910v1, §III, Eq. (3), for time-to-solution accounting with repeated attempts. Here $\delta = 0.01$; $p = 0.02$ gives 228 attempts and $p = 0.5$ gives seven. Times include each attempt’s verification. The requirement is a fixed success probability by the deadline, not the mean geometric waiting time $1/p$. An empirical unknown p needs its own uncertainty treatment; state-dependent errors, correlated restarts, adaptive stopping, or unrecognizable incorrect outputs invalidate this simple substitution. Primary time-to-solution account.

7. Equation (23.6) and Figure 23.3 are exact original calculations for independent outcomes $Y = -1$ or $Y = +1$ with $\mathrm{E}[Y] = \lambda\mu$. Since $Y^2 = 1$, its variance is $1 - \lambda^2\mu^2$. The mean is divided by a known nonzero attenuation. At $\mu = 0$, target standard uncertainty 0.02 gives $m = 2500/\lambda^2$. Independent state-

and outcome-independent retention $h = 1 / 4$ gives mean attempts m / h. No coverage probability or universal lower bound is inferred from that standard deviation. Biased selection or uncertain attenuation requires additional inference. The finite-sample distribution follows from a binomial count of positive outcomes, with success probability equal to one-half of one plus the attenuated mean.

8. Kristan Temme, Sergey Bravyi, and Jay M. Gambetta, "Error Mitigation for Short-Depth Quantum Circuits," Physical Review Letters 119, 180509 (2017), doi:10.1103/PhysRevLett.119.180509. The methods estimate expectation values through multiple noisy executions and classical processing, including noise extrapolation and probabilistic cancellation. Our attenuation example is a separate elementary estimator, not a full implementation of either method. Mitigating a final estimate does not itself provide an error-corrected intermediate quantum state. Calibration and sampling are resources rather than presumed free side information. Full author paper. Publication.

9. Ryuji Takagi, Suguru Endo, Shintaro Minagawa, and Mile Gu, "Fundamental limits of quantum error mitigation," npj Quantum Information 8, 114 (2022), doi:10.1038/s41534-022-00618-z. The author version is arXiv:2109.04457v5, revised 22 September 2022. Theorem 3, Eq. (9), bounds the maximum estimator spread under specified local depolarizing noise and protocol assumptions. The range-based Hoeffding budget in Corollary 2 is sufficient; a lower bound on range alone is not a necessary lower bound on actual sample count. The discussion explicitly allows fewer samples when variance is small. Figure 23.3 does not reproduce the paper's bounds or numerical data. Full author analysis. Published article.

10. Youngseok Kim and colleagues, "Evidence for the utility of quantum computing before fault tolerance," Nature 618, 500–505 (2023), doi:10.1038/s41586-023-06096-3; Joseph Tindall, Matthew Fishman, E. Miles Stoudenmire, and Dries Sels, "Efficient Tensor Network Simulation of IBM's Eagle Kicked Ising Experiment," PRX Quantum 5, 010308 (2024), doi:10.1103/PRXQuantum.5.010308. The original experiment's targeted-observable evidence and the later geometry-adapted tensor-network comparisons are different achievements. The exact comparisons are in §I.C, Fig. 3, and Fig. 4(a); Fig. 4(b–f) uses extrapolation and further cross-checks, with residual belief-propagation error. This does not establish efficient classical simulation of arbitrary quantum computation. Neither study is represented by the chapter's hypothetical timing numbers. Published experiment. Full classical analysis. Classical publication.

11. Seunghoon Lee and colleagues, "Evaluating the evidence for exponential quantum advantage in ground-state quantum chemistry," Nature Communications 14, 1952 (2023), doi:10.1038/s41467-023-37587-6. The author manuscript, arXiv:2208.02199v3, revised 14 November 2022, is titled "Is there evidence for exponential quantum advantage in quantum chemistry?" It analyzes ground-state energy estimation and the evidence for classical and quantum costs. Its conclusions do not assert that no useful quantum chemistry advantage exists. Figure 23.4 is an original comparison design: a quantum component should be tested against credible replacements within the hybrid workflow and the whole workflow against independently designed solvers. Benchmark quality, tuning, input access, and complete cost must remain comparable. Full author analysis. Publication.

12. Equation (23.7) is elementary interval reasoning for the ground energy of one fixed Hermitian Hamiltonian. A normalized physical trial state supplies a true expectation upper bound, not automatically a valid upper bound from a fluctuating point estimate. A separately justified lower bound and all relevant numerical uncertainties must enter the interval. For a small Hermitian matrix, one possible lower bound is obtained by subtracting the sum of absolute off-diagonal entries in each row from its diagonal entry and taking the smallest result. This Gershgorin bound may be loose. It does not provide a tight general molecular certificate or establish the accuracy of interactions omitted from the Hamiltonian. This continues Chapter 22's variational argument.

13. Thorsten Koch and colleagues, "The Quantum Optimization Benchmarking Library," Nature Computational Science 6, 653–671 (2026), doi:10.1038/s43588-026-00991-1, published 23 June 2026. The published full text supplies ten model-independent classes, instances, and reporting examples. Its classical baselines are not asserted to be the best attainable results for every instance. Only the comparison-design and reporting contribution is used here; no blanket complexity lower bound is inferred from the fact that a problem is NP-hard. Finding feasibility, improving an objective, and certifying optimality are distinct output specifications. Published full resource.

14. Maria Schuld and Nathan Killoran, “Quantum machine learning in feature Hilbert spaces,” Physical Review Letters 122, 040504 (2019), doi:10.1103/PhysRevLett.122.040504. For feature maps and hybrid kernel methods, see also author version arXiv:1803.07128v1, submitted 19 March 2018. Equations (23.8)–(23.10) are the book’s elementary one-qubit specialization. The classical three-feature identity is derived directly; the source is not cited as an experimental advantage for this toy map. Author paper Publication.

15. The kernel-execution count, score-error bound, and held-out-test examples are original calculations. The matrix-count example fixes 100 training and 1,000 test records and a single entry-by-entry kernel evaluation plan; diagonal entries use the known ideal value. The score bound applies to fixed coefficients, not to retraining on a perturbed matrix. For an independent fixed-size test, Hoeffding’s one-sided inequality gives a lower bound $\hat{p}-\sqrt{\frac{\ln\left(\frac{1}{\delta}\right)}{2m}}$ where $\hat{p}$ is the observed correct fraction, m is the number of independent test records, and δ is the one-sided failure probability. The coverage is at least $1-\delta$; use $m=1000$ and $\delta=0.05$. The resulting lower bounds are 0.8662977 and 0.9012977. The classifier and all tuning must be fixed independently of the test. No real dataset, trained device, hardware timing, or model-comparison significance result is claimed. See also Chapter 16, notes 10–11, for the sampling assumptions. The primary result is Wassily Hoeffding, “Probability Inequalities for Sums of Bounded Random Variables,” Journal of the American Statistical Association 58, 13–30 (1963), doi:10.1080/01621459.1963.10500830, Theorem 1, Eq. (2.3). Published article copy.

16. Joseph Bowles, Shahnawaz Ahmed, and Maria Schuld, “Better than classical? The subtle art of benchmarking quantum machine learning models,” arXiv:2403.07059v2, submitted 14 March 2024. See especially §§I and IV–VII of the author text. The study uses classical simulations of selected quantum models on specified small classification tasks, with classical baselines and model ablations. Its observations are not a theorem that all quantum learning fails, or an end-to-end hardware timing benchmark. This cited version remains identified as a preprint. Full author text.

17. Zheng-Hao Liu and colleagues, “Quantum learning advantage on a scalable photonic platform,” Science 389, 1332–1335 (2025), doi:10.1126/science.adv2560. The account here follows author manuscript arXiv:2502.07770v2, submitted 16 February 2025. The experiment concerns learning a multimode bosonic displacement process with entangled signal/reference resources; its sample-complexity comparison is against the stated entanglement-free access class, which excludes a retained reference entangled with the probe while permitting the specified adaptive strategies and entanglement within a multimode probe. This comparison establishes neither a wall-clock advantage for classification from classical inputs nor a universal learning advantage. Full author text Journal-linked record.

18. Elliot C. Eklund, Arkin Tikku, Patrick Sinnott, William J. Huggins, Guang Hao Low, Dominic W. Berry, and Ivan Kassal, “End-to-End Simulation of Chemical Dynamics on a Quantum Computer,” arXiv:2603.19007v1, submitted 19 March 2026. It analyzes a specified initial wavefunction, electronic and nuclear dynamics under a nonrelativistic Coulomb model, and extraction of chemical observables. Logical estimates and error bounds retain spatial-grid, truncation, access, and regularity assumptions; avoiding a Born–Oppenheimer separation does not mean omitting every other approximation. No experiment, final journal status, or unrestricted state-preparation guarantee is asserted. Versioned record. Full preprint. Section IV.A assumes that the momentum cutoff remains adequate during evolution and that periodic-image interactions are negligible; the preprint does not bound these two controllable errors.

CHAPTER 24

What Does Information Tell Us About Reality?

A detector leaves a mark in a notebook. A formula assigns that mark a probability. A picture of the world explains what, if anything, was definite before the mark appeared. These are three different achievements. A successful formula connects them, but does not make them identical.

We have followed that connection from a dark output of an interferometer to molecular predictions and tests of computational advantage. The equipment changed; the discipline did not. Specify the preparation, the permitted operations, and the recorded outcome. Then ask which conclusions the evidence supports.

The final question reaches further. Does a quantum state describe something physically present, or what someone can expect to learn? Can properties exist before we measure them? Does treating an observer as another physical system change the meaning of a fact?

These questions are not answered by saying that quantum mechanics is strange. Nor are they answered by the fact that a quantum computer works. We can do better: identify pictures that the mathematics rules out, show where an experiment could distinguish competing accounts, and locate the questions that remain after the calculation is complete.

A state is not a complete preparation history

Imagine two sources. The first tosses a fair coin and prepares either zero or one. The second tosses a fair coin and prepares either plus or minus. Each sends a single qubit while keeping its coin record. Can a receiver identify which source was used?

Not by measuring the delivered qubit alone. The two sources give the same density matrix:

$$\rho = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|$$
$$\rho = \frac{1}{2}|+\rangle\langle +| + \frac{1}{2}|-\rangle\langle -| = I_2/2. \tag{24.1}$$

This is more than agreement in the Z setting. Every possible measurement confined to that qubit has the same outcome probabilities for the two preparations. The equality says

what the receiver can predict with the stated access; it does not say that the two sources performed the same physical procedure.[1]

The retained record changes what can be learned. Given the first source's coin result, the receiver can predict Z with certainty. Given the second source's result, the corresponding certain prediction is in X. The local state alone does not reconstruct these different relationships. Figure 24.1 keeps the histories and the delivered system separate.

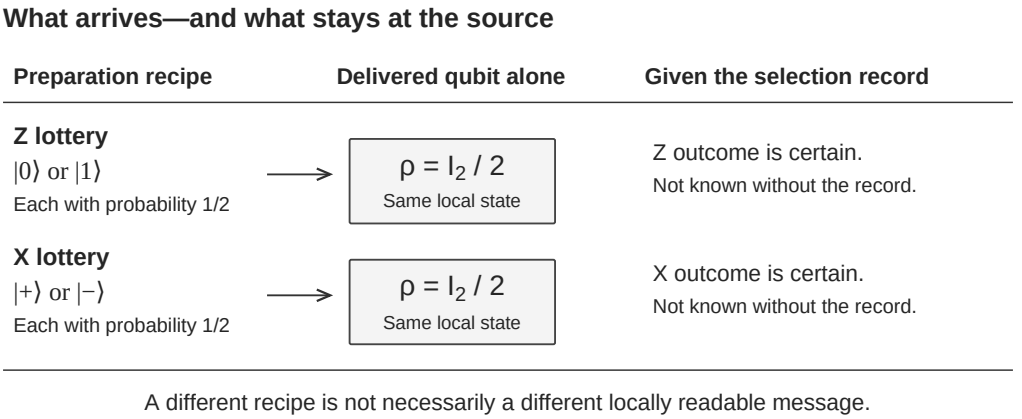


Figure 24.1. Different histories can give the same local state. Each source independently repeats its fair choice and retains a classical selection record. Both delivered qubit ensembles have density matrix $I_2/2$, so no measurement on a delivered qubit alone identifies the source. The last column conditions on the source and its actual selection record; it does not describe extra information available without that record. No claim that the complete source–record states are identical is made.

Now replace either lottery by a source that always prepares plus. Its Z results are still fair, but its X result is certain. A pure superposition is therefore not interchangeable with a lottery merely because one measurement gives the same frequencies. An information-based interpretation must explain that distinction too.

Even "the state is information" can mean several things. It might mean a predictive summary, an observer's probability assignment, or an ordinary probability distribution over hidden physical conditions. These are different proposals. In particular, the last proposal can be constrained mathematically.

Pusey, Barrett, and Rudolph considered models with underlying physical states that fix outcome probabilities for each specified measurement. Suppose different pure quantum states can correspond to overlapping distributions over those underlying states. Add preparation independence: independently prepared systems have independent underlying states. Their theorem shows that this combination cannot reproduce all quantum predictions.[2]

Overlap has a concrete meaning in this proposal: could exactly the same underlying physical condition occur when either of two different pure states is prepared? If so, that condition would have to supply the same measurement-response rule, whichever

preparation produced it. Preparation independence then constrains how such shared possibilities combine for separately prepared systems. It is this conjunction of premises that the theorem tests, not every use of probability in describing a quantum state.

The result concerns distinct pure states, not the equivalent mixtures in Eq. (24.1). Within its framework, the underlying condition must distinguish the pure-state labels. It does not prove that the wavefunction is a material wave in ordinary three-dimensional space, that it is a complete description, or that every interpretation using probabilities accepts the theorem's premises.

Nine questions that will not fit on one answer sheet

Bell's argument already excluded a broad shared-instruction explanation under its locality and setting-independence conditions. A different argument asks whether every possible observable could carry a pre-existing answer independent of which compatible questions accompany it. We can test that picture with two qubits, without sending them to distant laboratories.

Use the X, Y, and Z operators already familiar from gates and error correction. Each observable has outcomes $+1$ and -1. Write XI for X on the first qubit and identity on the second; XY means the product of X on the first and Y on the second. Two adjacent letters here describe a joint observable, not successive operations on one qubit.

Arrange nine observables as in Figure 24.2. The three in any one row commute, as do the three in any one column. Each such triple is a context: a set of questions that can be measured compatibly. The entire square is not one jointly compatible set. This is a form of the Peres–Mermin construction.[3]

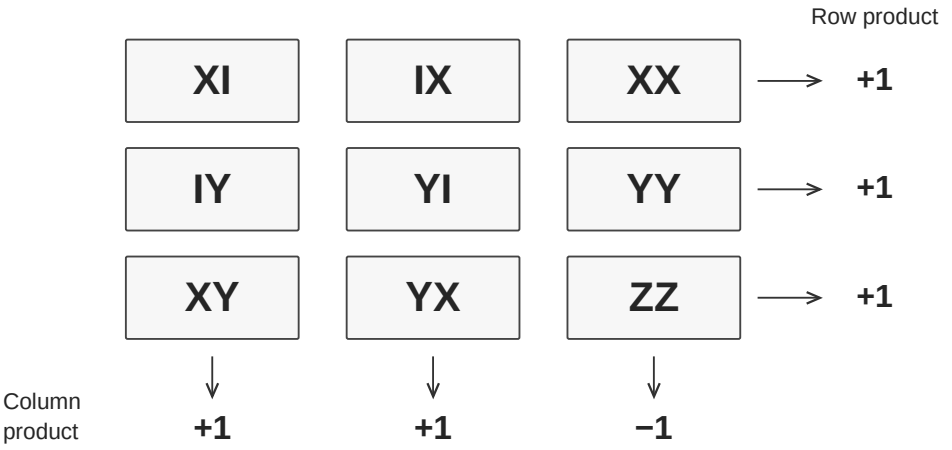


Figure 24.2. The same nine entries cannot carry context-independent answers. Each row or column is one compatible measurement context on two qubits. The operator products are $+I_4$ for every row and the first two columns, and $-I_4$ for the last column. Thus the corresponding products of ideal recorded signs are fixed even though individual signs may fluctuate. Multiplying all six requirements would make each assigned cell value occur twice. A single context-independent assignment would give $+1$, while the required signs multiply to -1. Separate fresh trials test different contexts.

The first row is straightforward: doing $X\,I$ and $I\,X$ has the combined action $X\,X$, and squaring $X\,X$ gives identity. The product of its three recorded signs must therefore be positive. The same is true for the second row and the first two columns.

A compatible joint measurement must preserve the distinctions that the other questions need. For example, measuring XX as a parity is not generally the same physical operation as separately reading both X values and forgetting them. The more detailed readout can disturb YY. As in error correction, the apparatus must genuinely ask the less revealing question.

The two remaining products reveal the distinction. On one qubit, $X\,Y = i\,Z$, whereas $Y\,X = -i\,Z$. Apply these rules separately to the two positions:

$$\begin{aligned}(X\,Y)(Y\,X) &= (i\,Z)\otimes(-i\,Z) = Z\,Z,\\ (X\,X)(Y\,Y) &= (i\,Z)\otimes(i\,Z) = -Z\,Z.\end{aligned} \tag{24.2}$$

Multiplying each result by the final $Z\,Z$ gives identity in the bottom row and minus identity in the last column. Opposite imaginary factors cancel in the first calculation; matching ones give a minus sign in the second. The products are fixed for every input state of the pair, not only an entangled input.

Try assigning one fixed sign to each cell. Require that a cell retain that sign whether it is read in its row or its column, and that a compatible product have the product of its assigned signs. Multiply the six row-and-column requirements. Every cell appears twice,

so its sign squares to one. The left side must be positive. But five required products are positive and one is negative. The right side must be negative. No assignment works.

The contradiction also gives a statistical test. Let r_j and c_j be the products of the three outcomes in row j and column j, measured on separate trials. Define the score

$$K = \langle r_1 \rangle + \langle r_2 \rangle + \langle r_3 \rangle + \langle c_1 \rangle + \langle c_2 \rangle - \langle c_3 \rangle . \tag{24.3}$$

For a fixed noncontextual answer sheet, at least one of the six signed contributions is wrong. Their sum is at most four. Randomly choosing sheets cannot raise that bound. Ideal quantum measurements instead give six:

$$K_{\text{noncontextual}} \leq 4 , K_{\text{quantum}} = 6 . \tag{24.4}$$

Cabello developed this state-independent inequality as an experimentally testable counterpart of the contradiction. Its force is specific: the outcomes cannot all be represented by the same predetermined, context-independent values with the stated compatibility relations. It is not a disproof of every possible account of an observer-independent world.[4]

A classical program told the entire row or column can simply choose a permitted triple. Such a program depends on context and lies outside the excluded class. Likewise, efficiently simulating these quantum measurements on a classical computer does not create the forbidden answer sheet. The simulator is given the experiment being performed.

The square has also been tested in the laboratory. In 2021, Qu and colleagues encoded two qubits in one photon's path and polarization and tested 26 input states. Their reported scores ranged from 5.784(5) to 5.877(3), above the noncontextual bound of four and below the ideal value of six. The parentheses give statistical uncertainties based on a Poisson counting model. The experiment also investigated measurement disturbance, while retaining a fair-sampling assumption for detected photons.[4]

An actual experiment must establish more than a large score. Measurements intended to represent the same observable in different contexts must genuinely be comparable. Earlier measurements must not introduce an unaccounted disturbance, and losses and finite statistics must be treated. A 2026 optical preprint by Liu and colleagues addresses this issue with sequential measurements designed to retain the same physical implementation across contexts. It tests a different contextuality inequality from our square; the shared issue is whether the apparatus really asks the same question.[5]

When the recorder joins the quantum system

Contextuality constrains an answer sheet. Measurement raises another question: what turns a physical interaction into one recorded answer?

Replace a complicated laboratory by two controlled qubits. S is the system and M a memory initially in zero. The memory records S's Z label by controlled NOT. For an input plus state, linearity gives

$$|+\rangle_S|0\rangle_M \to |\Phi^+\rangle_{SM} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}. \tag{24.5}$$

The two possible memory labels agree perfectly with the corresponding system labels. Yet the equation still contains both terms. Compare it with a source that genuinely prepares 00 or 11 by an ordinary fair lottery. Every measurement on M alone agrees between the two descriptions. That agreement does not make the complete preparations identical.

We can distinguish them with a joint test. Apply the same controlled NOT again, then Hadamard on S, and measure S in Z. The coherent preparation returns S to zero with certainty. For the lottery, 00 becomes 00 and 11 becomes 10 before the Hadamard. Either branch then gives equal Z probabilities. The zero probability is only one-half.

This experiment reverses a controlled correlation; it does not reverse a human experience. It is deliberately small enough to identify every operation. A memory register capable of participating coherently in a circuit has not thereby been shown to possess awareness.

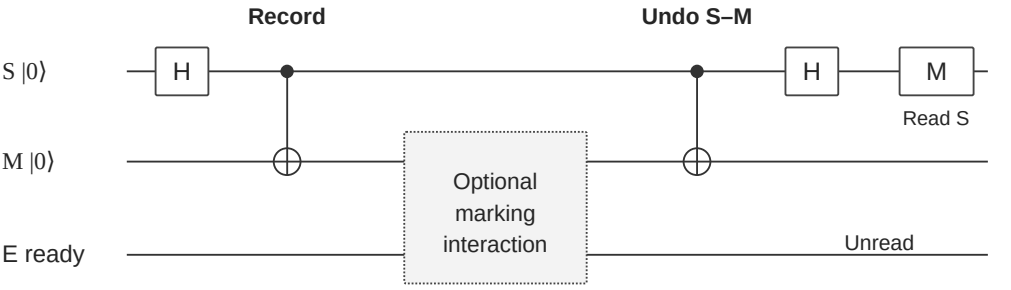
Now allow an extra system E to acquire a record of M before the reversal. The two alternatives leave E in normalized states $|e_0\rangle$ and $|e_1\rangle$. Call their overlap $\gamma = \langle e_1|e_0\rangle$. Undo only the S–M interaction, not the escaped record. The final probability is

$$\mathrm{P}(0\,\text{after reversal}) = \frac{1 + \mathrm{Re}\,\gamma}{2}. \tag{24.6}$$

Identical records give one; orthogonal records give one-half. The system and memory may be perfectly controlled while the unaddressed surroundings retain the distinction that prevents their interference. Figure 24.3 shows where this extra record enters.[6]

Reverse the interaction—but which records are included?

S: system M: controlled memory E: additional record



State before the S–M reversal	P(0)
Coherent S–M pair; no marking interaction	1
Coherent S–M–E state; orthogonal records in E	1/2
Ordinary 00/11 lottery on S–M instead	1/2

No intermediate result is selected. Every final readout is counted.

Figure 24.3. A reversible recorder and a record left outside the reversal. S begins in zero and is prepared in plus. The first controlled NOT writes its label coherently into M. An optional controlled interaction then correlates M with E; it leaves M's zero/one labels unchanged. The second controlled NOT reverses only the S–M interaction. Hadamard and Z readout test the remaining phase relationship. No intermediate outcome is selected. The tabulated results are ideal probabilities; E is unread. The ordinary 00/11 lottery is a different full preparation, not what taking a partial trace physically does. M denotes measurement in the Z basis.

For the general overlap in Eq. (24.6), the zero probability can lie anywhere between zero and one. A phase difference can shift this particular readout even when the two environmental vectors represent the same physical state. Varying a compensating phase would distinguish that shift from a loss of fringe contrast.

Writing the reduced state as a mixture therefore has an exact predictive use: it gives the probabilities available without the excluded systems. But that mathematical step has not chosen one term of the larger state. Decoherence explains suppressed interference and robust records; the unitary calculation alone has not derived a unique experienced outcome. We have also used the Born rule to obtain every probability, not derived that rule from the calculation.

What changes when the recorder is called an observer?

Wigner's-friend scenarios extend this tension by placing an observer and a record inside a laboratory that an outside experimenter treats quantum mechanically. A crucial physical question is whether the outside experimenter can still control the relevant alternatives coherently. Opening the laboratory to read a durable record and coherently recombining the laboratory's alternatives are different experiments.

Bong and colleagues formulated extended scenarios in which quantum predictions conflict with a combination of assumptions about observer-independent events, locality,

and freedom of measurement choices. The result constrains that combination; it does not identify, without further argument, which particular premise fails. Our two-qubit reversal is not itself a test of their inequality.[7]

Zeng, Labib, and Russo reported quantum-processor implementations of related tests in 2025, using increasingly elaborate quantum registers in the role assigned to the "friend." They explicitly identified experimental loopholes. Such work extends the controlled record-making that can be tested. It does not establish that the registers have subjective experiences or that two people can read the same intact notebook and disagree about its marks.[8]

The difficult question is the consistency of an account across different possible operations and different placements of the system boundary. A declaration that everything is subjective does not answer it. Neither does an instruction to call a sufficiently large register classical without saying which predictions that changes.

Different answers to the remaining question

There are several serious ways to complete the story. They should be compared at the place where they differ, rather than treated as rival explanations of why one Hadamard gate works. The brief map in Table 24.1 is not exhaustive; each entry represents a family with substantial internal development.

Everett's relative-state formulation retains unitary evolution for the complete isolated system, including recorders. It relates a definite record to a branch of the joint description rather than selecting a single branch by collapse. Connecting branch structure and probability weights to an observer's expectations is part of that account, not a consequence of merely drawing two terms on a page.[9]

Bohm's approach adds an actual particle configuration guided by the wavefunction. In its quantum-equilibrium formulation, the distribution of configurations reproduces the usual measurement statistics. Its guidance for entangled systems is nonlocal. Bell's theorem does not rule it out by declaring all hidden variables impossible; it lies outside the Bell-local class. It also does not attach context-independent values to every measured observable.[10]

Relational quantum mechanics, proposed by Rovelli, treats states and events as relative to interacting physical systems rather than as one absolute inventory. "Observer" here need not mean a conscious person. QBism instead centers an agent's probability assignments and experiences, with the Born rule constraining those assignments. These are distinct approaches; neither is the claim that wishing for an outcome forces a detector to produce it.[11,12]

Table 24.1**. Different proposals change different parts of the account.**

Approach	Central move	What an assessment must examine
Everettian	Retain universal unitary evolution; interpret records relative to branches.	The emergence of usable branches and the meaning of probability.
Bohmian	Add an actual configuration and its wavefunction-guided dynamics.	The guidance law, quantum equilibrium, and nonlocal measurement dynamics.
Relational	Make state and event descriptions relative to physical interactions.	How perspectives and later comparisons of records are related.
QBist	Treat quantum probabilities as an agent's constrained expectations.	The role of experience and why the Born rule is the required constraint.
Dynamical collapse	Modify evolution with a physical stochastic localization process.	The law, its parameters, and measurable departures from ordinary dynamics.

These are short descriptions of proposed accounts, not experimentally established rankings. The last row changes the dynamics and can change predictions. Different formulations of the other rows seek to recover ordinary quantum predictions in their applicable regimes. The table does not claim that every variant has the same scope or that these exhaust the alternatives.

Dynamical-collapse theories take a different step. They change the evolution law so that localization is an actual stochastic process, rather than only an update after a recorded result. Ghirardi, Rimini, and Weber supplied a concrete model of this kind. A specified modification can predict effects that ordinary isolated Schrödinger evolution does not.[13]

Calling all these proposals "interpretations" can hide that difference. Two accounts that assign the same probabilities to every experiment under discussion cannot be distinguished by repeating those experiments more accurately. A modified dynamics that assigns different probabilities can be tested. Conceptual coherence matters in both cases; empirical discrimination additionally needs a difference in predicted records.

A new cause must make a new prediction

A fading fringe alone cannot identify a new law. Consider an illustrative model in which ordinary environmental dephasing has rate Γ, and a proposed additional mechanism has rate λ. Suppose their independent effects give visibility

$$V(t) = e^{-(\Gamma+\lambda)t}, \Gamma, \lambda \geq 0. \tag{24.7}$$

One account sets the rates to 0.30 and zero per second. Another sets them to 0.10 and 0.20 per second. They give exactly the same curve at every waiting time. Collecting more points along that curve cannot decide how to split its total decay between the two causes.

Now stipulate a separately calibrated control that reduces only the environmental rate by a factor of four, leaving the proposed extra mechanism unchanged. At one second, the first account predicts visibility about 0.928; the second predicts about 0.799. Figure 24.4 makes the change in the question visible. The control, not additional decimal places in the old curve, creates the distinguishing prediction.[14]

The first curve cannot tell the two causes apart

V(t) = exp[−(Γ + λ)t] All rates below are in inverse seconds.

Rate account	Γ	λ	Visibility at 1 s
A: environmental only	0.30	0	**0.741**
B: two contributions	0.10	0.20	**0.741**

↓

Calibrated change: Γ → Γ/4, with λ unchanged

A: total rate 0.075	**0.928**
B: total rate 0.225	**0.799**

Invented rate example—not a collapse-law fit or experimental data.

Figure 24.4. An intervention can distinguish causes that one decay curve cannot. The two stipulated rate accounts both give $V(1\,\mathrm{s}) = e^{-0.30} \approx 0.741$. A calibrated intervention is assumed to replace Γ by $\Gamma / 4$ and leave λ fixed. Their new predictions are $e^{-0.075} \approx 0.928$ and $e^{-0.225} \approx 0.799$. The numerical rates and the selective control are invented for this inference example. They are not a full collapse model, a fit to an experiment, or parameters from the XENONnT analysis.

The qualification is physical. If the control also changes the second mechanism, or adds an unrecognized source of dephasing, the comparison must change. In this toy model, an extra exponential by itself does not explain why individual outcomes occur. It only illustrates how a proposed cause becomes distinguishable.

Real collapse tests use a specified dynamical law to predict additional effects. A 2026 XENON Collaboration study searched the XENONnT detector's low-energy data for spontaneous radiation associated with two collapse-model families. Its analysis included cancellation between radiation contributions from opposite charges. No significant excess was found, and the study constrained parameters of the stated models with temporally uncorrelated noise. It did not exclude every possible modification of quantum mechanics.[15]

A detector built for another purpose can thus test a foundational proposal. The route to that test is an emission spectrum, a background model, and a statistical comparison—not a judgment about how appealing a story sounds. The experiment and its interpretation remain connected by a calculable prediction.

Can information supply the rules themselves?

We have mostly begun with quantum rules and asked what information processing they permit. The direction can be reversed. Specify principles about preparation,

distinguishability, composition, and transformations, and ask what mathematical theory follows.

Chiribella, D'Ariano, and Perinotti derived finite-dimensional quantum theory from five operational axioms together with a purification postulate. Purification represents each mixed state as part of a pure joint state. With the purifying system fixed, different purifications of the same state must be related by a reversible operation on that system. These are substantive constraints on possible physical processes, not consequences of the word information.[16]

One of the premises, causality, can be read operationally: probabilities for a preparation outcome cannot depend on a later choice of measurement when all later outcomes are included. Conditioning afterward on one selected result is a different probability question. This familiar example gives one axiom a concrete meaning; the reconstruction requires the full set of premises, not this condition alone.

One example uses mathematics we already know. For $0 \le q \le 1$, prepare

$$|\Psi_q\rangle = \sqrt{q}\,|00\rangle + \sqrt{1-q}\,|11\rangle,$$
$$\rho_S = q\,|0\rangle\langle 0| + (1-q)\,|1\rangle\langle 1|. \tag{24.8}$$

The whole is pure; the second line describes one part. At an interior value of q, its mixedness need not come from a classical lottery selecting a pure state for that part. A unitary applied only to the other part can change the purification without changing this local matrix. The information perspective makes the relation between these descriptions a starting principle, rather than an awkward exception.[17]

Such reconstructions reveal how the rules fit together. They do not show that matter is literally made of binary digits, or that the universe is executing software. Even after a state-space framework has been obtained, predicting a particular atom or device still requires interactions, physical constants, and a preparation. An abstract permission to perform a transformation does not manufacture the apparatus.

What we can now say about the world

A record needs a carrier. Creating it requires an interaction; preserving it depends on what its surroundings can learn. These are physical relationships that experiments can test, not evidence that the world is merely a notebook.

The limits are as informative as the capabilities. An unknown pure state is not an unlimited readable message. It cannot be copied perfectly by a universal device. Entanglement requires a joint description. The Bell pair and measurements we used cannot be explained by shared local instructions under the stated test conditions. The square in this chapter rules out another tempting inventory: one context-independent answer to every question. Each conclusion identifies the model it excludes.

None requires us to stop asking what exists. They require better questions about existence. Does a proposed property belong to one system or a relationship? Is it invariant under a change of description? Does claiming that it was present earlier impose a measurable constraint? Which part of an argument supplies the probabilities, and which part tells a story about their meaning?

We can understand the physical relations expressed by the equations without treating them as a complete picture of reality. A coherent interpretation is an intellectual achievement; a distinct prediction supplies an additional opportunity for experiment. Neither should borrow the other's evidence.

From interference to chemistry, computation, and communication, the same rules have done more than limit our knowledge of an otherwise familiar world. They constrain how physical distinctions can be created, combined, preserved, and made available at all.

What information tells us about reality is therefore not that reality is whatever we choose to say about it. Possible preparations, interactions, and measurements constrain what can be learned and done. We can prepare a question, build the means of asking it, and discover that some answers no familiar picture could accommodate are nevertheless the answers the world gives.

Notes and sources

1. John Preskill, Quantum Information, Chapter 2, “Foundations I: States and Ensembles,” §§2.3 and 2.5. Equation (24.1) and Figure 24.1 apply the density-matrix rule to two fair preparation ensembles. For any single-qubit measurement effect F, its probability is $\mathrm{Tr}(F\rho)$, so equal matrices give equal local statistics. Correlated selection records are additional physical systems, not silently supplied to the receiver. Authored lecture chapter.

2. Matthew F. Pusey, Jonathan Barrett, and Terry Rudolph, “On the reality of the quantum state,” Nature Physics 8, 475–478 (2012), doi:10.1038/nphys2309. The author version is arXiv:1111.3328v3. The conclusion concerns overlapping ontic distributions for distinct pure states under preparation independence and the stated response framework; it is not an interpretation-independent identification of a material wave. Published record. Full author paper.

3. N. David Mermin, “Simple unified form for the major no-hidden-variables theorems,” Physical Review Letters 65, 3373–3376 (1990), doi:10.1103/PhysRevLett.65.3373. For the explicit square and its operator products in the convention used here, see the primary treatments in note 4. XY in a cell means $X \otimes Y$; within Eq. (24.2), juxtaposed parentheses mean operator multiplication. Original publication record.

4. Adán Cabello, “Experimentally Testable State-Independent Quantum Contextuality,” Physical Review Letters 101, 210401 (2008), doi:10.1103/PhysRevLett.101.210401, especially Eqs. (4)–(5); Dengke Qu, Kunkun Wang, Lei Xiao, Xiang Zhan, and Peng Xue, “State-independent test of quantum contextuality with either single photons or coherent light,” npj Quantum Information 7, 154 (2021), doi:10.1038/s41534-021-00492-1, Eqs. (1)–(2). Our square uses Qu and colleagues’ Pauli convention. Figures and derivations here are newly composed. The bound assumes predetermined context-independent outcomes with the stated compatible measurement structure; it is not a bound on all contextual classical simulations. Different contexts use fresh trials and independently chosen settings. Cabello, full author paper. Qu and colleagues, published full text. Qu and colleagues’ photon experiment leaves a detection-efficiency loophole and assumes fair sampling.

5. Jiaqi Liu, Bita Olamaei, Lijian Zhang, Ali Asadian, and Saleh Rahimi-Keshari, “Linear-optical test of quantum contextuality with sequential measurements,” arXiv:2605.18112v1, submitted 18 May 2026. Its single-photon sequential test uses a KCBS-type scenario, not the two-qubit square developed here. The narrow connection is context-consistent physical measurement implementation and treatment of losses. No journal publication, all-loopholes closure, or data from that experiment is asserted for our ideal score. Version record. Full author manuscript. The main single-photon test assumes fair sampling of detected events. A separate monitored-loss study varies the input vacuum component; this does not remove every experimental loophole.

6. Equations (24.5)–(24.6) and Figure 24.3 are original finite-register calculations using the environment-record and partial-trace rules developed in Chapters 9 and 12. Register order is S,M,E. A label-preserving isometry takes $|j\rangle_M|e_{\mathrm{r}}\rangle_E$ to $|j\rangle_M|e_j\rangle_E$ for $j=0,1$. The reversal acts only on S,M. For a general input, it leaves S with upper-right density entry multiplied by $\gamma=\langle e_1|e_0\rangle$. With a plus input, final Hadamard and Z readout give Eq. (24.6). No collapse or outcome-selection law is added. See Wojciech H. Zurek, “Environment-induced superselection rules,” Physical Review D 26, 1862–1880 (1982), doi:10.1103/PhysRevD.26.1862, and Preskill, Chapter 3, on the corresponding reduced-state mechanism. Original mechanism. Quantum operations. For the explicit reduced-state calculation, see Preskill, §3.4.2, Eqs. (3.97)–(3.105). For general complex overlap, the fixed-setting zero probability spans zero to one; the magnitude of the overlap sets the phase-optimized fringe contrast.

7. Kok-Wei Bong, Aníbal Utreras-Alarcón, Farzad Ghafari, Yeong-Cherng Liang, Nora Tischler, Eric G. Cavalcanti, Geoff J. Pryde, and Howard M. Wiseman, “A strong no-go theorem on the Wigner’s friend paradox,” Nature Physics 16, 1199–1205 (2020), doi:10.1038/s41567-020-0990-x. Author version arXiv:1907.05607v4, revised 15 March 2023, is identified as matching the published article. Its assumptions are Absoluteness of Observed Events, No-Superdeterminism, and Locality in the stated extended observer

scenario. It is not inferred merely from the controlled-NOT reversal in this chapter. A physical record register and a conscious observer are not identified by that two-qubit calculation. Full author paper, v4.

8. William J. Zeng, Farrokh Labib, and Vincent Russo, "Towards violations of Local Friendliness with quantum computers," Quantum 9, 1851 (2025), doi:10.22331/q-2025-09-05-1851. Published 5 September 2025; full author version arXiv:2409.15302v3. The paper reports proof-of-concept processor tests with acknowledged loopholes, including the absence of spacelike separation and proposes a measure of record-branch complexity. We use the implemented-record perspective, not the unestablished claim that increasing that measure establishes consciousness or a universal definition of observerhood. Published article. Full author text.

9. Hugh Everett III, " 'Relative State' Formulation of Quantum Mechanics," Reviews of Modern Physics 29, 454–462 (1957), doi:10.1103/RevModPhys.29.454. The linked transcription presents the short 1957 relative-state text with editorial annotations; those annotations are separate from Everett's original argument. The summary identifies universal unitary evolution and relative records, not an experimental proof of many worlds or a new derivation here of the Born rule. Later Everettian formulations develop the branch and probability questions differently. Publication record. Original text, annotated transcription.

10. David Bohm, "A Suggested Interpretation of the Quantum Theory in Terms of 'Hidden' Variables. I," Physical Review 85, 166–179 (1952), doi:10.1103/PhysRev.85.166. A university-hosted copy of the original article is linked below. Its configuration-plus-wavefunction dynamics and quantum-equilibrium statistics are the scope used here. Nonlocal guidance does not supply controllable faster-than-light communication in that equilibrium account. Definite positions do not supply a noncontextual answer sheet for every operator in Figure 24.2. Published record. University-hosted original paper. See also Part II, Physical Review 85, 180–193 (1952), doi:10.1103/PhysRev.85.180, §§2–3 and 8–9, for the measurement analysis, equilibrium statistics and nonlocal influence. University-hosted original Part II.

11. Carlo Rovelli, "Relational Quantum Mechanics," International Journal of Theoretical Physics 35, 1637–1678 (1996), doi:10.1007/BF02302261. The revised author version arXiv:quant-ph/9609002v2 (24 February 1997) explicitly uses observer for an interacting physical system, not necessarily a conscious agent. The account of relative states/events is presented as the author's proposal. It is not substituted for a mathematical consequence of Eq. (24.1), and the short summary does not cover every later relational formulation. Full revised author manuscript.

12. Christopher A. Fuchs, N. David Mermin, and Rüdiger Schack, "An introduction to QBism with an application to the locality of quantum mechanics," American Journal of Physics 82, 749–754 (2014), doi:10.1119/1.4874855. The linked author version is arXiv:1311.5253v1 (20 November 2013). The summary attributes the agent-centered probability and experience account to QBism; it does not endorse every philosophical conclusion in the paper as an experimental result. Born-rule constraints do not grant voluntary control over outcomes. Full author paper.

13. Gian Carlo Ghirardi, Alberto Rimini, and Tullio Weber, "Unified dynamics for microscopic and macroscopic systems," Physical Review D 34, 470–491 (1986), doi:10.1103/PhysRevD.34.470. The publication record and the explicit account in the modern primary test in note 15 support the distinction between modified stochastic dynamics and unitary decoherence. The chapter does not simulate GRW trajectories. Continuous spontaneous localization and Diósi–Penrose models in the later experiment are different specified collapse constructions. Original publication.

14. Equation (24.7) and Figure 24.4 are an original identifiability example, not a complete collapse dynamics. Independent exponential dephasing factors multiply, so only the sum of rates is determined before the intervention. The rate pairs in inverse seconds are $(0.30, 0)$ and $(0.10, 0.20)$. A separately assumed selective intervention gives totals 0.075 and 0.225. At one second the visibilities are 0.9277434863 and 0.7985162188, versus 0.7408182207 initially. The same reduced map can have different physical realizations. We do not infer individual outcome selection from this visibility model or claim that the selective control exists for every proposed mechanism.

15. E. Aprile and colleagues (XENON Collaboration), "Challenging Spontaneous Quantum Collapse with the XENONnT Dark Matter Detector," Physical Review Letters 136, 120201 (2026), doi:10.1103/2jm3-4976; published 23 March 2026. It analyzes 1–140 keV electronic-recoil data and modeled spontaneous radiation in

Markovian continuous-spontaneous-localization and Diósi–Penrose families, including opposite-charge cancellation and backgrounds. No significant excess was reported. Exclusions concern specified parameters and spectra, not all collapse theories. Its radiation model and data are not represented by our toy dephasing rates. Published full paper. Publication record.

16. Giulio Chiribella, Giacomo Mauro D'Ariano, and Paolo Perinotti, "Informational derivation of quantum theory," Physical Review A 84, 012311 (2011), doi:10.1103/PhysRevA.84.012311. See also author version arXiv:1011.6451v3. The five axioms are causality, perfect distinguishability, ideal compression, local distinguishability, and pure conditioning. For a fixed purifying system, purification includes uniqueness up to a reversible transformation on that system. The finite-dimensional reconstruction has these premises; it is not derivation from logic or ignorance alone, nor a specification of a particular molecular Hamiltonian. Full author reconstruction. Published record.

17. Equation (24.8) is the explicit two-qubit purification of the diagonal state with eigenvalues q and 1−q. Tracing either part gives that spectrum. An arbitrary local unitary on the purifying part leaves the other reduced state unchanged. This is a representation theorem within quantum theory, not evidence that a chosen laboratory's mixed state was historically prepared by this particular entanglement procedure. Normalization, reduced-state eigenvalues, and local-unitary invariance can be verified directly from the two terms in Eq. (24.8); the reduced-state eigenvalues also determine its entropy. States, ensembles, and purification.

EPILOGUE

The Same Experiment, Seen Differently

The detector is still dark.

Not because we have stopped sending light. In the ideal apparatus of the prologue, the source prepares one photon on each trial, both routes remain open, and detector A registers every arrival. Detector B records none. The beam splitters and mirrors are where we left them. We have returned to the same experiment with a different understanding.

At the beginning, closing a route seemed certain to make arrival less likely. Yet blocking the lower route made B respond. The difficulty was not the arithmetic. It was deciding what the arithmetic should describe.

We can now name the two contributions to B: probability amplitudes, with a relative sign fixed by the preparation, propagation, and recombination. They are not two independently occurring detection events. Before calculating the probability of the one final event, we add them:

$$\mathrm{P}(B) = \left(\frac{1}{2} - \frac{1}{2}\right)^2 = 0. \tag{E.1}$$

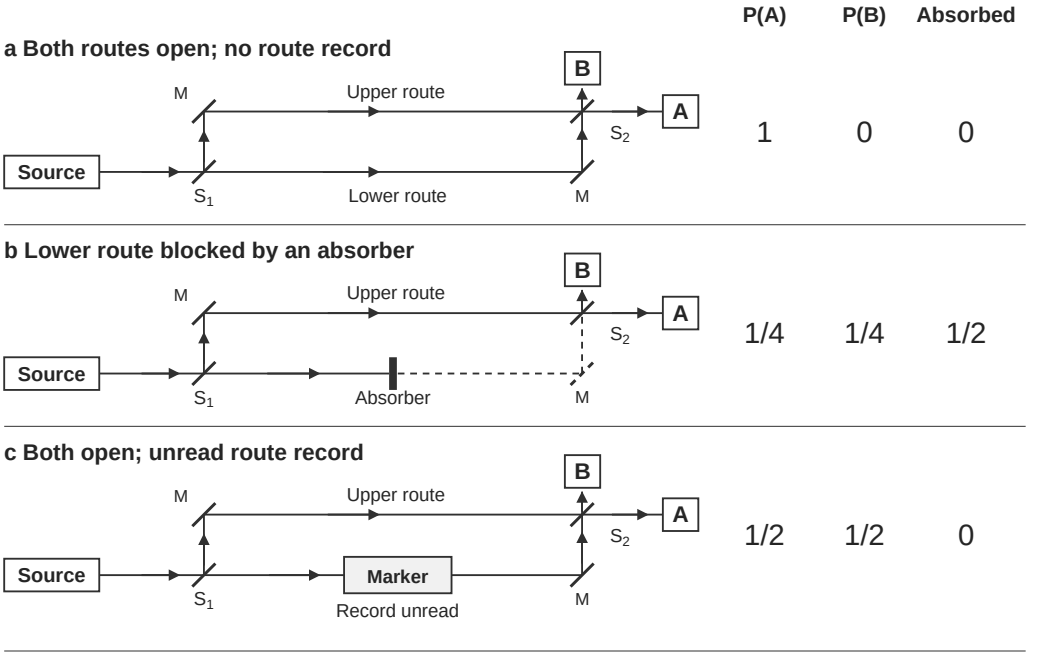
The zero is not missing probability. At A, the same apparatus makes the contributions reinforce. The complete prediction still accounts for every photon. A minus sign has not made any probability negative; it has changed which outcome the amplitudes permit.[1]

Put the absorber back in the lower route. Half the emitted photons are now absorbed on average. One-quarter reach A and one-quarter reach B. Among the survivors alone, the two detectors divide the arrivals equally. Those are different denominators, and we know why both belong in the account. The absorber has brightened one output while reducing the total arrival rate.

There is a second way to change the dark output. Leave both routes open, but let an additional system acquire perfectly distinguishable records of the two route labels, without absorbing the photon or otherwise changing those labels. Do not read that extra system. In this ideal marking model, A and B each receive half the photons. No photon is lost, yet the interference is absent from their separate counts. Figure E.1 puts the three complete predictions beside one another.

One apparatus, three complete predictions

Original phase setting; probabilities per photon emitted.



S_1, S_2: balanced splitters. M: mirrors. No outcomes discarded.

Figure E.1. The same destination is part of three different experiments. The optical routes retain the layout and A/B labels of the prologue. Each row is a separate ideal preparation and arrangement, at the original dark-output phase setting. The lower-route absorber removes half the emitted photons; the lossless marker instead leaves an orthogonal route record in an additional system that is not read. Every probability is per emitted photon. Mirrors and splitters are schematic; route lines are not measured photon tracks. The numbers are calculated predictions, not experimental frequencies.

These outcomes are no longer three versions of a paradox. Blocking a contribution and correlating it with an unread record are different physical operations. The first removes arrivals; the second changes which contributions can interfere in a measurement confined to the output ports. Neither operation waits for a person to notice it.

And the distinction does not end at those ports. With a controlled marker, a suitable joint operation can recover the original coherence. With records dispersed through many inaccessible surroundings, that recovery is a different and often impractical undertaking. Taking a partial trace tells us what we can predict without those surroundings. It does not announce that the complete state has become an ordinary lottery.[2]

We have also learned not to ask this apparatus to establish more than it can. Classical light waves can make the same bright and dark interference pattern. Individual registered pulses alone do not settle the character of the source. The single-photon account must be supported by appropriate photon statistics as well. Jacques and colleagues' experiment separated interference imaging from a test of anticorrelation between detections; those tests addressed different parts of the physical description.[3]

Now remove the absorber and the marker. Keep the routes matched, but give the lower-route contribution an additional phase ϕ. We no longer need to confine the prediction to complete darkness or complete brightness:

$$\mathrm{P}(B)=\frac{|1-e^{i\phi}|^2}{4}=\sin^2\left(\frac{\phi}{2}\right),\ \mathrm{P}(A)=1-\mathrm{P}(B). \tag{E.2}$$

At a quarter-cycle shift, the two outputs are equally probable. At a half-cycle shift, B receives every photon and A is dark. This is the same balanced, lossless calculation as before, with one controlled change. The equation describes separately prepared trials at each setting, not a photon gradually dividing itself among the detectors.

That phase can do more than satisfy our curiosity. If an unknown influence supplies it, the apparatus becomes a sensor. If a specified computation supplies it, recombination can make a relationship between function values readable. The probability rule is unchanged. What changes is the physical origin of the phase and the question attached to the result.

Our three programs made that progression concrete. One arranged cancellation to distinguish agreement from disagreement. Another used repeated reflections to increase the chance of an acceptable candidate. A third made contributions reinforce at labels related to an arithmetic period. None returned an independent answer from every term of a superposition. Each arranged a distribution from which the requested information could be recovered.[4]

The same change of perspective followed us into chemistry. The electron did not become a thread holding two nuclei together. We compared allowed joint states and their total energies with the separated fragments. Later, a better electronic calculation still had to pass through nuclear motion, thermal statistics, and a kinetic model before it could predict a product ratio. A useful number acquired its meaning from that chain.

This connection does not require the entire chain to run on one kind of computer. Quantum hardware may supply a state or a measurement that makes a classical calculation more informative. Classical computation may prepare the controls, select a model, decode records, or show that a proposed quantum contribution was unnecessary. The scientific question, rather than the proportion of quantum machinery, determines what counts as success.[5]

Records have had two roles throughout this journey. An unwanted record of a route can hide the phase we want to use. A deliberately limited record of an error can help preserve an unknown encoded state. The crucial difference is what the record distinguishes. A parity check can reveal a disturbed relation without identifying the logical alternative whose amplitudes must survive.

Teleportation gave a complementary lesson. Its two classical bits did not describe the input amplitudes. They selected a correction to the receiver’s member of a previously shared entangled pair. Sending those bits without that resource would not transfer the

state; sharing the resource without using the bits would not give the receiver the recovered input. The complete protocol succeeded where either resource alone did not.[6]

This is not merely a distinction that lives in ideal diagrams. In the 2026 experiment by Sivak and colleagues, error-detection records also supplied feedback for adjusting the controls of an error-corrected processor. The physical tests used repeated, freshly prepared memory experiments; the extension to uninterrupted long computations was supported by simulations. Even with that boundary retained, the achievement is concrete: records designed not to reveal the logical amplitudes helped the machine compensate for drifting conditions.[7]

Such progress is more interesting than a claim that the difficulties have vanished. It identifies a difficulty well enough to change it. A fragile correlation becomes something that can be prepared, tested, protected, or made deliberately sensitive to one influence rather than another. A limit becomes a design requirement. A failed comparison can identify the next useful experiment.

The foundations remain part of this practical work. The reduced state of either member of an entangled pair does not, by itself, specify the pair's joint state. Bell's comparison established a further limit: under specified locality and setting-independence assumptions, shared instructions cannot reproduce the tested quantum correlations. That is a stronger conclusion than finding two matching records, and a narrower one than declaring every possible physical account of reality impossible.[8]

Nor did we settle the measurement problem merely by explaining why interference becomes inaccessible. In the unitary description of a system, apparatus, and environment, suppressing a cross term in a reduced state does not select one term of the complete state. We used the Born rule to calculate probabilities. We did not derive it by using it.

What happens in an individual run, and what kind of thing a quantum state describes, therefore remain questions worth asking precisely. An interpretation can offer a coherent account of the same predictions. A modified law can offer different predictions to test. Calling either one an explanation does not remove the need to identify what it adds and what evidence bears on it.[9]

There is room here for conviction without pretending that every question is closed. The inability of an ordinary lottery to replace a coherent preparation is something we can demonstrate. The failure of a proposed copier is something we can prove under the quantum rules. The usefulness of a particular machine is something we can compare. Those achievements need not wait for one universally accepted picture of everything.

What should remain after the notation becomes less familiar? Not a collection of astonishing slogans. A way to return to the physical situation. Identify the alternatives. Specify the operation. Follow the records. Ask which measurement would distinguish the

proposed explanations. Then include the uncertainty and the work required to obtain its answer.

The question is demanding, but it is also an invitation. The rules do not merely tell us what cannot be known or done. They tell us how a new experiment might be possible. We can protect an unknown state without copying it. We can learn a field through a phase that once counted as an error. We can use one controlled quantum system to investigate another, provided the correspondence and the requested prediction are kept intact.

Return, finally, to the arrangement with both routes open and no marker. The phase has been restored to the setting at which B is dark. We can predict that result, explain how a controlled change would alter it, and name the tests that could expose a failure of our account. The prediction is exact for the ideal model; a real apparatus must supply evidence for how closely it meets the conditions.

The equipment is familiar now. Its behavior has not become the behavior of an ordinary lottery, and its mathematics has not become a picture of every hidden event. We have gained something more useful than familiarity alone: a connected explanation we can calculate with, challenge, and extend.

A photon is prepared. The two routes are open. Detector A responds, and B remains silent.

At the beginning, that silence was the puzzle. Now we know how to make it part of an answer.

Notes and sources

1. Equations (E.1)–(E.2) and Figure E.1 return to the prologue's ideal single-photon, balanced Mach–Zehnder model. A is the bright output and B the dark output at zero added phase. Amplitudes include propagation and reflection conventions. Probabilities are per emitted photon; the absorber's lost branch is included. See Richard P. Feynman, Robert B. Leighton, and Matthew Sands, The Feynman Lectures on Physics, Vol. III, Chapter 3, §3–1, for coherent addition and the distinction between amplitudes and probabilities. The epilogue's equations and figure are original evaluations, not laboratory data. Amplitude rules.

2. The marker comparison uses the label-preserving, initially independent pure-environment construction developed in Chapter 12. Orthogonal environmental records remove the route off-diagonal terms from the reduced state; tracing them out is not a physical choice of one global branch. Loss, marking, and a coherent phase shift are separate channels. See John Preskill, Quantum Information, Chapter 3, "Foundations II: Measurement and Evolution," §§3.1–3.2 and 3.4.2. The probabilities follow by applying each channel separately to the same input and summing over unobserved records, while retaining absorption as an outcome. Authored treatment of quantum operations.

3. V. Jacques, E. Wu, T. Toury, F. Treussart, A. Aspect, P. Grangier, and J.-F. Roch, "Single photon wavefront-splitting interference: An illustration of the light quantum in action," European Physical Journal D 35, 561–565 (2005), doi:10.1140/epjd/e2005-00201-y. The full author paper was archived as arXiv:2011.12664 in 2020; that deposit date is not the experiment's publication year. It uses a Fresnel biprism rather than our two-beam-splitter apparatus, and independently tests detection anticorrelation. The epilogue does not identify ordinary detector pulses or a classical interference fringe alone with proof of a single-photon source. Published record. Archived original paper.

4. The three algorithmic callbacks refer to Chapters 13–15, with their original task and access assumptions retained. Richard Cleve, Artur Ekert, Chiara Macchiavello, and Michele Mosca, "Quantum Algorithms Revisited," Proceedings of the Royal Society A 454, 339–354 (1998), doi:10.1098/rspa.1998.0164, develops the phase-kickback and Fourier mechanisms. Lov K. Grover, "A fast quantum mechanical algorithm for database search" (STOC 1996), and Peter W. Shor, "Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer," SIAM Journal on Computing 26, 1484–1509 (1997), supply the search and arithmetic constructions. No hardware-speed comparison is added in this epilogue. Controlled-phase and Fourier mechanisms. Search. Factoring.

5. The molecular and performance discussion synthesizes the explicitly worked models of Chapters 6 and 21–23. Binding was an energy comparison with defined fragments, not a classical tether. A product ratio additionally required a thermal and kinetic model. For a primary computational treatment of the steps from electronic energies to thermal rate coefficients, see Kevin Spiekermann, Lagnajit Pattanaik, and William H. Green, "High accuracy barrier heights, enthalpies, and rate coefficients for chemical reactions," Scientific Data 9, 417 (2022), doi:10.1038/s41597-022-01529-6. No new chemical result or contemporary quantum-advantage claim is made here. Published computational study.

6. Charles H. Bennett, Gilles Brassard, Claude Crépeau, Richard Jozsa, Asher Peres, and William K. Wootters, "Teleporting an unknown quantum state via dual classical and Einstein-Podolsky-Rosen channels," Physical Review Letters 70, 1895–1899 (1993), doi:10.1103/PhysRevLett.70.1895. In the ideal protocol, the shared pair is initially independent of the arbitrary unknown input. The four messages are equally likely and do not encode its amplitudes; averaging over an unavailable message leaves the receiver's state maximally mixed. The complete three-register calculation and reference-system preservation were developed in Chapter 11. The sender does not retain an independent usable copy. Error correction in Chapter 18 likewise records correctable damage without learning the logical amplitudes; the elementary parity circuit is not a universal fault-tolerant implementation. Original teleportation research record.

7. Volodymyr Sivak and colleagues, "Reinforcement learning control of quantum error correction," Nature 655, 879–884 (2026), doi:10.1038/s41586-026-10759-2, published 8 July 2026. Error-detection events guided control-parameter steering in repeated memory experiments; each shot re-prepared the state. The compatibility of exploration with uninterrupted longer computations and larger codes was examined through simulations. Decoder steering using logical-error estimates is an additional procedure, not silently included in

the detection-event-only claim. This is a callback to Chapter 18, not a claim of an indefinitely operating universal quantum computer. Published full study.

8. John S. Bell, “On the Einstein Podolsky Rosen paradox,” Physics Physique Fizika 1, 195–200 (1964), doi:10.1103/PhysicsPhysiqueFizika.1.195. The operational comparison developed in Chapter 10 was the later CHSH scenario, with independent settings, locality during the response interval, and explicit trial accounting. The claim excludes that defined model class rather than every hidden-variable or realist account. See also John Preskill, Chapter 4, “Quantum Entanglement” (2 November 2001), §4.1.1 for reduced states and §§4.2–4.3 for the Bell/CHSH treatment used in the book. Original publication record. Authored Bell/CHSH treatment.

9. The interpretive boundary is the same one established by the explicit measurement-chain calculations in Chapters 12 and 24. Unitary evolution gives a joint state; partial tracing restricts predictions and is not an outcome-selection operation. An empirically distinct modification requires its own probabilities and physical premises. This epilogue adds neither a Born-rule derivation nor a selected interpretation. For a primary unitary measurement account, see Hugh Everett III, “‘Relative State’ Formulation of Quantum Mechanics,” Reviews of Modern Physics 29, 454–462 (1957), doi:10.1103/RevModPhys.29.454. Everett’s proposal is a proposed interpretation, not an experimental conclusion of our interferometer. Original publication record. The linked transcription includes editorial annotations, which are separate from Everett’s original argument. Annotated transcription.

Glossary of Key Terms

These definitions recall the meanings used in this book. Chapter references lead to the main explanation; related entries connect ideas that are useful to compare. The notation guide at the front explains the conventions used for symbols and equations.

A

Ablation. A comparison that removes or replaces one component of a procedure to test what it contributes. In a hybrid calculation, replacing quantum candidate generation with a credible classical method, with fair retuning and total-cost accounting, tests the quantum component's value. A separate classical approach to the whole task remains another necessary comparator. *Chapter 23*. *See also* Quantum advantage.

Activation free energy. The free-energy difference between a reactant and a specified transition region at a stated temperature and standard state. It includes the relevant energetic and statistical contributions, beyond a fixed-nuclei electronic barrier. Its connection to a reaction rate also depends on the kinetic model and its transmission factor. *Chapter 22*. *See also* Transition state; Transmission factor.

Active space. A selected set of electrons and orbitals within which an electronic-structure calculation allows the retained electrons to rearrange, while treating the remaining possibilities more restrictively. An exact solution within that space does not recover correlations excluded from it. Consistent selection matters when comparing changing molecular structures. *Chapter 22*. *See also* Full configuration interaction; Spin orbital.

Algorithm. A specified procedure for solving a family of inputs, with a stated criterion for the answer. The task may require an exact result, an approximation, a sample, or a bounded error probability. The procedure, its physical implementation, and the resources required to run it are distinct parts of the account. *Chapter 8*. *See also* Logic gate; Quantum circuit; Uncomputation.

Amplitude amplification. A procedure that uses a coherent acceptance test and reflections to increase the probability of a desired result from a controllable preparation. It generalizes Grover's search beyond uniform starting states. The preparation, its inverse, and the acceptance test must all be implemented and counted; ordinary repetition does not provide the same operation. *Chapter 14*. *See also* Grover’s algorithm; Oracle.

Amplitude estimation. A family of quantum procedures for estimating the success probability associated with a coherent preparation. With the required controls, inverses, and preparation access, coherent methods can improve the dependence of query cost on precision compared with independent sampling. This query improvement does not make those stronger operations free to implement. *Chapter 16*. *See also* Amplitude amplification; Query complexity.

Analogue simulation. Quantum simulation in which an engineered Hamiltonian directly represents the target's intended interactions. The correspondence must include preparation, measurement, and any conversion between target and laboratory time. A faster physical evolution in the simulator is not, by itself, a computational speedup over a classical calculation. *Chapter 21*. *See also* Digital simulation; Quantum simulation.

Anharmonicity. The departure of a system's energy levels from equal spacing. For the three-level example, it is the difference between neighboring transition angular frequencies. Separating those frequencies helps a control pulse address the chosen qubit transition without exciting another level, although finite pulse strength can still cause leakage. *Chapter 17*. *See also* Leakage; Physical qubit.

Authentication. Protection that lets an accepted message be attributed to the intended participant, with a specified failure guarantee. An authenticated channel may still be public and may still be blocked. Quantum key distribution requires an authenticated classical conversation; quantum signal statistics alone do not establish a sender's identity. *Chapter 19*. *See also* Quantum key distribution; Post-quantum cryptography.

B

Basis. A complete set of reference states used to express a state’s components. For the orthonormal qubit bases used here, the two references are normalized and perfectly distinguishable by a suitable ideal measurement. Rewriting a state in a different basis changes its description; performing an operation or measurement changes the experiment. *Chapter 4*. *See also* Orthogonal states; Superposition; Inner product; Measurement.

BB84. The Bennett–Brassard quantum key-distribution protocol introduced in 1984. In the version developed here, one pair of matching preparation and measurement bases supplies prospective key bits, while the complementary pair supplies a test. Authenticated discussion, reconciliation, privacy amplification, and a matching security analysis are also required. *Chapter 19*. *See also* Quantum key distribution; Privacy amplification; Finite-key analysis.

Bell inequality. A bound on joint outcome statistics obeyed by Bell-local models under stated setting and trial assumptions. The book develops CHSH, which combines four correlations and has a local bound of two in magnitude. A justified violation excludes that model class; it does not select a unique interpretation of quantum theory. *Chapter 10*. *See also* Bell-local model; Measurement independence; Entanglement witness; No-signalling.

Bell-local model. A model in which, once shared information and relevant history are specified, each laboratory’s response depends only on its own current measurement setting. The shared-information distribution is independent of those current settings. The class permits complicated local devices and stochastic responses; it does not require simple or deterministic machinery. *Chapter 10*. *See also* Bell inequality; Measurement independence; No-signalling.

Bell state. One of a standard set of four mutually orthogonal, maximally entangled two-qubit states. Each member has maximally mixed single-qubit reduced states, while its joint correlations depend on which member was prepared. The Bell-plus state used throughout the book combines zero–zero and one–one with equal positive amplitudes. *Chapter 9*. *See also* Entanglement; Reduced state; Quantum teleportation.

Binary entropy. The Shannon entropy of two outcomes with probabilities p and one minus p. It is one bit when the outcomes are equally likely and zero when either is certain. Fractional entropy describes an average information rate, not a fraction of a physical binary digit on an individual trial. *Chapter 7*. *See also* Shannon entropy; Bit; Quantum compression.

Bit. A binary digit: one of two logical alternatives, conventionally labeled 0 and 1, represented by distinguishable physical states. Bit is also the unit of information obtained with base-two logarithms. A stored binary position does not necessarily supply one bit of new information; that depends on the source probabilities and available records. *Chapter 1*. *See also* Code; Shannon entropy; Qubit.

Bloch sphere. A geometric representation of a qubit's state using three coordinates obtained from X, Y, and Z measurement probabilities. Pure states lie on the unit sphere; mixed states lie inside the corresponding ball. The center is the maximally mixed state, and opposite surface points represent orthogonal pure states. *Chapter 4*. *See also* Qubit; Pure state; Mixed state; Quantum state tomography.

Bohmian mechanics. An approach that adds an actual particle configuration guided by a wavefunction. Its quantum-equilibrium formulation reproduces the usual measurement statistics, while guidance for entangled systems is nonlocal. It therefore lies outside the Bell-local class and does not assign context-independent values to every measured observable. *Chapter 24*. *See also* Bell-local model; Contextuality.

Born–Oppenheimer approximation. An approximation that separates electronic and nuclear motion by first solving for electrons at fixed nuclear positions. The resulting electronic energies, with nuclear repulsion included, supply potential-energy curves or surfaces for nuclear motion. Couplings between electronic states can require a more complete treatment. *Chapter 6*. *See also* Hamiltonian; Orbital; Zero-point energy.

Born rule. The quantum rule connecting a state and a specified measurement to outcome probabilities. For a pure state measured against a normalized basis reference, the probability is the squared magnitude of their overlap. Using the rule to derive a successful prediction does not derive the rule itself. *Chapter 2*. *See also* Probability amplitude; Inner product; Measurement.

C

Chemical accuracy. A conventional energy-error scale, often taken as about one kilocalorie per mole. It is not a universal guarantee of an adequate chemical prediction. The required tolerance depends on the final quantity: an error acceptable for one energy comparison can cause a large error in a predicted reaction rate. *Chapter 22*. *See also* Activation free energy; Uncertainty budget.

Classical simulation. Using classical computation to reproduce specified properties or measurement statistics of a quantum model or circuit. A simulator may exploit a compact representation without storing every amplitude. Efficient simulation of one circuit family or one requested observable does not establish efficient simulation of arbitrary quantum computations. *Chapter 16*. *See also* Clifford gate; Stabilizer; Tensor network.

Clifford gate. A gate that transforms Pauli operators into Pauli operators under conjugation, allowing stabilizer constraints to retain their compact form. Hadamard, controlled NOT, and the quarter-cycle phase gate generate the Clifford circuits discussed here. With suitable inputs and measurements, those circuits admit efficient classical simulation even when they create entanglement. *Chapter 16*. *See also* Stabilizer; Classical simulation; T gate.

Code. A rule assigning physical or symbolic representations to messages, together with a way to read them. A code need not be secret. Its usefulness depends on whether the assigned alternatives can actually be prepared, transmitted, and distinguished with the required reliability. *Chapter 1*. *See also* Bit; Correlation.

Code distance. For the stabilizer codes considered here, the smallest number of physical positions on which an undetectable, nontrivial logical Pauli operation can act. A larger distance permits correction of more errors under ideal checks. Distance alone does not specify the overhead or reliability of noisy gates, measurements, and decoding. *Chapter 18*. *See also* Logical qubit; Erasure (error correction); Fault tolerance.

Coherence. The phase relationships between specified state components that can affect interference. In a chosen basis, a density matrix's off-diagonal entries describe these relationships. Equal probabilities in one measurement do not establish equal coherence, and a loss of fringe contrast does not by itself identify its physical cause. *Chapter 2*. *See also* Relative phase; Density matrix; Interference; Dephasing.

Coherence decay time (T2). The decay constant for off-diagonal coherence in a specified exponential model after known free phase evolution is accounted for. It can include both energy relaxation and additional dephasing. Different experimental sequences, such as Ramsey and echo, can yield different decay times; the measurement procedure belongs with the quoted value. *Chapter 17*. *See also* Energy relaxation time (T1); Dephasing; Ramsey interferometry.

Complex conjugate. The number obtained by reversing the sign of a complex number's imaginary part. Geometrically, this reflects its arrow across the real axis. Multiplying a number by its conjugate gives its squared magnitude, which is real and nonnegative. *Chapter 3*. *See also* Complex number; Inner product; Born rule.

Complexity class. A family of problems grouped by permitted computation, resource growth, and success guarantee. For decision problems, P uses deterministic polynomial time; BPP allows classical randomness; BQP allows efficiently constructible quantum circuits with polynomial resources. BPP and BQP require bounded error on every input. Their definitions do not prove a strict separation between them. *Chapter 16*. *See also* NP; Polynomial time; Query complexity.

Complex number. A number with real and imaginary parts, represented by two coordinates on a plane. Its magnitude is the length of the corresponding arrow; a nonzero number's phase is its angle. Both parts enter amplitude calculations, and the word imaginary does not identify an unreal physical outcome. *Chapter 3*. *See also* Complex conjugate; Probability amplitude; Relative phase.

Conditional entropy. The uncertainty remaining about one variable after another record is supplied, averaged over the possible records. Classical conditional entropy cannot exceed the original entropy on average, although a particular record can increase uncertainty. Its value depends on the joint distribution, not just the separate totals. *Chapter 7*. *See also* Conditional probability; Shannon entropy; Mutual information.

Conditional probability. The probability assigned to an event when specified information is given. The probability of no click given no pulse need not equal the probability of no pulse given no click: missed detections can produce the same record. Inferences therefore depend on both preparation probabilities and the detection model. *Chapter 1*. *See also* Measurement; Conditional entropy.

Contextuality. Failure of a proposed assignment of measurement outcomes to be independent of the compatible measurements accompanying them. In the Peres–Mermin example, no predetermined signs for the nine observables satisfy every compatible row and column requirement. The conclusion concerns that specified noncontextual model and its measurement-compatibility assumptions. *Chapter 24*. *See also* Measurement; Bell inequality.

Continued fraction. A representation of a number through successive integer divisions that produces a sequence of rational approximations called convergents. In the factoring example, it converts a Fourier readout into candidate denominators related to the arithmetic order. Those candidates still need arithmetic tests; a prominent measurement outcome need not reveal the full order. *Chapter 15*. *See also* Order (modular arithmetic); Shor's algorithm.

Controlled-NOT gate. A two-system operation that flips the target when the control is one and otherwise leaves it unchanged. Its coherent quantum action is linear on superpositions. It copies orthogonal binary labels into a zero target, but can entangle a superposition with that target instead of producing two copies of the input state. *Chapter 11*. *See also* Logic gate; Entanglement; No-cloning theorem.

Correlation. A statistical relationship between variables or records, so that their joint probabilities do not factor into independent probabilities. Matching totals viewed separately need not reveal it; paired records can. Correlation by itself establishes neither a causal direction nor quantum entanglement. *Chapter 1*. *See also* Mutual information; Entanglement; Separable state.

D

Decoder. A procedure that uses error-check results, their history, the checking circuit, and a noise model to choose a recovery. It need not identify the actual physical error history. Different histories can support the same successful correction, while an incorrect inference can leave a logical error despite satisfactory final checks. *Chapter 18*. *See also* Syndrome; Quantum error correction; Surface code.

Decoherence. Loss of accessible interference between specified alternatives, described by suppression of relevant off-diagonal terms in a reduced density matrix. Environmental decoherence follows from correlations with unretained degrees of freedom; uncontrolled phase variation can produce the same reduced description. Decoherence alone does not select a single measurement outcome. *Chapter 12*. *See also* Coherence; Environment; Dephasing; Measurement problem.

Density matrix. A matrix describing a quantum state, including pure and mixed states. In a chosen orthonormal basis, diagonal entries give basis-measurement probabilities; off-diagonal entries retain relationships needed for other measurements. Its eigenvalues are nonnegative and sum to one. Different preparation procedures can yield the same density matrix. *Chapter 7*. *See also* Quantum state; Mixed state; Coherence; von Neumann entropy.

Dephasing. Reduction of off-diagonal coherence in a specified basis while its populations remain unchanged. Random phase variation and interactions with other degrees of freedom can produce the same dephasing map. When the basis consists of energy eigenstates, dephasing need not involve loss of the system's excitation. *Chapter 12*. *See also* Decoherence; Relative phase; Density matrix; Energy eigenstate.

Detuning. A frequency difference between a transition and the drive or reference used to address it. The sign follows the convention specified for the model. Detuning can change a controlled evolution or supply the phase measured by a sensor; a reference-frequency error can also imitate the signal being sought. *Chapter 20*. *See also* Ramsey interferometry; Quantum sensing.

Deutsch's algorithm. An algorithm that determines whether the two values of an unknown binary function agree. Its exact form uses one coherent reversible function query, phase kickback, and interference to produce the answer with certainty in the ideal model. The comparison is with two classical value queries for an exact answer under the stated access restriction. *Chapter 13*. *See also* Oracle; Phase kickback; Query complexity.

Device-independent certification. Establishing specified properties from observed input–output statistics without trusting a detailed model of a device's internal states and measurements. Bell-based protocols still require assumptions about settings, allowed communication, trial records, and statistical analysis. Device independence removes particular internal assumptions, rather than every assumption. *Chapter 10*. *See also* Bell inequality; Measurement independence; Entanglement witness.

Digital simulation. Quantum simulation implemented through a sequence of gates representing the target dynamics. Product formulas, for example, approximate simultaneous interactions by shorter ordered evolutions. The approximation error, gate implementation error, initial-state preparation, and measurement cost must each be assessed for the requested prediction. *Chapter 21*. *See also* Analogue simulation; Product formula; Quantum simulation.

Dissociation energy. The energy needed to separate a molecule into specified fragment states. The electronic well depth D_e is the difference between the separated-fragment energy and the minimum of a clamped-nuclei potential curve. In the zero-rotation nuclear model used here, D_0 starts from the lowest vibrational level and equals D_e minus its zero-point energy. Comparisons require matching initial levels and fragment references, with physical approximations specified. *Chapters 6 and 22*. *See also* Born–Oppenheimer approximation; Zero-point energy.

Dynamical collapse. A proposed modification of quantum evolution in which localization occurs through a physical stochastic process. A specified collapse law can predict records different from ordinary isolated Schrödinger evolution and can therefore be tested. A fading interference fringe alone does not identify collapse without distinguishing ordinary environmental effects. *Chapter 24*. *See also* Decoherence; Measurement problem.

E

Eigenvalue. A number associated with a nonzero vector that an operator maps to that same vector multiplied by the number. Hamiltonian eigenvalues give possible energy outcomes; density-matrix eigenvalues are nonnegative weights summing to one. The same mathematical term therefore need not denote the same physical quantity. *Chapter 5*. *See also* Operator; Energy eigenstate; Density matrix.

Electron correlation. Features of a many-electron joint state that an independent-electron approximation fails to capture. They matter when electron configurations must be described together, as in stretched molecular hydrogen, where a single doubly occupied bonding orbital gives the wrong separated-fragment picture. A one-electron density alone does not specify the complete joint state. *Chapter 6*. *See also* Orbital; Pauli exclusion principle; Wavefunction.

Electron volt. An energy unit equal to the energy gained by a charge of magnitude equal to the electron's charge through a potential difference of one volt. One electron volt is exactly $1.602176634 \times 10^{-19}$ joule. It is an energy, not a voltage. *Chapter 6*. See also Energy eigenstate; Hamiltonian.

Energy eigenstate. A state for which the Hamiltonian returns the same state vector multiplied by a number, its energy eigenvalue. Under a fixed isolated-system Hamiltonian, such a state changes only by a global phase, so its fixed-measurement probabilities remain constant. Stationarity does not mean that an electron sits motionless. *Chapter 5*. *See also* Hamiltonian; Global phase; Schrödinger equation.

Energy relaxation time (T1). The characteristic time for excited-state population to decay toward a lower state in a specified model. In the chapter's exponential, negligible-temperature model, relaxation also reduces coherence. A measured relaxation time does not by itself characterize dephasing, leakage, readout errors, or the performance of a complete circuit. *Chapter 17*. *See also* Coherence decay time (T2); Leakage.

Entanglement. A property of a joint quantum state relative to a specified division into subsystems: it cannot be expressed as a mixture of product states across that division. For a pure state, this means its state vector cannot factor. Matching outcomes alone do not establish entanglement, and entanglement does not enable controllable faster-than-light communication. *Chapter 9*. *See also* Product state; Separable state; Entanglement witness; No-signalling.

Entanglement swapping. Using a joint measurement on members of previously distributed entangled pairs to establish entanglement between their other members, which need not have interacted directly. The outcome determines the conditional state and any required correction. Prior distribution, local operations, and communicated outcome records remain part of the protocol. *Chapter 11*. *See also* Entanglement; Bell state; Quantum teleportation; No-signalling.

Entanglement witness. A measured quantity with a bound obeyed by every separable state under specified assumptions. A justified violation certifies entanglement. Failure to violate a particular witness is inconclusive, and a witness using calibrated qubit measurements does not have the same device assumptions as a Bell test. *Chapter 9*. *See also* Entanglement; Separable state; Bell inequality; Device-independent certification.

Environment. Degrees of freedom whose full description is not retained when predicting the behavior of a chosen system. An environment can be as small as one auxiliary qubit; it need not be a hot reservoir. Its interactions and any resulting records determine their effects on the reduced system. *Chapter 12*. *See also* Reduced state; Decoherence; Pointer states.

Erasure (error correction). A failure whose affected location is reliably identified, without revealing or restoring the lost quantum state. A photon-loss channel can supply a distinguishable vacuum flag. Knowing the location makes correction easier than for an arbitrary unannounced error, but the flag's reliability and any effects on other carriers still matter. *Chapter 18*. *See also* Code distance; Leakage; Quantum error correction; Memory reset.

Error cancellation. Reduction of error in a difference because the errors in its contributing quantities partly or wholly offset each other. More accurate individual energies can nevertheless yield a worse barrier if that cancellation changes. Shared contributions and statistically characterized covariance must be distinguished from unquantified common model omissions. *Chapter 22*. *See also* Activation free energy; Uncertainty budget.

Error mitigation. Methods that combine or correct noisy experimental records to estimate an intended observable under a specified noise account. They can increase sampling and calibration costs and need not produce a protected quantum state for later operations. Correcting an estimated mean does not automatically remove bias from an inaccurate noise model. *Chapter 23*. *See also* Quantum error correction; Uncertainty budget.

Everettian interpretation. An approach retaining unitary evolution for the complete isolated system, including physical recorders. Definite records are described relative to branches of the joint state rather than selected by a collapse process. Explaining usable branch structure and the role of probability is part of the account, beyond simply writing a superposition. *Chapter 24*. *See also* Measurement problem; Dynamical collapse.

F

Fault tolerance. Designing preparations, gates, measurements, and recovery so that faults remain within the code's correction capability under a specified noise model. It must account for faults introduced or spread by the checking apparatus itself. A correct ideal circuit or a good encoded memory is not automatically a fault-tolerant computation. *Chapter 18*. *See also* Quantum error correction; Code distance; Logical qubit.

Fidelity. A measure of agreement between an intended state and an actual state. For a pure target, the convention used here is the probability that the actual state passes an ideal projection test for that target. A value of 0.9 need not mean that ninety percent of trials are exact copies. *Chapter 11*. *See also* Projective measurement; Quantum state tomography; Quantum teleportation.

Finite-key analysis. A security calculation for a finite quantum key-distribution record. It accounts for limited test data, reconciliation leakage, verification, and a chosen security guarantee under a matching device and protocol model. An asymptotic secret fraction or a small observed error rate alone does not determine the usable finite key. *Chapter 19*. *See also* Quantum key distribution; Privacy amplification; Authentication.

Full configuration interaction. An electronic-structure calculation retaining every permitted many-electron configuration within a specified finite orbital basis and sector. It can solve that finite model without a further configuration truncation. It does not remove the finite-basis approximation or include interactions and nuclear effects excluded from the Hamiltonian. *Chapter 22*. *See also* Active space; Slater determinant; Spin orbital.

G

Global phase. A common factor of unit magnitude multiplying every amplitude of a complete state. It changes the state vector’s representation without changing its measurement predictions. Applying a phase to only one branch relative to another is a relative phase change in the larger system, not a global one. *Chapter 3*. *See also* Relative phase; Pure state; Unitary transformation.

Grover's algorithm. A search algorithm that alternates a phase change on acceptable candidates with a reflection about the prepared starting state. In the stated unstructured-search model it gives an optimal square-root query improvement. The usual rounds can overshoot their best stopping point, and a query advantage need not become an elapsed-time advantage. *Chapter 14*. *See also* Amplitude amplification; Oracle; Query complexity.

H

Hadamard gate. A one-qubit unitary that forms normalized sums and differences of the input amplitudes. It maps the zero/one basis to the plus/minus basis and applying it twice restores the input. Its coherent action can convert a relative sign into a readable outcome; it is not equivalent to a fair random choice. *Chapter 13*. *See also* Clifford gate; Deutsch's algorithm.

Hamiltonian. The operator containing the energy terms and interactions of a specified model. It determines coherent time evolution through the Schrödinger equation, and its eigenvalues give possible ideal energy-measurement results. A Hamiltonian matrix is not a probability table; its entries depend on the chosen basis. *Chapter 5*. *See also* Energy eigenstate; Schrödinger equation; Unitary transformation.

Heisenberg scaling. Inverse-linear improvement of local standard uncertainty with a declared sensing resource, such as the number of equally exposed probes in the ideal entangled example. Its meaning depends on what is counted and on preparation, readout, noise, and prior knowledge of the fringe branch. It is not an unrestricted precision guarantee. *Chapter 20*. *See also* Standard quantum limit; Quantum sensing.

Heralding. A record announcing that a specified preparation, link, or transfer has succeeded without supplying the unknown logical state itself. Its usefulness depends on the reliability of the announcement and on what remains available afterward. The quality of heralded successes and the probability and cost of obtaining them are separate quantities. *Chapter 19*. *See also* Erasure (error correction); Quantum repeater; Fidelity.

Hermitian. A matrix or operator equal to its adjoint. Its eigenvalues are real. A Hermitian Hamiltonian generates probability-preserving evolution in the finite closed systems used here. In matrix form, transposing and complex-conjugating it leaves it unchanged. *Chapter 5*. See also Hamiltonian; Unitary transformation.

Holevo bound. An upper bound on the classical information a receiver can obtain about an encoded label by measuring the delivered quantum system. For the book's pure signal states, that information cannot exceed the entropy of their average state. The bound need not be attained by a single-use measurement. *Chapter 7*. *See also* Mutual information; von Neumann entropy; Orthogonal states.

I

Inner product. A rule comparing state vectors by conjugating the components of one, multiplying corresponding components, and adding. The resulting complex number is their overlap. For normalized pure states, its squared magnitude supplies the probability of the corresponding outcome in a suitable projective measurement. *Chapter 4*. *See also* Complex conjugate; Orthogonal states; Born rule.

Interference. Reinforcement or cancellation when coherent amplitude contributions combine, changing the probabilities of resulting outcomes. The complete alternatives, including any retained distinguishing records, determine which contributions can combine coherently. Classical waves also interfere, so an interference pattern alone does not establish single-photon behavior or a computational advantage. *Chapter 2*. *See also* Probability amplitude; Coherence; Interferometer; Decoherence.

Interferometer. An arrangement that brings coherent contributions together so that their relative phase affects measured outputs. The book's opening example splits light into two routes and recombines them. Bright and dark outputs demonstrate interference; establishing a single-photon source requires additional preparation and detection tests. *Prologue*. *See also* Interference; Relative phase; Ramsey interferometry.

L

Landauer's bound. For the book's symmetric cyclic memory model, perfectly resetting a fair unknown bit exports at least Boltzmann's constant times reservoir temperature times the natural logarithm of two as mean heat. The account requires an initially uncorrelated thermal reservoir and all auxiliaries included. It concerns this entropy-reducing reset, not every gate, measurement, or individual fluctuating trial. *Chapter 8*. *See also* Bit; Shannon entropy; Logical reversibility; Uncomputation.

Leakage. Population leaving the physical subspace chosen to encode a qubit. Excitation of an unwanted third level is leakage, whereas a bit flip remains within the zero/one space. A detector may misassign leaked events to ordinary labels, and a result conditioned on remaining inside the encoding must also report retention. *Chapter 17*. *See also* Anharmonicity; Erasure (error correction); Physical qubit.

Logical qubit. One qubit of information encoded in a chosen subspace of a larger physical system. Its protection depends on the encoding, error model, check operations, recovery, and desired reliability. The carriers do not hold independent copies of the unknown state, and no universal conversion factor relates physical-qubit and logical-qubit counts. *Chapter 18*. *See also* Physical qubit; Quantum error correction; Code distance.

Logical reversibility. The property that the complete output of an operation uniquely determines its input. Retaining input or work records can make a computation reversible even when its answer alone merges alternatives. Logical reversibility differs from thermodynamic reversibility and does not guarantee negligible physical dissipation. *Chapter 8*. *See also* Unitary transformation; Uncomputation; Landauer's bound.

Logic gate. An operation defined by an elementary logical input–output rule, such as AND, XOR, or NOT. A physical device can implement that rule using different materials. Gate counts depend on which operations are designated elementary and do not directly specify transistor counts, running time, or heat. *Chapter 8*. *See also* Algorithm; Controlled-NOT gate; Toffoli gate.

M

Measurement. A physical procedure that produces an outcome record with probabilities determined by the input state and the apparatus. It may also change the state available afterward. The recorded outcome is one result of that procedure, not a complete description of the incoming state, and no human inspection is required. *Chapter 4*. *See also* Born rule; Projective measurement; Quantum state tomography; Measurement problem.

Measurement independence. The assumption that the current measurement settings do not select different ensembles of the shared hidden information used in a Bell-local model. It concerns statistical and causal relationships among setting sources and tested systems. It is distinct from spacelike separation and from claims about metaphysical freedom. *Chapter 10*. *See also* Bell-local model; Bell inequality; Device-independent certification.

Measurement problem. The question of how the quantum account of a measurement relates to the definite outcome recorded on an individual run. Unitary system–apparatus–environment evolution can suppress observable interference between records while retaining a joint superposition. Taking a partial trace does not itself select one outcome or derive the Born rule. *Chapter 12*. *See also* Measurement; Born rule; Decoherence; Partial trace.

Memory reset. Preparing a memory in a specified standard state regardless of its initial stored value. Resetting a fair unknown bit removes a distinction from that memory; a complete physical account tracks where its entropy and any correlated records go. This differs from a flagged loss of quantum information, called an erasure error. *Chapter 8*. *See also* Landauer's bound; Logical reversibility; Erasure (error correction).

Mixed state. A quantum state that cannot be represented by one normalized state vector. It may arise from a random preparation whose selection record is unavailable, or describe one part of an entangled whole. These situations can have identical local predictions while differing in their wider correlations. *Chapter 9*. *See also* Pure state; Density matrix; Reduced state.

Mutual information. The average reduction in uncertainty about one classical variable supplied by another. It measures their statistical relationship and is nonnegative and symmetric, rather than identifying a causal direction. An unpredictable receiver output can have zero mutual information about the sender's message. *Chapter 7*. *See also* Correlation; Conditional entropy; Quantum mutual information.

N

No-cloning theorem. The prohibition on one deterministic physical procedure making perfect copies of every arbitrary unknown quantum state from a single supplied specimen and input-independent auxiliaries. Orthogonal alternatives can be copied, and a known recipe can be used repeatedly. Approximate copying and restricted heralded copying are different tasks with different guarantees. *Chapter 11*. *See also* Orthogonal states; Controlled-NOT gate; Fidelity; Quantum teleportation.

Normalization. The condition that a state description gives total probability one. In an orthonormal basis, a pure state's squared amplitude magnitudes sum to one. When reference functions overlap, their inner products must also enter the normalization; summing squared coefficient magnitudes alone is generally insufficient. *Chapter 4*. *See also* Probability amplitude; Inner product; Orbital.

No-signalling. The constraint that an operation confined to one subsystem leaves another subsystem's reduced state unchanged when all outcomes are included and no system or message is transmitted to it. Conditional states may differ when an outcome is known, but identifying the corresponding trials requires that outcome record. *Chapter 9*. *See also* Reduced state; Conditional probability; Bell-local model; Quantum teleportation.

NP. The class of decision problems whose yes answers have polynomial-length classical certificates checkable in polynomial time. NP includes P and does not mean non-polynomial. An NP-complete problem is in NP and can represent every NP problem through a polynomial-cost translation. Shor's factoring algorithm does not supply a general efficient algorithm for NP-complete problems. *Chapter 16*. *See also* Complexity class; Polynomial time; Shor's algorithm.

O

One-time pad. Encryption that combines a binary message with an equally long, uniformly random, independent secret key using XOR. The same operation with the key recovers the message. Its secrecy requires the stated key conditions and no reuse; the pad alone does not authenticate the message or prevent undetected alteration. *Chapter 19*. *See also* Authentication; Quantum key distribution.

Operator. A mathematical rule that acts on a state vector. The Hamiltonian supplies energy and evolution information, while a unitary operator represents a reversible transformation. A matrix is a representation of an operator in a chosen basis, not necessarily a table of probabilities. *Chapter 5*. *See also* Hamiltonian; Eigenvalue; Unitary transformation.

Oracle. A specified operation or query interface whose uses are counted separately in an algorithmic comparison. A quantum function oracle must act coherently on permitted superpositions and preserve the required phases. Its implementation can contain substantial computation; an ordinary service returning measured classical values does not automatically supply that interface. *Chapter 13*. *See also* Phase kickback; Query complexity; Uncomputation.

Orbital. A one-electron spatial state used to describe an atom or molecule. It is not an electron's orbit or trajectory. Different atomic orbitals can overlap, so a coefficient in an expansion using them cannot automatically be squared and read as a probability of belonging to one atom. *Chapter 6*. *See also* Wavefunction; Inner product; Electron correlation.

Order (modular arithmetic). For an integer coprime to a modulus, the smallest positive exponent whose modular power returns one. That order is the repetition Shor's algorithm seeks. An even order can expose factors through its halfway power, but unsuitable halfway values or incompletely recovered denominators require further checks or another attempt. *Chapter 15*. *See also* Continued fraction; Shor’s algorithm.

Orthogonal states. States whose inner product is zero. Specified orthogonal pure states can be distinguished without error and with a conclusive answer on every trial by a suitable ideal measurement. Distinct nonorthogonal pure states cannot meet both requirements from one supplied copy without additional information-bearing resources. *Chapter 4*. *See also* Inner product; Basis; No-cloning theorem.

P

Partial trace. The mathematical operation that obtains a subsystem’s density matrix by matching and summing over the reference labels of the other systems. It restricts which predictions the description supplies. It does not require physically measuring or discarding a partner, and it differs from conditioning on a communicated outcome. *Chapter 9*. *See also* Reduced state; Density matrix; Conditional probability.

Pauli exclusion principle. The restriction that identical fermions, including electrons, cannot occupy the same complete one-particle state. For electrons it follows from antisymmetry of the full spatial-and-spin state. Two electrons can share a spatial orbital when the joint spin state supplies the required antisymmetry. *Chapter 6*. *See also* Spin; Orbital; Electron correlation.

Phase estimation. A quantum procedure for inferring an eigenvalue's phase from controlled applications of a unitary transformation. For time evolution, the phase is related to energy. Obtaining the desired energy requires suitable initial-state overlap, sufficient controlled evolution and precision, and appropriate phase-range interpretation; the procedure does not automatically prepare the ground state. *Chapter 22*. *See also* Phase kickback; Quantum Fourier transform; Variational quantum eigensolver.

Phase kickback. The appearance of an eigenvalue-dependent relative phase in a control register when a controlled operation acts on a target eigenstate. The target returns to the same physical state while different control components can acquire different phase factors. The effect describes the algebra of the joint operation, not a signal travelling backward along a wire. *Chapter 13*. *See also* Oracle; Deutsch’s algorithm; Phase estimation.

Physical qubit. A selected two-state encoding in a physical carrier, together with the means to prepare, control, preserve, and read it. Atoms, electrical modes, spins, and photons can supply different encodings. Other available states, loss, control hardware, and readout behavior remain part of the device's implementation account. *Chapter 17*. *See also* Logical qubit; Leakage; Readout.

Pointer states. States that remain comparatively stable while an interaction records their distinguishing properties in an apparatus or environment. Their selection depends on the dynamics, not simply on diagonalizing a density matrix. The book's ideal marker preserves its zero and one reference states while correlating their superpositions with the marker. *Chapter 12*. *See also* Environment; Decoherence; Quantum Darwinism.

Polynomial time. A running-time bound that grows no faster than a fixed power of the specified input length, up to constant factors. The length must be distinguished from the numerical value of an input. Polynomial scaling is an asymptotic property and does not by itself establish a practical advantage at a particular size. *Chapter 16*. *See also* Complexity class; Query complexity; Quantum advantage.

Post-quantum cryptography. Classical cryptographic methods designed to resist attacks by quantum as well as classical computers. They use ordinary classical communication and rely on the assumed difficulty of specified problems and correct implementation. Key establishment, signatures, authentication, and encryption remain distinct functions; no quantum transmission channel is required. *Chapter 19*. *See also* Authentication; Quantum key distribution.

Preparation uncertainty. A constraint on the outcome distributions available from a prepared state across alternative measurements on fresh copies. For a qubit, making Z certain forces each of the X and Y measurements to give equal probabilities for its two outcomes. This is distinct from uncertainty in an estimated probability and from a previous measurement changing the state. *Chapter 4*. See also Bloch sphere; Measurement; Quantum state tomography.

Privacy amplification. Turning partially secret material into a shorter key that is nearly uniform and nearly independent of an adversary's information. Suitable randomly selected hash functions can achieve this even with quantum side information when the required entropy bound holds. The independent selection seed may be public; hiding the shortening rule is not the source of secrecy. *Chapter 19*. *See also* Quantum key distribution; Finite-key analysis.

Probability amplitude. A generally complex number used to calculate a quantum outcome probability. For a specified discrete outcome, the probability is the squared magnitude of its total amplitude. Coherent contributions to the same complete outcome are added before that squaring; amplitudes for specified successive stages multiply. *Chapter 2*. *See also* Born rule; Complex number; Interference.

Probability density. A quantity that yields a probability when added over a specified interval or volume. For a normalized spatial wavefunction, the squared magnitude is the position probability density and its integral over all space is one. A density value at a point is not itself the probability of that exact point. *Chapter 6*. *See also* Wavefunction; Born rule.

Product formula. An approximation to evolution under a sum of Hamiltonian terms, constructed from ordered evolutions under individual terms. Shorter steps reduce the ordering error under stated bounds; symmetric sequences can cancel a leading error. Increasing the number of steps also increases the operations whose hardware errors and costs must be counted. *Chapter 21*. *See also* Digital simulation; Hamiltonian.

Product state. A joint state that factors into one state for each specified subsystem. Local measurement probabilities then factor into independent probabilities. A mixture of different product states can still be correlated, because the classical selection can coordinate which local states were prepared. *Chapter 9*. *See also* Tensor product; Separable state; Correlation; Entanglement.

Projection noise. Statistical fluctuations in recorded populations arising from the quantum probabilities of repeated measurements. For independent two-outcome trials, the recorded fraction fluctuates even with ideal preparation and detection. Translating that spread into uncertainty in a field or frequency also requires the calibrated response and its local slope. *Chapter 20*. *See also* Quantum sensing; Standard quantum limit.

Projective measurement. A measurement that separates mutually orthogonal alternatives in state space. For the ideal repeatable qubit measurements used here, each outcome leaves its corresponding basis state, so repeating the same measurement immediately gives the same answer. Intervening evolution or a different measurement can change that prediction. *Chapter 4*. *See also* Measurement; Basis; Orthogonal states.

Pure state. A quantum state representable by one normalized state vector, with vectors differing only by global phase representing the same state. Pure does not mean that every measurement has a certain result. A whole system can be pure while the reduced states of its entangled parts are mixed. *Chapter 9*. *See also* Global phase; Mixed state; Entanglement; Reduced state.

Purification. A pure state of a larger joint system whose reduced state on one part is the specified mixed state. Different purifications can have the same local predictions while differing in their correlations with the additional system. The construction does not mean that a partial trace physically selected a member of a classical lottery. *Chapter 24*. *See also* Mixed state; Partial trace; Reduced state.

Pusey–Barrett–Rudolph theorem. A theorem constraining models in which distinct pure quantum states correspond to overlapping probability distributions over underlying physical states. With preparation independence and the other stated model premises, those overlapping descriptions cannot reproduce all quantum predictions. The result does not prove that the wavefunction is a material wave in ordinary space or select one interpretation. *Chapter 24*. *See also* Pure state; Quantum state.

Q

QBism. An approach that interprets quantum probabilities through an agent's probability assignments and experiences, with the Born rule constraining those assignments. It differs from relational quantum mechanics and from treating quantum states as ordinary uncertainty over hidden physical conditions. It does not assert that wishing for a result determines a detector's outcome. *Chapter 24*. *See also* Born rule; Relational quantum mechanics.

Quantum advantage. An improvement by a specified quantum-assisted procedure over an appropriate classical alternative on the same task, with input access, output quality, success requirement, and resource measure stated. A proved scaling separation, a projected resource advantage, and an experimentally demonstrated improvement require different evidence and should be identified accordingly. *Chapter 23*. *See also* Query complexity; Resource estimate; Ablation.

Quantum circuit. A specified sequence of quantum operations and measurements on registers, with classical controls or communicated outcomes included where needed. Circuit wires identify systems and operation order, not necessarily spatial paths or equal durations. A gate diagram must still be connected to an implementable construction and resource account. *Chapter 11*. *See also* Register; Unitary transformation; Controlled-NOT gate; Algorithm.

Quantum compression. Encoding quantum signals into a smaller quantum memory and later recovering them while preserving the required quantum relationships. For an independently repeated pure-state source, the asymptotic rate is the von Neumann entropy of its average state. Matching only the average output density matrix, or compressing a classical measurement file, does not meet the recovery requirement. *Chapter 7*. *See also* von Neumann entropy; Fidelity; Density matrix.

Quantum Darwinism. A research program examining how repeated environmental records make selected properties accessible to multiple observers. It concerns which information persists and spreads under the dynamics, rather than biological reproduction or an environment choosing an outcome. Redundant readable records require more than a plateau in total quantum mutual information alone. *Chapter 12*. *See also* Pointer states; Environment; Quantum mutual information; Measurement problem.

Quantum eraser. A procedure that measures a distinguishing marker in a complementary way to reveal phase-related conditional groups or provide a result for active correction. Ignoring the marker does not restore an unknown state. Sorting already recorded outcomes reveals correlations without changing those past outcomes or supplying a recovered quantum system for later use. *Chapter 12*. *See also* Interference; Decoherence; No-signalling; Measurement.

Quantum error correction. Encoding and recovery that preserve an unknown quantum state, including its external correlations, against a specified class of errors. Checks reveal relationships needed for recovery without revealing the protected logical alternatives. The method does not clone the input and succeeds only while its code, noise assumptions, and implementation support the required correction. *Chapter 18*. *See also* Logical qubit; Syndrome; Fault tolerance; Error mitigation.

Quantum Fourier transform. A unitary transformation that combines input amplitudes with specified complex phase patterns indexed by output labels. It can make periodic structure appear in measurement probabilities without reading the input amplitude list. A compact transform circuit does not produce an exponentially long classical table of Fourier coefficients in one measurement. *Chapter 15*. *See also* Shor's algorithm; Phase estimation.

Quantum kernel. A similarity function between records defined through quantum-state preparations and measurements, often used by a classical learning procedure. Its value depends on the chosen encoding. A kernel obtained from a quantum circuit can still have an efficient exact classical expression, so quantum implementation alone does not establish a learning or cost advantage. *Chapter 23*. *See also* Quantum advantage; Ablation.

Quantum key distribution. A protocol for establishing shared classical secret-key material using quantum signals and authenticated classical discussion. Its security claim depends on the protocol, device model, observed evidence, finite-data analysis, and endpoint assumptions. It does not itself encrypt the later message or transfer an arbitrary unknown quantum state for storage. *Chapter 19*. *See also* BB84; Authentication; Privacy amplification; Post-quantum cryptography.

Quantum mutual information. A measure of total correlations in a joint quantum state: the sum of the two reduced-state entropies minus the joint-state entropy. It differs from mutual information between particular classical measurement records. Its value does not state how many classical bits one local readout can recover. *Chapter 9*. *See also* Mutual information; von Neumann entropy; Reduced state; Entanglement.

Quantum repeater. An intermediate system that supports quantum communication by preparing and joining shorter quantum links, often using memories and entanglement swapping. It does not universally copy and retransmit an unknown state. Storage quality, heralding, local failures, classical latency, and consumed entanglement determine the useful rate of a particular protocol. *Chapter 19*. *See also* Entanglement swapping; Heralding; Quantum teleportation.

Quantum sensing. Using a controlled quantum preparation and its response to an influence, such as a field or force, to estimate that influence through calibrated measurements. A single probe can perform quantum sensing without multiprobe entanglement. Claiming quantum enhancement additionally requires a comparison with a specified alternative at matched resources and conditions. *Chapter 20*. *See also* Ramsey interferometry; Projection noise; Spin squeezing.

Quantum simulation. Using one controlled quantum system to represent specified preparations, dynamics, and measurements of another model. An explicit encoding connects device records to the target question. Faithful execution of the model, adequacy of that model for the physical system, and computational advantage over alternatives are separate claims requiring their own evidence. *Chapter 21*. *See also* Analogue simulation; Digital simulation; Uncertainty budget.

Quantum state. A mathematical description used to predict outcomes of specified measurements on a quantum system. Pure states can be represented by normalized state vectors; density matrices also describe mixed states. The description need not reveal a unique preparation history, and one outcome does not disclose the entire state. *Chapter 4*. *See also* Pure state; Mixed state; Density matrix; Measurement.

Quantum state tomography. Reconstruction of a state description from measurements on many independently prepared copies, using enough measurement settings to determine the relevant state parameters. For one qubit, X, Y, and Z statistics suffice under the calibrated model. Finite data, source changes, and control errors must be included in the uncertainty. *Chapter 4*. *See also* Quantum state; Bloch sphere; Density matrix; Measurement.

Quantum teleportation. A protocol transferring a quantum state's role to another carrier using shared entanglement, local operations, and a communicated measurement result. In the ideal qubit protocol, one Bell pair is consumed and two classical bits select the receiver's correction. The original usable state is not retained, and communication is needed to recover the output. *Chapter 11*. *See also* Bell state; No-cloning theorem; No-signalling; Entanglement swapping.

Qubit. A quantum information system with a two-dimensional state space, such as two selected atomic levels or one photon restricted to two modes. A pure qubit state has two complex amplitudes in a chosen basis. Continuously many possible preparations do not make one qubit an unlimited readable classical message. *Chapter 4*. *See also* Bit; Basis; Bloch sphere; Orthogonal states.

Query complexity. The number of uses of a specified input-access operation needed to perform a task with a stated success guarantee. The interface and any restrictions on inspecting the input are part of the problem. Fewer queries need not mean less elapsed time, because constructing and executing each query can carry different costs. *Chapter 16*. *See also* Oracle; Quantum advantage; Complexity class.

R

Rabi oscillation. Coherent variation of the probabilities for occupying coupled quantum levels under an applied drive. In the ideal resonant two-level model, pulse duration controls population transfer and the preparation of superpositions. The Rabi frequency measures the coupling strength, rather than the undriven energy-level separation. *Chapter 5*. *See also* Hamiltonian; Qubit; Ramsey interferometry.

Ramsey interferometry. A two-pulse procedure in which the first pulse prepares coherent alternatives, a waiting interval accumulates relative phase, and the second pulse converts that phase into readout probabilities. It compares a system's transition frequency with a control reference. Longer waiting can increase sensitivity while also increasing noise and phase ambiguity. *Chapter 5*. *See also* Interferometer; Relative phase; Rabi oscillation.

Readout. The physical conversion of a chosen quantum measurement into an accessible record, together with the rule assigning that record an outcome. Photon counts or voltage traces need not directly print a state label. Assignment errors, missed events, calibration, and any state changes during acquisition affect what the record establishes. *Chapter 17*. *See also* Measurement; State-preparation-and-measurement errors.

Reduced state. The density matrix that predicts measurements restricted to one part of a joint system. It is obtained by taking the partial trace over the other parts. It need not be pure when the whole is pure, and separate reduced states do not determine the complete joint correlations. *Chapter 9*. *See also* Partial trace; Density matrix; Entanglement.

Register. A collection of storage positions used to hold an input, intermediate work, or an output. A quantum register consists of specified quantum systems whose joint state may be entangled. Drawing a wire or calling a register blank does not remove the costs of its physical carrier, preparation, and reuse. *Chapter 8*. *See also* Bit; Qubit; Quantum circuit; Entanglement.

Relational quantum mechanics. An approach treating states and events as relative to interactions between physical systems rather than as one absolute inventory. The observing system need not be conscious. An assessment must explain how such relative descriptions connect when systems later interact and compare records. *Chapter 24*. *See also* QBism; Measurement problem.

Relative phase. The difference in phase between amplitude components. Changing it can change the result of recombination while leaving the components' separate measurement probabilities unchanged. Its physical significance lies in comparisons between components; it is distinct from a common phase multiplying the complete state. *Chapter 3*. *See also* Global phase; Probability amplitude; Interference.

Resource estimate. A calculation of the resources a proposed implementation would require under stated algorithm, hardware, error, and workload assumptions. It can include physical qubits, gates, control time, auxiliary preparation, repeated attempts, and classical processing. Such an estimate is a conditional projection, not a performed computation or a forecast of availability. *Chapter 15*. *See also* Quantum advantage; Physical qubit; Logical qubit.

S

Sample-based quantum diagonalization. A hybrid electronic-structure method in which measured quantum occupation strings help select a reduced configuration space, whose Hamiltonian is then diagonalized classically. The samples do not reveal all wavefunction amplitudes. Its value must be assessed with the classical work and suitable classical configuration-selection alternatives included. *Chapter 22*. *See also* Active space; Slater determinant; Ablation.

Schrödinger equation. The evolution law connecting a quantum state's rate of change to the Hamiltonian. It determines how amplitudes evolve when the initial state and relevant interactions are specified. A mismatch between prediction and observation can involve the chosen model or its approximations, as well as the implemented controls. *Chapter 5*. *See also* Hamiltonian; Unitary transformation; Wavefunction.

Separable state. A joint state expressible as a probability-weighted mixture of product states across specified subsystems. The shared selection can produce ordinary correlations, so separable does not mean independent. A mixed state is entangled only if no such product-state mixture can describe it. *Chapter 9*. *See also* Product state; Entanglement; Correlation.

Shannon entropy. The probability-weighted average surprisal of a classical source: a measure of uncertainty about its possible outputs. With base-two logarithms it is measured in bits. It depends on the source model and any dependencies among outputs, rather than on a message's importance or the physical length of its record. *Chapter 7*. *See also* Surprisal; Bit; Conditional entropy; von Neumann entropy.

Shor's algorithm. A factoring algorithm that uses coherent modular arithmetic and Fourier interference to obtain records related to an arithmetic order, followed by classical processing and verification. Its resources grow polynomially with input bit length under the computational model. This does not prove that polynomial-time classical factoring is impossible or specify a ready-built machine. *Chapter 15*. *See also* Order (modular arithmetic); Quantum Fourier transform; Continued fraction.

Slater determinant. An antisymmetric many-electron basis state specified by a set of occupied spin orbitals. It implements the fermionic exchange rule; the occupation list does not assign a named electron to each orbital. Superpositions of determinants can describe electronic relationships unavailable to a single configuration. *Chapter 22*. *See also* Spin orbital; Full configuration interaction; Electron correlation.

Spin. An intrinsic quantum degree of freedom of a particle. It is not the rotation of a tiny material ball. For electrons, spin must be included with position when constructing the antisymmetric joint state; different spin values do not make identical electrons distinguishable species. *Chapter 6*. *See also* Pauli exclusion principle; Qubit; Entanglement.

Spin orbital. A spatial orbital together with a spin label, used as one of the available states in an electronic occupation description. Occupied spin orbitals define many-electron configurations subject to antisymmetry. The list describes joint alternatives and does not assign distinguishable identities to the electrons. *Chapter 22*. *See also* Orbital; Slater determinant; Pauli exclusion principle.

Spin squeezing. Redistributing collective-spin fluctuations to reduce noise in a chosen measurement while retaining enough response to a signal. Metrological squeezing compares the reduced variance with the remaining mean spin response. A smaller variance alone need not improve precision if the useful response or the complete experimental efficiency deteriorates. *Chapter 20*. *See also* Quantum sensing; Standard quantum limit.

Stabilizer. An operator that leaves the specified state or every state in an encoded subspace unchanged. Independent commuting stabilizer constraints can give a compact state description or define error checks that do not distinguish the protected logical alternatives. The constraints are properties of the encoding, not measurements of each constituent's individual state. *Chapter 16*. *See also* Clifford gate; Syndrome; Quantum error correction.

Standard quantum limit. A precision benchmark defined for a specified measurement and resource model. In the chapter's ideal comparison of equally exposed, independent probes, local standard uncertainty improves as the inverse square root of probe number. Different signal interactions, correlations, noise, and resource counts require a corresponding definition of the benchmark. *Chapter 20*. *See also* Heisenberg scaling; Projection noise; Spin squeezing.

State-preparation-and-measurement errors. Errors arising in the preparation of a test state and in the conversion of its final measurement into a recorded outcome; often abbreviated SPAM. A combined preparation-and-measurement score cannot be assigned entirely to a detector or a gate. Different characterization protocols separate these contributions only under their stated assumptions. *Chapter 17*. *See also* Readout; Fidelity.

Superposition. A state expressed as a combination of reference states with probability amplitudes. The reference alternatives must be specified: a state can be a superposition in one basis and a single basis state in another. A coherent superposition generally differs from a classical lottery over those alternatives. *Chapter 4*. *See also* Basis; Probability amplitude; Coherence; Mixed state.

Surface code. A quantum error-correcting code using local parity checks whose spatial and temporal patterns help a decoder infer recovery. Certain error chains implement logical operations without a remaining endpoint syndrome. Repeated noisy checks, the chosen decoder, code distance, and the physical error model determine the protection achieved. *Chapter 18*. *See also* Quantum error correction; Decoder; Code distance.

Surprisal. The information measure assigned to one positive-probability outcome: the negative base-two logarithm of its probability. Less probable outcomes have greater surprisal. Surprisals add for independent events; repeating a guaranteed copy of an already known result supplies no new surprisal. *Chapter 7*. *See also* Shannon entropy; Conditional probability; Bit.

Syndrome. The collection of error-check results identifying which encoded relationships disagree with their expected values. A suitable syndrome supplies recovery information without revealing the unknown logical state. It need not identify a unique physical error history, and an all-zero syndrome does not prove that no logical error occurred. *Chapter 18*. *See also* Stabilizer; Decoder; Quantum error correction.

T

Tensor network. A representation formed from linked arrays that can encode a quantum state or calculation without explicitly storing every amplitude. Its efficiency depends on the state's structure, the geometry, the measurements requested, and the approximation accuracy. Successful tensor-network simulation of selected problems is not a universal simulation guarantee. *Chapter 13*. *See also* Classical simulation; Quantum advantage.

Tensor product. The mathematical construction combining the state spaces of specified subsystems into a joint state space. For independently prepared pure states, joint amplitudes are products of the individual amplitudes. The tensor product specifies how descriptions combine; it is not itself a physical interaction. *Chapter 9*. *See also* Product state; Entanglement; Register.

T gate. A one-qubit gate that leaves the zero component unchanged and advances the phase of the one component by one-eighth of a cycle. It lies outside the Clifford gate set. Its presence can defeat the simple stabilizer update rule, but does not alone establish classical difficulty or useful quantum computation. *Chapter 16*. *See also* Clifford gate; Classical simulation.

Toffoli gate. A reversible three-bit operation that flips a target exactly when both control bits are one, retaining the controls. With an initially zero target, it records their AND result. The coherent quantum version extends the same basis-state rule linearly to superpositions; applying the ideal gate twice restores the input. *Chapter 8*. *See also* Logic gate; Controlled-NOT gate; Logical reversibility.

Trace distance. A measure of distinguishability between quantum states: it gives the largest possible difference in the probability of a single measurement event. Zero distance means the same physical state. A small value bounds errors for every measurement on those states, while agreement in one chosen measurement can conceal a larger state difference. *Chapter 21*. *See also* Fidelity; Quantum state; Measurement.

Transition state. The transition region used to describe passage from reactants toward products; in the conventional energy-surface treatment, its associated transition structure is a saddle with one unstable direction. The connecting path must be checked. A barrier along nuclear coordinates does not require the molecule to occupy an electronically excited state. *Chapter 22*. *See also* Activation free energy; Transmission factor.

Transmission factor. A factor in a specified transition-state rate model that accounts for departures from its simplest crossing prescription, such as recrossing or an included tunnelling correction. Its value and meaning depend on the dynamical model. Setting it to one is an explicit assumption, not a general consequence of knowing the barrier. *Chapter 22*. *See also* Activation free energy; Transition state.

U

Uncertainty budget. An account of the contributions limiting a stated prediction, such as sampling fluctuations, numerical approximation, calibration, and physical-model discrepancy. Their combination depends on whether they are justified bounds or statistically characterized errors, including correlations. More samples reduce sampling uncertainty under its assumptions; they do not automatically bound omitted mechanisms or remove bias. *Chapter 23*. *See also* Error cancellation; Chemical accuracy; Quantum advantage.

Uncomputation. Reversing working operations after preserving the desired result, so temporary registers return to their prepared states. The records needed to reverse those operations remain available. It clears work without blindly resetting every possible unknown value, but requires additional operations and an explicit memory account. *Chapter 8*. *See also* Logical reversibility; Register; Unitary transformation; Landauer's bound.

Unitary transformation. A reversible quantum transformation that preserves state normalization and inner products. Its inverse can restore an input without first learning the amplitudes, provided the required control is available. A unitary acting on a system alone cannot reset every possible input to one fixed state. *Chapter 5*. *See also* Inner product; Logical reversibility; Hamiltonian; Uncomputation.

V

Variational principle. The statement that a normalized trial state's average energy cannot fall below the true ground-state energy of the same Hamiltonian. Expanding or improving the trial space can lower that estimate. The bound does not automatically extend to predicted bond lengths, energy differences, or physical effects absent from the model. *Chapter 6*. *See also* Hamiltonian; Orbital; Born–Oppenheimer approximation.

Variational quantum eigensolver. A hybrid method that prepares parameterized quantum trial states, estimates their energy expectations, and uses classical optimization to change the parameters. Its accessible state family can restrict the best energy found. Finite-sample estimates need not obey the exact variational bound, and selecting a favorable noisy estimate can bias the reported result. *Chapter 22*. *See also* Phase estimation; Uncertainty budget; Active space.

von Neumann entropy. The Shannon entropy of a density matrix's eigenvalues, giving a basis-independent measure for the specified quantum state. A pure state has zero entropy even when particular measurements are random. The state entropy is distinct from the entropy of one measurement's outcomes and from the difficulty of calculating the state. *Chapter 7*. *See also* Density matrix; Shannon entropy; Pure state; Quantum compression.

W

Wavefunction. An amplitude representation of a pure quantum state. A spatial wavefunction assigns a complex amplitude density over position; its squared magnitude is a probability density. It describes predictions for measurements, rather than a drawn trajectory of a particle. *Chapter 6*. *See also* Probability amplitude; Probability density; Orbital; Pure state.

Z

Zero-point energy. The energy remaining above a potential minimum in the lowest allowed quantum state of a specified motion. For molecular vibration, it places the lowest vibrational level above the bottom of the electronic potential well, so dissociation from that level requires less energy than the full well depth. *Chapter 6*. *See also* Energy eigenstate; Born–Oppenheimer approximation; Wavefunction; Dissociation energy.

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